

hawkes_2pt_pipeline_demo

March 20, 2026

1 Hawkes 2-Point Function — Full Pipeline Demo

Model: Nonlinear Hawkes process, 2 populations, delta-function synaptic filters, quadratic non-linearity

Observable: 2-point correlation function $\langle \delta \dot{n}_1(t_1) \delta \dot{n}_2(t_2) \rangle$

Loop levels: Tree-level ($\ell = 0$) and 1-loop ($\ell = 1$)

1.1 What this notebook does

This notebook walks through the full automated Feynman diagram pipeline (Phases A–G) for a concrete MSRJD field theory.

1.1.1 Section 1 — Theory / Model Information

Loads the 2-population nonlinear Hawkes model and prints all of its defining data: response fields ($\tilde{n}_i$), physical fluctuation fields ($\delta \dot{n}_i$), parameters, synaptic kernels, operators, and the nonlinear transfer function ϕ . Then expands the MSRJD action to the required Taylor order and extracts the free-action kernel matrix $\mathbf{K}$, its Fourier transform $\hat{\mathbf{K}}(\omega)$, the retarded propagator $\hat{\mathbf{G}}(\omega) = \hat{\mathbf{K}}^{-1}(\omega)$, its poles, residue matrices, and the time-domain propagator $\mathbf{G}(t)$.

1.1.2 Section 2 — Prediagram Enumeration

For the 2-point function ($k = 2$), enumerates all labeled trees, undirected topologies, and directed prediagrams at tree level ($\ell = 0$) and 1-loop ($\ell = 1$). Plots each topology and prediagram with color-coded vertices: black = external legs, red = source vertices (noise insertions, in-degree 0), light blue = internal interaction vertices.

1.1.3 Section 3 — Vertex Extraction

Reads off all interaction vertex types (**VertexType**) and noise source types (**SourceType**) from the expanded action. Each vertex type has a bigrade (n, n_{phys}) , a list of response and physical legs, and a coefficient. Checks that the current Taylor order is sufficient to cover the maximum vertex degree appearing in the prediagrams.

1.1.4 Section 4 — Prediagram Filtering

Discards any prediagram whose internal vertex degree signatures don't match any available vertex or source type. Reports how many prediagrams survive at each loop level and plots the survivors.

1.1.5 Section 5 — Type Assignment (Fully Labeled Diagrams)

Sets up the external fields for the 2-point function ($\delta\dot{n}_1, \delta\dot{n}_2$) and builds the field-to-matrix-index maps. Then runs the constraint-satisfaction engine on each surviving prediagram: for every valid assignment of vertex types to internal vertices and field types to edges (consistent with the propagator matrix $\hat{\mathbf{G}}$), it produces a `TypedDiagram`. Each typed diagram is printed showing its external leg assignments, vertex types with coefficients, and edge propagator labels $\hat{G}_{ij}$.

1.1.6 Section 6 — Diagram Isomorphism Check

Groups the enumerated typed diagrams into equivalence classes by canonical signature. Two diagrams are isomorphic if they have identical external leg assignments, vertex types, and propagator indices on every edge. Reports the multiplicity of each unique diagram — this multiplicity is the symmetry factor that cancels the $1/n!$ from the Taylor expansion coefficients.

```
[1]: %display latex
import sys, os
sys.path.insert(0, os.path.join(os.path.dirname(os.path.abspath('__file__')), '..'))

from IPython.display import display, Math
from sage.all import (
    SR, matrix, latex, I, pi, exp, diff, solve, integrate, oo,
    dirac_delta, function, var
)

from msrjd.core.field_theory import FieldTheory, fourier_transform,
    inverse_fourier_transform
from msrjd.core.vertices import extract_vertex_types, extract_source_types,
    available_degrees
from msrjd.enumeration.loop_diagram_enumeration import enumerate_all
from msrjd.diagrams.filter import filter_prediagrams,
    classify_prediagram_vertices
from msrjd.diagrams.type_assignment import (
    enumerate_typed_diagrams, enumerate_all as enumerate_all_typed,
    build_field_index_map, TypedDiagram
)
from msrjd.diagrams.causality import filter_causal
from msrjd.diagrams.symmetry import symmetry_factor,
    compute_all_symmetry_factors

from models.hawkes_sage import HAWKES_MODEL
```

```
print('Imports loaded.')
```

Imports loaded.

1.2 1. Theory / Model Information

```
[2]: m = HAWKES_MODEL
print(f"Model: {m['name']}")
print(f"Convention: {m['background_rate_convention']}")
print()

print('  Index sets  ')
for name, vals in m['index_sets'].items():
    print(f'    {name}: {list(vals)}')
print()

print('  Response fields (not expanded - full integration variables)  ')
for f in m['response_fields']:
    print(f"    {f['name']} (latex: ${f.get('latex', f['name'])}$) - {f.
    ↪get('description', '')}")
print()

print('  Physical fields (expanded around MF background)  ')
for f in m['physical_fields']:
    print(f"    {f['name']} (latex: ${f.get('latex', f['name'])}$) - {f.
    ↪get('description', '')}")
print()

print('  Parameters  ')
for p in m.get('parameters', []):
    idx = '(indexed)' if p.get('indexed') else '(scalar)'
    dom = f", domain={p['domain']}" if p.get('domain') else ''
    print(f"    {p['name']} {idx}{dom} - {p.get('description', '')}")
print()

print('  Kernels  ')
for k in m.get('kernels', []):
    print(f"    {k['name']} - {k.get('description', '')}")
print()

print('  Operators  ')
for o in m.get('operators', []):
    print(f"    {o['name']} (latex: ${o.get('latex_name', o['name'])}$) - {o.
    ↪get('description', '')}")
print()
```

```

print('  Nonlinear functions  ')
for f in m.get('functions', []):
    print(f"    {f['name']} (latex: ${f.get('latex', f['name'])})$) - {f.
    get('description', '')}")
print()

print('  Specializations  ')
print('    quadratic (cubic and higher coefficients = 0)')
print('    g = (t) (instantaneous synaptic coupling)')

```

Model: Nonlinear Hawkes 2-population

Convention: $\dot{n}_i = -_i(v_i)$ [positive Poisson term: $+ \tilde{n}_i \dot{n}_i + (e^{\tilde{n}_i} - 1)$
 $_i]$

Index sets

pop: [0, 1]

Response fields (not expanded - full integration variables)

nt (latex: $\tilde{n}$) - response field conjugate to spike train

vt (latex: $\tilde{v}$) - response field conjugate to voltage

Physical fields (expanded around MF background)

dn (latex: $\delta \dot{n}$) - spike-train fluctuation around MF background

dv (latex: δv) - voltage fluctuation around MF background

Parameters

nstar (indexed), domain=positive - background firing rate $nstar_i =$
 $\phi_i(vstar_i)$

vstar (indexed) - background voltage

E (indexed) - external drive

tau (scalar), domain=positive - membrane time constant

Kernels

g - synaptic filter kernel $g(t)$

Operators

Dt (latex: ∂_t) - $_t$ (algebraic placeholder for the time-
derivative operator)

Nonlinear functions

phi (latex: φ) - nonlinear gain function $_i(v_i)$

Specializations

quadratic (cubic and higher coefficients = 0)

g = (t) (instantaneous synaptic coupling)

1.2.1 1.1 Expand the action and show all sectors

```
[3]: ft = FieldTheory(HAWKES_MODEL, taylor_order=4)
ft.expand()

R = ft.ring()
ns = ft._ns

print(f'Taylor order: {ft.taylor_order}')
print(f'Polynomial ring: {R}')
print(f'Ring generators: {R.gens()}')
print(f'Response generators (n_tilde={ft._n_tilde}): {R.gens()[ft._n_tilde:]}')
print(f'Physical generators: {R.gens()[ft._n_tilde:]}')
print()

passed = ft.sanity_check()
print()
ft.summary()
```

Taylor order: 4

Polynomial ring: Multivariate Polynomial Ring in nt1, nt2, vt1, vt2, dn1, dn2, dv1, dv2 over Symbolic Ring

Ring generators: (nt1, nt2, vt1, vt2, dn1, dn2, dv1, dv2)

Response generators (n_tilde=4): (nt1, nt2, vt1, vt2)

Physical generators: (dn1, dn2, dv1, dv2)

=== Sanity checks ===

[PASS] (n_tilde=0, n_phys=0) constant term

[PASS] (n_tilde=1, n_phys=0) tadpole - must vanish at MF saddle

[PASS] (n_tilde=0, n_phys=1) linear physical-only - must vanish at EOM

=== Action sectors ===

(n_tilde=1, n_phys=1) [free action]:

$$nt_1 dn_1 + (-w_{11}\delta) vt_1 dn_1 + (-w_{21}\delta) vt_2 dn_1 + nt_2 dn_2 + (-w_{12}\delta) vt_1 dn_2 + (-w_{22}\delta) vt_2 dn_2 + \varphi'_1 nt_1 dv_1 + vt_1 dv_1 + \varphi'_2 nt_2 dv_2 + vt_2 dv_2$$

(n_tilde=1, n_phys=2) [vertex (order 3)]:

$$\left(\frac{1}{2} \varphi''_1\right) nt_1 dv_1^2 + \left(\frac{1}{2} \varphi''_2\right) nt_2 dv_2^2$$

(n_tilde=2, n_phys=0) [noise kernel]:

$$nt_1^2 + nt_2^2$$

(n_tilde=2, n_phys=1) [vertex (order 3)]:

$$\left(\frac{1}{2} \varphi'_1\right) nt_1^2 dv_1 + \left(\frac{1}{2} \varphi'_2\right) nt_2^2 dv_2$$

(n_tilde=2, n_phys=2) [vertex (order 4)]:

$$\left(\frac{1}{4}\varphi_1''\right)nt_1^2dv_1^2 + \left(\frac{1}{4}\varphi_2''\right)nt_2^2dv_2^2$$

(n_tilde=3, n_phys=0) [noise kernel]:

$$nt_1^3 + nt_2^3$$

(n_tilde=3, n_phys=1) [vertex (order 4)]:

$$\left(\frac{1}{6}\varphi_1'\right)nt_1^3dv_1 + \left(\frac{1}{6}\varphi_2'\right)nt_2^3dv_2$$

(n_tilde=3, n_phys=2) [vertex (order 5)]:

$$\left(\frac{1}{12}\varphi_1''\right)nt_1^3dv_1^2 + \left(\frac{1}{12}\varphi_2''\right)nt_2^3dv_2^2$$

(n_tilde=4, n_phys=0) [noise kernel]:

$$nt_1^4 + nt_2^4$$

(n_tilde=4, n_phys=1) [vertex (order 5)]:

$$\left(\frac{1}{24}\varphi_1'\right)nt_1^4dv_1 + \left(\frac{1}{24}\varphi_2'\right)nt_2^4dv_2$$

(n_tilde=4, n_phys=2) [vertex (order 6)]:

$$\left(\frac{1}{48}\varphi_1''\right)nt_1^4dv_1^2 + \left(\frac{1}{48}\varphi_2''\right)nt_2^4dv_2^2$$

1.2.2 1.2 Propagator

Extract the kernel matrix $\mathbf{K}$ from the free action, Fourier transform, and invert.

```
[4]: import signal

S_free = ft.free_action()
ring_gen_names = [str(g) for g in R.gens()]

resp_names = [f'vt{i+1}' for i in ns.pop] + [f'nt{i+1}' for i in ns.pop]
phys_names = [f'dv{i+1}' for i in ns.pop] + [f'dn{i+1}' for i in ns.pop]

pos_to_row = {ring_gen_names.index(nm): i for i, nm in enumerate(resp_names)}
pos_to_col = {ring_gen_names.index(nm): j for j, nm in enumerate(phys_names)}

nf = len(resp_names)
K_data = [[SR(0)] * nf for _ in range(nf)]
for exp_vec, coeff in S_free.dict().items():
    row = col = None
    for k in range(len(ring_gen_names)):
        if exp_vec[k] > 0:
            if k in pos_to_row: row = pos_to_row[k]
            if k in pos_to_col: col = pos_to_col[k]
    if row is not None and col is not None:
```

```

        K_data[row][col] += SR(coeff)

K_mat = matrix(SR, K_data)

# Convert to kernel form
Dt      = ns.Dt
delta_D = ns.delta_D
delta_Dp = ns.delta_Dp

def _to_kernel(c):
    c = SR(c)
    if c.has(delta_D) or c.has(ns.g):
        return c
    p0 = c.subs({Dt: 0})
    rest = (c - p0).subs({Dt: delta_Dp})
    return p0 * delta_D + rest

K_ker = matrix(SR, [[_to_kernel(K_mat[i, j]) for j in range(nf)] for i in
    ↪range(nf)])

resp_sr = [ns.vt[i] for i in ns.pop] + [ns.nt[i] for i in ns.pop]
phys_sr = [ns.dv[i] for i in ns.pop] + [ns.dn[i] for i in ns.pop]

print('Field ordering:')
display(Math(r'\tilde{\mathbf{a}} = ' + latex(vector(resp_sr))
    + r', \quad \mathbf{a} = ' + latex(vector(phys_sr))))
print()
print('Kernel matrix K (time domain):')
display(Math(r'\mathbf{K} = ' + latex(K_ker)))

# Fourier transform
t_var = SR.var('t')
omega = SR.var('omega', latex_name=r'\omega')

time_domain = {
    delta_D: dirac_delta(t_var),
    delta_Dp: diff(dirac_delta(t_var), t_var),
    ns.g: function('g')(t_var),
}

K_ft_data = [[SR(0)] * nf for _ in range(nf)]
for i in range(nf):
    for j in range(nf):
        c = K_ker[i, j]
        if not c.is_zero():
            K_ft_data[i][j] = fourier_transform(SR(c).subs(time_domain), t_var,
    ↪omega)

```

```

K_ft = matrix(SR, K_ft_data)
print('Fourier-domain kernel:')
display(Math(r'\hat{\mathbf{K}}(\omega) = ' + latex(K_ft)))

# Propagator inverse
G_ft = K_ft.inverse().apply_map(lambda e: e.factor())
G_ft_explicit = True
print('Propagator:')
display(Math(r'\hat{\mathbf{G}}(\omega) = ' + latex(G_ft)))

# Adjugate, determinant, poles
adj_ft = K_ft.adjugate()
D_omega = K_ft.det().expand()
D_prime = diff(D_omega, omega)

pole_eqs = solve(D_omega == 0, omega)
pole_vals = [eq.rhs() if hasattr(eq, 'rhs') else eq for eq in pole_eqs]

print(f'\ndet(K) = {D_omega}')
print(f'Poles ({len(pole_vals)}):')
for k, p in enumerate(pole_vals):
    display(Math(r'\omega_{' + str(k+1) + '} = ' + latex(p)))

# Residue matrices
C_mats = []
for omega_k in pole_vals:
    C_data = [[SR(0)] * nf for _ in range(nf)]
    for i in range(nf):
        for j in range(nf):
            n_ij = adj_ft[i, j]
            if not n_ij.is_zero():
                num = n_ij.subs({omega: omega_k})
                den = D_prime.subs({omega: omega_k})
                C_data[i][j] = (I * num / den).factor()
    C_mats.append(matrix(SR, C_data))

print('Residue matrices:')
for k, C in enumerate(C_mats):
    display(Math(r'\mathbf{C}_{' + str(k+1) + r'} = ' + latex(C)))

# Time-domain propagator
G_t = sum(C_mats[k] * exp(I * pole_vals[k] * t_var) for k in
    range(len(pole_vals)))
G_t = G_t.apply_map(lambda e: e.simplify_full())
print('Time-domain propagator G(t) for t > 0:')
display(Math(r'\mathbf{G}(t) = ' + latex(G_t)))

```


Field ordering:

$$\tilde{\mathbf{a}} = (\tilde{v}_1, \tilde{v}_2, \tilde{n}_1, \tilde{n}_2), \quad \mathbf{a} = (\delta v_1, \delta v_2, \delta \dot{n}_1, \delta \dot{n}_2)$$

Kernel matrix K (time domain):

$$\mathbf{K} = \begin{pmatrix} \tau\delta' + \delta & 0 & -w_{11}\delta & -w_{12}\delta \\ 0 & \tau\delta' + \delta & -w_{21}\delta & -w_{22}\delta \\ \varphi'_1\delta & 0 & \delta & 0 \\ 0 & \varphi'_2\delta & 0 & \delta \end{pmatrix}$$

Fourier-domain kernel:

$$\hat{\mathbf{K}}(\omega) = \begin{pmatrix} i\omega\tau + 1 & 0 & -w_{11} & -w_{12} \\ 0 & i\omega\tau + 1 & -w_{21} & -w_{22} \\ \varphi'_1 & 0 & 1 & 0 \\ 0 & \varphi'_2 & 0 & 1 \end{pmatrix}$$

Propagator:

$$\hat{\mathbf{G}}(\omega) = \begin{pmatrix} \frac{i\omega\tau + \varphi'_2 w_{22} + 1}{\omega^2 \tau^2 - i\omega\varphi'_1 \tau w_{11} + \varphi'_1 \varphi'_2 w_{12} w_{21} - i\omega\varphi'_2 \tau w_{22} - \varphi'_1 \varphi'_2 w_{11} w_{22} - 2i\omega\tau - \varphi'_1 w_{11} - \varphi'_2 w_{22} - 1} & \frac{\omega^2 \tau^2 - i\omega\varphi'_1 \tau w_{11} + \varphi'_1 \varphi'_2 w_{12} w_{21} - i\omega\varphi'_2 \tau w_{22} - \varphi'_1 \varphi'_2 w_{11} w_{22} - 2i\omega\tau - \varphi'_1 w_{11} - \varphi'_2 w_{22} - 1}{i\omega\tau} \\ \frac{\omega^2 \tau^2 - i\omega\varphi'_1 \tau w_{11} + \varphi'_1 \varphi'_2 w_{12} w_{21} - i\omega\varphi'_2 \tau w_{22} - \varphi'_1 \varphi'_2 w_{11} w_{22} - 2i\omega\tau - \varphi'_1 w_{11} - \varphi'_2 w_{22} - 1}{(-i\omega\tau - \varphi'_2 w_{22} - 1)\varphi'_1} & -\frac{\omega^2 \tau^2 - i\omega\varphi'_1 \tau w_{11} + \varphi'_1 \varphi'_2 w_{12} w_{21} - i\omega\varphi'_2 \tau w_{22} - \varphi'_1 \varphi'_2 w_{11} w_{22} - 2i\omega\tau - \varphi'_1 w_{11} - \varphi'_2 w_{22} - 1}{\varphi'_1} \\ -\frac{\omega^2 \tau^2 - i\omega\varphi'_1 \tau w_{11} + \varphi'_1 \varphi'_2 w_{12} w_{21} - i\omega\varphi'_2 \tau w_{22} - \varphi'_1 \varphi'_2 w_{11} w_{22} - 2i\omega\tau - \varphi'_1 w_{11} - \varphi'_2 w_{22} - 1}{\varphi'_1 \varphi'_2 w_{21}} & -\frac{\omega^2 \tau^2 - i\omega\varphi'_1 \tau w_{11} + \varphi'_1 \varphi'_2 w_{12} w_{21} - i\omega\varphi'_2 \tau w_{22} - \varphi'_1 \varphi'_2 w_{11} w_{22} - 2i\omega\tau - \varphi'_1 w_{11} - \varphi'_2 w_{22} - 1}{(i\omega\tau + \varphi'_2 w_{22} + 1)} \\ -\frac{\omega^2 \tau^2 - i\omega\varphi'_1 \tau w_{11} + \varphi'_1 \varphi'_2 w_{12} w_{21} - i\omega\varphi'_2 \tau w_{22} - \varphi'_1 \varphi'_2 w_{11} w_{22} - 2i\omega\tau - \varphi'_1 w_{11} - \varphi'_2 w_{22} - 1}{\varphi'_1 \varphi'_2 w_{21}} & \frac{\omega^2 \tau^2 - i\omega\varphi'_1 \tau w_{11} + \varphi'_1 \varphi'_2 w_{12} w_{21} - i\omega\varphi'_2 \tau w_{22} - \varphi'_1 \varphi'_2 w_{11} w_{22} - 2i\omega\tau - \varphi'_1 w_{11} - \varphi'_2 w_{22} - 1}{(i\omega\tau + \varphi'_2 w_{22} + 1)} \end{pmatrix}$$

$$\det(\mathbf{K}) = -\omega^2 \tau^2 + I\omega\phi_{11}\tau w_{11} - \phi_{11}\phi_{22}w_{12}w_{21} + I\omega\phi_{12}\tau w_{22} + \phi_{11}\phi_{22}w_{11}w_{22} + 2I\omega\tau + \phi_{11}w_{11} + \phi_{12}w_{22} + 1$$

Poles (2):

$$\omega_1 = -\frac{-i\varphi'_1 w_{11} - i\varphi'_2 w_{22} + \sqrt{-\varphi'^2_{11} w_{11}^2 - 4\varphi'_1 \varphi'_2 w_{12} w_{21} + 2\varphi'_1 \varphi'_2 w_{11} w_{22} - \varphi'^2_{22} w_{22}^2} - 2i}{2\tau}$$

$$\omega_2 = \frac{i\varphi'_1 w_{11} + i\varphi'_2 w_{22} + \sqrt{-\varphi'^2_{11} w_{11}^2 - 4\varphi'_1 \varphi'_2 w_{12} w_{21} + 2\varphi'_1 \varphi'_2 w_{11} w_{22} - \varphi'^2_{22} w_{22}^2} + 2i}{2\tau}$$

Residue matrices:

$$\mathbf{C}_1 = \begin{pmatrix} \frac{-i\varphi'_1 w_{11} + i\varphi'_2 w_{22} + \sqrt{-\varphi'^2_{11} w_{11}^2 - 4\varphi'_1 \varphi'_2 w_{12} w_{21} + 2\varphi'_1 \varphi'_2 w_{11} w_{22} - \varphi'^2_{22} w_{22}^2}}{2\sqrt{-\varphi'^2_{11} w_{11}^2 - 4\varphi'_1 \varphi'_2 w_{12} w_{21} + 2\varphi'_1 \varphi'_2 w_{11} w_{22} - \varphi'^2_{22} w_{22}^2} \tau} & -\frac{i\varphi'_2 w_{22}}{\sqrt{-\varphi'^2_{11} w_{11}^2 - 4\varphi'_1 \varphi'_2 w_{12} w_{21} + 2\varphi'_1 \varphi'_2 w_{11} w_{22} - \varphi'^2_{22} w_{22}^2} \tau} \\ -\frac{i\varphi'_1 w_{21}}{\sqrt{-\varphi'^2_{11} w_{11}^2 - 4\varphi'_1 \varphi'_2 w_{12} w_{21} + 2\varphi'_1 \varphi'_2 w_{11} w_{22} - \varphi'^2_{22} w_{22}^2} \tau} & \frac{i\varphi'_1 w_{11} - i\varphi'_2 w_{22} + \sqrt{-\varphi'^2_{11} w_{11}^2 - 4\varphi'_1 \varphi'_2 w_{12} w_{21} + 2\varphi'_1 \varphi'_2 w_{11} w_{22} - \varphi'^2_{22} w_{22}^2}}{2\sqrt{-\varphi'^2_{11} w_{11}^2 - 4\varphi'_1 \varphi'_2 w_{12} w_{21} + 2\varphi'_1 \varphi'_2 w_{11} w_{22} - \varphi'^2_{22} w_{22}^2} \tau} \\ i\left(\frac{\varphi'_1 w_{11} - \varphi'_2 w_{22} + i\sqrt{-\varphi'^2_{11} w_{11}^2 - 4\varphi'_1 \varphi'_2 w_{12} w_{21} + 2\varphi'_1 \varphi'_2 w_{11} w_{22} - \varphi'^2_{22} w_{22}^2}}{2\sqrt{-\varphi'^2_{11} w_{11}^2 - 4\varphi'_1 \varphi'_2 w_{12} w_{21} + 2\varphi'_1 \varphi'_2 w_{11} w_{22} - \varphi'^2_{22} w_{22}^2} \tau}\right)\varphi'_1 & \frac{i\varphi'_1 \varphi'_2}{\sqrt{-\varphi'^2_{11} w_{11}^2 - 4\varphi'_1 \varphi'_2 w_{12} w_{21} + 2\varphi'_1 \varphi'_2 w_{11} w_{22} - \varphi'^2_{22} w_{22}^2} \tau} \\ \frac{i\varphi'_1 \varphi'_2 w_{21}}{\sqrt{-\varphi'^2_{11} w_{11}^2 - 4\varphi'_1 \varphi'_2 w_{12} w_{21} + 2\varphi'_1 \varphi'_2 w_{11} w_{22} - \varphi'^2_{22} w_{22}^2} \tau} & -\frac{i\left(\varphi'_1 w_{11} - \varphi'_2 w_{22} - i\sqrt{-\varphi'^2_{11} w_{11}^2 - 4\varphi'_1 \varphi'_2 w_{12} w_{21} + 2\varphi'_1 \varphi'_2 w_{11} w_{22} - \varphi'^2_{22} w_{22}^2}\right)}{2\sqrt{-\varphi'^2_{11} w_{11}^2 - 4\varphi'_1 \varphi'_2 w_{12} w_{21} + 2\varphi'_1 \varphi'_2 w_{11} w_{22} - \varphi'^2_{22} w_{22}^2} \tau} \end{pmatrix}$$

$$\mathbf{C}_2 = \begin{pmatrix} \frac{i\varphi'_1 w_{11} - i\varphi'_2 w_{22} + \sqrt{-\varphi_1'^2 w_{11}^2 - 4\varphi'_1 \varphi'_2 w_{12} w_{21} + 2\varphi'_1 \varphi'_2 w_{11} w_{22} - \varphi_2'^2 w_{22}^2}}{2\sqrt{-\varphi_1'^2 w_{11}^2 - 4\varphi'_1 \varphi'_2 w_{12} w_{21} + 2\varphi'_1 \varphi'_2 w_{11} w_{22} - \varphi_2'^2 w_{22}^2}} & \frac{i\varphi'_1 w_{21}}{\sqrt{-\varphi_1'^2 w_{11}^2 - 4\varphi'_1 \varphi'_2 w_{12} w_{21} + 2\varphi'_1 \varphi'_2 w_{11} w_{22} - \varphi_2'^2 w_{22}^2}} \\ -\frac{i\left(\varphi'_1 w_{11} - \varphi'_2 w_{22} - i\sqrt{-\varphi_1'^2 w_{11}^2 - 4\varphi'_1 \varphi'_2 w_{12} w_{21} + 2\varphi'_1 \varphi'_2 w_{11} w_{22} - \varphi_2'^2 w_{22}^2}\right)\varphi'_1}{2\sqrt{-\varphi_1'^2 w_{11}^2 - 4\varphi'_1 \varphi'_2 w_{12} w_{21} + 2\varphi'_1 \varphi'_2 w_{11} w_{22} - \varphi_2'^2 w_{22}^2}} & -\frac{i\varphi'_1 \varphi'_2}{\sqrt{-\varphi_1'^2 w_{11}^2 - 4\varphi'_1 \varphi'_2 w_{12} w_{21} + 2\varphi'_1 \varphi'_2 w_{11} w_{22} - \varphi_2'^2 w_{22}^2}} \end{pmatrix}$$

Time-domain propagator $G(t)$ for $t > 0$:

$$\mathbf{G}(t) = \begin{pmatrix} \frac{\left(\varphi_1'^2 w_{11}^2 + 4\varphi'_1 \varphi'_2 w_{12} w_{21} - 2\varphi'_1 \varphi'_2 w_{11} w_{22} + \varphi_2'^2 w_{22}^2 + \left(\varphi_1'^2 w_{11}^2 + 4\varphi'_1 \varphi'_2 w_{12} w_{21} - 2\varphi'_1 \varphi'_2 w_{11} w_{22} + \varphi_2'^2 w_{22}^2 \right) e^{\left(\frac{i\sqrt{-\varphi_1'^2 w_{11}^2 - 4\varphi'_1 \varphi'_2 w_{12} w_{21} + 2\varphi'_1 \varphi'_2 w_{11} w_{22} - \varphi_2'^2 w_{22}^2}}{2\sqrt{-\varphi_1'^2 w_{11}^2 - 4\varphi'_1 \varphi'_2 w_{12} w_{21} + 2\varphi'_1 \varphi'_2 w_{11} w_{22} - \varphi_2'^2 w_{22}^2}} \right)} \right)}{2\sqrt{-\varphi_1'^2 w_{11}^2 - 4\varphi'_1 \varphi'_2 w_{12} w_{21} + 2\varphi'_1 \varphi'_2 w_{11} w_{22} - \varphi_2'^2 w_{22}^2}} & \frac{i\varphi'_1 \varphi'_2}{\sqrt{-\varphi_1'^2 w_{11}^2 - 4\varphi'_1 \varphi'_2 w_{12} w_{21} + 2\varphi'_1 \varphi'_2 w_{11} w_{22} - \varphi_2'^2 w_{22}^2}} \\ -\frac{\left(\varphi_1'^3 w_{11}^2 + 4\varphi_1'^2 \varphi'_2 w_{12} w_{21} - 2\varphi_1'^2 \varphi'_2 w_{11} w_{22} + \varphi_1' \varphi_2'^2 w_{22}^2 + \left(\varphi_1'^3 w_{11}^2 + 4\varphi_1'^2 \varphi'_2 w_{12} w_{21} - 2\varphi_1'^2 \varphi'_2 w_{11} w_{22} + \varphi_1' \varphi_2'^2 w_{22}^2 \right) e^{\left(\frac{i\sqrt{-\varphi_1'^2 w_{11}^2 - 4\varphi'_1 \varphi'_2 w_{12} w_{21} + 2\varphi'_1 \varphi'_2 w_{11} w_{22} - \varphi_2'^2 w_{22}^2}}{2\sqrt{-\varphi_1'^2 w_{11}^2 - 4\varphi'_1 \varphi'_2 w_{12} w_{21} + 2\varphi'_1 \varphi'_2 w_{11} w_{22} - \varphi_2'^2 w_{22}^2}} \right)} \right)}{2\sqrt{-\varphi_1'^2 w_{11}^2 - 4\varphi'_1 \varphi'_2 w_{12} w_{21} + 2\varphi'_1 \varphi'_2 w_{11} w_{22} - \varphi_2'^2 w_{22}^2}} & \frac{i\varphi'_1 \varphi'_2}{\sqrt{-\varphi_1'^2 w_{11}^2 - 4\varphi'_1 \varphi'_2 w_{12} w_{21} + 2\varphi'_1 \varphi'_2 w_{11} w_{22} - \varphi_2'^2 w_{22}^2}} \end{pmatrix}$$

1.3 2. Prediagram Enumeration

Enumerate all trees, topologies, and prediagrams for the 2-point function at tree level ($\ell = 0$) and 1-loop ($\ell = 1$).

Color code: black = external legs, red = source vertices (in = 0), light blue = internal interaction vertices

```
[5]: from sage.plot.plot import graphics_array

def plot_prediagrams(pds, title_prefix=''):
    """Plot prediagrams with colored vertices."""
```

```

if not pds:
    print(' (none)')
    return
plots = []
for i, (D, G, leaves, internal) in enumerate(pds):
    leaves_set = set(leaves)
    color_map = {}
    for v in D.vertices():
        if v in leaves_set:
            color_map.setdefault('black', []).append(v)
        elif D.in_degree(v) == 0:
            color_map.setdefault('red', []).append(v)
        else:
            color_map.setdefault('lightblue', []).append(v)
    plots.append(D.plot(vertex_colors=color_map, vertex_size=400,
                        edge_thickness=2, title=f"{title_prefix}{i+1}"))
n_cols = min(4, len(plots))
n_rows = (len(plots) + n_cols - 1) // n_cols
graphics_array(plots, n_rows, n_cols).show(figsize=[5*n_cols, 4*n_rows])

def plot_topologies(topos, title_prefix=''):
    """Plot undirected topologies."""
    if not topos:
        print(' (none)')
        return
    plots = []
    for i, (G, leaves, internal) in enumerate(topos):
        leaves_set = set(leaves)
        color_map = {}
        for v in G.vertices():
            if v in leaves_set:
                color_map.setdefault('black', []).append(v)
            else:
                color_map.setdefault('lightgray', []).append(v)
        plots.append(G.plot(vertex_colors=color_map, vertex_size=400,
                            edge_thickness=2, title=f"{title_prefix}{i+1}"))
    n_cols = min(4, len(plots))
    n_rows = (len(plots) + n_cols - 1) // n_cols
    graphics_array(plots, n_rows, n_cols).show(figsize=[5*n_cols, 4*n_rows])

print('Plot helpers defined.')

```

Plot helpers defined.

1.3.1 2.1 Tree level ($\ell = 0$)

```
[6]: k = 2

trees_0, topos_0, pds_0, counts_0 = enumerate_all(k, 0, verbose=False)
print(f'k={k}, ell=0: {counts_0["n_trees"]} trees, {counts_0["n_topologies"]}┐
      ↳topologies, {counts_0["n_prediagrams"]} prediagrams')
print()

print('  Trees  ')
for i, (T, j, nl) in enumerate(trees_0):
    print(f'  Tree {i+1}: {T.num_verts()} vertices, {T.num_edges()} edges, {nl}┐
      ↳leaves, j={j}')
```

```
print()
print('  Topologies  ')
plot_topologies(topos_0, title_prefix='Topo ')

print('  Prediagrams  ')
plot_prediagrams(pds_0, title_prefix='PD ')

for i, (D, G, leaves, internal) in enumerate(pds_0):
    print(f'  PD {i+1}: edges={D.edges(labels=False)}')
```

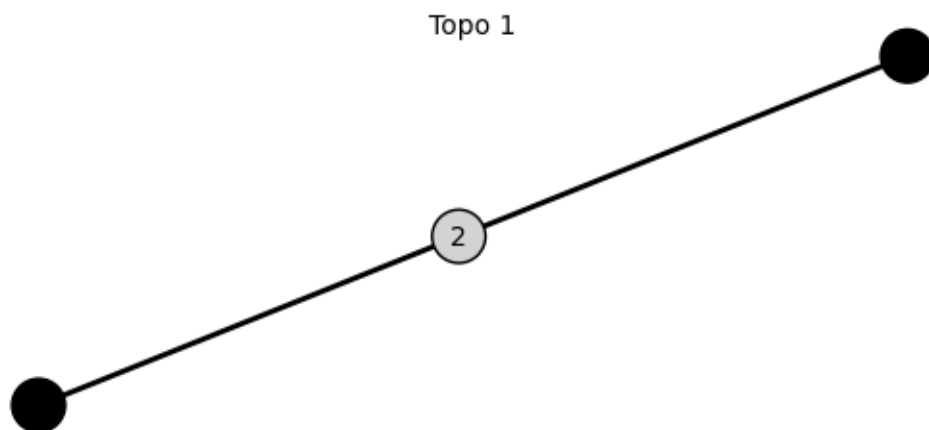
```
    for v in sorted(D.vertices()):
        role = 'leaf' if v in leaves else ('source' if D.in_degree(v)==0 else┐
      ↳'internal')
        print(f'      v{v}: in={D.in_degree(v)}, out={D.out_degree(v)} ({role})')
```

k=2, ell=0: 1 trees, 1 topologies, 1 prediagrams

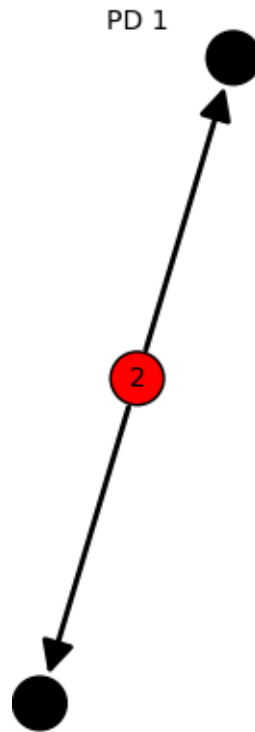
Trees

Tree 1: 3 vertices, 2 edges, 2 leaves, j=0

Topologies



Predigraphs



```
PD 1: edges=[(2, 0), (2, 1)]  
v0: in=1, out=0 (leaf)  
v1: in=1, out=0 (leaf)  
v2: in=0, out=2 (source)
```

1.3.2 2.2 One loop ($\ell = 1$)

```
[7]: trees_1, topos_1, pds_1, counts_1 = enumerate_all(k, 1, verbose=False)  
print(f'k={k}, ell=1: {counts_1["n_trees"]} trees, {counts_1["n_topologies"]} topologies, {counts_1["n_prediagrams"]} prediagrams')  
print()  
  
print(' Topologies ')  
plot_topologies(topos_1, title_prefix='Topo ')  
  
print(' Prediagrams ')  
plot_prediagrams(pds_1, title_prefix='PD ')  
  
for i, (D, G, leaves, internal) in enumerate(pds_1):
```

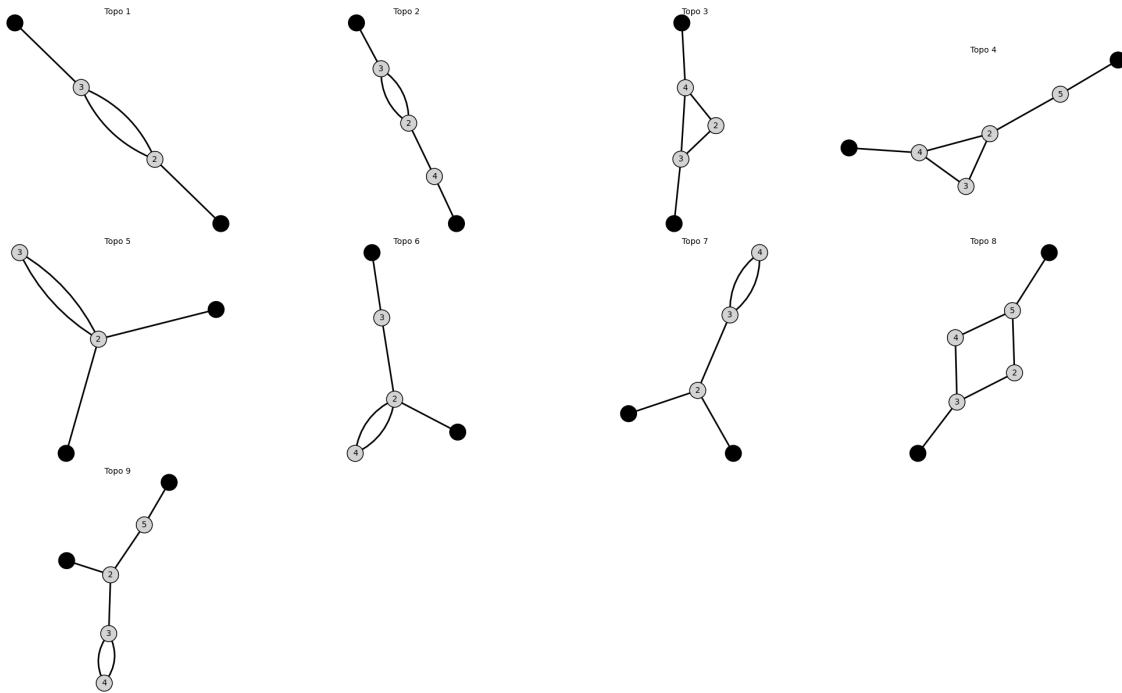
```

print(f' PD {i+1}: edges={D.edges(labels=False)}')
for v in sorted(D.vertices()):
    role = 'leaf' if v in leaves else ('source' if D.in_degree(v)==0 else
    ↪ 'internal')
    print(f'    v{v}: in={D.in_degree(v)}, out={D.out_degree(v)} ({role})')

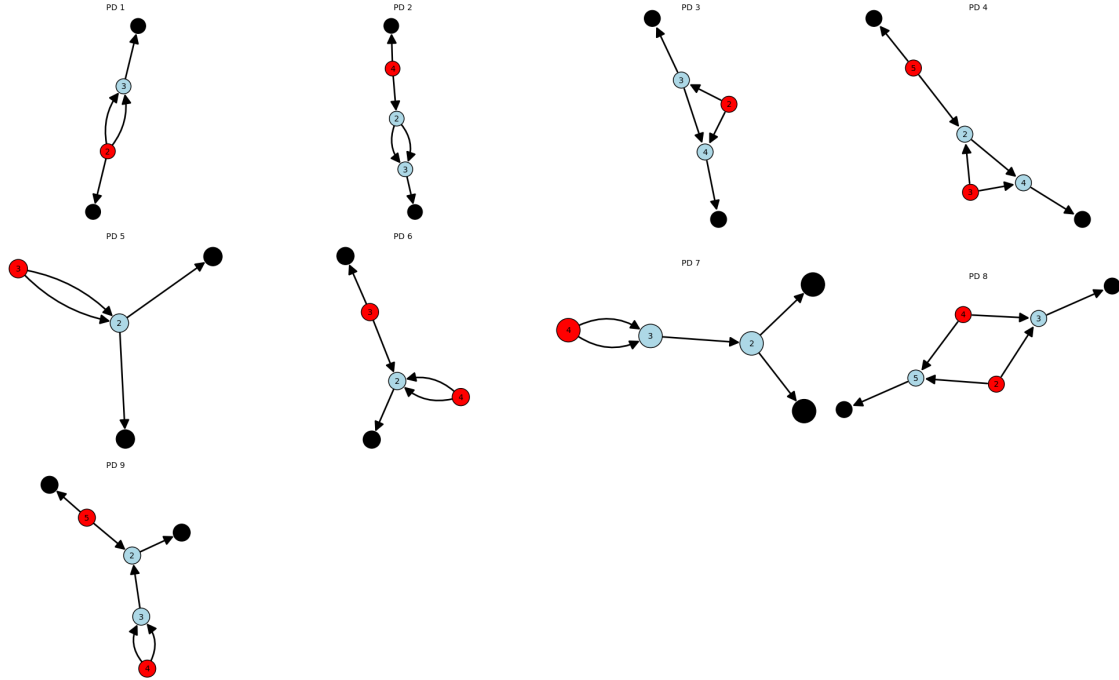
```

k=2, ell=1: 11 trees, 9 topologies, 9 prediagrams

Topologies



Prediagrams



PD 1: edges=[(2, 1), (2, 3), (2, 3), (3, 0)]
 v0: in=1, out=0 (leaf)
 v1: in=1, out=0 (leaf)
 v2: in=0, out=3 (source)
 v3: in=2, out=1 (internal)
 PD 2: edges=[(2, 3), (2, 3), (3, 0), (4, 1), (4, 2)]
 v0: in=1, out=0 (leaf)
 v1: in=1, out=0 (leaf)
 v2: in=1, out=2 (internal)
 v3: in=2, out=1 (internal)
 v4: in=0, out=2 (source)
 PD 3: edges=[(2, 3), (2, 4), (3, 0), (3, 4), (4, 1)]
 v0: in=1, out=0 (leaf)
 v1: in=1, out=0 (leaf)
 v2: in=0, out=2 (source)
 v3: in=1, out=2 (internal)
 v4: in=2, out=1 (internal)
 PD 4: edges=[(2, 4), (3, 2), (3, 4), (4, 0), (5, 1), (5, 2)]
 v0: in=1, out=0 (leaf)
 v1: in=1, out=0 (leaf)
 v2: in=2, out=1 (internal)
 v3: in=0, out=2 (source)
 v4: in=2, out=1 (internal)
 v5: in=0, out=2 (source)
 PD 5: edges=[(2, 0), (2, 1), (3, 2), (3, 2)]

```

v0: in=1, out=0 (leaf)
v1: in=1, out=0 (leaf)
v2: in=2, out=2 (internal)
v3: in=0, out=2 (source)
PD 6: edges=[(2, 1), (3, 0), (3, 2), (4, 2), (4, 2)]
v0: in=1, out=0 (leaf)
v1: in=1, out=0 (leaf)
v2: in=3, out=1 (internal)
v3: in=0, out=2 (source)
v4: in=0, out=2 (source)
PD 7: edges=[(2, 0), (2, 1), (3, 2), (4, 3), (4, 3)]
v0: in=1, out=0 (leaf)
v1: in=1, out=0 (leaf)
v2: in=1, out=2 (internal)
v3: in=2, out=1 (internal)
v4: in=0, out=2 (source)
PD 8: edges=[(2, 3), (2, 5), (3, 0), (4, 3), (4, 5), (5, 1)]
v0: in=1, out=0 (leaf)
v1: in=1, out=0 (leaf)
v2: in=0, out=2 (source)
v3: in=2, out=1 (internal)
v4: in=0, out=2 (source)
v5: in=2, out=1 (internal)
PD 9: edges=[(2, 1), (3, 2), (4, 3), (4, 3), (5, 0), (5, 2)]
v0: in=1, out=0 (leaf)
v1: in=1, out=0 (leaf)
v2: in=2, out=1 (internal)
v3: in=2, out=1 (internal)
v4: in=0, out=2 (source)
v5: in=0, out=2 (source)

```

1.4 3. Vertex Extraction

Extract typed vertices (`VertexType`, `SourceType`) from the expanded action, based on the maximum vertex degree required by the prediagrams.

```

[8]: from msrjd.enumeration.degree_scan import max_vertex_degree, \
      ↪ scan_source_vertices

all_pds = pds_0 + pds_1
max_deg = max_vertex_degree(all_pds)
src_degs = scan_source_vertices(all_pds)
print(f'Max vertex degree across all prediagrams: {max_deg}')
print(f'Source vertex out-degrees needed: {src_degs}')
print(f'Current Taylor order: {ft.taylor_order}')
print(f'Sufficient: {ft.taylor_order >= max_deg}')

```



```

print()

vtypes = extract_vertex_types(ft)
stypes = extract_source_types(ft)

print(f' Interaction vertex types ({len(vtypes)}) ')
for i, vt in enumerate(vtypes):
    print(f' V{i+1}: bigrade={vt.bigrade}, in_deg={vt.in_degree}, out_deg={vt.
    ↪out_degree}')
    print(f'      resp_legs={vt.response_legs}, phys_legs={vt.physical_legs}')
    display(Math(f'\\quad\\text{{coeff}} = {latex(vt.coefficient)}'))

print(f'\\n Source types ({len(stypes)}) ')
for i, st in enumerate(stypes):
    print(f' S{i+1}: bigrade={st.bigrade}, out_deg={st.out_degree}')
    print(f'      resp_legs={st.response_legs}')
    display(Math(f'\\quad\\text{{coeff}} = {latex(st.coefficient)}'))

int_degs, src_degs_avail = available_degrees(vtypes, stypes)
print(f'\\nAvailable interaction degrees (in, out): {sorted(int_degs)}')
print(f'Available source out-degrees: {sorted(src_degs_avail)}')

```

Max vertex degree across all prediagrams: 4
 Source vertex out-degrees needed: {2, 3}
 Current Taylor order: 4
 Sufficient: True

Interaction vertex types (14)
 V1: bigrade=(4, 2), in_deg=2, out_deg=4
 resp_legs=[('nt', 1), ('nt', 1), ('nt', 1), ('nt', 1)],
 phys_legs=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{48} \varphi_1''$$

V2: bigrade=(4, 2), in_deg=2, out_deg=4
 resp_legs=[('nt', 2), ('nt', 2), ('nt', 2), ('nt', 2)],
 phys_legs=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{48} \varphi_2''$$

V3: bigrade=(4, 1), in_deg=1, out_deg=4
 resp_legs=[('nt', 1), ('nt', 1), ('nt', 1), ('nt', 1)],
 phys_legs=[('dv', 1)]

$$\text{coeff} = \frac{1}{24} \varphi_1'$$

V4: bigrade=(4, 1), in_deg=1, out_deg=4
 resp_legs=[('nt', 2), ('nt', 2), ('nt', 2), ('nt', 2)],

phys_legs=[('dv', 2)]

$$\text{coeff} = \frac{1}{24} \varphi'_2$$

V5: bigrade=(3, 2), in_deg=2, out_deg=3
 resp_legs=[('nt', 1), ('nt', 1), ('nt', 1)], phys_legs=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{12} \varphi''_1$$

V6: bigrade=(3, 2), in_deg=2, out_deg=3
 resp_legs=[('nt', 2), ('nt', 2), ('nt', 2)], phys_legs=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{12} \varphi''_2$$

V7: bigrade=(3, 1), in_deg=1, out_deg=3
 resp_legs=[('nt', 1), ('nt', 1), ('nt', 1)], phys_legs=[('dv', 1)]

$$\text{coeff} = \frac{1}{6} \varphi'_1$$

V8: bigrade=(3, 1), in_deg=1, out_deg=3
 resp_legs=[('nt', 2), ('nt', 2), ('nt', 2)], phys_legs=[('dv', 2)]

$$\text{coeff} = \frac{1}{6} \varphi'_2$$

V9: bigrade=(2, 2), in_deg=2, out_deg=2
 resp_legs=[('nt', 1), ('nt', 1)], phys_legs=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{4} \varphi''_1$$

V10: bigrade=(2, 2), in_deg=2, out_deg=2
 resp_legs=[('nt', 2), ('nt', 2)], phys_legs=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{4} \varphi''_2$$

V11: bigrade=(2, 1), in_deg=1, out_deg=2
 resp_legs=[('nt', 1), ('nt', 1)], phys_legs=[('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi'_1$$

V12: bigrade=(2, 1), in_deg=1, out_deg=2
 resp_legs=[('nt', 2), ('nt', 2)], phys_legs=[('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi'_2$$

V13: bigrade=(1, 2), in_deg=2, out_deg=1
 resp_legs=[('nt', 1)], phys_legs=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

V14: bigrade=(1, 2), in_deg=2, out_deg=1
 resp_legs=[('nt', 2)], phys_legs=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

Source types (6)

S1: bigrade=(4, 0), out_deg=4
 resp_legs=[('nt', 1), ('nt', 1), ('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{24} nstar_1$$

S2: bigrade=(4, 0), out_deg=4
 resp_legs=[('nt', 2), ('nt', 2), ('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{24} nstar_2$$

S3: bigrade=(3, 0), out_deg=3
 resp_legs=[('nt', 1), ('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{6} nstar_1$$

S4: bigrade=(3, 0), out_deg=3
 resp_legs=[('nt', 2), ('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{6} nstar_2$$

S5: bigrade=(2, 0), out_deg=2
 resp_legs=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

S6: bigrade=(2, 0), out_deg=2
 resp_legs=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

Available interaction degrees (in, out): [(1, 2), (1, 3), (1, 4), (2, 1), (2, 2), (2, 3), (2, 4)]

Available source out-degrees: [2, 3, 4]

1.5 4. Prediagram Filtering

Remove prediagrams whose vertex degree signatures don't match any available vertex or source type.

```
[9]: print(' Tree level (ell=0) ')
kept_0, disc_0 = filter_prediagrams(pds_0, vtypes, stypes)
print(f' {len(pds_0)} prediagrams → {len(kept_0)} kept, {disc_0} discarded')

print()
print(' 1-loop (ell=1) ')
kept_1, disc_1 = filter_prediagrams(pds_1, vtypes, stypes)
print(f' {len(pds_1)} prediagrams → {len(kept_1)} kept, {disc_1} discarded')

if disc_1 > 0:
    print(f'\n Discarded prediagram indices:')
    kept_set = set(id(p) for p in kept_1)
    for i, pd in enumerate(pds_1):
        if id(pd) not in kept_set:
            D = pd[0]
            degs = [(D.in_degree(v), D.out_degree(v)) for v in pd[3]] #
            ↪internal verts
            print(f' PD {i+1}: internal vertex degrees = {degs}')

print()
print(' Surviving prediagrams (tree level) ')
plot_prediagrams(kept_0, title_prefix='Tree PD ')

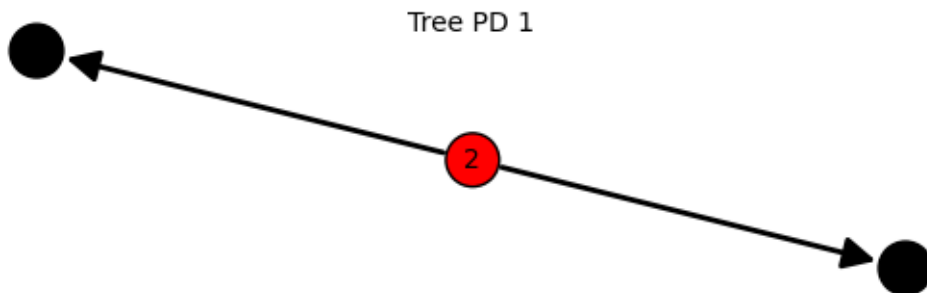
print(' Surviving prediagrams (1-loop) ')
plot_prediagrams(kept_1, title_prefix='1L PD ')
```

```
Tree level (ell=0)
1 prediagrams → 1 kept, 0 discarded

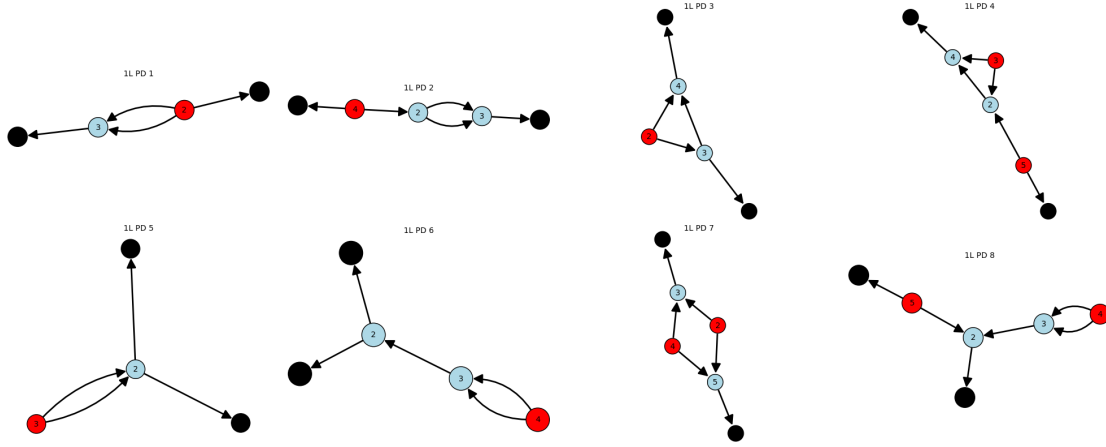
1-loop (ell=1)
9 prediagrams → 8 kept, 1 discarded

Discarded prediagram indices:
PD 6: internal vertex degrees = [(3, 1), (0, 2), (0, 2)]

Surviving prediagrams (tree level)
```



Surviving prediagrams (1-loop)



1.6 5. Type Assignment — Fully Labeled Diagrams

Enumerate all valid field-type assignments on each surviving prediagram.

External legs: $\delta\dot{n}_1$ and $\delta\dot{n}_2$ (the 2-point function fields).

Edge convention: each directed edge $u \rightarrow v$ carries propagator $\hat{G}_{ij}(\omega)$ where i is a response-field index from u and j is a physical-field index from v .

```
[10]: # External fields for the 2-point function <dn1 dn2>
# dn1, dn2 are physical fields - they sit at the incoming-edge end of leaves
external_fields = [('dn', 1), ('dn', 2)]

# Build field index maps
ring_var_names_list = list(ns._ring_var_names)
n_tilde = ft._n_tilde
resp_idx, phys_idx = build_field_index_map(ring_var_names_list, n_tilde)

print('Response field index map:')
for leg, idx in sorted(resp_idx.items(), key=lambda x: x[1]):
    print(f' {leg} → row {idx}')
print()
print('Physical field index map:')
for leg, idx in sorted(phys_idx.items(), key=lambda x: x[1]):
    print(f' {leg} → col {idx}')

# Helper to display typed diagrams nicely
def show_typed_diagram(td, idx):
    D = td.pred diagram[0]
```

```

leaves = td.prediagram[2]
print(f'\n{"="*60}')
print(f'Diagram {idx}')
print(f'{"="*60}')

print(' External legs:')
for lf, field in sorted(td.external_legs.items()):
    print(f'    leaf {lf} ← {field[0]}{field[1]}')

if td.vertex_assignments:
    print(' Vertex assignments:')
    for v, vtype in sorted(td.vertex_assignments.items()):
        tname = type(vtype).__name__
        print(f'    v{v} ({tname}): bigrade={vtype.bigrade}, '
              f'resp={vtype.response_legs}', end='')
        if hasattr(vtype, 'physical_legs'):
            print(f', phys={vtype.physical_legs}', end='')
        print()
        display(Math(f'\\quad\\text{{{coeff}}} = {latex(vtype.coefficient)}'))

print(' Edges (propagator assignments):')
for (u, v), (resp_leg, phys_leg) in sorted(td.edge_types.items()):
    ri, pi = td.propagator_indices.get((u, v), ('?', '?'))
    print(f'    {u} → {v}: {resp_leg[0]}{resp_leg[1]} →  

    ↪{phys_leg[0]}{phys_leg[1]} '
          f'G_hat[{ri},{pi}]')

```

Response field index map:

```

('nt', 1) → row 0
('nt', 2) → row 1
('vt', 1) → row 2
('vt', 2) → row 3

```

Physical field index map:

```

('dn', 1) → col 0
('dn', 2) → col 1
('dv', 1) → col 2
('dv', 2) → col 3

```

1.6.1 5.1 Tree-level typed diagrams

```

[11]: typed_tree = enumerate_all_typed(
        kept_0, external_fields, vtypes, stypes, G_ft, resp_idx, phys_idx
    )
print(f'Tree-level: {len(kept_0)} prediagrams → {len(typed_tree)} typed1  

    ↪diagrams')

```

```

for i, td in enumerate(typed_tree, 1):
    show_typed_diagram(td, f'Tree-{i}')

```

Tree-level: 1 prediagrams → 8 typed diagrams

=====
Diagram Tree-1
=====

External legs:
 leaf 0 ← dn1
 leaf 1 ← dn2
Vertex assignments:
 v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$
Edges (propagator assignments):
 2 → 0: nt1 → dn1 [G_hat[0,0]]
 2 → 1: nt1 → dn2 [G_hat[0,1]]

=====
Diagram Tree-2
=====

External legs:
 leaf 0 ← dn1
 leaf 1 ← dn2
Vertex assignments:
 v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$
Edges (propagator assignments):
 2 → 0: nt1 → dn1 [G_hat[0,0]]
 2 → 1: nt1 → dn2 [G_hat[0,1]]

=====
Diagram Tree-3
=====

External legs:
 leaf 0 ← dn1
 leaf 1 ← dn2
Vertex assignments:
 v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$
Edges (propagator assignments):
 2 → 0: nt2 → dn1 [G_hat[1,0]]

2 → 1: nt2 → dn2 [G_hat[1,1]]

Diagram Tree-4

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

Edges (propagator assignments):

2 → 0: nt2 → dn1 [G_hat[1,0]]

2 → 1: nt2 → dn2 [G_hat[1,1]]

Diagram Tree-5

External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

Edges (propagator assignments):

2 → 0: nt1 → dn2 [G_hat[0,1]]

2 → 1: nt1 → dn1 [G_hat[0,0]]

Diagram Tree-6

External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

Edges (propagator assignments):

2 → 0: nt1 → dn2 [G_hat[0,1]]

2 → 1: nt1 → dn1 [G_hat[0,0]]

Diagram Tree-7

```
=====
External legs:
  leaf 0 ← dn2
  leaf 1 ← dn1
Vertex assignments:
  v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

coeff =  $\frac{1}{2} nstar_2$ 

Edges (propagator assignments):
  2 → 0:  nt2 → dn2  [G_hat[1,1]]
  2 → 1:  nt2 → dn1  [G_hat[1,0]]
=====
```

Diagram Tree-8

```
=====
External legs:
  leaf 0 ← dn2
  leaf 1 ← dn1
Vertex assignments:
  v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

coeff =  $\frac{1}{2} nstar_2$ 

Edges (propagator assignments):
  2 → 0:  nt2 → dn2  [G_hat[1,1]]
  2 → 1:  nt2 → dn1  [G_hat[1,0]]
=====
```

1.6.2 5.2 One-loop typed diagrams

```
[12]: typed_1loop = enumerate_all_typed(
      kept_1, external_fields, vtypes, stypes, G_ft, resp_idx, phys_idx
    )
print(f'1-loop: {len(kept_1)} prediagrams → {len(typed_1loop)} typed diagrams')

for i, td in enumerate(typed_1loop, 1):
    show_typed_diagram(td, f'1L-{i}')
```

1-loop: 8 prediagrams → 1152 typed diagrams

Diagram 1L-1

```
=====
External legs:
  leaf 0 ← dn1
  leaf 1 ← dn2
Vertex assignments:
```

```

    v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

    coeff =  $\frac{1}{2} nstar_1$ 

    v3 (VertexType): bigrade=(2, 1), resp=[('nt', 1), ('nt', 1)], phys=[('dv',
1)]

    coeff =  $\frac{1}{2} \varphi'_1$ 

    v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

    coeff =  $\frac{1}{2} \varphi''_1$ 

    Edges (propagator assignments):
    2 → 3:  nt1 → dv1  [G_hat[0,2]]
    2 → 4:  nt1 → dv1  [G_hat[0,2]]
    3 → 0:  nt1 → dn1  [G_hat[0,0]]
    3 → 4:  nt1 → dv1  [G_hat[0,2]]
    4 → 1:  nt1 → dn2  [G_hat[0,1]]

=====
Diagram 1L-2
=====

    External legs:
    leaf 0 ← dn1
    leaf 1 ← dn2
    Vertex assignments:
    v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

    coeff =  $\frac{1}{2} nstar_1$ 

    v3 (VertexType): bigrade=(2, 1), resp=[('nt', 1), ('nt', 1)], phys=[('dv',
1)]

    coeff =  $\frac{1}{2} \varphi'_1$ 

    v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

    coeff =  $\frac{1}{2} \varphi''_1$ 

    Edges (propagator assignments):
    2 → 3:  nt1 → dv1  [G_hat[0,2]]
    2 → 4:  nt1 → dv1  [G_hat[0,2]]
    3 → 0:  nt1 → dn1  [G_hat[0,0]]
    3 → 4:  nt1 → dv1  [G_hat[0,2]]
    4 → 1:  nt1 → dn2  [G_hat[0,1]]

```

=====

Diagram 1L-3

=====

External legs:
 leaf 0 ← dn1
 leaf 1 ← dn2

Vertex assignments:
 v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$\text{coeff} = \frac{1}{2} nstar_1$

v3 (VertexType): bigrade=(2, 1), resp=[('nt', 1), ('nt', 1)], phys=[('dv', 1)]

$\text{coeff} = \frac{1}{2} \varphi'_1$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$\text{coeff} = \frac{1}{2} \varphi''_1$

Edges (propagator assignments):
 2 → 3: nt1 → dv1 [G_hat[0,2]]
 2 → 4: nt1 → dv1 [G_hat[0,2]]
 3 → 0: nt1 → dn1 [G_hat[0,0]]
 3 → 4: nt1 → dv1 [G_hat[0,2]]
 4 → 1: nt1 → dn2 [G_hat[0,1]]

=====

Diagram 1L-4

=====

External legs:
 leaf 0 ← dn1
 leaf 1 ← dn2

Vertex assignments:
 v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$\text{coeff} = \frac{1}{2} nstar_1$

v3 (VertexType): bigrade=(2, 1), resp=[('nt', 1), ('nt', 1)], phys=[('dv', 1)]

$\text{coeff} = \frac{1}{2} \varphi'_1$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$\text{coeff} = \frac{1}{2} \varphi''_1$

Edges (propagator assignments):
 2 → 3: nt1 → dv1 [G_hat[0,2]]
 2 → 4: nt1 → dv1 [G_hat[0,2]]
 3 → 0: nt1 → dn1 [G_hat[0,0]]
 3 → 4: nt1 → dv1 [G_hat[0,2]]
 4 → 1: nt1 → dn2 [G_hat[0,1]]

=====
 Diagram 1L-5
 =====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v3 (VertexType): bigrade=(2, 1), resp=[('nt', 1), ('nt', 1)], phys=[('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi'_1$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi''_1$$

Edges (propagator assignments):

2 → 3: nt1 → dv1 [G_hat[0,2]]

2 → 4: nt1 → dv1 [G_hat[0,2]]

3 → 0: nt1 → dn1 [G_hat[0,0]]

3 → 4: nt1 → dv1 [G_hat[0,2]]

4 → 1: nt1 → dn2 [G_hat[0,1]]

=====
 Diagram 1L-6
 =====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v3 (VertexType): bigrade=(2, 1), resp=[('nt', 1), ('nt', 1)], phys=[('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi'_1$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi''_1$$

Edges (propagator assignments):
 2 → 3: nt1 → dv1 [G_hat[0,2]]
 2 → 4: nt1 → dv1 [G_hat[0,2]]
 3 → 0: nt1 → dn1 [G_hat[0,0]]
 3 → 4: nt1 → dv1 [G_hat[0,2]]
 4 → 1: nt1 → dn2 [G_hat[0,1]]

=====
 Diagram 1L-7
 =====

External legs:
 leaf 0 ← dn1
 leaf 1 ← dn2

Vertex assignments:
 v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v3 (VertexType): bigrade=(2, 1), resp=[('nt', 1), ('nt', 1)], phys=[('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi'_1$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi''_1$$

Edges (propagator assignments):
 2 → 3: nt1 → dv1 [G_hat[0,2]]
 2 → 4: nt1 → dv1 [G_hat[0,2]]
 3 → 0: nt1 → dn1 [G_hat[0,0]]
 3 → 4: nt1 → dv1 [G_hat[0,2]]
 4 → 1: nt1 → dn2 [G_hat[0,1]]

=====
 Diagram 1L-8
 =====

External legs:
 leaf 0 ← dn1
 leaf 1 ← dn2

```

Vertex assignments:
  v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

coeff =  $\frac{1}{2} nstar_1$ 

  v3 (VertexType): bigrade=(2, 1), resp=[('nt', 1), ('nt', 1)], phys=[('dv',
1)]

coeff =  $\frac{1}{2} \varphi'_1$ 

  v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

coeff =  $\frac{1}{2} \varphi''_1$ 

Edges (propagator assignments):
  2 → 3:  nt1 → dv1  [G_hat[0,2]]
  2 → 4:  nt1 → dv1  [G_hat[0,2]]
  3 → 0:  nt1 → dn1  [G_hat[0,0]]
  3 → 4:  nt1 → dv1  [G_hat[0,2]]
  4 → 1:  nt1 → dn2  [G_hat[0,1]]

```

=====

Diagram 1L-9

=====

```

External legs:
  leaf 0 ← dn1
  leaf 1 ← dn2
Vertex assignments:
  v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

coeff =  $\frac{1}{2} nstar_1$ 

  v3 (VertexType): bigrade=(2, 1), resp=[('nt', 1), ('nt', 1)], phys=[('dv',
1)]

coeff =  $\frac{1}{2} \varphi'_1$ 

  v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

coeff =  $\frac{1}{2} \varphi''_2$ 

Edges (propagator assignments):
  2 → 3:  nt1 → dv1  [G_hat[0,2]]
  2 → 4:  nt1 → dv2  [G_hat[0,3]]
  3 → 0:  nt1 → dn1  [G_hat[0,0]]
  3 → 4:  nt1 → dv2  [G_hat[0,3]]
  4 → 1:  nt2 → dn2  [G_hat[1,1]]

```

=====

Diagram 1L-10

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v3 (VertexType): bigrade=(2, 1), resp=[('nt', 1), ('nt', 1)], phys=[('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi'_1$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi''_2$$

Edges (propagator assignments):

2 → 3: nt1 → dv1 [G_hat[0,2]]

2 → 4: nt1 → dv2 [G_hat[0,3]]

3 → 0: nt1 → dn1 [G_hat[0,0]]

3 → 4: nt1 → dv2 [G_hat[0,3]]

4 → 1: nt2 → dn2 [G_hat[1,1]]

=====

Diagram 1L-11

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v3 (VertexType): bigrade=(2, 1), resp=[('nt', 1), ('nt', 1)], phys=[('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi'_1$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi''_2$$

```

Edges (propagator assignments):
  2 → 3:  nt1 → dv1  [G_hat[0,2]]
  2 → 4:  nt1 → dv2  [G_hat[0,3]]
  3 → 0:  nt1 → dn1  [G_hat[0,0]]
  3 → 4:  nt1 → dv2  [G_hat[0,3]]
  4 → 1:  nt2 → dn2  [G_hat[1,1]]

```

Diagram 1L-12

```

External legs:
  leaf 0 ← dn1
  leaf 1 ← dn2
Vertex assignments:
  v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

coeff =  $\frac{1}{2} nstar_1$ 

  v3 (VertexType): bigrade=(2, 1), resp=[('nt', 1), ('nt', 1)], phys=[('dv',
1)]

coeff =  $\frac{1}{2} \varphi'_1$ 

  v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

coeff =  $\frac{1}{2} \varphi''_2$ 

Edges (propagator assignments):
  2 → 3:  nt1 → dv1  [G_hat[0,2]]
  2 → 4:  nt1 → dv2  [G_hat[0,3]]
  3 → 0:  nt1 → dn1  [G_hat[0,0]]
  3 → 4:  nt1 → dv2  [G_hat[0,3]]
  4 → 1:  nt2 → dn2  [G_hat[1,1]]

```

Diagram 1L-13

```

External legs:
  leaf 0 ← dn1
  leaf 1 ← dn2
Vertex assignments:
  v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

coeff =  $\frac{1}{2} nstar_1$ 

  v3 (VertexType): bigrade=(2, 1), resp=[('nt', 1), ('nt', 1)], phys=[('dv',
1)]

```


$$\text{coeff} = \frac{1}{2} \varphi'_1$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi''_2$$

Edges (propagator assignments):
 2 → 3: nt1 → dv1 [G_hat[0,2]]
 2 → 4: nt1 → dv2 [G_hat[0,3]]
 3 → 0: nt1 → dn1 [G_hat[0,0]]
 3 → 4: nt1 → dv2 [G_hat[0,3]]
 4 → 1: nt2 → dn2 [G_hat[1,1]]

=====
 Diagram 1L-14
 =====

External legs:
 leaf 0 ← dn1
 leaf 1 ← dn2

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v3 (VertexType): bigrade=(2, 1), resp=[('nt', 1), ('nt', 1)], phys=[('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi'_1$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi''_2$$

Edges (propagator assignments):
 2 → 3: nt1 → dv1 [G_hat[0,2]]
 2 → 4: nt1 → dv2 [G_hat[0,3]]
 3 → 0: nt1 → dn1 [G_hat[0,0]]
 3 → 4: nt1 → dv2 [G_hat[0,3]]
 4 → 1: nt2 → dn2 [G_hat[1,1]]

=====
 Diagram 1L-15
 =====

External legs:
 leaf 0 ← dn1
 leaf 1 ← dn2

```

Vertex assignments:
  v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

coeff =  $\frac{1}{2} nstar_1$ 

  v3 (VertexType): bigrade=(2, 1), resp=[('nt', 1), ('nt', 1)], phys=[('dv',
1)]

coeff =  $\frac{1}{2} \varphi'_1$ 

  v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

coeff =  $\frac{1}{2} \varphi''_2$ 

Edges (propagator assignments):
  2 → 3:  nt1 → dv1  [G_hat[0,2]]
  2 → 4:  nt1 → dv2  [G_hat[0,3]]
  3 → 0:  nt1 → dn1  [G_hat[0,0]]
  3 → 4:  nt1 → dv2  [G_hat[0,3]]
  4 → 1:  nt2 → dn2  [G_hat[1,1]]

```

=====

Diagram 1L-16

=====

```

External legs:
  leaf 0 ← dn1
  leaf 1 ← dn2
Vertex assignments:
  v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

coeff =  $\frac{1}{2} nstar_1$ 

  v3 (VertexType): bigrade=(2, 1), resp=[('nt', 1), ('nt', 1)], phys=[('dv',
1)]

coeff =  $\frac{1}{2} \varphi'_1$ 

  v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

coeff =  $\frac{1}{2} \varphi''_2$ 

Edges (propagator assignments):
  2 → 3:  nt1 → dv1  [G_hat[0,2]]
  2 → 4:  nt1 → dv2  [G_hat[0,3]]
  3 → 0:  nt1 → dn1  [G_hat[0,0]]
  3 → 4:  nt1 → dv2  [G_hat[0,3]]
  4 → 1:  nt2 → dn2  [G_hat[1,1]]

```

=====

Diagram 1L-17

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v3 (VertexType): bigrade=(2, 1), resp=[('nt', 2), ('nt', 2)], phys=[('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi'_2$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi''_1$$

Edges (propagator assignments):

2 → 3: nt1 → dv2 [G_hat[0,3]]

2 → 4: nt1 → dv1 [G_hat[0,2]]

3 → 0: nt2 → dn1 [G_hat[1,0]]

3 → 4: nt2 → dv1 [G_hat[1,2]]

4 → 1: nt1 → dn2 [G_hat[0,1]]

=====

Diagram 1L-18

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v3 (VertexType): bigrade=(2, 1), resp=[('nt', 2), ('nt', 2)], phys=[('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi'_2$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi''_1$$

Edges (propagator assignments):
 2 → 3: nt1 → dv2 [G_hat[0,3]]
 2 → 4: nt1 → dv1 [G_hat[0,2]]
 3 → 0: nt2 → dn1 [G_hat[1,0]]
 3 → 4: nt2 → dv1 [G_hat[1,2]]
 4 → 1: nt1 → dn2 [G_hat[0,1]]

=====
 Diagram 1L-19
 =====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v3 (VertexType): bigrade=(2, 1), resp=[('nt', 2), ('nt', 2)], phys=[('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi'_2$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi''_1$$

Edges (propagator assignments):

2 → 3: nt1 → dv2 [G_hat[0,3]]

2 → 4: nt1 → dv1 [G_hat[0,2]]

3 → 0: nt2 → dn1 [G_hat[1,0]]

3 → 4: nt2 → dv1 [G_hat[1,2]]

4 → 1: nt1 → dn2 [G_hat[0,1]]

=====
 Diagram 1L-20
 =====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v3 (VertexType): bigrade=(2, 1), resp=[('nt', 2), ('nt', 2)], phys=[('dv', 2)]

$\text{coeff} = \frac{1}{2} \varphi'_2$
 v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]
 $\text{coeff} = \frac{1}{2} \varphi''_1$
 Edges (propagator assignments):
 2 → 3: nt1 → dv2 [G_hat[0,3]]
 2 → 4: nt1 → dv1 [G_hat[0,2]]
 3 → 0: nt2 → dn1 [G_hat[1,0]]
 3 → 4: nt2 → dv1 [G_hat[1,2]]
 4 → 1: nt1 → dn2 [G_hat[0,1]]

=====
 Diagram 1L-21
 =====

External legs:
 leaf 0 ← dn1
 leaf 1 ← dn2
 Vertex assignments:
 v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]
 $\text{coeff} = \frac{1}{2} nstar_1$
 v3 (VertexType): bigrade=(2, 1), resp=[('nt', 2), ('nt', 2)], phys=[('dv', 2)]
 $\text{coeff} = \frac{1}{2} \varphi'_2$
 v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]
 $\text{coeff} = \frac{1}{2} \varphi''_1$

Edges (propagator assignments):
 2 → 3: nt1 → dv2 [G_hat[0,3]]
 2 → 4: nt1 → dv1 [G_hat[0,2]]
 3 → 0: nt2 → dn1 [G_hat[1,0]]
 3 → 4: nt2 → dv1 [G_hat[1,2]]
 4 → 1: nt1 → dn2 [G_hat[0,1]]
 =====
 Diagram 1L-22
 =====
 External legs:
 leaf 0 ← dn1
 leaf 1 ← dn2

```

Vertex assignments:
  v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

coeff =  $\frac{1}{2} nstar_1$ 

  v3 (VertexType): bigrade=(2, 1), resp=[('nt', 2), ('nt', 2)], phys=[('dv',
2)]

coeff =  $\frac{1}{2} \varphi'_2$ 

  v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

coeff =  $\frac{1}{2} \varphi''_1$ 

Edges (propagator assignments):
  2 → 3:  nt1 → dv2  [G_hat[0,3]]
  2 → 4:  nt1 → dv1  [G_hat[0,2]]
  3 → 0:  nt2 → dn1  [G_hat[1,0]]
  3 → 4:  nt2 → dv1  [G_hat[1,2]]
  4 → 1:  nt1 → dn2  [G_hat[0,1]]

```

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Diagram 1L-23

=====

```

External legs:
  leaf 0 ← dn1
  leaf 1 ← dn2
Vertex assignments:
  v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

coeff =  $\frac{1}{2} nstar_1$ 

  v3 (VertexType): bigrade=(2, 1), resp=[('nt', 2), ('nt', 2)], phys=[('dv',
2)]

coeff =  $\frac{1}{2} \varphi'_2$ 

  v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

coeff =  $\frac{1}{2} \varphi''_1$ 

Edges (propagator assignments):
  2 → 3:  nt1 → dv2  [G_hat[0,3]]
  2 → 4:  nt1 → dv1  [G_hat[0,2]]
  3 → 0:  nt2 → dn1  [G_hat[1,0]]
  3 → 4:  nt2 → dv1  [G_hat[1,2]]
  4 → 1:  nt1 → dn2  [G_hat[0,1]]

```

=====

Diagram 1L-24

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v3 (VertexType): bigrade=(2, 1), resp=[('nt', 2), ('nt', 2)], phys=[('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi'_2$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi''_1$$

Edges (propagator assignments):

2 → 3: nt1 → dv2 [G_hat[0,3]]

2 → 4: nt1 → dv1 [G_hat[0,2]]

3 → 0: nt2 → dn1 [G_hat[1,0]]

3 → 4: nt2 → dv1 [G_hat[1,2]]

4 → 1: nt1 → dn2 [G_hat[0,1]]

=====

Diagram 1L-25

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v3 (VertexType): bigrade=(2, 1), resp=[('nt', 2), ('nt', 2)], phys=[('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi'_2$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi''_2$$

```

Edges (propagator assignments):
  2 → 3:  nt1 → dv2  [G_hat[0,3]]
  2 → 4:  nt1 → dv2  [G_hat[0,3]]
  3 → 0:  nt2 → dn1  [G_hat[1,0]]
  3 → 4:  nt2 → dv2  [G_hat[1,3]]
  4 → 1:  nt2 → dn2  [G_hat[1,1]]

```

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Diagram 1L-26

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v3 (VertexType): bigrade=(2, 1), resp=[('nt', 2), ('nt', 2)], phys=[('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi'_2$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi''_2$$

Edges (propagator assignments):

2 → 3: nt1 → dv2 [G_hat[0,3]]

2 → 4: nt1 → dv2 [G_hat[0,3]]

3 → 0: nt2 → dn1 [G_hat[1,0]]

3 → 4: nt2 → dv2 [G_hat[1,3]]

4 → 1: nt2 → dn2 [G_hat[1,1]]

=====

Diagram 1L-27

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v3 (VertexType): bigrade=(2, 1), resp=[('nt', 2), ('nt', 2)], phys=[('dv', 2)]


```

coeff =  $\frac{1}{2} \varphi'_2$ 

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

coeff =  $\frac{1}{2} \varphi''_2$ 

Edges (propagator assignments):
  2 → 3:  nt1 → dv2  [G_hat[0,3]]
  2 → 4:  nt1 → dv2  [G_hat[0,3]]
  3 → 0:  nt2 → dn1  [G_hat[1,0]]
  3 → 4:  nt2 → dv2  [G_hat[1,3]]
  4 → 1:  nt2 → dn2  [G_hat[1,1]]

=====
Diagram 1L-28
=====

External legs:
  leaf 0 ← dn1
  leaf 1 ← dn2
Vertex assignments:
  v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

coeff =  $\frac{1}{2} nstar_1$ 

v3 (VertexType): bigrade=(2, 1), resp=[('nt', 2), ('nt', 2)], phys=[('dv',
2)]

coeff =  $\frac{1}{2} \varphi'_2$ 

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

coeff =  $\frac{1}{2} \varphi''_2$ 

Edges (propagator assignments):
  2 → 3:  nt1 → dv2  [G_hat[0,3]]
  2 → 4:  nt1 → dv2  [G_hat[0,3]]
  3 → 0:  nt2 → dn1  [G_hat[1,0]]
  3 → 4:  nt2 → dv2  [G_hat[1,3]]
  4 → 1:  nt2 → dn2  [G_hat[1,1]]

=====
Diagram 1L-29
=====

External legs:
  leaf 0 ← dn1
  leaf 1 ← dn2

```

```

Vertex assignments:
  v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

coeff =  $\frac{1}{2} nstar_1$ 

  v3 (VertexType): bigrade=(2, 1), resp=[('nt', 2), ('nt', 2)], phys=[('dv',
2)]

coeff =  $\frac{1}{2} \varphi'_2$ 

  v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

coeff =  $\frac{1}{2} \varphi''_2$ 

Edges (propagator assignments):
  2 → 3:  nt1 → dv2  [G_hat[0,3]]
  2 → 4:  nt1 → dv2  [G_hat[0,3]]
  3 → 0:  nt2 → dn1  [G_hat[1,0]]
  3 → 4:  nt2 → dv2  [G_hat[1,3]]
  4 → 1:  nt2 → dn2  [G_hat[1,1]]

```

=====

Diagram 1L-30

=====

```

External legs:
  leaf 0 ← dn1
  leaf 1 ← dn2
Vertex assignments:
  v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

coeff =  $\frac{1}{2} nstar_1$ 

  v3 (VertexType): bigrade=(2, 1), resp=[('nt', 2), ('nt', 2)], phys=[('dv',
2)]

coeff =  $\frac{1}{2} \varphi'_2$ 

  v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

coeff =  $\frac{1}{2} \varphi''_2$ 

Edges (propagator assignments):
  2 → 3:  nt1 → dv2  [G_hat[0,3]]
  2 → 4:  nt1 → dv2  [G_hat[0,3]]
  3 → 0:  nt2 → dn1  [G_hat[1,0]]
  3 → 4:  nt2 → dv2  [G_hat[1,3]]
  4 → 1:  nt2 → dn2  [G_hat[1,1]]

```

=====

Diagram 1L-31

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v3 (VertexType): bigrade=(2, 1), resp=[('nt', 2), ('nt', 2)], phys=[('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi'_2$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi''_2$$

Edges (propagator assignments):

2 → 3: nt1 → dv2 [G_hat[0,3]]

2 → 4: nt1 → dv2 [G_hat[0,3]]

3 → 0: nt2 → dn1 [G_hat[1,0]]

3 → 4: nt2 → dv2 [G_hat[1,3]]

4 → 1: nt2 → dn2 [G_hat[1,1]]

=====

Diagram 1L-32

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v3 (VertexType): bigrade=(2, 1), resp=[('nt', 2), ('nt', 2)], phys=[('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi'_2$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi''_2$$

Edges (propagator assignments):
 2 → 3: nt1 → dv2 [G_hat[0,3]]
 2 → 4: nt1 → dv2 [G_hat[0,3]]
 3 → 0: nt2 → dn1 [G_hat[1,0]]
 3 → 4: nt2 → dv2 [G_hat[1,3]]
 4 → 1: nt2 → dn2 [G_hat[1,1]]

=====
 Diagram 1L-33
 =====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v3 (VertexType): bigrade=(2, 1), resp=[('nt', 1), ('nt', 1)], phys=[('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi'_1$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi''_1$$

Edges (propagator assignments):

2 → 3: nt2 → dv1 [G_hat[1,2]]

2 → 4: nt2 → dv1 [G_hat[1,2]]

3 → 0: nt1 → dn1 [G_hat[0,0]]

3 → 4: nt1 → dv1 [G_hat[0,2]]

4 → 1: nt1 → dn2 [G_hat[0,1]]

=====
 Diagram 1L-34
 =====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v3 (VertexType): bigrade=(2, 1), resp=[('nt', 1), ('nt', 1)], phys=[('dv', 1)]

$\text{coeff} = \frac{1}{2} \varphi'_1$
 v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]
 $\text{coeff} = \frac{1}{2} \varphi''_1$
 Edges (propagator assignments):
 2 → 3: nt2 → dv1 [G_hat[1,2]]
 2 → 4: nt2 → dv1 [G_hat[1,2]]
 3 → 0: nt1 → dn1 [G_hat[0,0]]
 3 → 4: nt1 → dv1 [G_hat[0,2]]
 4 → 1: nt1 → dn2 [G_hat[0,1]]

=====
 Diagram 1L-35
 =====

External legs:
 leaf 0 ← dn1
 leaf 1 ← dn2
 Vertex assignments:
 v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]
 $\text{coeff} = \frac{1}{2} nstar_2$
 v3 (VertexType): bigrade=(2, 1), resp=[('nt', 1), ('nt', 1)], phys=[('dv', 1), ('dv', 1)]
 $\text{coeff} = \frac{1}{2} \varphi'_1$
 v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]
 $\text{coeff} = \frac{1}{2} \varphi''_1$

Edges (propagator assignments):
 2 → 3: nt2 → dv1 [G_hat[1,2]]
 2 → 4: nt2 → dv1 [G_hat[1,2]]
 3 → 0: nt1 → dn1 [G_hat[0,0]]
 3 → 4: nt1 → dv1 [G_hat[0,2]]
 4 → 1: nt1 → dn2 [G_hat[0,1]]
 =====
 Diagram 1L-36
 =====
 External legs:
 leaf 0 ← dn1
 leaf 1 ← dn2

```

Vertex assignments:
  v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

coeff =  $\frac{1}{2} nstar_2$ 

  v3 (VertexType): bigrade=(2, 1), resp=[('nt', 1), ('nt', 1)], phys=[('dv',
1)]

coeff =  $\frac{1}{2} \varphi'_1$ 

  v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

coeff =  $\frac{1}{2} \varphi''_1$ 

Edges (propagator assignments):
  2 → 3:  nt2 → dv1  [G_hat[1,2]]
  2 → 4:  nt2 → dv1  [G_hat[1,2]]
  3 → 0:  nt1 → dn1  [G_hat[0,0]]
  3 → 4:  nt1 → dv1  [G_hat[0,2]]
  4 → 1:  nt1 → dn2  [G_hat[0,1]]

```

=====

Diagram 1L-37

=====

```

External legs:
  leaf 0 ← dn1
  leaf 1 ← dn2
Vertex assignments:
  v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

coeff =  $\frac{1}{2} nstar_2$ 

  v3 (VertexType): bigrade=(2, 1), resp=[('nt', 1), ('nt', 1)], phys=[('dv',
1)]

coeff =  $\frac{1}{2} \varphi'_1$ 

  v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

coeff =  $\frac{1}{2} \varphi''_1$ 

Edges (propagator assignments):
  2 → 3:  nt2 → dv1  [G_hat[1,2]]
  2 → 4:  nt2 → dv1  [G_hat[1,2]]
  3 → 0:  nt1 → dn1  [G_hat[0,0]]
  3 → 4:  nt1 → dv1  [G_hat[0,2]]
  4 → 1:  nt1 → dn2  [G_hat[0,1]]

```

=====

Diagram 1L-38

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v3 (VertexType): bigrade=(2, 1), resp=[('nt', 1), ('nt', 1)], phys=[('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi'_1$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi''_1$$

Edges (propagator assignments):

2 → 3: nt2 → dv1 [G_hat[1,2]]

2 → 4: nt2 → dv1 [G_hat[1,2]]

3 → 0: nt1 → dn1 [G_hat[0,0]]

3 → 4: nt1 → dv1 [G_hat[0,2]]

4 → 1: nt1 → dn2 [G_hat[0,1]]

=====

Diagram 1L-39

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v3 (VertexType): bigrade=(2, 1), resp=[('nt', 1), ('nt', 1)], phys=[('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi'_1$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi''_1$$

Edges (propagator assignments):
 2 → 3: nt2 → dv1 [G_hat[1,2]]
 2 → 4: nt2 → dv1 [G_hat[1,2]]
 3 → 0: nt1 → dn1 [G_hat[0,0]]
 3 → 4: nt1 → dv1 [G_hat[0,2]]
 4 → 1: nt1 → dn2 [G_hat[0,1]]

=====
 Diagram 1L-40
 =====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v3 (VertexType): bigrade=(2, 1), resp=[('nt', 1), ('nt', 1)], phys=[('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi'_1$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi''_1$$

Edges (propagator assignments):

2 → 3: nt2 → dv1 [G_hat[1,2]]

2 → 4: nt2 → dv1 [G_hat[1,2]]

3 → 0: nt1 → dn1 [G_hat[0,0]]

3 → 4: nt1 → dv1 [G_hat[0,2]]

4 → 1: nt1 → dn2 [G_hat[0,1]]

=====
 Diagram 1L-41
 =====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v3 (VertexType): bigrade=(2, 1), resp=[('nt', 1), ('nt', 1)], phys=[('dv', 1)]

$\text{coeff} = \frac{1}{2} \varphi'_1$
 v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]
 $\text{coeff} = \frac{1}{2} \varphi''_2$
 Edges (propagator assignments):
 2 → 3: nt2 → dv1 [G_hat[1,2]]
 2 → 4: nt2 → dv2 [G_hat[1,3]]
 3 → 0: nt1 → dn1 [G_hat[0,0]]
 3 → 4: nt1 → dv2 [G_hat[0,3]]
 4 → 1: nt2 → dn2 [G_hat[1,1]]

=====
 Diagram 1L-42
 =====

External legs:
 leaf 0 ← dn1
 leaf 1 ← dn2
 Vertex assignments:
 v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]
 $\text{coeff} = \frac{1}{2} nstar_2$
 v3 (VertexType): bigrade=(2, 1), resp=[('nt', 1), ('nt', 1)], phys=[('dv', 1)]
 $\text{coeff} = \frac{1}{2} \varphi'_1$
 v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]
 $\text{coeff} = \frac{1}{2} \varphi''_2$

Edges (propagator assignments):
 2 → 3: nt2 → dv1 [G_hat[1,2]]
 2 → 4: nt2 → dv2 [G_hat[1,3]]
 3 → 0: nt1 → dn1 [G_hat[0,0]]
 3 → 4: nt1 → dv2 [G_hat[0,3]]
 4 → 1: nt2 → dn2 [G_hat[1,1]]
 =====
 Diagram 1L-43
 =====
 External legs:
 leaf 0 ← dn1
 leaf 1 ← dn2

```

Vertex assignments:
  v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

coeff =  $\frac{1}{2} nstar_2$ 

  v3 (VertexType): bigrade=(2, 1), resp=[('nt', 1), ('nt', 1)], phys=[('dv',
1)]

coeff =  $\frac{1}{2} \varphi'_1$ 

  v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

coeff =  $\frac{1}{2} \varphi''_2$ 

Edges (propagator assignments):
  2 → 3:  nt2 → dv1  [G_hat[1,2]]
  2 → 4:  nt2 → dv2  [G_hat[1,3]]
  3 → 0:  nt1 → dn1  [G_hat[0,0]]
  3 → 4:  nt1 → dv2  [G_hat[0,3]]
  4 → 1:  nt2 → dn2  [G_hat[1,1]]

```

=====

Diagram 1L-44

=====

```

External legs:
  leaf 0 ← dn1
  leaf 1 ← dn2
Vertex assignments:
  v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

coeff =  $\frac{1}{2} nstar_2$ 

  v3 (VertexType): bigrade=(2, 1), resp=[('nt', 1), ('nt', 1)], phys=[('dv',
1)]

coeff =  $\frac{1}{2} \varphi'_1$ 

  v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

coeff =  $\frac{1}{2} \varphi''_2$ 

Edges (propagator assignments):
  2 → 3:  nt2 → dv1  [G_hat[1,2]]
  2 → 4:  nt2 → dv2  [G_hat[1,3]]
  3 → 0:  nt1 → dn1  [G_hat[0,0]]
  3 → 4:  nt1 → dv2  [G_hat[0,3]]
  4 → 1:  nt2 → dn2  [G_hat[1,1]]

```

=====

Diagram 1L-45

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v3 (VertexType): bigrade=(2, 1), resp=[('nt', 1), ('nt', 1)], phys=[('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi'_1$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi''_2$$

Edges (propagator assignments):

2 → 3: nt2 → dv1 [G_hat[1,2]]

2 → 4: nt2 → dv2 [G_hat[1,3]]

3 → 0: nt1 → dn1 [G_hat[0,0]]

3 → 4: nt1 → dv2 [G_hat[0,3]]

4 → 1: nt2 → dn2 [G_hat[1,1]]

=====

Diagram 1L-46

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v3 (VertexType): bigrade=(2, 1), resp=[('nt', 1), ('nt', 1)], phys=[('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi'_1$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi''_2$$

Edges (propagator assignments):
 2 → 3: nt2 → dv1 [G_hat[1,2]]
 2 → 4: nt2 → dv2 [G_hat[1,3]]
 3 → 0: nt1 → dn1 [G_hat[0,0]]
 3 → 4: nt1 → dv2 [G_hat[0,3]]
 4 → 1: nt2 → dn2 [G_hat[1,1]]

=====
 Diagram 1L-47
 =====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v3 (VertexType): bigrade=(2, 1), resp=[('nt', 1), ('nt', 1)], phys=[('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi'_1$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi''_2$$

Edges (propagator assignments):

2 → 3: nt2 → dv1 [G_hat[1,2]]

2 → 4: nt2 → dv2 [G_hat[1,3]]

3 → 0: nt1 → dn1 [G_hat[0,0]]

3 → 4: nt1 → dv2 [G_hat[0,3]]

4 → 1: nt2 → dn2 [G_hat[1,1]]

=====
 Diagram 1L-48
 =====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v3 (VertexType): bigrade=(2, 1), resp=[('nt', 1), ('nt', 1)], phys=[('dv', 1)]

$\text{coeff} = \frac{1}{2} \varphi'_1$
 v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]
 $\text{coeff} = \frac{1}{2} \varphi''_2$
 Edges (propagator assignments):
 2 → 3: nt2 → dv1 [G_hat[1,2]]
 2 → 4: nt2 → dv2 [G_hat[1,3]]
 3 → 0: nt1 → dn1 [G_hat[0,0]]
 3 → 4: nt1 → dv2 [G_hat[0,3]]
 4 → 1: nt2 → dn2 [G_hat[1,1]]

=====
 Diagram 1L-49
 =====

External legs:
 leaf 0 ← dn1
 leaf 1 ← dn2
 Vertex assignments:
 v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]
 $\text{coeff} = \frac{1}{2} nstar_2$
 v3 (VertexType): bigrade=(2, 1), resp=[('nt', 2), ('nt', 2)], phys=[('dv', 2), ('dv', 2)]
 $\text{coeff} = \frac{1}{2} \varphi'_2$
 v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]
 $\text{coeff} = \frac{1}{2} \varphi''_1$

Edges (propagator assignments):
 2 → 3: nt2 → dv2 [G_hat[1,3]]
 2 → 4: nt2 → dv1 [G_hat[1,2]]
 3 → 0: nt2 → dn1 [G_hat[1,0]]
 3 → 4: nt2 → dv1 [G_hat[1,2]]
 4 → 1: nt1 → dn2 [G_hat[0,1]]
 =====
 Diagram 1L-50
 =====
 External legs:
 leaf 0 ← dn1
 leaf 1 ← dn2

```

Vertex assignments:
  v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

coeff =  $\frac{1}{2} nstar_2$ 

  v3 (VertexType): bigrade=(2, 1), resp=[('nt', 2), ('nt', 2)], phys=[('dv',
2)]

coeff =  $\frac{1}{2} \varphi'_2$ 

  v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

coeff =  $\frac{1}{2} \varphi''_1$ 

Edges (propagator assignments):
  2 → 3:  nt2 → dv2  [G_hat[1,3]]
  2 → 4:  nt2 → dv1  [G_hat[1,2]]
  3 → 0:  nt2 → dn1  [G_hat[1,0]]
  3 → 4:  nt2 → dv1  [G_hat[1,2]]
  4 → 1:  nt1 → dn2  [G_hat[0,1]]

```

=====
Diagram 1L-51
=====

```

External legs:
  leaf 0 ← dn1
  leaf 1 ← dn2
Vertex assignments:
  v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

coeff =  $\frac{1}{2} nstar_2$ 

  v3 (VertexType): bigrade=(2, 1), resp=[('nt', 2), ('nt', 2)], phys=[('dv',
2)]

coeff =  $\frac{1}{2} \varphi'_2$ 

  v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

coeff =  $\frac{1}{2} \varphi''_1$ 

Edges (propagator assignments):
  2 → 3:  nt2 → dv2  [G_hat[1,3]]
  2 → 4:  nt2 → dv1  [G_hat[1,2]]
  3 → 0:  nt2 → dn1  [G_hat[1,0]]
  3 → 4:  nt2 → dv1  [G_hat[1,2]]
  4 → 1:  nt1 → dn2  [G_hat[0,1]]

```

=====

Diagram 1L-52

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v3 (VertexType): bigrade=(2, 1), resp=[('nt', 2), ('nt', 2)], phys=[('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi'_2$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi''_1$$

Edges (propagator assignments):

2 → 3: nt2 → dv2 [G_hat[1,3]]

2 → 4: nt2 → dv1 [G_hat[1,2]]

3 → 0: nt2 → dn1 [G_hat[1,0]]

3 → 4: nt2 → dv1 [G_hat[1,2]]

4 → 1: nt1 → dn2 [G_hat[0,1]]

=====

Diagram 1L-53

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v3 (VertexType): bigrade=(2, 1), resp=[('nt', 2), ('nt', 2)], phys=[('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi'_2$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi''_1$$

Edges (propagator assignments):
 2 → 3: nt2 → dv2 [G_hat[1,3]]
 2 → 4: nt2 → dv1 [G_hat[1,2]]
 3 → 0: nt2 → dn1 [G_hat[1,0]]
 3 → 4: nt2 → dv1 [G_hat[1,2]]
 4 → 1: nt1 → dn2 [G_hat[0,1]]

=====
 Diagram 1L-54
 =====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v3 (VertexType): bigrade=(2, 1), resp=[('nt', 2), ('nt', 2)], phys=[('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi'_2$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi''_1$$

Edges (propagator assignments):

2 → 3: nt2 → dv2 [G_hat[1,3]]

2 → 4: nt2 → dv1 [G_hat[1,2]]

3 → 0: nt2 → dn1 [G_hat[1,0]]

3 → 4: nt2 → dv1 [G_hat[1,2]]

4 → 1: nt1 → dn2 [G_hat[0,1]]

=====
 Diagram 1L-55
 =====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v3 (VertexType): bigrade=(2, 1), resp=[('nt', 2), ('nt', 2)], phys=[('dv', 2)]

$\text{coeff} = \frac{1}{2} \varphi'_2$
 v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]
 $\text{coeff} = \frac{1}{2} \varphi''_1$
 Edges (propagator assignments):
 2 → 3: nt2 → dv2 [G_hat[1,3]]
 2 → 4: nt2 → dv1 [G_hat[1,2]]
 3 → 0: nt2 → dn1 [G_hat[1,0]]
 3 → 4: nt2 → dv1 [G_hat[1,2]]
 4 → 1: nt1 → dn2 [G_hat[0,1]]

=====
 Diagram 1L-56
 =====

External legs:
 leaf 0 ← dn1
 leaf 1 ← dn2
 Vertex assignments:
 v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]
 $\text{coeff} = \frac{1}{2} nstar_2$
 v3 (VertexType): bigrade=(2, 1), resp=[('nt', 2), ('nt', 2)], phys=[('dv', 2)]
 $\text{coeff} = \frac{1}{2} \varphi'_2$
 v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]
 $\text{coeff} = \frac{1}{2} \varphi''_1$

Edges (propagator assignments):
 2 → 3: nt2 → dv2 [G_hat[1,3]]
 2 → 4: nt2 → dv1 [G_hat[1,2]]
 3 → 0: nt2 → dn1 [G_hat[1,0]]
 3 → 4: nt2 → dv1 [G_hat[1,2]]
 4 → 1: nt1 → dn2 [G_hat[0,1]]
 =====
 Diagram 1L-57
 =====
 External legs:
 leaf 0 ← dn1
 leaf 1 ← dn2

```

Vertex assignments:
  v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

coeff =  $\frac{1}{2} nstar_2$ 

  v3 (VertexType): bigrade=(2, 1), resp=[('nt', 2), ('nt', 2)], phys=[('dv',
2)]

coeff =  $\frac{1}{2} \varphi'_2$ 

  v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

coeff =  $\frac{1}{2} \varphi''_2$ 

Edges (propagator assignments):
  2 → 3:  nt2 → dv2  [G_hat[1,3]]
  2 → 4:  nt2 → dv2  [G_hat[1,3]]
  3 → 0:  nt2 → dn1  [G_hat[1,0]]
  3 → 4:  nt2 → dv2  [G_hat[1,3]]
  4 → 1:  nt2 → dn2  [G_hat[1,1]]

```

=====

Diagram 1L-58

=====

```

External legs:
  leaf 0 ← dn1
  leaf 1 ← dn2
Vertex assignments:
  v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

coeff =  $\frac{1}{2} nstar_2$ 

  v3 (VertexType): bigrade=(2, 1), resp=[('nt', 2), ('nt', 2)], phys=[('dv',
2)]

coeff =  $\frac{1}{2} \varphi'_2$ 

  v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

coeff =  $\frac{1}{2} \varphi''_2$ 

Edges (propagator assignments):
  2 → 3:  nt2 → dv2  [G_hat[1,3]]
  2 → 4:  nt2 → dv2  [G_hat[1,3]]
  3 → 0:  nt2 → dn1  [G_hat[1,0]]
  3 → 4:  nt2 → dv2  [G_hat[1,3]]
  4 → 1:  nt2 → dn2  [G_hat[1,1]]

```

=====

Diagram 1L-59

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v3 (VertexType): bigrade=(2, 1), resp=[('nt', 2), ('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi'_2$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi''_2$$

Edges (propagator assignments):

2 → 3: nt2 → dv2 [G_hat[1,3]]

2 → 4: nt2 → dv2 [G_hat[1,3]]

3 → 0: nt2 → dn1 [G_hat[1,0]]

3 → 4: nt2 → dv2 [G_hat[1,3]]

4 → 1: nt2 → dn2 [G_hat[1,1]]

=====

Diagram 1L-60

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v3 (VertexType): bigrade=(2, 1), resp=[('nt', 2), ('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi'_2$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi''_2$$

```

Edges (propagator assignments):
  2 → 3:  nt2 → dv2  [G_hat[1,3]]
  2 → 4:  nt2 → dv2  [G_hat[1,3]]
  3 → 0:  nt2 → dn1  [G_hat[1,0]]
  3 → 4:  nt2 → dv2  [G_hat[1,3]]
  4 → 1:  nt2 → dn2  [G_hat[1,1]]

```

=====

Diagram 1L-61

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v3 (VertexType): bigrade=(2, 1), resp=[('nt', 2), ('nt', 2)], phys=[('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi'_2$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi''_2$$

Edges (propagator assignments):

2 → 3: nt2 → dv2 [G_hat[1,3]]

2 → 4: nt2 → dv2 [G_hat[1,3]]

3 → 0: nt2 → dn1 [G_hat[1,0]]

3 → 4: nt2 → dv2 [G_hat[1,3]]

4 → 1: nt2 → dn2 [G_hat[1,1]]

=====

Diagram 1L-62

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v3 (VertexType): bigrade=(2, 1), resp=[('nt', 2), ('nt', 2)], phys=[('dv', 2)]

$\text{coeff} = \frac{1}{2} \varphi'_2$
 v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]
 $\text{coeff} = \frac{1}{2} \varphi''_2$
 Edges (propagator assignments):
 2 → 3: nt2 → dv2 [G_hat[1,3]]
 2 → 4: nt2 → dv2 [G_hat[1,3]]
 3 → 0: nt2 → dn1 [G_hat[1,0]]
 3 → 4: nt2 → dv2 [G_hat[1,3]]
 4 → 1: nt2 → dn2 [G_hat[1,1]]

=====
 Diagram 1L-63
 =====

External legs:
 leaf 0 ← dn1
 leaf 1 ← dn2
 Vertex assignments:
 v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]
 $\text{coeff} = \frac{1}{2} nstar_2$
 v3 (VertexType): bigrade=(2, 1), resp=[('nt', 2), ('nt', 2)], phys=[('dv', 2), ('dv', 2)]
 $\text{coeff} = \frac{1}{2} \varphi'_2$
 v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]
 $\text{coeff} = \frac{1}{2} \varphi''_2$

Edges (propagator assignments):
 2 → 3: nt2 → dv2 [G_hat[1,3]]
 2 → 4: nt2 → dv2 [G_hat[1,3]]
 3 → 0: nt2 → dn1 [G_hat[1,0]]
 3 → 4: nt2 → dv2 [G_hat[1,3]]
 4 → 1: nt2 → dn2 [G_hat[1,1]]
 =====
 Diagram 1L-64
 =====
 External legs:
 leaf 0 ← dn1
 leaf 1 ← dn2

```

Vertex assignments:
  v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

coeff =  $\frac{1}{2} nstar_2$ 

  v3 (VertexType): bigrade=(2, 1), resp=[('nt', 2), ('nt', 2)], phys=[('dv',
2)]

coeff =  $\frac{1}{2} \varphi'_2$ 

  v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

coeff =  $\frac{1}{2} \varphi''_2$ 

Edges (propagator assignments):
  2 → 3:  nt2 → dv2  [G_hat[1,3]]
  2 → 4:  nt2 → dv2  [G_hat[1,3]]
  3 → 0:  nt2 → dn1  [G_hat[1,0]]
  3 → 4:  nt2 → dv2  [G_hat[1,3]]
  4 → 1:  nt2 → dn2  [G_hat[1,1]]

```

=====

Diagram 1L-65

=====

```

External legs:
  leaf 0 ← dn2
  leaf 1 ← dn1
Vertex assignments:
  v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

coeff =  $\frac{1}{2} nstar_1$ 

  v3 (VertexType): bigrade=(2, 1), resp=[('nt', 1), ('nt', 1)], phys=[('dv',
1)]

coeff =  $\frac{1}{2} \varphi'_1$ 

  v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

coeff =  $\frac{1}{2} \varphi''_1$ 

Edges (propagator assignments):
  2 → 3:  nt1 → dv1  [G_hat[0,2]]
  2 → 4:  nt1 → dv1  [G_hat[0,2]]
  3 → 0:  nt1 → dn2  [G_hat[0,1]]
  3 → 4:  nt1 → dv1  [G_hat[0,2]]
  4 → 1:  nt1 → dn1  [G_hat[0,0]]

```

=====

Diagram 1L-66

=====

External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v3 (VertexType): bigrade=(2, 1), resp=[('nt', 1), ('nt', 1)], phys=[('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi'_1$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi''_1$$

Edges (propagator assignments):

2 → 3: nt1 → dv1 [G_hat[0,2]]

2 → 4: nt1 → dv1 [G_hat[0,2]]

3 → 0: nt1 → dn2 [G_hat[0,1]]

3 → 4: nt1 → dv1 [G_hat[0,2]]

4 → 1: nt1 → dn1 [G_hat[0,0]]

=====

Diagram 1L-67

=====

External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v3 (VertexType): bigrade=(2, 1), resp=[('nt', 1), ('nt', 1)], phys=[('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi'_1$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi''_1$$

Edges (propagator assignments):
 2 → 3: nt1 → dv1 [G_hat[0,2]]
 2 → 4: nt1 → dv1 [G_hat[0,2]]
 3 → 0: nt1 → dn2 [G_hat[0,1]]
 3 → 4: nt1 → dv1 [G_hat[0,2]]
 4 → 1: nt1 → dn1 [G_hat[0,0]]

=====
 Diagram 1L-68
 =====

External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v3 (VertexType): bigrade=(2, 1), resp=[('nt', 1), ('nt', 1)], phys=[('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi'_1$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi''_1$$

Edges (propagator assignments):

2 → 3: nt1 → dv1 [G_hat[0,2]]

2 → 4: nt1 → dv1 [G_hat[0,2]]

3 → 0: nt1 → dn2 [G_hat[0,1]]

3 → 4: nt1 → dv1 [G_hat[0,2]]

4 → 1: nt1 → dn1 [G_hat[0,0]]

=====
 Diagram 1L-69
 =====

External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v3 (VertexType): bigrade=(2, 1), resp=[('nt', 1), ('nt', 1)], phys=[('dv', 1)]

$\text{coeff} = \frac{1}{2} \varphi'_1$
 v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]
 $\text{coeff} = \frac{1}{2} \varphi''_1$
 Edges (propagator assignments):
 2 → 3: nt1 → dv1 [G_hat[0,2]]
 2 → 4: nt1 → dv1 [G_hat[0,2]]
 3 → 0: nt1 → dn2 [G_hat[0,1]]
 3 → 4: nt1 → dv1 [G_hat[0,2]]
 4 → 1: nt1 → dn1 [G_hat[0,0]]

=====
 Diagram 1L-70
 =====

External legs:
 leaf 0 ← dn2
 leaf 1 ← dn1
 Vertex assignments:
 v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]
 $\text{coeff} = \frac{1}{2} nstar_1$
 v3 (VertexType): bigrade=(2, 1), resp=[('nt', 1), ('nt', 1)], phys=[('dv', 1), ('dv', 1)]
 $\text{coeff} = \frac{1}{2} \varphi'_1$
 v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]
 $\text{coeff} = \frac{1}{2} \varphi''_1$

Edges (propagator assignments):
 2 → 3: nt1 → dv1 [G_hat[0,2]]
 2 → 4: nt1 → dv1 [G_hat[0,2]]
 3 → 0: nt1 → dn2 [G_hat[0,1]]
 3 → 4: nt1 → dv1 [G_hat[0,2]]
 4 → 1: nt1 → dn1 [G_hat[0,0]]
 =====
 Diagram 1L-71
 =====
 External legs:
 leaf 0 ← dn2
 leaf 1 ← dn1

```

Vertex assignments:
  v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

coeff =  $\frac{1}{2} nstar_1$ 

  v3 (VertexType): bigrade=(2, 1), resp=[('nt', 1), ('nt', 1)], phys=[('dv',
1)]

coeff =  $\frac{1}{2} \varphi'_1$ 

  v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

coeff =  $\frac{1}{2} \varphi''_1$ 

Edges (propagator assignments):
  2 → 3:  nt1 → dv1  [G_hat[0,2]]
  2 → 4:  nt1 → dv1  [G_hat[0,2]]
  3 → 0:  nt1 → dn2  [G_hat[0,1]]
  3 → 4:  nt1 → dv1  [G_hat[0,2]]
  4 → 1:  nt1 → dn1  [G_hat[0,0]]

```

=====

Diagram 1L-72

=====

```

External legs:
  leaf 0 ← dn2
  leaf 1 ← dn1
Vertex assignments:
  v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

coeff =  $\frac{1}{2} nstar_1$ 

  v3 (VertexType): bigrade=(2, 1), resp=[('nt', 1), ('nt', 1)], phys=[('dv',
1)]

coeff =  $\frac{1}{2} \varphi'_1$ 

  v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

coeff =  $\frac{1}{2} \varphi''_1$ 

Edges (propagator assignments):
  2 → 3:  nt1 → dv1  [G_hat[0,2]]
  2 → 4:  nt1 → dv1  [G_hat[0,2]]
  3 → 0:  nt1 → dn2  [G_hat[0,1]]
  3 → 4:  nt1 → dv1  [G_hat[0,2]]
  4 → 1:  nt1 → dn1  [G_hat[0,0]]

```

=====

Diagram 1L-73

=====

External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v3 (VertexType): bigrade=(2, 1), resp=[('nt', 1), ('nt', 1)], phys=[('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi'_1$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi''_2$$

Edges (propagator assignments):

2 → 3: nt1 → dv1 [G_hat[0,2]]

2 → 4: nt1 → dv2 [G_hat[0,3]]

3 → 0: nt1 → dn2 [G_hat[0,1]]

3 → 4: nt1 → dv2 [G_hat[0,3]]

4 → 1: nt2 → dn1 [G_hat[1,0]]

=====

Diagram 1L-74

=====

External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v3 (VertexType): bigrade=(2, 1), resp=[('nt', 1), ('nt', 1)], phys=[('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi'_1$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi''_2$$

Edges (propagator assignments):
 2 → 3: nt1 → dv1 [G_hat[0,2]]
 2 → 4: nt1 → dv2 [G_hat[0,3]]
 3 → 0: nt1 → dn2 [G_hat[0,1]]
 3 → 4: nt1 → dv2 [G_hat[0,3]]
 4 → 1: nt2 → dn1 [G_hat[1,0]]

=====
 Diagram 1L-75
 =====

External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v3 (VertexType): bigrade=(2, 1), resp=[('nt', 1), ('nt', 1)], phys=[('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi'_1$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi''_2$$

Edges (propagator assignments):

2 → 3: nt1 → dv1 [G_hat[0,2]]

2 → 4: nt1 → dv2 [G_hat[0,3]]

3 → 0: nt1 → dn2 [G_hat[0,1]]

3 → 4: nt1 → dv2 [G_hat[0,3]]

4 → 1: nt2 → dn1 [G_hat[1,0]]

=====
 Diagram 1L-76
 =====

External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v3 (VertexType): bigrade=(2, 1), resp=[('nt', 1), ('nt', 1)], phys=[('dv', 1)]

$\text{coeff} = \frac{1}{2} \varphi'_1$
 v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]
 $\text{coeff} = \frac{1}{2} \varphi''_2$
 Edges (propagator assignments):
 2 → 3: nt1 → dv1 [G_hat[0,2]]
 2 → 4: nt1 → dv2 [G_hat[0,3]]
 3 → 0: nt1 → dn2 [G_hat[0,1]]
 3 → 4: nt1 → dv2 [G_hat[0,3]]
 4 → 1: nt2 → dn1 [G_hat[1,0]]

=====
 Diagram 1L-77
 =====

External legs:
 leaf 0 ← dn2
 leaf 1 ← dn1
 Vertex assignments:
 v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]
 $\text{coeff} = \frac{1}{2} nstar_1$
 v3 (VertexType): bigrade=(2, 1), resp=[('nt', 1), ('nt', 1)], phys=[('dv', 1)]
 $\text{coeff} = \frac{1}{2} \varphi'_1$
 v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]
 $\text{coeff} = \frac{1}{2} \varphi''_2$

Edges (propagator assignments):
 2 → 3: nt1 → dv1 [G_hat[0,2]]
 2 → 4: nt1 → dv2 [G_hat[0,3]]
 3 → 0: nt1 → dn2 [G_hat[0,1]]
 3 → 4: nt1 → dv2 [G_hat[0,3]]
 4 → 1: nt2 → dn1 [G_hat[1,0]]
 =====
 Diagram 1L-78
 =====
 External legs:
 leaf 0 ← dn2
 leaf 1 ← dn1

```

Vertex assignments:
  v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

coeff =  $\frac{1}{2} nstar_1$ 

  v3 (VertexType): bigrade=(2, 1), resp=[('nt', 1), ('nt', 1)], phys=[('dv',
1)]

coeff =  $\frac{1}{2} \varphi'_1$ 

  v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

coeff =  $\frac{1}{2} \varphi''_2$ 

Edges (propagator assignments):
  2 → 3:  nt1 → dv1  [G_hat[0,2]]
  2 → 4:  nt1 → dv2  [G_hat[0,3]]
  3 → 0:  nt1 → dn2  [G_hat[0,1]]
  3 → 4:  nt1 → dv2  [G_hat[0,3]]
  4 → 1:  nt2 → dn1  [G_hat[1,0]]

```

=====

Diagram 1L-79

=====

```

External legs:
  leaf 0 ← dn2
  leaf 1 ← dn1
Vertex assignments:
  v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

coeff =  $\frac{1}{2} nstar_1$ 

  v3 (VertexType): bigrade=(2, 1), resp=[('nt', 1), ('nt', 1)], phys=[('dv',
1)]

coeff =  $\frac{1}{2} \varphi'_1$ 

  v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

coeff =  $\frac{1}{2} \varphi''_2$ 

Edges (propagator assignments):
  2 → 3:  nt1 → dv1  [G_hat[0,2]]
  2 → 4:  nt1 → dv2  [G_hat[0,3]]
  3 → 0:  nt1 → dn2  [G_hat[0,1]]
  3 → 4:  nt1 → dv2  [G_hat[0,3]]
  4 → 1:  nt2 → dn1  [G_hat[1,0]]

```

=====

Diagram 1L-80

=====

External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v3 (VertexType): bigrade=(2, 1), resp=[('nt', 1), ('nt', 1)], phys=[('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi'_1$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi''_2$$

Edges (propagator assignments):

2 → 3: nt1 → dv1 [G_hat[0,2]]

2 → 4: nt1 → dv2 [G_hat[0,3]]

3 → 0: nt1 → dn2 [G_hat[0,1]]

3 → 4: nt1 → dv2 [G_hat[0,3]]

4 → 1: nt2 → dn1 [G_hat[1,0]]

=====

Diagram 1L-81

=====

External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v3 (VertexType): bigrade=(2, 1), resp=[('nt', 2), ('nt', 2)], phys=[('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi'_2$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi''_1$$

```

Edges (propagator assignments):
  2 → 3:  nt1 → dv2  [G_hat[0,3]]
  2 → 4:  nt1 → dv1  [G_hat[0,2]]
  3 → 0:  nt2 → dn2  [G_hat[1,1]]
  3 → 4:  nt2 → dv1  [G_hat[1,2]]
  4 → 1:  nt1 → dn1  [G_hat[0,0]]

```

=====

Diagram 1L-82

=====

External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v3 (VertexType): bigrade=(2, 1), resp=[('nt', 2), ('nt', 2)], phys=[('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi'_2$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi''_1$$

Edges (propagator assignments):

2 → 3: nt1 → dv2 [G_hat[0,3]]

2 → 4: nt1 → dv1 [G_hat[0,2]]

3 → 0: nt2 → dn2 [G_hat[1,1]]

3 → 4: nt2 → dv1 [G_hat[1,2]]

4 → 1: nt1 → dn1 [G_hat[0,0]]

=====

Diagram 1L-83

=====

External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v3 (VertexType): bigrade=(2, 1), resp=[('nt', 2), ('nt', 2)], phys=[('dv', 2)]

$\text{coeff} = \frac{1}{2} \varphi'_2$
 v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]
 $\text{coeff} = \frac{1}{2} \varphi''_1$
 Edges (propagator assignments):
 2 → 3: nt1 → dv2 [G_hat[0,3]]
 2 → 4: nt1 → dv1 [G_hat[0,2]]
 3 → 0: nt2 → dn2 [G_hat[1,1]]
 3 → 4: nt2 → dv1 [G_hat[1,2]]
 4 → 1: nt1 → dn1 [G_hat[0,0]]

=====
 Diagram 1L-84
 =====

External legs:
 leaf 0 ← dn2
 leaf 1 ← dn1
 Vertex assignments:
 v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]
 $\text{coeff} = \frac{1}{2} nstar_1$
 v3 (VertexType): bigrade=(2, 1), resp=[('nt', 2), ('nt', 2)], phys=[('dv', 2)]
 $\text{coeff} = \frac{1}{2} \varphi'_2$
 v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]
 $\text{coeff} = \frac{1}{2} \varphi''_1$

Edges (propagator assignments):
 2 → 3: nt1 → dv2 [G_hat[0,3]]
 2 → 4: nt1 → dv1 [G_hat[0,2]]
 3 → 0: nt2 → dn2 [G_hat[1,1]]
 3 → 4: nt2 → dv1 [G_hat[1,2]]
 4 → 1: nt1 → dn1 [G_hat[0,0]]
 =====
 Diagram 1L-85
 =====
 External legs:
 leaf 0 ← dn2
 leaf 1 ← dn1

```

Vertex assignments:
  v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

coeff =  $\frac{1}{2} nstar_1$ 

  v3 (VertexType): bigrade=(2, 1), resp=[('nt', 2), ('nt', 2)], phys=[('dv',
2)]

coeff =  $\frac{1}{2} \varphi'_2$ 

  v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

coeff =  $\frac{1}{2} \varphi''_1$ 

Edges (propagator assignments):
  2 → 3:  nt1 → dv2  [G_hat[0,3]]
  2 → 4:  nt1 → dv1  [G_hat[0,2]]
  3 → 0:  nt2 → dn2  [G_hat[1,1]]
  3 → 4:  nt2 → dv1  [G_hat[1,2]]
  4 → 1:  nt1 → dn1  [G_hat[0,0]]

```

=====

Diagram 1L-86

=====

```

External legs:
  leaf 0 ← dn2
  leaf 1 ← dn1
Vertex assignments:
  v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

coeff =  $\frac{1}{2} nstar_1$ 

  v3 (VertexType): bigrade=(2, 1), resp=[('nt', 2), ('nt', 2)], phys=[('dv',
2)]

coeff =  $\frac{1}{2} \varphi'_2$ 

  v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

coeff =  $\frac{1}{2} \varphi''_1$ 

Edges (propagator assignments):
  2 → 3:  nt1 → dv2  [G_hat[0,3]]
  2 → 4:  nt1 → dv1  [G_hat[0,2]]
  3 → 0:  nt2 → dn2  [G_hat[1,1]]
  3 → 4:  nt2 → dv1  [G_hat[1,2]]
  4 → 1:  nt1 → dn1  [G_hat[0,0]]

```

=====

Diagram 1L-87

=====

External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v3 (VertexType): bigrade=(2, 1), resp=[('nt', 2), ('nt', 2)], phys=[('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi'_2$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi''_1$$

Edges (propagator assignments):

2 → 3: nt1 → dv2 [G_hat[0,3]]

2 → 4: nt1 → dv1 [G_hat[0,2]]

3 → 0: nt2 → dn2 [G_hat[1,1]]

3 → 4: nt2 → dv1 [G_hat[1,2]]

4 → 1: nt1 → dn1 [G_hat[0,0]]

=====

Diagram 1L-88

=====

External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v3 (VertexType): bigrade=(2, 1), resp=[('nt', 2), ('nt', 2)], phys=[('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi'_2$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi''_1$$

```

Edges (propagator assignments):
  2 → 3:  nt1 → dv2  [G_hat[0,3]]
  2 → 4:  nt1 → dv1  [G_hat[0,2]]
  3 → 0:  nt2 → dn2  [G_hat[1,1]]
  3 → 4:  nt2 → dv1  [G_hat[1,2]]
  4 → 1:  nt1 → dn1  [G_hat[0,0]]

```

=====

Diagram 1L-89

=====

External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v3 (VertexType): bigrade=(2, 1), resp=[('nt', 2), ('nt', 2)], phys=[('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi'_2$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi''_2$$

Edges (propagator assignments):

2 → 3: nt1 → dv2 [G_hat[0,3]]

2 → 4: nt1 → dv2 [G_hat[0,3]]

3 → 0: nt2 → dn2 [G_hat[1,1]]

3 → 4: nt2 → dv2 [G_hat[1,3]]

4 → 1: nt2 → dn1 [G_hat[1,0]]

=====

Diagram 1L-90

=====

External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v3 (VertexType): bigrade=(2, 1), resp=[('nt', 2), ('nt', 2)], phys=[('dv', 2)]

$\text{coeff} = \frac{1}{2} \varphi'_2$
 v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]
 $\text{coeff} = \frac{1}{2} \varphi''_2$
 Edges (propagator assignments):
 2 → 3: nt1 → dv2 [G_hat[0,3]]
 2 → 4: nt1 → dv2 [G_hat[0,3]]
 3 → 0: nt2 → dn2 [G_hat[1,1]]
 3 → 4: nt2 → dv2 [G_hat[1,3]]
 4 → 1: nt2 → dn1 [G_hat[1,0]]

=====
 Diagram 1L-91
 =====

External legs:
 leaf 0 ← dn2
 leaf 1 ← dn1
 Vertex assignments:
 v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]
 $\text{coeff} = \frac{1}{2} nstar_1$
 v3 (VertexType): bigrade=(2, 1), resp=[('nt', 2), ('nt', 2)], phys=[('dv', 2)]
 $\text{coeff} = \frac{1}{2} \varphi'_2$
 v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]
 $\text{coeff} = \frac{1}{2} \varphi''_2$

Edges (propagator assignments):
 2 → 3: nt1 → dv2 [G_hat[0,3]]
 2 → 4: nt1 → dv2 [G_hat[0,3]]
 3 → 0: nt2 → dn2 [G_hat[1,1]]
 3 → 4: nt2 → dv2 [G_hat[1,3]]
 4 → 1: nt2 → dn1 [G_hat[1,0]]
 =====
 Diagram 1L-92
 =====
 External legs:
 leaf 0 ← dn2
 leaf 1 ← dn1

```

Vertex assignments:
  v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

coeff =  $\frac{1}{2} nstar_1$ 

  v3 (VertexType): bigrade=(2, 1), resp=[('nt', 2), ('nt', 2)], phys=[('dv',
2)]

coeff =  $\frac{1}{2} \varphi'_2$ 

  v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

coeff =  $\frac{1}{2} \varphi''_2$ 

Edges (propagator assignments):
  2 → 3:  nt1 → dv2  [G_hat[0,3]]
  2 → 4:  nt1 → dv2  [G_hat[0,3]]
  3 → 0:  nt2 → dn2  [G_hat[1,1]]
  3 → 4:  nt2 → dv2  [G_hat[1,3]]
  4 → 1:  nt2 → dn1  [G_hat[1,0]]

```

=====

Diagram 1L-93

=====

```

External legs:
  leaf 0 ← dn2
  leaf 1 ← dn1
Vertex assignments:
  v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

coeff =  $\frac{1}{2} nstar_1$ 

  v3 (VertexType): bigrade=(2, 1), resp=[('nt', 2), ('nt', 2)], phys=[('dv',
2)]

coeff =  $\frac{1}{2} \varphi'_2$ 

  v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

coeff =  $\frac{1}{2} \varphi''_2$ 

Edges (propagator assignments):
  2 → 3:  nt1 → dv2  [G_hat[0,3]]
  2 → 4:  nt1 → dv2  [G_hat[0,3]]
  3 → 0:  nt2 → dn2  [G_hat[1,1]]
  3 → 4:  nt2 → dv2  [G_hat[1,3]]
  4 → 1:  nt2 → dn1  [G_hat[1,0]]

```

=====

Diagram 1L-94

=====

External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v3 (VertexType): bigrade=(2, 1), resp=[('nt', 2), ('nt', 2)], phys=[('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi'_2$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi''_2$$

Edges (propagator assignments):

2 → 3: nt1 → dv2 [G_hat[0,3]]

2 → 4: nt1 → dv2 [G_hat[0,3]]

3 → 0: nt2 → dn2 [G_hat[1,1]]

3 → 4: nt2 → dv2 [G_hat[1,3]]

4 → 1: nt2 → dn1 [G_hat[1,0]]

=====

Diagram 1L-95

=====

External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v3 (VertexType): bigrade=(2, 1), resp=[('nt', 2), ('nt', 2)], phys=[('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi'_2$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi''_2$$

```

Edges (propagator assignments):
  2 → 3:  nt1 → dv2  [G_hat[0,3]]
  2 → 4:  nt1 → dv2  [G_hat[0,3]]
  3 → 0:  nt2 → dn2  [G_hat[1,1]]
  3 → 4:  nt2 → dv2  [G_hat[1,3]]
  4 → 1:  nt2 → dn1  [G_hat[1,0]]

```

=====

Diagram 1L-96

=====

External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v3 (VertexType): bigrade=(2, 1), resp=[('nt', 2), ('nt', 2)], phys=[('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi'_2$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi''_2$$

Edges (propagator assignments):

2 → 3: nt1 → dv2 [G_hat[0,3]]

2 → 4: nt1 → dv2 [G_hat[0,3]]

3 → 0: nt2 → dn2 [G_hat[1,1]]

3 → 4: nt2 → dv2 [G_hat[1,3]]

4 → 1: nt2 → dn1 [G_hat[1,0]]

=====

Diagram 1L-97

=====

External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v3 (VertexType): bigrade=(2, 1), resp=[('nt', 1), ('nt', 1)], phys=[('dv', 1)]

$\text{coeff} = \frac{1}{2} \varphi'_1$
 v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]
 $\text{coeff} = \frac{1}{2} \varphi''_1$
 Edges (propagator assignments):
 2 → 3: nt2 → dv1 [G_hat[1,2]]
 2 → 4: nt2 → dv1 [G_hat[1,2]]
 3 → 0: nt1 → dn2 [G_hat[0,1]]
 3 → 4: nt1 → dv1 [G_hat[0,2]]
 4 → 1: nt1 → dn1 [G_hat[0,0]]

=====
 Diagram 1L-98
 =====

External legs:
 leaf 0 ← dn2
 leaf 1 ← dn1
 Vertex assignments:
 v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]
 $\text{coeff} = \frac{1}{2} nstar_2$
 v3 (VertexType): bigrade=(2, 1), resp=[('nt', 1), ('nt', 1)], phys=[('dv', 1), ('dv', 1)]
 $\text{coeff} = \frac{1}{2} \varphi'_1$
 v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]
 $\text{coeff} = \frac{1}{2} \varphi''_1$

Edges (propagator assignments):
 2 → 3: nt2 → dv1 [G_hat[1,2]]
 2 → 4: nt2 → dv1 [G_hat[1,2]]
 3 → 0: nt1 → dn2 [G_hat[0,1]]
 3 → 4: nt1 → dv1 [G_hat[0,2]]
 4 → 1: nt1 → dn1 [G_hat[0,0]]
 =====
 Diagram 1L-99
 =====
 External legs:
 leaf 0 ← dn2
 leaf 1 ← dn1

```

Vertex assignments:
  v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

coeff =  $\frac{1}{2} nstar_2$ 

  v3 (VertexType): bigrade=(2, 1), resp=[('nt', 1), ('nt', 1)], phys=[('dv',
1)]

coeff =  $\frac{1}{2} \varphi'_1$ 

  v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

coeff =  $\frac{1}{2} \varphi''_1$ 

Edges (propagator assignments):
  2 → 3:  nt2 → dv1  [G_hat[1,2]]
  2 → 4:  nt2 → dv1  [G_hat[1,2]]
  3 → 0:  nt1 → dn2  [G_hat[0,1]]
  3 → 4:  nt1 → dv1  [G_hat[0,2]]
  4 → 1:  nt1 → dn1  [G_hat[0,0]]

```

=====
Diagram 1L-100
=====

```

External legs:
  leaf 0 ← dn2
  leaf 1 ← dn1
Vertex assignments:
  v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

coeff =  $\frac{1}{2} nstar_2$ 

  v3 (VertexType): bigrade=(2, 1), resp=[('nt', 1), ('nt', 1)], phys=[('dv',
1)]

coeff =  $\frac{1}{2} \varphi'_1$ 

  v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

coeff =  $\frac{1}{2} \varphi''_1$ 

Edges (propagator assignments):
  2 → 3:  nt2 → dv1  [G_hat[1,2]]
  2 → 4:  nt2 → dv1  [G_hat[1,2]]
  3 → 0:  nt1 → dn2  [G_hat[0,1]]
  3 → 4:  nt1 → dv1  [G_hat[0,2]]
  4 → 1:  nt1 → dn1  [G_hat[0,0]]

```

=====

Diagram 1L-101

=====

External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v3 (VertexType): bigrade=(2, 1), resp=[('nt', 1), ('nt', 1)], phys=[('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi'_1$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi''_1$$

Edges (propagator assignments):

2 → 3: nt2 → dv1 [G_hat[1,2]]

2 → 4: nt2 → dv1 [G_hat[1,2]]

3 → 0: nt1 → dn2 [G_hat[0,1]]

3 → 4: nt1 → dv1 [G_hat[0,2]]

4 → 1: nt1 → dn1 [G_hat[0,0]]

=====

Diagram 1L-102

=====

External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v3 (VertexType): bigrade=(2, 1), resp=[('nt', 1), ('nt', 1)], phys=[('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi'_1$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi''_1$$

Edges (propagator assignments):
 2 → 3: nt2 → dv1 [G_hat[1,2]]
 2 → 4: nt2 → dv1 [G_hat[1,2]]
 3 → 0: nt1 → dn2 [G_hat[0,1]]
 3 → 4: nt1 → dv1 [G_hat[0,2]]
 4 → 1: nt1 → dn1 [G_hat[0,0]]

=====
 Diagram 1L-103
 =====

External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v3 (VertexType): bigrade=(2, 1), resp=[('nt', 1), ('nt', 1)], phys=[('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi'_1$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi''_1$$

Edges (propagator assignments):

2 → 3: nt2 → dv1 [G_hat[1,2]]

2 → 4: nt2 → dv1 [G_hat[1,2]]

3 → 0: nt1 → dn2 [G_hat[0,1]]

3 → 4: nt1 → dv1 [G_hat[0,2]]

4 → 1: nt1 → dn1 [G_hat[0,0]]

=====
 Diagram 1L-104
 =====

External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v3 (VertexType): bigrade=(2, 1), resp=[('nt', 1), ('nt', 1)], phys=[('dv', 1)]

$\text{coeff} = \frac{1}{2} \varphi'_1$
 v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]
 $\text{coeff} = \frac{1}{2} \varphi''_1$
 Edges (propagator assignments):
 2 → 3: nt2 → dv1 [G_hat[1,2]]
 2 → 4: nt2 → dv1 [G_hat[1,2]]
 3 → 0: nt1 → dn2 [G_hat[0,1]]
 3 → 4: nt1 → dv1 [G_hat[0,2]]
 4 → 1: nt1 → dn1 [G_hat[0,0]]

=====
 Diagram 1L-105
 =====

External legs:
 leaf 0 ← dn2
 leaf 1 ← dn1
 Vertex assignments:
 v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]
 $\text{coeff} = \frac{1}{2} nstar_2$
 v3 (VertexType): bigrade=(2, 1), resp=[('nt', 1), ('nt', 1)], phys=[('dv', 1)]
 $\text{coeff} = \frac{1}{2} \varphi'_1$
 v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]
 $\text{coeff} = \frac{1}{2} \varphi''_2$

Edges (propagator assignments):
 2 → 3: nt2 → dv1 [G_hat[1,2]]
 2 → 4: nt2 → dv2 [G_hat[1,3]]
 3 → 0: nt1 → dn2 [G_hat[0,1]]
 3 → 4: nt1 → dv2 [G_hat[0,3]]
 4 → 1: nt2 → dn1 [G_hat[1,0]]
 =====
 Diagram 1L-106
 =====
 External legs:
 leaf 0 ← dn2
 leaf 1 ← dn1

```

Vertex assignments:
  v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

coeff =  $\frac{1}{2} nstar_2$ 

  v3 (VertexType): bigrade=(2, 1), resp=[('nt', 1), ('nt', 1)], phys=[('dv',
1)]

coeff =  $\frac{1}{2} \varphi'_1$ 

  v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

coeff =  $\frac{1}{2} \varphi''_2$ 

Edges (propagator assignments):
  2 → 3:  nt2 → dv1  [G_hat[1,2]]
  2 → 4:  nt2 → dv2  [G_hat[1,3]]
  3 → 0:  nt1 → dn2  [G_hat[0,1]]
  3 → 4:  nt1 → dv2  [G_hat[0,3]]
  4 → 1:  nt2 → dn1  [G_hat[1,0]]

```

=====

Diagram 1L-107

=====

```

External legs:
  leaf 0 ← dn2
  leaf 1 ← dn1
Vertex assignments:
  v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

coeff =  $\frac{1}{2} nstar_2$ 

  v3 (VertexType): bigrade=(2, 1), resp=[('nt', 1), ('nt', 1)], phys=[('dv',
1)]

coeff =  $\frac{1}{2} \varphi'_1$ 

  v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

coeff =  $\frac{1}{2} \varphi''_2$ 

Edges (propagator assignments):
  2 → 3:  nt2 → dv1  [G_hat[1,2]]
  2 → 4:  nt2 → dv2  [G_hat[1,3]]
  3 → 0:  nt1 → dn2  [G_hat[0,1]]
  3 → 4:  nt1 → dv2  [G_hat[0,3]]
  4 → 1:  nt2 → dn1  [G_hat[1,0]]

```

=====

Diagram 1L-108

=====

External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v3 (VertexType): bigrade=(2, 1), resp=[('nt', 1), ('nt', 1)], phys=[('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi'_1$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi''_2$$

Edges (propagator assignments):

2 → 3: nt2 → dv1 [G_hat[1,2]]

2 → 4: nt2 → dv2 [G_hat[1,3]]

3 → 0: nt1 → dn2 [G_hat[0,1]]

3 → 4: nt1 → dv2 [G_hat[0,3]]

4 → 1: nt2 → dn1 [G_hat[1,0]]

=====

Diagram 1L-109

=====

External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v3 (VertexType): bigrade=(2, 1), resp=[('nt', 1), ('nt', 1)], phys=[('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi'_1$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi''_2$$

Edges (propagator assignments):
 2 → 3: nt2 → dv1 [G_hat[1,2]]
 2 → 4: nt2 → dv2 [G_hat[1,3]]
 3 → 0: nt1 → dn2 [G_hat[0,1]]
 3 → 4: nt1 → dv2 [G_hat[0,3]]
 4 → 1: nt2 → dn1 [G_hat[1,0]]

=====
 Diagram 1L-110
 =====

External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v3 (VertexType): bigrade=(2, 1), resp=[('nt', 1), ('nt', 1)], phys=[('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi'_1$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi''_2$$

Edges (propagator assignments):

2 → 3: nt2 → dv1 [G_hat[1,2]]

2 → 4: nt2 → dv2 [G_hat[1,3]]

3 → 0: nt1 → dn2 [G_hat[0,1]]

3 → 4: nt1 → dv2 [G_hat[0,3]]

4 → 1: nt2 → dn1 [G_hat[1,0]]

=====
 Diagram 1L-111
 =====

External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v3 (VertexType): bigrade=(2, 1), resp=[('nt', 1), ('nt', 1)], phys=[('dv', 1)]

$\text{coeff} = \frac{1}{2} \varphi'_1$
 v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]
 $\text{coeff} = \frac{1}{2} \varphi''_2$
 Edges (propagator assignments):
 2 → 3: nt2 → dv1 [G_hat[1,2]]
 2 → 4: nt2 → dv2 [G_hat[1,3]]
 3 → 0: nt1 → dn2 [G_hat[0,1]]
 3 → 4: nt1 → dv2 [G_hat[0,3]]
 4 → 1: nt2 → dn1 [G_hat[1,0]]

=====
 Diagram 1L-112
 =====

External legs:
 leaf 0 ← dn2
 leaf 1 ← dn1
 Vertex assignments:
 v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]
 $\text{coeff} = \frac{1}{2} nstar_2$
 v3 (VertexType): bigrade=(2, 1), resp=[('nt', 1), ('nt', 1)], phys=[('dv', 1)]
 $\text{coeff} = \frac{1}{2} \varphi'_1$
 v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]
 $\text{coeff} = \frac{1}{2} \varphi''_2$

Edges (propagator assignments):
 2 → 3: nt2 → dv1 [G_hat[1,2]]
 2 → 4: nt2 → dv2 [G_hat[1,3]]
 3 → 0: nt1 → dn2 [G_hat[0,1]]
 3 → 4: nt1 → dv2 [G_hat[0,3]]
 4 → 1: nt2 → dn1 [G_hat[1,0]]
 =====
 Diagram 1L-113
 =====
 External legs:
 leaf 0 ← dn2
 leaf 1 ← dn1

```

Vertex assignments:
  v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

coeff =  $\frac{1}{2} nstar_2$ 

  v3 (VertexType): bigrade=(2, 1), resp=[('nt', 2), ('nt', 2)], phys=[('dv',
2)]

coeff =  $\frac{1}{2} \varphi'_2$ 

  v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

coeff =  $\frac{1}{2} \varphi''_1$ 

Edges (propagator assignments):
  2 → 3:  nt2 → dv2  [G_hat[1,3]]
  2 → 4:  nt2 → dv1  [G_hat[1,2]]
  3 → 0:  nt2 → dn2  [G_hat[1,1]]
  3 → 4:  nt2 → dv1  [G_hat[1,2]]
  4 → 1:  nt1 → dn1  [G_hat[0,0]]

```

=====

Diagram 1L-114

=====

```

External legs:
  leaf 0 ← dn2
  leaf 1 ← dn1
Vertex assignments:
  v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

coeff =  $\frac{1}{2} nstar_2$ 

  v3 (VertexType): bigrade=(2, 1), resp=[('nt', 2), ('nt', 2)], phys=[('dv',
2)]

coeff =  $\frac{1}{2} \varphi'_2$ 

  v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

coeff =  $\frac{1}{2} \varphi''_1$ 

Edges (propagator assignments):
  2 → 3:  nt2 → dv2  [G_hat[1,3]]
  2 → 4:  nt2 → dv1  [G_hat[1,2]]
  3 → 0:  nt2 → dn2  [G_hat[1,1]]
  3 → 4:  nt2 → dv1  [G_hat[1,2]]
  4 → 1:  nt1 → dn1  [G_hat[0,0]]

```

=====

Diagram 1L-115

=====

External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v3 (VertexType): bigrade=(2, 1), resp=[('nt', 2), ('nt', 2)], phys=[('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi'_2$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi''_1$$

Edges (propagator assignments):

2 → 3: nt2 → dv2 [G_hat[1,3]]

2 → 4: nt2 → dv1 [G_hat[1,2]]

3 → 0: nt2 → dn2 [G_hat[1,1]]

3 → 4: nt2 → dv1 [G_hat[1,2]]

4 → 1: nt1 → dn1 [G_hat[0,0]]

=====

Diagram 1L-116

=====

External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v3 (VertexType): bigrade=(2, 1), resp=[('nt', 2), ('nt', 2)], phys=[('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi'_2$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi''_1$$

Edges (propagator assignments):
 2 → 3: nt2 → dv2 [G_hat[1,3]]
 2 → 4: nt2 → dv1 [G_hat[1,2]]
 3 → 0: nt2 → dn2 [G_hat[1,1]]
 3 → 4: nt2 → dv1 [G_hat[1,2]]
 4 → 1: nt1 → dn1 [G_hat[0,0]]

=====
 Diagram 1L-117
 =====

External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v3 (VertexType): bigrade=(2, 1), resp=[('nt', 2), ('nt', 2)], phys=[('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi'_2$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi''_1$$

Edges (propagator assignments):

2 → 3: nt2 → dv2 [G_hat[1,3]]

2 → 4: nt2 → dv1 [G_hat[1,2]]

3 → 0: nt2 → dn2 [G_hat[1,1]]

3 → 4: nt2 → dv1 [G_hat[1,2]]

4 → 1: nt1 → dn1 [G_hat[0,0]]

=====
 Diagram 1L-118
 =====

External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v3 (VertexType): bigrade=(2, 1), resp=[('nt', 2), ('nt', 2)], phys=[('dv', 2)]

$\text{coeff} = \frac{1}{2} \varphi'_2$
 v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]
 $\text{coeff} = \frac{1}{2} \varphi''_1$
 Edges (propagator assignments):
 2 → 3: nt2 → dv2 [G_hat[1,3]]
 2 → 4: nt2 → dv1 [G_hat[1,2]]
 3 → 0: nt2 → dn2 [G_hat[1,1]]
 3 → 4: nt2 → dv1 [G_hat[1,2]]
 4 → 1: nt1 → dn1 [G_hat[0,0]]

=====
 Diagram 1L-119
 =====

External legs:
 leaf 0 ← dn2
 leaf 1 ← dn1
 Vertex assignments:
 v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]
 $\text{coeff} = \frac{1}{2} nstar_2$
 v3 (VertexType): bigrade=(2, 1), resp=[('nt', 2), ('nt', 2)], phys=[('dv', 2)]
 $\text{coeff} = \frac{1}{2} \varphi'_2$
 v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]
 $\text{coeff} = \frac{1}{2} \varphi''_1$

Edges (propagator assignments):
 2 → 3: nt2 → dv2 [G_hat[1,3]]
 2 → 4: nt2 → dv1 [G_hat[1,2]]
 3 → 0: nt2 → dn2 [G_hat[1,1]]
 3 → 4: nt2 → dv1 [G_hat[1,2]]
 4 → 1: nt1 → dn1 [G_hat[0,0]]
 =====
 Diagram 1L-120
 =====
 External legs:
 leaf 0 ← dn2
 leaf 1 ← dn1

```

Vertex assignments:
  v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

coeff =  $\frac{1}{2} nstar_2$ 

  v3 (VertexType): bigrade=(2, 1), resp=[('nt', 2), ('nt', 2)], phys=[('dv',
2)]

coeff =  $\frac{1}{2} \varphi'_2$ 

  v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

coeff =  $\frac{1}{2} \varphi''_1$ 

Edges (propagator assignments):
  2 → 3:  nt2 → dv2  [G_hat[1,3]]
  2 → 4:  nt2 → dv1  [G_hat[1,2]]
  3 → 0:  nt2 → dn2  [G_hat[1,1]]
  3 → 4:  nt2 → dv1  [G_hat[1,2]]
  4 → 1:  nt1 → dn1  [G_hat[0,0]]

```

=====

Diagram 1L-121

=====

```

External legs:
  leaf 0 ← dn2
  leaf 1 ← dn1
Vertex assignments:
  v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

coeff =  $\frac{1}{2} nstar_2$ 

  v3 (VertexType): bigrade=(2, 1), resp=[('nt', 2), ('nt', 2)], phys=[('dv',
2)]

coeff =  $\frac{1}{2} \varphi'_2$ 

  v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

coeff =  $\frac{1}{2} \varphi''_2$ 

Edges (propagator assignments):
  2 → 3:  nt2 → dv2  [G_hat[1,3]]
  2 → 4:  nt2 → dv2  [G_hat[1,3]]
  3 → 0:  nt2 → dn2  [G_hat[1,1]]
  3 → 4:  nt2 → dv2  [G_hat[1,3]]
  4 → 1:  nt2 → dn1  [G_hat[1,0]]

```

=====

Diagram 1L-122

=====

External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v3 (VertexType): bigrade=(2, 1), resp=[('nt', 2), ('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi'_2$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi''_2$$

Edges (propagator assignments):

2 → 3: nt2 → dv2 [G_hat[1,3]]

2 → 4: nt2 → dv2 [G_hat[1,3]]

3 → 0: nt2 → dn2 [G_hat[1,1]]

3 → 4: nt2 → dv2 [G_hat[1,3]]

4 → 1: nt2 → dn1 [G_hat[1,0]]

=====

Diagram 1L-123

=====

External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v3 (VertexType): bigrade=(2, 1), resp=[('nt', 2), ('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi'_2$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi''_2$$

Edges (propagator assignments):
 2 → 3: nt2 → dv2 [G_hat[1,3]]
 2 → 4: nt2 → dv2 [G_hat[1,3]]
 3 → 0: nt2 → dn2 [G_hat[1,1]]
 3 → 4: nt2 → dv2 [G_hat[1,3]]
 4 → 1: nt2 → dn1 [G_hat[1,0]]

=====
 Diagram 1L-124
 =====

External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v3 (VertexType): bigrade=(2, 1), resp=[('nt', 2), ('nt', 2)], phys=[('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi'_2$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi''_2$$

Edges (propagator assignments):

2 → 3: nt2 → dv2 [G_hat[1,3]]

2 → 4: nt2 → dv2 [G_hat[1,3]]

3 → 0: nt2 → dn2 [G_hat[1,1]]

3 → 4: nt2 → dv2 [G_hat[1,3]]

4 → 1: nt2 → dn1 [G_hat[1,0]]

=====
 Diagram 1L-125
 =====

External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v3 (VertexType): bigrade=(2, 1), resp=[('nt', 2), ('nt', 2)], phys=[('dv', 2)]

$\text{coeff} = \frac{1}{2} \varphi'_2$
 v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]
 $\text{coeff} = \frac{1}{2} \varphi''_2$
 Edges (propagator assignments):
 2 → 3: nt2 → dv2 [G_hat[1,3]]
 2 → 4: nt2 → dv2 [G_hat[1,3]]
 3 → 0: nt2 → dn2 [G_hat[1,1]]
 3 → 4: nt2 → dv2 [G_hat[1,3]]
 4 → 1: nt2 → dn1 [G_hat[1,0]]

=====
 Diagram 1L-126
 =====

External legs:
 leaf 0 ← dn2
 leaf 1 ← dn1
 Vertex assignments:
 v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]
 $\text{coeff} = \frac{1}{2} nstar_2$
 v3 (VertexType): bigrade=(2, 1), resp=[('nt', 2), ('nt', 2)], phys=[('dv', 2), ('dv', 2)]
 $\text{coeff} = \frac{1}{2} \varphi'_2$
 v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]
 $\text{coeff} = \frac{1}{2} \varphi''_2$

Edges (propagator assignments):
 2 → 3: nt2 → dv2 [G_hat[1,3]]
 2 → 4: nt2 → dv2 [G_hat[1,3]]
 3 → 0: nt2 → dn2 [G_hat[1,1]]
 3 → 4: nt2 → dv2 [G_hat[1,3]]
 4 → 1: nt2 → dn1 [G_hat[1,0]]
 =====
 Diagram 1L-127
 =====
 External legs:
 leaf 0 ← dn2
 leaf 1 ← dn1

```

Vertex assignments:
  v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

coeff =  $\frac{1}{2} nstar_2$ 

  v3 (VertexType): bigrade=(2, 1), resp=[('nt', 2), ('nt', 2)], phys=[('dv',
2)]

coeff =  $\frac{1}{2} \varphi'_2$ 

  v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

coeff =  $\frac{1}{2} \varphi''_2$ 

Edges (propagator assignments):
  2 → 3:  nt2 → dv2  [G_hat[1,3]]
  2 → 4:  nt2 → dv2  [G_hat[1,3]]
  3 → 0:  nt2 → dn2  [G_hat[1,1]]
  3 → 4:  nt2 → dv2  [G_hat[1,3]]
  4 → 1:  nt2 → dn1  [G_hat[1,0]]

```

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Diagram 1L-128

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External legs:
  leaf 0 ← dn2
  leaf 1 ← dn1
Vertex assignments:
  v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

coeff =  $\frac{1}{2} nstar_2$ 

  v3 (VertexType): bigrade=(2, 1), resp=[('nt', 2), ('nt', 2)], phys=[('dv',
2)]

coeff =  $\frac{1}{2} \varphi'_2$ 

  v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

coeff =  $\frac{1}{2} \varphi''_2$ 

Edges (propagator assignments):
  2 → 3:  nt2 → dv2  [G_hat[1,3]]
  2 → 4:  nt2 → dv2  [G_hat[1,3]]
  3 → 0:  nt2 → dn2  [G_hat[1,1]]
  3 → 4:  nt2 → dv2  [G_hat[1,3]]
  4 → 1:  nt2 → dn1  [G_hat[1,0]]

```

=====

Diagram 1L-129

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

Edges (propagator assignments):

2 → 4: nt1 → dv1 [G_hat[0,2]]

3 → 2: nt1 → dv1 [G_hat[0,2]]

3 → 4: nt1 → dv1 [G_hat[0,2]]

4 → 0: nt1 → dn1 [G_hat[0,0]]

5 → 1: nt1 → dn2 [G_hat[0,1]]

5 → 2: nt1 → dv1 [G_hat[0,2]]

=====

Diagram 1L-130

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

Edges (propagator assignments):

2 → 4: nt1 → dv1 [G_hat[0,2]]
 3 → 2: nt1 → dv1 [G_hat[0,2]]
 3 → 4: nt1 → dv1 [G_hat[0,2]]
 4 → 0: nt1 → dn1 [G_hat[0,0]]
 5 → 1: nt1 → dn2 [G_hat[0,1]]
 5 → 2: nt1 → dv1 [G_hat[0,2]]

=====
 Diagram 1L-131
 =====

External legs:

leaf 0 ← dn1
 leaf 1 ← dn2

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

Edges (propagator assignments):

2 → 4: nt1 → dv1 [G_hat[0,2]]
 3 → 2: nt1 → dv1 [G_hat[0,2]]
 3 → 4: nt1 → dv1 [G_hat[0,2]]
 4 → 0: nt1 → dn1 [G_hat[0,0]]
 5 → 1: nt1 → dn2 [G_hat[0,1]]
 5 → 2: nt1 → dv1 [G_hat[0,2]]

=====

Diagram 1L-132

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External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

Edges (propagator assignments):

2 → 4: nt1 → dv1 [G_hat[0,2]]

3 → 2: nt1 → dv1 [G_hat[0,2]]

3 → 4: nt1 → dv1 [G_hat[0,2]]

4 → 0: nt1 → dn1 [G_hat[0,0]]

5 → 1: nt1 → dn2 [G_hat[0,1]]

5 → 2: nt1 → dv1 [G_hat[0,2]]

=====

Diagram 1L-133

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

Edges (propagator assignments):

2 → 4: nt1 → dv1 [G_hat[0,2]]
 3 → 2: nt1 → dv1 [G_hat[0,2]]
 3 → 4: nt1 → dv1 [G_hat[0,2]]
 4 → 0: nt1 → dn1 [G_hat[0,0]]
 5 → 1: nt1 → dn2 [G_hat[0,1]]
 5 → 2: nt1 → dv1 [G_hat[0,2]]

=====
 Diagram 1L-134
 =====

External legs:

leaf 0 ← dn1
 leaf 1 ← dn2

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

Edges (propagator assignments):

2 → 4: nt1 → dv1 [G_hat[0,2]]
 3 → 2: nt1 → dv1 [G_hat[0,2]]
 3 → 4: nt1 → dv1 [G_hat[0,2]]
 4 → 0: nt1 → dn1 [G_hat[0,0]]
 5 → 1: nt1 → dn2 [G_hat[0,1]]
 5 → 2: nt1 → dv1 [G_hat[0,2]]

=====

Diagram 1L-135

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

Edges (propagator assignments):

2 → 4: nt1 → dv1 [G_hat[0,2]]

3 → 2: nt1 → dv1 [G_hat[0,2]]

3 → 4: nt1 → dv1 [G_hat[0,2]]

4 → 0: nt1 → dn1 [G_hat[0,0]]

5 → 1: nt1 → dn2 [G_hat[0,1]]

5 → 2: nt1 → dv1 [G_hat[0,2]]

=====

Diagram 1L-136

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

Edges (propagator assignments):

2 → 4: nt1 → dv1 [G_hat[0,2]]
 3 → 2: nt1 → dv1 [G_hat[0,2]]
 3 → 4: nt1 → dv1 [G_hat[0,2]]
 4 → 0: nt1 → dn1 [G_hat[0,0]]
 5 → 1: nt1 → dn2 [G_hat[0,1]]
 5 → 2: nt1 → dv1 [G_hat[0,2]]

=====
 Diagram 1L-137
 =====

External legs:

leaf 0 ← dn1
 leaf 1 ← dn2

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

Edges (propagator assignments):

2 → 4: nt1 → dv1 [G_hat[0,2]]
 3 → 2: nt1 → dv1 [G_hat[0,2]]
 3 → 4: nt1 → dv1 [G_hat[0,2]]
 4 → 0: nt1 → dn1 [G_hat[0,0]]
 5 → 1: nt1 → dn2 [G_hat[0,1]]
 5 → 2: nt1 → dv1 [G_hat[0,2]]

=====

Diagram 1L-138

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

Edges (propagator assignments):

2 → 4: nt1 → dv1 [G_hat[0,2]]

3 → 2: nt1 → dv1 [G_hat[0,2]]

3 → 4: nt1 → dv1 [G_hat[0,2]]

4 → 0: nt1 → dn1 [G_hat[0,0]]

5 → 1: nt1 → dn2 [G_hat[0,1]]

5 → 2: nt1 → dv1 [G_hat[0,2]]

=====

Diagram 1L-139

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

Edges (propagator assignments):

2 → 4: nt1 → dv1 [G_hat[0,2]]
 3 → 2: nt1 → dv1 [G_hat[0,2]]
 3 → 4: nt1 → dv1 [G_hat[0,2]]
 4 → 0: nt1 → dn1 [G_hat[0,0]]
 5 → 1: nt1 → dn2 [G_hat[0,1]]
 5 → 2: nt1 → dv1 [G_hat[0,2]]

=====
 Diagram 1L-140
 =====

External legs:

leaf 0 ← dn1
 leaf 1 ← dn2

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

Edges (propagator assignments):

2 → 4: nt1 → dv1 [G_hat[0,2]]
 3 → 2: nt1 → dv1 [G_hat[0,2]]
 3 → 4: nt1 → dv1 [G_hat[0,2]]
 4 → 0: nt1 → dn1 [G_hat[0,0]]
 5 → 1: nt1 → dn2 [G_hat[0,1]]
 5 → 2: nt1 → dv1 [G_hat[0,2]]

=====

Diagram 1L-141

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

Edges (propagator assignments):

2 → 4: nt1 → dv1 [G_hat[0,2]]

3 → 2: nt1 → dv1 [G_hat[0,2]]

3 → 4: nt1 → dv1 [G_hat[0,2]]

4 → 0: nt1 → dn1 [G_hat[0,0]]

5 → 1: nt1 → dn2 [G_hat[0,1]]

5 → 2: nt1 → dv1 [G_hat[0,2]]

=====

Diagram 1L-142

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

Edges (propagator assignments):

2 → 4: nt1 → dv1 [G_hat[0,2]]
 3 → 2: nt1 → dv1 [G_hat[0,2]]
 3 → 4: nt1 → dv1 [G_hat[0,2]]
 4 → 0: nt1 → dn1 [G_hat[0,0]]
 5 → 1: nt1 → dn2 [G_hat[0,1]]
 5 → 2: nt1 → dv1 [G_hat[0,2]]

=====
 Diagram 1L-143
 =====

External legs:

leaf 0 ← dn1
 leaf 1 ← dn2

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

Edges (propagator assignments):

2 → 4: nt1 → dv1 [G_hat[0,2]]
 3 → 2: nt1 → dv1 [G_hat[0,2]]
 3 → 4: nt1 → dv1 [G_hat[0,2]]
 4 → 0: nt1 → dn1 [G_hat[0,0]]
 5 → 1: nt1 → dn2 [G_hat[0,1]]
 5 → 2: nt1 → dv1 [G_hat[0,2]]

=====

Diagram 1L-144

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

Edges (propagator assignments):

2 → 4: nt1 → dv1 [G_hat[0,2]]

3 → 2: nt1 → dv1 [G_hat[0,2]]

3 → 4: nt1 → dv1 [G_hat[0,2]]

4 → 0: nt1 → dn1 [G_hat[0,0]]

5 → 1: nt1 → dn2 [G_hat[0,1]]

5 → 2: nt1 → dv1 [G_hat[0,2]]

=====

Diagram 1L-145

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

Edges (propagator assignments):

2 → 4: nt1 → dv2 [G_hat[0,3]]
 3 → 2: nt1 → dv1 [G_hat[0,2]]
 3 → 4: nt1 → dv2 [G_hat[0,3]]
 4 → 0: nt2 → dn1 [G_hat[1,0]]
 5 → 1: nt1 → dn2 [G_hat[0,1]]
 5 → 2: nt1 → dv1 [G_hat[0,2]]

=====
 Diagram 1L-146
 =====

External legs:

leaf 0 ← dn1
 leaf 1 ← dn2

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

Edges (propagator assignments):

2 → 4: nt1 → dv2 [G_hat[0,3]]
 3 → 2: nt1 → dv1 [G_hat[0,2]]
 3 → 4: nt1 → dv2 [G_hat[0,3]]
 4 → 0: nt2 → dn1 [G_hat[1,0]]
 5 → 1: nt1 → dn2 [G_hat[0,1]]
 5 → 2: nt1 → dv1 [G_hat[0,2]]

=====

Diagram 1L-147

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

Edges (propagator assignments):

2 → 4: nt1 → dv2 [G_hat[0,3]]

3 → 2: nt1 → dv1 [G_hat[0,2]]

3 → 4: nt1 → dv2 [G_hat[0,3]]

4 → 0: nt2 → dn1 [G_hat[1,0]]

5 → 1: nt1 → dn2 [G_hat[0,1]]

5 → 2: nt1 → dv1 [G_hat[0,2]]

=====

Diagram 1L-148

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External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

Edges (propagator assignments):

2 → 4: nt1 → dv2 [G_hat[0,3]]
 3 → 2: nt1 → dv1 [G_hat[0,2]]
 3 → 4: nt1 → dv2 [G_hat[0,3]]
 4 → 0: nt2 → dn1 [G_hat[1,0]]
 5 → 1: nt1 → dn2 [G_hat[0,1]]
 5 → 2: nt1 → dv1 [G_hat[0,2]]

=====
 Diagram 1L-149
 =====

External legs:

leaf 0 ← dn1
 leaf 1 ← dn2

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

Edges (propagator assignments):

2 → 4: nt1 → dv2 [G_hat[0,3]]
 3 → 2: nt1 → dv1 [G_hat[0,2]]
 3 → 4: nt1 → dv2 [G_hat[0,3]]
 4 → 0: nt2 → dn1 [G_hat[1,0]]
 5 → 1: nt1 → dn2 [G_hat[0,1]]
 5 → 2: nt1 → dv1 [G_hat[0,2]]


```

=====
Diagram 1L-150
=====
External legs:
  leaf 0 ← dn1
  leaf 1 ← dn2
Vertex assignments:
  v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

coeff =  $\frac{1}{2} \varphi_1''$ 

  v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

coeff =  $\frac{1}{2} nstar_1$ 

  v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

coeff =  $\frac{1}{2} \varphi_2''$ 

  v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

coeff =  $\frac{1}{2} nstar_1$ 

Edges (propagator assignments):
  2 → 4:  nt1 → dv2  [G_hat[0,3]]
  3 → 2:  nt1 → dv1  [G_hat[0,2]]
  3 → 4:  nt1 → dv2  [G_hat[0,3]]
  4 → 0:  nt2 → dn1  [G_hat[1,0]]
  5 → 1:  nt1 → dn2  [G_hat[0,1]]
  5 → 2:  nt1 → dv1  [G_hat[0,2]]

```

```

=====
Diagram 1L-151
=====
External legs:
  leaf 0 ← dn1
  leaf 1 ← dn2
Vertex assignments:
  v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

coeff =  $\frac{1}{2} \varphi_1''$ 

  v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

coeff =  $\frac{1}{2} nstar_1$ 

```

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

Edges (propagator assignments):

2 → 4: nt1 → dv2 [G_hat[0,3]]
 3 → 2: nt1 → dv1 [G_hat[0,2]]
 3 → 4: nt1 → dv2 [G_hat[0,3]]
 4 → 0: nt2 → dn1 [G_hat[1,0]]
 5 → 1: nt1 → dn2 [G_hat[0,1]]
 5 → 2: nt1 → dv1 [G_hat[0,2]]

=====
 Diagram 1L-152
 =====

External legs:

leaf 0 ← dn1
 leaf 1 ← dn2

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

Edges (propagator assignments):

2 → 4: nt1 → dv2 [G_hat[0,3]]
 3 → 2: nt1 → dv1 [G_hat[0,2]]
 3 → 4: nt1 → dv2 [G_hat[0,3]]
 4 → 0: nt2 → dn1 [G_hat[1,0]]
 5 → 1: nt1 → dn2 [G_hat[0,1]]
 5 → 2: nt1 → dv1 [G_hat[0,2]]

=====

Diagram 1L-153

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

Edges (propagator assignments):

2 → 4: nt1 → dv2 [G_hat[0,3]]

3 → 2: nt1 → dv1 [G_hat[0,2]]

3 → 4: nt1 → dv2 [G_hat[0,3]]

4 → 0: nt2 → dn1 [G_hat[1,0]]

5 → 1: nt1 → dn2 [G_hat[0,1]]

5 → 2: nt1 → dv1 [G_hat[0,2]]

=====

Diagram 1L-154

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

Edges (propagator assignments):

2 → 4: nt1 → dv2 [G_hat[0,3]]
 3 → 2: nt1 → dv1 [G_hat[0,2]]
 3 → 4: nt1 → dv2 [G_hat[0,3]]
 4 → 0: nt2 → dn1 [G_hat[1,0]]
 5 → 1: nt1 → dn2 [G_hat[0,1]]
 5 → 2: nt1 → dv1 [G_hat[0,2]]

=====
 Diagram 1L-155
 =====

External legs:

leaf 0 ← dn1
 leaf 1 ← dn2

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

Edges (propagator assignments):

2 → 4: nt1 → dv2 [G_hat[0,3]]
 3 → 2: nt1 → dv1 [G_hat[0,2]]
 3 → 4: nt1 → dv2 [G_hat[0,3]]
 4 → 0: nt2 → dn1 [G_hat[1,0]]
 5 → 1: nt1 → dn2 [G_hat[0,1]]
 5 → 2: nt1 → dv1 [G_hat[0,2]]

=====

Diagram 1L-156

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

Edges (propagator assignments):

2 → 4: nt1 → dv2 [G_hat[0,3]]

3 → 2: nt1 → dv1 [G_hat[0,2]]

3 → 4: nt1 → dv2 [G_hat[0,3]]

4 → 0: nt2 → dn1 [G_hat[1,0]]

5 → 1: nt1 → dn2 [G_hat[0,1]]

5 → 2: nt1 → dv1 [G_hat[0,2]]

=====

Diagram 1L-157

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

Edges (propagator assignments):

2 → 4: nt1 → dv2 [G_hat[0,3]]
 3 → 2: nt1 → dv1 [G_hat[0,2]]
 3 → 4: nt1 → dv2 [G_hat[0,3]]
 4 → 0: nt2 → dn1 [G_hat[1,0]]
 5 → 1: nt1 → dn2 [G_hat[0,1]]
 5 → 2: nt1 → dv1 [G_hat[0,2]]

=====
 Diagram 1L-158
 =====

External legs:

leaf 0 ← dn1
 leaf 1 ← dn2

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

Edges (propagator assignments):

2 → 4: nt1 → dv2 [G_hat[0,3]]
 3 → 2: nt1 → dv1 [G_hat[0,2]]
 3 → 4: nt1 → dv2 [G_hat[0,3]]
 4 → 0: nt2 → dn1 [G_hat[1,0]]
 5 → 1: nt1 → dn2 [G_hat[0,1]]
 5 → 2: nt1 → dv1 [G_hat[0,2]]

=====

Diagram 1L-159

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

Edges (propagator assignments):

2 → 4: nt1 → dv2 [G_hat[0,3]]

3 → 2: nt1 → dv1 [G_hat[0,2]]

3 → 4: nt1 → dv2 [G_hat[0,3]]

4 → 0: nt2 → dn1 [G_hat[1,0]]

5 → 1: nt1 → dn2 [G_hat[0,1]]

5 → 2: nt1 → dv1 [G_hat[0,2]]

=====

Diagram 1L-160

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

Edges (propagator assignments):

2 → 4: nt1 → dv2 [G_hat[0,3]]
 3 → 2: nt1 → dv1 [G_hat[0,2]]
 3 → 4: nt1 → dv2 [G_hat[0,3]]
 4 → 0: nt2 → dn1 [G_hat[1,0]]
 5 → 1: nt1 → dn2 [G_hat[0,1]]
 5 → 2: nt1 → dv1 [G_hat[0,2]]

=====
 Diagram 1L-161
 =====

External legs:

leaf 0 ← dn1
 leaf 1 ← dn2

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

Edges (propagator assignments):

2 → 4: nt2 → dv1 [G_hat[1,2]]
 3 → 2: nt1 → dv2 [G_hat[0,3]]
 3 → 4: nt1 → dv1 [G_hat[0,2]]
 4 → 0: nt1 → dn1 [G_hat[0,0]]
 5 → 1: nt1 → dn2 [G_hat[0,1]]
 5 → 2: nt1 → dv2 [G_hat[0,3]]

=====

Diagram 1L-162

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

Edges (propagator assignments):

2 → 4: nt2 → dv1 [G_hat[1,2]]

3 → 2: nt1 → dv2 [G_hat[0,3]]

3 → 4: nt1 → dv1 [G_hat[0,2]]

4 → 0: nt1 → dn1 [G_hat[0,0]]

5 → 1: nt1 → dn2 [G_hat[0,1]]

5 → 2: nt1 → dv2 [G_hat[0,3]]

=====

Diagram 1L-163

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

Edges (propagator assignments):

2 → 4: nt2 → dv1 [G_hat[1,2]]
 3 → 2: nt1 → dv2 [G_hat[0,3]]
 3 → 4: nt1 → dv1 [G_hat[0,2]]
 4 → 0: nt1 → dn1 [G_hat[0,0]]
 5 → 1: nt1 → dn2 [G_hat[0,1]]
 5 → 2: nt1 → dv2 [G_hat[0,3]]

=====
 Diagram 1L-164
 =====

External legs:

leaf 0 ← dn1
 leaf 1 ← dn2

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

Edges (propagator assignments):

2 → 4: nt2 → dv1 [G_hat[1,2]]
 3 → 2: nt1 → dv2 [G_hat[0,3]]
 3 → 4: nt1 → dv1 [G_hat[0,2]]
 4 → 0: nt1 → dn1 [G_hat[0,0]]
 5 → 1: nt1 → dn2 [G_hat[0,1]]
 5 → 2: nt1 → dv2 [G_hat[0,3]]

=====

Diagram 1L-165

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

Edges (propagator assignments):

2 → 4: nt2 → dv1 [G_hat[1,2]]

3 → 2: nt1 → dv2 [G_hat[0,3]]

3 → 4: nt1 → dv1 [G_hat[0,2]]

4 → 0: nt1 → dn1 [G_hat[0,0]]

5 → 1: nt1 → dn2 [G_hat[0,1]]

5 → 2: nt1 → dv2 [G_hat[0,3]]

=====

Diagram 1L-166

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

Edges (propagator assignments):

2 → 4: nt2 → dv1 [G_hat[1,2]]
 3 → 2: nt1 → dv2 [G_hat[0,3]]
 3 → 4: nt1 → dv1 [G_hat[0,2]]
 4 → 0: nt1 → dn1 [G_hat[0,0]]
 5 → 1: nt1 → dn2 [G_hat[0,1]]
 5 → 2: nt1 → dv2 [G_hat[0,3]]

=====
 Diagram 1L-167
 =====

External legs:

leaf 0 ← dn1
 leaf 1 ← dn2

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

Edges (propagator assignments):

2 → 4: nt2 → dv1 [G_hat[1,2]]
 3 → 2: nt1 → dv2 [G_hat[0,3]]
 3 → 4: nt1 → dv1 [G_hat[0,2]]
 4 → 0: nt1 → dn1 [G_hat[0,0]]
 5 → 1: nt1 → dn2 [G_hat[0,1]]
 5 → 2: nt1 → dv2 [G_hat[0,3]]

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Diagram 1L-168

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External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

Edges (propagator assignments):

2 → 4: nt2 → dv1 [G_hat[1,2]]

3 → 2: nt1 → dv2 [G_hat[0,3]]

3 → 4: nt1 → dv1 [G_hat[0,2]]

4 → 0: nt1 → dn1 [G_hat[0,0]]

5 → 1: nt1 → dn2 [G_hat[0,1]]

5 → 2: nt1 → dv2 [G_hat[0,3]]

=====

Diagram 1L-169

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

Edges (propagator assignments):

2 → 4: nt2 → dv1 [G_hat[1,2]]
 3 → 2: nt1 → dv2 [G_hat[0,3]]
 3 → 4: nt1 → dv1 [G_hat[0,2]]
 4 → 0: nt1 → dn1 [G_hat[0,0]]
 5 → 1: nt1 → dn2 [G_hat[0,1]]
 5 → 2: nt1 → dv2 [G_hat[0,3]]

=====
 Diagram 1L-170
 =====

External legs:

leaf 0 ← dn1
 leaf 1 ← dn2

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

Edges (propagator assignments):

2 → 4: nt2 → dv1 [G_hat[1,2]]
 3 → 2: nt1 → dv2 [G_hat[0,3]]
 3 → 4: nt1 → dv1 [G_hat[0,2]]
 4 → 0: nt1 → dn1 [G_hat[0,0]]
 5 → 1: nt1 → dn2 [G_hat[0,1]]
 5 → 2: nt1 → dv2 [G_hat[0,3]]

=====

Diagram 1L-171

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

Edges (propagator assignments):

2 → 4: nt2 → dv1 [G_hat[1,2]]

3 → 2: nt1 → dv2 [G_hat[0,3]]

3 → 4: nt1 → dv1 [G_hat[0,2]]

4 → 0: nt1 → dn1 [G_hat[0,0]]

5 → 1: nt1 → dn2 [G_hat[0,1]]

5 → 2: nt1 → dv2 [G_hat[0,3]]

=====

Diagram 1L-172

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

Edges (propagator assignments):

2 → 4: nt2 → dv1 [G_hat[1,2]]
 3 → 2: nt1 → dv2 [G_hat[0,3]]
 3 → 4: nt1 → dv1 [G_hat[0,2]]
 4 → 0: nt1 → dn1 [G_hat[0,0]]
 5 → 1: nt1 → dn2 [G_hat[0,1]]
 5 → 2: nt1 → dv2 [G_hat[0,3]]

=====
 Diagram 1L-173
 =====

External legs:

leaf 0 ← dn1
 leaf 1 ← dn2

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

Edges (propagator assignments):

2 → 4: nt2 → dv1 [G_hat[1,2]]
 3 → 2: nt1 → dv2 [G_hat[0,3]]
 3 → 4: nt1 → dv1 [G_hat[0,2]]
 4 → 0: nt1 → dn1 [G_hat[0,0]]
 5 → 1: nt1 → dn2 [G_hat[0,1]]
 5 → 2: nt1 → dv2 [G_hat[0,3]]

=====

Diagram 1L-174

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

Edges (propagator assignments):

2 → 4: nt2 → dv1 [G_hat[1,2]]

3 → 2: nt1 → dv2 [G_hat[0,3]]

3 → 4: nt1 → dv1 [G_hat[0,2]]

4 → 0: nt1 → dn1 [G_hat[0,0]]

5 → 1: nt1 → dn2 [G_hat[0,1]]

5 → 2: nt1 → dv2 [G_hat[0,3]]

=====

Diagram 1L-175

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

Edges (propagator assignments):

2 → 4: nt2 → dv1 [G_hat[1,2]]
 3 → 2: nt1 → dv2 [G_hat[0,3]]
 3 → 4: nt1 → dv1 [G_hat[0,2]]
 4 → 0: nt1 → dn1 [G_hat[0,0]]
 5 → 1: nt1 → dn2 [G_hat[0,1]]
 5 → 2: nt1 → dv2 [G_hat[0,3]]

=====
 Diagram 1L-176
 =====

External legs:

leaf 0 ← dn1
 leaf 1 ← dn2

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

Edges (propagator assignments):

2 → 4: nt2 → dv1 [G_hat[1,2]]
 3 → 2: nt1 → dv2 [G_hat[0,3]]
 3 → 4: nt1 → dv1 [G_hat[0,2]]
 4 → 0: nt1 → dn1 [G_hat[0,0]]
 5 → 1: nt1 → dn2 [G_hat[0,1]]
 5 → 2: nt1 → dv2 [G_hat[0,3]]

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=====
Diagram 1L-177
=====
External legs:
  leaf 0 ← dn1
  leaf 1 ← dn2
Vertex assignments:
  v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

coeff =  $\frac{1}{2} \varphi_2''$ 

  v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

coeff =  $\frac{1}{2} nstar_1$ 

  v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

coeff =  $\frac{1}{2} \varphi_2''$ 

  v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

coeff =  $\frac{1}{2} nstar_1$ 

Edges (propagator assignments):
  2 → 4:  nt2 → dv2  [G_hat[1,3]]
  3 → 2:  nt1 → dv2  [G_hat[0,3]]
  3 → 4:  nt1 → dv2  [G_hat[0,3]]
  4 → 0:  nt2 → dn1  [G_hat[1,0]]
  5 → 1:  nt1 → dn2  [G_hat[0,1]]
  5 → 2:  nt1 → dv2  [G_hat[0,3]]

```

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=====
Diagram 1L-178
=====
External legs:
  leaf 0 ← dn1
  leaf 1 ← dn2
Vertex assignments:
  v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

coeff =  $\frac{1}{2} \varphi_2''$ 

  v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

coeff =  $\frac{1}{2} nstar_1$ 

```

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

Edges (propagator assignments):

2 → 4: nt2 → dv2 [G_hat[1,3]]
 3 → 2: nt1 → dv2 [G_hat[0,3]]
 3 → 4: nt1 → dv2 [G_hat[0,3]]
 4 → 0: nt2 → dn1 [G_hat[1,0]]
 5 → 1: nt1 → dn2 [G_hat[0,1]]
 5 → 2: nt1 → dv2 [G_hat[0,3]]

=====
 Diagram 1L-179
 =====

External legs:

leaf 0 ← dn1
 leaf 1 ← dn2

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

Edges (propagator assignments):

2 → 4: nt2 → dv2 [G_hat[1,3]]
 3 → 2: nt1 → dv2 [G_hat[0,3]]
 3 → 4: nt1 → dv2 [G_hat[0,3]]
 4 → 0: nt2 → dn1 [G_hat[1,0]]
 5 → 1: nt1 → dn2 [G_hat[0,1]]
 5 → 2: nt1 → dv2 [G_hat[0,3]]

=====

Diagram 1L-180

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

Edges (propagator assignments):

2 → 4: nt2 → dv2 [G_hat[1,3]]

3 → 2: nt1 → dv2 [G_hat[0,3]]

3 → 4: nt1 → dv2 [G_hat[0,3]]

4 → 0: nt2 → dn1 [G_hat[1,0]]

5 → 1: nt1 → dn2 [G_hat[0,1]]

5 → 2: nt1 → dv2 [G_hat[0,3]]

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Diagram 1L-181

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

Edges (propagator assignments):

2 → 4: nt2 → dv2 [G_hat[1,3]]
 3 → 2: nt1 → dv2 [G_hat[0,3]]
 3 → 4: nt1 → dv2 [G_hat[0,3]]
 4 → 0: nt2 → dn1 [G_hat[1,0]]
 5 → 1: nt1 → dn2 [G_hat[0,1]]
 5 → 2: nt1 → dv2 [G_hat[0,3]]

=====
 Diagram 1L-182
 =====

External legs:

leaf 0 ← dn1
 leaf 1 ← dn2

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

Edges (propagator assignments):

2 → 4: nt2 → dv2 [G_hat[1,3]]
 3 → 2: nt1 → dv2 [G_hat[0,3]]
 3 → 4: nt1 → dv2 [G_hat[0,3]]
 4 → 0: nt2 → dn1 [G_hat[1,0]]
 5 → 1: nt1 → dn2 [G_hat[0,1]]
 5 → 2: nt1 → dv2 [G_hat[0,3]]

=====

Diagram 1L-183

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

Edges (propagator assignments):

2 → 4: nt2 → dv2 [G_hat[1,3]]

3 → 2: nt1 → dv2 [G_hat[0,3]]

3 → 4: nt1 → dv2 [G_hat[0,3]]

4 → 0: nt2 → dn1 [G_hat[1,0]]

5 → 1: nt1 → dn2 [G_hat[0,1]]

5 → 2: nt1 → dv2 [G_hat[0,3]]

=====

Diagram 1L-184

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

Edges (propagator assignments):

2 → 4: nt2 → dv2 [G_hat[1,3]]
 3 → 2: nt1 → dv2 [G_hat[0,3]]
 3 → 4: nt1 → dv2 [G_hat[0,3]]
 4 → 0: nt2 → dn1 [G_hat[1,0]]
 5 → 1: nt1 → dn2 [G_hat[0,1]]
 5 → 2: nt1 → dv2 [G_hat[0,3]]

=====
 Diagram 1L-185
 =====

External legs:

leaf 0 ← dn1
 leaf 1 ← dn2

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

Edges (propagator assignments):

2 → 4: nt2 → dv2 [G_hat[1,3]]
 3 → 2: nt1 → dv2 [G_hat[0,3]]
 3 → 4: nt1 → dv2 [G_hat[0,3]]
 4 → 0: nt2 → dn1 [G_hat[1,0]]
 5 → 1: nt1 → dn2 [G_hat[0,1]]
 5 → 2: nt1 → dv2 [G_hat[0,3]]

=====

Diagram 1L-186

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

Edges (propagator assignments):

2 → 4: nt2 → dv2 [G_hat[1,3]]

3 → 2: nt1 → dv2 [G_hat[0,3]]

3 → 4: nt1 → dv2 [G_hat[0,3]]

4 → 0: nt2 → dn1 [G_hat[1,0]]

5 → 1: nt1 → dn2 [G_hat[0,1]]

5 → 2: nt1 → dv2 [G_hat[0,3]]

=====

Diagram 1L-187

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

Edges (propagator assignments):

2 → 4: nt2 → dv2 [G_hat[1,3]]
 3 → 2: nt1 → dv2 [G_hat[0,3]]
 3 → 4: nt1 → dv2 [G_hat[0,3]]
 4 → 0: nt2 → dn1 [G_hat[1,0]]
 5 → 1: nt1 → dn2 [G_hat[0,1]]
 5 → 2: nt1 → dv2 [G_hat[0,3]]

=====
 Diagram 1L-188
 =====

External legs:

leaf 0 ← dn1
 leaf 1 ← dn2

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

Edges (propagator assignments):

2 → 4: nt2 → dv2 [G_hat[1,3]]
 3 → 2: nt1 → dv2 [G_hat[0,3]]
 3 → 4: nt1 → dv2 [G_hat[0,3]]
 4 → 0: nt2 → dn1 [G_hat[1,0]]
 5 → 1: nt1 → dn2 [G_hat[0,1]]
 5 → 2: nt1 → dv2 [G_hat[0,3]]

=====

Diagram 1L-189

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

Edges (propagator assignments):

2 → 4: nt2 → dv2 [G_hat[1,3]]

3 → 2: nt1 → dv2 [G_hat[0,3]]

3 → 4: nt1 → dv2 [G_hat[0,3]]

4 → 0: nt2 → dn1 [G_hat[1,0]]

5 → 1: nt1 → dn2 [G_hat[0,1]]

5 → 2: nt1 → dv2 [G_hat[0,3]]

=====

Diagram 1L-190

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

Edges (propagator assignments):

2 → 4: nt2 → dv2 [G_hat[1,3]]
 3 → 2: nt1 → dv2 [G_hat[0,3]]
 3 → 4: nt1 → dv2 [G_hat[0,3]]
 4 → 0: nt2 → dn1 [G_hat[1,0]]
 5 → 1: nt1 → dn2 [G_hat[0,1]]
 5 → 2: nt1 → dv2 [G_hat[0,3]]

=====
 Diagram 1L-191
 =====

External legs:

leaf 0 ← dn1
 leaf 1 ← dn2

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

Edges (propagator assignments):

2 → 4: nt2 → dv2 [G_hat[1,3]]
 3 → 2: nt1 → dv2 [G_hat[0,3]]
 3 → 4: nt1 → dv2 [G_hat[0,3]]
 4 → 0: nt2 → dn1 [G_hat[1,0]]
 5 → 1: nt1 → dn2 [G_hat[0,1]]
 5 → 2: nt1 → dv2 [G_hat[0,3]]

```

=====
Diagram 1L-192
=====
External legs:
  leaf 0 ← dn1
  leaf 1 ← dn2
Vertex assignments:
  v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

coeff =  $\frac{1}{2} \varphi_2''$ 

  v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

coeff =  $\frac{1}{2} nstar_1$ 

  v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

coeff =  $\frac{1}{2} \varphi_2''$ 

  v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

coeff =  $\frac{1}{2} nstar_1$ 

Edges (propagator assignments):
  2 → 4:  nt2 → dv2  [G_hat[1,3]]
  3 → 2:  nt1 → dv2  [G_hat[0,3]]
  3 → 4:  nt1 → dv2  [G_hat[0,3]]
  4 → 0:  nt2 → dn1  [G_hat[1,0]]
  5 → 1:  nt1 → dn2  [G_hat[0,1]]
  5 → 2:  nt1 → dv2  [G_hat[0,3]]

```

```

=====
Diagram 1L-193
=====
External legs:
  leaf 0 ← dn1
  leaf 1 ← dn2
Vertex assignments:
  v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

coeff =  $\frac{1}{2} \varphi_1''$ 

  v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

coeff =  $\frac{1}{2} nstar_1$ 

```

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

Edges (propagator assignments):

2 → 4: nt1 → dv1 [G_hat[0,2]]
 3 → 2: nt1 → dv1 [G_hat[0,2]]
 3 → 4: nt1 → dv1 [G_hat[0,2]]
 4 → 0: nt1 → dn1 [G_hat[0,0]]
 5 → 1: nt2 → dn2 [G_hat[1,1]]
 5 → 2: nt2 → dv1 [G_hat[1,2]]

=====
 Diagram 1L-194
 =====

External legs:

leaf 0 ← dn1
 leaf 1 ← dn2

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

Edges (propagator assignments):

2 → 4: nt1 → dv1 [G_hat[0,2]]
 3 → 2: nt1 → dv1 [G_hat[0,2]]
 3 → 4: nt1 → dv1 [G_hat[0,2]]
 4 → 0: nt1 → dn1 [G_hat[0,0]]
 5 → 1: nt2 → dn2 [G_hat[1,1]]
 5 → 2: nt2 → dv1 [G_hat[1,2]]

=====

Diagram 1L-195

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

Edges (propagator assignments):

2 → 4: nt1 → dv1 [G_hat[0,2]]

3 → 2: nt1 → dv1 [G_hat[0,2]]

3 → 4: nt1 → dv1 [G_hat[0,2]]

4 → 0: nt1 → dn1 [G_hat[0,0]]

5 → 1: nt2 → dn2 [G_hat[1,1]]

5 → 2: nt2 → dv1 [G_hat[1,2]]

=====

Diagram 1L-196

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

Edges (propagator assignments):

2 → 4: nt1 → dv1 [G_hat[0,2]]
 3 → 2: nt1 → dv1 [G_hat[0,2]]
 3 → 4: nt1 → dv1 [G_hat[0,2]]
 4 → 0: nt1 → dn1 [G_hat[0,0]]
 5 → 1: nt2 → dn2 [G_hat[1,1]]
 5 → 2: nt2 → dv1 [G_hat[1,2]]

=====
 Diagram 1L-197
 =====

External legs:

leaf 0 ← dn1
 leaf 1 ← dn2

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

Edges (propagator assignments):

2 → 4: nt1 → dv1 [G_hat[0,2]]
 3 → 2: nt1 → dv1 [G_hat[0,2]]
 3 → 4: nt1 → dv1 [G_hat[0,2]]
 4 → 0: nt1 → dn1 [G_hat[0,0]]
 5 → 1: nt2 → dn2 [G_hat[1,1]]
 5 → 2: nt2 → dv1 [G_hat[1,2]]

=====

Diagram 1L-198

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

Edges (propagator assignments):

2 → 4: nt1 → dv1 [G_hat[0,2]]

3 → 2: nt1 → dv1 [G_hat[0,2]]

3 → 4: nt1 → dv1 [G_hat[0,2]]

4 → 0: nt1 → dn1 [G_hat[0,0]]

5 → 1: nt2 → dn2 [G_hat[1,1]]

5 → 2: nt2 → dv1 [G_hat[1,2]]

=====

Diagram 1L-199

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

Edges (propagator assignments):

2 → 4: nt1 → dv1 [G_hat[0,2]]
 3 → 2: nt1 → dv1 [G_hat[0,2]]
 3 → 4: nt1 → dv1 [G_hat[0,2]]
 4 → 0: nt1 → dn1 [G_hat[0,0]]
 5 → 1: nt2 → dn2 [G_hat[1,1]]
 5 → 2: nt2 → dv1 [G_hat[1,2]]

=====
 Diagram 1L-200
 =====

External legs:

leaf 0 ← dn1
 leaf 1 ← dn2

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

Edges (propagator assignments):

2 → 4: nt1 → dv1 [G_hat[0,2]]
 3 → 2: nt1 → dv1 [G_hat[0,2]]
 3 → 4: nt1 → dv1 [G_hat[0,2]]
 4 → 0: nt1 → dn1 [G_hat[0,0]]
 5 → 1: nt2 → dn2 [G_hat[1,1]]
 5 → 2: nt2 → dv1 [G_hat[1,2]]

=====

Diagram 1L-201

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

Edges (propagator assignments):

2 → 4: nt1 → dv1 [G_hat[0,2]]

3 → 2: nt1 → dv1 [G_hat[0,2]]

3 → 4: nt1 → dv1 [G_hat[0,2]]

4 → 0: nt1 → dn1 [G_hat[0,0]]

5 → 1: nt2 → dn2 [G_hat[1,1]]

5 → 2: nt2 → dv1 [G_hat[1,2]]

=====

Diagram 1L-202

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

Edges (propagator assignments):

2 → 4: nt1 → dv1 [G_hat[0,2]]
 3 → 2: nt1 → dv1 [G_hat[0,2]]
 3 → 4: nt1 → dv1 [G_hat[0,2]]
 4 → 0: nt1 → dn1 [G_hat[0,0]]
 5 → 1: nt2 → dn2 [G_hat[1,1]]
 5 → 2: nt2 → dv1 [G_hat[1,2]]

=====
 Diagram 1L-203
 =====

External legs:

leaf 0 ← dn1
 leaf 1 ← dn2

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

Edges (propagator assignments):

2 → 4: nt1 → dv1 [G_hat[0,2]]
 3 → 2: nt1 → dv1 [G_hat[0,2]]
 3 → 4: nt1 → dv1 [G_hat[0,2]]
 4 → 0: nt1 → dn1 [G_hat[0,0]]
 5 → 1: nt2 → dn2 [G_hat[1,1]]
 5 → 2: nt2 → dv1 [G_hat[1,2]]

=====

Diagram 1L-204

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

Edges (propagator assignments):

2 → 4: nt1 → dv1 [G_hat[0,2]]

3 → 2: nt1 → dv1 [G_hat[0,2]]

3 → 4: nt1 → dv1 [G_hat[0,2]]

4 → 0: nt1 → dn1 [G_hat[0,0]]

5 → 1: nt2 → dn2 [G_hat[1,1]]

5 → 2: nt2 → dv1 [G_hat[1,2]]

=====

Diagram 1L-205

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

Edges (propagator assignments):

2 → 4: nt1 → dv1 [G_hat[0,2]]
 3 → 2: nt1 → dv1 [G_hat[0,2]]
 3 → 4: nt1 → dv1 [G_hat[0,2]]
 4 → 0: nt1 → dn1 [G_hat[0,0]]
 5 → 1: nt2 → dn2 [G_hat[1,1]]
 5 → 2: nt2 → dv1 [G_hat[1,2]]

=====
 Diagram 1L-206
 =====

External legs:

leaf 0 ← dn1
 leaf 1 ← dn2

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

Edges (propagator assignments):

2 → 4: nt1 → dv1 [G_hat[0,2]]
 3 → 2: nt1 → dv1 [G_hat[0,2]]
 3 → 4: nt1 → dv1 [G_hat[0,2]]
 4 → 0: nt1 → dn1 [G_hat[0,0]]
 5 → 1: nt2 → dn2 [G_hat[1,1]]
 5 → 2: nt2 → dv1 [G_hat[1,2]]

=====

Diagram 1L-207

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

Edges (propagator assignments):

2 → 4: nt1 → dv1 [G_hat[0,2]]

3 → 2: nt1 → dv1 [G_hat[0,2]]

3 → 4: nt1 → dv1 [G_hat[0,2]]

4 → 0: nt1 → dn1 [G_hat[0,0]]

5 → 1: nt2 → dn2 [G_hat[1,1]]

5 → 2: nt2 → dv1 [G_hat[1,2]]

=====

Diagram 1L-208

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

Edges (propagator assignments):

2 → 4: nt1 → dv1 [G_hat[0,2]]
 3 → 2: nt1 → dv1 [G_hat[0,2]]
 3 → 4: nt1 → dv1 [G_hat[0,2]]
 4 → 0: nt1 → dn1 [G_hat[0,0]]
 5 → 1: nt2 → dn2 [G_hat[1,1]]
 5 → 2: nt2 → dv1 [G_hat[1,2]]

=====
 Diagram 1L-209
 =====

External legs:

leaf 0 ← dn1
 leaf 1 ← dn2

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

Edges (propagator assignments):

2 → 4: nt1 → dv2 [G_hat[0,3]]
 3 → 2: nt1 → dv1 [G_hat[0,2]]
 3 → 4: nt1 → dv2 [G_hat[0,3]]
 4 → 0: nt2 → dn1 [G_hat[1,0]]
 5 → 1: nt2 → dn2 [G_hat[1,1]]
 5 → 2: nt2 → dv1 [G_hat[1,2]]

=====

Diagram 1L-210

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

Edges (propagator assignments):

2 → 4: nt1 → dv2 [G_hat[0,3]]

3 → 2: nt1 → dv1 [G_hat[0,2]]

3 → 4: nt1 → dv2 [G_hat[0,3]]

4 → 0: nt2 → dn1 [G_hat[1,0]]

5 → 1: nt2 → dn2 [G_hat[1,1]]

5 → 2: nt2 → dv1 [G_hat[1,2]]

=====

Diagram 1L-211

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

Edges (propagator assignments):

2 → 4: nt1 → dv2 [G_hat[0,3]]
 3 → 2: nt1 → dv1 [G_hat[0,2]]
 3 → 4: nt1 → dv2 [G_hat[0,3]]
 4 → 0: nt2 → dn1 [G_hat[1,0]]
 5 → 1: nt2 → dn2 [G_hat[1,1]]
 5 → 2: nt2 → dv1 [G_hat[1,2]]

=====
 Diagram 1L-212
 =====

External legs:

leaf 0 ← dn1
 leaf 1 ← dn2

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

Edges (propagator assignments):

2 → 4: nt1 → dv2 [G_hat[0,3]]
 3 → 2: nt1 → dv1 [G_hat[0,2]]
 3 → 4: nt1 → dv2 [G_hat[0,3]]
 4 → 0: nt2 → dn1 [G_hat[1,0]]
 5 → 1: nt2 → dn2 [G_hat[1,1]]
 5 → 2: nt2 → dv1 [G_hat[1,2]]

```

=====
Diagram 1L-213
=====

External legs:
  leaf 0 ← dn1
  leaf 1 ← dn2
Vertex assignments:
  v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

coeff =  $\frac{1}{2} \varphi_1''$ 

  v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

coeff =  $\frac{1}{2} nstar_1$ 

  v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

coeff =  $\frac{1}{2} \varphi_2''$ 

  v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

coeff =  $\frac{1}{2} nstar_2$ 

Edges (propagator assignments):
  2 → 4:  nt1 → dv2  [G_hat[0,3]]
  3 → 2:  nt1 → dv1  [G_hat[0,2]]
  3 → 4:  nt1 → dv2  [G_hat[0,3]]
  4 → 0:  nt2 → dn1  [G_hat[1,0]]
  5 → 1:  nt2 → dn2  [G_hat[1,1]]
  5 → 2:  nt2 → dv1  [G_hat[1,2]]

```

```

=====
Diagram 1L-214
=====

External legs:
  leaf 0 ← dn1
  leaf 1 ← dn2
Vertex assignments:
  v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

coeff =  $\frac{1}{2} \varphi_1''$ 

  v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

coeff =  $\frac{1}{2} nstar_1$ 

```

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

Edges (propagator assignments):

2 → 4: nt1 → dv2 [G_hat[0,3]]
 3 → 2: nt1 → dv1 [G_hat[0,2]]
 3 → 4: nt1 → dv2 [G_hat[0,3]]
 4 → 0: nt2 → dn1 [G_hat[1,0]]
 5 → 1: nt2 → dn2 [G_hat[1,1]]
 5 → 2: nt2 → dv1 [G_hat[1,2]]

=====
 Diagram 1L-215
 =====

External legs:

leaf 0 ← dn1
 leaf 1 ← dn2

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

Edges (propagator assignments):

2 → 4: nt1 → dv2 [G_hat[0,3]]
 3 → 2: nt1 → dv1 [G_hat[0,2]]
 3 → 4: nt1 → dv2 [G_hat[0,3]]
 4 → 0: nt2 → dn1 [G_hat[1,0]]
 5 → 1: nt2 → dn2 [G_hat[1,1]]
 5 → 2: nt2 → dv1 [G_hat[1,2]]

=====

Diagram 1L-216

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

Edges (propagator assignments):

2 → 4: nt1 → dv2 [G_hat[0,3]]

3 → 2: nt1 → dv1 [G_hat[0,2]]

3 → 4: nt1 → dv2 [G_hat[0,3]]

4 → 0: nt2 → dn1 [G_hat[1,0]]

5 → 1: nt2 → dn2 [G_hat[1,1]]

5 → 2: nt2 → dv1 [G_hat[1,2]]

=====

Diagram 1L-217

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

Edges (propagator assignments):

2 → 4: nt1 → dv2 [G_hat[0,3]]
 3 → 2: nt1 → dv1 [G_hat[0,2]]
 3 → 4: nt1 → dv2 [G_hat[0,3]]
 4 → 0: nt2 → dn1 [G_hat[1,0]]
 5 → 1: nt2 → dn2 [G_hat[1,1]]
 5 → 2: nt2 → dv1 [G_hat[1,2]]

=====
 Diagram 1L-218
 =====

External legs:

leaf 0 ← dn1
 leaf 1 ← dn2

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

Edges (propagator assignments):

2 → 4: nt1 → dv2 [G_hat[0,3]]
 3 → 2: nt1 → dv1 [G_hat[0,2]]
 3 → 4: nt1 → dv2 [G_hat[0,3]]
 4 → 0: nt2 → dn1 [G_hat[1,0]]
 5 → 1: nt2 → dn2 [G_hat[1,1]]
 5 → 2: nt2 → dv1 [G_hat[1,2]]

=====

Diagram 1L-219

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

Edges (propagator assignments):

2 → 4: nt1 → dv2 [G_hat[0,3]]

3 → 2: nt1 → dv1 [G_hat[0,2]]

3 → 4: nt1 → dv2 [G_hat[0,3]]

4 → 0: nt2 → dn1 [G_hat[1,0]]

5 → 1: nt2 → dn2 [G_hat[1,1]]

5 → 2: nt2 → dv1 [G_hat[1,2]]

=====

Diagram 1L-220

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

Edges (propagator assignments):

2 → 4: nt1 → dv2 [G_hat[0,3]]
 3 → 2: nt1 → dv1 [G_hat[0,2]]
 3 → 4: nt1 → dv2 [G_hat[0,3]]
 4 → 0: nt2 → dn1 [G_hat[1,0]]
 5 → 1: nt2 → dn2 [G_hat[1,1]]
 5 → 2: nt2 → dv1 [G_hat[1,2]]

=====
 Diagram 1L-221
 =====

External legs:

leaf 0 ← dn1
 leaf 1 ← dn2

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

Edges (propagator assignments):

2 → 4: nt1 → dv2 [G_hat[0,3]]
 3 → 2: nt1 → dv1 [G_hat[0,2]]
 3 → 4: nt1 → dv2 [G_hat[0,3]]
 4 → 0: nt2 → dn1 [G_hat[1,0]]
 5 → 1: nt2 → dn2 [G_hat[1,1]]
 5 → 2: nt2 → dv1 [G_hat[1,2]]


```

=====
Diagram 1L-222
=====
External legs:
  leaf 0 ← dn1
  leaf 1 ← dn2
Vertex assignments:
  v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

coeff =  $\frac{1}{2} \varphi_1''$ 

  v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

coeff =  $\frac{1}{2} nstar_1$ 

  v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

coeff =  $\frac{1}{2} \varphi_2''$ 

  v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

coeff =  $\frac{1}{2} nstar_2$ 

Edges (propagator assignments):
  2 → 4:  nt1 → dv2  [G_hat[0,3]]
  3 → 2:  nt1 → dv1  [G_hat[0,2]]
  3 → 4:  nt1 → dv2  [G_hat[0,3]]
  4 → 0:  nt2 → dn1  [G_hat[1,0]]
  5 → 1:  nt2 → dn2  [G_hat[1,1]]
  5 → 2:  nt2 → dv1  [G_hat[1,2]]

```

```

=====
Diagram 1L-223
=====
External legs:
  leaf 0 ← dn1
  leaf 1 ← dn2
Vertex assignments:
  v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

coeff =  $\frac{1}{2} \varphi_1''$ 

  v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

coeff =  $\frac{1}{2} nstar_1$ 

```

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

Edges (propagator assignments):

2 → 4: nt1 → dv2 [G_hat[0,3]]
 3 → 2: nt1 → dv1 [G_hat[0,2]]
 3 → 4: nt1 → dv2 [G_hat[0,3]]
 4 → 0: nt2 → dn1 [G_hat[1,0]]
 5 → 1: nt2 → dn2 [G_hat[1,1]]
 5 → 2: nt2 → dv1 [G_hat[1,2]]

=====
 Diagram 1L-224
 =====

External legs:

leaf 0 ← dn1
 leaf 1 ← dn2

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

Edges (propagator assignments):

2 → 4: nt1 → dv2 [G_hat[0,3]]
 3 → 2: nt1 → dv1 [G_hat[0,2]]
 3 → 4: nt1 → dv2 [G_hat[0,3]]
 4 → 0: nt2 → dn1 [G_hat[1,0]]
 5 → 1: nt2 → dn2 [G_hat[1,1]]
 5 → 2: nt2 → dv1 [G_hat[1,2]]

=====

Diagram 1L-225

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

Edges (propagator assignments):

2 → 4: nt2 → dv1 [G_hat[1,2]]

3 → 2: nt1 → dv2 [G_hat[0,3]]

3 → 4: nt1 → dv1 [G_hat[0,2]]

4 → 0: nt1 → dn1 [G_hat[0,0]]

5 → 1: nt2 → dn2 [G_hat[1,1]]

5 → 2: nt2 → dv2 [G_hat[1,3]]

=====

Diagram 1L-226

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

Edges (propagator assignments):

2 → 4: nt2 → dv1 [G_hat[1,2]]
 3 → 2: nt1 → dv2 [G_hat[0,3]]
 3 → 4: nt1 → dv1 [G_hat[0,2]]
 4 → 0: nt1 → dn1 [G_hat[0,0]]
 5 → 1: nt2 → dn2 [G_hat[1,1]]
 5 → 2: nt2 → dv2 [G_hat[1,3]]

=====
 Diagram 1L-227
 =====

External legs:

leaf 0 ← dn1
 leaf 1 ← dn2

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

Edges (propagator assignments):

2 → 4: nt2 → dv1 [G_hat[1,2]]
 3 → 2: nt1 → dv2 [G_hat[0,3]]
 3 → 4: nt1 → dv1 [G_hat[0,2]]
 4 → 0: nt1 → dn1 [G_hat[0,0]]
 5 → 1: nt2 → dn2 [G_hat[1,1]]
 5 → 2: nt2 → dv2 [G_hat[1,3]]

```

=====
Diagram 1L-228
=====

External legs:
  leaf 0 ← dn1
  leaf 1 ← dn2
Vertex assignments:
  v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

coeff =  $\frac{1}{2} \varphi_2''$ 

  v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

coeff =  $\frac{1}{2} nstar_1$ 

  v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

coeff =  $\frac{1}{2} \varphi_1''$ 

  v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

coeff =  $\frac{1}{2} nstar_2$ 

Edges (propagator assignments):
  2 → 4:  nt2 → dv1  [G_hat[1,2]]
  3 → 2:  nt1 → dv2  [G_hat[0,3]]
  3 → 4:  nt1 → dv1  [G_hat[0,2]]
  4 → 0:  nt1 → dn1  [G_hat[0,0]]
  5 → 1:  nt2 → dn2  [G_hat[1,1]]
  5 → 2:  nt2 → dv2  [G_hat[1,3]]

```

```

=====
Diagram 1L-229
=====

External legs:
  leaf 0 ← dn1
  leaf 1 ← dn2
Vertex assignments:
  v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

coeff =  $\frac{1}{2} \varphi_2''$ 

  v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

coeff =  $\frac{1}{2} nstar_1$ 

```

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

Edges (propagator assignments):

2 → 4: nt2 → dv1 [G_hat[1,2]]
 3 → 2: nt1 → dv2 [G_hat[0,3]]
 3 → 4: nt1 → dv1 [G_hat[0,2]]
 4 → 0: nt1 → dn1 [G_hat[0,0]]
 5 → 1: nt2 → dn2 [G_hat[1,1]]
 5 → 2: nt2 → dv2 [G_hat[1,3]]

=====
 Diagram 1L-230
 =====

External legs:

leaf 0 ← dn1
 leaf 1 ← dn2

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

Edges (propagator assignments):

2 → 4: nt2 → dv1 [G_hat[1,2]]
 3 → 2: nt1 → dv2 [G_hat[0,3]]
 3 → 4: nt1 → dv1 [G_hat[0,2]]
 4 → 0: nt1 → dn1 [G_hat[0,0]]
 5 → 1: nt2 → dn2 [G_hat[1,1]]
 5 → 2: nt2 → dv2 [G_hat[1,3]]

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Diagram 1L-231

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External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

Edges (propagator assignments):

2 → 4: nt2 → dv1 [G_hat[1,2]]

3 → 2: nt1 → dv2 [G_hat[0,3]]

3 → 4: nt1 → dv1 [G_hat[0,2]]

4 → 0: nt1 → dn1 [G_hat[0,0]]

5 → 1: nt2 → dn2 [G_hat[1,1]]

5 → 2: nt2 → dv2 [G_hat[1,3]]

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Diagram 1L-232

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External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

Edges (propagator assignments):

2 → 4: nt2 → dv1 [G_hat[1,2]]
 3 → 2: nt1 → dv2 [G_hat[0,3]]
 3 → 4: nt1 → dv1 [G_hat[0,2]]
 4 → 0: nt1 → dn1 [G_hat[0,0]]
 5 → 1: nt2 → dn2 [G_hat[1,1]]
 5 → 2: nt2 → dv2 [G_hat[1,3]]

=====
 Diagram 1L-233
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External legs:

leaf 0 ← dn1
 leaf 1 ← dn2

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

Edges (propagator assignments):

2 → 4: nt2 → dv1 [G_hat[1,2]]
 3 → 2: nt1 → dv2 [G_hat[0,3]]
 3 → 4: nt1 → dv1 [G_hat[0,2]]
 4 → 0: nt1 → dn1 [G_hat[0,0]]
 5 → 1: nt2 → dn2 [G_hat[1,1]]
 5 → 2: nt2 → dv2 [G_hat[1,3]]

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Diagram 1L-234

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External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

Edges (propagator assignments):

2 → 4: nt2 → dv1 [G_hat[1,2]]

3 → 2: nt1 → dv2 [G_hat[0,3]]

3 → 4: nt1 → dv1 [G_hat[0,2]]

4 → 0: nt1 → dn1 [G_hat[0,0]]

5 → 1: nt2 → dn2 [G_hat[1,1]]

5 → 2: nt2 → dv2 [G_hat[1,3]]

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Diagram 1L-235

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External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

Edges (propagator assignments):

2 → 4: nt2 → dv1 [G_hat[1,2]]
 3 → 2: nt1 → dv2 [G_hat[0,3]]
 3 → 4: nt1 → dv1 [G_hat[0,2]]
 4 → 0: nt1 → dn1 [G_hat[0,0]]
 5 → 1: nt2 → dn2 [G_hat[1,1]]
 5 → 2: nt2 → dv2 [G_hat[1,3]]

=====
 Diagram 1L-236
 =====

External legs:

leaf 0 ← dn1
 leaf 1 ← dn2

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

Edges (propagator assignments):

2 → 4: nt2 → dv1 [G_hat[1,2]]
 3 → 2: nt1 → dv2 [G_hat[0,3]]
 3 → 4: nt1 → dv1 [G_hat[0,2]]
 4 → 0: nt1 → dn1 [G_hat[0,0]]
 5 → 1: nt2 → dn2 [G_hat[1,1]]
 5 → 2: nt2 → dv2 [G_hat[1,3]]

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Diagram 1L-237

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External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

Edges (propagator assignments):

2 → 4: nt2 → dv1 [G_hat[1,2]]

3 → 2: nt1 → dv2 [G_hat[0,3]]

3 → 4: nt1 → dv1 [G_hat[0,2]]

4 → 0: nt1 → dn1 [G_hat[0,0]]

5 → 1: nt2 → dn2 [G_hat[1,1]]

5 → 2: nt2 → dv2 [G_hat[1,3]]

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Diagram 1L-238

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External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

Edges (propagator assignments):

2 → 4: nt2 → dv1 [G_hat[1,2]]
 3 → 2: nt1 → dv2 [G_hat[0,3]]
 3 → 4: nt1 → dv1 [G_hat[0,2]]
 4 → 0: nt1 → dn1 [G_hat[0,0]]
 5 → 1: nt2 → dn2 [G_hat[1,1]]
 5 → 2: nt2 → dv2 [G_hat[1,3]]

=====
 Diagram 1L-239
 =====

External legs:

leaf 0 ← dn1
 leaf 1 ← dn2

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

Edges (propagator assignments):

2 → 4: nt2 → dv1 [G_hat[1,2]]
 3 → 2: nt1 → dv2 [G_hat[0,3]]
 3 → 4: nt1 → dv1 [G_hat[0,2]]
 4 → 0: nt1 → dn1 [G_hat[0,0]]
 5 → 1: nt2 → dn2 [G_hat[1,1]]
 5 → 2: nt2 → dv2 [G_hat[1,3]]

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Diagram 1L-240

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External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

Edges (propagator assignments):

2 → 4: nt2 → dv1 [G_hat[1,2]]

3 → 2: nt1 → dv2 [G_hat[0,3]]

3 → 4: nt1 → dv1 [G_hat[0,2]]

4 → 0: nt1 → dn1 [G_hat[0,0]]

5 → 1: nt2 → dn2 [G_hat[1,1]]

5 → 2: nt2 → dv2 [G_hat[1,3]]

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Diagram 1L-241

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External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

Edges (propagator assignments):

2 → 4: nt2 → dv2 [G_hat[1,3]]
 3 → 2: nt1 → dv2 [G_hat[0,3]]
 3 → 4: nt1 → dv2 [G_hat[0,3]]
 4 → 0: nt2 → dn1 [G_hat[1,0]]
 5 → 1: nt2 → dn2 [G_hat[1,1]]
 5 → 2: nt2 → dv2 [G_hat[1,3]]

=====
 Diagram 1L-242
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External legs:

leaf 0 ← dn1
 leaf 1 ← dn2

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

Edges (propagator assignments):

2 → 4: nt2 → dv2 [G_hat[1,3]]
 3 → 2: nt1 → dv2 [G_hat[0,3]]
 3 → 4: nt1 → dv2 [G_hat[0,3]]
 4 → 0: nt2 → dn1 [G_hat[1,0]]
 5 → 1: nt2 → dn2 [G_hat[1,1]]
 5 → 2: nt2 → dv2 [G_hat[1,3]]

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Diagram 1L-243

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External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

Edges (propagator assignments):

2 → 4: nt2 → dv2 [G_hat[1,3]]

3 → 2: nt1 → dv2 [G_hat[0,3]]

3 → 4: nt1 → dv2 [G_hat[0,3]]

4 → 0: nt2 → dn1 [G_hat[1,0]]

5 → 1: nt2 → dn2 [G_hat[1,1]]

5 → 2: nt2 → dv2 [G_hat[1,3]]

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Diagram 1L-244

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External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

Edges (propagator assignments):

2 → 4: nt2 → dv2 [G_hat[1,3]]
 3 → 2: nt1 → dv2 [G_hat[0,3]]
 3 → 4: nt1 → dv2 [G_hat[0,3]]
 4 → 0: nt2 → dn1 [G_hat[1,0]]
 5 → 1: nt2 → dn2 [G_hat[1,1]]
 5 → 2: nt2 → dv2 [G_hat[1,3]]

=====
 Diagram 1L-245
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External legs:

leaf 0 ← dn1
 leaf 1 ← dn2

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

Edges (propagator assignments):

2 → 4: nt2 → dv2 [G_hat[1,3]]
 3 → 2: nt1 → dv2 [G_hat[0,3]]
 3 → 4: nt1 → dv2 [G_hat[0,3]]
 4 → 0: nt2 → dn1 [G_hat[1,0]]
 5 → 1: nt2 → dn2 [G_hat[1,1]]
 5 → 2: nt2 → dv2 [G_hat[1,3]]

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Diagram 1L-246

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External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

Edges (propagator assignments):

2 → 4: nt2 → dv2 [G_hat[1,3]]

3 → 2: nt1 → dv2 [G_hat[0,3]]

3 → 4: nt1 → dv2 [G_hat[0,3]]

4 → 0: nt2 → dn1 [G_hat[1,0]]

5 → 1: nt2 → dn2 [G_hat[1,1]]

5 → 2: nt2 → dv2 [G_hat[1,3]]

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Diagram 1L-247

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External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

Edges (propagator assignments):

2 → 4: nt2 → dv2 [G_hat[1,3]]
 3 → 2: nt1 → dv2 [G_hat[0,3]]
 3 → 4: nt1 → dv2 [G_hat[0,3]]
 4 → 0: nt2 → dn1 [G_hat[1,0]]
 5 → 1: nt2 → dn2 [G_hat[1,1]]
 5 → 2: nt2 → dv2 [G_hat[1,3]]

=====
 Diagram 1L-248
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External legs:

leaf 0 ← dn1
 leaf 1 ← dn2

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

Edges (propagator assignments):

2 → 4: nt2 → dv2 [G_hat[1,3]]
 3 → 2: nt1 → dv2 [G_hat[0,3]]
 3 → 4: nt1 → dv2 [G_hat[0,3]]
 4 → 0: nt2 → dn1 [G_hat[1,0]]
 5 → 1: nt2 → dn2 [G_hat[1,1]]
 5 → 2: nt2 → dv2 [G_hat[1,3]]

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Diagram 1L-249

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External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

Edges (propagator assignments):

2 → 4: nt2 → dv2 [G_hat[1,3]]

3 → 2: nt1 → dv2 [G_hat[0,3]]

3 → 4: nt1 → dv2 [G_hat[0,3]]

4 → 0: nt2 → dn1 [G_hat[1,0]]

5 → 1: nt2 → dn2 [G_hat[1,1]]

5 → 2: nt2 → dv2 [G_hat[1,3]]

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Diagram 1L-250

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External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

Edges (propagator assignments):

2 → 4: nt2 → dv2 [G_hat[1,3]]
 3 → 2: nt1 → dv2 [G_hat[0,3]]
 3 → 4: nt1 → dv2 [G_hat[0,3]]
 4 → 0: nt2 → dn1 [G_hat[1,0]]
 5 → 1: nt2 → dn2 [G_hat[1,1]]
 5 → 2: nt2 → dv2 [G_hat[1,3]]

=====
 Diagram 1L-251
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External legs:

leaf 0 ← dn1
 leaf 1 ← dn2

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

Edges (propagator assignments):

2 → 4: nt2 → dv2 [G_hat[1,3]]
 3 → 2: nt1 → dv2 [G_hat[0,3]]
 3 → 4: nt1 → dv2 [G_hat[0,3]]
 4 → 0: nt2 → dn1 [G_hat[1,0]]
 5 → 1: nt2 → dn2 [G_hat[1,1]]
 5 → 2: nt2 → dv2 [G_hat[1,3]]

=====

Diagram 1L-252

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External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

Edges (propagator assignments):

2 → 4: nt2 → dv2 [G_hat[1,3]]

3 → 2: nt1 → dv2 [G_hat[0,3]]

3 → 4: nt1 → dv2 [G_hat[0,3]]

4 → 0: nt2 → dn1 [G_hat[1,0]]

5 → 1: nt2 → dn2 [G_hat[1,1]]

5 → 2: nt2 → dv2 [G_hat[1,3]]

=====

Diagram 1L-253

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External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

Edges (propagator assignments):

2 → 4: nt2 → dv2 [G_hat[1,3]]
 3 → 2: nt1 → dv2 [G_hat[0,3]]
 3 → 4: nt1 → dv2 [G_hat[0,3]]
 4 → 0: nt2 → dn1 [G_hat[1,0]]
 5 → 1: nt2 → dn2 [G_hat[1,1]]
 5 → 2: nt2 → dv2 [G_hat[1,3]]

=====
 Diagram 1L-254
 =====

External legs:

leaf 0 ← dn1
 leaf 1 ← dn2

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

Edges (propagator assignments):

2 → 4: nt2 → dv2 [G_hat[1,3]]
 3 → 2: nt1 → dv2 [G_hat[0,3]]
 3 → 4: nt1 → dv2 [G_hat[0,3]]
 4 → 0: nt2 → dn1 [G_hat[1,0]]
 5 → 1: nt2 → dn2 [G_hat[1,1]]
 5 → 2: nt2 → dv2 [G_hat[1,3]]

```

=====
Diagram 1L-255
=====
External legs:
  leaf 0 ← dn1
  leaf 1 ← dn2
Vertex assignments:
  v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

coeff =  $\frac{1}{2} \varphi_2''$ 

  v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

coeff =  $\frac{1}{2} nstar_1$ 

  v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

coeff =  $\frac{1}{2} \varphi_2''$ 

  v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

coeff =  $\frac{1}{2} nstar_2$ 

Edges (propagator assignments):
  2 → 4:  nt2 → dv2  [G_hat[1,3]]
  3 → 2:  nt1 → dv2  [G_hat[0,3]]
  3 → 4:  nt1 → dv2  [G_hat[0,3]]
  4 → 0:  nt2 → dn1  [G_hat[1,0]]
  5 → 1:  nt2 → dn2  [G_hat[1,1]]
  5 → 2:  nt2 → dv2  [G_hat[1,3]]

```

```

=====
Diagram 1L-256
=====
External legs:
  leaf 0 ← dn1
  leaf 1 ← dn2
Vertex assignments:
  v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

coeff =  $\frac{1}{2} \varphi_2''$ 

  v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

coeff =  $\frac{1}{2} nstar_1$ 

```

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

Edges (propagator assignments):

2 → 4: nt2 → dv2 [G_hat[1,3]]
 3 → 2: nt1 → dv2 [G_hat[0,3]]
 3 → 4: nt1 → dv2 [G_hat[0,3]]
 4 → 0: nt2 → dn1 [G_hat[1,0]]
 5 → 1: nt2 → dn2 [G_hat[1,1]]
 5 → 2: nt2 → dv2 [G_hat[1,3]]

=====
 Diagram 1L-257
 =====

External legs:

leaf 0 ← dn1
 leaf 1 ← dn2

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

Edges (propagator assignments):

2 → 4: nt1 → dv1 [G_hat[0,2]]
 3 → 2: nt2 → dv1 [G_hat[1,2]]
 3 → 4: nt2 → dv1 [G_hat[1,2]]
 4 → 0: nt1 → dn1 [G_hat[0,0]]
 5 → 1: nt1 → dn2 [G_hat[0,1]]
 5 → 2: nt1 → dv1 [G_hat[0,2]]

=====

Diagram 1L-258

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

Edges (propagator assignments):

2 → 4: nt1 → dv1 [G_hat[0,2]]

3 → 2: nt2 → dv1 [G_hat[1,2]]

3 → 4: nt2 → dv1 [G_hat[1,2]]

4 → 0: nt1 → dn1 [G_hat[0,0]]

5 → 1: nt1 → dn2 [G_hat[0,1]]

5 → 2: nt1 → dv1 [G_hat[0,2]]

=====

Diagram 1L-259

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

Edges (propagator assignments):

2 → 4: nt1 → dv1 [G_hat[0,2]]
 3 → 2: nt2 → dv1 [G_hat[1,2]]
 3 → 4: nt2 → dv1 [G_hat[1,2]]
 4 → 0: nt1 → dn1 [G_hat[0,0]]
 5 → 1: nt1 → dn2 [G_hat[0,1]]
 5 → 2: nt1 → dv1 [G_hat[0,2]]

=====
 Diagram 1L-260
 =====

External legs:

leaf 0 ← dn1
 leaf 1 ← dn2

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

Edges (propagator assignments):

2 → 4: nt1 → dv1 [G_hat[0,2]]
 3 → 2: nt2 → dv1 [G_hat[1,2]]
 3 → 4: nt2 → dv1 [G_hat[1,2]]
 4 → 0: nt1 → dn1 [G_hat[0,0]]
 5 → 1: nt1 → dn2 [G_hat[0,1]]
 5 → 2: nt1 → dv1 [G_hat[0,2]]

=====

Diagram 1L-261

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

Edges (propagator assignments):

2 → 4: nt1 → dv1 [G_hat[0,2]]

3 → 2: nt2 → dv1 [G_hat[1,2]]

3 → 4: nt2 → dv1 [G_hat[1,2]]

4 → 0: nt1 → dn1 [G_hat[0,0]]

5 → 1: nt1 → dn2 [G_hat[0,1]]

5 → 2: nt1 → dv1 [G_hat[0,2]]

=====

Diagram 1L-262

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

Edges (propagator assignments):

2 → 4: nt1 → dv1 [G_hat[0,2]]
 3 → 2: nt2 → dv1 [G_hat[1,2]]
 3 → 4: nt2 → dv1 [G_hat[1,2]]
 4 → 0: nt1 → dn1 [G_hat[0,0]]
 5 → 1: nt1 → dn2 [G_hat[0,1]]
 5 → 2: nt1 → dv1 [G_hat[0,2]]

=====
 Diagram 1L-263
 =====

External legs:

leaf 0 ← dn1
 leaf 1 ← dn2

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

Edges (propagator assignments):

2 → 4: nt1 → dv1 [G_hat[0,2]]
 3 → 2: nt2 → dv1 [G_hat[1,2]]
 3 → 4: nt2 → dv1 [G_hat[1,2]]
 4 → 0: nt1 → dn1 [G_hat[0,0]]
 5 → 1: nt1 → dn2 [G_hat[0,1]]
 5 → 2: nt1 → dv1 [G_hat[0,2]]

=====

Diagram 1L-264

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

Edges (propagator assignments):

2 → 4: nt1 → dv1 [G_hat[0,2]]

3 → 2: nt2 → dv1 [G_hat[1,2]]

3 → 4: nt2 → dv1 [G_hat[1,2]]

4 → 0: nt1 → dn1 [G_hat[0,0]]

5 → 1: nt1 → dn2 [G_hat[0,1]]

5 → 2: nt1 → dv1 [G_hat[0,2]]

=====

Diagram 1L-265

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

Edges (propagator assignments):

2 → 4: nt1 → dv1 [G_hat[0,2]]
 3 → 2: nt2 → dv1 [G_hat[1,2]]
 3 → 4: nt2 → dv1 [G_hat[1,2]]
 4 → 0: nt1 → dn1 [G_hat[0,0]]
 5 → 1: nt1 → dn2 [G_hat[0,1]]
 5 → 2: nt1 → dv1 [G_hat[0,2]]

=====
 Diagram 1L-266
 =====

External legs:

leaf 0 ← dn1
 leaf 1 ← dn2

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

Edges (propagator assignments):

2 → 4: nt1 → dv1 [G_hat[0,2]]
 3 → 2: nt2 → dv1 [G_hat[1,2]]
 3 → 4: nt2 → dv1 [G_hat[1,2]]
 4 → 0: nt1 → dn1 [G_hat[0,0]]
 5 → 1: nt1 → dn2 [G_hat[0,1]]
 5 → 2: nt1 → dv1 [G_hat[0,2]]

=====

Diagram 1L-267

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

Edges (propagator assignments):

2 → 4: nt1 → dv1 [G_hat[0,2]]

3 → 2: nt2 → dv1 [G_hat[1,2]]

3 → 4: nt2 → dv1 [G_hat[1,2]]

4 → 0: nt1 → dn1 [G_hat[0,0]]

5 → 1: nt1 → dn2 [G_hat[0,1]]

5 → 2: nt1 → dv1 [G_hat[0,2]]

=====

Diagram 1L-268

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

Edges (propagator assignments):

2 → 4: nt1 → dv1 [G_hat[0,2]]
 3 → 2: nt2 → dv1 [G_hat[1,2]]
 3 → 4: nt2 → dv1 [G_hat[1,2]]
 4 → 0: nt1 → dn1 [G_hat[0,0]]
 5 → 1: nt1 → dn2 [G_hat[0,1]]
 5 → 2: nt1 → dv1 [G_hat[0,2]]

=====
 Diagram 1L-269
 =====

External legs:

leaf 0 ← dn1
 leaf 1 ← dn2

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

Edges (propagator assignments):

2 → 4: nt1 → dv1 [G_hat[0,2]]
 3 → 2: nt2 → dv1 [G_hat[1,2]]
 3 → 4: nt2 → dv1 [G_hat[1,2]]
 4 → 0: nt1 → dn1 [G_hat[0,0]]
 5 → 1: nt1 → dn2 [G_hat[0,1]]
 5 → 2: nt1 → dv1 [G_hat[0,2]]

=====

Diagram 1L-270

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

Edges (propagator assignments):

2 → 4: nt1 → dv1 [G_hat[0,2]]

3 → 2: nt2 → dv1 [G_hat[1,2]]

3 → 4: nt2 → dv1 [G_hat[1,2]]

4 → 0: nt1 → dn1 [G_hat[0,0]]

5 → 1: nt1 → dn2 [G_hat[0,1]]

5 → 2: nt1 → dv1 [G_hat[0,2]]

=====

Diagram 1L-271

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

Edges (propagator assignments):

2 → 4: nt1 → dv1 [G_hat[0,2]]
 3 → 2: nt2 → dv1 [G_hat[1,2]]
 3 → 4: nt2 → dv1 [G_hat[1,2]]
 4 → 0: nt1 → dn1 [G_hat[0,0]]
 5 → 1: nt1 → dn2 [G_hat[0,1]]
 5 → 2: nt1 → dv1 [G_hat[0,2]]

=====
 Diagram 1L-272
 =====

External legs:

leaf 0 ← dn1
 leaf 1 ← dn2

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

Edges (propagator assignments):

2 → 4: nt1 → dv1 [G_hat[0,2]]
 3 → 2: nt2 → dv1 [G_hat[1,2]]
 3 → 4: nt2 → dv1 [G_hat[1,2]]
 4 → 0: nt1 → dn1 [G_hat[0,0]]
 5 → 1: nt1 → dn2 [G_hat[0,1]]
 5 → 2: nt1 → dv1 [G_hat[0,2]]

=====

Diagram 1L-273

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

Edges (propagator assignments):

2 → 4: nt1 → dv2 [G_hat[0,3]]

3 → 2: nt2 → dv1 [G_hat[1,2]]

3 → 4: nt2 → dv2 [G_hat[1,3]]

4 → 0: nt2 → dn1 [G_hat[1,0]]

5 → 1: nt1 → dn2 [G_hat[0,1]]

5 → 2: nt1 → dv1 [G_hat[0,2]]

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Diagram 1L-274

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External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

Edges (propagator assignments):

2 → 4: nt1 → dv2 [G_hat[0,3]]
 3 → 2: nt2 → dv1 [G_hat[1,2]]
 3 → 4: nt2 → dv2 [G_hat[1,3]]
 4 → 0: nt2 → dn1 [G_hat[1,0]]
 5 → 1: nt1 → dn2 [G_hat[0,1]]
 5 → 2: nt1 → dv1 [G_hat[0,2]]

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 Diagram 1L-275
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External legs:

leaf 0 ← dn1
 leaf 1 ← dn2

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

Edges (propagator assignments):

2 → 4: nt1 → dv2 [G_hat[0,3]]
 3 → 2: nt2 → dv1 [G_hat[1,2]]
 3 → 4: nt2 → dv2 [G_hat[1,3]]
 4 → 0: nt2 → dn1 [G_hat[1,0]]
 5 → 1: nt1 → dn2 [G_hat[0,1]]
 5 → 2: nt1 → dv1 [G_hat[0,2]]

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Diagram 1L-276

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External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

Edges (propagator assignments):

2 → 4: nt1 → dv2 [G_hat[0,3]]

3 → 2: nt2 → dv1 [G_hat[1,2]]

3 → 4: nt2 → dv2 [G_hat[1,3]]

4 → 0: nt2 → dn1 [G_hat[1,0]]

5 → 1: nt1 → dn2 [G_hat[0,1]]

5 → 2: nt1 → dv1 [G_hat[0,2]]

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Diagram 1L-277

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External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

Edges (propagator assignments):

2 → 4: nt1 → dv2 [G_hat[0,3]]
 3 → 2: nt2 → dv1 [G_hat[1,2]]
 3 → 4: nt2 → dv2 [G_hat[1,3]]
 4 → 0: nt2 → dn1 [G_hat[1,0]]
 5 → 1: nt1 → dn2 [G_hat[0,1]]
 5 → 2: nt1 → dv1 [G_hat[0,2]]

Diagram 1L-278

External legs:

leaf 0 ← dn1
 leaf 1 ← dn2

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

Edges (propagator assignments):

2 → 4: nt1 → dv2 [G_hat[0,3]]
 3 → 2: nt2 → dv1 [G_hat[1,2]]
 3 → 4: nt2 → dv2 [G_hat[1,3]]
 4 → 0: nt2 → dn1 [G_hat[1,0]]
 5 → 1: nt1 → dn2 [G_hat[0,1]]
 5 → 2: nt1 → dv1 [G_hat[0,2]]

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Diagram 1L-279

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External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

Edges (propagator assignments):

2 → 4: nt1 → dv2 [G_hat[0,3]]

3 → 2: nt2 → dv1 [G_hat[1,2]]

3 → 4: nt2 → dv2 [G_hat[1,3]]

4 → 0: nt2 → dn1 [G_hat[1,0]]

5 → 1: nt1 → dn2 [G_hat[0,1]]

5 → 2: nt1 → dv1 [G_hat[0,2]]

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Diagram 1L-280

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External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

Edges (propagator assignments):

2 → 4: nt1 → dv2 [G_hat[0,3]]
 3 → 2: nt2 → dv1 [G_hat[1,2]]
 3 → 4: nt2 → dv2 [G_hat[1,3]]
 4 → 0: nt2 → dn1 [G_hat[1,0]]
 5 → 1: nt1 → dn2 [G_hat[0,1]]
 5 → 2: nt1 → dv1 [G_hat[0,2]]

=====
 Diagram 1L-281
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External legs:

leaf 0 ← dn1
 leaf 1 ← dn2

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

Edges (propagator assignments):

2 → 4: nt1 → dv2 [G_hat[0,3]]
 3 → 2: nt2 → dv1 [G_hat[1,2]]
 3 → 4: nt2 → dv2 [G_hat[1,3]]
 4 → 0: nt2 → dn1 [G_hat[1,0]]
 5 → 1: nt1 → dn2 [G_hat[0,1]]
 5 → 2: nt1 → dv1 [G_hat[0,2]]

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Diagram 1L-282

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External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

Edges (propagator assignments):

2 → 4: nt1 → dv2 [G_hat[0,3]]

3 → 2: nt2 → dv1 [G_hat[1,2]]

3 → 4: nt2 → dv2 [G_hat[1,3]]

4 → 0: nt2 → dn1 [G_hat[1,0]]

5 → 1: nt1 → dn2 [G_hat[0,1]]

5 → 2: nt1 → dv1 [G_hat[0,2]]

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Diagram 1L-283

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External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

Edges (propagator assignments):

2 → 4: nt1 → dv2 [G_hat[0,3]]
 3 → 2: nt2 → dv1 [G_hat[1,2]]
 3 → 4: nt2 → dv2 [G_hat[1,3]]
 4 → 0: nt2 → dn1 [G_hat[1,0]]
 5 → 1: nt1 → dn2 [G_hat[0,1]]
 5 → 2: nt1 → dv1 [G_hat[0,2]]

=====
 Diagram 1L-284
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External legs:

leaf 0 ← dn1
 leaf 1 ← dn2

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

Edges (propagator assignments):

2 → 4: nt1 → dv2 [G_hat[0,3]]
 3 → 2: nt2 → dv1 [G_hat[1,2]]
 3 → 4: nt2 → dv2 [G_hat[1,3]]
 4 → 0: nt2 → dn1 [G_hat[1,0]]
 5 → 1: nt1 → dn2 [G_hat[0,1]]
 5 → 2: nt1 → dv1 [G_hat[0,2]]

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Diagram 1L-285

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External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

Edges (propagator assignments):

2 → 4: nt1 → dv2 [G_hat[0,3]]

3 → 2: nt2 → dv1 [G_hat[1,2]]

3 → 4: nt2 → dv2 [G_hat[1,3]]

4 → 0: nt2 → dn1 [G_hat[1,0]]

5 → 1: nt1 → dn2 [G_hat[0,1]]

5 → 2: nt1 → dv1 [G_hat[0,2]]

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Diagram 1L-286

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External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

Edges (propagator assignments):

2 → 4: nt1 → dv2 [G_hat[0,3]]
 3 → 2: nt2 → dv1 [G_hat[1,2]]
 3 → 4: nt2 → dv2 [G_hat[1,3]]
 4 → 0: nt2 → dn1 [G_hat[1,0]]
 5 → 1: nt1 → dn2 [G_hat[0,1]]
 5 → 2: nt1 → dv1 [G_hat[0,2]]

=====
 Diagram 1L-287
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External legs:

leaf 0 ← dn1
 leaf 1 ← dn2

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

Edges (propagator assignments):

2 → 4: nt1 → dv2 [G_hat[0,3]]
 3 → 2: nt2 → dv1 [G_hat[1,2]]
 3 → 4: nt2 → dv2 [G_hat[1,3]]
 4 → 0: nt2 → dn1 [G_hat[1,0]]
 5 → 1: nt1 → dn2 [G_hat[0,1]]
 5 → 2: nt1 → dv1 [G_hat[0,2]]

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Diagram 1L-288

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External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

Edges (propagator assignments):

2 → 4: nt1 → dv2 [G_hat[0,3]]

3 → 2: nt2 → dv1 [G_hat[1,2]]

3 → 4: nt2 → dv2 [G_hat[1,3]]

4 → 0: nt2 → dn1 [G_hat[1,0]]

5 → 1: nt1 → dn2 [G_hat[0,1]]

5 → 2: nt1 → dv1 [G_hat[0,2]]

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Diagram 1L-289

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External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

Edges (propagator assignments):

2 → 4: nt2 → dv1 [G_hat[1,2]]
 3 → 2: nt2 → dv2 [G_hat[1,3]]
 3 → 4: nt2 → dv1 [G_hat[1,2]]
 4 → 0: nt1 → dn1 [G_hat[0,0]]
 5 → 1: nt1 → dn2 [G_hat[0,1]]
 5 → 2: nt1 → dv2 [G_hat[0,3]]

=====
 Diagram 1L-290
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External legs:

leaf 0 ← dn1
 leaf 1 ← dn2

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

Edges (propagator assignments):

2 → 4: nt2 → dv1 [G_hat[1,2]]
 3 → 2: nt2 → dv2 [G_hat[1,3]]
 3 → 4: nt2 → dv1 [G_hat[1,2]]
 4 → 0: nt1 → dn1 [G_hat[0,0]]
 5 → 1: nt1 → dn2 [G_hat[0,1]]
 5 → 2: nt1 → dv2 [G_hat[0,3]]

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Diagram 1L-291

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External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

Edges (propagator assignments):

2 → 4: nt2 → dv1 [G_hat[1,2]]

3 → 2: nt2 → dv2 [G_hat[1,3]]

3 → 4: nt2 → dv1 [G_hat[1,2]]

4 → 0: nt1 → dn1 [G_hat[0,0]]

5 → 1: nt1 → dn2 [G_hat[0,1]]

5 → 2: nt1 → dv2 [G_hat[0,3]]

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Diagram 1L-292

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External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

Edges (propagator assignments):

2 → 4: nt2 → dv1 [G_hat[1,2]]
 3 → 2: nt2 → dv2 [G_hat[1,3]]
 3 → 4: nt2 → dv1 [G_hat[1,2]]
 4 → 0: nt1 → dn1 [G_hat[0,0]]
 5 → 1: nt1 → dn2 [G_hat[0,1]]
 5 → 2: nt1 → dv2 [G_hat[0,3]]

=====
 Diagram 1L-293
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External legs:

leaf 0 ← dn1
 leaf 1 ← dn2

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

Edges (propagator assignments):

2 → 4: nt2 → dv1 [G_hat[1,2]]
 3 → 2: nt2 → dv2 [G_hat[1,3]]
 3 → 4: nt2 → dv1 [G_hat[1,2]]
 4 → 0: nt1 → dn1 [G_hat[0,0]]
 5 → 1: nt1 → dn2 [G_hat[0,1]]
 5 → 2: nt1 → dv2 [G_hat[0,3]]

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Diagram 1L-294

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External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

Edges (propagator assignments):

2 → 4: nt2 → dv1 [G_hat[1,2]]

3 → 2: nt2 → dv2 [G_hat[1,3]]

3 → 4: nt2 → dv1 [G_hat[1,2]]

4 → 0: nt1 → dn1 [G_hat[0,0]]

5 → 1: nt1 → dn2 [G_hat[0,1]]

5 → 2: nt1 → dv2 [G_hat[0,3]]

=====

Diagram 1L-295

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External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

Edges (propagator assignments):

2 → 4: nt2 → dv1 [G_hat[1,2]]
 3 → 2: nt2 → dv2 [G_hat[1,3]]
 3 → 4: nt2 → dv1 [G_hat[1,2]]
 4 → 0: nt1 → dn1 [G_hat[0,0]]
 5 → 1: nt1 → dn2 [G_hat[0,1]]
 5 → 2: nt1 → dv2 [G_hat[0,3]]

=====
 Diagram 1L-296
 =====

External legs:

leaf 0 ← dn1
 leaf 1 ← dn2

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

Edges (propagator assignments):

2 → 4: nt2 → dv1 [G_hat[1,2]]
 3 → 2: nt2 → dv2 [G_hat[1,3]]
 3 → 4: nt2 → dv1 [G_hat[1,2]]
 4 → 0: nt1 → dn1 [G_hat[0,0]]
 5 → 1: nt1 → dn2 [G_hat[0,1]]
 5 → 2: nt1 → dv2 [G_hat[0,3]]

=====

Diagram 1L-297

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External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

Edges (propagator assignments):

2 → 4: nt2 → dv1 [G_hat[1,2]]

3 → 2: nt2 → dv2 [G_hat[1,3]]

3 → 4: nt2 → dv1 [G_hat[1,2]]

4 → 0: nt1 → dn1 [G_hat[0,0]]

5 → 1: nt1 → dn2 [G_hat[0,1]]

5 → 2: nt1 → dv2 [G_hat[0,3]]

=====

Diagram 1L-298

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

Edges (propagator assignments):

2 → 4: nt2 → dv1 [G_hat[1,2]]
 3 → 2: nt2 → dv2 [G_hat[1,3]]
 3 → 4: nt2 → dv1 [G_hat[1,2]]
 4 → 0: nt1 → dn1 [G_hat[0,0]]
 5 → 1: nt1 → dn2 [G_hat[0,1]]
 5 → 2: nt1 → dv2 [G_hat[0,3]]

=====
 Diagram 1L-299
 =====

External legs:

leaf 0 ← dn1
 leaf 1 ← dn2

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

Edges (propagator assignments):

2 → 4: nt2 → dv1 [G_hat[1,2]]
 3 → 2: nt2 → dv2 [G_hat[1,3]]
 3 → 4: nt2 → dv1 [G_hat[1,2]]
 4 → 0: nt1 → dn1 [G_hat[0,0]]
 5 → 1: nt1 → dn2 [G_hat[0,1]]
 5 → 2: nt1 → dv2 [G_hat[0,3]]

=====

Diagram 1L-300

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

Edges (propagator assignments):

2 → 4: nt2 → dv1 [G_hat[1,2]]

3 → 2: nt2 → dv2 [G_hat[1,3]]

3 → 4: nt2 → dv1 [G_hat[1,2]]

4 → 0: nt1 → dn1 [G_hat[0,0]]

5 → 1: nt1 → dn2 [G_hat[0,1]]

5 → 2: nt1 → dv2 [G_hat[0,3]]

=====

Diagram 1L-301

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

Edges (propagator assignments):

2 → 4: nt2 → dv1 [G_hat[1,2]]
 3 → 2: nt2 → dv2 [G_hat[1,3]]
 3 → 4: nt2 → dv1 [G_hat[1,2]]
 4 → 0: nt1 → dn1 [G_hat[0,0]]
 5 → 1: nt1 → dn2 [G_hat[0,1]]
 5 → 2: nt1 → dv2 [G_hat[0,3]]

=====
 Diagram 1L-302
 =====

External legs:

leaf 0 ← dn1
 leaf 1 ← dn2

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

Edges (propagator assignments):

2 → 4: nt2 → dv1 [G_hat[1,2]]
 3 → 2: nt2 → dv2 [G_hat[1,3]]
 3 → 4: nt2 → dv1 [G_hat[1,2]]
 4 → 0: nt1 → dn1 [G_hat[0,0]]
 5 → 1: nt1 → dn2 [G_hat[0,1]]
 5 → 2: nt1 → dv2 [G_hat[0,3]]

=====

Diagram 1L-303

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

Edges (propagator assignments):

2 → 4: nt2 → dv1 [G_hat[1,2]]

3 → 2: nt2 → dv2 [G_hat[1,3]]

3 → 4: nt2 → dv1 [G_hat[1,2]]

4 → 0: nt1 → dn1 [G_hat[0,0]]

5 → 1: nt1 → dn2 [G_hat[0,1]]

5 → 2: nt1 → dv2 [G_hat[0,3]]

=====

Diagram 1L-304

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External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

Edges (propagator assignments):

2 → 4: nt2 → dv1 [G_hat[1,2]]
 3 → 2: nt2 → dv2 [G_hat[1,3]]
 3 → 4: nt2 → dv1 [G_hat[1,2]]
 4 → 0: nt1 → dn1 [G_hat[0,0]]
 5 → 1: nt1 → dn2 [G_hat[0,1]]
 5 → 2: nt1 → dv2 [G_hat[0,3]]

=====
 Diagram 1L-305
 =====

External legs:

leaf 0 ← dn1
 leaf 1 ← dn2

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

Edges (propagator assignments):

2 → 4: nt2 → dv2 [G_hat[1,3]]
 3 → 2: nt2 → dv2 [G_hat[1,3]]
 3 → 4: nt2 → dv2 [G_hat[1,3]]
 4 → 0: nt2 → dn1 [G_hat[1,0]]
 5 → 1: nt1 → dn2 [G_hat[0,1]]
 5 → 2: nt1 → dv2 [G_hat[0,3]]

=====

Diagram 1L-306

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

Edges (propagator assignments):

2 → 4: nt2 → dv2 [G_hat[1,3]]

3 → 2: nt2 → dv2 [G_hat[1,3]]

3 → 4: nt2 → dv2 [G_hat[1,3]]

4 → 0: nt2 → dn1 [G_hat[1,0]]

5 → 1: nt1 → dn2 [G_hat[0,1]]

5 → 2: nt1 → dv2 [G_hat[0,3]]

=====

Diagram 1L-307

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External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

Edges (propagator assignments):

2 → 4: nt2 → dv2 [G_hat[1,3]]
 3 → 2: nt2 → dv2 [G_hat[1,3]]
 3 → 4: nt2 → dv2 [G_hat[1,3]]
 4 → 0: nt2 → dn1 [G_hat[1,0]]
 5 → 1: nt1 → dn2 [G_hat[0,1]]
 5 → 2: nt1 → dv2 [G_hat[0,3]]

=====
 Diagram 1L-308
 =====

External legs:

leaf 0 ← dn1
 leaf 1 ← dn2

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

Edges (propagator assignments):

2 → 4: nt2 → dv2 [G_hat[1,3]]
 3 → 2: nt2 → dv2 [G_hat[1,3]]
 3 → 4: nt2 → dv2 [G_hat[1,3]]
 4 → 0: nt2 → dn1 [G_hat[1,0]]
 5 → 1: nt1 → dn2 [G_hat[0,1]]
 5 → 2: nt1 → dv2 [G_hat[0,3]]

=====

Diagram 1L-309

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

Edges (propagator assignments):

2 → 4: nt2 → dv2 [G_hat[1,3]]

3 → 2: nt2 → dv2 [G_hat[1,3]]

3 → 4: nt2 → dv2 [G_hat[1,3]]

4 → 0: nt2 → dn1 [G_hat[1,0]]

5 → 1: nt1 → dn2 [G_hat[0,1]]

5 → 2: nt1 → dv2 [G_hat[0,3]]

=====

Diagram 1L-310

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

Edges (propagator assignments):

2 → 4: nt2 → dv2 [G_hat[1,3]]
 3 → 2: nt2 → dv2 [G_hat[1,3]]
 3 → 4: nt2 → dv2 [G_hat[1,3]]
 4 → 0: nt2 → dn1 [G_hat[1,0]]
 5 → 1: nt1 → dn2 [G_hat[0,1]]
 5 → 2: nt1 → dv2 [G_hat[0,3]]

=====
 Diagram 1L-311
 =====

External legs:

leaf 0 ← dn1
 leaf 1 ← dn2

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

Edges (propagator assignments):

2 → 4: nt2 → dv2 [G_hat[1,3]]
 3 → 2: nt2 → dv2 [G_hat[1,3]]
 3 → 4: nt2 → dv2 [G_hat[1,3]]
 4 → 0: nt2 → dn1 [G_hat[1,0]]
 5 → 1: nt1 → dn2 [G_hat[0,1]]
 5 → 2: nt1 → dv2 [G_hat[0,3]]

=====

Diagram 1L-312

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

Edges (propagator assignments):

2 → 4: nt2 → dv2 [G_hat[1,3]]

3 → 2: nt2 → dv2 [G_hat[1,3]]

3 → 4: nt2 → dv2 [G_hat[1,3]]

4 → 0: nt2 → dn1 [G_hat[1,0]]

5 → 1: nt1 → dn2 [G_hat[0,1]]

5 → 2: nt1 → dv2 [G_hat[0,3]]

=====

Diagram 1L-313

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

Edges (propagator assignments):

2 → 4: nt2 → dv2 [G_hat[1,3]]
 3 → 2: nt2 → dv2 [G_hat[1,3]]
 3 → 4: nt2 → dv2 [G_hat[1,3]]
 4 → 0: nt2 → dn1 [G_hat[1,0]]
 5 → 1: nt1 → dn2 [G_hat[0,1]]
 5 → 2: nt1 → dv2 [G_hat[0,3]]

=====
 Diagram 1L-314
 =====

External legs:

leaf 0 ← dn1
 leaf 1 ← dn2

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

Edges (propagator assignments):

2 → 4: nt2 → dv2 [G_hat[1,3]]
 3 → 2: nt2 → dv2 [G_hat[1,3]]
 3 → 4: nt2 → dv2 [G_hat[1,3]]
 4 → 0: nt2 → dn1 [G_hat[1,0]]
 5 → 1: nt1 → dn2 [G_hat[0,1]]
 5 → 2: nt1 → dv2 [G_hat[0,3]]

```

=====
Diagram 1L-315
=====
External legs:
  leaf 0 ← dn1
  leaf 1 ← dn2
Vertex assignments:
  v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

coeff =  $\frac{1}{2} \varphi_2''$ 

  v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

coeff =  $\frac{1}{2} nstar_2$ 

  v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

coeff =  $\frac{1}{2} \varphi_2''$ 

  v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

coeff =  $\frac{1}{2} nstar_1$ 

Edges (propagator assignments):
  2 → 4:  nt2 → dv2  [G_hat[1,3]]
  3 → 2:  nt2 → dv2  [G_hat[1,3]]
  3 → 4:  nt2 → dv2  [G_hat[1,3]]
  4 → 0:  nt2 → dn1  [G_hat[1,0]]
  5 → 1:  nt1 → dn2  [G_hat[0,1]]
  5 → 2:  nt1 → dv2  [G_hat[0,3]]

```

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=====
Diagram 1L-316
=====
External legs:
  leaf 0 ← dn1
  leaf 1 ← dn2
Vertex assignments:
  v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

coeff =  $\frac{1}{2} \varphi_2''$ 

  v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

coeff =  $\frac{1}{2} nstar_2$ 

```

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

Edges (propagator assignments):

2 → 4: nt2 → dv2 [G_hat[1,3]]
 3 → 2: nt2 → dv2 [G_hat[1,3]]
 3 → 4: nt2 → dv2 [G_hat[1,3]]
 4 → 0: nt2 → dn1 [G_hat[1,0]]
 5 → 1: nt1 → dn2 [G_hat[0,1]]
 5 → 2: nt1 → dv2 [G_hat[0,3]]

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 Diagram 1L-317
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External legs:

leaf 0 ← dn1
 leaf 1 ← dn2

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

Edges (propagator assignments):

2 → 4: nt2 → dv2 [G_hat[1,3]]
 3 → 2: nt2 → dv2 [G_hat[1,3]]
 3 → 4: nt2 → dv2 [G_hat[1,3]]
 4 → 0: nt2 → dn1 [G_hat[1,0]]
 5 → 1: nt1 → dn2 [G_hat[0,1]]
 5 → 2: nt1 → dv2 [G_hat[0,3]]

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Diagram 1L-318

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External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

Edges (propagator assignments):

2 → 4: nt2 → dv2 [G_hat[1,3]]

3 → 2: nt2 → dv2 [G_hat[1,3]]

3 → 4: nt2 → dv2 [G_hat[1,3]]

4 → 0: nt2 → dn1 [G_hat[1,0]]

5 → 1: nt1 → dn2 [G_hat[0,1]]

5 → 2: nt1 → dv2 [G_hat[0,3]]

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Diagram 1L-319

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External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

Edges (propagator assignments):

2 → 4: nt2 → dv2 [G_hat[1,3]]
 3 → 2: nt2 → dv2 [G_hat[1,3]]
 3 → 4: nt2 → dv2 [G_hat[1,3]]
 4 → 0: nt2 → dn1 [G_hat[1,0]]
 5 → 1: nt1 → dn2 [G_hat[0,1]]
 5 → 2: nt1 → dv2 [G_hat[0,3]]

=====
 Diagram 1L-320
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External legs:

leaf 0 ← dn1
 leaf 1 ← dn2

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

Edges (propagator assignments):

2 → 4: nt2 → dv2 [G_hat[1,3]]
 3 → 2: nt2 → dv2 [G_hat[1,3]]
 3 → 4: nt2 → dv2 [G_hat[1,3]]
 4 → 0: nt2 → dn1 [G_hat[1,0]]
 5 → 1: nt1 → dn2 [G_hat[0,1]]
 5 → 2: nt1 → dv2 [G_hat[0,3]]

=====

Diagram 1L-321

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External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

Edges (propagator assignments):

2 → 4: nt1 → dv1 [G_hat[0,2]]

3 → 2: nt2 → dv1 [G_hat[1,2]]

3 → 4: nt2 → dv1 [G_hat[1,2]]

4 → 0: nt1 → dn1 [G_hat[0,0]]

5 → 1: nt2 → dn2 [G_hat[1,1]]

5 → 2: nt2 → dv1 [G_hat[1,2]]

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Diagram 1L-322

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External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

Edges (propagator assignments):

2 → 4: nt1 → dv1 [G_hat[0,2]]
 3 → 2: nt2 → dv1 [G_hat[1,2]]
 3 → 4: nt2 → dv1 [G_hat[1,2]]
 4 → 0: nt1 → dn1 [G_hat[0,0]]
 5 → 1: nt2 → dn2 [G_hat[1,1]]
 5 → 2: nt2 → dv1 [G_hat[1,2]]

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 Diagram 1L-323
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External legs:

leaf 0 ← dn1
 leaf 1 ← dn2

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

Edges (propagator assignments):

2 → 4: nt1 → dv1 [G_hat[0,2]]
 3 → 2: nt2 → dv1 [G_hat[1,2]]
 3 → 4: nt2 → dv1 [G_hat[1,2]]
 4 → 0: nt1 → dn1 [G_hat[0,0]]
 5 → 1: nt2 → dn2 [G_hat[1,1]]
 5 → 2: nt2 → dv1 [G_hat[1,2]]

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Diagram 1L-324

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External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

Edges (propagator assignments):

2 → 4: nt1 → dv1 [G_hat[0,2]]

3 → 2: nt2 → dv1 [G_hat[1,2]]

3 → 4: nt2 → dv1 [G_hat[1,2]]

4 → 0: nt1 → dn1 [G_hat[0,0]]

5 → 1: nt2 → dn2 [G_hat[1,1]]

5 → 2: nt2 → dv1 [G_hat[1,2]]

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Diagram 1L-325

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External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

Edges (propagator assignments):

2 → 4: nt1 → dv1 [G_hat[0,2]]
 3 → 2: nt2 → dv1 [G_hat[1,2]]
 3 → 4: nt2 → dv1 [G_hat[1,2]]
 4 → 0: nt1 → dn1 [G_hat[0,0]]
 5 → 1: nt2 → dn2 [G_hat[1,1]]
 5 → 2: nt2 → dv1 [G_hat[1,2]]

=====
 Diagram 1L-326
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External legs:

leaf 0 ← dn1
 leaf 1 ← dn2

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

Edges (propagator assignments):

2 → 4: nt1 → dv1 [G_hat[0,2]]
 3 → 2: nt2 → dv1 [G_hat[1,2]]
 3 → 4: nt2 → dv1 [G_hat[1,2]]
 4 → 0: nt1 → dn1 [G_hat[0,0]]
 5 → 1: nt2 → dn2 [G_hat[1,1]]
 5 → 2: nt2 → dv1 [G_hat[1,2]]

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Diagram 1L-327

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External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

Edges (propagator assignments):

2 → 4: nt1 → dv1 [G_hat[0,2]]

3 → 2: nt2 → dv1 [G_hat[1,2]]

3 → 4: nt2 → dv1 [G_hat[1,2]]

4 → 0: nt1 → dn1 [G_hat[0,0]]

5 → 1: nt2 → dn2 [G_hat[1,1]]

5 → 2: nt2 → dv1 [G_hat[1,2]]

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Diagram 1L-328

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External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

Edges (propagator assignments):

2 → 4: nt1 → dv1 [G_hat[0,2]]
 3 → 2: nt2 → dv1 [G_hat[1,2]]
 3 → 4: nt2 → dv1 [G_hat[1,2]]
 4 → 0: nt1 → dn1 [G_hat[0,0]]
 5 → 1: nt2 → dn2 [G_hat[1,1]]
 5 → 2: nt2 → dv1 [G_hat[1,2]]

=====
 Diagram 1L-329
 =====

External legs:

leaf 0 ← dn1
 leaf 1 ← dn2

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

Edges (propagator assignments):

2 → 4: nt1 → dv1 [G_hat[0,2]]
 3 → 2: nt2 → dv1 [G_hat[1,2]]
 3 → 4: nt2 → dv1 [G_hat[1,2]]
 4 → 0: nt1 → dn1 [G_hat[0,0]]
 5 → 1: nt2 → dn2 [G_hat[1,1]]
 5 → 2: nt2 → dv1 [G_hat[1,2]]

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Diagram 1L-330

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External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

Edges (propagator assignments):

2 → 4: nt1 → dv1 [G_hat[0,2]]

3 → 2: nt2 → dv1 [G_hat[1,2]]

3 → 4: nt2 → dv1 [G_hat[1,2]]

4 → 0: nt1 → dn1 [G_hat[0,0]]

5 → 1: nt2 → dn2 [G_hat[1,1]]

5 → 2: nt2 → dv1 [G_hat[1,2]]

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Diagram 1L-331

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External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

Edges (propagator assignments):

2 → 4: nt1 → dv1 [G_hat[0,2]]
 3 → 2: nt2 → dv1 [G_hat[1,2]]
 3 → 4: nt2 → dv1 [G_hat[1,2]]
 4 → 0: nt1 → dn1 [G_hat[0,0]]
 5 → 1: nt2 → dn2 [G_hat[1,1]]
 5 → 2: nt2 → dv1 [G_hat[1,2]]

=====
 Diagram 1L-332
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External legs:

leaf 0 ← dn1
 leaf 1 ← dn2

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

Edges (propagator assignments):

2 → 4: nt1 → dv1 [G_hat[0,2]]
 3 → 2: nt2 → dv1 [G_hat[1,2]]
 3 → 4: nt2 → dv1 [G_hat[1,2]]
 4 → 0: nt1 → dn1 [G_hat[0,0]]
 5 → 1: nt2 → dn2 [G_hat[1,1]]
 5 → 2: nt2 → dv1 [G_hat[1,2]]

=====

Diagram 1L-333

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External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

Edges (propagator assignments):

2 → 4: nt1 → dv1 [G_hat[0,2]]

3 → 2: nt2 → dv1 [G_hat[1,2]]

3 → 4: nt2 → dv1 [G_hat[1,2]]

4 → 0: nt1 → dn1 [G_hat[0,0]]

5 → 1: nt2 → dn2 [G_hat[1,1]]

5 → 2: nt2 → dv1 [G_hat[1,2]]

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Diagram 1L-334

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External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

Edges (propagator assignments):

2 → 4: nt1 → dv1 [G_hat[0,2]]
 3 → 2: nt2 → dv1 [G_hat[1,2]]
 3 → 4: nt2 → dv1 [G_hat[1,2]]
 4 → 0: nt1 → dn1 [G_hat[0,0]]
 5 → 1: nt2 → dn2 [G_hat[1,1]]
 5 → 2: nt2 → dv1 [G_hat[1,2]]

=====
 Diagram 1L-335
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External legs:

leaf 0 ← dn1
 leaf 1 ← dn2

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

Edges (propagator assignments):

2 → 4: nt1 → dv1 [G_hat[0,2]]
 3 → 2: nt2 → dv1 [G_hat[1,2]]
 3 → 4: nt2 → dv1 [G_hat[1,2]]
 4 → 0: nt1 → dn1 [G_hat[0,0]]
 5 → 1: nt2 → dn2 [G_hat[1,1]]
 5 → 2: nt2 → dv1 [G_hat[1,2]]

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Diagram 1L-336

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External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

Edges (propagator assignments):

2 → 4: nt1 → dv1 [G_hat[0,2]]

3 → 2: nt2 → dv1 [G_hat[1,2]]

3 → 4: nt2 → dv1 [G_hat[1,2]]

4 → 0: nt1 → dn1 [G_hat[0,0]]

5 → 1: nt2 → dn2 [G_hat[1,1]]

5 → 2: nt2 → dv1 [G_hat[1,2]]

=====

Diagram 1L-337

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External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

Edges (propagator assignments):

2 → 4: nt1 → dv2 [G_hat[0,3]]
 3 → 2: nt2 → dv1 [G_hat[1,2]]
 3 → 4: nt2 → dv2 [G_hat[1,3]]
 4 → 0: nt2 → dn1 [G_hat[1,0]]
 5 → 1: nt2 → dn2 [G_hat[1,1]]
 5 → 2: nt2 → dv1 [G_hat[1,2]]

=====
 Diagram 1L-338
 =====

External legs:

leaf 0 ← dn1
 leaf 1 ← dn2

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

Edges (propagator assignments):

2 → 4: nt1 → dv2 [G_hat[0,3]]
 3 → 2: nt2 → dv1 [G_hat[1,2]]
 3 → 4: nt2 → dv2 [G_hat[1,3]]
 4 → 0: nt2 → dn1 [G_hat[1,0]]
 5 → 1: nt2 → dn2 [G_hat[1,1]]
 5 → 2: nt2 → dv1 [G_hat[1,2]]

=====

Diagram 1L-339

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

Edges (propagator assignments):

2 → 4: nt1 → dv2 [G_hat[0,3]]

3 → 2: nt2 → dv1 [G_hat[1,2]]

3 → 4: nt2 → dv2 [G_hat[1,3]]

4 → 0: nt2 → dn1 [G_hat[1,0]]

5 → 1: nt2 → dn2 [G_hat[1,1]]

5 → 2: nt2 → dv1 [G_hat[1,2]]

=====

Diagram 1L-340

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External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

Edges (propagator assignments):

2 → 4: nt1 → dv2 [G_hat[0,3]]
 3 → 2: nt2 → dv1 [G_hat[1,2]]
 3 → 4: nt2 → dv2 [G_hat[1,3]]
 4 → 0: nt2 → dn1 [G_hat[1,0]]
 5 → 1: nt2 → dn2 [G_hat[1,1]]
 5 → 2: nt2 → dv1 [G_hat[1,2]]

=====
 Diagram 1L-341
 =====

External legs:

leaf 0 ← dn1
 leaf 1 ← dn2

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

Edges (propagator assignments):

2 → 4: nt1 → dv2 [G_hat[0,3]]
 3 → 2: nt2 → dv1 [G_hat[1,2]]
 3 → 4: nt2 → dv2 [G_hat[1,3]]
 4 → 0: nt2 → dn1 [G_hat[1,0]]
 5 → 1: nt2 → dn2 [G_hat[1,1]]
 5 → 2: nt2 → dv1 [G_hat[1,2]]


```

=====
Diagram 1L-342
=====

External legs:
  leaf 0 ← dn1
  leaf 1 ← dn2
Vertex assignments:
  v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

coeff =  $\frac{1}{2} \varphi_1''$ 

  v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

coeff =  $\frac{1}{2} nstar_2$ 

  v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

coeff =  $\frac{1}{2} \varphi_2''$ 

  v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

coeff =  $\frac{1}{2} nstar_2$ 

Edges (propagator assignments):
  2 → 4:  nt1 → dv2  [G_hat[0,3]]
  3 → 2:  nt2 → dv1  [G_hat[1,2]]
  3 → 4:  nt2 → dv2  [G_hat[1,3]]
  4 → 0:  nt2 → dn1  [G_hat[1,0]]
  5 → 1:  nt2 → dn2  [G_hat[1,1]]
  5 → 2:  nt2 → dv1  [G_hat[1,2]]

```

```

=====
Diagram 1L-343
=====

External legs:
  leaf 0 ← dn1
  leaf 1 ← dn2
Vertex assignments:
  v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

coeff =  $\frac{1}{2} \varphi_1''$ 

  v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

coeff =  $\frac{1}{2} nstar_2$ 

```

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

Edges (propagator assignments):

2 → 4: nt1 → dv2 [G_hat[0,3]]
 3 → 2: nt2 → dv1 [G_hat[1,2]]
 3 → 4: nt2 → dv2 [G_hat[1,3]]
 4 → 0: nt2 → dn1 [G_hat[1,0]]
 5 → 1: nt2 → dn2 [G_hat[1,1]]
 5 → 2: nt2 → dv1 [G_hat[1,2]]

Diagram 1L-344

External legs:

leaf 0 ← dn1
 leaf 1 ← dn2

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

Edges (propagator assignments):

2 → 4: nt1 → dv2 [G_hat[0,3]]
 3 → 2: nt2 → dv1 [G_hat[1,2]]
 3 → 4: nt2 → dv2 [G_hat[1,3]]
 4 → 0: nt2 → dn1 [G_hat[1,0]]
 5 → 1: nt2 → dn2 [G_hat[1,1]]
 5 → 2: nt2 → dv1 [G_hat[1,2]]

=====

Diagram 1L-345

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

Edges (propagator assignments):

2 → 4: nt1 → dv2 [G_hat[0,3]]

3 → 2: nt2 → dv1 [G_hat[1,2]]

3 → 4: nt2 → dv2 [G_hat[1,3]]

4 → 0: nt2 → dn1 [G_hat[1,0]]

5 → 1: nt2 → dn2 [G_hat[1,1]]

5 → 2: nt2 → dv1 [G_hat[1,2]]

=====

Diagram 1L-346

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

Edges (propagator assignments):

2 → 4: nt1 → dv2 [G_hat[0,3]]
 3 → 2: nt2 → dv1 [G_hat[1,2]]
 3 → 4: nt2 → dv2 [G_hat[1,3]]
 4 → 0: nt2 → dn1 [G_hat[1,0]]
 5 → 1: nt2 → dn2 [G_hat[1,1]]
 5 → 2: nt2 → dv1 [G_hat[1,2]]

Diagram 1L-347

External legs:

leaf 0 ← dn1
 leaf 1 ← dn2

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

Edges (propagator assignments):

2 → 4: nt1 → dv2 [G_hat[0,3]]
 3 → 2: nt2 → dv1 [G_hat[1,2]]
 3 → 4: nt2 → dv2 [G_hat[1,3]]
 4 → 0: nt2 → dn1 [G_hat[1,0]]
 5 → 1: nt2 → dn2 [G_hat[1,1]]
 5 → 2: nt2 → dv1 [G_hat[1,2]]

=====

Diagram 1L-348

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

Edges (propagator assignments):

2 → 4: nt1 → dv2 [G_hat[0,3]]

3 → 2: nt2 → dv1 [G_hat[1,2]]

3 → 4: nt2 → dv2 [G_hat[1,3]]

4 → 0: nt2 → dn1 [G_hat[1,0]]

5 → 1: nt2 → dn2 [G_hat[1,1]]

5 → 2: nt2 → dv1 [G_hat[1,2]]

=====

Diagram 1L-349

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

Edges (propagator assignments):

2 → 4: nt1 → dv2 [G_hat[0,3]]
 3 → 2: nt2 → dv1 [G_hat[1,2]]
 3 → 4: nt2 → dv2 [G_hat[1,3]]
 4 → 0: nt2 → dn1 [G_hat[1,0]]
 5 → 1: nt2 → dn2 [G_hat[1,1]]
 5 → 2: nt2 → dv1 [G_hat[1,2]]

=====
 Diagram 1L-350
 =====

External legs:

leaf 0 ← dn1
 leaf 1 ← dn2

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

Edges (propagator assignments):

2 → 4: nt1 → dv2 [G_hat[0,3]]
 3 → 2: nt2 → dv1 [G_hat[1,2]]
 3 → 4: nt2 → dv2 [G_hat[1,3]]
 4 → 0: nt2 → dn1 [G_hat[1,0]]
 5 → 1: nt2 → dn2 [G_hat[1,1]]
 5 → 2: nt2 → dv1 [G_hat[1,2]]

```

=====
Diagram 1L-351
=====

External legs:
  leaf 0 ← dn1
  leaf 1 ← dn2
Vertex assignments:
  v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

coeff =  $\frac{1}{2} \varphi_1''$ 

  v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

coeff =  $\frac{1}{2} nstar_2$ 

  v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

coeff =  $\frac{1}{2} \varphi_2''$ 

  v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

coeff =  $\frac{1}{2} nstar_2$ 

Edges (propagator assignments):
  2 → 4:  nt1 → dv2  [G_hat[0,3]]
  3 → 2:  nt2 → dv1  [G_hat[1,2]]
  3 → 4:  nt2 → dv2  [G_hat[1,3]]
  4 → 0:  nt2 → dn1  [G_hat[1,0]]
  5 → 1:  nt2 → dn2  [G_hat[1,1]]
  5 → 2:  nt2 → dv1  [G_hat[1,2]]

```

```

=====
Diagram 1L-352
=====

External legs:
  leaf 0 ← dn1
  leaf 1 ← dn2
Vertex assignments:
  v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

coeff =  $\frac{1}{2} \varphi_1''$ 

  v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

coeff =  $\frac{1}{2} nstar_2$ 

```

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

Edges (propagator assignments):

2 → 4: nt1 → dv2 [G_hat[0,3]]
 3 → 2: nt2 → dv1 [G_hat[1,2]]
 3 → 4: nt2 → dv2 [G_hat[1,3]]
 4 → 0: nt2 → dn1 [G_hat[1,0]]
 5 → 1: nt2 → dn2 [G_hat[1,1]]
 5 → 2: nt2 → dv1 [G_hat[1,2]]

=====
 Diagram 1L-353
 =====

External legs:

leaf 0 ← dn1
 leaf 1 ← dn2

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

Edges (propagator assignments):

2 → 4: nt2 → dv1 [G_hat[1,2]]
 3 → 2: nt2 → dv2 [G_hat[1,3]]
 3 → 4: nt2 → dv1 [G_hat[1,2]]
 4 → 0: nt1 → dn1 [G_hat[0,0]]
 5 → 1: nt2 → dn2 [G_hat[1,1]]
 5 → 2: nt2 → dv2 [G_hat[1,3]]

=====

Diagram 1L-354

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

Edges (propagator assignments):

2 → 4: nt2 → dv1 [G_hat[1,2]]

3 → 2: nt2 → dv2 [G_hat[1,3]]

3 → 4: nt2 → dv1 [G_hat[1,2]]

4 → 0: nt1 → dn1 [G_hat[0,0]]

5 → 1: nt2 → dn2 [G_hat[1,1]]

5 → 2: nt2 → dv2 [G_hat[1,3]]

=====

Diagram 1L-355

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

Edges (propagator assignments):

2 → 4: nt2 → dv1 [G_hat[1,2]]
 3 → 2: nt2 → dv2 [G_hat[1,3]]
 3 → 4: nt2 → dv1 [G_hat[1,2]]
 4 → 0: nt1 → dn1 [G_hat[0,0]]
 5 → 1: nt2 → dn2 [G_hat[1,1]]
 5 → 2: nt2 → dv2 [G_hat[1,3]]

=====
 Diagram 1L-356
 =====

External legs:

leaf 0 ← dn1
 leaf 1 ← dn2

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

Edges (propagator assignments):

2 → 4: nt2 → dv1 [G_hat[1,2]]
 3 → 2: nt2 → dv2 [G_hat[1,3]]
 3 → 4: nt2 → dv1 [G_hat[1,2]]
 4 → 0: nt1 → dn1 [G_hat[0,0]]
 5 → 1: nt2 → dn2 [G_hat[1,1]]
 5 → 2: nt2 → dv2 [G_hat[1,3]]

=====

Diagram 1L-357

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

Edges (propagator assignments):

2 → 4: nt2 → dv1 [G_hat[1,2]]

3 → 2: nt2 → dv2 [G_hat[1,3]]

3 → 4: nt2 → dv1 [G_hat[1,2]]

4 → 0: nt1 → dn1 [G_hat[0,0]]

5 → 1: nt2 → dn2 [G_hat[1,1]]

5 → 2: nt2 → dv2 [G_hat[1,3]]

=====

Diagram 1L-358

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

Edges (propagator assignments):

2 → 4: nt2 → dv1 [G_hat[1,2]]
 3 → 2: nt2 → dv2 [G_hat[1,3]]
 3 → 4: nt2 → dv1 [G_hat[1,2]]
 4 → 0: nt1 → dn1 [G_hat[0,0]]
 5 → 1: nt2 → dn2 [G_hat[1,1]]
 5 → 2: nt2 → dv2 [G_hat[1,3]]

=====
 Diagram 1L-359
 =====

External legs:

leaf 0 ← dn1
 leaf 1 ← dn2

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

Edges (propagator assignments):

2 → 4: nt2 → dv1 [G_hat[1,2]]
 3 → 2: nt2 → dv2 [G_hat[1,3]]
 3 → 4: nt2 → dv1 [G_hat[1,2]]
 4 → 0: nt1 → dn1 [G_hat[0,0]]
 5 → 1: nt2 → dn2 [G_hat[1,1]]
 5 → 2: nt2 → dv2 [G_hat[1,3]]

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Diagram 1L-360

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External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

Edges (propagator assignments):

2 → 4: nt2 → dv1 [G_hat[1,2]]

3 → 2: nt2 → dv2 [G_hat[1,3]]

3 → 4: nt2 → dv1 [G_hat[1,2]]

4 → 0: nt1 → dn1 [G_hat[0,0]]

5 → 1: nt2 → dn2 [G_hat[1,1]]

5 → 2: nt2 → dv2 [G_hat[1,3]]

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Diagram 1L-361

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External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

Edges (propagator assignments):

2 → 4: nt2 → dv1 [G_hat[1,2]]
 3 → 2: nt2 → dv2 [G_hat[1,3]]
 3 → 4: nt2 → dv1 [G_hat[1,2]]
 4 → 0: nt1 → dn1 [G_hat[0,0]]
 5 → 1: nt2 → dn2 [G_hat[1,1]]
 5 → 2: nt2 → dv2 [G_hat[1,3]]

=====
 Diagram 1L-362
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External legs:

leaf 0 ← dn1
 leaf 1 ← dn2

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

Edges (propagator assignments):

2 → 4: nt2 → dv1 [G_hat[1,2]]
 3 → 2: nt2 → dv2 [G_hat[1,3]]
 3 → 4: nt2 → dv1 [G_hat[1,2]]
 4 → 0: nt1 → dn1 [G_hat[0,0]]
 5 → 1: nt2 → dn2 [G_hat[1,1]]
 5 → 2: nt2 → dv2 [G_hat[1,3]]

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Diagram 1L-363

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External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

Edges (propagator assignments):

2 → 4: nt2 → dv1 [G_hat[1,2]]

3 → 2: nt2 → dv2 [G_hat[1,3]]

3 → 4: nt2 → dv1 [G_hat[1,2]]

4 → 0: nt1 → dn1 [G_hat[0,0]]

5 → 1: nt2 → dn2 [G_hat[1,1]]

5 → 2: nt2 → dv2 [G_hat[1,3]]

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Diagram 1L-364

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External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

Edges (propagator assignments):

2 → 4: nt2 → dv1 [G_hat[1,2]]
 3 → 2: nt2 → dv2 [G_hat[1,3]]
 3 → 4: nt2 → dv1 [G_hat[1,2]]
 4 → 0: nt1 → dn1 [G_hat[0,0]]
 5 → 1: nt2 → dn2 [G_hat[1,1]]
 5 → 2: nt2 → dv2 [G_hat[1,3]]

=====
 Diagram 1L-365
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External legs:

leaf 0 ← dn1
 leaf 1 ← dn2

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

Edges (propagator assignments):

2 → 4: nt2 → dv1 [G_hat[1,2]]
 3 → 2: nt2 → dv2 [G_hat[1,3]]
 3 → 4: nt2 → dv1 [G_hat[1,2]]
 4 → 0: nt1 → dn1 [G_hat[0,0]]
 5 → 1: nt2 → dn2 [G_hat[1,1]]
 5 → 2: nt2 → dv2 [G_hat[1,3]]

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Diagram 1L-366

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External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

Edges (propagator assignments):

2 → 4: nt2 → dv1 [G_hat[1,2]]

3 → 2: nt2 → dv2 [G_hat[1,3]]

3 → 4: nt2 → dv1 [G_hat[1,2]]

4 → 0: nt1 → dn1 [G_hat[0,0]]

5 → 1: nt2 → dn2 [G_hat[1,1]]

5 → 2: nt2 → dv2 [G_hat[1,3]]

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Diagram 1L-367

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External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

Edges (propagator assignments):

2 → 4: nt2 → dv1 [G_hat[1,2]]
 3 → 2: nt2 → dv2 [G_hat[1,3]]
 3 → 4: nt2 → dv1 [G_hat[1,2]]
 4 → 0: nt1 → dn1 [G_hat[0,0]]
 5 → 1: nt2 → dn2 [G_hat[1,1]]
 5 → 2: nt2 → dv2 [G_hat[1,3]]

=====
 Diagram 1L-368
 =====

External legs:

leaf 0 ← dn1
 leaf 1 ← dn2

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

Edges (propagator assignments):

2 → 4: nt2 → dv1 [G_hat[1,2]]
 3 → 2: nt2 → dv2 [G_hat[1,3]]
 3 → 4: nt2 → dv1 [G_hat[1,2]]
 4 → 0: nt1 → dn1 [G_hat[0,0]]
 5 → 1: nt2 → dn2 [G_hat[1,1]]
 5 → 2: nt2 → dv2 [G_hat[1,3]]

=====

Diagram 1L-369

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External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

Edges (propagator assignments):

2 → 4: nt2 → dv2 [G_hat[1,3]]

3 → 2: nt2 → dv2 [G_hat[1,3]]

3 → 4: nt2 → dv2 [G_hat[1,3]]

4 → 0: nt2 → dn1 [G_hat[1,0]]

5 → 1: nt2 → dn2 [G_hat[1,1]]

5 → 2: nt2 → dv2 [G_hat[1,3]]

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Diagram 1L-370

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External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

Edges (propagator assignments):

2 → 4: nt2 → dv2 [G_hat[1,3]]
 3 → 2: nt2 → dv2 [G_hat[1,3]]
 3 → 4: nt2 → dv2 [G_hat[1,3]]
 4 → 0: nt2 → dn1 [G_hat[1,0]]
 5 → 1: nt2 → dn2 [G_hat[1,1]]
 5 → 2: nt2 → dv2 [G_hat[1,3]]

=====
 Diagram 1L-371
 =====

External legs:

leaf 0 ← dn1
 leaf 1 ← dn2

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

Edges (propagator assignments):

2 → 4: nt2 → dv2 [G_hat[1,3]]
 3 → 2: nt2 → dv2 [G_hat[1,3]]
 3 → 4: nt2 → dv2 [G_hat[1,3]]
 4 → 0: nt2 → dn1 [G_hat[1,0]]
 5 → 1: nt2 → dn2 [G_hat[1,1]]
 5 → 2: nt2 → dv2 [G_hat[1,3]]

=====

Diagram 1L-372

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External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

Edges (propagator assignments):

2 → 4: nt2 → dv2 [G_hat[1,3]]

3 → 2: nt2 → dv2 [G_hat[1,3]]

3 → 4: nt2 → dv2 [G_hat[1,3]]

4 → 0: nt2 → dn1 [G_hat[1,0]]

5 → 1: nt2 → dn2 [G_hat[1,1]]

5 → 2: nt2 → dv2 [G_hat[1,3]]

=====

Diagram 1L-373

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External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

Edges (propagator assignments):

2 → 4: nt2 → dv2 [G_hat[1,3]]
 3 → 2: nt2 → dv2 [G_hat[1,3]]
 3 → 4: nt2 → dv2 [G_hat[1,3]]
 4 → 0: nt2 → dn1 [G_hat[1,0]]
 5 → 1: nt2 → dn2 [G_hat[1,1]]
 5 → 2: nt2 → dv2 [G_hat[1,3]]

Diagram 1L-374

External legs:

leaf 0 ← dn1
 leaf 1 ← dn2

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

Edges (propagator assignments):

2 → 4: nt2 → dv2 [G_hat[1,3]]
 3 → 2: nt2 → dv2 [G_hat[1,3]]
 3 → 4: nt2 → dv2 [G_hat[1,3]]
 4 → 0: nt2 → dn1 [G_hat[1,0]]
 5 → 1: nt2 → dn2 [G_hat[1,1]]
 5 → 2: nt2 → dv2 [G_hat[1,3]]

```

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Diagram 1L-375
=====
External legs:
  leaf 0 ← dn1
  leaf 1 ← dn2
Vertex assignments:
  v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

coeff =  $\frac{1}{2} \varphi_2''$ 

  v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

coeff =  $\frac{1}{2} nstar_2$ 

  v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

coeff =  $\frac{1}{2} \varphi_2''$ 

  v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

coeff =  $\frac{1}{2} nstar_2$ 

Edges (propagator assignments):
  2 → 4:  nt2 → dv2  [G_hat[1,3]]
  3 → 2:  nt2 → dv2  [G_hat[1,3]]
  3 → 4:  nt2 → dv2  [G_hat[1,3]]
  4 → 0:  nt2 → dn1  [G_hat[1,0]]
  5 → 1:  nt2 → dn2  [G_hat[1,1]]
  5 → 2:  nt2 → dv2  [G_hat[1,3]]

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=====
Diagram 1L-376
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External legs:
  leaf 0 ← dn1
  leaf 1 ← dn2
Vertex assignments:
  v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

coeff =  $\frac{1}{2} \varphi_2''$ 

  v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

coeff =  $\frac{1}{2} nstar_2$ 

```

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

Edges (propagator assignments):

2 → 4: nt2 → dv2 [G_hat[1,3]]
 3 → 2: nt2 → dv2 [G_hat[1,3]]
 3 → 4: nt2 → dv2 [G_hat[1,3]]
 4 → 0: nt2 → dn1 [G_hat[1,0]]
 5 → 1: nt2 → dn2 [G_hat[1,1]]
 5 → 2: nt2 → dv2 [G_hat[1,3]]

=====
 Diagram 1L-377
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External legs:

leaf 0 ← dn1
 leaf 1 ← dn2

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

Edges (propagator assignments):

2 → 4: nt2 → dv2 [G_hat[1,3]]
 3 → 2: nt2 → dv2 [G_hat[1,3]]
 3 → 4: nt2 → dv2 [G_hat[1,3]]
 4 → 0: nt2 → dn1 [G_hat[1,0]]
 5 → 1: nt2 → dn2 [G_hat[1,1]]
 5 → 2: nt2 → dv2 [G_hat[1,3]]

=====

Diagram 1L-378

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External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

Edges (propagator assignments):

2 → 4: nt2 → dv2 [G_hat[1,3]]

3 → 2: nt2 → dv2 [G_hat[1,3]]

3 → 4: nt2 → dv2 [G_hat[1,3]]

4 → 0: nt2 → dn1 [G_hat[1,0]]

5 → 1: nt2 → dn2 [G_hat[1,1]]

5 → 2: nt2 → dv2 [G_hat[1,3]]

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Diagram 1L-379

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External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

Edges (propagator assignments):

2 → 4: nt2 → dv2 [G_hat[1,3]]
 3 → 2: nt2 → dv2 [G_hat[1,3]]
 3 → 4: nt2 → dv2 [G_hat[1,3]]
 4 → 0: nt2 → dn1 [G_hat[1,0]]
 5 → 1: nt2 → dn2 [G_hat[1,1]]
 5 → 2: nt2 → dv2 [G_hat[1,3]]

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 Diagram 1L-380
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External legs:

leaf 0 ← dn1
 leaf 1 ← dn2

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

Edges (propagator assignments):

2 → 4: nt2 → dv2 [G_hat[1,3]]
 3 → 2: nt2 → dv2 [G_hat[1,3]]
 3 → 4: nt2 → dv2 [G_hat[1,3]]
 4 → 0: nt2 → dn1 [G_hat[1,0]]
 5 → 1: nt2 → dn2 [G_hat[1,1]]
 5 → 2: nt2 → dv2 [G_hat[1,3]]

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Diagram 1L-381
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External legs:
  leaf 0 ← dn1
  leaf 1 ← dn2
Vertex assignments:
  v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

coeff =  $\frac{1}{2} \varphi_2''$ 

  v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

coeff =  $\frac{1}{2} nstar_2$ 

  v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

coeff =  $\frac{1}{2} \varphi_2''$ 

  v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

coeff =  $\frac{1}{2} nstar_2$ 

Edges (propagator assignments):
  2 → 4:  nt2 → dv2  [G_hat[1,3]]
  3 → 2:  nt2 → dv2  [G_hat[1,3]]
  3 → 4:  nt2 → dv2  [G_hat[1,3]]
  4 → 0:  nt2 → dn1  [G_hat[1,0]]
  5 → 1:  nt2 → dn2  [G_hat[1,1]]
  5 → 2:  nt2 → dv2  [G_hat[1,3]]

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Diagram 1L-382
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External legs:
  leaf 0 ← dn1
  leaf 1 ← dn2
Vertex assignments:
  v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

coeff =  $\frac{1}{2} \varphi_2''$ 

  v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

coeff =  $\frac{1}{2} nstar_2$ 

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v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

Edges (propagator assignments):

2 → 4: nt2 → dv2 [G_hat[1,3]]
 3 → 2: nt2 → dv2 [G_hat[1,3]]
 3 → 4: nt2 → dv2 [G_hat[1,3]]
 4 → 0: nt2 → dn1 [G_hat[1,0]]
 5 → 1: nt2 → dn2 [G_hat[1,1]]
 5 → 2: nt2 → dv2 [G_hat[1,3]]

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 Diagram 1L-383
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External legs:

leaf 0 ← dn1
 leaf 1 ← dn2

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

Edges (propagator assignments):

2 → 4: nt2 → dv2 [G_hat[1,3]]
 3 → 2: nt2 → dv2 [G_hat[1,3]]
 3 → 4: nt2 → dv2 [G_hat[1,3]]
 4 → 0: nt2 → dn1 [G_hat[1,0]]
 5 → 1: nt2 → dn2 [G_hat[1,1]]
 5 → 2: nt2 → dv2 [G_hat[1,3]]

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Diagram 1L-384

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External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

Edges (propagator assignments):

2 → 4: nt2 → dv2 [G_hat[1,3]]

3 → 2: nt2 → dv2 [G_hat[1,3]]

3 → 4: nt2 → dv2 [G_hat[1,3]]

4 → 0: nt2 → dn1 [G_hat[1,0]]

5 → 1: nt2 → dn2 [G_hat[1,1]]

5 → 2: nt2 → dv2 [G_hat[1,3]]

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Diagram 1L-385

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External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

Edges (propagator assignments):

2 → 4: nt1 → dv1 [G_hat[0,2]]
 3 → 2: nt1 → dv1 [G_hat[0,2]]
 3 → 4: nt1 → dv1 [G_hat[0,2]]
 4 → 0: nt1 → dn2 [G_hat[0,1]]
 5 → 1: nt1 → dn1 [G_hat[0,0]]
 5 → 2: nt1 → dv1 [G_hat[0,2]]

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 Diagram 1L-386
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External legs:

leaf 0 ← dn2
 leaf 1 ← dn1

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

Edges (propagator assignments):

2 → 4: nt1 → dv1 [G_hat[0,2]]
 3 → 2: nt1 → dv1 [G_hat[0,2]]
 3 → 4: nt1 → dv1 [G_hat[0,2]]
 4 → 0: nt1 → dn2 [G_hat[0,1]]
 5 → 1: nt1 → dn1 [G_hat[0,0]]
 5 → 2: nt1 → dv1 [G_hat[0,2]]

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Diagram 1L-387

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External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

Edges (propagator assignments):

2 → 4: nt1 → dv1 [G_hat[0,2]]

3 → 2: nt1 → dv1 [G_hat[0,2]]

3 → 4: nt1 → dv1 [G_hat[0,2]]

4 → 0: nt1 → dn2 [G_hat[0,1]]

5 → 1: nt1 → dn1 [G_hat[0,0]]

5 → 2: nt1 → dv1 [G_hat[0,2]]

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Diagram 1L-388

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External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

Edges (propagator assignments):

2 → 4: nt1 → dv1 [G_hat[0,2]]
 3 → 2: nt1 → dv1 [G_hat[0,2]]
 3 → 4: nt1 → dv1 [G_hat[0,2]]
 4 → 0: nt1 → dn2 [G_hat[0,1]]
 5 → 1: nt1 → dn1 [G_hat[0,0]]
 5 → 2: nt1 → dv1 [G_hat[0,2]]

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 Diagram 1L-389
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External legs:

leaf 0 ← dn2
 leaf 1 ← dn1

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

Edges (propagator assignments):

2 → 4: nt1 → dv1 [G_hat[0,2]]
 3 → 2: nt1 → dv1 [G_hat[0,2]]
 3 → 4: nt1 → dv1 [G_hat[0,2]]
 4 → 0: nt1 → dn2 [G_hat[0,1]]
 5 → 1: nt1 → dn1 [G_hat[0,0]]
 5 → 2: nt1 → dv1 [G_hat[0,2]]

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Diagram 1L-390

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External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

Edges (propagator assignments):

2 → 4: nt1 → dv1 [G_hat[0,2]]

3 → 2: nt1 → dv1 [G_hat[0,2]]

3 → 4: nt1 → dv1 [G_hat[0,2]]

4 → 0: nt1 → dn2 [G_hat[0,1]]

5 → 1: nt1 → dn1 [G_hat[0,0]]

5 → 2: nt1 → dv1 [G_hat[0,2]]

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Diagram 1L-391

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External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

Edges (propagator assignments):

2 → 4: nt1 → dv1 [G_hat[0,2]]
 3 → 2: nt1 → dv1 [G_hat[0,2]]
 3 → 4: nt1 → dv1 [G_hat[0,2]]
 4 → 0: nt1 → dn2 [G_hat[0,1]]
 5 → 1: nt1 → dn1 [G_hat[0,0]]
 5 → 2: nt1 → dv1 [G_hat[0,2]]

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 Diagram 1L-392
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External legs:

leaf 0 ← dn2
 leaf 1 ← dn1

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

Edges (propagator assignments):

2 → 4: nt1 → dv1 [G_hat[0,2]]
 3 → 2: nt1 → dv1 [G_hat[0,2]]
 3 → 4: nt1 → dv1 [G_hat[0,2]]
 4 → 0: nt1 → dn2 [G_hat[0,1]]
 5 → 1: nt1 → dn1 [G_hat[0,0]]
 5 → 2: nt1 → dv1 [G_hat[0,2]]

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Diagram 1L-393

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External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

Edges (propagator assignments):

2 → 4: nt1 → dv1 [G_hat[0,2]]

3 → 2: nt1 → dv1 [G_hat[0,2]]

3 → 4: nt1 → dv1 [G_hat[0,2]]

4 → 0: nt1 → dn2 [G_hat[0,1]]

5 → 1: nt1 → dn1 [G_hat[0,0]]

5 → 2: nt1 → dv1 [G_hat[0,2]]

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Diagram 1L-394

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External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

Edges (propagator assignments):

2 → 4: nt1 → dv1 [G_hat[0,2]]
 3 → 2: nt1 → dv1 [G_hat[0,2]]
 3 → 4: nt1 → dv1 [G_hat[0,2]]
 4 → 0: nt1 → dn2 [G_hat[0,1]]
 5 → 1: nt1 → dn1 [G_hat[0,0]]
 5 → 2: nt1 → dv1 [G_hat[0,2]]

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 Diagram 1L-395
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External legs:

leaf 0 ← dn2
 leaf 1 ← dn1

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

Edges (propagator assignments):

2 → 4: nt1 → dv1 [G_hat[0,2]]
 3 → 2: nt1 → dv1 [G_hat[0,2]]
 3 → 4: nt1 → dv1 [G_hat[0,2]]
 4 → 0: nt1 → dn2 [G_hat[0,1]]
 5 → 1: nt1 → dn1 [G_hat[0,0]]
 5 → 2: nt1 → dv1 [G_hat[0,2]]

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Diagram 1L-396

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External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

Edges (propagator assignments):

2 → 4: nt1 → dv1 [G_hat[0,2]]

3 → 2: nt1 → dv1 [G_hat[0,2]]

3 → 4: nt1 → dv1 [G_hat[0,2]]

4 → 0: nt1 → dn2 [G_hat[0,1]]

5 → 1: nt1 → dn1 [G_hat[0,0]]

5 → 2: nt1 → dv1 [G_hat[0,2]]

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Diagram 1L-397

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External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

Edges (propagator assignments):

2 → 4: nt1 → dv1 [G_hat[0,2]]
 3 → 2: nt1 → dv1 [G_hat[0,2]]
 3 → 4: nt1 → dv1 [G_hat[0,2]]
 4 → 0: nt1 → dn2 [G_hat[0,1]]
 5 → 1: nt1 → dn1 [G_hat[0,0]]
 5 → 2: nt1 → dv1 [G_hat[0,2]]

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 Diagram 1L-398
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External legs:

leaf 0 ← dn2
 leaf 1 ← dn1

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

Edges (propagator assignments):

2 → 4: nt1 → dv1 [G_hat[0,2]]
 3 → 2: nt1 → dv1 [G_hat[0,2]]
 3 → 4: nt1 → dv1 [G_hat[0,2]]
 4 → 0: nt1 → dn2 [G_hat[0,1]]
 5 → 1: nt1 → dn1 [G_hat[0,0]]
 5 → 2: nt1 → dv1 [G_hat[0,2]]

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Diagram 1L-399

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External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

Edges (propagator assignments):

2 → 4: nt1 → dv1 [G_hat[0,2]]

3 → 2: nt1 → dv1 [G_hat[0,2]]

3 → 4: nt1 → dv1 [G_hat[0,2]]

4 → 0: nt1 → dn2 [G_hat[0,1]]

5 → 1: nt1 → dn1 [G_hat[0,0]]

5 → 2: nt1 → dv1 [G_hat[0,2]]

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Diagram 1L-400

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External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

Edges (propagator assignments):

2 → 4: nt1 → dv1 [G_hat[0,2]]
 3 → 2: nt1 → dv1 [G_hat[0,2]]
 3 → 4: nt1 → dv1 [G_hat[0,2]]
 4 → 0: nt1 → dn2 [G_hat[0,1]]
 5 → 1: nt1 → dn1 [G_hat[0,0]]
 5 → 2: nt1 → dv1 [G_hat[0,2]]

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 Diagram 1L-401
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External legs:

leaf 0 ← dn2
 leaf 1 ← dn1

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

Edges (propagator assignments):

2 → 4: nt1 → dv2 [G_hat[0,3]]
 3 → 2: nt1 → dv1 [G_hat[0,2]]
 3 → 4: nt1 → dv2 [G_hat[0,3]]
 4 → 0: nt2 → dn2 [G_hat[1,1]]
 5 → 1: nt1 → dn1 [G_hat[0,0]]
 5 → 2: nt1 → dv1 [G_hat[0,2]]

=====

Diagram 1L-402

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External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

Edges (propagator assignments):

2 → 4: nt1 → dv2 [G_hat[0,3]]

3 → 2: nt1 → dv1 [G_hat[0,2]]

3 → 4: nt1 → dv2 [G_hat[0,3]]

4 → 0: nt2 → dn2 [G_hat[1,1]]

5 → 1: nt1 → dn1 [G_hat[0,0]]

5 → 2: nt1 → dv1 [G_hat[0,2]]

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Diagram 1L-403

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External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

Edges (propagator assignments):

2 → 4: nt1 → dv2 [G_hat[0,3]]
 3 → 2: nt1 → dv1 [G_hat[0,2]]
 3 → 4: nt1 → dv2 [G_hat[0,3]]
 4 → 0: nt2 → dn2 [G_hat[1,1]]
 5 → 1: nt1 → dn1 [G_hat[0,0]]
 5 → 2: nt1 → dv1 [G_hat[0,2]]

=====
 Diagram 1L-404
 =====

External legs:

leaf 0 ← dn2
 leaf 1 ← dn1

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

Edges (propagator assignments):

2 → 4: nt1 → dv2 [G_hat[0,3]]
 3 → 2: nt1 → dv1 [G_hat[0,2]]
 3 → 4: nt1 → dv2 [G_hat[0,3]]
 4 → 0: nt2 → dn2 [G_hat[1,1]]
 5 → 1: nt1 → dn1 [G_hat[0,0]]
 5 → 2: nt1 → dv1 [G_hat[0,2]]

=====

Diagram 1L-405

=====

External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

Edges (propagator assignments):

2 → 4: nt1 → dv2 [G_hat[0,3]]

3 → 2: nt1 → dv1 [G_hat[0,2]]

3 → 4: nt1 → dv2 [G_hat[0,3]]

4 → 0: nt2 → dn2 [G_hat[1,1]]

5 → 1: nt1 → dn1 [G_hat[0,0]]

5 → 2: nt1 → dv1 [G_hat[0,2]]

=====

Diagram 1L-406

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External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

Edges (propagator assignments):

2 → 4: nt1 → dv2 [G_hat[0,3]]
 3 → 2: nt1 → dv1 [G_hat[0,2]]
 3 → 4: nt1 → dv2 [G_hat[0,3]]
 4 → 0: nt2 → dn2 [G_hat[1,1]]
 5 → 1: nt1 → dn1 [G_hat[0,0]]
 5 → 2: nt1 → dv1 [G_hat[0,2]]

=====
 Diagram 1L-407
 =====

External legs:

leaf 0 ← dn2
 leaf 1 ← dn1

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

Edges (propagator assignments):

2 → 4: nt1 → dv2 [G_hat[0,3]]
 3 → 2: nt1 → dv1 [G_hat[0,2]]
 3 → 4: nt1 → dv2 [G_hat[0,3]]
 4 → 0: nt2 → dn2 [G_hat[1,1]]
 5 → 1: nt1 → dn1 [G_hat[0,0]]
 5 → 2: nt1 → dv1 [G_hat[0,2]]

=====

Diagram 1L-408

=====

External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

Edges (propagator assignments):

2 → 4: nt1 → dv2 [G_hat[0,3]]

3 → 2: nt1 → dv1 [G_hat[0,2]]

3 → 4: nt1 → dv2 [G_hat[0,3]]

4 → 0: nt2 → dn2 [G_hat[1,1]]

5 → 1: nt1 → dn1 [G_hat[0,0]]

5 → 2: nt1 → dv1 [G_hat[0,2]]

=====

Diagram 1L-409

=====

External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

Edges (propagator assignments):

2 → 4: nt1 → dv2 [G_hat[0,3]]
 3 → 2: nt1 → dv1 [G_hat[0,2]]
 3 → 4: nt1 → dv2 [G_hat[0,3]]
 4 → 0: nt2 → dn2 [G_hat[1,1]]
 5 → 1: nt1 → dn1 [G_hat[0,0]]
 5 → 2: nt1 → dv1 [G_hat[0,2]]

=====
 Diagram 1L-410
 =====

External legs:

leaf 0 ← dn2
 leaf 1 ← dn1

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

Edges (propagator assignments):

2 → 4: nt1 → dv2 [G_hat[0,3]]
 3 → 2: nt1 → dv1 [G_hat[0,2]]
 3 → 4: nt1 → dv2 [G_hat[0,3]]
 4 → 0: nt2 → dn2 [G_hat[1,1]]
 5 → 1: nt1 → dn1 [G_hat[0,0]]
 5 → 2: nt1 → dv1 [G_hat[0,2]]

```

=====
Diagram 1L-411
=====
External legs:
  leaf 0 ← dn2
  leaf 1 ← dn1
Vertex assignments:
  v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

coeff =  $\frac{1}{2} \varphi_1''$ 

  v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

coeff =  $\frac{1}{2} nstar_1$ 

  v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

coeff =  $\frac{1}{2} \varphi_2''$ 

  v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

coeff =  $\frac{1}{2} nstar_1$ 

Edges (propagator assignments):
  2 → 4:  nt1 → dv2  [G_hat[0,3]]
  3 → 2:  nt1 → dv1  [G_hat[0,2]]
  3 → 4:  nt1 → dv2  [G_hat[0,3]]
  4 → 0:  nt2 → dn2  [G_hat[1,1]]
  5 → 1:  nt1 → dn1  [G_hat[0,0]]
  5 → 2:  nt1 → dv1  [G_hat[0,2]]

```

```

=====
Diagram 1L-412
=====
External legs:
  leaf 0 ← dn2
  leaf 1 ← dn1
Vertex assignments:
  v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

coeff =  $\frac{1}{2} \varphi_1''$ 

  v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

coeff =  $\frac{1}{2} nstar_1$ 

```

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

Edges (propagator assignments):

2 → 4: nt1 → dv2 [G_hat[0,3]]
 3 → 2: nt1 → dv1 [G_hat[0,2]]
 3 → 4: nt1 → dv2 [G_hat[0,3]]
 4 → 0: nt2 → dn2 [G_hat[1,1]]
 5 → 1: nt1 → dn1 [G_hat[0,0]]
 5 → 2: nt1 → dv1 [G_hat[0,2]]

=====
 Diagram 1L-413
 =====

External legs:

leaf 0 ← dn2
 leaf 1 ← dn1

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

Edges (propagator assignments):

2 → 4: nt1 → dv2 [G_hat[0,3]]
 3 → 2: nt1 → dv1 [G_hat[0,2]]
 3 → 4: nt1 → dv2 [G_hat[0,3]]
 4 → 0: nt2 → dn2 [G_hat[1,1]]
 5 → 1: nt1 → dn1 [G_hat[0,0]]
 5 → 2: nt1 → dv1 [G_hat[0,2]]

=====

Diagram 1L-414

=====

External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

Edges (propagator assignments):

2 → 4: nt1 → dv2 [G_hat[0,3]]

3 → 2: nt1 → dv1 [G_hat[0,2]]

3 → 4: nt1 → dv2 [G_hat[0,3]]

4 → 0: nt2 → dn2 [G_hat[1,1]]

5 → 1: nt1 → dn1 [G_hat[0,0]]

5 → 2: nt1 → dv1 [G_hat[0,2]]

=====

Diagram 1L-415

=====

External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

Edges (propagator assignments):

2 → 4: nt1 → dv2 [G_hat[0,3]]
 3 → 2: nt1 → dv1 [G_hat[0,2]]
 3 → 4: nt1 → dv2 [G_hat[0,3]]
 4 → 0: nt2 → dn2 [G_hat[1,1]]
 5 → 1: nt1 → dn1 [G_hat[0,0]]
 5 → 2: nt1 → dv1 [G_hat[0,2]]

=====
 Diagram 1L-416
 =====

External legs:

leaf 0 ← dn2
 leaf 1 ← dn1

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

Edges (propagator assignments):

2 → 4: nt1 → dv2 [G_hat[0,3]]
 3 → 2: nt1 → dv1 [G_hat[0,2]]
 3 → 4: nt1 → dv2 [G_hat[0,3]]
 4 → 0: nt2 → dn2 [G_hat[1,1]]
 5 → 1: nt1 → dn1 [G_hat[0,0]]
 5 → 2: nt1 → dv1 [G_hat[0,2]]

=====

Diagram 1L-417

=====

External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

Edges (propagator assignments):

2 → 4: nt2 → dv1 [G_hat[1,2]]

3 → 2: nt1 → dv2 [G_hat[0,3]]

3 → 4: nt1 → dv1 [G_hat[0,2]]

4 → 0: nt1 → dn2 [G_hat[0,1]]

5 → 1: nt1 → dn1 [G_hat[0,0]]

5 → 2: nt1 → dv2 [G_hat[0,3]]

=====

Diagram 1L-418

=====

External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

Edges (propagator assignments):

2 → 4: nt2 → dv1 [G_hat[1,2]]
 3 → 2: nt1 → dv2 [G_hat[0,3]]
 3 → 4: nt1 → dv1 [G_hat[0,2]]
 4 → 0: nt1 → dn2 [G_hat[0,1]]
 5 → 1: nt1 → dn1 [G_hat[0,0]]
 5 → 2: nt1 → dv2 [G_hat[0,3]]

=====
 Diagram 1L-419
 =====

External legs:

leaf 0 ← dn2
 leaf 1 ← dn1

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

Edges (propagator assignments):

2 → 4: nt2 → dv1 [G_hat[1,2]]
 3 → 2: nt1 → dv2 [G_hat[0,3]]
 3 → 4: nt1 → dv1 [G_hat[0,2]]
 4 → 0: nt1 → dn2 [G_hat[0,1]]
 5 → 1: nt1 → dn1 [G_hat[0,0]]
 5 → 2: nt1 → dv2 [G_hat[0,3]]

```

=====
Diagram 1L-420
=====
External legs:
  leaf 0 ← dn2
  leaf 1 ← dn1
Vertex assignments:
  v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

coeff =  $\frac{1}{2} \varphi_2''$ 

  v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

coeff =  $\frac{1}{2} nstar_1$ 

  v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

coeff =  $\frac{1}{2} \varphi_1''$ 

  v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

coeff =  $\frac{1}{2} nstar_1$ 

Edges (propagator assignments):
  2 → 4:  nt2 → dv1  [G_hat[1,2]]
  3 → 2:  nt1 → dv2  [G_hat[0,3]]
  3 → 4:  nt1 → dv1  [G_hat[0,2]]
  4 → 0:  nt1 → dn2  [G_hat[0,1]]
  5 → 1:  nt1 → dn1  [G_hat[0,0]]
  5 → 2:  nt1 → dv2  [G_hat[0,3]]

```

```

=====
Diagram 1L-421
=====
External legs:
  leaf 0 ← dn2
  leaf 1 ← dn1
Vertex assignments:
  v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

coeff =  $\frac{1}{2} \varphi_2''$ 

  v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

coeff =  $\frac{1}{2} nstar_1$ 

```

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

Edges (propagator assignments):

2 → 4: nt2 → dv1 [G_hat[1,2]]
 3 → 2: nt1 → dv2 [G_hat[0,3]]
 3 → 4: nt1 → dv1 [G_hat[0,2]]
 4 → 0: nt1 → dn2 [G_hat[0,1]]
 5 → 1: nt1 → dn1 [G_hat[0,0]]
 5 → 2: nt1 → dv2 [G_hat[0,3]]

=====
 Diagram 1L-422
 =====

External legs:

leaf 0 ← dn2
 leaf 1 ← dn1

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

Edges (propagator assignments):

2 → 4: nt2 → dv1 [G_hat[1,2]]
 3 → 2: nt1 → dv2 [G_hat[0,3]]
 3 → 4: nt1 → dv1 [G_hat[0,2]]
 4 → 0: nt1 → dn2 [G_hat[0,1]]
 5 → 1: nt1 → dn1 [G_hat[0,0]]
 5 → 2: nt1 → dv2 [G_hat[0,3]]

```

=====
Diagram 1L-423
=====
External legs:
  leaf 0 ← dn2
  leaf 1 ← dn1
Vertex assignments:
  v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

coeff =  $\frac{1}{2} \varphi_2''$ 

  v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

coeff =  $\frac{1}{2} nstar_1$ 

  v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

coeff =  $\frac{1}{2} \varphi_1''$ 

  v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

coeff =  $\frac{1}{2} nstar_1$ 

Edges (propagator assignments):
  2 → 4:  nt2 → dv1  [G_hat[1,2]]
  3 → 2:  nt1 → dv2  [G_hat[0,3]]
  3 → 4:  nt1 → dv1  [G_hat[0,2]]
  4 → 0:  nt1 → dn2  [G_hat[0,1]]
  5 → 1:  nt1 → dn1  [G_hat[0,0]]
  5 → 2:  nt1 → dv2  [G_hat[0,3]]

```

```

=====
Diagram 1L-424
=====
External legs:
  leaf 0 ← dn2
  leaf 1 ← dn1
Vertex assignments:
  v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

coeff =  $\frac{1}{2} \varphi_2''$ 

  v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

coeff =  $\frac{1}{2} nstar_1$ 

```

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

Edges (propagator assignments):

2 → 4: nt2 → dv1 [G_hat[1,2]]
 3 → 2: nt1 → dv2 [G_hat[0,3]]
 3 → 4: nt1 → dv1 [G_hat[0,2]]
 4 → 0: nt1 → dn2 [G_hat[0,1]]
 5 → 1: nt1 → dn1 [G_hat[0,0]]
 5 → 2: nt1 → dv2 [G_hat[0,3]]

=====
 Diagram 1L-425
 =====

External legs:

leaf 0 ← dn2
 leaf 1 ← dn1

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

Edges (propagator assignments):

2 → 4: nt2 → dv1 [G_hat[1,2]]
 3 → 2: nt1 → dv2 [G_hat[0,3]]
 3 → 4: nt1 → dv1 [G_hat[0,2]]
 4 → 0: nt1 → dn2 [G_hat[0,1]]
 5 → 1: nt1 → dn1 [G_hat[0,0]]
 5 → 2: nt1 → dv2 [G_hat[0,3]]

=====

Diagram 1L-426

=====

External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

Edges (propagator assignments):

2 → 4: nt2 → dv1 [G_hat[1,2]]

3 → 2: nt1 → dv2 [G_hat[0,3]]

3 → 4: nt1 → dv1 [G_hat[0,2]]

4 → 0: nt1 → dn2 [G_hat[0,1]]

5 → 1: nt1 → dn1 [G_hat[0,0]]

5 → 2: nt1 → dv2 [G_hat[0,3]]

=====

Diagram 1L-427

=====

External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

Edges (propagator assignments):

2 → 4: nt2 → dv1 [G_hat[1,2]]
 3 → 2: nt1 → dv2 [G_hat[0,3]]
 3 → 4: nt1 → dv1 [G_hat[0,2]]
 4 → 0: nt1 → dn2 [G_hat[0,1]]
 5 → 1: nt1 → dn1 [G_hat[0,0]]
 5 → 2: nt1 → dv2 [G_hat[0,3]]

=====
 Diagram 1L-428
 =====

External legs:

leaf 0 ← dn2
 leaf 1 ← dn1

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

Edges (propagator assignments):

2 → 4: nt2 → dv1 [G_hat[1,2]]
 3 → 2: nt1 → dv2 [G_hat[0,3]]
 3 → 4: nt1 → dv1 [G_hat[0,2]]
 4 → 0: nt1 → dn2 [G_hat[0,1]]
 5 → 1: nt1 → dn1 [G_hat[0,0]]
 5 → 2: nt1 → dv2 [G_hat[0,3]]

=====

Diagram 1L-429

=====

External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

Edges (propagator assignments):

2 → 4: nt2 → dv1 [G_hat[1,2]]

3 → 2: nt1 → dv2 [G_hat[0,3]]

3 → 4: nt1 → dv1 [G_hat[0,2]]

4 → 0: nt1 → dn2 [G_hat[0,1]]

5 → 1: nt1 → dn1 [G_hat[0,0]]

5 → 2: nt1 → dv2 [G_hat[0,3]]

=====

Diagram 1L-430

=====

External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

Edges (propagator assignments):

2 → 4: nt2 → dv1 [G_hat[1,2]]
 3 → 2: nt1 → dv2 [G_hat[0,3]]
 3 → 4: nt1 → dv1 [G_hat[0,2]]
 4 → 0: nt1 → dn2 [G_hat[0,1]]
 5 → 1: nt1 → dn1 [G_hat[0,0]]
 5 → 2: nt1 → dv2 [G_hat[0,3]]

=====
 Diagram 1L-431
 =====

External legs:

leaf 0 ← dn2
 leaf 1 ← dn1

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

Edges (propagator assignments):

2 → 4: nt2 → dv1 [G_hat[1,2]]
 3 → 2: nt1 → dv2 [G_hat[0,3]]
 3 → 4: nt1 → dv1 [G_hat[0,2]]
 4 → 0: nt1 → dn2 [G_hat[0,1]]
 5 → 1: nt1 → dn1 [G_hat[0,0]]
 5 → 2: nt1 → dv2 [G_hat[0,3]]

=====

Diagram 1L-432

=====

External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

Edges (propagator assignments):

2 → 4: nt2 → dv1 [G_hat[1,2]]

3 → 2: nt1 → dv2 [G_hat[0,3]]

3 → 4: nt1 → dv1 [G_hat[0,2]]

4 → 0: nt1 → dn2 [G_hat[0,1]]

5 → 1: nt1 → dn1 [G_hat[0,0]]

5 → 2: nt1 → dv2 [G_hat[0,3]]

=====

Diagram 1L-433

=====

External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

Edges (propagator assignments):

2 → 4: nt2 → dv2 [G_hat[1,3]]
 3 → 2: nt1 → dv2 [G_hat[0,3]]
 3 → 4: nt1 → dv2 [G_hat[0,3]]
 4 → 0: nt2 → dn2 [G_hat[1,1]]
 5 → 1: nt1 → dn1 [G_hat[0,0]]
 5 → 2: nt1 → dv2 [G_hat[0,3]]

=====
 Diagram 1L-434
 =====

External legs:

leaf 0 ← dn2
 leaf 1 ← dn1

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

Edges (propagator assignments):

2 → 4: nt2 → dv2 [G_hat[1,3]]
 3 → 2: nt1 → dv2 [G_hat[0,3]]
 3 → 4: nt1 → dv2 [G_hat[0,3]]
 4 → 0: nt2 → dn2 [G_hat[1,1]]
 5 → 1: nt1 → dn1 [G_hat[0,0]]
 5 → 2: nt1 → dv2 [G_hat[0,3]]

```

=====
Diagram 1L-435
=====

External legs:
  leaf 0 ← dn2
  leaf 1 ← dn1
Vertex assignments:
  v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

coeff =  $\frac{1}{2} \varphi_2''$ 

  v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

coeff =  $\frac{1}{2} nstar_1$ 

  v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

coeff =  $\frac{1}{2} \varphi_2''$ 

  v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

coeff =  $\frac{1}{2} nstar_1$ 

Edges (propagator assignments):
  2 → 4:  nt2 → dv2  [G_hat[1,3]]
  3 → 2:  nt1 → dv2  [G_hat[0,3]]
  3 → 4:  nt1 → dv2  [G_hat[0,3]]
  4 → 0:  nt2 → dn2  [G_hat[1,1]]
  5 → 1:  nt1 → dn1  [G_hat[0,0]]
  5 → 2:  nt1 → dv2  [G_hat[0,3]]

```

```

=====
Diagram 1L-436
=====

External legs:
  leaf 0 ← dn2
  leaf 1 ← dn1
Vertex assignments:
  v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

coeff =  $\frac{1}{2} \varphi_2''$ 

  v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

coeff =  $\frac{1}{2} nstar_1$ 

```

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

Edges (propagator assignments):

2 → 4: nt2 → dv2 [G_hat[1,3]]
 3 → 2: nt1 → dv2 [G_hat[0,3]]
 3 → 4: nt1 → dv2 [G_hat[0,3]]
 4 → 0: nt2 → dn2 [G_hat[1,1]]
 5 → 1: nt1 → dn1 [G_hat[0,0]]
 5 → 2: nt1 → dv2 [G_hat[0,3]]

=====
 Diagram 1L-437
 =====

External legs:

leaf 0 ← dn2
 leaf 1 ← dn1

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

Edges (propagator assignments):

2 → 4: nt2 → dv2 [G_hat[1,3]]
 3 → 2: nt1 → dv2 [G_hat[0,3]]
 3 → 4: nt1 → dv2 [G_hat[0,3]]
 4 → 0: nt2 → dn2 [G_hat[1,1]]
 5 → 1: nt1 → dn1 [G_hat[0,0]]
 5 → 2: nt1 → dv2 [G_hat[0,3]]


```

=====
Diagram 1L-438
=====
External legs:
  leaf 0 ← dn2
  leaf 1 ← dn1
Vertex assignments:
  v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

coeff =  $\frac{1}{2} \varphi_2''$ 

  v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

coeff =  $\frac{1}{2} nstar_1$ 

  v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

coeff =  $\frac{1}{2} \varphi_2''$ 

  v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

coeff =  $\frac{1}{2} nstar_1$ 

Edges (propagator assignments):
  2 → 4:  nt2 → dv2  [G_hat[1,3]]
  3 → 2:  nt1 → dv2  [G_hat[0,3]]
  3 → 4:  nt1 → dv2  [G_hat[0,3]]
  4 → 0:  nt2 → dn2  [G_hat[1,1]]
  5 → 1:  nt1 → dn1  [G_hat[0,0]]
  5 → 2:  nt1 → dv2  [G_hat[0,3]]

```

```

=====
Diagram 1L-439
=====
External legs:
  leaf 0 ← dn2
  leaf 1 ← dn1
Vertex assignments:
  v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

coeff =  $\frac{1}{2} \varphi_2''$ 

  v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

coeff =  $\frac{1}{2} nstar_1$ 

```

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

Edges (propagator assignments):

2 → 4: nt2 → dv2 [G_hat[1,3]]
 3 → 2: nt1 → dv2 [G_hat[0,3]]
 3 → 4: nt1 → dv2 [G_hat[0,3]]
 4 → 0: nt2 → dn2 [G_hat[1,1]]
 5 → 1: nt1 → dn1 [G_hat[0,0]]
 5 → 2: nt1 → dv2 [G_hat[0,3]]

Diagram 1L-440

External legs:

leaf 0 ← dn2
 leaf 1 ← dn1

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

Edges (propagator assignments):

2 → 4: nt2 → dv2 [G_hat[1,3]]
 3 → 2: nt1 → dv2 [G_hat[0,3]]
 3 → 4: nt1 → dv2 [G_hat[0,3]]
 4 → 0: nt2 → dn2 [G_hat[1,1]]
 5 → 1: nt1 → dn1 [G_hat[0,0]]
 5 → 2: nt1 → dv2 [G_hat[0,3]]

```

=====
Diagram 1L-441
=====
External legs:
  leaf 0 ← dn2
  leaf 1 ← dn1
Vertex assignments:
  v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

coeff =  $\frac{1}{2} \varphi_2''$ 

  v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

coeff =  $\frac{1}{2} nstar_1$ 

  v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

coeff =  $\frac{1}{2} \varphi_2''$ 

  v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

coeff =  $\frac{1}{2} nstar_1$ 

Edges (propagator assignments):
  2 → 4:  nt2 → dv2  [G_hat[1,3]]
  3 → 2:  nt1 → dv2  [G_hat[0,3]]
  3 → 4:  nt1 → dv2  [G_hat[0,3]]
  4 → 0:  nt2 → dn2  [G_hat[1,1]]
  5 → 1:  nt1 → dn1  [G_hat[0,0]]
  5 → 2:  nt1 → dv2  [G_hat[0,3]]

```

```

=====
Diagram 1L-442
=====
External legs:
  leaf 0 ← dn2
  leaf 1 ← dn1
Vertex assignments:
  v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

coeff =  $\frac{1}{2} \varphi_2''$ 

  v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

coeff =  $\frac{1}{2} nstar_1$ 

```

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

Edges (propagator assignments):

2 → 4: nt2 → dv2 [G_hat[1,3]]
 3 → 2: nt1 → dv2 [G_hat[0,3]]
 3 → 4: nt1 → dv2 [G_hat[0,3]]
 4 → 0: nt2 → dn2 [G_hat[1,1]]
 5 → 1: nt1 → dn1 [G_hat[0,0]]
 5 → 2: nt1 → dv2 [G_hat[0,3]]

=====
 Diagram 1L-443
 =====

External legs:

leaf 0 ← dn2
 leaf 1 ← dn1

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

Edges (propagator assignments):

2 → 4: nt2 → dv2 [G_hat[1,3]]
 3 → 2: nt1 → dv2 [G_hat[0,3]]
 3 → 4: nt1 → dv2 [G_hat[0,3]]
 4 → 0: nt2 → dn2 [G_hat[1,1]]
 5 → 1: nt1 → dn1 [G_hat[0,0]]
 5 → 2: nt1 → dv2 [G_hat[0,3]]

```

=====
Diagram 1L-444
=====
External legs:
  leaf 0 ← dn2
  leaf 1 ← dn1
Vertex assignments:
  v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

coeff =  $\frac{1}{2} \varphi_2''$ 

  v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

coeff =  $\frac{1}{2} nstar_1$ 

  v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

coeff =  $\frac{1}{2} \varphi_2''$ 

  v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

coeff =  $\frac{1}{2} nstar_1$ 

Edges (propagator assignments):
  2 → 4:  nt2 → dv2  [G_hat[1,3]]
  3 → 2:  nt1 → dv2  [G_hat[0,3]]
  3 → 4:  nt1 → dv2  [G_hat[0,3]]
  4 → 0:  nt2 → dn2  [G_hat[1,1]]
  5 → 1:  nt1 → dn1  [G_hat[0,0]]
  5 → 2:  nt1 → dv2  [G_hat[0,3]]

```

```

=====
Diagram 1L-445
=====
External legs:
  leaf 0 ← dn2
  leaf 1 ← dn1
Vertex assignments:
  v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

coeff =  $\frac{1}{2} \varphi_2''$ 

  v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

coeff =  $\frac{1}{2} nstar_1$ 

```

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

Edges (propagator assignments):

2 → 4: nt2 → dv2 [G_hat[1,3]]
 3 → 2: nt1 → dv2 [G_hat[0,3]]
 3 → 4: nt1 → dv2 [G_hat[0,3]]
 4 → 0: nt2 → dn2 [G_hat[1,1]]
 5 → 1: nt1 → dn1 [G_hat[0,0]]
 5 → 2: nt1 → dv2 [G_hat[0,3]]

=====
 Diagram 1L-446
 =====

External legs:

leaf 0 ← dn2
 leaf 1 ← dn1

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

Edges (propagator assignments):

2 → 4: nt2 → dv2 [G_hat[1,3]]
 3 → 2: nt1 → dv2 [G_hat[0,3]]
 3 → 4: nt1 → dv2 [G_hat[0,3]]
 4 → 0: nt2 → dn2 [G_hat[1,1]]
 5 → 1: nt1 → dn1 [G_hat[0,0]]
 5 → 2: nt1 → dv2 [G_hat[0,3]]

```

=====
Diagram 1L-447
=====
External legs:
  leaf 0 ← dn2
  leaf 1 ← dn1
Vertex assignments:
  v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

coeff =  $\frac{1}{2} \varphi_2''$ 

  v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

coeff =  $\frac{1}{2} nstar_1$ 

  v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

coeff =  $\frac{1}{2} \varphi_2''$ 

  v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

coeff =  $\frac{1}{2} nstar_1$ 

Edges (propagator assignments):
  2 → 4:  nt2 → dv2  [G_hat[1,3]]
  3 → 2:  nt1 → dv2  [G_hat[0,3]]
  3 → 4:  nt1 → dv2  [G_hat[0,3]]
  4 → 0:  nt2 → dn2  [G_hat[1,1]]
  5 → 1:  nt1 → dn1  [G_hat[0,0]]
  5 → 2:  nt1 → dv2  [G_hat[0,3]]

```

```

=====
Diagram 1L-448
=====
External legs:
  leaf 0 ← dn2
  leaf 1 ← dn1
Vertex assignments:
  v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

coeff =  $\frac{1}{2} \varphi_2''$ 

  v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

coeff =  $\frac{1}{2} nstar_1$ 

```

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

Edges (propagator assignments):

2 → 4: nt2 → dv2 [G_hat[1,3]]
 3 → 2: nt1 → dv2 [G_hat[0,3]]
 3 → 4: nt1 → dv2 [G_hat[0,3]]
 4 → 0: nt2 → dn2 [G_hat[1,1]]
 5 → 1: nt1 → dn1 [G_hat[0,0]]
 5 → 2: nt1 → dv2 [G_hat[0,3]]

=====
 Diagram 1L-449
 =====

External legs:

leaf 0 ← dn2
 leaf 1 ← dn1

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

Edges (propagator assignments):

2 → 4: nt1 → dv1 [G_hat[0,2]]
 3 → 2: nt1 → dv1 [G_hat[0,2]]
 3 → 4: nt1 → dv1 [G_hat[0,2]]
 4 → 0: nt1 → dn2 [G_hat[0,1]]
 5 → 1: nt2 → dn1 [G_hat[1,0]]
 5 → 2: nt2 → dv1 [G_hat[1,2]]


```

=====
Diagram 1L-450
=====
External legs:
  leaf 0 ← dn2
  leaf 1 ← dn1
Vertex assignments:
  v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

coeff =  $\frac{1}{2} \varphi_1''$ 

  v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

coeff =  $\frac{1}{2} nstar_1$ 

  v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

coeff =  $\frac{1}{2} \varphi_1''$ 

  v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

coeff =  $\frac{1}{2} nstar_2$ 

Edges (propagator assignments):
  2 → 4:  nt1 → dv1  [G_hat[0,2]]
  3 → 2:  nt1 → dv1  [G_hat[0,2]]
  3 → 4:  nt1 → dv1  [G_hat[0,2]]
  4 → 0:  nt1 → dn2  [G_hat[0,1]]
  5 → 1:  nt2 → dn1  [G_hat[1,0]]
  5 → 2:  nt2 → dv1  [G_hat[1,2]]

```

```

=====
Diagram 1L-451
=====
External legs:
  leaf 0 ← dn2
  leaf 1 ← dn1
Vertex assignments:
  v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

coeff =  $\frac{1}{2} \varphi_1''$ 

  v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

coeff =  $\frac{1}{2} nstar_1$ 

```

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

Edges (propagator assignments):

2 → 4: nt1 → dv1 [G_hat[0,2]]
 3 → 2: nt1 → dv1 [G_hat[0,2]]
 3 → 4: nt1 → dv1 [G_hat[0,2]]
 4 → 0: nt1 → dn2 [G_hat[0,1]]
 5 → 1: nt2 → dn1 [G_hat[1,0]]
 5 → 2: nt2 → dv1 [G_hat[1,2]]

=====
 Diagram 1L-452
 =====

External legs:

leaf 0 ← dn2
 leaf 1 ← dn1

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

Edges (propagator assignments):

2 → 4: nt1 → dv1 [G_hat[0,2]]
 3 → 2: nt1 → dv1 [G_hat[0,2]]
 3 → 4: nt1 → dv1 [G_hat[0,2]]
 4 → 0: nt1 → dn2 [G_hat[0,1]]
 5 → 1: nt2 → dn1 [G_hat[1,0]]
 5 → 2: nt2 → dv1 [G_hat[1,2]]

```

=====
Diagram 1L-453
=====

External legs:
  leaf 0 ← dn2
  leaf 1 ← dn1
Vertex assignments:
  v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

coeff =  $\frac{1}{2} \varphi_1''$ 

  v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

coeff =  $\frac{1}{2} nstar_1$ 

  v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

coeff =  $\frac{1}{2} \varphi_1''$ 

  v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

coeff =  $\frac{1}{2} nstar_2$ 

Edges (propagator assignments):
  2 → 4:  nt1 → dv1  [G_hat[0,2]]
  3 → 2:  nt1 → dv1  [G_hat[0,2]]
  3 → 4:  nt1 → dv1  [G_hat[0,2]]
  4 → 0:  nt1 → dn2  [G_hat[0,1]]
  5 → 1:  nt2 → dn1  [G_hat[1,0]]
  5 → 2:  nt2 → dv1  [G_hat[1,2]]

```

```

=====
Diagram 1L-454
=====

External legs:
  leaf 0 ← dn2
  leaf 1 ← dn1
Vertex assignments:
  v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

coeff =  $\frac{1}{2} \varphi_1''$ 

  v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

coeff =  $\frac{1}{2} nstar_1$ 

```

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

Edges (propagator assignments):

2 → 4: nt1 → dv1 [G_hat[0,2]]
 3 → 2: nt1 → dv1 [G_hat[0,2]]
 3 → 4: nt1 → dv1 [G_hat[0,2]]
 4 → 0: nt1 → dn2 [G_hat[0,1]]
 5 → 1: nt2 → dn1 [G_hat[1,0]]
 5 → 2: nt2 → dv1 [G_hat[1,2]]

=====
 Diagram 1L-455
 =====

External legs:

leaf 0 ← dn2
 leaf 1 ← dn1

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

Edges (propagator assignments):

2 → 4: nt1 → dv1 [G_hat[0,2]]
 3 → 2: nt1 → dv1 [G_hat[0,2]]
 3 → 4: nt1 → dv1 [G_hat[0,2]]
 4 → 0: nt1 → dn2 [G_hat[0,1]]
 5 → 1: nt2 → dn1 [G_hat[1,0]]
 5 → 2: nt2 → dv1 [G_hat[1,2]]

```

=====
Diagram 1L-456
=====
External legs:
  leaf 0 ← dn2
  leaf 1 ← dn1
Vertex assignments:
  v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

coeff =  $\frac{1}{2} \varphi_1''$ 

  v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

coeff =  $\frac{1}{2} nstar_1$ 

  v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

coeff =  $\frac{1}{2} \varphi_1''$ 

  v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

coeff =  $\frac{1}{2} nstar_2$ 

Edges (propagator assignments):
  2 → 4:  nt1 → dv1  [G_hat[0,2]]
  3 → 2:  nt1 → dv1  [G_hat[0,2]]
  3 → 4:  nt1 → dv1  [G_hat[0,2]]
  4 → 0:  nt1 → dn2  [G_hat[0,1]]
  5 → 1:  nt2 → dn1  [G_hat[1,0]]
  5 → 2:  nt2 → dv1  [G_hat[1,2]]

```

```

=====
Diagram 1L-457
=====
External legs:
  leaf 0 ← dn2
  leaf 1 ← dn1
Vertex assignments:
  v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

coeff =  $\frac{1}{2} \varphi_1''$ 

  v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

coeff =  $\frac{1}{2} nstar_1$ 

```

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

Edges (propagator assignments):

2 → 4: nt1 → dv1 [G_hat[0,2]]
 3 → 2: nt1 → dv1 [G_hat[0,2]]
 3 → 4: nt1 → dv1 [G_hat[0,2]]
 4 → 0: nt1 → dn2 [G_hat[0,1]]
 5 → 1: nt2 → dn1 [G_hat[1,0]]
 5 → 2: nt2 → dv1 [G_hat[1,2]]

=====
 Diagram 1L-458
 =====

External legs:

leaf 0 ← dn2
 leaf 1 ← dn1

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

Edges (propagator assignments):

2 → 4: nt1 → dv1 [G_hat[0,2]]
 3 → 2: nt1 → dv1 [G_hat[0,2]]
 3 → 4: nt1 → dv1 [G_hat[0,2]]
 4 → 0: nt1 → dn2 [G_hat[0,1]]
 5 → 1: nt2 → dn1 [G_hat[1,0]]
 5 → 2: nt2 → dv1 [G_hat[1,2]]

=====

Diagram 1L-459

=====

External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

Edges (propagator assignments):

2 → 4: nt1 → dv1 [G_hat[0,2]]

3 → 2: nt1 → dv1 [G_hat[0,2]]

3 → 4: nt1 → dv1 [G_hat[0,2]]

4 → 0: nt1 → dn2 [G_hat[0,1]]

5 → 1: nt2 → dn1 [G_hat[1,0]]

5 → 2: nt2 → dv1 [G_hat[1,2]]

=====

Diagram 1L-460

=====

External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

Edges (propagator assignments):

2 → 4: nt1 → dv1 [G_hat[0,2]]
 3 → 2: nt1 → dv1 [G_hat[0,2]]
 3 → 4: nt1 → dv1 [G_hat[0,2]]
 4 → 0: nt1 → dn2 [G_hat[0,1]]
 5 → 1: nt2 → dn1 [G_hat[1,0]]
 5 → 2: nt2 → dv1 [G_hat[1,2]]

=====
 Diagram 1L-461
 =====

External legs:

leaf 0 ← dn2
 leaf 1 ← dn1

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

Edges (propagator assignments):

2 → 4: nt1 → dv1 [G_hat[0,2]]
 3 → 2: nt1 → dv1 [G_hat[0,2]]
 3 → 4: nt1 → dv1 [G_hat[0,2]]
 4 → 0: nt1 → dn2 [G_hat[0,1]]
 5 → 1: nt2 → dn1 [G_hat[1,0]]
 5 → 2: nt2 → dv1 [G_hat[1,2]]

=====

Diagram 1L-462

=====

External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

Edges (propagator assignments):

2 → 4: nt1 → dv1 [G_hat[0,2]]

3 → 2: nt1 → dv1 [G_hat[0,2]]

3 → 4: nt1 → dv1 [G_hat[0,2]]

4 → 0: nt1 → dn2 [G_hat[0,1]]

5 → 1: nt2 → dn1 [G_hat[1,0]]

5 → 2: nt2 → dv1 [G_hat[1,2]]

=====

Diagram 1L-463

=====

External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

Edges (propagator assignments):

2 → 4: nt1 → dv1 [G_hat[0,2]]
 3 → 2: nt1 → dv1 [G_hat[0,2]]
 3 → 4: nt1 → dv1 [G_hat[0,2]]
 4 → 0: nt1 → dn2 [G_hat[0,1]]
 5 → 1: nt2 → dn1 [G_hat[1,0]]
 5 → 2: nt2 → dv1 [G_hat[1,2]]

=====
 Diagram 1L-464
 =====

External legs:

leaf 0 ← dn2
 leaf 1 ← dn1

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

Edges (propagator assignments):

2 → 4: nt1 → dv1 [G_hat[0,2]]
 3 → 2: nt1 → dv1 [G_hat[0,2]]
 3 → 4: nt1 → dv1 [G_hat[0,2]]
 4 → 0: nt1 → dn2 [G_hat[0,1]]
 5 → 1: nt2 → dn1 [G_hat[1,0]]
 5 → 2: nt2 → dv1 [G_hat[1,2]]

```

=====
Diagram 1L-465
=====
External legs:
  leaf 0 ← dn2
  leaf 1 ← dn1
Vertex assignments:
  v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

coeff =  $\frac{1}{2} \varphi_1''$ 

  v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

coeff =  $\frac{1}{2} nstar_1$ 

  v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

coeff =  $\frac{1}{2} \varphi_2''$ 

  v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

coeff =  $\frac{1}{2} nstar_2$ 

Edges (propagator assignments):
  2 → 4:  nt1 → dv2  [G_hat[0,3]]
  3 → 2:  nt1 → dv1  [G_hat[0,2]]
  3 → 4:  nt1 → dv2  [G_hat[0,3]]
  4 → 0:  nt2 → dn2  [G_hat[1,1]]
  5 → 1:  nt2 → dn1  [G_hat[1,0]]
  5 → 2:  nt2 → dv1  [G_hat[1,2]]

```

```

=====
Diagram 1L-466
=====
External legs:
  leaf 0 ← dn2
  leaf 1 ← dn1
Vertex assignments:
  v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

coeff =  $\frac{1}{2} \varphi_1''$ 

  v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

coeff =  $\frac{1}{2} nstar_1$ 

```

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

Edges (propagator assignments):

2 → 4: nt1 → dv2 [G_hat[0,3]]
 3 → 2: nt1 → dv1 [G_hat[0,2]]
 3 → 4: nt1 → dv2 [G_hat[0,3]]
 4 → 0: nt2 → dn2 [G_hat[1,1]]
 5 → 1: nt2 → dn1 [G_hat[1,0]]
 5 → 2: nt2 → dv1 [G_hat[1,2]]

=====
 Diagram 1L-467
 =====

External legs:

leaf 0 ← dn2
 leaf 1 ← dn1

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

Edges (propagator assignments):

2 → 4: nt1 → dv2 [G_hat[0,3]]
 3 → 2: nt1 → dv1 [G_hat[0,2]]
 3 → 4: nt1 → dv2 [G_hat[0,3]]
 4 → 0: nt2 → dn2 [G_hat[1,1]]
 5 → 1: nt2 → dn1 [G_hat[1,0]]
 5 → 2: nt2 → dv1 [G_hat[1,2]]

```

=====
Diagram 1L-468
=====
External legs:
  leaf 0 ← dn2
  leaf 1 ← dn1
Vertex assignments:
  v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

coeff =  $\frac{1}{2} \varphi_1''$ 

  v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

coeff =  $\frac{1}{2} nstar_1$ 

  v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

coeff =  $\frac{1}{2} \varphi_2''$ 

  v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

coeff =  $\frac{1}{2} nstar_2$ 

Edges (propagator assignments):
  2 → 4:  nt1 → dv2  [G_hat[0,3]]
  3 → 2:  nt1 → dv1  [G_hat[0,2]]
  3 → 4:  nt1 → dv2  [G_hat[0,3]]
  4 → 0:  nt2 → dn2  [G_hat[1,1]]
  5 → 1:  nt2 → dn1  [G_hat[1,0]]
  5 → 2:  nt2 → dv1  [G_hat[1,2]]

```

```

=====
Diagram 1L-469
=====
External legs:
  leaf 0 ← dn2
  leaf 1 ← dn1
Vertex assignments:
  v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

coeff =  $\frac{1}{2} \varphi_1''$ 

  v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

coeff =  $\frac{1}{2} nstar_1$ 

```

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

Edges (propagator assignments):

2 → 4: nt1 → dv2 [G_hat[0,3]]
 3 → 2: nt1 → dv1 [G_hat[0,2]]
 3 → 4: nt1 → dv2 [G_hat[0,3]]
 4 → 0: nt2 → dn2 [G_hat[1,1]]
 5 → 1: nt2 → dn1 [G_hat[1,0]]
 5 → 2: nt2 → dv1 [G_hat[1,2]]

=====
 Diagram 1L-470
 =====

External legs:

leaf 0 ← dn2
 leaf 1 ← dn1

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

Edges (propagator assignments):

2 → 4: nt1 → dv2 [G_hat[0,3]]
 3 → 2: nt1 → dv1 [G_hat[0,2]]
 3 → 4: nt1 → dv2 [G_hat[0,3]]
 4 → 0: nt2 → dn2 [G_hat[1,1]]
 5 → 1: nt2 → dn1 [G_hat[1,0]]
 5 → 2: nt2 → dv1 [G_hat[1,2]]

```

=====
Diagram 1L-471
=====
External legs:
  leaf 0 ← dn2
  leaf 1 ← dn1
Vertex assignments:
  v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

coeff =  $\frac{1}{2} \varphi_1''$ 

  v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

coeff =  $\frac{1}{2} nstar_1$ 

  v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

coeff =  $\frac{1}{2} \varphi_2''$ 

  v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

coeff =  $\frac{1}{2} nstar_2$ 

Edges (propagator assignments):
  2 → 4:  nt1 → dv2  [G_hat[0,3]]
  3 → 2:  nt1 → dv1  [G_hat[0,2]]
  3 → 4:  nt1 → dv2  [G_hat[0,3]]
  4 → 0:  nt2 → dn2  [G_hat[1,1]]
  5 → 1:  nt2 → dn1  [G_hat[1,0]]
  5 → 2:  nt2 → dv1  [G_hat[1,2]]

```

```

=====
Diagram 1L-472
=====
External legs:
  leaf 0 ← dn2
  leaf 1 ← dn1
Vertex assignments:
  v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

coeff =  $\frac{1}{2} \varphi_1''$ 

  v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

coeff =  $\frac{1}{2} nstar_1$ 

```

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

Edges (propagator assignments):

2 → 4: nt1 → dv2 [G_hat[0,3]]
 3 → 2: nt1 → dv1 [G_hat[0,2]]
 3 → 4: nt1 → dv2 [G_hat[0,3]]
 4 → 0: nt2 → dn2 [G_hat[1,1]]
 5 → 1: nt2 → dn1 [G_hat[1,0]]
 5 → 2: nt2 → dv1 [G_hat[1,2]]

=====
 Diagram 1L-473
 =====

External legs:

leaf 0 ← dn2
 leaf 1 ← dn1

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

Edges (propagator assignments):

2 → 4: nt1 → dv2 [G_hat[0,3]]
 3 → 2: nt1 → dv1 [G_hat[0,2]]
 3 → 4: nt1 → dv2 [G_hat[0,3]]
 4 → 0: nt2 → dn2 [G_hat[1,1]]
 5 → 1: nt2 → dn1 [G_hat[1,0]]
 5 → 2: nt2 → dv1 [G_hat[1,2]]

=====

Diagram 1L-474

=====

External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

Edges (propagator assignments):

2 → 4: nt1 → dv2 [G_hat[0,3]]

3 → 2: nt1 → dv1 [G_hat[0,2]]

3 → 4: nt1 → dv2 [G_hat[0,3]]

4 → 0: nt2 → dn2 [G_hat[1,1]]

5 → 1: nt2 → dn1 [G_hat[1,0]]

5 → 2: nt2 → dv1 [G_hat[1,2]]

=====

Diagram 1L-475

=====

External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

Edges (propagator assignments):

2 → 4: nt1 → dv2 [G_hat[0,3]]
 3 → 2: nt1 → dv1 [G_hat[0,2]]
 3 → 4: nt1 → dv2 [G_hat[0,3]]
 4 → 0: nt2 → dn2 [G_hat[1,1]]
 5 → 1: nt2 → dn1 [G_hat[1,0]]
 5 → 2: nt2 → dv1 [G_hat[1,2]]

=====
 Diagram 1L-476
 =====

External legs:

leaf 0 ← dn2
 leaf 1 ← dn1

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

Edges (propagator assignments):

2 → 4: nt1 → dv2 [G_hat[0,3]]
 3 → 2: nt1 → dv1 [G_hat[0,2]]
 3 → 4: nt1 → dv2 [G_hat[0,3]]
 4 → 0: nt2 → dn2 [G_hat[1,1]]
 5 → 1: nt2 → dn1 [G_hat[1,0]]
 5 → 2: nt2 → dv1 [G_hat[1,2]]

=====

Diagram 1L-477

=====

External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

Edges (propagator assignments):

2 → 4: nt1 → dv2 [G_hat[0,3]]

3 → 2: nt1 → dv1 [G_hat[0,2]]

3 → 4: nt1 → dv2 [G_hat[0,3]]

4 → 0: nt2 → dn2 [G_hat[1,1]]

5 → 1: nt2 → dn1 [G_hat[1,0]]

5 → 2: nt2 → dv1 [G_hat[1,2]]

=====

Diagram 1L-478

=====

External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

Edges (propagator assignments):

2 → 4: nt1 → dv2 [G_hat[0,3]]
 3 → 2: nt1 → dv1 [G_hat[0,2]]
 3 → 4: nt1 → dv2 [G_hat[0,3]]
 4 → 0: nt2 → dn2 [G_hat[1,1]]
 5 → 1: nt2 → dn1 [G_hat[1,0]]
 5 → 2: nt2 → dv1 [G_hat[1,2]]

=====
 Diagram 1L-479
 =====

External legs:

leaf 0 ← dn2
 leaf 1 ← dn1

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

Edges (propagator assignments):

2 → 4: nt1 → dv2 [G_hat[0,3]]
 3 → 2: nt1 → dv1 [G_hat[0,2]]
 3 → 4: nt1 → dv2 [G_hat[0,3]]
 4 → 0: nt2 → dn2 [G_hat[1,1]]
 5 → 1: nt2 → dn1 [G_hat[1,0]]
 5 → 2: nt2 → dv1 [G_hat[1,2]]

```

=====
Diagram 1L-480
=====
External legs:
  leaf 0 ← dn2
  leaf 1 ← dn1
Vertex assignments:
  v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

coeff =  $\frac{1}{2} \varphi_1''$ 

  v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

coeff =  $\frac{1}{2} nstar_1$ 

  v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

coeff =  $\frac{1}{2} \varphi_2''$ 

  v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

coeff =  $\frac{1}{2} nstar_2$ 

Edges (propagator assignments):
  2 → 4:  nt1 → dv2  [G_hat[0,3]]
  3 → 2:  nt1 → dv1  [G_hat[0,2]]
  3 → 4:  nt1 → dv2  [G_hat[0,3]]
  4 → 0:  nt2 → dn2  [G_hat[1,1]]
  5 → 1:  nt2 → dn1  [G_hat[1,0]]
  5 → 2:  nt2 → dv1  [G_hat[1,2]]

```

```

=====
Diagram 1L-481
=====
External legs:
  leaf 0 ← dn2
  leaf 1 ← dn1
Vertex assignments:
  v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

coeff =  $\frac{1}{2} \varphi_2''$ 

  v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

coeff =  $\frac{1}{2} nstar_1$ 

```

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

Edges (propagator assignments):

2 → 4: nt2 → dv1 [G_hat[1,2]]
 3 → 2: nt1 → dv2 [G_hat[0,3]]
 3 → 4: nt1 → dv1 [G_hat[0,2]]
 4 → 0: nt1 → dn2 [G_hat[0,1]]
 5 → 1: nt2 → dn1 [G_hat[1,0]]
 5 → 2: nt2 → dv2 [G_hat[1,3]]

=====
 Diagram 1L-482
 =====

External legs:

leaf 0 ← dn2
 leaf 1 ← dn1

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

Edges (propagator assignments):

2 → 4: nt2 → dv1 [G_hat[1,2]]
 3 → 2: nt1 → dv2 [G_hat[0,3]]
 3 → 4: nt1 → dv1 [G_hat[0,2]]
 4 → 0: nt1 → dn2 [G_hat[0,1]]
 5 → 1: nt2 → dn1 [G_hat[1,0]]
 5 → 2: nt2 → dv2 [G_hat[1,3]]

=====

Diagram 1L-483

=====

External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

Edges (propagator assignments):

2 → 4: nt2 → dv1 [G_hat[1,2]]

3 → 2: nt1 → dv2 [G_hat[0,3]]

3 → 4: nt1 → dv1 [G_hat[0,2]]

4 → 0: nt1 → dn2 [G_hat[0,1]]

5 → 1: nt2 → dn1 [G_hat[1,0]]

5 → 2: nt2 → dv2 [G_hat[1,3]]

=====

Diagram 1L-484

=====

External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

Edges (propagator assignments):

2 → 4: nt2 → dv1 [G_hat[1,2]]
 3 → 2: nt1 → dv2 [G_hat[0,3]]
 3 → 4: nt1 → dv1 [G_hat[0,2]]
 4 → 0: nt1 → dn2 [G_hat[0,1]]
 5 → 1: nt2 → dn1 [G_hat[1,0]]
 5 → 2: nt2 → dv2 [G_hat[1,3]]

=====
 Diagram 1L-485
 =====

External legs:

leaf 0 ← dn2
 leaf 1 ← dn1

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

Edges (propagator assignments):

2 → 4: nt2 → dv1 [G_hat[1,2]]
 3 → 2: nt1 → dv2 [G_hat[0,3]]
 3 → 4: nt1 → dv1 [G_hat[0,2]]
 4 → 0: nt1 → dn2 [G_hat[0,1]]
 5 → 1: nt2 → dn1 [G_hat[1,0]]
 5 → 2: nt2 → dv2 [G_hat[1,3]]

=====

Diagram 1L-486

=====

External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

Edges (propagator assignments):

2 → 4: nt2 → dv1 [G_hat[1,2]]

3 → 2: nt1 → dv2 [G_hat[0,3]]

3 → 4: nt1 → dv1 [G_hat[0,2]]

4 → 0: nt1 → dn2 [G_hat[0,1]]

5 → 1: nt2 → dn1 [G_hat[1,0]]

5 → 2: nt2 → dv2 [G_hat[1,3]]

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Diagram 1L-487

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External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

Edges (propagator assignments):

2 → 4: nt2 → dv1 [G_hat[1,2]]
 3 → 2: nt1 → dv2 [G_hat[0,3]]
 3 → 4: nt1 → dv1 [G_hat[0,2]]
 4 → 0: nt1 → dn2 [G_hat[0,1]]
 5 → 1: nt2 → dn1 [G_hat[1,0]]
 5 → 2: nt2 → dv2 [G_hat[1,3]]

=====
 Diagram 1L-488
 =====

External legs:

leaf 0 ← dn2
 leaf 1 ← dn1

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

Edges (propagator assignments):

2 → 4: nt2 → dv1 [G_hat[1,2]]
 3 → 2: nt1 → dv2 [G_hat[0,3]]
 3 → 4: nt1 → dv1 [G_hat[0,2]]
 4 → 0: nt1 → dn2 [G_hat[0,1]]
 5 → 1: nt2 → dn1 [G_hat[1,0]]
 5 → 2: nt2 → dv2 [G_hat[1,3]]

=====

Diagram 1L-489

=====

External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

Edges (propagator assignments):

2 → 4: nt2 → dv1 [G_hat[1,2]]

3 → 2: nt1 → dv2 [G_hat[0,3]]

3 → 4: nt1 → dv1 [G_hat[0,2]]

4 → 0: nt1 → dn2 [G_hat[0,1]]

5 → 1: nt2 → dn1 [G_hat[1,0]]

5 → 2: nt2 → dv2 [G_hat[1,3]]

=====

Diagram 1L-490

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External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

Edges (propagator assignments):

2 → 4: nt2 → dv1 [G_hat[1,2]]
 3 → 2: nt1 → dv2 [G_hat[0,3]]
 3 → 4: nt1 → dv1 [G_hat[0,2]]
 4 → 0: nt1 → dn2 [G_hat[0,1]]
 5 → 1: nt2 → dn1 [G_hat[1,0]]
 5 → 2: nt2 → dv2 [G_hat[1,3]]

=====
 Diagram 1L-491
 =====

External legs:

leaf 0 ← dn2
 leaf 1 ← dn1

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

Edges (propagator assignments):

2 → 4: nt2 → dv1 [G_hat[1,2]]
 3 → 2: nt1 → dv2 [G_hat[0,3]]
 3 → 4: nt1 → dv1 [G_hat[0,2]]
 4 → 0: nt1 → dn2 [G_hat[0,1]]
 5 → 1: nt2 → dn1 [G_hat[1,0]]
 5 → 2: nt2 → dv2 [G_hat[1,3]]

=====

Diagram 1L-492

=====

External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

Edges (propagator assignments):

2 → 4: nt2 → dv1 [G_hat[1,2]]

3 → 2: nt1 → dv2 [G_hat[0,3]]

3 → 4: nt1 → dv1 [G_hat[0,2]]

4 → 0: nt1 → dn2 [G_hat[0,1]]

5 → 1: nt2 → dn1 [G_hat[1,0]]

5 → 2: nt2 → dv2 [G_hat[1,3]]

=====

Diagram 1L-493

=====

External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

Edges (propagator assignments):

2 → 4: nt2 → dv1 [G_hat[1,2]]
 3 → 2: nt1 → dv2 [G_hat[0,3]]
 3 → 4: nt1 → dv1 [G_hat[0,2]]
 4 → 0: nt1 → dn2 [G_hat[0,1]]
 5 → 1: nt2 → dn1 [G_hat[1,0]]
 5 → 2: nt2 → dv2 [G_hat[1,3]]

=====
 Diagram 1L-494
 =====

External legs:

leaf 0 ← dn2
 leaf 1 ← dn1

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

Edges (propagator assignments):

2 → 4: nt2 → dv1 [G_hat[1,2]]
 3 → 2: nt1 → dv2 [G_hat[0,3]]
 3 → 4: nt1 → dv1 [G_hat[0,2]]
 4 → 0: nt1 → dn2 [G_hat[0,1]]
 5 → 1: nt2 → dn1 [G_hat[1,0]]
 5 → 2: nt2 → dv2 [G_hat[1,3]]

=====

Diagram 1L-495

=====

External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

Edges (propagator assignments):

2 → 4: nt2 → dv1 [G_hat[1,2]]

3 → 2: nt1 → dv2 [G_hat[0,3]]

3 → 4: nt1 → dv1 [G_hat[0,2]]

4 → 0: nt1 → dn2 [G_hat[0,1]]

5 → 1: nt2 → dn1 [G_hat[1,0]]

5 → 2: nt2 → dv2 [G_hat[1,3]]

=====

Diagram 1L-496

=====

External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

Edges (propagator assignments):

2 → 4: nt2 → dv1 [G_hat[1,2]]
 3 → 2: nt1 → dv2 [G_hat[0,3]]
 3 → 4: nt1 → dv1 [G_hat[0,2]]
 4 → 0: nt1 → dn2 [G_hat[0,1]]
 5 → 1: nt2 → dn1 [G_hat[1,0]]
 5 → 2: nt2 → dv2 [G_hat[1,3]]

Diagram 1L-497

External legs:

leaf 0 ← dn2
 leaf 1 ← dn1

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

Edges (propagator assignments):

2 → 4: nt2 → dv2 [G_hat[1,3]]
 3 → 2: nt1 → dv2 [G_hat[0,3]]
 3 → 4: nt1 → dv2 [G_hat[0,3]]
 4 → 0: nt2 → dn2 [G_hat[1,1]]
 5 → 1: nt2 → dn1 [G_hat[1,0]]
 5 → 2: nt2 → dv2 [G_hat[1,3]]


```

=====
Diagram 1L-498
=====

External legs:
  leaf 0 ← dn2
  leaf 1 ← dn1
Vertex assignments:
  v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

coeff =  $\frac{1}{2} \varphi_2''$ 

  v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

coeff =  $\frac{1}{2} nstar_1$ 

  v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

coeff =  $\frac{1}{2} \varphi_2''$ 

  v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

coeff =  $\frac{1}{2} nstar_2$ 

Edges (propagator assignments):
  2 → 4:  nt2 → dv2  [G_hat[1,3]]
  3 → 2:  nt1 → dv2  [G_hat[0,3]]
  3 → 4:  nt1 → dv2  [G_hat[0,3]]
  4 → 0:  nt2 → dn2  [G_hat[1,1]]
  5 → 1:  nt2 → dn1  [G_hat[1,0]]
  5 → 2:  nt2 → dv2  [G_hat[1,3]]

```

```

=====
Diagram 1L-499
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External legs:
  leaf 0 ← dn2
  leaf 1 ← dn1
Vertex assignments:
  v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

coeff =  $\frac{1}{2} \varphi_2''$ 

  v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

coeff =  $\frac{1}{2} nstar_1$ 

```

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

Edges (propagator assignments):

2 → 4: nt2 → dv2 [G_hat[1,3]]
 3 → 2: nt1 → dv2 [G_hat[0,3]]
 3 → 4: nt1 → dv2 [G_hat[0,3]]
 4 → 0: nt2 → dn2 [G_hat[1,1]]
 5 → 1: nt2 → dn1 [G_hat[1,0]]
 5 → 2: nt2 → dv2 [G_hat[1,3]]

=====
 Diagram 1L-500
 =====

External legs:

leaf 0 ← dn2
 leaf 1 ← dn1

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

Edges (propagator assignments):

2 → 4: nt2 → dv2 [G_hat[1,3]]
 3 → 2: nt1 → dv2 [G_hat[0,3]]
 3 → 4: nt1 → dv2 [G_hat[0,3]]
 4 → 0: nt2 → dn2 [G_hat[1,1]]
 5 → 1: nt2 → dn1 [G_hat[1,0]]
 5 → 2: nt2 → dv2 [G_hat[1,3]]

```

=====
Diagram 1L-501
=====

External legs:
  leaf 0 ← dn2
  leaf 1 ← dn1
Vertex assignments:
  v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

coeff =  $\frac{1}{2} \varphi_2''$ 

  v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

coeff =  $\frac{1}{2} nstar_1$ 

  v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

coeff =  $\frac{1}{2} \varphi_2''$ 

  v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

coeff =  $\frac{1}{2} nstar_2$ 

Edges (propagator assignments):
  2 → 4:  nt2 → dv2  [G_hat[1,3]]
  3 → 2:  nt1 → dv2  [G_hat[0,3]]
  3 → 4:  nt1 → dv2  [G_hat[0,3]]
  4 → 0:  nt2 → dn2  [G_hat[1,1]]
  5 → 1:  nt2 → dn1  [G_hat[1,0]]
  5 → 2:  nt2 → dv2  [G_hat[1,3]]

```

```

=====
Diagram 1L-502
=====

External legs:
  leaf 0 ← dn2
  leaf 1 ← dn1
Vertex assignments:
  v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

coeff =  $\frac{1}{2} \varphi_2''$ 

  v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

coeff =  $\frac{1}{2} nstar_1$ 

```

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

Edges (propagator assignments):

2 → 4: nt2 → dv2 [G_hat[1,3]]
 3 → 2: nt1 → dv2 [G_hat[0,3]]
 3 → 4: nt1 → dv2 [G_hat[0,3]]
 4 → 0: nt2 → dn2 [G_hat[1,1]]
 5 → 1: nt2 → dn1 [G_hat[1,0]]
 5 → 2: nt2 → dv2 [G_hat[1,3]]

=====
 Diagram 1L-503
 =====

External legs:

leaf 0 ← dn2
 leaf 1 ← dn1

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

Edges (propagator assignments):

2 → 4: nt2 → dv2 [G_hat[1,3]]
 3 → 2: nt1 → dv2 [G_hat[0,3]]
 3 → 4: nt1 → dv2 [G_hat[0,3]]
 4 → 0: nt2 → dn2 [G_hat[1,1]]
 5 → 1: nt2 → dn1 [G_hat[1,0]]
 5 → 2: nt2 → dv2 [G_hat[1,3]]

=====

Diagram 1L-504

=====

External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

Edges (propagator assignments):

2 → 4: nt2 → dv2 [G_hat[1,3]]

3 → 2: nt1 → dv2 [G_hat[0,3]]

3 → 4: nt1 → dv2 [G_hat[0,3]]

4 → 0: nt2 → dn2 [G_hat[1,1]]

5 → 1: nt2 → dn1 [G_hat[1,0]]

5 → 2: nt2 → dv2 [G_hat[1,3]]

=====

Diagram 1L-505

=====

External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

Edges (propagator assignments):

2 → 4: nt2 → dv2 [G_hat[1,3]]
 3 → 2: nt1 → dv2 [G_hat[0,3]]
 3 → 4: nt1 → dv2 [G_hat[0,3]]
 4 → 0: nt2 → dn2 [G_hat[1,1]]
 5 → 1: nt2 → dn1 [G_hat[1,0]]
 5 → 2: nt2 → dv2 [G_hat[1,3]]

=====
 Diagram 1L-506
 =====

External legs:

leaf 0 ← dn2
 leaf 1 ← dn1

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

Edges (propagator assignments):

2 → 4: nt2 → dv2 [G_hat[1,3]]
 3 → 2: nt1 → dv2 [G_hat[0,3]]
 3 → 4: nt1 → dv2 [G_hat[0,3]]
 4 → 0: nt2 → dn2 [G_hat[1,1]]
 5 → 1: nt2 → dn1 [G_hat[1,0]]
 5 → 2: nt2 → dv2 [G_hat[1,3]]

=====

Diagram 1L-507

=====

External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

Edges (propagator assignments):

2 → 4: nt2 → dv2 [G_hat[1,3]]

3 → 2: nt1 → dv2 [G_hat[0,3]]

3 → 4: nt1 → dv2 [G_hat[0,3]]

4 → 0: nt2 → dn2 [G_hat[1,1]]

5 → 1: nt2 → dn1 [G_hat[1,0]]

5 → 2: nt2 → dv2 [G_hat[1,3]]

=====

Diagram 1L-508

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External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

Edges (propagator assignments):

2 → 4: nt2 → dv2 [G_hat[1,3]]
 3 → 2: nt1 → dv2 [G_hat[0,3]]
 3 → 4: nt1 → dv2 [G_hat[0,3]]
 4 → 0: nt2 → dn2 [G_hat[1,1]]
 5 → 1: nt2 → dn1 [G_hat[1,0]]
 5 → 2: nt2 → dv2 [G_hat[1,3]]

=====
 Diagram 1L-509
 =====

External legs:

leaf 0 ← dn2
 leaf 1 ← dn1

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

Edges (propagator assignments):

2 → 4: nt2 → dv2 [G_hat[1,3]]
 3 → 2: nt1 → dv2 [G_hat[0,3]]
 3 → 4: nt1 → dv2 [G_hat[0,3]]
 4 → 0: nt2 → dn2 [G_hat[1,1]]
 5 → 1: nt2 → dn1 [G_hat[1,0]]
 5 → 2: nt2 → dv2 [G_hat[1,3]]


```

=====
Diagram 1L-510
=====

External legs:
  leaf 0 ← dn2
  leaf 1 ← dn1
Vertex assignments:
  v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

coeff =  $\frac{1}{2} \varphi_2''$ 

  v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

coeff =  $\frac{1}{2} nstar_1$ 

  v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

coeff =  $\frac{1}{2} \varphi_2''$ 

  v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

coeff =  $\frac{1}{2} nstar_2$ 

Edges (propagator assignments):
  2 → 4:  nt2 → dv2  [G_hat[1,3]]
  3 → 2:  nt1 → dv2  [G_hat[0,3]]
  3 → 4:  nt1 → dv2  [G_hat[0,3]]
  4 → 0:  nt2 → dn2  [G_hat[1,1]]
  5 → 1:  nt2 → dn1  [G_hat[1,0]]
  5 → 2:  nt2 → dv2  [G_hat[1,3]]

```

```

=====
Diagram 1L-511
=====

External legs:
  leaf 0 ← dn2
  leaf 1 ← dn1
Vertex assignments:
  v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

coeff =  $\frac{1}{2} \varphi_2''$ 

  v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

coeff =  $\frac{1}{2} nstar_1$ 

```

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

Edges (propagator assignments):

2 → 4: nt2 → dv2 [G_hat[1,3]]
 3 → 2: nt1 → dv2 [G_hat[0,3]]
 3 → 4: nt1 → dv2 [G_hat[0,3]]
 4 → 0: nt2 → dn2 [G_hat[1,1]]
 5 → 1: nt2 → dn1 [G_hat[1,0]]
 5 → 2: nt2 → dv2 [G_hat[1,3]]

=====
 Diagram 1L-512
 =====

External legs:

leaf 0 ← dn2
 leaf 1 ← dn1

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

Edges (propagator assignments):

2 → 4: nt2 → dv2 [G_hat[1,3]]
 3 → 2: nt1 → dv2 [G_hat[0,3]]
 3 → 4: nt1 → dv2 [G_hat[0,3]]
 4 → 0: nt2 → dn2 [G_hat[1,1]]
 5 → 1: nt2 → dn1 [G_hat[1,0]]
 5 → 2: nt2 → dv2 [G_hat[1,3]]

```

=====
Diagram 1L-513
=====
External legs:
  leaf 0 ← dn2
  leaf 1 ← dn1
Vertex assignments:
  v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

coeff =  $\frac{1}{2} \varphi_1''$ 

  v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

coeff =  $\frac{1}{2} nstar_2$ 

  v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

coeff =  $\frac{1}{2} \varphi_1''$ 

  v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

coeff =  $\frac{1}{2} nstar_1$ 

Edges (propagator assignments):
  2 → 4:  nt1 → dv1  [G_hat[0,2]]
  3 → 2:  nt2 → dv1  [G_hat[1,2]]
  3 → 4:  nt2 → dv1  [G_hat[1,2]]
  4 → 0:  nt1 → dn2  [G_hat[0,1]]
  5 → 1:  nt1 → dn1  [G_hat[0,0]]
  5 → 2:  nt1 → dv1  [G_hat[0,2]]

```

```

=====
Diagram 1L-514
=====
External legs:
  leaf 0 ← dn2
  leaf 1 ← dn1
Vertex assignments:
  v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

coeff =  $\frac{1}{2} \varphi_1''$ 

  v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

coeff =  $\frac{1}{2} nstar_2$ 

```

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

Edges (propagator assignments):

2 → 4: nt1 → dv1 [G_hat[0,2]]
 3 → 2: nt2 → dv1 [G_hat[1,2]]
 3 → 4: nt2 → dv1 [G_hat[1,2]]
 4 → 0: nt1 → dn2 [G_hat[0,1]]
 5 → 1: nt1 → dn1 [G_hat[0,0]]
 5 → 2: nt1 → dv1 [G_hat[0,2]]

=====
 Diagram 1L-515
 =====

External legs:

leaf 0 ← dn2
 leaf 1 ← dn1

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

Edges (propagator assignments):

2 → 4: nt1 → dv1 [G_hat[0,2]]
 3 → 2: nt2 → dv1 [G_hat[1,2]]
 3 → 4: nt2 → dv1 [G_hat[1,2]]
 4 → 0: nt1 → dn2 [G_hat[0,1]]
 5 → 1: nt1 → dn1 [G_hat[0,0]]
 5 → 2: nt1 → dv1 [G_hat[0,2]]

=====

Diagram 1L-516

=====

External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

Edges (propagator assignments):

2 → 4: nt1 → dv1 [G_hat[0,2]]

3 → 2: nt2 → dv1 [G_hat[1,2]]

3 → 4: nt2 → dv1 [G_hat[1,2]]

4 → 0: nt1 → dn2 [G_hat[0,1]]

5 → 1: nt1 → dn1 [G_hat[0,0]]

5 → 2: nt1 → dv1 [G_hat[0,2]]

=====

Diagram 1L-517

=====

External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

Edges (propagator assignments):

2 → 4: nt1 → dv1 [G_hat[0,2]]
 3 → 2: nt2 → dv1 [G_hat[1,2]]
 3 → 4: nt2 → dv1 [G_hat[1,2]]
 4 → 0: nt1 → dn2 [G_hat[0,1]]
 5 → 1: nt1 → dn1 [G_hat[0,0]]
 5 → 2: nt1 → dv1 [G_hat[0,2]]

=====
 Diagram 1L-518
 =====

External legs:

leaf 0 ← dn2
 leaf 1 ← dn1

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

Edges (propagator assignments):

2 → 4: nt1 → dv1 [G_hat[0,2]]
 3 → 2: nt2 → dv1 [G_hat[1,2]]
 3 → 4: nt2 → dv1 [G_hat[1,2]]
 4 → 0: nt1 → dn2 [G_hat[0,1]]
 5 → 1: nt1 → dn1 [G_hat[0,0]]
 5 → 2: nt1 → dv1 [G_hat[0,2]]

=====

Diagram 1L-519

=====

External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

Edges (propagator assignments):

2 → 4: nt1 → dv1 [G_hat[0,2]]

3 → 2: nt2 → dv1 [G_hat[1,2]]

3 → 4: nt2 → dv1 [G_hat[1,2]]

4 → 0: nt1 → dn2 [G_hat[0,1]]

5 → 1: nt1 → dn1 [G_hat[0,0]]

5 → 2: nt1 → dv1 [G_hat[0,2]]

=====

Diagram 1L-520

=====

External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

Edges (propagator assignments):

2 → 4: nt1 → dv1 [G_hat[0,2]]
 3 → 2: nt2 → dv1 [G_hat[1,2]]
 3 → 4: nt2 → dv1 [G_hat[1,2]]
 4 → 0: nt1 → dn2 [G_hat[0,1]]
 5 → 1: nt1 → dn1 [G_hat[0,0]]
 5 → 2: nt1 → dv1 [G_hat[0,2]]

=====
 Diagram 1L-521
 =====

External legs:

leaf 0 ← dn2
 leaf 1 ← dn1

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

Edges (propagator assignments):

2 → 4: nt1 → dv1 [G_hat[0,2]]
 3 → 2: nt2 → dv1 [G_hat[1,2]]
 3 → 4: nt2 → dv1 [G_hat[1,2]]
 4 → 0: nt1 → dn2 [G_hat[0,1]]
 5 → 1: nt1 → dn1 [G_hat[0,0]]
 5 → 2: nt1 → dv1 [G_hat[0,2]]


```

=====
Diagram 1L-522
=====

External legs:
  leaf 0 ← dn2
  leaf 1 ← dn1
Vertex assignments:
  v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

coeff =  $\frac{1}{2} \varphi_1''$ 

  v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

coeff =  $\frac{1}{2} nstar_2$ 

  v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

coeff =  $\frac{1}{2} \varphi_1''$ 

  v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

coeff =  $\frac{1}{2} nstar_1$ 

Edges (propagator assignments):
  2 → 4:  nt1 → dv1  [G_hat[0,2]]
  3 → 2:  nt2 → dv1  [G_hat[1,2]]
  3 → 4:  nt2 → dv1  [G_hat[1,2]]
  4 → 0:  nt1 → dn2  [G_hat[0,1]]
  5 → 1:  nt1 → dn1  [G_hat[0,0]]
  5 → 2:  nt1 → dv1  [G_hat[0,2]]

```

```

=====
Diagram 1L-523
=====

External legs:
  leaf 0 ← dn2
  leaf 1 ← dn1
Vertex assignments:
  v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

coeff =  $\frac{1}{2} \varphi_1''$ 

  v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

coeff =  $\frac{1}{2} nstar_2$ 

```

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

Edges (propagator assignments):

2 → 4: nt1 → dv1 [G_hat[0,2]]
 3 → 2: nt2 → dv1 [G_hat[1,2]]
 3 → 4: nt2 → dv1 [G_hat[1,2]]
 4 → 0: nt1 → dn2 [G_hat[0,1]]
 5 → 1: nt1 → dn1 [G_hat[0,0]]
 5 → 2: nt1 → dv1 [G_hat[0,2]]

=====
 Diagram 1L-524
 =====

External legs:

leaf 0 ← dn2
 leaf 1 ← dn1

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

Edges (propagator assignments):

2 → 4: nt1 → dv1 [G_hat[0,2]]
 3 → 2: nt2 → dv1 [G_hat[1,2]]
 3 → 4: nt2 → dv1 [G_hat[1,2]]
 4 → 0: nt1 → dn2 [G_hat[0,1]]
 5 → 1: nt1 → dn1 [G_hat[0,0]]
 5 → 2: nt1 → dv1 [G_hat[0,2]]

```

=====
Diagram 1L-525
=====

External legs:
  leaf 0 ← dn2
  leaf 1 ← dn1
Vertex assignments:
  v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

coeff =  $\frac{1}{2} \varphi_1''$ 

  v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

coeff =  $\frac{1}{2} nstar_2$ 

  v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

coeff =  $\frac{1}{2} \varphi_1''$ 

  v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

coeff =  $\frac{1}{2} nstar_1$ 

Edges (propagator assignments):
  2 → 4:  nt1 → dv1  [G_hat[0,2]]
  3 → 2:  nt2 → dv1  [G_hat[1,2]]
  3 → 4:  nt2 → dv1  [G_hat[1,2]]
  4 → 0:  nt1 → dn2  [G_hat[0,1]]
  5 → 1:  nt1 → dn1  [G_hat[0,0]]
  5 → 2:  nt1 → dv1  [G_hat[0,2]]

```

```

=====
Diagram 1L-526
=====

External legs:
  leaf 0 ← dn2
  leaf 1 ← dn1
Vertex assignments:
  v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

coeff =  $\frac{1}{2} \varphi_1''$ 

  v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

coeff =  $\frac{1}{2} nstar_2$ 

```

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

Edges (propagator assignments):

2 → 4: nt1 → dv1 [G_hat[0,2]]
 3 → 2: nt2 → dv1 [G_hat[1,2]]
 3 → 4: nt2 → dv1 [G_hat[1,2]]
 4 → 0: nt1 → dn2 [G_hat[0,1]]
 5 → 1: nt1 → dn1 [G_hat[0,0]]
 5 → 2: nt1 → dv1 [G_hat[0,2]]

=====
 Diagram 1L-527
 =====

External legs:

leaf 0 ← dn2
 leaf 1 ← dn1

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

Edges (propagator assignments):

2 → 4: nt1 → dv1 [G_hat[0,2]]
 3 → 2: nt2 → dv1 [G_hat[1,2]]
 3 → 4: nt2 → dv1 [G_hat[1,2]]
 4 → 0: nt1 → dn2 [G_hat[0,1]]
 5 → 1: nt1 → dn1 [G_hat[0,0]]
 5 → 2: nt1 → dv1 [G_hat[0,2]]

```

=====
Diagram 1L-528
=====

External legs:
  leaf 0 ← dn2
  leaf 1 ← dn1
Vertex assignments:
  v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

coeff =  $\frac{1}{2} \varphi_1''$ 

  v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

coeff =  $\frac{1}{2} nstar_2$ 

  v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

coeff =  $\frac{1}{2} \varphi_1''$ 

  v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

coeff =  $\frac{1}{2} nstar_1$ 

Edges (propagator assignments):
  2 → 4:  nt1 → dv1  [G_hat[0,2]]
  3 → 2:  nt2 → dv1  [G_hat[1,2]]
  3 → 4:  nt2 → dv1  [G_hat[1,2]]
  4 → 0:  nt1 → dn2  [G_hat[0,1]]
  5 → 1:  nt1 → dn1  [G_hat[0,0]]
  5 → 2:  nt1 → dv1  [G_hat[0,2]]

```

```

=====
Diagram 1L-529
=====

External legs:
  leaf 0 ← dn2
  leaf 1 ← dn1
Vertex assignments:
  v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

coeff =  $\frac{1}{2} \varphi_1''$ 

  v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

coeff =  $\frac{1}{2} nstar_2$ 

```

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

Edges (propagator assignments):

2 → 4: nt1 → dv2 [G_hat[0,3]]
 3 → 2: nt2 → dv1 [G_hat[1,2]]
 3 → 4: nt2 → dv2 [G_hat[1,3]]
 4 → 0: nt2 → dn2 [G_hat[1,1]]
 5 → 1: nt1 → dn1 [G_hat[0,0]]
 5 → 2: nt1 → dv1 [G_hat[0,2]]

=====
 Diagram 1L-530
 =====

External legs:

leaf 0 ← dn2
 leaf 1 ← dn1

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

Edges (propagator assignments):

2 → 4: nt1 → dv2 [G_hat[0,3]]
 3 → 2: nt2 → dv1 [G_hat[1,2]]
 3 → 4: nt2 → dv2 [G_hat[1,3]]
 4 → 0: nt2 → dn2 [G_hat[1,1]]
 5 → 1: nt1 → dn1 [G_hat[0,0]]
 5 → 2: nt1 → dv1 [G_hat[0,2]]

```

=====
Diagram 1L-531
=====
External legs:
  leaf 0 ← dn2
  leaf 1 ← dn1
Vertex assignments:
  v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

coeff =  $\frac{1}{2} \varphi_1''$ 

  v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

coeff =  $\frac{1}{2} nstar_2$ 

  v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

coeff =  $\frac{1}{2} \varphi_2''$ 

  v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

coeff =  $\frac{1}{2} nstar_1$ 

Edges (propagator assignments):
  2 → 4:  nt1 → dv2  [G_hat[0,3]]
  3 → 2:  nt2 → dv1  [G_hat[1,2]]
  3 → 4:  nt2 → dv2  [G_hat[1,3]]
  4 → 0:  nt2 → dn2  [G_hat[1,1]]
  5 → 1:  nt1 → dn1  [G_hat[0,0]]
  5 → 2:  nt1 → dv1  [G_hat[0,2]]

```

```

=====
Diagram 1L-532
=====
External legs:
  leaf 0 ← dn2
  leaf 1 ← dn1
Vertex assignments:
  v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

coeff =  $\frac{1}{2} \varphi_1''$ 

  v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

coeff =  $\frac{1}{2} nstar_2$ 

```

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

Edges (propagator assignments):

2 → 4: nt1 → dv2 [G_hat[0,3]]
 3 → 2: nt2 → dv1 [G_hat[1,2]]
 3 → 4: nt2 → dv2 [G_hat[1,3]]
 4 → 0: nt2 → dn2 [G_hat[1,1]]
 5 → 1: nt1 → dn1 [G_hat[0,0]]
 5 → 2: nt1 → dv1 [G_hat[0,2]]

=====
 Diagram 1L-533
 =====

External legs:

leaf 0 ← dn2
 leaf 1 ← dn1

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

Edges (propagator assignments):

2 → 4: nt1 → dv2 [G_hat[0,3]]
 3 → 2: nt2 → dv1 [G_hat[1,2]]
 3 → 4: nt2 → dv2 [G_hat[1,3]]
 4 → 0: nt2 → dn2 [G_hat[1,1]]
 5 → 1: nt1 → dn1 [G_hat[0,0]]
 5 → 2: nt1 → dv1 [G_hat[0,2]]

=====

Diagram 1L-534

=====

External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

Edges (propagator assignments):

2 → 4: nt1 → dv2 [G_hat[0,3]]

3 → 2: nt2 → dv1 [G_hat[1,2]]

3 → 4: nt2 → dv2 [G_hat[1,3]]

4 → 0: nt2 → dn2 [G_hat[1,1]]

5 → 1: nt1 → dn1 [G_hat[0,0]]

5 → 2: nt1 → dv1 [G_hat[0,2]]

=====

Diagram 1L-535

=====

External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

Edges (propagator assignments):

2 → 4: nt1 → dv2 [G_hat[0,3]]
 3 → 2: nt2 → dv1 [G_hat[1,2]]
 3 → 4: nt2 → dv2 [G_hat[1,3]]
 4 → 0: nt2 → dn2 [G_hat[1,1]]
 5 → 1: nt1 → dn1 [G_hat[0,0]]
 5 → 2: nt1 → dv1 [G_hat[0,2]]

=====
 Diagram 1L-536
 =====

External legs:

leaf 0 ← dn2
 leaf 1 ← dn1

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

Edges (propagator assignments):

2 → 4: nt1 → dv2 [G_hat[0,3]]
 3 → 2: nt2 → dv1 [G_hat[1,2]]
 3 → 4: nt2 → dv2 [G_hat[1,3]]
 4 → 0: nt2 → dn2 [G_hat[1,1]]
 5 → 1: nt1 → dn1 [G_hat[0,0]]
 5 → 2: nt1 → dv1 [G_hat[0,2]]

=====

Diagram 1L-537

=====

External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

Edges (propagator assignments):

2 → 4: nt1 → dv2 [G_hat[0,3]]

3 → 2: nt2 → dv1 [G_hat[1,2]]

3 → 4: nt2 → dv2 [G_hat[1,3]]

4 → 0: nt2 → dn2 [G_hat[1,1]]

5 → 1: nt1 → dn1 [G_hat[0,0]]

5 → 2: nt1 → dv1 [G_hat[0,2]]

=====

Diagram 1L-538

=====

External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

Edges (propagator assignments):

2 → 4: nt1 → dv2 [G_hat[0,3]]
 3 → 2: nt2 → dv1 [G_hat[1,2]]
 3 → 4: nt2 → dv2 [G_hat[1,3]]
 4 → 0: nt2 → dn2 [G_hat[1,1]]
 5 → 1: nt1 → dn1 [G_hat[0,0]]
 5 → 2: nt1 → dv1 [G_hat[0,2]]

=====
 Diagram 1L-539
 =====

External legs:

leaf 0 ← dn2
 leaf 1 ← dn1

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

Edges (propagator assignments):

2 → 4: nt1 → dv2 [G_hat[0,3]]
 3 → 2: nt2 → dv1 [G_hat[1,2]]
 3 → 4: nt2 → dv2 [G_hat[1,3]]
 4 → 0: nt2 → dn2 [G_hat[1,1]]
 5 → 1: nt1 → dn1 [G_hat[0,0]]
 5 → 2: nt1 → dv1 [G_hat[0,2]]

```

=====
Diagram 1L-540
=====

External legs:
  leaf 0 ← dn2
  leaf 1 ← dn1
Vertex assignments:
  v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

coeff =  $\frac{1}{2} \varphi_1''$ 

  v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

coeff =  $\frac{1}{2} nstar_2$ 

  v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

coeff =  $\frac{1}{2} \varphi_2''$ 

  v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

coeff =  $\frac{1}{2} nstar_1$ 

Edges (propagator assignments):
  2 → 4:  nt1 → dv2  [G_hat[0,3]]
  3 → 2:  nt2 → dv1  [G_hat[1,2]]
  3 → 4:  nt2 → dv2  [G_hat[1,3]]
  4 → 0:  nt2 → dn2  [G_hat[1,1]]
  5 → 1:  nt1 → dn1  [G_hat[0,0]]
  5 → 2:  nt1 → dv1  [G_hat[0,2]]

```

```

=====
Diagram 1L-541
=====

External legs:
  leaf 0 ← dn2
  leaf 1 ← dn1
Vertex assignments:
  v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

coeff =  $\frac{1}{2} \varphi_1''$ 

  v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

coeff =  $\frac{1}{2} nstar_2$ 

```

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

Edges (propagator assignments):

2 → 4: nt1 → dv2 [G_hat[0,3]]
 3 → 2: nt2 → dv1 [G_hat[1,2]]
 3 → 4: nt2 → dv2 [G_hat[1,3]]
 4 → 0: nt2 → dn2 [G_hat[1,1]]
 5 → 1: nt1 → dn1 [G_hat[0,0]]
 5 → 2: nt1 → dv1 [G_hat[0,2]]

=====
 Diagram 1L-542
 =====

External legs:

leaf 0 ← dn2
 leaf 1 ← dn1

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

Edges (propagator assignments):

2 → 4: nt1 → dv2 [G_hat[0,3]]
 3 → 2: nt2 → dv1 [G_hat[1,2]]
 3 → 4: nt2 → dv2 [G_hat[1,3]]
 4 → 0: nt2 → dn2 [G_hat[1,1]]
 5 → 1: nt1 → dn1 [G_hat[0,0]]
 5 → 2: nt1 → dv1 [G_hat[0,2]]

=====

Diagram 1L-543

=====

External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

Edges (propagator assignments):

2 → 4: nt1 → dv2 [G_hat[0,3]]

3 → 2: nt2 → dv1 [G_hat[1,2]]

3 → 4: nt2 → dv2 [G_hat[1,3]]

4 → 0: nt2 → dn2 [G_hat[1,1]]

5 → 1: nt1 → dn1 [G_hat[0,0]]

5 → 2: nt1 → dv1 [G_hat[0,2]]

=====

Diagram 1L-544

=====

External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

Edges (propagator assignments):

2 → 4: nt1 → dv2 [G_hat[0,3]]
 3 → 2: nt2 → dv1 [G_hat[1,2]]
 3 → 4: nt2 → dv2 [G_hat[1,3]]
 4 → 0: nt2 → dn2 [G_hat[1,1]]
 5 → 1: nt1 → dn1 [G_hat[0,0]]
 5 → 2: nt1 → dv1 [G_hat[0,2]]

=====
 Diagram 1L-545
 =====

External legs:

leaf 0 ← dn2
 leaf 1 ← dn1

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

Edges (propagator assignments):

2 → 4: nt2 → dv1 [G_hat[1,2]]
 3 → 2: nt2 → dv2 [G_hat[1,3]]
 3 → 4: nt2 → dv1 [G_hat[1,2]]
 4 → 0: nt1 → dn2 [G_hat[0,1]]
 5 → 1: nt1 → dn1 [G_hat[0,0]]
 5 → 2: nt1 → dv2 [G_hat[0,3]]

=====

Diagram 1L-546

=====

External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

Edges (propagator assignments):

2 → 4: nt2 → dv1 [G_hat[1,2]]

3 → 2: nt2 → dv2 [G_hat[1,3]]

3 → 4: nt2 → dv1 [G_hat[1,2]]

4 → 0: nt1 → dn2 [G_hat[0,1]]

5 → 1: nt1 → dn1 [G_hat[0,0]]

5 → 2: nt1 → dv2 [G_hat[0,3]]

=====

Diagram 1L-547

=====

External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

Edges (propagator assignments):

2 → 4: nt2 → dv1 [G_hat[1,2]]
 3 → 2: nt2 → dv2 [G_hat[1,3]]
 3 → 4: nt2 → dv1 [G_hat[1,2]]
 4 → 0: nt1 → dn2 [G_hat[0,1]]
 5 → 1: nt1 → dn1 [G_hat[0,0]]
 5 → 2: nt1 → dv2 [G_hat[0,3]]

=====
 Diagram 1L-548
 =====

External legs:

leaf 0 ← dn2
 leaf 1 ← dn1

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

Edges (propagator assignments):

2 → 4: nt2 → dv1 [G_hat[1,2]]
 3 → 2: nt2 → dv2 [G_hat[1,3]]
 3 → 4: nt2 → dv1 [G_hat[1,2]]
 4 → 0: nt1 → dn2 [G_hat[0,1]]
 5 → 1: nt1 → dn1 [G_hat[0,0]]
 5 → 2: nt1 → dv2 [G_hat[0,3]]

=====

Diagram 1L-549

=====

External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

Edges (propagator assignments):

2 → 4: nt2 → dv1 [G_hat[1,2]]

3 → 2: nt2 → dv2 [G_hat[1,3]]

3 → 4: nt2 → dv1 [G_hat[1,2]]

4 → 0: nt1 → dn2 [G_hat[0,1]]

5 → 1: nt1 → dn1 [G_hat[0,0]]

5 → 2: nt1 → dv2 [G_hat[0,3]]

=====

Diagram 1L-550

=====

External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

Edges (propagator assignments):

2 → 4: nt2 → dv1 [G_hat[1,2]]
 3 → 2: nt2 → dv2 [G_hat[1,3]]
 3 → 4: nt2 → dv1 [G_hat[1,2]]
 4 → 0: nt1 → dn2 [G_hat[0,1]]
 5 → 1: nt1 → dn1 [G_hat[0,0]]
 5 → 2: nt1 → dv2 [G_hat[0,3]]

=====
 Diagram 1L-551
 =====

External legs:

leaf 0 ← dn2
 leaf 1 ← dn1

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

Edges (propagator assignments):

2 → 4: nt2 → dv1 [G_hat[1,2]]
 3 → 2: nt2 → dv2 [G_hat[1,3]]
 3 → 4: nt2 → dv1 [G_hat[1,2]]
 4 → 0: nt1 → dn2 [G_hat[0,1]]
 5 → 1: nt1 → dn1 [G_hat[0,0]]
 5 → 2: nt1 → dv2 [G_hat[0,3]]

```

=====
Diagram 1L-552
=====
External legs:
  leaf 0 ← dn2
  leaf 1 ← dn1
Vertex assignments:
  v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

coeff =  $\frac{1}{2} \varphi_2''$ 

  v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

coeff =  $\frac{1}{2} nstar_2$ 

  v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

coeff =  $\frac{1}{2} \varphi_1''$ 

  v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

coeff =  $\frac{1}{2} nstar_1$ 

Edges (propagator assignments):
  2 → 4:  nt2 → dv1  [G_hat[1,2]]
  3 → 2:  nt2 → dv2  [G_hat[1,3]]
  3 → 4:  nt2 → dv1  [G_hat[1,2]]
  4 → 0:  nt1 → dn2  [G_hat[0,1]]
  5 → 1:  nt1 → dn1  [G_hat[0,0]]
  5 → 2:  nt1 → dv2  [G_hat[0,3]]

```

```

=====
Diagram 1L-553
=====
External legs:
  leaf 0 ← dn2
  leaf 1 ← dn1
Vertex assignments:
  v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

coeff =  $\frac{1}{2} \varphi_2''$ 

  v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

coeff =  $\frac{1}{2} nstar_2$ 

```

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

Edges (propagator assignments):

2 → 4: nt2 → dv1 [G_hat[1,2]]
 3 → 2: nt2 → dv2 [G_hat[1,3]]
 3 → 4: nt2 → dv1 [G_hat[1,2]]
 4 → 0: nt1 → dn2 [G_hat[0,1]]
 5 → 1: nt1 → dn1 [G_hat[0,0]]
 5 → 2: nt1 → dv2 [G_hat[0,3]]

=====
 Diagram 1L-554
 =====

External legs:

leaf 0 ← dn2
 leaf 1 ← dn1

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

Edges (propagator assignments):

2 → 4: nt2 → dv1 [G_hat[1,2]]
 3 → 2: nt2 → dv2 [G_hat[1,3]]
 3 → 4: nt2 → dv1 [G_hat[1,2]]
 4 → 0: nt1 → dn2 [G_hat[0,1]]
 5 → 1: nt1 → dn1 [G_hat[0,0]]
 5 → 2: nt1 → dv2 [G_hat[0,3]]

```

=====
Diagram 1L-555
=====
External legs:
  leaf 0 ← dn2
  leaf 1 ← dn1
Vertex assignments:
  v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

coeff =  $\frac{1}{2} \varphi_2''$ 

  v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

coeff =  $\frac{1}{2} nstar_2$ 

  v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

coeff =  $\frac{1}{2} \varphi_1''$ 

  v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

coeff =  $\frac{1}{2} nstar_1$ 

Edges (propagator assignments):
  2 → 4:  nt2 → dv1  [G_hat[1,2]]
  3 → 2:  nt2 → dv2  [G_hat[1,3]]
  3 → 4:  nt2 → dv1  [G_hat[1,2]]
  4 → 0:  nt1 → dn2  [G_hat[0,1]]
  5 → 1:  nt1 → dn1  [G_hat[0,0]]
  5 → 2:  nt1 → dv2  [G_hat[0,3]]

```

```

=====
Diagram 1L-556
=====
External legs:
  leaf 0 ← dn2
  leaf 1 ← dn1
Vertex assignments:
  v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

coeff =  $\frac{1}{2} \varphi_2''$ 

  v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

coeff =  $\frac{1}{2} nstar_2$ 

```

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

Edges (propagator assignments):

2 → 4: nt2 → dv1 [G_hat[1,2]]
 3 → 2: nt2 → dv2 [G_hat[1,3]]
 3 → 4: nt2 → dv1 [G_hat[1,2]]
 4 → 0: nt1 → dn2 [G_hat[0,1]]
 5 → 1: nt1 → dn1 [G_hat[0,0]]
 5 → 2: nt1 → dv2 [G_hat[0,3]]

=====
 Diagram 1L-557
 =====

External legs:

leaf 0 ← dn2
 leaf 1 ← dn1

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

Edges (propagator assignments):

2 → 4: nt2 → dv1 [G_hat[1,2]]
 3 → 2: nt2 → dv2 [G_hat[1,3]]
 3 → 4: nt2 → dv1 [G_hat[1,2]]
 4 → 0: nt1 → dn2 [G_hat[0,1]]
 5 → 1: nt1 → dn1 [G_hat[0,0]]
 5 → 2: nt1 → dv2 [G_hat[0,3]]

=====

Diagram 1L-558

=====

External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

Edges (propagator assignments):

2 → 4: nt2 → dv1 [G_hat[1,2]]

3 → 2: nt2 → dv2 [G_hat[1,3]]

3 → 4: nt2 → dv1 [G_hat[1,2]]

4 → 0: nt1 → dn2 [G_hat[0,1]]

5 → 1: nt1 → dn1 [G_hat[0,0]]

5 → 2: nt1 → dv2 [G_hat[0,3]]

=====

Diagram 1L-559

=====

External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

Edges (propagator assignments):

2 → 4: nt2 → dv1 [G_hat[1,2]]
 3 → 2: nt2 → dv2 [G_hat[1,3]]
 3 → 4: nt2 → dv1 [G_hat[1,2]]
 4 → 0: nt1 → dn2 [G_hat[0,1]]
 5 → 1: nt1 → dn1 [G_hat[0,0]]
 5 → 2: nt1 → dv2 [G_hat[0,3]]

=====
 Diagram 1L-560
 =====

External legs:

leaf 0 ← dn2
 leaf 1 ← dn1

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

Edges (propagator assignments):

2 → 4: nt2 → dv1 [G_hat[1,2]]
 3 → 2: nt2 → dv2 [G_hat[1,3]]
 3 → 4: nt2 → dv1 [G_hat[1,2]]
 4 → 0: nt1 → dn2 [G_hat[0,1]]
 5 → 1: nt1 → dn1 [G_hat[0,0]]
 5 → 2: nt1 → dv2 [G_hat[0,3]]

```

=====
Diagram 1L-561
=====

External legs:
  leaf 0 ← dn2
  leaf 1 ← dn1
Vertex assignments:
  v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

coeff =  $\frac{1}{2} \varphi_2''$ 

  v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

coeff =  $\frac{1}{2} nstar_2$ 

  v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

coeff =  $\frac{1}{2} \varphi_2''$ 

  v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

coeff =  $\frac{1}{2} nstar_1$ 

Edges (propagator assignments):
  2 → 4:  nt2 → dv2  [G_hat[1,3]]
  3 → 2:  nt2 → dv2  [G_hat[1,3]]
  3 → 4:  nt2 → dv2  [G_hat[1,3]]
  4 → 0:  nt2 → dn2  [G_hat[1,1]]
  5 → 1:  nt1 → dn1  [G_hat[0,0]]
  5 → 2:  nt1 → dv2  [G_hat[0,3]]

```

```

=====
Diagram 1L-562
=====

External legs:
  leaf 0 ← dn2
  leaf 1 ← dn1
Vertex assignments:
  v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

coeff =  $\frac{1}{2} \varphi_2''$ 

  v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

coeff =  $\frac{1}{2} nstar_2$ 

```

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

Edges (propagator assignments):

2 → 4: nt2 → dv2 [G_hat[1,3]]
 3 → 2: nt2 → dv2 [G_hat[1,3]]
 3 → 4: nt2 → dv2 [G_hat[1,3]]
 4 → 0: nt2 → dn2 [G_hat[1,1]]
 5 → 1: nt1 → dn1 [G_hat[0,0]]
 5 → 2: nt1 → dv2 [G_hat[0,3]]

=====
 Diagram 1L-563
 =====

External legs:

leaf 0 ← dn2
 leaf 1 ← dn1

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

Edges (propagator assignments):

2 → 4: nt2 → dv2 [G_hat[1,3]]
 3 → 2: nt2 → dv2 [G_hat[1,3]]
 3 → 4: nt2 → dv2 [G_hat[1,3]]
 4 → 0: nt2 → dn2 [G_hat[1,1]]
 5 → 1: nt1 → dn1 [G_hat[0,0]]
 5 → 2: nt1 → dv2 [G_hat[0,3]]

```

=====
Diagram 1L-564
=====

External legs:
  leaf 0 ← dn2
  leaf 1 ← dn1
Vertex assignments:
  v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

coeff =  $\frac{1}{2} \varphi_2''$ 

  v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

coeff =  $\frac{1}{2} nstar_2$ 

  v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

coeff =  $\frac{1}{2} \varphi_2''$ 

  v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

coeff =  $\frac{1}{2} nstar_1$ 

Edges (propagator assignments):
  2 → 4:  nt2 → dv2  [G_hat[1,3]]
  3 → 2:  nt2 → dv2  [G_hat[1,3]]
  3 → 4:  nt2 → dv2  [G_hat[1,3]]
  4 → 0:  nt2 → dn2  [G_hat[1,1]]
  5 → 1:  nt1 → dn1  [G_hat[0,0]]
  5 → 2:  nt1 → dv2  [G_hat[0,3]]

```

```

=====
Diagram 1L-565
=====

External legs:
  leaf 0 ← dn2
  leaf 1 ← dn1
Vertex assignments:
  v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

coeff =  $\frac{1}{2} \varphi_2''$ 

  v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

coeff =  $\frac{1}{2} nstar_2$ 

```

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

Edges (propagator assignments):

2 → 4: nt2 → dv2 [G_hat[1,3]]
 3 → 2: nt2 → dv2 [G_hat[1,3]]
 3 → 4: nt2 → dv2 [G_hat[1,3]]
 4 → 0: nt2 → dn2 [G_hat[1,1]]
 5 → 1: nt1 → dn1 [G_hat[0,0]]
 5 → 2: nt1 → dv2 [G_hat[0,3]]

=====
 Diagram 1L-566
 =====

External legs:

leaf 0 ← dn2
 leaf 1 ← dn1

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

Edges (propagator assignments):

2 → 4: nt2 → dv2 [G_hat[1,3]]
 3 → 2: nt2 → dv2 [G_hat[1,3]]
 3 → 4: nt2 → dv2 [G_hat[1,3]]
 4 → 0: nt2 → dn2 [G_hat[1,1]]
 5 → 1: nt1 → dn1 [G_hat[0,0]]
 5 → 2: nt1 → dv2 [G_hat[0,3]]

```

=====
Diagram 1L-567
=====
External legs:
  leaf 0 ← dn2
  leaf 1 ← dn1
Vertex assignments:
  v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

coeff =  $\frac{1}{2} \varphi_2''$ 

  v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

coeff =  $\frac{1}{2} nstar_2$ 

  v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

coeff =  $\frac{1}{2} \varphi_2''$ 

  v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

coeff =  $\frac{1}{2} nstar_1$ 

Edges (propagator assignments):
  2 → 4:  nt2 → dv2  [G_hat[1,3]]
  3 → 2:  nt2 → dv2  [G_hat[1,3]]
  3 → 4:  nt2 → dv2  [G_hat[1,3]]
  4 → 0:  nt2 → dn2  [G_hat[1,1]]
  5 → 1:  nt1 → dn1  [G_hat[0,0]]
  5 → 2:  nt1 → dv2  [G_hat[0,3]]

```

```

=====
Diagram 1L-568
=====
External legs:
  leaf 0 ← dn2
  leaf 1 ← dn1
Vertex assignments:
  v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

coeff =  $\frac{1}{2} \varphi_2''$ 

  v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

coeff =  $\frac{1}{2} nstar_2$ 

```

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

Edges (propagator assignments):

2 → 4: nt2 → dv2 [G_hat[1,3]]
 3 → 2: nt2 → dv2 [G_hat[1,3]]
 3 → 4: nt2 → dv2 [G_hat[1,3]]
 4 → 0: nt2 → dn2 [G_hat[1,1]]
 5 → 1: nt1 → dn1 [G_hat[0,0]]
 5 → 2: nt1 → dv2 [G_hat[0,3]]

=====
 Diagram 1L-569
 =====

External legs:

leaf 0 ← dn2
 leaf 1 ← dn1

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

Edges (propagator assignments):

2 → 4: nt2 → dv2 [G_hat[1,3]]
 3 → 2: nt2 → dv2 [G_hat[1,3]]
 3 → 4: nt2 → dv2 [G_hat[1,3]]
 4 → 0: nt2 → dn2 [G_hat[1,1]]
 5 → 1: nt1 → dn1 [G_hat[0,0]]
 5 → 2: nt1 → dv2 [G_hat[0,3]]

=====

Diagram 1L-570

=====

External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

Edges (propagator assignments):

2 → 4: nt2 → dv2 [G_hat[1,3]]

3 → 2: nt2 → dv2 [G_hat[1,3]]

3 → 4: nt2 → dv2 [G_hat[1,3]]

4 → 0: nt2 → dn2 [G_hat[1,1]]

5 → 1: nt1 → dn1 [G_hat[0,0]]

5 → 2: nt1 → dv2 [G_hat[0,3]]

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Diagram 1L-571

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External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

Edges (propagator assignments):

2 → 4: nt2 → dv2 [G_hat[1,3]]
 3 → 2: nt2 → dv2 [G_hat[1,3]]
 3 → 4: nt2 → dv2 [G_hat[1,3]]
 4 → 0: nt2 → dn2 [G_hat[1,1]]
 5 → 1: nt1 → dn1 [G_hat[0,0]]
 5 → 2: nt1 → dv2 [G_hat[0,3]]

=====
 Diagram 1L-572
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External legs:

leaf 0 ← dn2
 leaf 1 ← dn1

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

Edges (propagator assignments):

2 → 4: nt2 → dv2 [G_hat[1,3]]
 3 → 2: nt2 → dv2 [G_hat[1,3]]
 3 → 4: nt2 → dv2 [G_hat[1,3]]
 4 → 0: nt2 → dn2 [G_hat[1,1]]
 5 → 1: nt1 → dn1 [G_hat[0,0]]
 5 → 2: nt1 → dv2 [G_hat[0,3]]

```

=====
Diagram 1L-573
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External legs:
  leaf 0 ← dn2
  leaf 1 ← dn1
Vertex assignments:
  v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

coeff =  $\frac{1}{2} \varphi_2''$ 

  v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

coeff =  $\frac{1}{2} nstar_2$ 

  v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

coeff =  $\frac{1}{2} \varphi_2''$ 

  v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

coeff =  $\frac{1}{2} nstar_1$ 

Edges (propagator assignments):
  2 → 4:  nt2 → dv2  [G_hat[1,3]]
  3 → 2:  nt2 → dv2  [G_hat[1,3]]
  3 → 4:  nt2 → dv2  [G_hat[1,3]]
  4 → 0:  nt2 → dn2  [G_hat[1,1]]
  5 → 1:  nt1 → dn1  [G_hat[0,0]]
  5 → 2:  nt1 → dv2  [G_hat[0,3]]

```

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=====
Diagram 1L-574
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External legs:
  leaf 0 ← dn2
  leaf 1 ← dn1
Vertex assignments:
  v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

coeff =  $\frac{1}{2} \varphi_2''$ 

  v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

coeff =  $\frac{1}{2} nstar_2$ 

```

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

Edges (propagator assignments):

2 → 4: nt2 → dv2 [G_hat[1,3]]
 3 → 2: nt2 → dv2 [G_hat[1,3]]
 3 → 4: nt2 → dv2 [G_hat[1,3]]
 4 → 0: nt2 → dn2 [G_hat[1,1]]
 5 → 1: nt1 → dn1 [G_hat[0,0]]
 5 → 2: nt1 → dv2 [G_hat[0,3]]

=====
 Diagram 1L-575
 =====

External legs:

leaf 0 ← dn2
 leaf 1 ← dn1

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

Edges (propagator assignments):

2 → 4: nt2 → dv2 [G_hat[1,3]]
 3 → 2: nt2 → dv2 [G_hat[1,3]]
 3 → 4: nt2 → dv2 [G_hat[1,3]]
 4 → 0: nt2 → dn2 [G_hat[1,1]]
 5 → 1: nt1 → dn1 [G_hat[0,0]]
 5 → 2: nt1 → dv2 [G_hat[0,3]]

=====

Diagram 1L-576

=====

External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

Edges (propagator assignments):

2 → 4: nt2 → dv2 [G_hat[1,3]]

3 → 2: nt2 → dv2 [G_hat[1,3]]

3 → 4: nt2 → dv2 [G_hat[1,3]]

4 → 0: nt2 → dn2 [G_hat[1,1]]

5 → 1: nt1 → dn1 [G_hat[0,0]]

5 → 2: nt1 → dv2 [G_hat[0,3]]

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Diagram 1L-577

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External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

Edges (propagator assignments):

2 → 4: nt1 → dv1 [G_hat[0,2]]
 3 → 2: nt2 → dv1 [G_hat[1,2]]
 3 → 4: nt2 → dv1 [G_hat[1,2]]
 4 → 0: nt1 → dn2 [G_hat[0,1]]
 5 → 1: nt2 → dn1 [G_hat[1,0]]
 5 → 2: nt2 → dv1 [G_hat[1,2]]

=====
 Diagram 1L-578
 =====

External legs:

leaf 0 ← dn2
 leaf 1 ← dn1

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

Edges (propagator assignments):

2 → 4: nt1 → dv1 [G_hat[0,2]]
 3 → 2: nt2 → dv1 [G_hat[1,2]]
 3 → 4: nt2 → dv1 [G_hat[1,2]]
 4 → 0: nt1 → dn2 [G_hat[0,1]]
 5 → 1: nt2 → dn1 [G_hat[1,0]]
 5 → 2: nt2 → dv1 [G_hat[1,2]]

=====

Diagram 1L-579

=====

External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

Edges (propagator assignments):

2 → 4: nt1 → dv1 [G_hat[0,2]]

3 → 2: nt2 → dv1 [G_hat[1,2]]

3 → 4: nt2 → dv1 [G_hat[1,2]]

4 → 0: nt1 → dn2 [G_hat[0,1]]

5 → 1: nt2 → dn1 [G_hat[1,0]]

5 → 2: nt2 → dv1 [G_hat[1,2]]

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Diagram 1L-580

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External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

Edges (propagator assignments):

2 → 4: nt1 → dv1 [G_hat[0,2]]
 3 → 2: nt2 → dv1 [G_hat[1,2]]
 3 → 4: nt2 → dv1 [G_hat[1,2]]
 4 → 0: nt1 → dn2 [G_hat[0,1]]
 5 → 1: nt2 → dn1 [G_hat[1,0]]
 5 → 2: nt2 → dv1 [G_hat[1,2]]

=====
 Diagram 1L-581
 =====

External legs:

leaf 0 ← dn2
 leaf 1 ← dn1

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

Edges (propagator assignments):

2 → 4: nt1 → dv1 [G_hat[0,2]]
 3 → 2: nt2 → dv1 [G_hat[1,2]]
 3 → 4: nt2 → dv1 [G_hat[1,2]]
 4 → 0: nt1 → dn2 [G_hat[0,1]]
 5 → 1: nt2 → dn1 [G_hat[1,0]]
 5 → 2: nt2 → dv1 [G_hat[1,2]]

=====

Diagram 1L-582

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External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

Edges (propagator assignments):

2 → 4: nt1 → dv1 [G_hat[0,2]]

3 → 2: nt2 → dv1 [G_hat[1,2]]

3 → 4: nt2 → dv1 [G_hat[1,2]]

4 → 0: nt1 → dn2 [G_hat[0,1]]

5 → 1: nt2 → dn1 [G_hat[1,0]]

5 → 2: nt2 → dv1 [G_hat[1,2]]

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Diagram 1L-583

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External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

Edges (propagator assignments):

2 → 4: nt1 → dv1 [G_hat[0,2]]

3 → 2: nt2 → dv1 [G_hat[1,2]]

3 → 4: nt2 → dv1 [G_hat[1,2]]

4 → 0: nt1 → dn2 [G_hat[0,1]]

5 → 1: nt2 → dn1 [G_hat[1,0]]

5 → 2: nt2 → dv1 [G_hat[1,2]]

=====
Diagram 1L-584
=====

External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

Edges (propagator assignments):

2 → 4: nt1 → dv1 [G_hat[0,2]]

3 → 2: nt2 → dv1 [G_hat[1,2]]

3 → 4: nt2 → dv1 [G_hat[1,2]]

4 → 0: nt1 → dn2 [G_hat[0,1]]

5 → 1: nt2 → dn1 [G_hat[1,0]]

5 → 2: nt2 → dv1 [G_hat[1,2]]

=====

Diagram 1L-585

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External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

Edges (propagator assignments):

2 → 4: nt1 → dv1 [G_hat[0,2]]

3 → 2: nt2 → dv1 [G_hat[1,2]]

3 → 4: nt2 → dv1 [G_hat[1,2]]

4 → 0: nt1 → dn2 [G_hat[0,1]]

5 → 1: nt2 → dn1 [G_hat[1,0]]

5 → 2: nt2 → dv1 [G_hat[1,2]]

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Diagram 1L-586

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External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

Edges (propagator assignments):

2 → 4: nt1 → dv1 [G_hat[0,2]]
 3 → 2: nt2 → dv1 [G_hat[1,2]]
 3 → 4: nt2 → dv1 [G_hat[1,2]]
 4 → 0: nt1 → dn2 [G_hat[0,1]]
 5 → 1: nt2 → dn1 [G_hat[1,0]]
 5 → 2: nt2 → dv1 [G_hat[1,2]]

=====
 Diagram 1L-587
 =====

External legs:

leaf 0 ← dn2
 leaf 1 ← dn1

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

Edges (propagator assignments):

2 → 4: nt1 → dv1 [G_hat[0,2]]
 3 → 2: nt2 → dv1 [G_hat[1,2]]
 3 → 4: nt2 → dv1 [G_hat[1,2]]
 4 → 0: nt1 → dn2 [G_hat[0,1]]
 5 → 1: nt2 → dn1 [G_hat[1,0]]
 5 → 2: nt2 → dv1 [G_hat[1,2]]

=====

Diagram 1L-588

=====

External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

Edges (propagator assignments):

2 → 4: nt1 → dv1 [G_hat[0,2]]

3 → 2: nt2 → dv1 [G_hat[1,2]]

3 → 4: nt2 → dv1 [G_hat[1,2]]

4 → 0: nt1 → dn2 [G_hat[0,1]]

5 → 1: nt2 → dn1 [G_hat[1,0]]

5 → 2: nt2 → dv1 [G_hat[1,2]]

=====

Diagram 1L-589

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External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

Edges (propagator assignments):

2 → 4: nt1 → dv1 [G_hat[0,2]]
 3 → 2: nt2 → dv1 [G_hat[1,2]]
 3 → 4: nt2 → dv1 [G_hat[1,2]]
 4 → 0: nt1 → dn2 [G_hat[0,1]]
 5 → 1: nt2 → dn1 [G_hat[1,0]]
 5 → 2: nt2 → dv1 [G_hat[1,2]]

=====
 Diagram 1L-590
 =====

External legs:

leaf 0 ← dn2
 leaf 1 ← dn1

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

Edges (propagator assignments):

2 → 4: nt1 → dv1 [G_hat[0,2]]
 3 → 2: nt2 → dv1 [G_hat[1,2]]
 3 → 4: nt2 → dv1 [G_hat[1,2]]
 4 → 0: nt1 → dn2 [G_hat[0,1]]
 5 → 1: nt2 → dn1 [G_hat[1,0]]
 5 → 2: nt2 → dv1 [G_hat[1,2]]

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=====
Diagram 1L-591
=====

External legs:
  leaf 0 ← dn2
  leaf 1 ← dn1
Vertex assignments:
  v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

coeff =  $\frac{1}{2} \varphi_1''$ 

  v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

coeff =  $\frac{1}{2} nstar_2$ 

  v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

coeff =  $\frac{1}{2} \varphi_1''$ 

  v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

coeff =  $\frac{1}{2} nstar_2$ 

Edges (propagator assignments):
  2 → 4:  nt1 → dv1  [G_hat[0,2]]
  3 → 2:  nt2 → dv1  [G_hat[1,2]]
  3 → 4:  nt2 → dv1  [G_hat[1,2]]
  4 → 0:  nt1 → dn2  [G_hat[0,1]]
  5 → 1:  nt2 → dn1  [G_hat[1,0]]
  5 → 2:  nt2 → dv1  [G_hat[1,2]]

```

```

=====
Diagram 1L-592
=====

External legs:
  leaf 0 ← dn2
  leaf 1 ← dn1
Vertex assignments:
  v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

coeff =  $\frac{1}{2} \varphi_1''$ 

  v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

coeff =  $\frac{1}{2} nstar_2$ 

```

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

Edges (propagator assignments):

2 → 4: nt1 → dv1 [G_hat[0,2]]
 3 → 2: nt2 → dv1 [G_hat[1,2]]
 3 → 4: nt2 → dv1 [G_hat[1,2]]
 4 → 0: nt1 → dn2 [G_hat[0,1]]
 5 → 1: nt2 → dn1 [G_hat[1,0]]
 5 → 2: nt2 → dv1 [G_hat[1,2]]

=====
 Diagram 1L-593
 =====

External legs:

leaf 0 ← dn2
 leaf 1 ← dn1

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

Edges (propagator assignments):

2 → 4: nt1 → dv2 [G_hat[0,3]]
 3 → 2: nt2 → dv1 [G_hat[1,2]]
 3 → 4: nt2 → dv2 [G_hat[1,3]]
 4 → 0: nt2 → dn2 [G_hat[1,1]]
 5 → 1: nt2 → dn1 [G_hat[1,0]]
 5 → 2: nt2 → dv1 [G_hat[1,2]]

=====

Diagram 1L-594

=====

External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

Edges (propagator assignments):

2 → 4: nt1 → dv2 [G_hat[0,3]]

3 → 2: nt2 → dv1 [G_hat[1,2]]

3 → 4: nt2 → dv2 [G_hat[1,3]]

4 → 0: nt2 → dn2 [G_hat[1,1]]

5 → 1: nt2 → dn1 [G_hat[1,0]]

5 → 2: nt2 → dv1 [G_hat[1,2]]

=====

Diagram 1L-595

=====

External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

Edges (propagator assignments):

2 → 4: nt1 → dv2 [G_hat[0,3]]
 3 → 2: nt2 → dv1 [G_hat[1,2]]
 3 → 4: nt2 → dv2 [G_hat[1,3]]
 4 → 0: nt2 → dn2 [G_hat[1,1]]
 5 → 1: nt2 → dn1 [G_hat[1,0]]
 5 → 2: nt2 → dv1 [G_hat[1,2]]

=====
 Diagram 1L-596
 =====

External legs:

leaf 0 ← dn2
 leaf 1 ← dn1

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

Edges (propagator assignments):

2 → 4: nt1 → dv2 [G_hat[0,3]]
 3 → 2: nt2 → dv1 [G_hat[1,2]]
 3 → 4: nt2 → dv2 [G_hat[1,3]]
 4 → 0: nt2 → dn2 [G_hat[1,1]]
 5 → 1: nt2 → dn1 [G_hat[1,0]]
 5 → 2: nt2 → dv1 [G_hat[1,2]]

=====

Diagram 1L-597

=====

External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

Edges (propagator assignments):

2 → 4: nt1 → dv2 [G_hat[0,3]]

3 → 2: nt2 → dv1 [G_hat[1,2]]

3 → 4: nt2 → dv2 [G_hat[1,3]]

4 → 0: nt2 → dn2 [G_hat[1,1]]

5 → 1: nt2 → dn1 [G_hat[1,0]]

5 → 2: nt2 → dv1 [G_hat[1,2]]

=====

Diagram 1L-598

=====

External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

Edges (propagator assignments):

2 → 4: nt1 → dv2 [G_hat[0,3]]
 3 → 2: nt2 → dv1 [G_hat[1,2]]
 3 → 4: nt2 → dv2 [G_hat[1,3]]
 4 → 0: nt2 → dn2 [G_hat[1,1]]
 5 → 1: nt2 → dn1 [G_hat[1,0]]
 5 → 2: nt2 → dv1 [G_hat[1,2]]

=====
 Diagram 1L-599
 =====

External legs:

leaf 0 ← dn2
 leaf 1 ← dn1

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

Edges (propagator assignments):

2 → 4: nt1 → dv2 [G_hat[0,3]]
 3 → 2: nt2 → dv1 [G_hat[1,2]]
 3 → 4: nt2 → dv2 [G_hat[1,3]]
 4 → 0: nt2 → dn2 [G_hat[1,1]]
 5 → 1: nt2 → dn1 [G_hat[1,0]]
 5 → 2: nt2 → dv1 [G_hat[1,2]]

```

=====
Diagram 1L-600
=====

External legs:
  leaf 0 ← dn2
  leaf 1 ← dn1
Vertex assignments:
  v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

coeff =  $\frac{1}{2} \varphi_1''$ 

  v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

coeff =  $\frac{1}{2} nstar_2$ 

  v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

coeff =  $\frac{1}{2} \varphi_2''$ 

  v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

coeff =  $\frac{1}{2} nstar_2$ 

Edges (propagator assignments):
  2 → 4:  nt1 → dv2  [G_hat[0,3]]
  3 → 2:  nt2 → dv1  [G_hat[1,2]]
  3 → 4:  nt2 → dv2  [G_hat[1,3]]
  4 → 0:  nt2 → dn2  [G_hat[1,1]]
  5 → 1:  nt2 → dn1  [G_hat[1,0]]
  5 → 2:  nt2 → dv1  [G_hat[1,2]]

```

```

=====
Diagram 1L-601
=====

External legs:
  leaf 0 ← dn2
  leaf 1 ← dn1
Vertex assignments:
  v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

coeff =  $\frac{1}{2} \varphi_1''$ 

  v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

coeff =  $\frac{1}{2} nstar_2$ 

```

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

Edges (propagator assignments):

2 → 4: nt1 → dv2 [G_hat[0,3]]
 3 → 2: nt2 → dv1 [G_hat[1,2]]
 3 → 4: nt2 → dv2 [G_hat[1,3]]
 4 → 0: nt2 → dn2 [G_hat[1,1]]
 5 → 1: nt2 → dn1 [G_hat[1,0]]
 5 → 2: nt2 → dv1 [G_hat[1,2]]

=====
 Diagram 1L-602
 =====

External legs:

leaf 0 ← dn2
 leaf 1 ← dn1

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

Edges (propagator assignments):

2 → 4: nt1 → dv2 [G_hat[0,3]]
 3 → 2: nt2 → dv1 [G_hat[1,2]]
 3 → 4: nt2 → dv2 [G_hat[1,3]]
 4 → 0: nt2 → dn2 [G_hat[1,1]]
 5 → 1: nt2 → dn1 [G_hat[1,0]]
 5 → 2: nt2 → dv1 [G_hat[1,2]]

```

=====
Diagram 1L-603
=====

External legs:
  leaf 0 ← dn2
  leaf 1 ← dn1
Vertex assignments:
  v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

coeff =  $\frac{1}{2} \varphi_1''$ 

  v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

coeff =  $\frac{1}{2} nstar_2$ 

  v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

coeff =  $\frac{1}{2} \varphi_2''$ 

  v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

coeff =  $\frac{1}{2} nstar_2$ 

Edges (propagator assignments):
  2 → 4:  nt1 → dv2  [G_hat[0,3]]
  3 → 2:  nt2 → dv1  [G_hat[1,2]]
  3 → 4:  nt2 → dv2  [G_hat[1,3]]
  4 → 0:  nt2 → dn2  [G_hat[1,1]]
  5 → 1:  nt2 → dn1  [G_hat[1,0]]
  5 → 2:  nt2 → dv1  [G_hat[1,2]]

```

```

=====
Diagram 1L-604
=====

External legs:
  leaf 0 ← dn2
  leaf 1 ← dn1
Vertex assignments:
  v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

coeff =  $\frac{1}{2} \varphi_1''$ 

  v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

coeff =  $\frac{1}{2} nstar_2$ 

```

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

Edges (propagator assignments):

2 → 4: nt1 → dv2 [G_hat[0,3]]
 3 → 2: nt2 → dv1 [G_hat[1,2]]
 3 → 4: nt2 → dv2 [G_hat[1,3]]
 4 → 0: nt2 → dn2 [G_hat[1,1]]
 5 → 1: nt2 → dn1 [G_hat[1,0]]
 5 → 2: nt2 → dv1 [G_hat[1,2]]

=====
 Diagram 1L-605
 =====

External legs:

leaf 0 ← dn2
 leaf 1 ← dn1

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

Edges (propagator assignments):

2 → 4: nt1 → dv2 [G_hat[0,3]]
 3 → 2: nt2 → dv1 [G_hat[1,2]]
 3 → 4: nt2 → dv2 [G_hat[1,3]]
 4 → 0: nt2 → dn2 [G_hat[1,1]]
 5 → 1: nt2 → dn1 [G_hat[1,0]]
 5 → 2: nt2 → dv1 [G_hat[1,2]]


```

=====
Diagram 1L-606
=====

External legs:
  leaf 0 ← dn2
  leaf 1 ← dn1
Vertex assignments:
  v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

coeff =  $\frac{1}{2} \varphi_1''$ 

  v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

coeff =  $\frac{1}{2} nstar_2$ 

  v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

coeff =  $\frac{1}{2} \varphi_2''$ 

  v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

coeff =  $\frac{1}{2} nstar_2$ 

Edges (propagator assignments):
  2 → 4:  nt1 → dv2  [G_hat[0,3]]
  3 → 2:  nt2 → dv1  [G_hat[1,2]]
  3 → 4:  nt2 → dv2  [G_hat[1,3]]
  4 → 0:  nt2 → dn2  [G_hat[1,1]]
  5 → 1:  nt2 → dn1  [G_hat[1,0]]
  5 → 2:  nt2 → dv1  [G_hat[1,2]]

```

```

=====
Diagram 1L-607
=====

External legs:
  leaf 0 ← dn2
  leaf 1 ← dn1
Vertex assignments:
  v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

coeff =  $\frac{1}{2} \varphi_1''$ 

  v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

coeff =  $\frac{1}{2} nstar_2$ 

```

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

Edges (propagator assignments):

2 → 4: nt1 → dv2 [G_hat[0,3]]
 3 → 2: nt2 → dv1 [G_hat[1,2]]
 3 → 4: nt2 → dv2 [G_hat[1,3]]
 4 → 0: nt2 → dn2 [G_hat[1,1]]
 5 → 1: nt2 → dn1 [G_hat[1,0]]
 5 → 2: nt2 → dv1 [G_hat[1,2]]

=====
 Diagram 1L-608
 =====

External legs:

leaf 0 ← dn2
 leaf 1 ← dn1

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

Edges (propagator assignments):

2 → 4: nt1 → dv2 [G_hat[0,3]]
 3 → 2: nt2 → dv1 [G_hat[1,2]]
 3 → 4: nt2 → dv2 [G_hat[1,3]]
 4 → 0: nt2 → dn2 [G_hat[1,1]]
 5 → 1: nt2 → dn1 [G_hat[1,0]]
 5 → 2: nt2 → dv1 [G_hat[1,2]]

=====

Diagram 1L-609

=====

External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

Edges (propagator assignments):

2 → 4: nt2 → dv1 [G_hat[1,2]]

3 → 2: nt2 → dv2 [G_hat[1,3]]

3 → 4: nt2 → dv1 [G_hat[1,2]]

4 → 0: nt1 → dn2 [G_hat[0,1]]

5 → 1: nt2 → dn1 [G_hat[1,0]]

5 → 2: nt2 → dv2 [G_hat[1,3]]

=====

Diagram 1L-610

=====

External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

Edges (propagator assignments):

2 → 4: nt2 → dv1 [G_hat[1,2]]
 3 → 2: nt2 → dv2 [G_hat[1,3]]
 3 → 4: nt2 → dv1 [G_hat[1,2]]
 4 → 0: nt1 → dn2 [G_hat[0,1]]
 5 → 1: nt2 → dn1 [G_hat[1,0]]
 5 → 2: nt2 → dv2 [G_hat[1,3]]

=====
 Diagram 1L-611
 =====

External legs:

leaf 0 ← dn2
 leaf 1 ← dn1

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

Edges (propagator assignments):

2 → 4: nt2 → dv1 [G_hat[1,2]]
 3 → 2: nt2 → dv2 [G_hat[1,3]]
 3 → 4: nt2 → dv1 [G_hat[1,2]]
 4 → 0: nt1 → dn2 [G_hat[0,1]]
 5 → 1: nt2 → dn1 [G_hat[1,0]]
 5 → 2: nt2 → dv2 [G_hat[1,3]]

```

=====
Diagram 1L-612
=====
External legs:
  leaf 0 ← dn2
  leaf 1 ← dn1
Vertex assignments:
  v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

coeff =  $\frac{1}{2} \varphi_2''$ 

  v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

coeff =  $\frac{1}{2} nstar_2$ 

  v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

coeff =  $\frac{1}{2} \varphi_1''$ 

  v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

coeff =  $\frac{1}{2} nstar_2$ 

Edges (propagator assignments):
  2 → 4:  nt2 → dv1  [G_hat[1,2]]
  3 → 2:  nt2 → dv2  [G_hat[1,3]]
  3 → 4:  nt2 → dv1  [G_hat[1,2]]
  4 → 0:  nt1 → dn2  [G_hat[0,1]]
  5 → 1:  nt2 → dn1  [G_hat[1,0]]
  5 → 2:  nt2 → dv2  [G_hat[1,3]]

```

```

=====
Diagram 1L-613
=====
External legs:
  leaf 0 ← dn2
  leaf 1 ← dn1
Vertex assignments:
  v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

coeff =  $\frac{1}{2} \varphi_2''$ 

  v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

coeff =  $\frac{1}{2} nstar_2$ 

```

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

Edges (propagator assignments):

2 → 4: nt2 → dv1 [G_hat[1,2]]
 3 → 2: nt2 → dv2 [G_hat[1,3]]
 3 → 4: nt2 → dv1 [G_hat[1,2]]
 4 → 0: nt1 → dn2 [G_hat[0,1]]
 5 → 1: nt2 → dn1 [G_hat[1,0]]
 5 → 2: nt2 → dv2 [G_hat[1,3]]

=====
 Diagram 1L-614
 =====

External legs:

leaf 0 ← dn2
 leaf 1 ← dn1

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

Edges (propagator assignments):

2 → 4: nt2 → dv1 [G_hat[1,2]]
 3 → 2: nt2 → dv2 [G_hat[1,3]]
 3 → 4: nt2 → dv1 [G_hat[1,2]]
 4 → 0: nt1 → dn2 [G_hat[0,1]]
 5 → 1: nt2 → dn1 [G_hat[1,0]]
 5 → 2: nt2 → dv2 [G_hat[1,3]]

```

=====
Diagram 1L-615
=====

External legs:
  leaf 0 ← dn2
  leaf 1 ← dn1
Vertex assignments:
  v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

coeff =  $\frac{1}{2} \varphi_2''$ 

  v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

coeff =  $\frac{1}{2} nstar_2$ 

  v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

coeff =  $\frac{1}{2} \varphi_1''$ 

  v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

coeff =  $\frac{1}{2} nstar_2$ 

Edges (propagator assignments):
  2 → 4:  nt2 → dv1  [G_hat[1,2]]
  3 → 2:  nt2 → dv2  [G_hat[1,3]]
  3 → 4:  nt2 → dv1  [G_hat[1,2]]
  4 → 0:  nt1 → dn2  [G_hat[0,1]]
  5 → 1:  nt2 → dn1  [G_hat[1,0]]
  5 → 2:  nt2 → dv2  [G_hat[1,3]]

```

```

=====
Diagram 1L-616
=====

External legs:
  leaf 0 ← dn2
  leaf 1 ← dn1
Vertex assignments:
  v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

coeff =  $\frac{1}{2} \varphi_2''$ 

  v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

coeff =  $\frac{1}{2} nstar_2$ 

```

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

Edges (propagator assignments):

2 → 4: nt2 → dv1 [G_hat[1,2]]
 3 → 2: nt2 → dv2 [G_hat[1,3]]
 3 → 4: nt2 → dv1 [G_hat[1,2]]
 4 → 0: nt1 → dn2 [G_hat[0,1]]
 5 → 1: nt2 → dn1 [G_hat[1,0]]
 5 → 2: nt2 → dv2 [G_hat[1,3]]

=====
 Diagram 1L-617
 =====

External legs:

leaf 0 ← dn2
 leaf 1 ← dn1

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

Edges (propagator assignments):

2 → 4: nt2 → dv1 [G_hat[1,2]]
 3 → 2: nt2 → dv2 [G_hat[1,3]]
 3 → 4: nt2 → dv1 [G_hat[1,2]]
 4 → 0: nt1 → dn2 [G_hat[0,1]]
 5 → 1: nt2 → dn1 [G_hat[1,0]]
 5 → 2: nt2 → dv2 [G_hat[1,3]]


```

=====
Diagram 1L-618
=====
External legs:
  leaf 0 ← dn2
  leaf 1 ← dn1
Vertex assignments:
  v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

coeff =  $\frac{1}{2} \varphi_2''$ 

  v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

coeff =  $\frac{1}{2} nstar_2$ 

  v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

coeff =  $\frac{1}{2} \varphi_1''$ 

  v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

coeff =  $\frac{1}{2} nstar_2$ 

Edges (propagator assignments):
  2 → 4:  nt2 → dv1  [G_hat[1,2]]
  3 → 2:  nt2 → dv2  [G_hat[1,3]]
  3 → 4:  nt2 → dv1  [G_hat[1,2]]
  4 → 0:  nt1 → dn2  [G_hat[0,1]]
  5 → 1:  nt2 → dn1  [G_hat[1,0]]
  5 → 2:  nt2 → dv2  [G_hat[1,3]]

```

```

=====
Diagram 1L-619
=====
External legs:
  leaf 0 ← dn2
  leaf 1 ← dn1
Vertex assignments:
  v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

coeff =  $\frac{1}{2} \varphi_2''$ 

  v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

coeff =  $\frac{1}{2} nstar_2$ 

```

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

Edges (propagator assignments):

2 → 4: nt2 → dv1 [G_hat[1,2]]
 3 → 2: nt2 → dv2 [G_hat[1,3]]
 3 → 4: nt2 → dv1 [G_hat[1,2]]
 4 → 0: nt1 → dn2 [G_hat[0,1]]
 5 → 1: nt2 → dn1 [G_hat[1,0]]
 5 → 2: nt2 → dv2 [G_hat[1,3]]

Diagram 1L-620

External legs:

leaf 0 ← dn2
 leaf 1 ← dn1

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

Edges (propagator assignments):

2 → 4: nt2 → dv1 [G_hat[1,2]]
 3 → 2: nt2 → dv2 [G_hat[1,3]]
 3 → 4: nt2 → dv1 [G_hat[1,2]]
 4 → 0: nt1 → dn2 [G_hat[0,1]]
 5 → 1: nt2 → dn1 [G_hat[1,0]]
 5 → 2: nt2 → dv2 [G_hat[1,3]]

```

=====
Diagram 1L-621
=====
External legs:
  leaf 0 ← dn2
  leaf 1 ← dn1
Vertex assignments:
  v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

coeff =  $\frac{1}{2} \varphi_2''$ 

  v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

coeff =  $\frac{1}{2} nstar_2$ 

  v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

coeff =  $\frac{1}{2} \varphi_1''$ 

  v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

coeff =  $\frac{1}{2} nstar_2$ 

Edges (propagator assignments):
  2 → 4:  nt2 → dv1  [G_hat[1,2]]
  3 → 2:  nt2 → dv2  [G_hat[1,3]]
  3 → 4:  nt2 → dv1  [G_hat[1,2]]
  4 → 0:  nt1 → dn2  [G_hat[0,1]]
  5 → 1:  nt2 → dn1  [G_hat[1,0]]
  5 → 2:  nt2 → dv2  [G_hat[1,3]]

```

```

=====
Diagram 1L-622
=====
External legs:
  leaf 0 ← dn2
  leaf 1 ← dn1
Vertex assignments:
  v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

coeff =  $\frac{1}{2} \varphi_2''$ 

  v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

coeff =  $\frac{1}{2} nstar_2$ 

```

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

Edges (propagator assignments):

2 → 4: nt2 → dv1 [G_hat[1,2]]
 3 → 2: nt2 → dv2 [G_hat[1,3]]
 3 → 4: nt2 → dv1 [G_hat[1,2]]
 4 → 0: nt1 → dn2 [G_hat[0,1]]
 5 → 1: nt2 → dn1 [G_hat[1,0]]
 5 → 2: nt2 → dv2 [G_hat[1,3]]

=====
 Diagram 1L-623
 =====

External legs:

leaf 0 ← dn2
 leaf 1 ← dn1

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

Edges (propagator assignments):

2 → 4: nt2 → dv1 [G_hat[1,2]]
 3 → 2: nt2 → dv2 [G_hat[1,3]]
 3 → 4: nt2 → dv1 [G_hat[1,2]]
 4 → 0: nt1 → dn2 [G_hat[0,1]]
 5 → 1: nt2 → dn1 [G_hat[1,0]]
 5 → 2: nt2 → dv2 [G_hat[1,3]]

```

=====
Diagram 1L-624
=====
External legs:
  leaf 0 ← dn2
  leaf 1 ← dn1
Vertex assignments:
  v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

coeff =  $\frac{1}{2} \varphi_2''$ 

  v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

coeff =  $\frac{1}{2} nstar_2$ 

  v4 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

coeff =  $\frac{1}{2} \varphi_1''$ 

  v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

coeff =  $\frac{1}{2} nstar_2$ 

Edges (propagator assignments):
  2 → 4:  nt2 → dv1  [G_hat[1,2]]
  3 → 2:  nt2 → dv2  [G_hat[1,3]]
  3 → 4:  nt2 → dv1  [G_hat[1,2]]
  4 → 0:  nt1 → dn2  [G_hat[0,1]]
  5 → 1:  nt2 → dn1  [G_hat[1,0]]
  5 → 2:  nt2 → dv2  [G_hat[1,3]]

```

```

=====
Diagram 1L-625
=====
External legs:
  leaf 0 ← dn2
  leaf 1 ← dn1
Vertex assignments:
  v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

coeff =  $\frac{1}{2} \varphi_2''$ 

  v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

coeff =  $\frac{1}{2} nstar_2$ 

```

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

Edges (propagator assignments):

2 → 4: nt2 → dv2 [G_hat[1,3]]
 3 → 2: nt2 → dv2 [G_hat[1,3]]
 3 → 4: nt2 → dv2 [G_hat[1,3]]
 4 → 0: nt2 → dn2 [G_hat[1,1]]
 5 → 1: nt2 → dn1 [G_hat[1,0]]
 5 → 2: nt2 → dv2 [G_hat[1,3]]

=====
 Diagram 1L-626
 =====

External legs:

leaf 0 ← dn2
 leaf 1 ← dn1

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

Edges (propagator assignments):

2 → 4: nt2 → dv2 [G_hat[1,3]]
 3 → 2: nt2 → dv2 [G_hat[1,3]]
 3 → 4: nt2 → dv2 [G_hat[1,3]]
 4 → 0: nt2 → dn2 [G_hat[1,1]]
 5 → 1: nt2 → dn1 [G_hat[1,0]]
 5 → 2: nt2 → dv2 [G_hat[1,3]]

```

=====
Diagram 1L-627
=====

External legs:
  leaf 0 ← dn2
  leaf 1 ← dn1
Vertex assignments:
  v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

coeff =  $\frac{1}{2} \varphi_2''$ 

  v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

coeff =  $\frac{1}{2} nstar_2$ 

  v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

coeff =  $\frac{1}{2} \varphi_2''$ 

  v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

coeff =  $\frac{1}{2} nstar_2$ 

Edges (propagator assignments):
  2 → 4:  nt2 → dv2  [G_hat[1,3]]
  3 → 2:  nt2 → dv2  [G_hat[1,3]]
  3 → 4:  nt2 → dv2  [G_hat[1,3]]
  4 → 0:  nt2 → dn2  [G_hat[1,1]]
  5 → 1:  nt2 → dn1  [G_hat[1,0]]
  5 → 2:  nt2 → dv2  [G_hat[1,3]]

```

```

=====
Diagram 1L-628
=====

External legs:
  leaf 0 ← dn2
  leaf 1 ← dn1
Vertex assignments:
  v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

coeff =  $\frac{1}{2} \varphi_2''$ 

  v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

coeff =  $\frac{1}{2} nstar_2$ 

```

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

Edges (propagator assignments):

2 → 4: nt2 → dv2 [G_hat[1,3]]
 3 → 2: nt2 → dv2 [G_hat[1,3]]
 3 → 4: nt2 → dv2 [G_hat[1,3]]
 4 → 0: nt2 → dn2 [G_hat[1,1]]
 5 → 1: nt2 → dn1 [G_hat[1,0]]
 5 → 2: nt2 → dv2 [G_hat[1,3]]

Diagram 1L-629

External legs:

leaf 0 ← dn2
 leaf 1 ← dn1

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

Edges (propagator assignments):

2 → 4: nt2 → dv2 [G_hat[1,3]]
 3 → 2: nt2 → dv2 [G_hat[1,3]]
 3 → 4: nt2 → dv2 [G_hat[1,3]]
 4 → 0: nt2 → dn2 [G_hat[1,1]]
 5 → 1: nt2 → dn1 [G_hat[1,0]]
 5 → 2: nt2 → dv2 [G_hat[1,3]]


```

=====
Diagram 1L-630
=====
External legs:
  leaf 0 ← dn2
  leaf 1 ← dn1
Vertex assignments:
  v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

coeff =  $\frac{1}{2} \varphi_2''$ 

  v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

coeff =  $\frac{1}{2} nstar_2$ 

  v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

coeff =  $\frac{1}{2} \varphi_2''$ 

  v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

coeff =  $\frac{1}{2} nstar_2$ 

Edges (propagator assignments):
  2 → 4:  nt2 → dv2  [G_hat[1,3]]
  3 → 2:  nt2 → dv2  [G_hat[1,3]]
  3 → 4:  nt2 → dv2  [G_hat[1,3]]
  4 → 0:  nt2 → dn2  [G_hat[1,1]]
  5 → 1:  nt2 → dn1  [G_hat[1,0]]
  5 → 2:  nt2 → dv2  [G_hat[1,3]]

```

```

=====
Diagram 1L-631
=====
External legs:
  leaf 0 ← dn2
  leaf 1 ← dn1
Vertex assignments:
  v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

coeff =  $\frac{1}{2} \varphi_2''$ 

  v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

coeff =  $\frac{1}{2} nstar_2$ 

```

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

Edges (propagator assignments):

2 → 4: nt2 → dv2 [G_hat[1,3]]
 3 → 2: nt2 → dv2 [G_hat[1,3]]
 3 → 4: nt2 → dv2 [G_hat[1,3]]
 4 → 0: nt2 → dn2 [G_hat[1,1]]
 5 → 1: nt2 → dn1 [G_hat[1,0]]
 5 → 2: nt2 → dv2 [G_hat[1,3]]

=====
 Diagram 1L-632
 =====

External legs:

leaf 0 ← dn2
 leaf 1 ← dn1

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

Edges (propagator assignments):

2 → 4: nt2 → dv2 [G_hat[1,3]]
 3 → 2: nt2 → dv2 [G_hat[1,3]]
 3 → 4: nt2 → dv2 [G_hat[1,3]]
 4 → 0: nt2 → dn2 [G_hat[1,1]]
 5 → 1: nt2 → dn1 [G_hat[1,0]]
 5 → 2: nt2 → dv2 [G_hat[1,3]]

```

=====
Diagram 1L-633
=====

External legs:
  leaf 0 ← dn2
  leaf 1 ← dn1
Vertex assignments:
  v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

coeff =  $\frac{1}{2} \varphi_2''$ 

  v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

coeff =  $\frac{1}{2} nstar_2$ 

  v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

coeff =  $\frac{1}{2} \varphi_2''$ 

  v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

coeff =  $\frac{1}{2} nstar_2$ 

Edges (propagator assignments):
  2 → 4:  nt2 → dv2  [G_hat[1,3]]
  3 → 2:  nt2 → dv2  [G_hat[1,3]]
  3 → 4:  nt2 → dv2  [G_hat[1,3]]
  4 → 0:  nt2 → dn2  [G_hat[1,1]]
  5 → 1:  nt2 → dn1  [G_hat[1,0]]
  5 → 2:  nt2 → dv2  [G_hat[1,3]]

```

```

=====
Diagram 1L-634
=====

External legs:
  leaf 0 ← dn2
  leaf 1 ← dn1
Vertex assignments:
  v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

coeff =  $\frac{1}{2} \varphi_2''$ 

  v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

coeff =  $\frac{1}{2} nstar_2$ 

```

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

Edges (propagator assignments):

2 → 4: nt2 → dv2 [G_hat[1,3]]
 3 → 2: nt2 → dv2 [G_hat[1,3]]
 3 → 4: nt2 → dv2 [G_hat[1,3]]
 4 → 0: nt2 → dn2 [G_hat[1,1]]
 5 → 1: nt2 → dn1 [G_hat[1,0]]
 5 → 2: nt2 → dv2 [G_hat[1,3]]

=====
 Diagram 1L-635
 =====

External legs:

leaf 0 ← dn2
 leaf 1 ← dn1

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

Edges (propagator assignments):

2 → 4: nt2 → dv2 [G_hat[1,3]]
 3 → 2: nt2 → dv2 [G_hat[1,3]]
 3 → 4: nt2 → dv2 [G_hat[1,3]]
 4 → 0: nt2 → dn2 [G_hat[1,1]]
 5 → 1: nt2 → dn1 [G_hat[1,0]]
 5 → 2: nt2 → dv2 [G_hat[1,3]]

```

=====
Diagram 1L-636
=====
External legs:
  leaf 0 ← dn2
  leaf 1 ← dn1
Vertex assignments:
  v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

coeff =  $\frac{1}{2} \varphi_2''$ 

  v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

coeff =  $\frac{1}{2} nstar_2$ 

  v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

coeff =  $\frac{1}{2} \varphi_2''$ 

  v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

coeff =  $\frac{1}{2} nstar_2$ 

Edges (propagator assignments):
  2 → 4:  nt2 → dv2  [G_hat[1,3]]
  3 → 2:  nt2 → dv2  [G_hat[1,3]]
  3 → 4:  nt2 → dv2  [G_hat[1,3]]
  4 → 0:  nt2 → dn2  [G_hat[1,1]]
  5 → 1:  nt2 → dn1  [G_hat[1,0]]
  5 → 2:  nt2 → dv2  [G_hat[1,3]]

```

```

=====
Diagram 1L-637
=====
External legs:
  leaf 0 ← dn2
  leaf 1 ← dn1
Vertex assignments:
  v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

coeff =  $\frac{1}{2} \varphi_2''$ 

  v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

coeff =  $\frac{1}{2} nstar_2$ 

```

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

Edges (propagator assignments):

2 → 4: nt2 → dv2 [G_hat[1,3]]
 3 → 2: nt2 → dv2 [G_hat[1,3]]
 3 → 4: nt2 → dv2 [G_hat[1,3]]
 4 → 0: nt2 → dn2 [G_hat[1,1]]
 5 → 1: nt2 → dn1 [G_hat[1,0]]
 5 → 2: nt2 → dv2 [G_hat[1,3]]

=====
 Diagram 1L-638
 =====

External legs:

leaf 0 ← dn2
 leaf 1 ← dn1

Vertex assignments:

v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

Edges (propagator assignments):

2 → 4: nt2 → dv2 [G_hat[1,3]]
 3 → 2: nt2 → dv2 [G_hat[1,3]]
 3 → 4: nt2 → dv2 [G_hat[1,3]]
 4 → 0: nt2 → dn2 [G_hat[1,1]]
 5 → 1: nt2 → dn1 [G_hat[1,0]]
 5 → 2: nt2 → dv2 [G_hat[1,3]]

```

=====
Diagram 1L-639
=====

External legs:
  leaf 0 ← dn2
  leaf 1 ← dn1
Vertex assignments:
  v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

coeff =  $\frac{1}{2} \varphi_2''$ 

  v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

coeff =  $\frac{1}{2} nstar_2$ 

  v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

coeff =  $\frac{1}{2} \varphi_2''$ 

  v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

coeff =  $\frac{1}{2} nstar_2$ 

Edges (propagator assignments):
  2 → 4:  nt2 → dv2  [G_hat[1,3]]
  3 → 2:  nt2 → dv2  [G_hat[1,3]]
  3 → 4:  nt2 → dv2  [G_hat[1,3]]
  4 → 0:  nt2 → dn2  [G_hat[1,1]]
  5 → 1:  nt2 → dn1  [G_hat[1,0]]
  5 → 2:  nt2 → dv2  [G_hat[1,3]]

```

```

=====
Diagram 1L-640
=====

External legs:
  leaf 0 ← dn2
  leaf 1 ← dn1
Vertex assignments:
  v2 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

coeff =  $\frac{1}{2} \varphi_2''$ 

  v3 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

coeff =  $\frac{1}{2} nstar_2$ 

```

v4 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v5 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

Edges (propagator assignments):

2 → 4: nt2 → dv2 [G_hat[1,3]]
 3 → 2: nt2 → dv2 [G_hat[1,3]]
 3 → 4: nt2 → dv2 [G_hat[1,3]]
 4 → 0: nt2 → dn2 [G_hat[1,1]]
 5 → 1: nt2 → dn1 [G_hat[1,0]]
 5 → 2: nt2 → dv2 [G_hat[1,3]]

=====
 Diagram 1L-641
 =====

External legs:

leaf 0 ← dn1
 leaf 1 ← dn2

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

Edges (propagator assignments):

2 → 3: nt1 → dv1 [G_hat[0,2]]
 2 → 5: nt1 → dv1 [G_hat[0,2]]
 3 → 0: nt1 → dn1 [G_hat[0,0]]
 4 → 3: nt1 → dv1 [G_hat[0,2]]
 4 → 5: nt1 → dv1 [G_hat[0,2]]
 5 → 1: nt1 → dn2 [G_hat[0,1]]

=====

Diagram 1L-642

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

Edges (propagator assignments):

2 → 3: nt1 → dv1 [G_hat[0,2]]

2 → 5: nt1 → dv1 [G_hat[0,2]]

3 → 0: nt1 → dn1 [G_hat[0,0]]

4 → 3: nt1 → dv1 [G_hat[0,2]]

4 → 5: nt1 → dv1 [G_hat[0,2]]

5 → 1: nt1 → dn2 [G_hat[0,1]]

=====

Diagram 1L-643

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

```

    v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

    coeff =  $\frac{1}{2} nstar_1$ 

    v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

    coeff =  $\frac{1}{2} \varphi_1''$ 

    Edges (propagator assignments):
      2 → 3:  nt1 → dv1  [G_hat[0,2]]
      2 → 5:  nt1 → dv1  [G_hat[0,2]]
      3 → 0:  nt1 → dn1  [G_hat[0,0]]
      4 → 3:  nt1 → dv1  [G_hat[0,2]]
      4 → 5:  nt1 → dv1  [G_hat[0,2]]
      5 → 1:  nt1 → dn2  [G_hat[0,1]]

=====
Diagram 1L-644
=====

    External legs:
      leaf 0 ← dn1
      leaf 1 ← dn2
    Vertex assignments:
      v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

      coeff =  $\frac{1}{2} nstar_1$ 

      v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

      coeff =  $\frac{1}{2} \varphi_1''$ 

      v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

      coeff =  $\frac{1}{2} nstar_1$ 

      v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

      coeff =  $\frac{1}{2} \varphi_1''$ 

    Edges (propagator assignments):
      2 → 3:  nt1 → dv1  [G_hat[0,2]]
      2 → 5:  nt1 → dv1  [G_hat[0,2]]
      3 → 0:  nt1 → dn1  [G_hat[0,0]]
      4 → 3:  nt1 → dv1  [G_hat[0,2]]
      4 → 5:  nt1 → dv1  [G_hat[0,2]]
      5 → 1:  nt1 → dn2  [G_hat[0,1]]

```

=====

Diagram 1L-645

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

Edges (propagator assignments):

2 → 3: nt1 → dv1 [G_hat[0,2]]

2 → 5: nt1 → dv1 [G_hat[0,2]]

3 → 0: nt1 → dn1 [G_hat[0,0]]

4 → 3: nt1 → dv1 [G_hat[0,2]]

4 → 5: nt1 → dv1 [G_hat[0,2]]

5 → 1: nt1 → dn2 [G_hat[0,1]]

=====

Diagram 1L-646

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

```

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

coeff =  $\frac{1}{2} nstar_1$ 

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

coeff =  $\frac{1}{2} \varphi_1''$ 

Edges (propagator assignments):
  2 → 3:  nt1 → dv1  [G_hat[0,2]]
  2 → 5:  nt1 → dv1  [G_hat[0,2]]
  3 → 0:  nt1 → dn1  [G_hat[0,0]]
  4 → 3:  nt1 → dv1  [G_hat[0,2]]
  4 → 5:  nt1 → dv1  [G_hat[0,2]]
  5 → 1:  nt1 → dn2  [G_hat[0,1]]

=====
Diagram 1L-647
=====

External legs:
  leaf 0 ← dn1
  leaf 1 ← dn2
Vertex assignments:
  v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

coeff =  $\frac{1}{2} nstar_1$ 

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

coeff =  $\frac{1}{2} \varphi_1''$ 

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

coeff =  $\frac{1}{2} nstar_1$ 

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

coeff =  $\frac{1}{2} \varphi_1''$ 

Edges (propagator assignments):
  2 → 3:  nt1 → dv1  [G_hat[0,2]]
  2 → 5:  nt1 → dv1  [G_hat[0,2]]
  3 → 0:  nt1 → dn1  [G_hat[0,0]]
  4 → 3:  nt1 → dv1  [G_hat[0,2]]
  4 → 5:  nt1 → dv1  [G_hat[0,2]]
  5 → 1:  nt1 → dn2  [G_hat[0,1]]

```

=====

Diagram 1L-648

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

Edges (propagator assignments):

2 → 3: nt1 → dv1 [G_hat[0,2]]

2 → 5: nt1 → dv1 [G_hat[0,2]]

3 → 0: nt1 → dn1 [G_hat[0,0]]

4 → 3: nt1 → dv1 [G_hat[0,2]]

4 → 5: nt1 → dv1 [G_hat[0,2]]

5 → 1: nt1 → dn2 [G_hat[0,1]]

=====

Diagram 1L-649

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

```

    v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

    coeff =  $\frac{1}{2} nstar_1$ 

    v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

    coeff =  $\frac{1}{2} \varphi_1''$ 

    Edges (propagator assignments):
      2 → 3:  nt1 → dv1  [G_hat[0,2]]
      2 → 5:  nt1 → dv1  [G_hat[0,2]]
      3 → 0:  nt1 → dn1  [G_hat[0,0]]
      4 → 3:  nt1 → dv1  [G_hat[0,2]]
      4 → 5:  nt1 → dv1  [G_hat[0,2]]
      5 → 1:  nt1 → dn2  [G_hat[0,1]]

=====
Diagram 1L-650
=====

    External legs:
      leaf 0 ← dn1
      leaf 1 ← dn2
    Vertex assignments:
      v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

      coeff =  $\frac{1}{2} nstar_1$ 

      v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

      coeff =  $\frac{1}{2} \varphi_1''$ 

      v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

      coeff =  $\frac{1}{2} nstar_1$ 

      v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

      coeff =  $\frac{1}{2} \varphi_1''$ 

    Edges (propagator assignments):
      2 → 3:  nt1 → dv1  [G_hat[0,2]]
      2 → 5:  nt1 → dv1  [G_hat[0,2]]
      3 → 0:  nt1 → dn1  [G_hat[0,0]]
      4 → 3:  nt1 → dv1  [G_hat[0,2]]
      4 → 5:  nt1 → dv1  [G_hat[0,2]]
      5 → 1:  nt1 → dn2  [G_hat[0,1]]

```

=====

Diagram 1L-651

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

Edges (propagator assignments):

2 → 3: nt1 → dv1 [G_hat[0,2]]

2 → 5: nt1 → dv1 [G_hat[0,2]]

3 → 0: nt1 → dn1 [G_hat[0,0]]

4 → 3: nt1 → dv1 [G_hat[0,2]]

4 → 5: nt1 → dv1 [G_hat[0,2]]

5 → 1: nt1 → dn2 [G_hat[0,1]]

=====

Diagram 1L-652

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

```

    v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

    coeff =  $\frac{1}{2} nstar_1$ 

    v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

    coeff =  $\frac{1}{2} \varphi_1''$ 

    Edges (propagator assignments):
      2 → 3:  nt1 → dv1  [G_hat[0,2]]
      2 → 5:  nt1 → dv1  [G_hat[0,2]]
      3 → 0:  nt1 → dn1  [G_hat[0,0]]
      4 → 3:  nt1 → dv1  [G_hat[0,2]]
      4 → 5:  nt1 → dv1  [G_hat[0,2]]
      5 → 1:  nt1 → dn2  [G_hat[0,1]]

=====
Diagram 1L-653
=====

    External legs:
      leaf 0 ← dn1
      leaf 1 ← dn2
    Vertex assignments:
      v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

      coeff =  $\frac{1}{2} nstar_1$ 

      v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

      coeff =  $\frac{1}{2} \varphi_1''$ 

      v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

      coeff =  $\frac{1}{2} nstar_1$ 

      v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

      coeff =  $\frac{1}{2} \varphi_1''$ 

    Edges (propagator assignments):
      2 → 3:  nt1 → dv1  [G_hat[0,2]]
      2 → 5:  nt1 → dv1  [G_hat[0,2]]
      3 → 0:  nt1 → dn1  [G_hat[0,0]]
      4 → 3:  nt1 → dv1  [G_hat[0,2]]
      4 → 5:  nt1 → dv1  [G_hat[0,2]]
      5 → 1:  nt1 → dn2  [G_hat[0,1]]

```


=====

Diagram 1L-654

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

Edges (propagator assignments):

2 → 3: nt1 → dv1 [G_hat[0,2]]

2 → 5: nt1 → dv1 [G_hat[0,2]]

3 → 0: nt1 → dn1 [G_hat[0,0]]

4 → 3: nt1 → dv1 [G_hat[0,2]]

4 → 5: nt1 → dv1 [G_hat[0,2]]

5 → 1: nt1 → dn2 [G_hat[0,1]]

=====

Diagram 1L-655

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

```

    v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

    coeff =  $\frac{1}{2} nstar_1$ 

    v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

    coeff =  $\frac{1}{2} \varphi_1''$ 

    Edges (propagator assignments):
      2 → 3:  nt1 → dv1  [G_hat[0,2]]
      2 → 5:  nt1 → dv1  [G_hat[0,2]]
      3 → 0:  nt1 → dn1  [G_hat[0,0]]
      4 → 3:  nt1 → dv1  [G_hat[0,2]]
      4 → 5:  nt1 → dv1  [G_hat[0,2]]
      5 → 1:  nt1 → dn2  [G_hat[0,1]]

=====
Diagram 1L-656
=====

    External legs:
      leaf 0 ← dn1
      leaf 1 ← dn2
    Vertex assignments:
      v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

      coeff =  $\frac{1}{2} nstar_1$ 

      v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

      coeff =  $\frac{1}{2} \varphi_1''$ 

      v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

      coeff =  $\frac{1}{2} nstar_1$ 

      v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

      coeff =  $\frac{1}{2} \varphi_1''$ 

    Edges (propagator assignments):
      2 → 3:  nt1 → dv1  [G_hat[0,2]]
      2 → 5:  nt1 → dv1  [G_hat[0,2]]
      3 → 0:  nt1 → dn1  [G_hat[0,0]]
      4 → 3:  nt1 → dv1  [G_hat[0,2]]
      4 → 5:  nt1 → dv1  [G_hat[0,2]]
      5 → 1:  nt1 → dn2  [G_hat[0,1]]

```

=====

Diagram 1L-657

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

Edges (propagator assignments):

2 → 3: nt1 → dv1 [G_hat[0,2]]

2 → 5: nt1 → dv2 [G_hat[0,3]]

3 → 0: nt1 → dn1 [G_hat[0,0]]

4 → 3: nt1 → dv1 [G_hat[0,2]]

4 → 5: nt1 → dv2 [G_hat[0,3]]

5 → 1: nt2 → dn2 [G_hat[1,1]]

=====

Diagram 1L-658

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

```

    v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

    coeff =  $\frac{1}{2} nstar_1$ 

    v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

    coeff =  $\frac{1}{2} \varphi_2''$ 

    Edges (propagator assignments):
      2 → 3:  nt1 → dv1  [G_hat[0,2]]
      2 → 5:  nt1 → dv2  [G_hat[0,3]]
      3 → 0:  nt1 → dn1  [G_hat[0,0]]
      4 → 3:  nt1 → dv1  [G_hat[0,2]]
      4 → 5:  nt1 → dv2  [G_hat[0,3]]
      5 → 1:  nt2 → dn2  [G_hat[1,1]]

=====
Diagram 1L-659
=====

    External legs:
      leaf 0 ← dn1
      leaf 1 ← dn2
    Vertex assignments:
      v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

      coeff =  $\frac{1}{2} nstar_1$ 

      v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

      coeff =  $\frac{1}{2} \varphi_1''$ 

      v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

      coeff =  $\frac{1}{2} nstar_1$ 

      v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

      coeff =  $\frac{1}{2} \varphi_2''$ 

    Edges (propagator assignments):
      2 → 3:  nt1 → dv1  [G_hat[0,2]]
      2 → 5:  nt1 → dv2  [G_hat[0,3]]
      3 → 0:  nt1 → dn1  [G_hat[0,0]]
      4 → 3:  nt1 → dv1  [G_hat[0,2]]
      4 → 5:  nt1 → dv2  [G_hat[0,3]]
      5 → 1:  nt2 → dn2  [G_hat[1,1]]

```

=====

Diagram 1L-660

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

Edges (propagator assignments):

2 → 3: nt1 → dv1 [G_hat[0,2]]

2 → 5: nt1 → dv2 [G_hat[0,3]]

3 → 0: nt1 → dn1 [G_hat[0,0]]

4 → 3: nt1 → dv1 [G_hat[0,2]]

4 → 5: nt1 → dv2 [G_hat[0,3]]

5 → 1: nt2 → dn2 [G_hat[1,1]]

=====

Diagram 1L-661

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

```

    v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

coeff =  $\frac{1}{2} nstar_1$ 

    v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

coeff =  $\frac{1}{2} \varphi_2''$ 

Edges (propagator assignments):
    2 → 3:  nt1 → dv1  [G_hat[0,2]]
    2 → 5:  nt1 → dv2  [G_hat[0,3]]
    3 → 0:  nt1 → dn1  [G_hat[0,0]]
    4 → 3:  nt1 → dv1  [G_hat[0,2]]
    4 → 5:  nt1 → dv2  [G_hat[0,3]]
    5 → 1:  nt2 → dn2  [G_hat[1,1]]

=====
Diagram 1L-662
=====

External legs:
    leaf 0 ← dn1
    leaf 1 ← dn2
Vertex assignments:
    v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

coeff =  $\frac{1}{2} nstar_1$ 

    v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

coeff =  $\frac{1}{2} \varphi_1''$ 

    v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

coeff =  $\frac{1}{2} nstar_1$ 

    v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

coeff =  $\frac{1}{2} \varphi_2''$ 

Edges (propagator assignments):
    2 → 3:  nt1 → dv1  [G_hat[0,2]]
    2 → 5:  nt1 → dv2  [G_hat[0,3]]
    3 → 0:  nt1 → dn1  [G_hat[0,0]]
    4 → 3:  nt1 → dv1  [G_hat[0,2]]
    4 → 5:  nt1 → dv2  [G_hat[0,3]]
    5 → 1:  nt2 → dn2  [G_hat[1,1]]

```

=====

Diagram 1L-663

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

Edges (propagator assignments):

2 → 3: nt1 → dv1 [G_hat[0,2]]

2 → 5: nt1 → dv2 [G_hat[0,3]]

3 → 0: nt1 → dn1 [G_hat[0,0]]

4 → 3: nt1 → dv1 [G_hat[0,2]]

4 → 5: nt1 → dv2 [G_hat[0,3]]

5 → 1: nt2 → dn2 [G_hat[1,1]]

=====

Diagram 1L-664

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

```

    v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

coeff =  $\frac{1}{2} nstar_1$ 

    v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

coeff =  $\frac{1}{2} \varphi_2''$ 

Edges (propagator assignments):
    2 → 3:  nt1 → dv1  [G_hat[0,2]]
    2 → 5:  nt1 → dv2  [G_hat[0,3]]
    3 → 0:  nt1 → dn1  [G_hat[0,0]]
    4 → 3:  nt1 → dv1  [G_hat[0,2]]
    4 → 5:  nt1 → dv2  [G_hat[0,3]]
    5 → 1:  nt2 → dn2  [G_hat[1,1]]

=====
Diagram 1L-665
=====

External legs:
    leaf 0 ← dn1
    leaf 1 ← dn2
Vertex assignments:
    v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

coeff =  $\frac{1}{2} nstar_1$ 

    v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

coeff =  $\frac{1}{2} \varphi_1''$ 

    v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

coeff =  $\frac{1}{2} nstar_1$ 

    v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

coeff =  $\frac{1}{2} \varphi_2''$ 

Edges (propagator assignments):
    2 → 3:  nt1 → dv1  [G_hat[0,2]]
    2 → 5:  nt1 → dv2  [G_hat[0,3]]
    3 → 0:  nt1 → dn1  [G_hat[0,0]]
    4 → 3:  nt1 → dv1  [G_hat[0,2]]
    4 → 5:  nt1 → dv2  [G_hat[0,3]]
    5 → 1:  nt2 → dn2  [G_hat[1,1]]

```


=====

Diagram 1L-666

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

Edges (propagator assignments):

2 → 3: nt1 → dv1 [G_hat[0,2]]

2 → 5: nt1 → dv2 [G_hat[0,3]]

3 → 0: nt1 → dn1 [G_hat[0,0]]

4 → 3: nt1 → dv1 [G_hat[0,2]]

4 → 5: nt1 → dv2 [G_hat[0,3]]

5 → 1: nt2 → dn2 [G_hat[1,1]]

=====

Diagram 1L-667

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

```

    v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

    coeff =  $\frac{1}{2} nstar_1$ 

    v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

    coeff =  $\frac{1}{2} \varphi_2''$ 

    Edges (propagator assignments):
    2 → 3:  nt1 → dv1  [G_hat[0,2]]
    2 → 5:  nt1 → dv2  [G_hat[0,3]]
    3 → 0:  nt1 → dn1  [G_hat[0,0]]
    4 → 3:  nt1 → dv1  [G_hat[0,2]]
    4 → 5:  nt1 → dv2  [G_hat[0,3]]
    5 → 1:  nt2 → dn2  [G_hat[1,1]]

=====
Diagram 1L-668
=====

    External legs:
    leaf 0 ← dn1
    leaf 1 ← dn2
    Vertex assignments:
    v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

    coeff =  $\frac{1}{2} nstar_1$ 

    v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

    coeff =  $\frac{1}{2} \varphi_1''$ 

    v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

    coeff =  $\frac{1}{2} nstar_1$ 

    v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

    coeff =  $\frac{1}{2} \varphi_2''$ 

    Edges (propagator assignments):
    2 → 3:  nt1 → dv1  [G_hat[0,2]]
    2 → 5:  nt1 → dv2  [G_hat[0,3]]
    3 → 0:  nt1 → dn1  [G_hat[0,0]]
    4 → 3:  nt1 → dv1  [G_hat[0,2]]
    4 → 5:  nt1 → dv2  [G_hat[0,3]]
    5 → 1:  nt2 → dn2  [G_hat[1,1]]

```

=====

Diagram 1L-669

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

Edges (propagator assignments):

2 → 3: nt1 → dv1 [G_hat[0,2]]

2 → 5: nt1 → dv2 [G_hat[0,3]]

3 → 0: nt1 → dn1 [G_hat[0,0]]

4 → 3: nt1 → dv1 [G_hat[0,2]]

4 → 5: nt1 → dv2 [G_hat[0,3]]

5 → 1: nt2 → dn2 [G_hat[1,1]]

=====

Diagram 1L-670

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

```

    v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

coeff =  $\frac{1}{2} nstar_1$ 

    v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

coeff =  $\frac{1}{2} \varphi_2''$ 

Edges (propagator assignments):
    2 → 3:  nt1 → dv1  [G_hat[0,2]]
    2 → 5:  nt1 → dv2  [G_hat[0,3]]
    3 → 0:  nt1 → dn1  [G_hat[0,0]]
    4 → 3:  nt1 → dv1  [G_hat[0,2]]
    4 → 5:  nt1 → dv2  [G_hat[0,3]]
    5 → 1:  nt2 → dn2  [G_hat[1,1]]

=====
Diagram 1L-671
=====

External legs:
    leaf 0 ← dn1
    leaf 1 ← dn2
Vertex assignments:
    v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

coeff =  $\frac{1}{2} nstar_1$ 

    v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

coeff =  $\frac{1}{2} \varphi_1''$ 

    v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

coeff =  $\frac{1}{2} nstar_1$ 

    v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

coeff =  $\frac{1}{2} \varphi_2''$ 

Edges (propagator assignments):
    2 → 3:  nt1 → dv1  [G_hat[0,2]]
    2 → 5:  nt1 → dv2  [G_hat[0,3]]
    3 → 0:  nt1 → dn1  [G_hat[0,0]]
    4 → 3:  nt1 → dv1  [G_hat[0,2]]
    4 → 5:  nt1 → dv2  [G_hat[0,3]]
    5 → 1:  nt2 → dn2  [G_hat[1,1]]

```

=====

Diagram 1L-672

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

Edges (propagator assignments):

2 → 3: nt1 → dv1 [G_hat[0,2]]

2 → 5: nt1 → dv2 [G_hat[0,3]]

3 → 0: nt1 → dn1 [G_hat[0,0]]

4 → 3: nt1 → dv1 [G_hat[0,2]]

4 → 5: nt1 → dv2 [G_hat[0,3]]

5 → 1: nt2 → dn2 [G_hat[1,1]]

=====

Diagram 1L-673

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

```

    v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

    coeff =  $\frac{1}{2} nstar_1$ 

    v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

    coeff =  $\frac{1}{2} \varphi_1''$ 

    Edges (propagator assignments):
      2 → 3:  nt1 → dv2  [G_hat[0,3]]
      2 → 5:  nt1 → dv1  [G_hat[0,2]]
      3 → 0:  nt2 → dn1  [G_hat[1,0]]
      4 → 3:  nt1 → dv2  [G_hat[0,3]]
      4 → 5:  nt1 → dv1  [G_hat[0,2]]
      5 → 1:  nt1 → dn2  [G_hat[0,1]]

=====
Diagram 1L-674
=====

    External legs:
      leaf 0 ← dn1
      leaf 1 ← dn2
    Vertex assignments:
      v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

      coeff =  $\frac{1}{2} nstar_1$ 

      v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

      coeff =  $\frac{1}{2} \varphi_2''$ 

      v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

      coeff =  $\frac{1}{2} nstar_1$ 

      v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

      coeff =  $\frac{1}{2} \varphi_1''$ 

    Edges (propagator assignments):
      2 → 3:  nt1 → dv2  [G_hat[0,3]]
      2 → 5:  nt1 → dv1  [G_hat[0,2]]
      3 → 0:  nt2 → dn1  [G_hat[1,0]]
      4 → 3:  nt1 → dv2  [G_hat[0,3]]
      4 → 5:  nt1 → dv1  [G_hat[0,2]]
      5 → 1:  nt1 → dn2  [G_hat[0,1]]

```

=====

Diagram 1L-675

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

Edges (propagator assignments):

2 → 3: nt1 → dv2 [G_hat[0,3]]

2 → 5: nt1 → dv1 [G_hat[0,2]]

3 → 0: nt2 → dn1 [G_hat[1,0]]

4 → 3: nt1 → dv2 [G_hat[0,3]]

4 → 5: nt1 → dv1 [G_hat[0,2]]

5 → 1: nt1 → dn2 [G_hat[0,1]]

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Diagram 1L-676

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

```

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

coeff =  $\frac{1}{2} nstar_1$ 

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

coeff =  $\frac{1}{2} \varphi_1''$ 

Edges (propagator assignments):
  2 → 3:  nt1 → dv2  [G_hat[0,3]]
  2 → 5:  nt1 → dv1  [G_hat[0,2]]
  3 → 0:  nt2 → dn1  [G_hat[1,0]]
  4 → 3:  nt1 → dv2  [G_hat[0,3]]
  4 → 5:  nt1 → dv1  [G_hat[0,2]]
  5 → 1:  nt1 → dn2  [G_hat[0,1]]

=====
Diagram 1L-677
=====

External legs:
  leaf 0 ← dn1
  leaf 1 ← dn2
Vertex assignments:
  v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

coeff =  $\frac{1}{2} nstar_1$ 

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

coeff =  $\frac{1}{2} \varphi_2''$ 

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

coeff =  $\frac{1}{2} nstar_1$ 

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

coeff =  $\frac{1}{2} \varphi_1''$ 

Edges (propagator assignments):
  2 → 3:  nt1 → dv2  [G_hat[0,3]]
  2 → 5:  nt1 → dv1  [G_hat[0,2]]
  3 → 0:  nt2 → dn1  [G_hat[1,0]]
  4 → 3:  nt1 → dv2  [G_hat[0,3]]
  4 → 5:  nt1 → dv1  [G_hat[0,2]]
  5 → 1:  nt1 → dn2  [G_hat[0,1]]

```


=====

Diagram 1L-678

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

Edges (propagator assignments):

2 → 3: nt1 → dv2 [G_hat[0,3]]

2 → 5: nt1 → dv1 [G_hat[0,2]]

3 → 0: nt2 → dn1 [G_hat[1,0]]

4 → 3: nt1 → dv2 [G_hat[0,3]]

4 → 5: nt1 → dv1 [G_hat[0,2]]

5 → 1: nt1 → dn2 [G_hat[0,1]]

=====

Diagram 1L-679

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

```

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

coeff =  $\frac{1}{2} nstar_1$ 

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

coeff =  $\frac{1}{2} \varphi_1''$ 

Edges (propagator assignments):
  2 → 3:  nt1 → dv2  [G_hat[0,3]]
  2 → 5:  nt1 → dv1  [G_hat[0,2]]
  3 → 0:  nt2 → dn1  [G_hat[1,0]]
  4 → 3:  nt1 → dv2  [G_hat[0,3]]
  4 → 5:  nt1 → dv1  [G_hat[0,2]]
  5 → 1:  nt1 → dn2  [G_hat[0,1]]

=====
Diagram 1L-680
=====

External legs:
  leaf 0 ← dn1
  leaf 1 ← dn2
Vertex assignments:
  v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

coeff =  $\frac{1}{2} nstar_1$ 

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

coeff =  $\frac{1}{2} \varphi_2''$ 

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

coeff =  $\frac{1}{2} nstar_1$ 

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

coeff =  $\frac{1}{2} \varphi_1''$ 

Edges (propagator assignments):
  2 → 3:  nt1 → dv2  [G_hat[0,3]]
  2 → 5:  nt1 → dv1  [G_hat[0,2]]
  3 → 0:  nt2 → dn1  [G_hat[1,0]]
  4 → 3:  nt1 → dv2  [G_hat[0,3]]
  4 → 5:  nt1 → dv1  [G_hat[0,2]]
  5 → 1:  nt1 → dn2  [G_hat[0,1]]

```

=====

Diagram 1L-681

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

Edges (propagator assignments):

2 → 3: nt1 → dv2 [G_hat[0,3]]

2 → 5: nt1 → dv1 [G_hat[0,2]]

3 → 0: nt2 → dn1 [G_hat[1,0]]

4 → 3: nt1 → dv2 [G_hat[0,3]]

4 → 5: nt1 → dv1 [G_hat[0,2]]

5 → 1: nt1 → dn2 [G_hat[0,1]]

=====

Diagram 1L-682

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

```

    v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

    coeff =  $\frac{1}{2} nstar_1$ 

    v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

    coeff =  $\frac{1}{2} \varphi_1''$ 

    Edges (propagator assignments):
      2 → 3:  nt1 → dv2  [G_hat[0,3]]
      2 → 5:  nt1 → dv1  [G_hat[0,2]]
      3 → 0:  nt2 → dn1  [G_hat[1,0]]
      4 → 3:  nt1 → dv2  [G_hat[0,3]]
      4 → 5:  nt1 → dv1  [G_hat[0,2]]
      5 → 1:  nt1 → dn2  [G_hat[0,1]]

=====
Diagram 1L-683
=====

    External legs:
      leaf 0 ← dn1
      leaf 1 ← dn2
    Vertex assignments:
      v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

      coeff =  $\frac{1}{2} nstar_1$ 

      v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

      coeff =  $\frac{1}{2} \varphi_2''$ 

      v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

      coeff =  $\frac{1}{2} nstar_1$ 

      v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

      coeff =  $\frac{1}{2} \varphi_1''$ 

    Edges (propagator assignments):
      2 → 3:  nt1 → dv2  [G_hat[0,3]]
      2 → 5:  nt1 → dv1  [G_hat[0,2]]
      3 → 0:  nt2 → dn1  [G_hat[1,0]]
      4 → 3:  nt1 → dv2  [G_hat[0,3]]
      4 → 5:  nt1 → dv1  [G_hat[0,2]]
      5 → 1:  nt1 → dn2  [G_hat[0,1]]

```

=====

Diagram 1L-684

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

Edges (propagator assignments):

2 → 3: nt1 → dv2 [G_hat[0,3]]

2 → 5: nt1 → dv1 [G_hat[0,2]]

3 → 0: nt2 → dn1 [G_hat[1,0]]

4 → 3: nt1 → dv2 [G_hat[0,3]]

4 → 5: nt1 → dv1 [G_hat[0,2]]

5 → 1: nt1 → dn2 [G_hat[0,1]]

=====

Diagram 1L-685

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

```

    v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

    coeff =  $\frac{1}{2} nstar_1$ 

    v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

    coeff =  $\frac{1}{2} \varphi_1''$ 

    Edges (propagator assignments):
      2 → 3:  nt1 → dv2  [G_hat[0,3]]
      2 → 5:  nt1 → dv1  [G_hat[0,2]]
      3 → 0:  nt2 → dn1  [G_hat[1,0]]
      4 → 3:  nt1 → dv2  [G_hat[0,3]]
      4 → 5:  nt1 → dv1  [G_hat[0,2]]
      5 → 1:  nt1 → dn2  [G_hat[0,1]]

=====
Diagram 1L-686
=====

    External legs:
      leaf 0 ← dn1
      leaf 1 ← dn2
    Vertex assignments:
      v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

      coeff =  $\frac{1}{2} nstar_1$ 

      v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

      coeff =  $\frac{1}{2} \varphi_2''$ 

      v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

      coeff =  $\frac{1}{2} nstar_1$ 

      v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

      coeff =  $\frac{1}{2} \varphi_1''$ 

    Edges (propagator assignments):
      2 → 3:  nt1 → dv2  [G_hat[0,3]]
      2 → 5:  nt1 → dv1  [G_hat[0,2]]
      3 → 0:  nt2 → dn1  [G_hat[1,0]]
      4 → 3:  nt1 → dv2  [G_hat[0,3]]
      4 → 5:  nt1 → dv1  [G_hat[0,2]]
      5 → 1:  nt1 → dn2  [G_hat[0,1]]

```

=====

Diagram 1L-687

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

Edges (propagator assignments):

2 → 3: nt1 → dv2 [G_hat[0,3]]

2 → 5: nt1 → dv1 [G_hat[0,2]]

3 → 0: nt2 → dn1 [G_hat[1,0]]

4 → 3: nt1 → dv2 [G_hat[0,3]]

4 → 5: nt1 → dv1 [G_hat[0,2]]

5 → 1: nt1 → dn2 [G_hat[0,1]]

=====

Diagram 1L-688

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

```

    v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

    coeff =  $\frac{1}{2} nstar_1$ 

    v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

    coeff =  $\frac{1}{2} \varphi_1''$ 

    Edges (propagator assignments):
      2 → 3:  nt1 → dv2  [G_hat[0,3]]
      2 → 5:  nt1 → dv1  [G_hat[0,2]]
      3 → 0:  nt2 → dn1  [G_hat[1,0]]
      4 → 3:  nt1 → dv2  [G_hat[0,3]]
      4 → 5:  nt1 → dv1  [G_hat[0,2]]
      5 → 1:  nt1 → dn2  [G_hat[0,1]]

=====
Diagram 1L-689
=====

    External legs:
      leaf 0 ← dn1
      leaf 1 ← dn2
    Vertex assignments:
      v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

      coeff =  $\frac{1}{2} nstar_1$ 

      v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

      coeff =  $\frac{1}{2} \varphi_2''$ 

      v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

      coeff =  $\frac{1}{2} nstar_1$ 

      v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

      coeff =  $\frac{1}{2} \varphi_2''$ 

    Edges (propagator assignments):
      2 → 3:  nt1 → dv2  [G_hat[0,3]]
      2 → 5:  nt1 → dv2  [G_hat[0,3]]
      3 → 0:  nt2 → dn1  [G_hat[1,0]]
      4 → 3:  nt1 → dv2  [G_hat[0,3]]
      4 → 5:  nt1 → dv2  [G_hat[0,3]]
      5 → 1:  nt2 → dn2  [G_hat[1,1]]

```


=====

Diagram 1L-690

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

Edges (propagator assignments):

2 → 3: nt1 → dv2 [G_hat[0,3]]

2 → 5: nt1 → dv2 [G_hat[0,3]]

3 → 0: nt2 → dn1 [G_hat[1,0]]

4 → 3: nt1 → dv2 [G_hat[0,3]]

4 → 5: nt1 → dv2 [G_hat[0,3]]

5 → 1: nt2 → dn2 [G_hat[1,1]]

=====

Diagram 1L-691

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

```

    v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

    coeff =  $\frac{1}{2} nstar_1$ 

    v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

    coeff =  $\frac{1}{2} \varphi_2''$ 

    Edges (propagator assignments):
      2 → 3:  nt1 → dv2  [G_hat[0,3]]
      2 → 5:  nt1 → dv2  [G_hat[0,3]]
      3 → 0:  nt2 → dn1  [G_hat[1,0]]
      4 → 3:  nt1 → dv2  [G_hat[0,3]]
      4 → 5:  nt1 → dv2  [G_hat[0,3]]
      5 → 1:  nt2 → dn2  [G_hat[1,1]]

=====
Diagram 1L-692
=====

    External legs:
      leaf 0 ← dn1
      leaf 1 ← dn2
    Vertex assignments:
      v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

      coeff =  $\frac{1}{2} nstar_1$ 

      v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

      coeff =  $\frac{1}{2} \varphi_2''$ 

      v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

      coeff =  $\frac{1}{2} nstar_1$ 

      v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

      coeff =  $\frac{1}{2} \varphi_2''$ 

    Edges (propagator assignments):
      2 → 3:  nt1 → dv2  [G_hat[0,3]]
      2 → 5:  nt1 → dv2  [G_hat[0,3]]
      3 → 0:  nt2 → dn1  [G_hat[1,0]]
      4 → 3:  nt1 → dv2  [G_hat[0,3]]
      4 → 5:  nt1 → dv2  [G_hat[0,3]]
      5 → 1:  nt2 → dn2  [G_hat[1,1]]

```

=====

Diagram 1L-693

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

Edges (propagator assignments):

2 → 3: nt1 → dv2 [G_hat[0,3]]

2 → 5: nt1 → dv2 [G_hat[0,3]]

3 → 0: nt2 → dn1 [G_hat[1,0]]

4 → 3: nt1 → dv2 [G_hat[0,3]]

4 → 5: nt1 → dv2 [G_hat[0,3]]

5 → 1: nt2 → dn2 [G_hat[1,1]]

=====

Diagram 1L-694

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

```

    v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

    coeff =  $\frac{1}{2} nstar_1$ 

    v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

    coeff =  $\frac{1}{2} \varphi_2''$ 

    Edges (propagator assignments):
      2 → 3:  nt1 → dv2  [G_hat[0,3]]
      2 → 5:  nt1 → dv2  [G_hat[0,3]]
      3 → 0:  nt2 → dn1  [G_hat[1,0]]
      4 → 3:  nt1 → dv2  [G_hat[0,3]]
      4 → 5:  nt1 → dv2  [G_hat[0,3]]
      5 → 1:  nt2 → dn2  [G_hat[1,1]]

=====
Diagram 1L-695
=====

    External legs:
      leaf 0 ← dn1
      leaf 1 ← dn2
    Vertex assignments:
      v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

      coeff =  $\frac{1}{2} nstar_1$ 

      v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

      coeff =  $\frac{1}{2} \varphi_2''$ 

      v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

      coeff =  $\frac{1}{2} nstar_1$ 

      v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

      coeff =  $\frac{1}{2} \varphi_2''$ 

    Edges (propagator assignments):
      2 → 3:  nt1 → dv2  [G_hat[0,3]]
      2 → 5:  nt1 → dv2  [G_hat[0,3]]
      3 → 0:  nt2 → dn1  [G_hat[1,0]]
      4 → 3:  nt1 → dv2  [G_hat[0,3]]
      4 → 5:  nt1 → dv2  [G_hat[0,3]]
      5 → 1:  nt2 → dn2  [G_hat[1,1]]

```

=====

Diagram 1L-696

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

Edges (propagator assignments):

2 → 3: nt1 → dv2 [G_hat[0,3]]

2 → 5: nt1 → dv2 [G_hat[0,3]]

3 → 0: nt2 → dn1 [G_hat[1,0]]

4 → 3: nt1 → dv2 [G_hat[0,3]]

4 → 5: nt1 → dv2 [G_hat[0,3]]

5 → 1: nt2 → dn2 [G_hat[1,1]]

=====

Diagram 1L-697

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

```

    v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

    coeff =  $\frac{1}{2} nstar_1$ 

    v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

    coeff =  $\frac{1}{2} \varphi_2''$ 

    Edges (propagator assignments):
      2 → 3:  nt1 → dv2  [G_hat[0,3]]
      2 → 5:  nt1 → dv2  [G_hat[0,3]]
      3 → 0:  nt2 → dn1  [G_hat[1,0]]
      4 → 3:  nt1 → dv2  [G_hat[0,3]]
      4 → 5:  nt1 → dv2  [G_hat[0,3]]
      5 → 1:  nt2 → dn2  [G_hat[1,1]]

=====
Diagram 1L-698
=====

    External legs:
      leaf 0 ← dn1
      leaf 1 ← dn2
    Vertex assignments:
      v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

      coeff =  $\frac{1}{2} nstar_1$ 

      v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

      coeff =  $\frac{1}{2} \varphi_2''$ 

      v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

      coeff =  $\frac{1}{2} nstar_1$ 

      v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

      coeff =  $\frac{1}{2} \varphi_2''$ 

    Edges (propagator assignments):
      2 → 3:  nt1 → dv2  [G_hat[0,3]]
      2 → 5:  nt1 → dv2  [G_hat[0,3]]
      3 → 0:  nt2 → dn1  [G_hat[1,0]]
      4 → 3:  nt1 → dv2  [G_hat[0,3]]
      4 → 5:  nt1 → dv2  [G_hat[0,3]]
      5 → 1:  nt2 → dn2  [G_hat[1,1]]

```

=====

Diagram 1L-699

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

Edges (propagator assignments):

2 → 3: nt1 → dv2 [G_hat[0,3]]

2 → 5: nt1 → dv2 [G_hat[0,3]]

3 → 0: nt2 → dn1 [G_hat[1,0]]

4 → 3: nt1 → dv2 [G_hat[0,3]]

4 → 5: nt1 → dv2 [G_hat[0,3]]

5 → 1: nt2 → dn2 [G_hat[1,1]]

=====

Diagram 1L-700

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

```

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

coeff =  $\frac{1}{2} nstar_1$ 

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

coeff =  $\frac{1}{2} \varphi_2''$ 

Edges (propagator assignments):
  2 → 3:  nt1 → dv2  [G_hat[0,3]]
  2 → 5:  nt1 → dv2  [G_hat[0,3]]
  3 → 0:  nt2 → dn1  [G_hat[1,0]]
  4 → 3:  nt1 → dv2  [G_hat[0,3]]
  4 → 5:  nt1 → dv2  [G_hat[0,3]]
  5 → 1:  nt2 → dn2  [G_hat[1,1]]

=====
Diagram 1L-701
=====

External legs:
  leaf 0 ← dn1
  leaf 1 ← dn2
Vertex assignments:
  v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

coeff =  $\frac{1}{2} nstar_1$ 

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

coeff =  $\frac{1}{2} \varphi_2''$ 

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

coeff =  $\frac{1}{2} nstar_1$ 

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

coeff =  $\frac{1}{2} \varphi_2''$ 

Edges (propagator assignments):
  2 → 3:  nt1 → dv2  [G_hat[0,3]]
  2 → 5:  nt1 → dv2  [G_hat[0,3]]
  3 → 0:  nt2 → dn1  [G_hat[1,0]]
  4 → 3:  nt1 → dv2  [G_hat[0,3]]
  4 → 5:  nt1 → dv2  [G_hat[0,3]]
  5 → 1:  nt2 → dn2  [G_hat[1,1]]

```


=====

Diagram 1L-702

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

Edges (propagator assignments):

2 → 3: nt1 → dv2 [G_hat[0,3]]

2 → 5: nt1 → dv2 [G_hat[0,3]]

3 → 0: nt2 → dn1 [G_hat[1,0]]

4 → 3: nt1 → dv2 [G_hat[0,3]]

4 → 5: nt1 → dv2 [G_hat[0,3]]

5 → 1: nt2 → dn2 [G_hat[1,1]]

=====

Diagram 1L-703

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

```

    v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

    coeff =  $\frac{1}{2} nstar_1$ 

    v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

    coeff =  $\frac{1}{2} \varphi_2''$ 

    Edges (propagator assignments):
      2 → 3:  nt1 → dv2  [G_hat[0,3]]
      2 → 5:  nt1 → dv2  [G_hat[0,3]]
      3 → 0:  nt2 → dn1  [G_hat[1,0]]
      4 → 3:  nt1 → dv2  [G_hat[0,3]]
      4 → 5:  nt1 → dv2  [G_hat[0,3]]
      5 → 1:  nt2 → dn2  [G_hat[1,1]]

=====
Diagram 1L-704
=====

    External legs:
      leaf 0 ← dn1
      leaf 1 ← dn2
    Vertex assignments:
      v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

      coeff =  $\frac{1}{2} nstar_1$ 

      v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

      coeff =  $\frac{1}{2} \varphi_2''$ 

      v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

      coeff =  $\frac{1}{2} nstar_1$ 

      v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

      coeff =  $\frac{1}{2} \varphi_2''$ 

      Edges (propagator assignments):
        2 → 3:  nt1 → dv2  [G_hat[0,3]]
        2 → 5:  nt1 → dv2  [G_hat[0,3]]
        3 → 0:  nt2 → dn1  [G_hat[1,0]]
        4 → 3:  nt1 → dv2  [G_hat[0,3]]
        4 → 5:  nt1 → dv2  [G_hat[0,3]]
        5 → 1:  nt2 → dn2  [G_hat[1,1]]

```

=====

Diagram 1L-705

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

Edges (propagator assignments):

2 → 3: nt1 → dv1 [G_hat[0,2]]

2 → 5: nt1 → dv1 [G_hat[0,2]]

3 → 0: nt1 → dn1 [G_hat[0,0]]

4 → 3: nt2 → dv1 [G_hat[1,2]]

4 → 5: nt2 → dv1 [G_hat[1,2]]

5 → 1: nt1 → dn2 [G_hat[0,1]]

=====

Diagram 1L-706

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

```

    v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

    coeff =  $\frac{1}{2} nstar_2$ 

    v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

    coeff =  $\frac{1}{2} \varphi_1''$ 

    Edges (propagator assignments):
      2 → 3:  nt1 → dv1  [G_hat[0,2]]
      2 → 5:  nt1 → dv1  [G_hat[0,2]]
      3 → 0:  nt1 → dn1  [G_hat[0,0]]
      4 → 3:  nt2 → dv1  [G_hat[1,2]]
      4 → 5:  nt2 → dv1  [G_hat[1,2]]
      5 → 1:  nt1 → dn2  [G_hat[0,1]]

=====
Diagram 1L-707
=====

    External legs:
      leaf 0 ← dn1
      leaf 1 ← dn2
    Vertex assignments:
      v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

      coeff =  $\frac{1}{2} nstar_1$ 

      v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

      coeff =  $\frac{1}{2} \varphi_1''$ 

      v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

      coeff =  $\frac{1}{2} nstar_2$ 

      v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

      coeff =  $\frac{1}{2} \varphi_1''$ 

    Edges (propagator assignments):
      2 → 3:  nt1 → dv1  [G_hat[0,2]]
      2 → 5:  nt1 → dv1  [G_hat[0,2]]
      3 → 0:  nt1 → dn1  [G_hat[0,0]]
      4 → 3:  nt2 → dv1  [G_hat[1,2]]
      4 → 5:  nt2 → dv1  [G_hat[1,2]]
      5 → 1:  nt1 → dn2  [G_hat[0,1]]

```

=====

Diagram 1L-708

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

Edges (propagator assignments):

2 → 3: nt1 → dv1 [G_hat[0,2]]

2 → 5: nt1 → dv1 [G_hat[0,2]]

3 → 0: nt1 → dn1 [G_hat[0,0]]

4 → 3: nt2 → dv1 [G_hat[1,2]]

4 → 5: nt2 → dv1 [G_hat[1,2]]

5 → 1: nt1 → dn2 [G_hat[0,1]]

=====

Diagram 1L-709

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

```

    v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

    coeff =  $\frac{1}{2} nstar_2$ 

    v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

    coeff =  $\frac{1}{2} \varphi_1''$ 

    Edges (propagator assignments):
      2 → 3:  nt1 → dv1  [G_hat[0,2]]
      2 → 5:  nt1 → dv1  [G_hat[0,2]]
      3 → 0:  nt1 → dn1  [G_hat[0,0]]
      4 → 3:  nt2 → dv1  [G_hat[1,2]]
      4 → 5:  nt2 → dv1  [G_hat[1,2]]
      5 → 1:  nt1 → dn2  [G_hat[0,1]]

=====
Diagram 1L-710
=====

    External legs:
      leaf 0 ← dn1
      leaf 1 ← dn2
    Vertex assignments:
      v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

      coeff =  $\frac{1}{2} nstar_1$ 

      v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

      coeff =  $\frac{1}{2} \varphi_1''$ 

      v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

      coeff =  $\frac{1}{2} nstar_2$ 

      v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

      coeff =  $\frac{1}{2} \varphi_1''$ 

    Edges (propagator assignments):
      2 → 3:  nt1 → dv1  [G_hat[0,2]]
      2 → 5:  nt1 → dv1  [G_hat[0,2]]
      3 → 0:  nt1 → dn1  [G_hat[0,0]]
      4 → 3:  nt2 → dv1  [G_hat[1,2]]
      4 → 5:  nt2 → dv1  [G_hat[1,2]]
      5 → 1:  nt1 → dn2  [G_hat[0,1]]

```

=====

Diagram 1L-711

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

Edges (propagator assignments):

2 → 3: nt1 → dv1 [G_hat[0,2]]

2 → 5: nt1 → dv1 [G_hat[0,2]]

3 → 0: nt1 → dn1 [G_hat[0,0]]

4 → 3: nt2 → dv1 [G_hat[1,2]]

4 → 5: nt2 → dv1 [G_hat[1,2]]

5 → 1: nt1 → dn2 [G_hat[0,1]]

=====

Diagram 1L-712

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

```

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

coeff =  $\frac{1}{2} nstar_2$ 

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

coeff =  $\frac{1}{2} \varphi_1''$ 

Edges (propagator assignments):
  2 → 3:  nt1 → dv1  [G_hat[0,2]]
  2 → 5:  nt1 → dv1  [G_hat[0,2]]
  3 → 0:  nt1 → dn1  [G_hat[0,0]]
  4 → 3:  nt2 → dv1  [G_hat[1,2]]
  4 → 5:  nt2 → dv1  [G_hat[1,2]]
  5 → 1:  nt1 → dn2  [G_hat[0,1]]

=====
Diagram 1L-713
=====

External legs:
  leaf 0 ← dn1
  leaf 1 ← dn2
Vertex assignments:
  v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

coeff =  $\frac{1}{2} nstar_1$ 

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

coeff =  $\frac{1}{2} \varphi_1''$ 

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

coeff =  $\frac{1}{2} nstar_2$ 

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

coeff =  $\frac{1}{2} \varphi_1''$ 

Edges (propagator assignments):
  2 → 3:  nt1 → dv1  [G_hat[0,2]]
  2 → 5:  nt1 → dv1  [G_hat[0,2]]
  3 → 0:  nt1 → dn1  [G_hat[0,0]]
  4 → 3:  nt2 → dv1  [G_hat[1,2]]
  4 → 5:  nt2 → dv1  [G_hat[1,2]]
  5 → 1:  nt1 → dn2  [G_hat[0,1]]

```


=====

Diagram 1L-714

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

Edges (propagator assignments):

2 → 3: nt1 → dv1 [G_hat[0,2]]

2 → 5: nt1 → dv1 [G_hat[0,2]]

3 → 0: nt1 → dn1 [G_hat[0,0]]

4 → 3: nt2 → dv1 [G_hat[1,2]]

4 → 5: nt2 → dv1 [G_hat[1,2]]

5 → 1: nt1 → dn2 [G_hat[0,1]]

=====

Diagram 1L-715

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

```

    v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

    coeff =  $\frac{1}{2} nstar_2$ 

    v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

    coeff =  $\frac{1}{2} \varphi_1''$ 

    Edges (propagator assignments):
      2 → 3:  nt1 → dv1  [G_hat[0,2]]
      2 → 5:  nt1 → dv1  [G_hat[0,2]]
      3 → 0:  nt1 → dn1  [G_hat[0,0]]
      4 → 3:  nt2 → dv1  [G_hat[1,2]]
      4 → 5:  nt2 → dv1  [G_hat[1,2]]
      5 → 1:  nt1 → dn2  [G_hat[0,1]]

=====
Diagram 1L-716
=====

    External legs:
      leaf 0 ← dn1
      leaf 1 ← dn2
    Vertex assignments:
      v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

      coeff =  $\frac{1}{2} nstar_1$ 

      v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

      coeff =  $\frac{1}{2} \varphi_1''$ 

      v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

      coeff =  $\frac{1}{2} nstar_2$ 

      v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

      coeff =  $\frac{1}{2} \varphi_1''$ 

    Edges (propagator assignments):
      2 → 3:  nt1 → dv1  [G_hat[0,2]]
      2 → 5:  nt1 → dv1  [G_hat[0,2]]
      3 → 0:  nt1 → dn1  [G_hat[0,0]]
      4 → 3:  nt2 → dv1  [G_hat[1,2]]
      4 → 5:  nt2 → dv1  [G_hat[1,2]]
      5 → 1:  nt1 → dn2  [G_hat[0,1]]

```

=====

Diagram 1L-717

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

Edges (propagator assignments):

2 → 3: nt1 → dv1 [G_hat[0,2]]

2 → 5: nt1 → dv1 [G_hat[0,2]]

3 → 0: nt1 → dn1 [G_hat[0,0]]

4 → 3: nt2 → dv1 [G_hat[1,2]]

4 → 5: nt2 → dv1 [G_hat[1,2]]

5 → 1: nt1 → dn2 [G_hat[0,1]]

=====

Diagram 1L-718

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

```

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

coeff =  $\frac{1}{2} nstar_2$ 

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

coeff =  $\frac{1}{2} \varphi_1''$ 

Edges (propagator assignments):
  2 → 3:  nt1 → dv1  [G_hat[0,2]]
  2 → 5:  nt1 → dv1  [G_hat[0,2]]
  3 → 0:  nt1 → dn1  [G_hat[0,0]]
  4 → 3:  nt2 → dv1  [G_hat[1,2]]
  4 → 5:  nt2 → dv1  [G_hat[1,2]]
  5 → 1:  nt1 → dn2  [G_hat[0,1]]

=====
Diagram 1L-719
=====

External legs:
  leaf 0 ← dn1
  leaf 1 ← dn2
Vertex assignments:
  v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

coeff =  $\frac{1}{2} nstar_1$ 

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

coeff =  $\frac{1}{2} \varphi_1''$ 

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

coeff =  $\frac{1}{2} nstar_2$ 

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

coeff =  $\frac{1}{2} \varphi_1''$ 

Edges (propagator assignments):
  2 → 3:  nt1 → dv1  [G_hat[0,2]]
  2 → 5:  nt1 → dv1  [G_hat[0,2]]
  3 → 0:  nt1 → dn1  [G_hat[0,0]]
  4 → 3:  nt2 → dv1  [G_hat[1,2]]
  4 → 5:  nt2 → dv1  [G_hat[1,2]]
  5 → 1:  nt1 → dn2  [G_hat[0,1]]

```

=====

Diagram 1L-720

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

Edges (propagator assignments):

2 → 3: nt1 → dv1 [G_hat[0,2]]

2 → 5: nt1 → dv1 [G_hat[0,2]]

3 → 0: nt1 → dn1 [G_hat[0,0]]

4 → 3: nt2 → dv1 [G_hat[1,2]]

4 → 5: nt2 → dv1 [G_hat[1,2]]

5 → 1: nt1 → dn2 [G_hat[0,1]]

=====

Diagram 1L-721

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

```

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

coeff =  $\frac{1}{2} nstar_2$ 

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

coeff =  $\frac{1}{2} \varphi_2''$ 

Edges (propagator assignments):
  2 → 3:  nt1 → dv1  [G_hat[0,2]]
  2 → 5:  nt1 → dv2  [G_hat[0,3]]
  3 → 0:  nt1 → dn1  [G_hat[0,0]]
  4 → 3:  nt2 → dv1  [G_hat[1,2]]
  4 → 5:  nt2 → dv2  [G_hat[1,3]]
  5 → 1:  nt2 → dn2  [G_hat[1,1]]

=====
Diagram 1L-722
=====

External legs:
  leaf 0 ← dn1
  leaf 1 ← dn2
Vertex assignments:
  v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

coeff =  $\frac{1}{2} nstar_1$ 

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

coeff =  $\frac{1}{2} \varphi_1''$ 

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

coeff =  $\frac{1}{2} nstar_2$ 

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

coeff =  $\frac{1}{2} \varphi_2''$ 

Edges (propagator assignments):
  2 → 3:  nt1 → dv1  [G_hat[0,2]]
  2 → 5:  nt1 → dv2  [G_hat[0,3]]
  3 → 0:  nt1 → dn1  [G_hat[0,0]]
  4 → 3:  nt2 → dv1  [G_hat[1,2]]
  4 → 5:  nt2 → dv2  [G_hat[1,3]]
  5 → 1:  nt2 → dn2  [G_hat[1,1]]

```

=====

Diagram 1L-723

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

Edges (propagator assignments):

2 → 3: nt1 → dv1 [G_hat[0,2]]

2 → 5: nt1 → dv2 [G_hat[0,3]]

3 → 0: nt1 → dn1 [G_hat[0,0]]

4 → 3: nt2 → dv1 [G_hat[1,2]]

4 → 5: nt2 → dv2 [G_hat[1,3]]

5 → 1: nt2 → dn2 [G_hat[1,1]]

=====

Diagram 1L-724

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

```

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

coeff =  $\frac{1}{2} nstar_2$ 

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

coeff =  $\frac{1}{2} \varphi_2''$ 

Edges (propagator assignments):
  2 → 3:  nt1 → dv1  [G_hat[0,2]]
  2 → 5:  nt1 → dv2  [G_hat[0,3]]
  3 → 0:  nt1 → dn1  [G_hat[0,0]]
  4 → 3:  nt2 → dv1  [G_hat[1,2]]
  4 → 5:  nt2 → dv2  [G_hat[1,3]]
  5 → 1:  nt2 → dn2  [G_hat[1,1]]

=====
Diagram 1L-725
=====

External legs:
  leaf 0 ← dn1
  leaf 1 ← dn2
Vertex assignments:
  v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

coeff =  $\frac{1}{2} nstar_1$ 

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

coeff =  $\frac{1}{2} \varphi_1''$ 

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

coeff =  $\frac{1}{2} nstar_2$ 

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

coeff =  $\frac{1}{2} \varphi_2''$ 

Edges (propagator assignments):
  2 → 3:  nt1 → dv1  [G_hat[0,2]]
  2 → 5:  nt1 → dv2  [G_hat[0,3]]
  3 → 0:  nt1 → dn1  [G_hat[0,0]]
  4 → 3:  nt2 → dv1  [G_hat[1,2]]
  4 → 5:  nt2 → dv2  [G_hat[1,3]]
  5 → 1:  nt2 → dn2  [G_hat[1,1]]

```


=====

Diagram 1L-726

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

Edges (propagator assignments):

2 → 3: nt1 → dv1 [G_hat[0,2]]

2 → 5: nt1 → dv2 [G_hat[0,3]]

3 → 0: nt1 → dn1 [G_hat[0,0]]

4 → 3: nt2 → dv1 [G_hat[1,2]]

4 → 5: nt2 → dv2 [G_hat[1,3]]

5 → 1: nt2 → dn2 [G_hat[1,1]]

=====

Diagram 1L-727

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

```

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

coeff =  $\frac{1}{2} nstar_2$ 

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

coeff =  $\frac{1}{2} \varphi_2''$ 

Edges (propagator assignments):
  2 → 3:  nt1 → dv1  [G_hat[0,2]]
  2 → 5:  nt1 → dv2  [G_hat[0,3]]
  3 → 0:  nt1 → dn1  [G_hat[0,0]]
  4 → 3:  nt2 → dv1  [G_hat[1,2]]
  4 → 5:  nt2 → dv2  [G_hat[1,3]]
  5 → 1:  nt2 → dn2  [G_hat[1,1]]

=====
Diagram 1L-728
=====

External legs:
  leaf 0 ← dn1
  leaf 1 ← dn2
Vertex assignments:
  v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

coeff =  $\frac{1}{2} nstar_1$ 

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

coeff =  $\frac{1}{2} \varphi_1''$ 

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

coeff =  $\frac{1}{2} nstar_2$ 

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

coeff =  $\frac{1}{2} \varphi_2''$ 

Edges (propagator assignments):
  2 → 3:  nt1 → dv1  [G_hat[0,2]]
  2 → 5:  nt1 → dv2  [G_hat[0,3]]
  3 → 0:  nt1 → dn1  [G_hat[0,0]]
  4 → 3:  nt2 → dv1  [G_hat[1,2]]
  4 → 5:  nt2 → dv2  [G_hat[1,3]]
  5 → 1:  nt2 → dn2  [G_hat[1,1]]

```

=====

Diagram 1L-729

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

Edges (propagator assignments):

2 → 3: nt1 → dv1 [G_hat[0,2]]

2 → 5: nt1 → dv2 [G_hat[0,3]]

3 → 0: nt1 → dn1 [G_hat[0,0]]

4 → 3: nt2 → dv1 [G_hat[1,2]]

4 → 5: nt2 → dv2 [G_hat[1,3]]

5 → 1: nt2 → dn2 [G_hat[1,1]]

=====

Diagram 1L-730

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

```

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

coeff =  $\frac{1}{2} nstar_2$ 

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

coeff =  $\frac{1}{2} \varphi_2''$ 

Edges (propagator assignments):
  2 → 3:  nt1 → dv1  [G_hat[0,2]]
  2 → 5:  nt1 → dv2  [G_hat[0,3]]
  3 → 0:  nt1 → dn1  [G_hat[0,0]]
  4 → 3:  nt2 → dv1  [G_hat[1,2]]
  4 → 5:  nt2 → dv2  [G_hat[1,3]]
  5 → 1:  nt2 → dn2  [G_hat[1,1]]

=====
Diagram 1L-731
=====

External legs:
  leaf 0 ← dn1
  leaf 1 ← dn2
Vertex assignments:
  v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

coeff =  $\frac{1}{2} nstar_1$ 

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

coeff =  $\frac{1}{2} \varphi_1''$ 

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

coeff =  $\frac{1}{2} nstar_2$ 

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

coeff =  $\frac{1}{2} \varphi_2''$ 

Edges (propagator assignments):
  2 → 3:  nt1 → dv1  [G_hat[0,2]]
  2 → 5:  nt1 → dv2  [G_hat[0,3]]
  3 → 0:  nt1 → dn1  [G_hat[0,0]]
  4 → 3:  nt2 → dv1  [G_hat[1,2]]
  4 → 5:  nt2 → dv2  [G_hat[1,3]]
  5 → 1:  nt2 → dn2  [G_hat[1,1]]

```

=====

Diagram 1L-732

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

Edges (propagator assignments):

2 → 3: nt1 → dv1 [G_hat[0,2]]

2 → 5: nt1 → dv2 [G_hat[0,3]]

3 → 0: nt1 → dn1 [G_hat[0,0]]

4 → 3: nt2 → dv1 [G_hat[1,2]]

4 → 5: nt2 → dv2 [G_hat[1,3]]

5 → 1: nt2 → dn2 [G_hat[1,1]]

=====

Diagram 1L-733

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

```

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

coeff =  $\frac{1}{2} nstar_2$ 

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

coeff =  $\frac{1}{2} \varphi_2''$ 

Edges (propagator assignments):
  2 → 3:  nt1 → dv1  [G_hat[0,2]]
  2 → 5:  nt1 → dv2  [G_hat[0,3]]
  3 → 0:  nt1 → dn1  [G_hat[0,0]]
  4 → 3:  nt2 → dv1  [G_hat[1,2]]
  4 → 5:  nt2 → dv2  [G_hat[1,3]]
  5 → 1:  nt2 → dn2  [G_hat[1,1]]

=====
Diagram 1L-734
=====

External legs:
  leaf 0 ← dn1
  leaf 1 ← dn2
Vertex assignments:
  v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

coeff =  $\frac{1}{2} nstar_1$ 

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

coeff =  $\frac{1}{2} \varphi_1''$ 

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

coeff =  $\frac{1}{2} nstar_2$ 

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

coeff =  $\frac{1}{2} \varphi_2''$ 

Edges (propagator assignments):
  2 → 3:  nt1 → dv1  [G_hat[0,2]]
  2 → 5:  nt1 → dv2  [G_hat[0,3]]
  3 → 0:  nt1 → dn1  [G_hat[0,0]]
  4 → 3:  nt2 → dv1  [G_hat[1,2]]
  4 → 5:  nt2 → dv2  [G_hat[1,3]]
  5 → 1:  nt2 → dn2  [G_hat[1,1]]

```

=====

Diagram 1L-735

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

Edges (propagator assignments):

2 → 3: nt1 → dv1 [G_hat[0,2]]

2 → 5: nt1 → dv2 [G_hat[0,3]]

3 → 0: nt1 → dn1 [G_hat[0,0]]

4 → 3: nt2 → dv1 [G_hat[1,2]]

4 → 5: nt2 → dv2 [G_hat[1,3]]

5 → 1: nt2 → dn2 [G_hat[1,1]]

=====

Diagram 1L-736

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

```

    v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

    coeff =  $\frac{1}{2} nstar_2$ 

    v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

    coeff =  $\frac{1}{2} \varphi_2''$ 

    Edges (propagator assignments):
      2 → 3:  nt1 → dv1  [G_hat[0,2]]
      2 → 5:  nt1 → dv2  [G_hat[0,3]]
      3 → 0:  nt1 → dn1  [G_hat[0,0]]
      4 → 3:  nt2 → dv1  [G_hat[1,2]]
      4 → 5:  nt2 → dv2  [G_hat[1,3]]
      5 → 1:  nt2 → dn2  [G_hat[1,1]]

=====
Diagram 1L-737
=====

    External legs:
      leaf 0 ← dn1
      leaf 1 ← dn2
    Vertex assignments:
      v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

      coeff =  $\frac{1}{2} nstar_1$ 

      v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

      coeff =  $\frac{1}{2} \varphi_2''$ 

      v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

      coeff =  $\frac{1}{2} nstar_2$ 

      v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

      coeff =  $\frac{1}{2} \varphi_1''$ 

    Edges (propagator assignments):
      2 → 3:  nt1 → dv2  [G_hat[0,3]]
      2 → 5:  nt1 → dv1  [G_hat[0,2]]
      3 → 0:  nt2 → dn1  [G_hat[1,0]]
      4 → 3:  nt2 → dv2  [G_hat[1,3]]
      4 → 5:  nt2 → dv1  [G_hat[1,2]]
      5 → 1:  nt1 → dn2  [G_hat[0,1]]

```


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Diagram 1L-738

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External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

Edges (propagator assignments):

2 → 3: nt1 → dv2 [G_hat[0,3]]

2 → 5: nt1 → dv1 [G_hat[0,2]]

3 → 0: nt2 → dn1 [G_hat[1,0]]

4 → 3: nt2 → dv2 [G_hat[1,3]]

4 → 5: nt2 → dv1 [G_hat[1,2]]

5 → 1: nt1 → dn2 [G_hat[0,1]]

=====

Diagram 1L-739

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

```

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

coeff =  $\frac{1}{2} nstar_2$ 

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

coeff =  $\frac{1}{2} \varphi_1''$ 

Edges (propagator assignments):
  2 → 3:  nt1 → dv2  [G_hat[0,3]]
  2 → 5:  nt1 → dv1  [G_hat[0,2]]
  3 → 0:  nt2 → dn1  [G_hat[1,0]]
  4 → 3:  nt2 → dv2  [G_hat[1,3]]
  4 → 5:  nt2 → dv1  [G_hat[1,2]]
  5 → 1:  nt1 → dn2  [G_hat[0,1]]

=====
Diagram 1L-740
=====

External legs:
  leaf 0 ← dn1
  leaf 1 ← dn2
Vertex assignments:
  v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

coeff =  $\frac{1}{2} nstar_1$ 

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

coeff =  $\frac{1}{2} \varphi_2''$ 

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

coeff =  $\frac{1}{2} nstar_2$ 

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

coeff =  $\frac{1}{2} \varphi_1''$ 

Edges (propagator assignments):
  2 → 3:  nt1 → dv2  [G_hat[0,3]]
  2 → 5:  nt1 → dv1  [G_hat[0,2]]
  3 → 0:  nt2 → dn1  [G_hat[1,0]]
  4 → 3:  nt2 → dv2  [G_hat[1,3]]
  4 → 5:  nt2 → dv1  [G_hat[1,2]]
  5 → 1:  nt1 → dn2  [G_hat[0,1]]

```

=====

Diagram 1L-741

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

Edges (propagator assignments):

2 → 3: nt1 → dv2 [G_hat[0,3]]

2 → 5: nt1 → dv1 [G_hat[0,2]]

3 → 0: nt2 → dn1 [G_hat[1,0]]

4 → 3: nt2 → dv2 [G_hat[1,3]]

4 → 5: nt2 → dv1 [G_hat[1,2]]

5 → 1: nt1 → dn2 [G_hat[0,1]]

=====

Diagram 1L-742

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

```

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

coeff =  $\frac{1}{2} nstar_2$ 

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

coeff =  $\frac{1}{2} \varphi_1''$ 

Edges (propagator assignments):
  2 → 3:  nt1 → dv2  [G_hat[0,3]]
  2 → 5:  nt1 → dv1  [G_hat[0,2]]
  3 → 0:  nt2 → dn1  [G_hat[1,0]]
  4 → 3:  nt2 → dv2  [G_hat[1,3]]
  4 → 5:  nt2 → dv1  [G_hat[1,2]]
  5 → 1:  nt1 → dn2  [G_hat[0,1]]

=====
Diagram 1L-743
=====

External legs:
  leaf 0 ← dn1
  leaf 1 ← dn2
Vertex assignments:
  v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

coeff =  $\frac{1}{2} nstar_1$ 

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

coeff =  $\frac{1}{2} \varphi_2''$ 

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

coeff =  $\frac{1}{2} nstar_2$ 

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

coeff =  $\frac{1}{2} \varphi_1''$ 

Edges (propagator assignments):
  2 → 3:  nt1 → dv2  [G_hat[0,3]]
  2 → 5:  nt1 → dv1  [G_hat[0,2]]
  3 → 0:  nt2 → dn1  [G_hat[1,0]]
  4 → 3:  nt2 → dv2  [G_hat[1,3]]
  4 → 5:  nt2 → dv1  [G_hat[1,2]]
  5 → 1:  nt1 → dn2  [G_hat[0,1]]

```

=====

Diagram 1L-744

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

Edges (propagator assignments):

2 → 3: nt1 → dv2 [G_hat[0,3]]

2 → 5: nt1 → dv1 [G_hat[0,2]]

3 → 0: nt2 → dn1 [G_hat[1,0]]

4 → 3: nt2 → dv2 [G_hat[1,3]]

4 → 5: nt2 → dv1 [G_hat[1,2]]

5 → 1: nt1 → dn2 [G_hat[0,1]]

=====

Diagram 1L-745

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

```

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

coeff =  $\frac{1}{2} nstar_2$ 

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

coeff =  $\frac{1}{2} \varphi_1''$ 

Edges (propagator assignments):
  2 → 3:  nt1 → dv2  [G_hat[0,3]]
  2 → 5:  nt1 → dv1  [G_hat[0,2]]
  3 → 0:  nt2 → dn1  [G_hat[1,0]]
  4 → 3:  nt2 → dv2  [G_hat[1,3]]
  4 → 5:  nt2 → dv1  [G_hat[1,2]]
  5 → 1:  nt1 → dn2  [G_hat[0,1]]

=====
Diagram 1L-746
=====

External legs:
  leaf 0 ← dn1
  leaf 1 ← dn2
Vertex assignments:
  v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

coeff =  $\frac{1}{2} nstar_1$ 

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

coeff =  $\frac{1}{2} \varphi_2''$ 

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

coeff =  $\frac{1}{2} nstar_2$ 

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

coeff =  $\frac{1}{2} \varphi_1''$ 

Edges (propagator assignments):
  2 → 3:  nt1 → dv2  [G_hat[0,3]]
  2 → 5:  nt1 → dv1  [G_hat[0,2]]
  3 → 0:  nt2 → dn1  [G_hat[1,0]]
  4 → 3:  nt2 → dv2  [G_hat[1,3]]
  4 → 5:  nt2 → dv1  [G_hat[1,2]]
  5 → 1:  nt1 → dn2  [G_hat[0,1]]

```

=====

Diagram 1L-747

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

Edges (propagator assignments):

2 → 3: nt1 → dv2 [G_hat[0,3]]

2 → 5: nt1 → dv1 [G_hat[0,2]]

3 → 0: nt2 → dn1 [G_hat[1,0]]

4 → 3: nt2 → dv2 [G_hat[1,3]]

4 → 5: nt2 → dv1 [G_hat[1,2]]

5 → 1: nt1 → dn2 [G_hat[0,1]]

=====

Diagram 1L-748

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

```

    v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

    coeff =  $\frac{1}{2} nstar_2$ 

    v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

    coeff =  $\frac{1}{2} \varphi_1''$ 

    Edges (propagator assignments):
      2 → 3:  nt1 → dv2  [G_hat[0,3]]
      2 → 5:  nt1 → dv1  [G_hat[0,2]]
      3 → 0:  nt2 → dn1  [G_hat[1,0]]
      4 → 3:  nt2 → dv2  [G_hat[1,3]]
      4 → 5:  nt2 → dv1  [G_hat[1,2]]
      5 → 1:  nt1 → dn2  [G_hat[0,1]]

=====
Diagram 1L-749
=====

    External legs:
      leaf 0 ← dn1
      leaf 1 ← dn2
    Vertex assignments:
      v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

      coeff =  $\frac{1}{2} nstar_1$ 

      v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

      coeff =  $\frac{1}{2} \varphi_2''$ 

      v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

      coeff =  $\frac{1}{2} nstar_2$ 

      v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

      coeff =  $\frac{1}{2} \varphi_1''$ 

    Edges (propagator assignments):
      2 → 3:  nt1 → dv2  [G_hat[0,3]]
      2 → 5:  nt1 → dv1  [G_hat[0,2]]
      3 → 0:  nt2 → dn1  [G_hat[1,0]]
      4 → 3:  nt2 → dv2  [G_hat[1,3]]
      4 → 5:  nt2 → dv1  [G_hat[1,2]]
      5 → 1:  nt1 → dn2  [G_hat[0,1]]

```


=====

Diagram 1L-750

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

Edges (propagator assignments):

2 → 3: nt1 → dv2 [G_hat[0,3]]

2 → 5: nt1 → dv1 [G_hat[0,2]]

3 → 0: nt2 → dn1 [G_hat[1,0]]

4 → 3: nt2 → dv2 [G_hat[1,3]]

4 → 5: nt2 → dv1 [G_hat[1,2]]

5 → 1: nt1 → dn2 [G_hat[0,1]]

=====

Diagram 1L-751

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

```

    v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

    coeff =  $\frac{1}{2} nstar_2$ 

    v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

    coeff =  $\frac{1}{2} \varphi_1''$ 

    Edges (propagator assignments):
      2 → 3:  nt1 → dv2  [G_hat[0,3]]
      2 → 5:  nt1 → dv1  [G_hat[0,2]]
      3 → 0:  nt2 → dn1  [G_hat[1,0]]
      4 → 3:  nt2 → dv2  [G_hat[1,3]]
      4 → 5:  nt2 → dv1  [G_hat[1,2]]
      5 → 1:  nt1 → dn2  [G_hat[0,1]]

=====
Diagram 1L-752
=====

    External legs:
      leaf 0 ← dn1
      leaf 1 ← dn2
    Vertex assignments:
      v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

      coeff =  $\frac{1}{2} nstar_1$ 

      v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

      coeff =  $\frac{1}{2} \varphi_2''$ 

      v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

      coeff =  $\frac{1}{2} nstar_2$ 

      v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

      coeff =  $\frac{1}{2} \varphi_1''$ 

    Edges (propagator assignments):
      2 → 3:  nt1 → dv2  [G_hat[0,3]]
      2 → 5:  nt1 → dv1  [G_hat[0,2]]
      3 → 0:  nt2 → dn1  [G_hat[1,0]]
      4 → 3:  nt2 → dv2  [G_hat[1,3]]
      4 → 5:  nt2 → dv1  [G_hat[1,2]]
      5 → 1:  nt1 → dn2  [G_hat[0,1]]

```

=====

Diagram 1L-753

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

Edges (propagator assignments):

2 → 3: nt1 → dv2 [G_hat[0,3]]

2 → 5: nt1 → dv2 [G_hat[0,3]]

3 → 0: nt2 → dn1 [G_hat[1,0]]

4 → 3: nt2 → dv2 [G_hat[1,3]]

4 → 5: nt2 → dv2 [G_hat[1,3]]

5 → 1: nt2 → dn2 [G_hat[1,1]]

=====

Diagram 1L-754

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

```

    v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

    coeff =  $\frac{1}{2} nstar_2$ 

    v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

    coeff =  $\frac{1}{2} \varphi_2''$ 

    Edges (propagator assignments):
      2 → 3:  nt1 → dv2  [G_hat[0,3]]
      2 → 5:  nt1 → dv2  [G_hat[0,3]]
      3 → 0:  nt2 → dn1  [G_hat[1,0]]
      4 → 3:  nt2 → dv2  [G_hat[1,3]]
      4 → 5:  nt2 → dv2  [G_hat[1,3]]
      5 → 1:  nt2 → dn2  [G_hat[1,1]]

=====
Diagram 1L-755
=====

    External legs:
      leaf 0 ← dn1
      leaf 1 ← dn2
    Vertex assignments:
      v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

      coeff =  $\frac{1}{2} nstar_1$ 

      v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

      coeff =  $\frac{1}{2} \varphi_2''$ 

      v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

      coeff =  $\frac{1}{2} nstar_2$ 

      v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

      coeff =  $\frac{1}{2} \varphi_2''$ 

    Edges (propagator assignments):
      2 → 3:  nt1 → dv2  [G_hat[0,3]]
      2 → 5:  nt1 → dv2  [G_hat[0,3]]
      3 → 0:  nt2 → dn1  [G_hat[1,0]]
      4 → 3:  nt2 → dv2  [G_hat[1,3]]
      4 → 5:  nt2 → dv2  [G_hat[1,3]]
      5 → 1:  nt2 → dn2  [G_hat[1,1]]

```

=====

Diagram 1L-756

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

Edges (propagator assignments):

2 → 3: nt1 → dv2 [G_hat[0,3]]

2 → 5: nt1 → dv2 [G_hat[0,3]]

3 → 0: nt2 → dn1 [G_hat[1,0]]

4 → 3: nt2 → dv2 [G_hat[1,3]]

4 → 5: nt2 → dv2 [G_hat[1,3]]

5 → 1: nt2 → dn2 [G_hat[1,1]]

=====

Diagram 1L-757

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

```

    v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

    coeff =  $\frac{1}{2} nstar_2$ 

    v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

    coeff =  $\frac{1}{2} \varphi_2''$ 

    Edges (propagator assignments):
      2 → 3:  nt1 → dv2  [G_hat[0,3]]
      2 → 5:  nt1 → dv2  [G_hat[0,3]]
      3 → 0:  nt2 → dn1  [G_hat[1,0]]
      4 → 3:  nt2 → dv2  [G_hat[1,3]]
      4 → 5:  nt2 → dv2  [G_hat[1,3]]
      5 → 1:  nt2 → dn2  [G_hat[1,1]]

=====
Diagram 1L-758
=====

    External legs:
      leaf 0 ← dn1
      leaf 1 ← dn2
    Vertex assignments:
      v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

      coeff =  $\frac{1}{2} nstar_1$ 

      v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

      coeff =  $\frac{1}{2} \varphi_2''$ 

      v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

      coeff =  $\frac{1}{2} nstar_2$ 

      v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

      coeff =  $\frac{1}{2} \varphi_2''$ 

    Edges (propagator assignments):
      2 → 3:  nt1 → dv2  [G_hat[0,3]]
      2 → 5:  nt1 → dv2  [G_hat[0,3]]
      3 → 0:  nt2 → dn1  [G_hat[1,0]]
      4 → 3:  nt2 → dv2  [G_hat[1,3]]
      4 → 5:  nt2 → dv2  [G_hat[1,3]]
      5 → 1:  nt2 → dn2  [G_hat[1,1]]

```

=====

Diagram 1L-759

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

Edges (propagator assignments):

2 → 3: nt1 → dv2 [G_hat[0,3]]

2 → 5: nt1 → dv2 [G_hat[0,3]]

3 → 0: nt2 → dn1 [G_hat[1,0]]

4 → 3: nt2 → dv2 [G_hat[1,3]]

4 → 5: nt2 → dv2 [G_hat[1,3]]

5 → 1: nt2 → dn2 [G_hat[1,1]]

=====

Diagram 1L-760

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

```

    v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

    coeff =  $\frac{1}{2} nstar_2$ 

    v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

    coeff =  $\frac{1}{2} \varphi_2''$ 

    Edges (propagator assignments):
      2 → 3:  nt1 → dv2  [G_hat[0,3]]
      2 → 5:  nt1 → dv2  [G_hat[0,3]]
      3 → 0:  nt2 → dn1  [G_hat[1,0]]
      4 → 3:  nt2 → dv2  [G_hat[1,3]]
      4 → 5:  nt2 → dv2  [G_hat[1,3]]
      5 → 1:  nt2 → dn2  [G_hat[1,1]]

=====
Diagram 1L-761
=====

    External legs:
      leaf 0 ← dn1
      leaf 1 ← dn2
    Vertex assignments:
      v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

      coeff =  $\frac{1}{2} nstar_1$ 

      v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

      coeff =  $\frac{1}{2} \varphi_2''$ 

      v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

      coeff =  $\frac{1}{2} nstar_2$ 

      v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

      coeff =  $\frac{1}{2} \varphi_2''$ 

    Edges (propagator assignments):
      2 → 3:  nt1 → dv2  [G_hat[0,3]]
      2 → 5:  nt1 → dv2  [G_hat[0,3]]
      3 → 0:  nt2 → dn1  [G_hat[1,0]]
      4 → 3:  nt2 → dv2  [G_hat[1,3]]
      4 → 5:  nt2 → dv2  [G_hat[1,3]]
      5 → 1:  nt2 → dn2  [G_hat[1,1]]

```


=====

Diagram 1L-762

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

Edges (propagator assignments):

2 → 3: nt1 → dv2 [G_hat[0,3]]

2 → 5: nt1 → dv2 [G_hat[0,3]]

3 → 0: nt2 → dn1 [G_hat[1,0]]

4 → 3: nt2 → dv2 [G_hat[1,3]]

4 → 5: nt2 → dv2 [G_hat[1,3]]

5 → 1: nt2 → dn2 [G_hat[1,1]]

=====

Diagram 1L-763

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

```

    v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

    coeff =  $\frac{1}{2} nstar_2$ 

    v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

    coeff =  $\frac{1}{2} \varphi_2''$ 

    Edges (propagator assignments):
      2 → 3:  nt1 → dv2  [G_hat[0,3]]
      2 → 5:  nt1 → dv2  [G_hat[0,3]]
      3 → 0:  nt2 → dn1  [G_hat[1,0]]
      4 → 3:  nt2 → dv2  [G_hat[1,3]]
      4 → 5:  nt2 → dv2  [G_hat[1,3]]
      5 → 1:  nt2 → dn2  [G_hat[1,1]]

=====
Diagram 1L-764
=====

    External legs:
      leaf 0 ← dn1
      leaf 1 ← dn2
    Vertex assignments:
      v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

      coeff =  $\frac{1}{2} nstar_1$ 

      v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

      coeff =  $\frac{1}{2} \varphi_2''$ 

      v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

      coeff =  $\frac{1}{2} nstar_2$ 

      v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

      coeff =  $\frac{1}{2} \varphi_2''$ 

    Edges (propagator assignments):
      2 → 3:  nt1 → dv2  [G_hat[0,3]]
      2 → 5:  nt1 → dv2  [G_hat[0,3]]
      3 → 0:  nt2 → dn1  [G_hat[1,0]]
      4 → 3:  nt2 → dv2  [G_hat[1,3]]
      4 → 5:  nt2 → dv2  [G_hat[1,3]]
      5 → 1:  nt2 → dn2  [G_hat[1,1]]

```

=====

Diagram 1L-765

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

Edges (propagator assignments):

2 → 3: nt1 → dv2 [G_hat[0,3]]

2 → 5: nt1 → dv2 [G_hat[0,3]]

3 → 0: nt2 → dn1 [G_hat[1,0]]

4 → 3: nt2 → dv2 [G_hat[1,3]]

4 → 5: nt2 → dv2 [G_hat[1,3]]

5 → 1: nt2 → dn2 [G_hat[1,1]]

=====

Diagram 1L-766

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

```

    v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

    coeff =  $\frac{1}{2} nstar_2$ 

    v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

    coeff =  $\frac{1}{2} \varphi_2''$ 

    Edges (propagator assignments):
      2 → 3:  nt1 → dv2  [G_hat[0,3]]
      2 → 5:  nt1 → dv2  [G_hat[0,3]]
      3 → 0:  nt2 → dn1  [G_hat[1,0]]
      4 → 3:  nt2 → dv2  [G_hat[1,3]]
      4 → 5:  nt2 → dv2  [G_hat[1,3]]
      5 → 1:  nt2 → dn2  [G_hat[1,1]]

=====
Diagram 1L-767
=====

    External legs:
      leaf 0 ← dn1
      leaf 1 ← dn2
    Vertex assignments:
      v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

      coeff =  $\frac{1}{2} nstar_1$ 

      v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

      coeff =  $\frac{1}{2} \varphi_2''$ 

      v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

      coeff =  $\frac{1}{2} nstar_2$ 

      v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

      coeff =  $\frac{1}{2} \varphi_2''$ 

    Edges (propagator assignments):
      2 → 3:  nt1 → dv2  [G_hat[0,3]]
      2 → 5:  nt1 → dv2  [G_hat[0,3]]
      3 → 0:  nt2 → dn1  [G_hat[1,0]]
      4 → 3:  nt2 → dv2  [G_hat[1,3]]
      4 → 5:  nt2 → dv2  [G_hat[1,3]]
      5 → 1:  nt2 → dn2  [G_hat[1,1]]

```

=====

Diagram 1L-768

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

Edges (propagator assignments):

2 → 3: nt1 → dv2 [G_hat[0,3]]

2 → 5: nt1 → dv2 [G_hat[0,3]]

3 → 0: nt2 → dn1 [G_hat[1,0]]

4 → 3: nt2 → dv2 [G_hat[1,3]]

4 → 5: nt2 → dv2 [G_hat[1,3]]

5 → 1: nt2 → dn2 [G_hat[1,1]]

=====

Diagram 1L-769

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

```

    v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

    coeff =  $\frac{1}{2} nstar_1$ 

    v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

    coeff =  $\frac{1}{2} \varphi_1''$ 

    Edges (propagator assignments):
      2 → 3:  nt2 → dv1  [G_hat[1,2]]
      2 → 5:  nt2 → dv1  [G_hat[1,2]]
      3 → 0:  nt1 → dn1  [G_hat[0,0]]
      4 → 3:  nt1 → dv1  [G_hat[0,2]]
      4 → 5:  nt1 → dv1  [G_hat[0,2]]
      5 → 1:  nt1 → dn2  [G_hat[0,1]]

=====
Diagram 1L-770
=====

    External legs:
      leaf 0 ← dn1
      leaf 1 ← dn2
    Vertex assignments:
      v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

      coeff =  $\frac{1}{2} nstar_2$ 

      v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

      coeff =  $\frac{1}{2} \varphi_1''$ 

      v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

      coeff =  $\frac{1}{2} nstar_1$ 

      v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

      coeff =  $\frac{1}{2} \varphi_1''$ 

    Edges (propagator assignments):
      2 → 3:  nt2 → dv1  [G_hat[1,2]]
      2 → 5:  nt2 → dv1  [G_hat[1,2]]
      3 → 0:  nt1 → dn1  [G_hat[0,0]]
      4 → 3:  nt1 → dv1  [G_hat[0,2]]
      4 → 5:  nt1 → dv1  [G_hat[0,2]]
      5 → 1:  nt1 → dn2  [G_hat[0,1]]

```

=====

Diagram 1L-771

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

Edges (propagator assignments):

2 → 3: nt2 → dv1 [G_hat[1,2]]

2 → 5: nt2 → dv1 [G_hat[1,2]]

3 → 0: nt1 → dn1 [G_hat[0,0]]

4 → 3: nt1 → dv1 [G_hat[0,2]]

4 → 5: nt1 → dv1 [G_hat[0,2]]

5 → 1: nt1 → dn2 [G_hat[0,1]]

=====

Diagram 1L-772

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

```

    v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

    coeff =  $\frac{1}{2} nstar_1$ 

    v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

    coeff =  $\frac{1}{2} \varphi_1''$ 

    Edges (propagator assignments):
      2 → 3:  nt2 → dv1  [G_hat[1,2]]
      2 → 5:  nt2 → dv1  [G_hat[1,2]]
      3 → 0:  nt1 → dn1  [G_hat[0,0]]
      4 → 3:  nt1 → dv1  [G_hat[0,2]]
      4 → 5:  nt1 → dv1  [G_hat[0,2]]
      5 → 1:  nt1 → dn2  [G_hat[0,1]]

=====
Diagram 1L-773
=====

    External legs:
      leaf 0 ← dn1
      leaf 1 ← dn2
    Vertex assignments:
      v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

      coeff =  $\frac{1}{2} nstar_2$ 

      v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

      coeff =  $\frac{1}{2} \varphi_1''$ 

      v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

      coeff =  $\frac{1}{2} nstar_1$ 

      v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

      coeff =  $\frac{1}{2} \varphi_1''$ 

    Edges (propagator assignments):
      2 → 3:  nt2 → dv1  [G_hat[1,2]]
      2 → 5:  nt2 → dv1  [G_hat[1,2]]
      3 → 0:  nt1 → dn1  [G_hat[0,0]]
      4 → 3:  nt1 → dv1  [G_hat[0,2]]
      4 → 5:  nt1 → dv1  [G_hat[0,2]]
      5 → 1:  nt1 → dn2  [G_hat[0,1]]

```


=====

Diagram 1L-774

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

Edges (propagator assignments):

2 → 3: nt2 → dv1 [G_hat[1,2]]

2 → 5: nt2 → dv1 [G_hat[1,2]]

3 → 0: nt1 → dn1 [G_hat[0,0]]

4 → 3: nt1 → dv1 [G_hat[0,2]]

4 → 5: nt1 → dv1 [G_hat[0,2]]

5 → 1: nt1 → dn2 [G_hat[0,1]]

=====

Diagram 1L-775

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

```

    v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

    coeff =  $\frac{1}{2} nstar_1$ 

    v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

    coeff =  $\frac{1}{2} \varphi_1''$ 

    Edges (propagator assignments):
      2 → 3:  nt2 → dv1  [G_hat[1,2]]
      2 → 5:  nt2 → dv1  [G_hat[1,2]]
      3 → 0:  nt1 → dn1  [G_hat[0,0]]
      4 → 3:  nt1 → dv1  [G_hat[0,2]]
      4 → 5:  nt1 → dv1  [G_hat[0,2]]
      5 → 1:  nt1 → dn2  [G_hat[0,1]]

=====
Diagram 1L-776
=====

    External legs:
      leaf 0 ← dn1
      leaf 1 ← dn2
    Vertex assignments:
      v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

      coeff =  $\frac{1}{2} nstar_2$ 

      v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

      coeff =  $\frac{1}{2} \varphi_1''$ 

      v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

      coeff =  $\frac{1}{2} nstar_1$ 

      v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

      coeff =  $\frac{1}{2} \varphi_1''$ 

    Edges (propagator assignments):
      2 → 3:  nt2 → dv1  [G_hat[1,2]]
      2 → 5:  nt2 → dv1  [G_hat[1,2]]
      3 → 0:  nt1 → dn1  [G_hat[0,0]]
      4 → 3:  nt1 → dv1  [G_hat[0,2]]
      4 → 5:  nt1 → dv1  [G_hat[0,2]]
      5 → 1:  nt1 → dn2  [G_hat[0,1]]

```

=====

Diagram 1L-777

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

Edges (propagator assignments):

2 → 3: nt2 → dv1 [G_hat[1,2]]

2 → 5: nt2 → dv1 [G_hat[1,2]]

3 → 0: nt1 → dn1 [G_hat[0,0]]

4 → 3: nt1 → dv1 [G_hat[0,2]]

4 → 5: nt1 → dv1 [G_hat[0,2]]

5 → 1: nt1 → dn2 [G_hat[0,1]]

=====

Diagram 1L-778

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

```

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

coeff =  $\frac{1}{2} nstar_1$ 

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

coeff =  $\frac{1}{2} \varphi_1''$ 

Edges (propagator assignments):
  2 → 3:  nt2 → dv1  [G_hat[1,2]]
  2 → 5:  nt2 → dv1  [G_hat[1,2]]
  3 → 0:  nt1 → dn1  [G_hat[0,0]]
  4 → 3:  nt1 → dv1  [G_hat[0,2]]
  4 → 5:  nt1 → dv1  [G_hat[0,2]]
  5 → 1:  nt1 → dn2  [G_hat[0,1]]

=====
Diagram 1L-779
=====

External legs:
  leaf 0 ← dn1
  leaf 1 ← dn2
Vertex assignments:
  v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

coeff =  $\frac{1}{2} nstar_2$ 

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

coeff =  $\frac{1}{2} \varphi_1''$ 

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

coeff =  $\frac{1}{2} nstar_1$ 

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

coeff =  $\frac{1}{2} \varphi_1''$ 

Edges (propagator assignments):
  2 → 3:  nt2 → dv1  [G_hat[1,2]]
  2 → 5:  nt2 → dv1  [G_hat[1,2]]
  3 → 0:  nt1 → dn1  [G_hat[0,0]]
  4 → 3:  nt1 → dv1  [G_hat[0,2]]
  4 → 5:  nt1 → dv1  [G_hat[0,2]]
  5 → 1:  nt1 → dn2  [G_hat[0,1]]

```

=====

Diagram 1L-780

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

Edges (propagator assignments):

2 → 3: nt2 → dv1 [G_hat[1,2]]

2 → 5: nt2 → dv1 [G_hat[1,2]]

3 → 0: nt1 → dn1 [G_hat[0,0]]

4 → 3: nt1 → dv1 [G_hat[0,2]]

4 → 5: nt1 → dv1 [G_hat[0,2]]

5 → 1: nt1 → dn2 [G_hat[0,1]]

=====

Diagram 1L-781

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

```

    v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

    coeff =  $\frac{1}{2} nstar_1$ 

    v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

    coeff =  $\frac{1}{2} \varphi_1''$ 

    Edges (propagator assignments):
      2 → 3:  nt2 → dv1  [G_hat[1,2]]
      2 → 5:  nt2 → dv1  [G_hat[1,2]]
      3 → 0:  nt1 → dn1  [G_hat[0,0]]
      4 → 3:  nt1 → dv1  [G_hat[0,2]]
      4 → 5:  nt1 → dv1  [G_hat[0,2]]
      5 → 1:  nt1 → dn2  [G_hat[0,1]]

=====
Diagram 1L-782
=====

    External legs:
      leaf 0 ← dn1
      leaf 1 ← dn2
    Vertex assignments:
      v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

      coeff =  $\frac{1}{2} nstar_2$ 

      v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

      coeff =  $\frac{1}{2} \varphi_1''$ 

      v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

      coeff =  $\frac{1}{2} nstar_1$ 

      v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

      coeff =  $\frac{1}{2} \varphi_1''$ 

    Edges (propagator assignments):
      2 → 3:  nt2 → dv1  [G_hat[1,2]]
      2 → 5:  nt2 → dv1  [G_hat[1,2]]
      3 → 0:  nt1 → dn1  [G_hat[0,0]]
      4 → 3:  nt1 → dv1  [G_hat[0,2]]
      4 → 5:  nt1 → dv1  [G_hat[0,2]]
      5 → 1:  nt1 → dn2  [G_hat[0,1]]

```

=====

Diagram 1L-783

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

Edges (propagator assignments):

2 → 3: nt2 → dv1 [G_hat[1,2]]

2 → 5: nt2 → dv1 [G_hat[1,2]]

3 → 0: nt1 → dn1 [G_hat[0,0]]

4 → 3: nt1 → dv1 [G_hat[0,2]]

4 → 5: nt1 → dv1 [G_hat[0,2]]

5 → 1: nt1 → dn2 [G_hat[0,1]]

=====

Diagram 1L-784

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

```

    v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

    coeff =  $\frac{1}{2} nstar_1$ 

    v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

    coeff =  $\frac{1}{2} \varphi_1''$ 

    Edges (propagator assignments):
      2 → 3:  nt2 → dv1  [G_hat[1,2]]
      2 → 5:  nt2 → dv1  [G_hat[1,2]]
      3 → 0:  nt1 → dn1  [G_hat[0,0]]
      4 → 3:  nt1 → dv1  [G_hat[0,2]]
      4 → 5:  nt1 → dv1  [G_hat[0,2]]
      5 → 1:  nt1 → dn2  [G_hat[0,1]]

=====
Diagram 1L-785
=====

    External legs:
      leaf 0 ← dn1
      leaf 1 ← dn2
    Vertex assignments:
      v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

      coeff =  $\frac{1}{2} nstar_2$ 

      v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

      coeff =  $\frac{1}{2} \varphi_1''$ 

      v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

      coeff =  $\frac{1}{2} nstar_1$ 

      v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

      coeff =  $\frac{1}{2} \varphi_2''$ 

    Edges (propagator assignments):
      2 → 3:  nt2 → dv1  [G_hat[1,2]]
      2 → 5:  nt2 → dv2  [G_hat[1,3]]
      3 → 0:  nt1 → dn1  [G_hat[0,0]]
      4 → 3:  nt1 → dv1  [G_hat[0,2]]
      4 → 5:  nt1 → dv2  [G_hat[0,3]]
      5 → 1:  nt2 → dn2  [G_hat[1,1]]

```


=====

Diagram 1L-786

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

Edges (propagator assignments):

2 → 3: nt2 → dv1 [G_hat[1,2]]

2 → 5: nt2 → dv2 [G_hat[1,3]]

3 → 0: nt1 → dn1 [G_hat[0,0]]

4 → 3: nt1 → dv1 [G_hat[0,2]]

4 → 5: nt1 → dv2 [G_hat[0,3]]

5 → 1: nt2 → dn2 [G_hat[1,1]]

=====

Diagram 1L-787

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

```

    v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

    coeff =  $\frac{1}{2} nstar_1$ 

    v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

    coeff =  $\frac{1}{2} \varphi_2''$ 

    Edges (propagator assignments):
      2 → 3:  nt2 → dv1  [G_hat[1,2]]
      2 → 5:  nt2 → dv2  [G_hat[1,3]]
      3 → 0:  nt1 → dn1  [G_hat[0,0]]
      4 → 3:  nt1 → dv1  [G_hat[0,2]]
      4 → 5:  nt1 → dv2  [G_hat[0,3]]
      5 → 1:  nt2 → dn2  [G_hat[1,1]]

=====
Diagram 1L-788
=====

    External legs:
      leaf 0 ← dn1
      leaf 1 ← dn2
    Vertex assignments:
      v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

      coeff =  $\frac{1}{2} nstar_2$ 

      v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

      coeff =  $\frac{1}{2} \varphi_1''$ 

      v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

      coeff =  $\frac{1}{2} nstar_1$ 

      v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

      coeff =  $\frac{1}{2} \varphi_2''$ 

    Edges (propagator assignments):
      2 → 3:  nt2 → dv1  [G_hat[1,2]]
      2 → 5:  nt2 → dv2  [G_hat[1,3]]
      3 → 0:  nt1 → dn1  [G_hat[0,0]]
      4 → 3:  nt1 → dv1  [G_hat[0,2]]
      4 → 5:  nt1 → dv2  [G_hat[0,3]]
      5 → 1:  nt2 → dn2  [G_hat[1,1]]

```

=====

Diagram 1L-789

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

Edges (propagator assignments):

2 → 3: nt2 → dv1 [G_hat[1,2]]

2 → 5: nt2 → dv2 [G_hat[1,3]]

3 → 0: nt1 → dn1 [G_hat[0,0]]

4 → 3: nt1 → dv1 [G_hat[0,2]]

4 → 5: nt1 → dv2 [G_hat[0,3]]

5 → 1: nt2 → dn2 [G_hat[1,1]]

=====

Diagram 1L-790

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

```

    v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

    coeff =  $\frac{1}{2} nstar_1$ 

    v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

    coeff =  $\frac{1}{2} \varphi_2''$ 

    Edges (propagator assignments):
      2 → 3:  nt2 → dv1  [G_hat[1,2]]
      2 → 5:  nt2 → dv2  [G_hat[1,3]]
      3 → 0:  nt1 → dn1  [G_hat[0,0]]
      4 → 3:  nt1 → dv1  [G_hat[0,2]]
      4 → 5:  nt1 → dv2  [G_hat[0,3]]
      5 → 1:  nt2 → dn2  [G_hat[1,1]]

=====
Diagram 1L-791
=====

    External legs:
      leaf 0 ← dn1
      leaf 1 ← dn2
    Vertex assignments:
      v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

      coeff =  $\frac{1}{2} nstar_2$ 

      v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

      coeff =  $\frac{1}{2} \varphi_1''$ 

      v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

      coeff =  $\frac{1}{2} nstar_1$ 

      v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

      coeff =  $\frac{1}{2} \varphi_2''$ 

    Edges (propagator assignments):
      2 → 3:  nt2 → dv1  [G_hat[1,2]]
      2 → 5:  nt2 → dv2  [G_hat[1,3]]
      3 → 0:  nt1 → dn1  [G_hat[0,0]]
      4 → 3:  nt1 → dv1  [G_hat[0,2]]
      4 → 5:  nt1 → dv2  [G_hat[0,3]]
      5 → 1:  nt2 → dn2  [G_hat[1,1]]

```

=====

Diagram 1L-792

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

Edges (propagator assignments):

2 → 3: nt2 → dv1 [G_hat[1,2]]

2 → 5: nt2 → dv2 [G_hat[1,3]]

3 → 0: nt1 → dn1 [G_hat[0,0]]

4 → 3: nt1 → dv1 [G_hat[0,2]]

4 → 5: nt1 → dv2 [G_hat[0,3]]

5 → 1: nt2 → dn2 [G_hat[1,1]]

=====

Diagram 1L-793

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

```

    v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

coeff =  $\frac{1}{2} nstar_1$ 

    v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

coeff =  $\frac{1}{2} \varphi_2''$ 

Edges (propagator assignments):
    2 → 3:  nt2 → dv1  [G_hat[1,2]]
    2 → 5:  nt2 → dv2  [G_hat[1,3]]
    3 → 0:  nt1 → dn1  [G_hat[0,0]]
    4 → 3:  nt1 → dv1  [G_hat[0,2]]
    4 → 5:  nt1 → dv2  [G_hat[0,3]]
    5 → 1:  nt2 → dn2  [G_hat[1,1]]

=====
Diagram 1L-794
=====

External legs:
    leaf 0 ← dn1
    leaf 1 ← dn2
Vertex assignments:
    v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

coeff =  $\frac{1}{2} nstar_2$ 

    v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

coeff =  $\frac{1}{2} \varphi_1''$ 

    v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

coeff =  $\frac{1}{2} nstar_1$ 

    v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

coeff =  $\frac{1}{2} \varphi_2''$ 

Edges (propagator assignments):
    2 → 3:  nt2 → dv1  [G_hat[1,2]]
    2 → 5:  nt2 → dv2  [G_hat[1,3]]
    3 → 0:  nt1 → dn1  [G_hat[0,0]]
    4 → 3:  nt1 → dv1  [G_hat[0,2]]
    4 → 5:  nt1 → dv2  [G_hat[0,3]]
    5 → 1:  nt2 → dn2  [G_hat[1,1]]

```

=====

Diagram 1L-795

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

Edges (propagator assignments):

2 → 3: nt2 → dv1 [G_hat[1,2]]

2 → 5: nt2 → dv2 [G_hat[1,3]]

3 → 0: nt1 → dn1 [G_hat[0,0]]

4 → 3: nt1 → dv1 [G_hat[0,2]]

4 → 5: nt1 → dv2 [G_hat[0,3]]

5 → 1: nt2 → dn2 [G_hat[1,1]]

=====

Diagram 1L-796

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

```

    v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

    coeff =  $\frac{1}{2} nstar_1$ 

    v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

    coeff =  $\frac{1}{2} \varphi_2''$ 

    Edges (propagator assignments):
      2 → 3:  nt2 → dv1  [G_hat[1,2]]
      2 → 5:  nt2 → dv2  [G_hat[1,3]]
      3 → 0:  nt1 → dn1  [G_hat[0,0]]
      4 → 3:  nt1 → dv1  [G_hat[0,2]]
      4 → 5:  nt1 → dv2  [G_hat[0,3]]
      5 → 1:  nt2 → dn2  [G_hat[1,1]]

=====
Diagram 1L-797
=====

    External legs:
      leaf 0 ← dn1
      leaf 1 ← dn2
    Vertex assignments:
      v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

      coeff =  $\frac{1}{2} nstar_2$ 

      v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

      coeff =  $\frac{1}{2} \varphi_1''$ 

      v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

      coeff =  $\frac{1}{2} nstar_1$ 

      v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

      coeff =  $\frac{1}{2} \varphi_2''$ 

    Edges (propagator assignments):
      2 → 3:  nt2 → dv1  [G_hat[1,2]]
      2 → 5:  nt2 → dv2  [G_hat[1,3]]
      3 → 0:  nt1 → dn1  [G_hat[0,0]]
      4 → 3:  nt1 → dv1  [G_hat[0,2]]
      4 → 5:  nt1 → dv2  [G_hat[0,3]]
      5 → 1:  nt2 → dn2  [G_hat[1,1]]

```


=====

Diagram 1L-798

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

Edges (propagator assignments):

2 → 3: nt2 → dv1 [G_hat[1,2]]

2 → 5: nt2 → dv2 [G_hat[1,3]]

3 → 0: nt1 → dn1 [G_hat[0,0]]

4 → 3: nt1 → dv1 [G_hat[0,2]]

4 → 5: nt1 → dv2 [G_hat[0,3]]

5 → 1: nt2 → dn2 [G_hat[1,1]]

=====

Diagram 1L-799

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External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

```

    v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

    coeff =  $\frac{1}{2} nstar_1$ 

    v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

    coeff =  $\frac{1}{2} \varphi_2''$ 

    Edges (propagator assignments):
      2 → 3:  nt2 → dv1  [G_hat[1,2]]
      2 → 5:  nt2 → dv2  [G_hat[1,3]]
      3 → 0:  nt1 → dn1  [G_hat[0,0]]
      4 → 3:  nt1 → dv1  [G_hat[0,2]]
      4 → 5:  nt1 → dv2  [G_hat[0,3]]
      5 → 1:  nt2 → dn2  [G_hat[1,1]]

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Diagram 1L-800
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    External legs:
      leaf 0 ← dn1
      leaf 1 ← dn2
    Vertex assignments:
      v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

      coeff =  $\frac{1}{2} nstar_2$ 

      v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

      coeff =  $\frac{1}{2} \varphi_1''$ 

      v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

      coeff =  $\frac{1}{2} nstar_1$ 

      v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

      coeff =  $\frac{1}{2} \varphi_2''$ 

    Edges (propagator assignments):
      2 → 3:  nt2 → dv1  [G_hat[1,2]]
      2 → 5:  nt2 → dv2  [G_hat[1,3]]
      3 → 0:  nt1 → dn1  [G_hat[0,0]]
      4 → 3:  nt1 → dv1  [G_hat[0,2]]
      4 → 5:  nt1 → dv2  [G_hat[0,3]]
      5 → 1:  nt2 → dn2  [G_hat[1,1]]

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Diagram 1L-801

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External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

Edges (propagator assignments):

2 → 3: nt2 → dv2 [G_hat[1,3]]

2 → 5: nt2 → dv1 [G_hat[1,2]]

3 → 0: nt2 → dn1 [G_hat[1,0]]

4 → 3: nt1 → dv2 [G_hat[0,3]]

4 → 5: nt1 → dv1 [G_hat[0,2]]

5 → 1: nt1 → dn2 [G_hat[0,1]]

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Diagram 1L-802

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External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

```

    v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

    coeff =  $\frac{1}{2} nstar_1$ 

    v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

    coeff =  $\frac{1}{2} \varphi_1''$ 

    Edges (propagator assignments):
      2 → 3:  nt2 → dv2  [G_hat[1,3]]
      2 → 5:  nt2 → dv1  [G_hat[1,2]]
      3 → 0:  nt2 → dn1  [G_hat[1,0]]
      4 → 3:  nt1 → dv2  [G_hat[0,3]]
      4 → 5:  nt1 → dv1  [G_hat[0,2]]
      5 → 1:  nt1 → dn2  [G_hat[0,1]]

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Diagram 1L-803
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    External legs:
      leaf 0 ← dn1
      leaf 1 ← dn2
    Vertex assignments:
      v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

      coeff =  $\frac{1}{2} nstar_2$ 

      v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

      coeff =  $\frac{1}{2} \varphi_2''$ 

      v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

      coeff =  $\frac{1}{2} nstar_1$ 

      v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

      coeff =  $\frac{1}{2} \varphi_1''$ 

    Edges (propagator assignments):
      2 → 3:  nt2 → dv2  [G_hat[1,3]]
      2 → 5:  nt2 → dv1  [G_hat[1,2]]
      3 → 0:  nt2 → dn1  [G_hat[1,0]]
      4 → 3:  nt1 → dv2  [G_hat[0,3]]
      4 → 5:  nt1 → dv1  [G_hat[0,2]]
      5 → 1:  nt1 → dn2  [G_hat[0,1]]

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Diagram 1L-804

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External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

Edges (propagator assignments):

2 → 3: nt2 → dv2 [G_hat[1,3]]

2 → 5: nt2 → dv1 [G_hat[1,2]]

3 → 0: nt2 → dn1 [G_hat[1,0]]

4 → 3: nt1 → dv2 [G_hat[0,3]]

4 → 5: nt1 → dv1 [G_hat[0,2]]

5 → 1: nt1 → dn2 [G_hat[0,1]]

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Diagram 1L-805

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External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

```

    v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

    coeff =  $\frac{1}{2} nstar_1$ 

    v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

    coeff =  $\frac{1}{2} \varphi_1''$ 

    Edges (propagator assignments):
      2 → 3:  nt2 → dv2  [G_hat[1,3]]
      2 → 5:  nt2 → dv1  [G_hat[1,2]]
      3 → 0:  nt2 → dn1  [G_hat[1,0]]
      4 → 3:  nt1 → dv2  [G_hat[0,3]]
      4 → 5:  nt1 → dv1  [G_hat[0,2]]
      5 → 1:  nt1 → dn2  [G_hat[0,1]]

=====
Diagram 1L-806
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    External legs:
      leaf 0 ← dn1
      leaf 1 ← dn2
    Vertex assignments:
      v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

      coeff =  $\frac{1}{2} nstar_2$ 

      v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

      coeff =  $\frac{1}{2} \varphi_2''$ 

      v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

      coeff =  $\frac{1}{2} nstar_1$ 

      v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

      coeff =  $\frac{1}{2} \varphi_1''$ 

    Edges (propagator assignments):
      2 → 3:  nt2 → dv2  [G_hat[1,3]]
      2 → 5:  nt2 → dv1  [G_hat[1,2]]
      3 → 0:  nt2 → dn1  [G_hat[1,0]]
      4 → 3:  nt1 → dv2  [G_hat[0,3]]
      4 → 5:  nt1 → dv1  [G_hat[0,2]]
      5 → 1:  nt1 → dn2  [G_hat[0,1]]

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Diagram 1L-807

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External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

Edges (propagator assignments):

2 → 3: nt2 → dv2 [G_hat[1,3]]

2 → 5: nt2 → dv1 [G_hat[1,2]]

3 → 0: nt2 → dn1 [G_hat[1,0]]

4 → 3: nt1 → dv2 [G_hat[0,3]]

4 → 5: nt1 → dv1 [G_hat[0,2]]

5 → 1: nt1 → dn2 [G_hat[0,1]]

=====

Diagram 1L-808

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

```

    v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

    coeff =  $\frac{1}{2} nstar_1$ 

    v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

    coeff =  $\frac{1}{2} \varphi_1''$ 

    Edges (propagator assignments):
      2 → 3:  nt2 → dv2  [G_hat[1,3]]
      2 → 5:  nt2 → dv1  [G_hat[1,2]]
      3 → 0:  nt2 → dn1  [G_hat[1,0]]
      4 → 3:  nt1 → dv2  [G_hat[0,3]]
      4 → 5:  nt1 → dv1  [G_hat[0,2]]
      5 → 1:  nt1 → dn2  [G_hat[0,1]]

=====
Diagram 1L-809
=====

    External legs:
      leaf 0 ← dn1
      leaf 1 ← dn2
    Vertex assignments:
      v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

      coeff =  $\frac{1}{2} nstar_2$ 

      v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

      coeff =  $\frac{1}{2} \varphi_2''$ 

      v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

      coeff =  $\frac{1}{2} nstar_1$ 

      v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

      coeff =  $\frac{1}{2} \varphi_1''$ 

    Edges (propagator assignments):
      2 → 3:  nt2 → dv2  [G_hat[1,3]]
      2 → 5:  nt2 → dv1  [G_hat[1,2]]
      3 → 0:  nt2 → dn1  [G_hat[1,0]]
      4 → 3:  nt1 → dv2  [G_hat[0,3]]
      4 → 5:  nt1 → dv1  [G_hat[0,2]]
      5 → 1:  nt1 → dn2  [G_hat[0,1]]

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Diagram 1L-810

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

Edges (propagator assignments):

2 → 3: nt2 → dv2 [G_hat[1,3]]

2 → 5: nt2 → dv1 [G_hat[1,2]]

3 → 0: nt2 → dn1 [G_hat[1,0]]

4 → 3: nt1 → dv2 [G_hat[0,3]]

4 → 5: nt1 → dv1 [G_hat[0,2]]

5 → 1: nt1 → dn2 [G_hat[0,1]]

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Diagram 1L-811

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

```

    v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

    coeff =  $\frac{1}{2} nstar_1$ 

    v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

    coeff =  $\frac{1}{2} \varphi_1''$ 

    Edges (propagator assignments):
      2 → 3:  nt2 → dv2  [G_hat[1,3]]
      2 → 5:  nt2 → dv1  [G_hat[1,2]]
      3 → 0:  nt2 → dn1  [G_hat[1,0]]
      4 → 3:  nt1 → dv2  [G_hat[0,3]]
      4 → 5:  nt1 → dv1  [G_hat[0,2]]
      5 → 1:  nt1 → dn2  [G_hat[0,1]]

=====
Diagram 1L-812
=====

    External legs:
      leaf 0 ← dn1
      leaf 1 ← dn2
    Vertex assignments:
      v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

      coeff =  $\frac{1}{2} nstar_2$ 

      v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

      coeff =  $\frac{1}{2} \varphi_2''$ 

      v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

      coeff =  $\frac{1}{2} nstar_1$ 

      v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

      coeff =  $\frac{1}{2} \varphi_1''$ 

    Edges (propagator assignments):
      2 → 3:  nt2 → dv2  [G_hat[1,3]]
      2 → 5:  nt2 → dv1  [G_hat[1,2]]
      3 → 0:  nt2 → dn1  [G_hat[1,0]]
      4 → 3:  nt1 → dv2  [G_hat[0,3]]
      4 → 5:  nt1 → dv1  [G_hat[0,2]]
      5 → 1:  nt1 → dn2  [G_hat[0,1]]

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Diagram 1L-813

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

Edges (propagator assignments):

2 → 3: nt2 → dv2 [G_hat[1,3]]

2 → 5: nt2 → dv1 [G_hat[1,2]]

3 → 0: nt2 → dn1 [G_hat[1,0]]

4 → 3: nt1 → dv2 [G_hat[0,3]]

4 → 5: nt1 → dv1 [G_hat[0,2]]

5 → 1: nt1 → dn2 [G_hat[0,1]]

=====

Diagram 1L-814

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

```

    v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

    coeff =  $\frac{1}{2} nstar_1$ 

    v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

    coeff =  $\frac{1}{2} \varphi_1''$ 

    Edges (propagator assignments):
      2 → 3:  nt2 → dv2  [G_hat[1,3]]
      2 → 5:  nt2 → dv1  [G_hat[1,2]]
      3 → 0:  nt2 → dn1  [G_hat[1,0]]
      4 → 3:  nt1 → dv2  [G_hat[0,3]]
      4 → 5:  nt1 → dv1  [G_hat[0,2]]
      5 → 1:  nt1 → dn2  [G_hat[0,1]]

=====
Diagram 1L-815
=====

    External legs:
      leaf 0 ← dn1
      leaf 1 ← dn2
    Vertex assignments:
      v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

      coeff =  $\frac{1}{2} nstar_2$ 

      v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

      coeff =  $\frac{1}{2} \varphi_2''$ 

      v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

      coeff =  $\frac{1}{2} nstar_1$ 

      v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

      coeff =  $\frac{1}{2} \varphi_1''$ 

    Edges (propagator assignments):
      2 → 3:  nt2 → dv2  [G_hat[1,3]]
      2 → 5:  nt2 → dv1  [G_hat[1,2]]
      3 → 0:  nt2 → dn1  [G_hat[1,0]]
      4 → 3:  nt1 → dv2  [G_hat[0,3]]
      4 → 5:  nt1 → dv1  [G_hat[0,2]]
      5 → 1:  nt1 → dn2  [G_hat[0,1]]

```

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Diagram 1L-816

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

Edges (propagator assignments):

2 → 3: nt2 → dv2 [G_hat[1,3]]

2 → 5: nt2 → dv1 [G_hat[1,2]]

3 → 0: nt2 → dn1 [G_hat[1,0]]

4 → 3: nt1 → dv2 [G_hat[0,3]]

4 → 5: nt1 → dv1 [G_hat[0,2]]

5 → 1: nt1 → dn2 [G_hat[0,1]]

=====

Diagram 1L-817

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

```

    v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

coeff =  $\frac{1}{2} nstar_1$ 

    v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

coeff =  $\frac{1}{2} \varphi_2''$ 

Edges (propagator assignments):
    2 → 3:  nt2 → dv2  [G_hat[1,3]]
    2 → 5:  nt2 → dv2  [G_hat[1,3]]
    3 → 0:  nt2 → dn1  [G_hat[1,0]]
    4 → 3:  nt1 → dv2  [G_hat[0,3]]
    4 → 5:  nt1 → dv2  [G_hat[0,3]]
    5 → 1:  nt2 → dn2  [G_hat[1,1]]

=====
Diagram 1L-818
=====

External legs:
    leaf 0 ← dn1
    leaf 1 ← dn2
Vertex assignments:
    v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

coeff =  $\frac{1}{2} nstar_2$ 

    v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

coeff =  $\frac{1}{2} \varphi_2''$ 

    v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

coeff =  $\frac{1}{2} nstar_1$ 

    v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

coeff =  $\frac{1}{2} \varphi_2''$ 

Edges (propagator assignments):
    2 → 3:  nt2 → dv2  [G_hat[1,3]]
    2 → 5:  nt2 → dv2  [G_hat[1,3]]
    3 → 0:  nt2 → dn1  [G_hat[1,0]]
    4 → 3:  nt1 → dv2  [G_hat[0,3]]
    4 → 5:  nt1 → dv2  [G_hat[0,3]]
    5 → 1:  nt2 → dn2  [G_hat[1,1]]

```

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Diagram 1L-819

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

Edges (propagator assignments):

2 → 3: nt2 → dv2 [G_hat[1,3]]

2 → 5: nt2 → dv2 [G_hat[1,3]]

3 → 0: nt2 → dn1 [G_hat[1,0]]

4 → 3: nt1 → dv2 [G_hat[0,3]]

4 → 5: nt1 → dv2 [G_hat[0,3]]

5 → 1: nt2 → dn2 [G_hat[1,1]]

=====

Diagram 1L-820

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

```

    v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

coeff =  $\frac{1}{2} nstar_1$ 

    v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

coeff =  $\frac{1}{2} \varphi_2''$ 

Edges (propagator assignments):
    2 → 3:  nt2 → dv2  [G_hat[1,3]]
    2 → 5:  nt2 → dv2  [G_hat[1,3]]
    3 → 0:  nt2 → dn1  [G_hat[1,0]]
    4 → 3:  nt1 → dv2  [G_hat[0,3]]
    4 → 5:  nt1 → dv2  [G_hat[0,3]]
    5 → 1:  nt2 → dn2  [G_hat[1,1]]

=====
Diagram 1L-821
=====

External legs:
    leaf 0 ← dn1
    leaf 1 ← dn2
Vertex assignments:
    v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

coeff =  $\frac{1}{2} nstar_2$ 

    v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

coeff =  $\frac{1}{2} \varphi_2''$ 

    v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

coeff =  $\frac{1}{2} nstar_1$ 

    v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

coeff =  $\frac{1}{2} \varphi_2''$ 

Edges (propagator assignments):
    2 → 3:  nt2 → dv2  [G_hat[1,3]]
    2 → 5:  nt2 → dv2  [G_hat[1,3]]
    3 → 0:  nt2 → dn1  [G_hat[1,0]]
    4 → 3:  nt1 → dv2  [G_hat[0,3]]
    4 → 5:  nt1 → dv2  [G_hat[0,3]]
    5 → 1:  nt2 → dn2  [G_hat[1,1]]

```


=====

Diagram 1L-822

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

Edges (propagator assignments):

2 → 3: nt2 → dv2 [G_hat[1,3]]

2 → 5: nt2 → dv2 [G_hat[1,3]]

3 → 0: nt2 → dn1 [G_hat[1,0]]

4 → 3: nt1 → dv2 [G_hat[0,3]]

4 → 5: nt1 → dv2 [G_hat[0,3]]

5 → 1: nt2 → dn2 [G_hat[1,1]]

=====

Diagram 1L-823

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

```

    v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

    coeff =  $\frac{1}{2} nstar_1$ 

    v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

    coeff =  $\frac{1}{2} \varphi_2''$ 

    Edges (propagator assignments):
      2 → 3:  nt2 → dv2  [G_hat[1,3]]
      2 → 5:  nt2 → dv2  [G_hat[1,3]]
      3 → 0:  nt2 → dn1  [G_hat[1,0]]
      4 → 3:  nt1 → dv2  [G_hat[0,3]]
      4 → 5:  nt1 → dv2  [G_hat[0,3]]
      5 → 1:  nt2 → dn2  [G_hat[1,1]]

=====
Diagram 1L-824
=====

    External legs:
      leaf 0 ← dn1
      leaf 1 ← dn2
    Vertex assignments:
      v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

      coeff =  $\frac{1}{2} nstar_2$ 

      v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

      coeff =  $\frac{1}{2} \varphi_2''$ 

      v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

      coeff =  $\frac{1}{2} nstar_1$ 

      v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

      coeff =  $\frac{1}{2} \varphi_2''$ 

    Edges (propagator assignments):
      2 → 3:  nt2 → dv2  [G_hat[1,3]]
      2 → 5:  nt2 → dv2  [G_hat[1,3]]
      3 → 0:  nt2 → dn1  [G_hat[1,0]]
      4 → 3:  nt1 → dv2  [G_hat[0,3]]
      4 → 5:  nt1 → dv2  [G_hat[0,3]]
      5 → 1:  nt2 → dn2  [G_hat[1,1]]

```

=====

Diagram 1L-825

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

Edges (propagator assignments):

2 → 3: nt2 → dv2 [G_hat[1,3]]

2 → 5: nt2 → dv2 [G_hat[1,3]]

3 → 0: nt2 → dn1 [G_hat[1,0]]

4 → 3: nt1 → dv2 [G_hat[0,3]]

4 → 5: nt1 → dv2 [G_hat[0,3]]

5 → 1: nt2 → dn2 [G_hat[1,1]]

=====

Diagram 1L-826

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

```

    v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

    coeff =  $\frac{1}{2} nstar_1$ 

    v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

    coeff =  $\frac{1}{2} \varphi_2''$ 

    Edges (propagator assignments):
      2 → 3:  nt2 → dv2  [G_hat[1,3]]
      2 → 5:  nt2 → dv2  [G_hat[1,3]]
      3 → 0:  nt2 → dn1  [G_hat[1,0]]
      4 → 3:  nt1 → dv2  [G_hat[0,3]]
      4 → 5:  nt1 → dv2  [G_hat[0,3]]
      5 → 1:  nt2 → dn2  [G_hat[1,1]]

=====
Diagram 1L-827
=====

    External legs:
      leaf 0 ← dn1
      leaf 1 ← dn2
    Vertex assignments:
      v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

      coeff =  $\frac{1}{2} nstar_2$ 

      v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

      coeff =  $\frac{1}{2} \varphi_2''$ 

      v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

      coeff =  $\frac{1}{2} nstar_1$ 

      v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

      coeff =  $\frac{1}{2} \varphi_2''$ 

    Edges (propagator assignments):
      2 → 3:  nt2 → dv2  [G_hat[1,3]]
      2 → 5:  nt2 → dv2  [G_hat[1,3]]
      3 → 0:  nt2 → dn1  [G_hat[1,0]]
      4 → 3:  nt1 → dv2  [G_hat[0,3]]
      4 → 5:  nt1 → dv2  [G_hat[0,3]]
      5 → 1:  nt2 → dn2  [G_hat[1,1]]

```

=====

Diagram 1L-828

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

Edges (propagator assignments):

2 → 3: nt2 → dv2 [G_hat[1,3]]

2 → 5: nt2 → dv2 [G_hat[1,3]]

3 → 0: nt2 → dn1 [G_hat[1,0]]

4 → 3: nt1 → dv2 [G_hat[0,3]]

4 → 5: nt1 → dv2 [G_hat[0,3]]

5 → 1: nt2 → dn2 [G_hat[1,1]]

=====

Diagram 1L-829

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

```

    v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

    coeff =  $\frac{1}{2} nstar_1$ 

    v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

    coeff =  $\frac{1}{2} \varphi_2''$ 

    Edges (propagator assignments):
      2 → 3:  nt2 → dv2  [G_hat[1,3]]
      2 → 5:  nt2 → dv2  [G_hat[1,3]]
      3 → 0:  nt2 → dn1  [G_hat[1,0]]
      4 → 3:  nt1 → dv2  [G_hat[0,3]]
      4 → 5:  nt1 → dv2  [G_hat[0,3]]
      5 → 1:  nt2 → dn2  [G_hat[1,1]]

=====
Diagram 1L-830
=====

    External legs:
      leaf 0 ← dn1
      leaf 1 ← dn2
    Vertex assignments:
      v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

      coeff =  $\frac{1}{2} nstar_2$ 

      v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

      coeff =  $\frac{1}{2} \varphi_2''$ 

      v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

      coeff =  $\frac{1}{2} nstar_1$ 

      v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

      coeff =  $\frac{1}{2} \varphi_2''$ 

    Edges (propagator assignments):
      2 → 3:  nt2 → dv2  [G_hat[1,3]]
      2 → 5:  nt2 → dv2  [G_hat[1,3]]
      3 → 0:  nt2 → dn1  [G_hat[1,0]]
      4 → 3:  nt1 → dv2  [G_hat[0,3]]
      4 → 5:  nt1 → dv2  [G_hat[0,3]]
      5 → 1:  nt2 → dn2  [G_hat[1,1]]

```

=====

Diagram 1L-831

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

Edges (propagator assignments):

2 → 3: nt2 → dv2 [G_hat[1,3]]

2 → 5: nt2 → dv2 [G_hat[1,3]]

3 → 0: nt2 → dn1 [G_hat[1,0]]

4 → 3: nt1 → dv2 [G_hat[0,3]]

4 → 5: nt1 → dv2 [G_hat[0,3]]

5 → 1: nt2 → dn2 [G_hat[1,1]]

=====

Diagram 1L-832

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

```

    v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

    coeff =  $\frac{1}{2} nstar_1$ 

    v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

    coeff =  $\frac{1}{2} \varphi_2''$ 

    Edges (propagator assignments):
      2 → 3:  nt2 → dv2  [G_hat[1,3]]
      2 → 5:  nt2 → dv2  [G_hat[1,3]]
      3 → 0:  nt2 → dn1  [G_hat[1,0]]
      4 → 3:  nt1 → dv2  [G_hat[0,3]]
      4 → 5:  nt1 → dv2  [G_hat[0,3]]
      5 → 1:  nt2 → dn2  [G_hat[1,1]]

=====
Diagram 1L-833
=====

    External legs:
      leaf 0 ← dn1
      leaf 1 ← dn2
    Vertex assignments:
      v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

      coeff =  $\frac{1}{2} nstar_2$ 

      v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

      coeff =  $\frac{1}{2} \varphi_1''$ 

      v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

      coeff =  $\frac{1}{2} nstar_2$ 

      v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

      coeff =  $\frac{1}{2} \varphi_1''$ 

    Edges (propagator assignments):
      2 → 3:  nt2 → dv1  [G_hat[1,2]]
      2 → 5:  nt2 → dv1  [G_hat[1,2]]
      3 → 0:  nt1 → dn1  [G_hat[0,0]]
      4 → 3:  nt2 → dv1  [G_hat[1,2]]
      4 → 5:  nt2 → dv1  [G_hat[1,2]]
      5 → 1:  nt1 → dn2  [G_hat[0,1]]

```


=====

Diagram 1L-834

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

Edges (propagator assignments):

2 → 3: nt2 → dv1 [G_hat[1,2]]

2 → 5: nt2 → dv1 [G_hat[1,2]]

3 → 0: nt1 → dn1 [G_hat[0,0]]

4 → 3: nt2 → dv1 [G_hat[1,2]]

4 → 5: nt2 → dv1 [G_hat[1,2]]

5 → 1: nt1 → dn2 [G_hat[0,1]]

=====

Diagram 1L-835

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

```

    v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

    coeff =  $\frac{1}{2} nstar_2$ 

    v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

    coeff =  $\frac{1}{2} \varphi_1''$ 

    Edges (propagator assignments):
      2 → 3:  nt2 → dv1  [G_hat[1,2]]
      2 → 5:  nt2 → dv1  [G_hat[1,2]]
      3 → 0:  nt1 → dn1  [G_hat[0,0]]
      4 → 3:  nt2 → dv1  [G_hat[1,2]]
      4 → 5:  nt2 → dv1  [G_hat[1,2]]
      5 → 1:  nt1 → dn2  [G_hat[0,1]]

=====
Diagram 1L-836
=====

    External legs:
      leaf 0 ← dn1
      leaf 1 ← dn2
    Vertex assignments:
      v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

      coeff =  $\frac{1}{2} nstar_2$ 

      v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

      coeff =  $\frac{1}{2} \varphi_1''$ 

      v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

      coeff =  $\frac{1}{2} nstar_2$ 

      v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

      coeff =  $\frac{1}{2} \varphi_1''$ 

    Edges (propagator assignments):
      2 → 3:  nt2 → dv1  [G_hat[1,2]]
      2 → 5:  nt2 → dv1  [G_hat[1,2]]
      3 → 0:  nt1 → dn1  [G_hat[0,0]]
      4 → 3:  nt2 → dv1  [G_hat[1,2]]
      4 → 5:  nt2 → dv1  [G_hat[1,2]]
      5 → 1:  nt1 → dn2  [G_hat[0,1]]

```

=====

Diagram 1L-837

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

Edges (propagator assignments):

2 → 3: nt2 → dv1 [G_hat[1,2]]

2 → 5: nt2 → dv1 [G_hat[1,2]]

3 → 0: nt1 → dn1 [G_hat[0,0]]

4 → 3: nt2 → dv1 [G_hat[1,2]]

4 → 5: nt2 → dv1 [G_hat[1,2]]

5 → 1: nt1 → dn2 [G_hat[0,1]]

=====

Diagram 1L-838

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

```

    v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

    coeff =  $\frac{1}{2} nstar_2$ 

    v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

    coeff =  $\frac{1}{2} \varphi_1''$ 

    Edges (propagator assignments):
      2 → 3:  nt2 → dv1  [G_hat[1,2]]
      2 → 5:  nt2 → dv1  [G_hat[1,2]]
      3 → 0:  nt1 → dn1  [G_hat[0,0]]
      4 → 3:  nt2 → dv1  [G_hat[1,2]]
      4 → 5:  nt2 → dv1  [G_hat[1,2]]
      5 → 1:  nt1 → dn2  [G_hat[0,1]]

=====
Diagram 1L-839
=====

    External legs:
      leaf 0 ← dn1
      leaf 1 ← dn2
    Vertex assignments:
      v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

      coeff =  $\frac{1}{2} nstar_2$ 

      v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

      coeff =  $\frac{1}{2} \varphi_1''$ 

      v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

      coeff =  $\frac{1}{2} nstar_2$ 

      v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

      coeff =  $\frac{1}{2} \varphi_1''$ 

    Edges (propagator assignments):
      2 → 3:  nt2 → dv1  [G_hat[1,2]]
      2 → 5:  nt2 → dv1  [G_hat[1,2]]
      3 → 0:  nt1 → dn1  [G_hat[0,0]]
      4 → 3:  nt2 → dv1  [G_hat[1,2]]
      4 → 5:  nt2 → dv1  [G_hat[1,2]]
      5 → 1:  nt1 → dn2  [G_hat[0,1]]

```

=====

Diagram 1L-840

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

Edges (propagator assignments):

2 → 3: nt2 → dv1 [G_hat[1,2]]

2 → 5: nt2 → dv1 [G_hat[1,2]]

3 → 0: nt1 → dn1 [G_hat[0,0]]

4 → 3: nt2 → dv1 [G_hat[1,2]]

4 → 5: nt2 → dv1 [G_hat[1,2]]

5 → 1: nt1 → dn2 [G_hat[0,1]]

=====

Diagram 1L-841

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

```

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

coeff =  $\frac{1}{2} nstar_2$ 

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

coeff =  $\frac{1}{2} \varphi_1''$ 

Edges (propagator assignments):
  2 → 3:  nt2 → dv1  [G_hat[1,2]]
  2 → 5:  nt2 → dv1  [G_hat[1,2]]
  3 → 0:  nt1 → dn1  [G_hat[0,0]]
  4 → 3:  nt2 → dv1  [G_hat[1,2]]
  4 → 5:  nt2 → dv1  [G_hat[1,2]]
  5 → 1:  nt1 → dn2  [G_hat[0,1]]

=====
Diagram 1L-842
=====

External legs:
  leaf 0 ← dn1
  leaf 1 ← dn2
Vertex assignments:
  v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

coeff =  $\frac{1}{2} nstar_2$ 

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

coeff =  $\frac{1}{2} \varphi_1''$ 

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

coeff =  $\frac{1}{2} nstar_2$ 

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

coeff =  $\frac{1}{2} \varphi_1''$ 

Edges (propagator assignments):
  2 → 3:  nt2 → dv1  [G_hat[1,2]]
  2 → 5:  nt2 → dv1  [G_hat[1,2]]
  3 → 0:  nt1 → dn1  [G_hat[0,0]]
  4 → 3:  nt2 → dv1  [G_hat[1,2]]
  4 → 5:  nt2 → dv1  [G_hat[1,2]]
  5 → 1:  nt1 → dn2  [G_hat[0,1]]

```

=====

Diagram 1L-843

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

Edges (propagator assignments):

2 → 3: nt2 → dv1 [G_hat[1,2]]

2 → 5: nt2 → dv1 [G_hat[1,2]]

3 → 0: nt1 → dn1 [G_hat[0,0]]

4 → 3: nt2 → dv1 [G_hat[1,2]]

4 → 5: nt2 → dv1 [G_hat[1,2]]

5 → 1: nt1 → dn2 [G_hat[0,1]]

=====

Diagram 1L-844

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

```

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

coeff =  $\frac{1}{2} nstar_2$ 

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

coeff =  $\frac{1}{2} \varphi_1''$ 

Edges (propagator assignments):
  2 → 3:  nt2 → dv1  [G_hat[1,2]]
  2 → 5:  nt2 → dv1  [G_hat[1,2]]
  3 → 0:  nt1 → dn1  [G_hat[0,0]]
  4 → 3:  nt2 → dv1  [G_hat[1,2]]
  4 → 5:  nt2 → dv1  [G_hat[1,2]]
  5 → 1:  nt1 → dn2  [G_hat[0,1]]

=====
Diagram 1L-845
=====

External legs:
  leaf 0 ← dn1
  leaf 1 ← dn2
Vertex assignments:
  v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

coeff =  $\frac{1}{2} nstar_2$ 

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

coeff =  $\frac{1}{2} \varphi_1''$ 

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

coeff =  $\frac{1}{2} nstar_2$ 

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

coeff =  $\frac{1}{2} \varphi_1''$ 

Edges (propagator assignments):
  2 → 3:  nt2 → dv1  [G_hat[1,2]]
  2 → 5:  nt2 → dv1  [G_hat[1,2]]
  3 → 0:  nt1 → dn1  [G_hat[0,0]]
  4 → 3:  nt2 → dv1  [G_hat[1,2]]
  4 → 5:  nt2 → dv1  [G_hat[1,2]]
  5 → 1:  nt1 → dn2  [G_hat[0,1]]

```


=====

Diagram 1L-846

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

Edges (propagator assignments):

2 → 3: nt2 → dv1 [G_hat[1,2]]

2 → 5: nt2 → dv1 [G_hat[1,2]]

3 → 0: nt1 → dn1 [G_hat[0,0]]

4 → 3: nt2 → dv1 [G_hat[1,2]]

4 → 5: nt2 → dv1 [G_hat[1,2]]

5 → 1: nt1 → dn2 [G_hat[0,1]]

=====

Diagram 1L-847

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

```

    v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

    coeff =  $\frac{1}{2} nstar_2$ 

    v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

    coeff =  $\frac{1}{2} \varphi_1''$ 

    Edges (propagator assignments):
      2 → 3:  nt2 → dv1  [G_hat[1,2]]
      2 → 5:  nt2 → dv1  [G_hat[1,2]]
      3 → 0:  nt1 → dn1  [G_hat[0,0]]
      4 → 3:  nt2 → dv1  [G_hat[1,2]]
      4 → 5:  nt2 → dv1  [G_hat[1,2]]
      5 → 1:  nt1 → dn2  [G_hat[0,1]]

=====
Diagram 1L-848
=====

    External legs:
      leaf 0 ← dn1
      leaf 1 ← dn2
    Vertex assignments:
      v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

      coeff =  $\frac{1}{2} nstar_2$ 

      v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

      coeff =  $\frac{1}{2} \varphi_1''$ 

      v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

      coeff =  $\frac{1}{2} nstar_2$ 

      v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

      coeff =  $\frac{1}{2} \varphi_1''$ 

    Edges (propagator assignments):
      2 → 3:  nt2 → dv1  [G_hat[1,2]]
      2 → 5:  nt2 → dv1  [G_hat[1,2]]
      3 → 0:  nt1 → dn1  [G_hat[0,0]]
      4 → 3:  nt2 → dv1  [G_hat[1,2]]
      4 → 5:  nt2 → dv1  [G_hat[1,2]]
      5 → 1:  nt1 → dn2  [G_hat[0,1]]

```

=====

Diagram 1L-849

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

Edges (propagator assignments):

2 → 3: nt2 → dv1 [G_hat[1,2]]

2 → 5: nt2 → dv2 [G_hat[1,3]]

3 → 0: nt1 → dn1 [G_hat[0,0]]

4 → 3: nt2 → dv1 [G_hat[1,2]]

4 → 5: nt2 → dv2 [G_hat[1,3]]

5 → 1: nt2 → dn2 [G_hat[1,1]]

=====

Diagram 1L-850

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

```

    v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

    coeff =  $\frac{1}{2} nstar_2$ 

    v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

    coeff =  $\frac{1}{2} \varphi_2''$ 

    Edges (propagator assignments):
      2 → 3:  nt2 → dv1  [G_hat[1,2]]
      2 → 5:  nt2 → dv2  [G_hat[1,3]]
      3 → 0:  nt1 → dn1  [G_hat[0,0]]
      4 → 3:  nt2 → dv1  [G_hat[1,2]]
      4 → 5:  nt2 → dv2  [G_hat[1,3]]
      5 → 1:  nt2 → dn2  [G_hat[1,1]]

=====
Diagram 1L-851
=====

    External legs:
      leaf 0 ← dn1
      leaf 1 ← dn2
    Vertex assignments:
      v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

      coeff =  $\frac{1}{2} nstar_2$ 

      v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

      coeff =  $\frac{1}{2} \varphi_1''$ 

      v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

      coeff =  $\frac{1}{2} nstar_2$ 

      v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

      coeff =  $\frac{1}{2} \varphi_2''$ 

    Edges (propagator assignments):
      2 → 3:  nt2 → dv1  [G_hat[1,2]]
      2 → 5:  nt2 → dv2  [G_hat[1,3]]
      3 → 0:  nt1 → dn1  [G_hat[0,0]]
      4 → 3:  nt2 → dv1  [G_hat[1,2]]
      4 → 5:  nt2 → dv2  [G_hat[1,3]]
      5 → 1:  nt2 → dn2  [G_hat[1,1]]

```

=====

Diagram 1L-852

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

Edges (propagator assignments):

2 → 3: nt2 → dv1 [G_hat[1,2]]

2 → 5: nt2 → dv2 [G_hat[1,3]]

3 → 0: nt1 → dn1 [G_hat[0,0]]

4 → 3: nt2 → dv1 [G_hat[1,2]]

4 → 5: nt2 → dv2 [G_hat[1,3]]

5 → 1: nt2 → dn2 [G_hat[1,1]]

=====

Diagram 1L-853

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

```

    v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

    coeff =  $\frac{1}{2} nstar_2$ 

    v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

    coeff =  $\frac{1}{2} \varphi_2''$ 

    Edges (propagator assignments):
      2 → 3:  nt2 → dv1  [G_hat[1,2]]
      2 → 5:  nt2 → dv2  [G_hat[1,3]]
      3 → 0:  nt1 → dn1  [G_hat[0,0]]
      4 → 3:  nt2 → dv1  [G_hat[1,2]]
      4 → 5:  nt2 → dv2  [G_hat[1,3]]
      5 → 1:  nt2 → dn2  [G_hat[1,1]]

=====
Diagram 1L-854
=====

    External legs:
      leaf 0 ← dn1
      leaf 1 ← dn2
    Vertex assignments:
      v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

      coeff =  $\frac{1}{2} nstar_2$ 

      v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

      coeff =  $\frac{1}{2} \varphi_1''$ 

      v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

      coeff =  $\frac{1}{2} nstar_2$ 

      v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

      coeff =  $\frac{1}{2} \varphi_2''$ 

    Edges (propagator assignments):
      2 → 3:  nt2 → dv1  [G_hat[1,2]]
      2 → 5:  nt2 → dv2  [G_hat[1,3]]
      3 → 0:  nt1 → dn1  [G_hat[0,0]]
      4 → 3:  nt2 → dv1  [G_hat[1,2]]
      4 → 5:  nt2 → dv2  [G_hat[1,3]]
      5 → 1:  nt2 → dn2  [G_hat[1,1]]

```

=====

Diagram 1L-855

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

Edges (propagator assignments):

2 → 3: nt2 → dv1 [G_hat[1,2]]

2 → 5: nt2 → dv2 [G_hat[1,3]]

3 → 0: nt1 → dn1 [G_hat[0,0]]

4 → 3: nt2 → dv1 [G_hat[1,2]]

4 → 5: nt2 → dv2 [G_hat[1,3]]

5 → 1: nt2 → dn2 [G_hat[1,1]]

=====

Diagram 1L-856

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

```

    v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

    coeff =  $\frac{1}{2} nstar_2$ 

    v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

    coeff =  $\frac{1}{2} \varphi_2''$ 

    Edges (propagator assignments):
      2 → 3:  nt2 → dv1  [G_hat[1,2]]
      2 → 5:  nt2 → dv2  [G_hat[1,3]]
      3 → 0:  nt1 → dn1  [G_hat[0,0]]
      4 → 3:  nt2 → dv1  [G_hat[1,2]]
      4 → 5:  nt2 → dv2  [G_hat[1,3]]
      5 → 1:  nt2 → dn2  [G_hat[1,1]]

=====
Diagram 1L-857
=====

    External legs:
      leaf 0 ← dn1
      leaf 1 ← dn2
    Vertex assignments:
      v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

      coeff =  $\frac{1}{2} nstar_2$ 

      v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

      coeff =  $\frac{1}{2} \varphi_1''$ 

      v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

      coeff =  $\frac{1}{2} nstar_2$ 

      v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

      coeff =  $\frac{1}{2} \varphi_2''$ 

    Edges (propagator assignments):
      2 → 3:  nt2 → dv1  [G_hat[1,2]]
      2 → 5:  nt2 → dv2  [G_hat[1,3]]
      3 → 0:  nt1 → dn1  [G_hat[0,0]]
      4 → 3:  nt2 → dv1  [G_hat[1,2]]
      4 → 5:  nt2 → dv2  [G_hat[1,3]]
      5 → 1:  nt2 → dn2  [G_hat[1,1]]

```


=====

Diagram 1L-858

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

Edges (propagator assignments):

2 → 3: nt2 → dv1 [G_hat[1,2]]

2 → 5: nt2 → dv2 [G_hat[1,3]]

3 → 0: nt1 → dn1 [G_hat[0,0]]

4 → 3: nt2 → dv1 [G_hat[1,2]]

4 → 5: nt2 → dv2 [G_hat[1,3]]

5 → 1: nt2 → dn2 [G_hat[1,1]]

=====

Diagram 1L-859

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

```

    v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

    coeff =  $\frac{1}{2} nstar_2$ 

    v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

    coeff =  $\frac{1}{2} \varphi_2''$ 

    Edges (propagator assignments):
      2 → 3:  nt2 → dv1  [G_hat[1,2]]
      2 → 5:  nt2 → dv2  [G_hat[1,3]]
      3 → 0:  nt1 → dn1  [G_hat[0,0]]
      4 → 3:  nt2 → dv1  [G_hat[1,2]]
      4 → 5:  nt2 → dv2  [G_hat[1,3]]
      5 → 1:  nt2 → dn2  [G_hat[1,1]]

=====
Diagram 1L-860
=====

    External legs:
      leaf 0 ← dn1
      leaf 1 ← dn2
    Vertex assignments:
      v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

      coeff =  $\frac{1}{2} nstar_2$ 

      v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

      coeff =  $\frac{1}{2} \varphi_1''$ 

      v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

      coeff =  $\frac{1}{2} nstar_2$ 

      v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

      coeff =  $\frac{1}{2} \varphi_2''$ 

    Edges (propagator assignments):
      2 → 3:  nt2 → dv1  [G_hat[1,2]]
      2 → 5:  nt2 → dv2  [G_hat[1,3]]
      3 → 0:  nt1 → dn1  [G_hat[0,0]]
      4 → 3:  nt2 → dv1  [G_hat[1,2]]
      4 → 5:  nt2 → dv2  [G_hat[1,3]]
      5 → 1:  nt2 → dn2  [G_hat[1,1]]

```

=====

Diagram 1L-861

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

Edges (propagator assignments):

2 → 3: nt2 → dv1 [G_hat[1,2]]

2 → 5: nt2 → dv2 [G_hat[1,3]]

3 → 0: nt1 → dn1 [G_hat[0,0]]

4 → 3: nt2 → dv1 [G_hat[1,2]]

4 → 5: nt2 → dv2 [G_hat[1,3]]

5 → 1: nt2 → dn2 [G_hat[1,1]]

=====

Diagram 1L-862

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

```

    v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

    coeff =  $\frac{1}{2} nstar_2$ 

    v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

    coeff =  $\frac{1}{2} \varphi_2''$ 

    Edges (propagator assignments):
      2 → 3:  nt2 → dv1  [G_hat[1,2]]
      2 → 5:  nt2 → dv2  [G_hat[1,3]]
      3 → 0:  nt1 → dn1  [G_hat[0,0]]
      4 → 3:  nt2 → dv1  [G_hat[1,2]]
      4 → 5:  nt2 → dv2  [G_hat[1,3]]
      5 → 1:  nt2 → dn2  [G_hat[1,1]]

=====
Diagram 1L-863
=====

    External legs:
      leaf 0 ← dn1
      leaf 1 ← dn2
    Vertex assignments:
      v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

      coeff =  $\frac{1}{2} nstar_2$ 

      v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

      coeff =  $\frac{1}{2} \varphi_1''$ 

      v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

      coeff =  $\frac{1}{2} nstar_2$ 

      v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

      coeff =  $\frac{1}{2} \varphi_2''$ 

    Edges (propagator assignments):
      2 → 3:  nt2 → dv1  [G_hat[1,2]]
      2 → 5:  nt2 → dv2  [G_hat[1,3]]
      3 → 0:  nt1 → dn1  [G_hat[0,0]]
      4 → 3:  nt2 → dv1  [G_hat[1,2]]
      4 → 5:  nt2 → dv2  [G_hat[1,3]]
      5 → 1:  nt2 → dn2  [G_hat[1,1]]

```

=====

Diagram 1L-864

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

Edges (propagator assignments):

2 → 3: nt2 → dv1 [G_hat[1,2]]

2 → 5: nt2 → dv2 [G_hat[1,3]]

3 → 0: nt1 → dn1 [G_hat[0,0]]

4 → 3: nt2 → dv1 [G_hat[1,2]]

4 → 5: nt2 → dv2 [G_hat[1,3]]

5 → 1: nt2 → dn2 [G_hat[1,1]]

=====

Diagram 1L-865

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

```

    v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

    coeff =  $\frac{1}{2} nstar_2$ 

    v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

    coeff =  $\frac{1}{2} \varphi_1''$ 

    Edges (propagator assignments):
      2 → 3:  nt2 → dv2  [G_hat[1,3]]
      2 → 5:  nt2 → dv1  [G_hat[1,2]]
      3 → 0:  nt2 → dn1  [G_hat[1,0]]
      4 → 3:  nt2 → dv2  [G_hat[1,3]]
      4 → 5:  nt2 → dv1  [G_hat[1,2]]
      5 → 1:  nt1 → dn2  [G_hat[0,1]]

=====
Diagram 1L-866
=====

    External legs:
      leaf 0 ← dn1
      leaf 1 ← dn2
    Vertex assignments:
      v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

      coeff =  $\frac{1}{2} nstar_2$ 

      v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

      coeff =  $\frac{1}{2} \varphi_2''$ 

      v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

      coeff =  $\frac{1}{2} nstar_2$ 

      v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

      coeff =  $\frac{1}{2} \varphi_1''$ 

    Edges (propagator assignments):
      2 → 3:  nt2 → dv2  [G_hat[1,3]]
      2 → 5:  nt2 → dv1  [G_hat[1,2]]
      3 → 0:  nt2 → dn1  [G_hat[1,0]]
      4 → 3:  nt2 → dv2  [G_hat[1,3]]
      4 → 5:  nt2 → dv1  [G_hat[1,2]]
      5 → 1:  nt1 → dn2  [G_hat[0,1]]

```

=====

Diagram 1L-867

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

Edges (propagator assignments):

2 → 3: nt2 → dv2 [G_hat[1,3]]

2 → 5: nt2 → dv1 [G_hat[1,2]]

3 → 0: nt2 → dn1 [G_hat[1,0]]

4 → 3: nt2 → dv2 [G_hat[1,3]]

4 → 5: nt2 → dv1 [G_hat[1,2]]

5 → 1: nt1 → dn2 [G_hat[0,1]]

=====

Diagram 1L-868

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

```

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

coeff =  $\frac{1}{2} nstar_2$ 

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

coeff =  $\frac{1}{2} \varphi_1''$ 

Edges (propagator assignments):
  2 → 3:  nt2 → dv2  [G_hat[1,3]]
  2 → 5:  nt2 → dv1  [G_hat[1,2]]
  3 → 0:  nt2 → dn1  [G_hat[1,0]]
  4 → 3:  nt2 → dv2  [G_hat[1,3]]
  4 → 5:  nt2 → dv1  [G_hat[1,2]]
  5 → 1:  nt1 → dn2  [G_hat[0,1]]

=====
Diagram 1L-869
=====

External legs:
  leaf 0 ← dn1
  leaf 1 ← dn2
Vertex assignments:
  v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

coeff =  $\frac{1}{2} nstar_2$ 

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

coeff =  $\frac{1}{2} \varphi_2''$ 

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

coeff =  $\frac{1}{2} nstar_2$ 

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

coeff =  $\frac{1}{2} \varphi_1''$ 

Edges (propagator assignments):
  2 → 3:  nt2 → dv2  [G_hat[1,3]]
  2 → 5:  nt2 → dv1  [G_hat[1,2]]
  3 → 0:  nt2 → dn1  [G_hat[1,0]]
  4 → 3:  nt2 → dv2  [G_hat[1,3]]
  4 → 5:  nt2 → dv1  [G_hat[1,2]]
  5 → 1:  nt1 → dn2  [G_hat[0,1]]

```


=====

Diagram 1L-870

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

Edges (propagator assignments):

2 → 3: nt2 → dv2 [G_hat[1,3]]

2 → 5: nt2 → dv1 [G_hat[1,2]]

3 → 0: nt2 → dn1 [G_hat[1,0]]

4 → 3: nt2 → dv2 [G_hat[1,3]]

4 → 5: nt2 → dv1 [G_hat[1,2]]

5 → 1: nt1 → dn2 [G_hat[0,1]]

=====

Diagram 1L-871

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

```

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

coeff =  $\frac{1}{2} nstar_2$ 

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

coeff =  $\frac{1}{2} \varphi_1''$ 

Edges (propagator assignments):
  2 → 3:  nt2 → dv2  [G_hat[1,3]]
  2 → 5:  nt2 → dv1  [G_hat[1,2]]
  3 → 0:  nt2 → dn1  [G_hat[1,0]]
  4 → 3:  nt2 → dv2  [G_hat[1,3]]
  4 → 5:  nt2 → dv1  [G_hat[1,2]]
  5 → 1:  nt1 → dn2  [G_hat[0,1]]

=====
Diagram 1L-872
=====

External legs:
  leaf 0 ← dn1
  leaf 1 ← dn2
Vertex assignments:
  v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

coeff =  $\frac{1}{2} nstar_2$ 

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

coeff =  $\frac{1}{2} \varphi_2''$ 

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

coeff =  $\frac{1}{2} nstar_2$ 

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

coeff =  $\frac{1}{2} \varphi_1''$ 

Edges (propagator assignments):
  2 → 3:  nt2 → dv2  [G_hat[1,3]]
  2 → 5:  nt2 → dv1  [G_hat[1,2]]
  3 → 0:  nt2 → dn1  [G_hat[1,0]]
  4 → 3:  nt2 → dv2  [G_hat[1,3]]
  4 → 5:  nt2 → dv1  [G_hat[1,2]]
  5 → 1:  nt1 → dn2  [G_hat[0,1]]

```

=====

Diagram 1L-873

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

Edges (propagator assignments):

2 → 3: nt2 → dv2 [G_hat[1,3]]

2 → 5: nt2 → dv1 [G_hat[1,2]]

3 → 0: nt2 → dn1 [G_hat[1,0]]

4 → 3: nt2 → dv2 [G_hat[1,3]]

4 → 5: nt2 → dv1 [G_hat[1,2]]

5 → 1: nt1 → dn2 [G_hat[0,1]]

=====

Diagram 1L-874

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

```

    v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

    coeff =  $\frac{1}{2} nstar_2$ 

    v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

    coeff =  $\frac{1}{2} \varphi_1''$ 

    Edges (propagator assignments):
      2 → 3:  nt2 → dv2  [G_hat[1,3]]
      2 → 5:  nt2 → dv1  [G_hat[1,2]]
      3 → 0:  nt2 → dn1  [G_hat[1,0]]
      4 → 3:  nt2 → dv2  [G_hat[1,3]]
      4 → 5:  nt2 → dv1  [G_hat[1,2]]
      5 → 1:  nt1 → dn2  [G_hat[0,1]]

=====
Diagram 1L-875
=====

    External legs:
      leaf 0 ← dn1
      leaf 1 ← dn2
    Vertex assignments:
      v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

      coeff =  $\frac{1}{2} nstar_2$ 

      v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

      coeff =  $\frac{1}{2} \varphi_2''$ 

      v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

      coeff =  $\frac{1}{2} nstar_2$ 

      v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

      coeff =  $\frac{1}{2} \varphi_1''$ 

    Edges (propagator assignments):
      2 → 3:  nt2 → dv2  [G_hat[1,3]]
      2 → 5:  nt2 → dv1  [G_hat[1,2]]
      3 → 0:  nt2 → dn1  [G_hat[1,0]]
      4 → 3:  nt2 → dv2  [G_hat[1,3]]
      4 → 5:  nt2 → dv1  [G_hat[1,2]]
      5 → 1:  nt1 → dn2  [G_hat[0,1]]

```

=====

Diagram 1L-876

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

Edges (propagator assignments):

2 → 3: nt2 → dv2 [G_hat[1,3]]

2 → 5: nt2 → dv1 [G_hat[1,2]]

3 → 0: nt2 → dn1 [G_hat[1,0]]

4 → 3: nt2 → dv2 [G_hat[1,3]]

4 → 5: nt2 → dv1 [G_hat[1,2]]

5 → 1: nt1 → dn2 [G_hat[0,1]]

=====

Diagram 1L-877

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

```

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

coeff =  $\frac{1}{2} nstar_2$ 

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

coeff =  $\frac{1}{2} \varphi_1''$ 

Edges (propagator assignments):
  2 → 3:  nt2 → dv2  [G_hat[1,3]]
  2 → 5:  nt2 → dv1  [G_hat[1,2]]
  3 → 0:  nt2 → dn1  [G_hat[1,0]]
  4 → 3:  nt2 → dv2  [G_hat[1,3]]
  4 → 5:  nt2 → dv1  [G_hat[1,2]]
  5 → 1:  nt1 → dn2  [G_hat[0,1]]

=====
Diagram 1L-878
=====

External legs:
  leaf 0 ← dn1
  leaf 1 ← dn2
Vertex assignments:
  v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

coeff =  $\frac{1}{2} nstar_2$ 

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

coeff =  $\frac{1}{2} \varphi_2''$ 

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

coeff =  $\frac{1}{2} nstar_2$ 

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

coeff =  $\frac{1}{2} \varphi_1''$ 

Edges (propagator assignments):
  2 → 3:  nt2 → dv2  [G_hat[1,3]]
  2 → 5:  nt2 → dv1  [G_hat[1,2]]
  3 → 0:  nt2 → dn1  [G_hat[1,0]]
  4 → 3:  nt2 → dv2  [G_hat[1,3]]
  4 → 5:  nt2 → dv1  [G_hat[1,2]]
  5 → 1:  nt1 → dn2  [G_hat[0,1]]

```

=====

Diagram 1L-879

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

Edges (propagator assignments):

2 → 3: nt2 → dv2 [G_hat[1,3]]

2 → 5: nt2 → dv1 [G_hat[1,2]]

3 → 0: nt2 → dn1 [G_hat[1,0]]

4 → 3: nt2 → dv2 [G_hat[1,3]]

4 → 5: nt2 → dv1 [G_hat[1,2]]

5 → 1: nt1 → dn2 [G_hat[0,1]]

=====

Diagram 1L-880

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

```

    v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

    coeff =  $\frac{1}{2} nstar_2$ 

    v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

    coeff =  $\frac{1}{2} \varphi_1''$ 

    Edges (propagator assignments):
      2 → 3:  nt2 → dv2  [G_hat[1,3]]
      2 → 5:  nt2 → dv1  [G_hat[1,2]]
      3 → 0:  nt2 → dn1  [G_hat[1,0]]
      4 → 3:  nt2 → dv2  [G_hat[1,3]]
      4 → 5:  nt2 → dv1  [G_hat[1,2]]
      5 → 1:  nt1 → dn2  [G_hat[0,1]]

=====
Diagram 1L-881
=====

    External legs:
      leaf 0 ← dn1
      leaf 1 ← dn2
    Vertex assignments:
      v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

      coeff =  $\frac{1}{2} nstar_2$ 

      v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

      coeff =  $\frac{1}{2} \varphi_2''$ 

      v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

      coeff =  $\frac{1}{2} nstar_2$ 

      v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

      coeff =  $\frac{1}{2} \varphi_2''$ 

    Edges (propagator assignments):
      2 → 3:  nt2 → dv2  [G_hat[1,3]]
      2 → 5:  nt2 → dv2  [G_hat[1,3]]
      3 → 0:  nt2 → dn1  [G_hat[1,0]]
      4 → 3:  nt2 → dv2  [G_hat[1,3]]
      4 → 5:  nt2 → dv2  [G_hat[1,3]]
      5 → 1:  nt2 → dn2  [G_hat[1,1]]

```


=====

Diagram 1L-882

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

Edges (propagator assignments):

2 → 3: nt2 → dv2 [G_hat[1,3]]

2 → 5: nt2 → dv2 [G_hat[1,3]]

3 → 0: nt2 → dn1 [G_hat[1,0]]

4 → 3: nt2 → dv2 [G_hat[1,3]]

4 → 5: nt2 → dv2 [G_hat[1,3]]

5 → 1: nt2 → dn2 [G_hat[1,1]]

=====

Diagram 1L-883

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

```

    v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

    coeff =  $\frac{1}{2} nstar_2$ 

    v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

    coeff =  $\frac{1}{2} \varphi_2''$ 

    Edges (propagator assignments):
      2 → 3:  nt2 → dv2  [G_hat[1,3]]
      2 → 5:  nt2 → dv2  [G_hat[1,3]]
      3 → 0:  nt2 → dn1  [G_hat[1,0]]
      4 → 3:  nt2 → dv2  [G_hat[1,3]]
      4 → 5:  nt2 → dv2  [G_hat[1,3]]
      5 → 1:  nt2 → dn2  [G_hat[1,1]]

=====
Diagram 1L-884
=====

    External legs:
      leaf 0 ← dn1
      leaf 1 ← dn2
    Vertex assignments:
      v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

      coeff =  $\frac{1}{2} nstar_2$ 

      v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

      coeff =  $\frac{1}{2} \varphi_2''$ 

      v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

      coeff =  $\frac{1}{2} nstar_2$ 

      v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

      coeff =  $\frac{1}{2} \varphi_2''$ 

    Edges (propagator assignments):
      2 → 3:  nt2 → dv2  [G_hat[1,3]]
      2 → 5:  nt2 → dv2  [G_hat[1,3]]
      3 → 0:  nt2 → dn1  [G_hat[1,0]]
      4 → 3:  nt2 → dv2  [G_hat[1,3]]
      4 → 5:  nt2 → dv2  [G_hat[1,3]]
      5 → 1:  nt2 → dn2  [G_hat[1,1]]

```

=====

Diagram 1L-885

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

Edges (propagator assignments):

2 → 3: nt2 → dv2 [G_hat[1,3]]

2 → 5: nt2 → dv2 [G_hat[1,3]]

3 → 0: nt2 → dn1 [G_hat[1,0]]

4 → 3: nt2 → dv2 [G_hat[1,3]]

4 → 5: nt2 → dv2 [G_hat[1,3]]

5 → 1: nt2 → dn2 [G_hat[1,1]]

=====

Diagram 1L-886

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

```

    v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

    coeff =  $\frac{1}{2} nstar_2$ 

    v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

    coeff =  $\frac{1}{2} \varphi_2''$ 

    Edges (propagator assignments):
      2 → 3:  nt2 → dv2  [G_hat[1,3]]
      2 → 5:  nt2 → dv2  [G_hat[1,3]]
      3 → 0:  nt2 → dn1  [G_hat[1,0]]
      4 → 3:  nt2 → dv2  [G_hat[1,3]]
      4 → 5:  nt2 → dv2  [G_hat[1,3]]
      5 → 1:  nt2 → dn2  [G_hat[1,1]]

=====
Diagram 1L-887
=====

    External legs:
      leaf 0 ← dn1
      leaf 1 ← dn2
    Vertex assignments:
      v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

      coeff =  $\frac{1}{2} nstar_2$ 

      v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

      coeff =  $\frac{1}{2} \varphi_2''$ 

      v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

      coeff =  $\frac{1}{2} nstar_2$ 

      v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

      coeff =  $\frac{1}{2} \varphi_2''$ 

    Edges (propagator assignments):
      2 → 3:  nt2 → dv2  [G_hat[1,3]]
      2 → 5:  nt2 → dv2  [G_hat[1,3]]
      3 → 0:  nt2 → dn1  [G_hat[1,0]]
      4 → 3:  nt2 → dv2  [G_hat[1,3]]
      4 → 5:  nt2 → dv2  [G_hat[1,3]]
      5 → 1:  nt2 → dn2  [G_hat[1,1]]

```

=====

Diagram 1L-888

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

Edges (propagator assignments):

2 → 3: nt2 → dv2 [G_hat[1,3]]

2 → 5: nt2 → dv2 [G_hat[1,3]]

3 → 0: nt2 → dn1 [G_hat[1,0]]

4 → 3: nt2 → dv2 [G_hat[1,3]]

4 → 5: nt2 → dv2 [G_hat[1,3]]

5 → 1: nt2 → dn2 [G_hat[1,1]]

=====

Diagram 1L-889

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

```

    v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

    coeff =  $\frac{1}{2} nstar_2$ 

    v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

    coeff =  $\frac{1}{2} \varphi_2''$ 

    Edges (propagator assignments):
      2 → 3:  nt2 → dv2  [G_hat[1,3]]
      2 → 5:  nt2 → dv2  [G_hat[1,3]]
      3 → 0:  nt2 → dn1  [G_hat[1,0]]
      4 → 3:  nt2 → dv2  [G_hat[1,3]]
      4 → 5:  nt2 → dv2  [G_hat[1,3]]
      5 → 1:  nt2 → dn2  [G_hat[1,1]]

=====
Diagram 1L-890
=====

    External legs:
      leaf 0 ← dn1
      leaf 1 ← dn2
    Vertex assignments:
      v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

      coeff =  $\frac{1}{2} nstar_2$ 

      v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

      coeff =  $\frac{1}{2} \varphi_2''$ 

      v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

      coeff =  $\frac{1}{2} nstar_2$ 

      v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

      coeff =  $\frac{1}{2} \varphi_2''$ 

    Edges (propagator assignments):
      2 → 3:  nt2 → dv2  [G_hat[1,3]]
      2 → 5:  nt2 → dv2  [G_hat[1,3]]
      3 → 0:  nt2 → dn1  [G_hat[1,0]]
      4 → 3:  nt2 → dv2  [G_hat[1,3]]
      4 → 5:  nt2 → dv2  [G_hat[1,3]]
      5 → 1:  nt2 → dn2  [G_hat[1,1]]

```

=====

Diagram 1L-891

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

Edges (propagator assignments):

2 → 3: nt2 → dv2 [G_hat[1,3]]

2 → 5: nt2 → dv2 [G_hat[1,3]]

3 → 0: nt2 → dn1 [G_hat[1,0]]

4 → 3: nt2 → dv2 [G_hat[1,3]]

4 → 5: nt2 → dv2 [G_hat[1,3]]

5 → 1: nt2 → dn2 [G_hat[1,1]]

=====

Diagram 1L-892

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

```

    v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

    coeff =  $\frac{1}{2} nstar_2$ 

    v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

    coeff =  $\frac{1}{2} \varphi_2''$ 

    Edges (propagator assignments):
      2 → 3:  nt2 → dv2  [G_hat[1,3]]
      2 → 5:  nt2 → dv2  [G_hat[1,3]]
      3 → 0:  nt2 → dn1  [G_hat[1,0]]
      4 → 3:  nt2 → dv2  [G_hat[1,3]]
      4 → 5:  nt2 → dv2  [G_hat[1,3]]
      5 → 1:  nt2 → dn2  [G_hat[1,1]]

=====
Diagram 1L-893
=====

    External legs:
      leaf 0 ← dn1
      leaf 1 ← dn2
    Vertex assignments:
      v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

      coeff =  $\frac{1}{2} nstar_2$ 

      v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

      coeff =  $\frac{1}{2} \varphi_2''$ 

      v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

      coeff =  $\frac{1}{2} nstar_2$ 

      v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

      coeff =  $\frac{1}{2} \varphi_2''$ 

    Edges (propagator assignments):
      2 → 3:  nt2 → dv2  [G_hat[1,3]]
      2 → 5:  nt2 → dv2  [G_hat[1,3]]
      3 → 0:  nt2 → dn1  [G_hat[1,0]]
      4 → 3:  nt2 → dv2  [G_hat[1,3]]
      4 → 5:  nt2 → dv2  [G_hat[1,3]]
      5 → 1:  nt2 → dn2  [G_hat[1,1]]

```


=====

Diagram 1L-894

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

Edges (propagator assignments):

2 → 3: nt2 → dv2 [G_hat[1,3]]

2 → 5: nt2 → dv2 [G_hat[1,3]]

3 → 0: nt2 → dn1 [G_hat[1,0]]

4 → 3: nt2 → dv2 [G_hat[1,3]]

4 → 5: nt2 → dv2 [G_hat[1,3]]

5 → 1: nt2 → dn2 [G_hat[1,1]]

=====

Diagram 1L-895

=====

External legs:

leaf 0 ← dn1

leaf 1 ← dn2

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

```

    v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

    coeff =  $\frac{1}{2} nstar_2$ 

    v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

    coeff =  $\frac{1}{2} \varphi_2''$ 

    Edges (propagator assignments):
      2 → 3:  nt2 → dv2  [G_hat[1,3]]
      2 → 5:  nt2 → dv2  [G_hat[1,3]]
      3 → 0:  nt2 → dn1  [G_hat[1,0]]
      4 → 3:  nt2 → dv2  [G_hat[1,3]]
      4 → 5:  nt2 → dv2  [G_hat[1,3]]
      5 → 1:  nt2 → dn2  [G_hat[1,1]]

=====
Diagram 1L-896
=====

    External legs:
      leaf 0 ← dn1
      leaf 1 ← dn2
    Vertex assignments:
      v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

      coeff =  $\frac{1}{2} nstar_2$ 

      v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

      coeff =  $\frac{1}{2} \varphi_2''$ 

      v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

      coeff =  $\frac{1}{2} nstar_2$ 

      v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

      coeff =  $\frac{1}{2} \varphi_2''$ 

    Edges (propagator assignments):
      2 → 3:  nt2 → dv2  [G_hat[1,3]]
      2 → 5:  nt2 → dv2  [G_hat[1,3]]
      3 → 0:  nt2 → dn1  [G_hat[1,0]]
      4 → 3:  nt2 → dv2  [G_hat[1,3]]
      4 → 5:  nt2 → dv2  [G_hat[1,3]]
      5 → 1:  nt2 → dn2  [G_hat[1,1]]

```

=====

Diagram 1L-897

=====

External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

Edges (propagator assignments):

2 → 3: nt1 → dv1 [G_hat[0,2]]

2 → 5: nt1 → dv1 [G_hat[0,2]]

3 → 0: nt1 → dn2 [G_hat[0,1]]

4 → 3: nt1 → dv1 [G_hat[0,2]]

4 → 5: nt1 → dv1 [G_hat[0,2]]

5 → 1: nt1 → dn1 [G_hat[0,0]]

=====

Diagram 1L-898

=====

External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

```

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

coeff =  $\frac{1}{2} nstar_1$ 

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

coeff =  $\frac{1}{2} \varphi_1''$ 

Edges (propagator assignments):
  2 → 3:  nt1 → dv1  [G_hat[0,2]]
  2 → 5:  nt1 → dv1  [G_hat[0,2]]
  3 → 0:  nt1 → dn2  [G_hat[0,1]]
  4 → 3:  nt1 → dv1  [G_hat[0,2]]
  4 → 5:  nt1 → dv1  [G_hat[0,2]]
  5 → 1:  nt1 → dn1  [G_hat[0,0]]

=====
Diagram 1L-899
=====

External legs:
  leaf 0 ← dn2
  leaf 1 ← dn1
Vertex assignments:
  v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

coeff =  $\frac{1}{2} nstar_1$ 

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

coeff =  $\frac{1}{2} \varphi_1''$ 

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

coeff =  $\frac{1}{2} nstar_1$ 

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

coeff =  $\frac{1}{2} \varphi_1''$ 

Edges (propagator assignments):
  2 → 3:  nt1 → dv1  [G_hat[0,2]]
  2 → 5:  nt1 → dv1  [G_hat[0,2]]
  3 → 0:  nt1 → dn2  [G_hat[0,1]]
  4 → 3:  nt1 → dv1  [G_hat[0,2]]
  4 → 5:  nt1 → dv1  [G_hat[0,2]]
  5 → 1:  nt1 → dn1  [G_hat[0,0]]

```

=====

Diagram 1L-900

=====

External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

Edges (propagator assignments):

2 → 3: nt1 → dv1 [G_hat[0,2]]

2 → 5: nt1 → dv1 [G_hat[0,2]]

3 → 0: nt1 → dn2 [G_hat[0,1]]

4 → 3: nt1 → dv1 [G_hat[0,2]]

4 → 5: nt1 → dv1 [G_hat[0,2]]

5 → 1: nt1 → dn1 [G_hat[0,0]]

=====

Diagram 1L-901

=====

External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

```

    v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

    coeff =  $\frac{1}{2} nstar_1$ 

    v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

    coeff =  $\frac{1}{2} \varphi_1''$ 

    Edges (propagator assignments):
      2 → 3:  nt1 → dv1  [G_hat[0,2]]
      2 → 5:  nt1 → dv1  [G_hat[0,2]]
      3 → 0:  nt1 → dn2  [G_hat[0,1]]
      4 → 3:  nt1 → dv1  [G_hat[0,2]]
      4 → 5:  nt1 → dv1  [G_hat[0,2]]
      5 → 1:  nt1 → dn1  [G_hat[0,0]]

=====
Diagram 1L-902
=====

    External legs:
      leaf 0 ← dn2
      leaf 1 ← dn1
    Vertex assignments:
      v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

      coeff =  $\frac{1}{2} nstar_1$ 

      v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

      coeff =  $\frac{1}{2} \varphi_1''$ 

      v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

      coeff =  $\frac{1}{2} nstar_1$ 

      v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

      coeff =  $\frac{1}{2} \varphi_1''$ 

    Edges (propagator assignments):
      2 → 3:  nt1 → dv1  [G_hat[0,2]]
      2 → 5:  nt1 → dv1  [G_hat[0,2]]
      3 → 0:  nt1 → dn2  [G_hat[0,1]]
      4 → 3:  nt1 → dv1  [G_hat[0,2]]
      4 → 5:  nt1 → dv1  [G_hat[0,2]]
      5 → 1:  nt1 → dn1  [G_hat[0,0]]

```

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Diagram 1L-903

=====

External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

Edges (propagator assignments):

2 → 3: nt1 → dv1 [G_hat[0,2]]

2 → 5: nt1 → dv1 [G_hat[0,2]]

3 → 0: nt1 → dn2 [G_hat[0,1]]

4 → 3: nt1 → dv1 [G_hat[0,2]]

4 → 5: nt1 → dv1 [G_hat[0,2]]

5 → 1: nt1 → dn1 [G_hat[0,0]]

=====

Diagram 1L-904

=====

External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

```

    v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

    coeff =  $\frac{1}{2} nstar_1$ 

    v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

    coeff =  $\frac{1}{2} \varphi_1''$ 

    Edges (propagator assignments):
      2 → 3:  nt1 → dv1  [G_hat[0,2]]
      2 → 5:  nt1 → dv1  [G_hat[0,2]]
      3 → 0:  nt1 → dn2  [G_hat[0,1]]
      4 → 3:  nt1 → dv1  [G_hat[0,2]]
      4 → 5:  nt1 → dv1  [G_hat[0,2]]
      5 → 1:  nt1 → dn1  [G_hat[0,0]]

=====
Diagram 1L-905
=====

    External legs:
      leaf 0 ← dn2
      leaf 1 ← dn1
    Vertex assignments:
      v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

      coeff =  $\frac{1}{2} nstar_1$ 

      v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

      coeff =  $\frac{1}{2} \varphi_1''$ 

      v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

      coeff =  $\frac{1}{2} nstar_1$ 

      v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

      coeff =  $\frac{1}{2} \varphi_1''$ 

    Edges (propagator assignments):
      2 → 3:  nt1 → dv1  [G_hat[0,2]]
      2 → 5:  nt1 → dv1  [G_hat[0,2]]
      3 → 0:  nt1 → dn2  [G_hat[0,1]]
      4 → 3:  nt1 → dv1  [G_hat[0,2]]
      4 → 5:  nt1 → dv1  [G_hat[0,2]]
      5 → 1:  nt1 → dn1  [G_hat[0,0]]

```


=====

Diagram 1L-906

=====

External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

Edges (propagator assignments):

2 → 3: nt1 → dv1 [G_hat[0,2]]

2 → 5: nt1 → dv1 [G_hat[0,2]]

3 → 0: nt1 → dn2 [G_hat[0,1]]

4 → 3: nt1 → dv1 [G_hat[0,2]]

4 → 5: nt1 → dv1 [G_hat[0,2]]

5 → 1: nt1 → dn1 [G_hat[0,0]]

=====

Diagram 1L-907

=====

External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

```

    v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

    coeff =  $\frac{1}{2} nstar_1$ 

    v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

    coeff =  $\frac{1}{2} \varphi_1''$ 

    Edges (propagator assignments):
      2 → 3:  nt1 → dv1  [G_hat[0,2]]
      2 → 5:  nt1 → dv1  [G_hat[0,2]]
      3 → 0:  nt1 → dn2  [G_hat[0,1]]
      4 → 3:  nt1 → dv1  [G_hat[0,2]]
      4 → 5:  nt1 → dv1  [G_hat[0,2]]
      5 → 1:  nt1 → dn1  [G_hat[0,0]]

=====
Diagram 1L-908
=====

    External legs:
      leaf 0 ← dn2
      leaf 1 ← dn1

    Vertex assignments:
      v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

      coeff =  $\frac{1}{2} nstar_1$ 

      v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

      coeff =  $\frac{1}{2} \varphi_1''$ 

      v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

      coeff =  $\frac{1}{2} nstar_1$ 

      v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

      coeff =  $\frac{1}{2} \varphi_1''$ 

    Edges (propagator assignments):
      2 → 3:  nt1 → dv1  [G_hat[0,2]]
      2 → 5:  nt1 → dv1  [G_hat[0,2]]
      3 → 0:  nt1 → dn2  [G_hat[0,1]]
      4 → 3:  nt1 → dv1  [G_hat[0,2]]
      4 → 5:  nt1 → dv1  [G_hat[0,2]]
      5 → 1:  nt1 → dn1  [G_hat[0,0]]

```

=====

Diagram 1L-909

=====

External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

Edges (propagator assignments):

2 → 3: nt1 → dv1 [G_hat[0,2]]

2 → 5: nt1 → dv1 [G_hat[0,2]]

3 → 0: nt1 → dn2 [G_hat[0,1]]

4 → 3: nt1 → dv1 [G_hat[0,2]]

4 → 5: nt1 → dv1 [G_hat[0,2]]

5 → 1: nt1 → dn1 [G_hat[0,0]]

=====

Diagram 1L-910

=====

External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

```

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

coeff =  $\frac{1}{2} nstar_1$ 

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

coeff =  $\frac{1}{2} \varphi_1''$ 

Edges (propagator assignments):
  2 → 3:  nt1 → dv1  [G_hat[0,2]]
  2 → 5:  nt1 → dv1  [G_hat[0,2]]
  3 → 0:  nt1 → dn2  [G_hat[0,1]]
  4 → 3:  nt1 → dv1  [G_hat[0,2]]
  4 → 5:  nt1 → dv1  [G_hat[0,2]]
  5 → 1:  nt1 → dn1  [G_hat[0,0]]

=====
Diagram 1L-911
=====

External legs:
  leaf 0 ← dn2
  leaf 1 ← dn1
Vertex assignments:
  v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

coeff =  $\frac{1}{2} nstar_1$ 

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

coeff =  $\frac{1}{2} \varphi_1''$ 

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

coeff =  $\frac{1}{2} nstar_1$ 

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

coeff =  $\frac{1}{2} \varphi_1''$ 

Edges (propagator assignments):
  2 → 3:  nt1 → dv1  [G_hat[0,2]]
  2 → 5:  nt1 → dv1  [G_hat[0,2]]
  3 → 0:  nt1 → dn2  [G_hat[0,1]]
  4 → 3:  nt1 → dv1  [G_hat[0,2]]
  4 → 5:  nt1 → dv1  [G_hat[0,2]]
  5 → 1:  nt1 → dn1  [G_hat[0,0]]

```

=====

Diagram 1L-912

=====

External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

Edges (propagator assignments):

2 → 3: nt1 → dv1 [G_hat[0,2]]

2 → 5: nt1 → dv1 [G_hat[0,2]]

3 → 0: nt1 → dn2 [G_hat[0,1]]

4 → 3: nt1 → dv1 [G_hat[0,2]]

4 → 5: nt1 → dv1 [G_hat[0,2]]

5 → 1: nt1 → dn1 [G_hat[0,0]]

=====

Diagram 1L-913

=====

External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

```

    v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

    coeff =  $\frac{1}{2} nstar_1$ 

    v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

    coeff =  $\frac{1}{2} \varphi_2''$ 

    Edges (propagator assignments):
      2 → 3:  nt1 → dv1  [G_hat[0,2]]
      2 → 5:  nt1 → dv2  [G_hat[0,3]]
      3 → 0:  nt1 → dn2  [G_hat[0,1]]
      4 → 3:  nt1 → dv1  [G_hat[0,2]]
      4 → 5:  nt1 → dv2  [G_hat[0,3]]
      5 → 1:  nt2 → dn1  [G_hat[1,0]]

=====
Diagram 1L-914
=====

    External legs:
      leaf 0 ← dn2
      leaf 1 ← dn1
    Vertex assignments:
      v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

      coeff =  $\frac{1}{2} nstar_1$ 

      v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

      coeff =  $\frac{1}{2} \varphi_1''$ 

      v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

      coeff =  $\frac{1}{2} nstar_1$ 

      v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

      coeff =  $\frac{1}{2} \varphi_2''$ 

    Edges (propagator assignments):
      2 → 3:  nt1 → dv1  [G_hat[0,2]]
      2 → 5:  nt1 → dv2  [G_hat[0,3]]
      3 → 0:  nt1 → dn2  [G_hat[0,1]]
      4 → 3:  nt1 → dv1  [G_hat[0,2]]
      4 → 5:  nt1 → dv2  [G_hat[0,3]]
      5 → 1:  nt2 → dn1  [G_hat[1,0]]

```

=====

Diagram 1L-915

=====

External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

Edges (propagator assignments):

2 → 3: nt1 → dv1 [G_hat[0,2]]

2 → 5: nt1 → dv2 [G_hat[0,3]]

3 → 0: nt1 → dn2 [G_hat[0,1]]

4 → 3: nt1 → dv1 [G_hat[0,2]]

4 → 5: nt1 → dv2 [G_hat[0,3]]

5 → 1: nt2 → dn1 [G_hat[1,0]]

=====

Diagram 1L-916

=====

External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

```

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

coeff =  $\frac{1}{2} nstar_1$ 

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

coeff =  $\frac{1}{2} \varphi_2''$ 

Edges (propagator assignments):
  2 → 3:  nt1 → dv1  [G_hat[0,2]]
  2 → 5:  nt1 → dv2  [G_hat[0,3]]
  3 → 0:  nt1 → dn2  [G_hat[0,1]]
  4 → 3:  nt1 → dv1  [G_hat[0,2]]
  4 → 5:  nt1 → dv2  [G_hat[0,3]]
  5 → 1:  nt2 → dn1  [G_hat[1,0]]

=====
Diagram 1L-917
=====

External legs:
  leaf 0 ← dn2
  leaf 1 ← dn1
Vertex assignments:
  v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

coeff =  $\frac{1}{2} nstar_1$ 

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

coeff =  $\frac{1}{2} \varphi_1''$ 

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

coeff =  $\frac{1}{2} nstar_1$ 

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

coeff =  $\frac{1}{2} \varphi_2''$ 

Edges (propagator assignments):
  2 → 3:  nt1 → dv1  [G_hat[0,2]]
  2 → 5:  nt1 → dv2  [G_hat[0,3]]
  3 → 0:  nt1 → dn2  [G_hat[0,1]]
  4 → 3:  nt1 → dv1  [G_hat[0,2]]
  4 → 5:  nt1 → dv2  [G_hat[0,3]]
  5 → 1:  nt2 → dn1  [G_hat[1,0]]

```


=====

Diagram 1L-918

=====

External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

Edges (propagator assignments):

2 → 3: nt1 → dv1 [G_hat[0,2]]

2 → 5: nt1 → dv2 [G_hat[0,3]]

3 → 0: nt1 → dn2 [G_hat[0,1]]

4 → 3: nt1 → dv1 [G_hat[0,2]]

4 → 5: nt1 → dv2 [G_hat[0,3]]

5 → 1: nt2 → dn1 [G_hat[1,0]]

=====

Diagram 1L-919

=====

External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

```

    v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

    coeff =  $\frac{1}{2} nstar_1$ 

    v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

    coeff =  $\frac{1}{2} \varphi_2''$ 

    Edges (propagator assignments):
      2 → 3:  nt1 → dv1  [G_hat[0,2]]
      2 → 5:  nt1 → dv2  [G_hat[0,3]]
      3 → 0:  nt1 → dn2  [G_hat[0,1]]
      4 → 3:  nt1 → dv1  [G_hat[0,2]]
      4 → 5:  nt1 → dv2  [G_hat[0,3]]
      5 → 1:  nt2 → dn1  [G_hat[1,0]]

=====
Diagram 1L-920
=====

    External legs:
      leaf 0 ← dn2
      leaf 1 ← dn1
    Vertex assignments:
      v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

      coeff =  $\frac{1}{2} nstar_1$ 

      v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

      coeff =  $\frac{1}{2} \varphi_1''$ 

      v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

      coeff =  $\frac{1}{2} nstar_1$ 

      v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

      coeff =  $\frac{1}{2} \varphi_2''$ 

    Edges (propagator assignments):
      2 → 3:  nt1 → dv1  [G_hat[0,2]]
      2 → 5:  nt1 → dv2  [G_hat[0,3]]
      3 → 0:  nt1 → dn2  [G_hat[0,1]]
      4 → 3:  nt1 → dv1  [G_hat[0,2]]
      4 → 5:  nt1 → dv2  [G_hat[0,3]]
      5 → 1:  nt2 → dn1  [G_hat[1,0]]

```

=====

Diagram 1L-921

=====

External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

Edges (propagator assignments):

2 → 3: nt1 → dv1 [G_hat[0,2]]

2 → 5: nt1 → dv2 [G_hat[0,3]]

3 → 0: nt1 → dn2 [G_hat[0,1]]

4 → 3: nt1 → dv1 [G_hat[0,2]]

4 → 5: nt1 → dv2 [G_hat[0,3]]

5 → 1: nt2 → dn1 [G_hat[1,0]]

=====

Diagram 1L-922

=====

External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

```

    v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

    coeff =  $\frac{1}{2} nstar_1$ 

    v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

    coeff =  $\frac{1}{2} \varphi_2''$ 

    Edges (propagator assignments):
      2 → 3:  nt1 → dv1  [G_hat[0,2]]
      2 → 5:  nt1 → dv2  [G_hat[0,3]]
      3 → 0:  nt1 → dn2  [G_hat[0,1]]
      4 → 3:  nt1 → dv1  [G_hat[0,2]]
      4 → 5:  nt1 → dv2  [G_hat[0,3]]
      5 → 1:  nt2 → dn1  [G_hat[1,0]]

=====
Diagram 1L-923
=====

    External legs:
      leaf 0 ← dn2
      leaf 1 ← dn1
    Vertex assignments:
      v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

      coeff =  $\frac{1}{2} nstar_1$ 

      v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

      coeff =  $\frac{1}{2} \varphi_1''$ 

      v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

      coeff =  $\frac{1}{2} nstar_1$ 

      v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

      coeff =  $\frac{1}{2} \varphi_2''$ 

    Edges (propagator assignments):
      2 → 3:  nt1 → dv1  [G_hat[0,2]]
      2 → 5:  nt1 → dv2  [G_hat[0,3]]
      3 → 0:  nt1 → dn2  [G_hat[0,1]]
      4 → 3:  nt1 → dv1  [G_hat[0,2]]
      4 → 5:  nt1 → dv2  [G_hat[0,3]]
      5 → 1:  nt2 → dn1  [G_hat[1,0]]

```

=====

Diagram 1L-924

=====

External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

Edges (propagator assignments):

2 → 3: nt1 → dv1 [G_hat[0,2]]

2 → 5: nt1 → dv2 [G_hat[0,3]]

3 → 0: nt1 → dn2 [G_hat[0,1]]

4 → 3: nt1 → dv1 [G_hat[0,2]]

4 → 5: nt1 → dv2 [G_hat[0,3]]

5 → 1: nt2 → dn1 [G_hat[1,0]]

=====

Diagram 1L-925

=====

External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

```

    v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

coeff =  $\frac{1}{2} nstar_1$ 

    v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

coeff =  $\frac{1}{2} \varphi_2''$ 

Edges (propagator assignments):
    2 → 3:  nt1 → dv1  [G_hat[0,2]]
    2 → 5:  nt1 → dv2  [G_hat[0,3]]
    3 → 0:  nt1 → dn2  [G_hat[0,1]]
    4 → 3:  nt1 → dv1  [G_hat[0,2]]
    4 → 5:  nt1 → dv2  [G_hat[0,3]]
    5 → 1:  nt2 → dn1  [G_hat[1,0]]

=====
Diagram 1L-926
=====

External legs:
    leaf 0 ← dn2
    leaf 1 ← dn1
Vertex assignments:
    v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

coeff =  $\frac{1}{2} nstar_1$ 

    v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

coeff =  $\frac{1}{2} \varphi_1''$ 

    v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

coeff =  $\frac{1}{2} nstar_1$ 

    v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

coeff =  $\frac{1}{2} \varphi_2''$ 

Edges (propagator assignments):
    2 → 3:  nt1 → dv1  [G_hat[0,2]]
    2 → 5:  nt1 → dv2  [G_hat[0,3]]
    3 → 0:  nt1 → dn2  [G_hat[0,1]]
    4 → 3:  nt1 → dv1  [G_hat[0,2]]
    4 → 5:  nt1 → dv2  [G_hat[0,3]]
    5 → 1:  nt2 → dn1  [G_hat[1,0]]

```

=====

Diagram 1L-927

=====

External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

Edges (propagator assignments):

2 → 3: nt1 → dv1 [G_hat[0,2]]

2 → 5: nt1 → dv2 [G_hat[0,3]]

3 → 0: nt1 → dn2 [G_hat[0,1]]

4 → 3: nt1 → dv1 [G_hat[0,2]]

4 → 5: nt1 → dv2 [G_hat[0,3]]

5 → 1: nt2 → dn1 [G_hat[1,0]]

=====

Diagram 1L-928

=====

External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

```

    v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

    coeff =  $\frac{1}{2} nstar_1$ 

    v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

    coeff =  $\frac{1}{2} \varphi_2''$ 

    Edges (propagator assignments):
      2 → 3:  nt1 → dv1  [G_hat[0,2]]
      2 → 5:  nt1 → dv2  [G_hat[0,3]]
      3 → 0:  nt1 → dn2  [G_hat[0,1]]
      4 → 3:  nt1 → dv1  [G_hat[0,2]]
      4 → 5:  nt1 → dv2  [G_hat[0,3]]
      5 → 1:  nt2 → dn1  [G_hat[1,0]]

=====
Diagram 1L-929
=====

    External legs:
      leaf 0 ← dn2
      leaf 1 ← dn1
    Vertex assignments:
      v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

      coeff =  $\frac{1}{2} nstar_1$ 

      v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

      coeff =  $\frac{1}{2} \varphi_2''$ 

      v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

      coeff =  $\frac{1}{2} nstar_1$ 

      v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

      coeff =  $\frac{1}{2} \varphi_1''$ 

    Edges (propagator assignments):
      2 → 3:  nt1 → dv2  [G_hat[0,3]]
      2 → 5:  nt1 → dv1  [G_hat[0,2]]
      3 → 0:  nt2 → dn2  [G_hat[1,1]]
      4 → 3:  nt1 → dv2  [G_hat[0,3]]
      4 → 5:  nt1 → dv1  [G_hat[0,2]]
      5 → 1:  nt1 → dn1  [G_hat[0,0]]

```


=====

Diagram 1L-930

=====

External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

Edges (propagator assignments):

2 → 3: nt1 → dv2 [G_hat[0,3]]

2 → 5: nt1 → dv1 [G_hat[0,2]]

3 → 0: nt2 → dn2 [G_hat[1,1]]

4 → 3: nt1 → dv2 [G_hat[0,3]]

4 → 5: nt1 → dv1 [G_hat[0,2]]

5 → 1: nt1 → dn1 [G_hat[0,0]]

=====

Diagram 1L-931

=====

External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

```

    v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

    coeff =  $\frac{1}{2} nstar_1$ 

    v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

    coeff =  $\frac{1}{2} \varphi_1''$ 

    Edges (propagator assignments):
      2 → 3:  nt1 → dv2  [G_hat[0,3]]
      2 → 5:  nt1 → dv1  [G_hat[0,2]]
      3 → 0:  nt2 → dn2  [G_hat[1,1]]
      4 → 3:  nt1 → dv2  [G_hat[0,3]]
      4 → 5:  nt1 → dv1  [G_hat[0,2]]
      5 → 1:  nt1 → dn1  [G_hat[0,0]]

=====
Diagram 1L-932
=====

    External legs:
      leaf 0 ← dn2
      leaf 1 ← dn1
    Vertex assignments:
      v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

      coeff =  $\frac{1}{2} nstar_1$ 

      v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

      coeff =  $\frac{1}{2} \varphi_2''$ 

      v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

      coeff =  $\frac{1}{2} nstar_1$ 

      v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

      coeff =  $\frac{1}{2} \varphi_1''$ 

    Edges (propagator assignments):
      2 → 3:  nt1 → dv2  [G_hat[0,3]]
      2 → 5:  nt1 → dv1  [G_hat[0,2]]
      3 → 0:  nt2 → dn2  [G_hat[1,1]]
      4 → 3:  nt1 → dv2  [G_hat[0,3]]
      4 → 5:  nt1 → dv1  [G_hat[0,2]]
      5 → 1:  nt1 → dn1  [G_hat[0,0]]

```

=====

Diagram 1L-933

=====

External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

Edges (propagator assignments):

2 → 3: nt1 → dv2 [G_hat[0,3]]

2 → 5: nt1 → dv1 [G_hat[0,2]]

3 → 0: nt2 → dn2 [G_hat[1,1]]

4 → 3: nt1 → dv2 [G_hat[0,3]]

4 → 5: nt1 → dv1 [G_hat[0,2]]

5 → 1: nt1 → dn1 [G_hat[0,0]]

=====

Diagram 1L-934

=====

External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

```

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

coeff =  $\frac{1}{2} nstar_1$ 

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

coeff =  $\frac{1}{2} \varphi_1''$ 

Edges (propagator assignments):
  2 → 3:  nt1 → dv2  [G_hat[0,3]]
  2 → 5:  nt1 → dv1  [G_hat[0,2]]
  3 → 0:  nt2 → dn2  [G_hat[1,1]]
  4 → 3:  nt1 → dv2  [G_hat[0,3]]
  4 → 5:  nt1 → dv1  [G_hat[0,2]]
  5 → 1:  nt1 → dn1  [G_hat[0,0]]

=====
Diagram 1L-935
=====

External legs:
  leaf 0 ← dn2
  leaf 1 ← dn1
Vertex assignments:
  v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

coeff =  $\frac{1}{2} nstar_1$ 

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

coeff =  $\frac{1}{2} \varphi_2''$ 

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

coeff =  $\frac{1}{2} nstar_1$ 

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

coeff =  $\frac{1}{2} \varphi_1''$ 

Edges (propagator assignments):
  2 → 3:  nt1 → dv2  [G_hat[0,3]]
  2 → 5:  nt1 → dv1  [G_hat[0,2]]
  3 → 0:  nt2 → dn2  [G_hat[1,1]]
  4 → 3:  nt1 → dv2  [G_hat[0,3]]
  4 → 5:  nt1 → dv1  [G_hat[0,2]]
  5 → 1:  nt1 → dn1  [G_hat[0,0]]

```

=====

Diagram 1L-936

=====

External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

Edges (propagator assignments):

2 → 3: nt1 → dv2 [G_hat[0,3]]

2 → 5: nt1 → dv1 [G_hat[0,2]]

3 → 0: nt2 → dn2 [G_hat[1,1]]

4 → 3: nt1 → dv2 [G_hat[0,3]]

4 → 5: nt1 → dv1 [G_hat[0,2]]

5 → 1: nt1 → dn1 [G_hat[0,0]]

=====

Diagram 1L-937

=====

External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

```

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

coeff =  $\frac{1}{2} nstar_1$ 

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

coeff =  $\frac{1}{2} \varphi_1''$ 

Edges (propagator assignments):
  2 → 3:  nt1 → dv2  [G_hat[0,3]]
  2 → 5:  nt1 → dv1  [G_hat[0,2]]
  3 → 0:  nt2 → dn2  [G_hat[1,1]]
  4 → 3:  nt1 → dv2  [G_hat[0,3]]
  4 → 5:  nt1 → dv1  [G_hat[0,2]]
  5 → 1:  nt1 → dn1  [G_hat[0,0]]

=====
Diagram 1L-938
=====

External legs:
  leaf 0 ← dn2
  leaf 1 ← dn1
Vertex assignments:
  v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

coeff =  $\frac{1}{2} nstar_1$ 

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

coeff =  $\frac{1}{2} \varphi_2''$ 

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

coeff =  $\frac{1}{2} nstar_1$ 

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

coeff =  $\frac{1}{2} \varphi_1''$ 

Edges (propagator assignments):
  2 → 3:  nt1 → dv2  [G_hat[0,3]]
  2 → 5:  nt1 → dv1  [G_hat[0,2]]
  3 → 0:  nt2 → dn2  [G_hat[1,1]]
  4 → 3:  nt1 → dv2  [G_hat[0,3]]
  4 → 5:  nt1 → dv1  [G_hat[0,2]]
  5 → 1:  nt1 → dn1  [G_hat[0,0]]

```

=====

Diagram 1L-939

=====

External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

Edges (propagator assignments):

2 → 3: nt1 → dv2 [G_hat[0,3]]

2 → 5: nt1 → dv1 [G_hat[0,2]]

3 → 0: nt2 → dn2 [G_hat[1,1]]

4 → 3: nt1 → dv2 [G_hat[0,3]]

4 → 5: nt1 → dv1 [G_hat[0,2]]

5 → 1: nt1 → dn1 [G_hat[0,0]]

=====

Diagram 1L-940

=====

External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

```

    v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

    coeff =  $\frac{1}{2} nstar_1$ 

    v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

    coeff =  $\frac{1}{2} \varphi_1''$ 

    Edges (propagator assignments):
      2 → 3:  nt1 → dv2  [G_hat[0,3]]
      2 → 5:  nt1 → dv1  [G_hat[0,2]]
      3 → 0:  nt2 → dn2  [G_hat[1,1]]
      4 → 3:  nt1 → dv2  [G_hat[0,3]]
      4 → 5:  nt1 → dv1  [G_hat[0,2]]
      5 → 1:  nt1 → dn1  [G_hat[0,0]]

=====
Diagram 1L-941
=====

    External legs:
      leaf 0 ← dn2
      leaf 1 ← dn1
    Vertex assignments:
      v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

      coeff =  $\frac{1}{2} nstar_1$ 

      v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

      coeff =  $\frac{1}{2} \varphi_2''$ 

      v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

      coeff =  $\frac{1}{2} nstar_1$ 

      v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

      coeff =  $\frac{1}{2} \varphi_1''$ 

    Edges (propagator assignments):
      2 → 3:  nt1 → dv2  [G_hat[0,3]]
      2 → 5:  nt1 → dv1  [G_hat[0,2]]
      3 → 0:  nt2 → dn2  [G_hat[1,1]]
      4 → 3:  nt1 → dv2  [G_hat[0,3]]
      4 → 5:  nt1 → dv1  [G_hat[0,2]]
      5 → 1:  nt1 → dn1  [G_hat[0,0]]

```


=====

Diagram 1L-942

=====

External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

Edges (propagator assignments):

2 → 3: nt1 → dv2 [G_hat[0,3]]

2 → 5: nt1 → dv1 [G_hat[0,2]]

3 → 0: nt2 → dn2 [G_hat[1,1]]

4 → 3: nt1 → dv2 [G_hat[0,3]]

4 → 5: nt1 → dv1 [G_hat[0,2]]

5 → 1: nt1 → dn1 [G_hat[0,0]]

=====

Diagram 1L-943

=====

External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

```

    v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

    coeff =  $\frac{1}{2} nstar_1$ 

    v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

    coeff =  $\frac{1}{2} \varphi_1''$ 

    Edges (propagator assignments):
      2 → 3:  nt1 → dv2  [G_hat[0,3]]
      2 → 5:  nt1 → dv1  [G_hat[0,2]]
      3 → 0:  nt2 → dn2  [G_hat[1,1]]
      4 → 3:  nt1 → dv2  [G_hat[0,3]]
      4 → 5:  nt1 → dv1  [G_hat[0,2]]
      5 → 1:  nt1 → dn1  [G_hat[0,0]]

=====
Diagram 1L-944
=====

    External legs:
      leaf 0 ← dn2
      leaf 1 ← dn1
    Vertex assignments:
      v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

      coeff =  $\frac{1}{2} nstar_1$ 

      v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

      coeff =  $\frac{1}{2} \varphi_2''$ 

      v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

      coeff =  $\frac{1}{2} nstar_1$ 

      v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

      coeff =  $\frac{1}{2} \varphi_1''$ 

    Edges (propagator assignments):
      2 → 3:  nt1 → dv2  [G_hat[0,3]]
      2 → 5:  nt1 → dv1  [G_hat[0,2]]
      3 → 0:  nt2 → dn2  [G_hat[1,1]]
      4 → 3:  nt1 → dv2  [G_hat[0,3]]
      4 → 5:  nt1 → dv1  [G_hat[0,2]]
      5 → 1:  nt1 → dn1  [G_hat[0,0]]

```

=====

Diagram 1L-945

=====

External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

Edges (propagator assignments):

2 → 3: nt1 → dv2 [G_hat[0,3]]

2 → 5: nt1 → dv2 [G_hat[0,3]]

3 → 0: nt2 → dn2 [G_hat[1,1]]

4 → 3: nt1 → dv2 [G_hat[0,3]]

4 → 5: nt1 → dv2 [G_hat[0,3]]

5 → 1: nt2 → dn1 [G_hat[1,0]]

=====

Diagram 1L-946

=====

External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

```

    v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

coeff =  $\frac{1}{2} nstar_1$ 

    v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

coeff =  $\frac{1}{2} \varphi_2''$ 

Edges (propagator assignments):
    2 → 3:  nt1 → dv2  [G_hat[0,3]]
    2 → 5:  nt1 → dv2  [G_hat[0,3]]
    3 → 0:  nt2 → dn2  [G_hat[1,1]]
    4 → 3:  nt1 → dv2  [G_hat[0,3]]
    4 → 5:  nt1 → dv2  [G_hat[0,3]]
    5 → 1:  nt2 → dn1  [G_hat[1,0]]

=====
Diagram 1L-947
=====

External legs:
    leaf 0 ← dn2
    leaf 1 ← dn1
Vertex assignments:
    v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

coeff =  $\frac{1}{2} nstar_1$ 

    v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

coeff =  $\frac{1}{2} \varphi_2''$ 

    v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

coeff =  $\frac{1}{2} nstar_1$ 

    v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

coeff =  $\frac{1}{2} \varphi_2''$ 

Edges (propagator assignments):
    2 → 3:  nt1 → dv2  [G_hat[0,3]]
    2 → 5:  nt1 → dv2  [G_hat[0,3]]
    3 → 0:  nt2 → dn2  [G_hat[1,1]]
    4 → 3:  nt1 → dv2  [G_hat[0,3]]
    4 → 5:  nt1 → dv2  [G_hat[0,3]]
    5 → 1:  nt2 → dn1  [G_hat[1,0]]

```

=====

Diagram 1L-948

=====

External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

Edges (propagator assignments):

2 → 3: nt1 → dv2 [G_hat[0,3]]

2 → 5: nt1 → dv2 [G_hat[0,3]]

3 → 0: nt2 → dn2 [G_hat[1,1]]

4 → 3: nt1 → dv2 [G_hat[0,3]]

4 → 5: nt1 → dv2 [G_hat[0,3]]

5 → 1: nt2 → dn1 [G_hat[1,0]]

=====

Diagram 1L-949

=====

External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

```

    v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

    coeff =  $\frac{1}{2} nstar_1$ 

    v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

    coeff =  $\frac{1}{2} \varphi_2''$ 

    Edges (propagator assignments):
      2 → 3:  nt1 → dv2  [G_hat[0,3]]
      2 → 5:  nt1 → dv2  [G_hat[0,3]]
      3 → 0:  nt2 → dn2  [G_hat[1,1]]
      4 → 3:  nt1 → dv2  [G_hat[0,3]]
      4 → 5:  nt1 → dv2  [G_hat[0,3]]
      5 → 1:  nt2 → dn1  [G_hat[1,0]]

=====
Diagram 1L-950
=====

    External legs:
      leaf 0 ← dn2
      leaf 1 ← dn1
    Vertex assignments:
      v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

      coeff =  $\frac{1}{2} nstar_1$ 

      v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

      coeff =  $\frac{1}{2} \varphi_2''$ 

      v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

      coeff =  $\frac{1}{2} nstar_1$ 

      v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

      coeff =  $\frac{1}{2} \varphi_2''$ 

    Edges (propagator assignments):
      2 → 3:  nt1 → dv2  [G_hat[0,3]]
      2 → 5:  nt1 → dv2  [G_hat[0,3]]
      3 → 0:  nt2 → dn2  [G_hat[1,1]]
      4 → 3:  nt1 → dv2  [G_hat[0,3]]
      4 → 5:  nt1 → dv2  [G_hat[0,3]]
      5 → 1:  nt2 → dn1  [G_hat[1,0]]

```

=====

Diagram 1L-951

=====

External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

Edges (propagator assignments):

2 → 3: nt1 → dv2 [G_hat[0,3]]

2 → 5: nt1 → dv2 [G_hat[0,3]]

3 → 0: nt2 → dn2 [G_hat[1,1]]

4 → 3: nt1 → dv2 [G_hat[0,3]]

4 → 5: nt1 → dv2 [G_hat[0,3]]

5 → 1: nt2 → dn1 [G_hat[1,0]]

=====

Diagram 1L-952

=====

External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

```

    v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

coeff =  $\frac{1}{2} nstar_1$ 

    v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

coeff =  $\frac{1}{2} \varphi_2''$ 

Edges (propagator assignments):
    2 → 3:  nt1 → dv2  [G_hat[0,3]]
    2 → 5:  nt1 → dv2  [G_hat[0,3]]
    3 → 0:  nt2 → dn2  [G_hat[1,1]]
    4 → 3:  nt1 → dv2  [G_hat[0,3]]
    4 → 5:  nt1 → dv2  [G_hat[0,3]]
    5 → 1:  nt2 → dn1  [G_hat[1,0]]

=====
Diagram 1L-953
=====

External legs:
    leaf 0 ← dn2
    leaf 1 ← dn1
Vertex assignments:
    v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

coeff =  $\frac{1}{2} nstar_1$ 

    v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

coeff =  $\frac{1}{2} \varphi_2''$ 

    v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

coeff =  $\frac{1}{2} nstar_1$ 

    v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

coeff =  $\frac{1}{2} \varphi_2''$ 

Edges (propagator assignments):
    2 → 3:  nt1 → dv2  [G_hat[0,3]]
    2 → 5:  nt1 → dv2  [G_hat[0,3]]
    3 → 0:  nt2 → dn2  [G_hat[1,1]]
    4 → 3:  nt1 → dv2  [G_hat[0,3]]
    4 → 5:  nt1 → dv2  [G_hat[0,3]]
    5 → 1:  nt2 → dn1  [G_hat[1,0]]

```


=====

Diagram 1L-954

=====

External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

Edges (propagator assignments):

2 → 3: nt1 → dv2 [G_hat[0,3]]

2 → 5: nt1 → dv2 [G_hat[0,3]]

3 → 0: nt2 → dn2 [G_hat[1,1]]

4 → 3: nt1 → dv2 [G_hat[0,3]]

4 → 5: nt1 → dv2 [G_hat[0,3]]

5 → 1: nt2 → dn1 [G_hat[1,0]]

=====

Diagram 1L-955

=====

External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

```

    v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

coeff =  $\frac{1}{2} nstar_1$ 

    v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

coeff =  $\frac{1}{2} \varphi_2''$ 

Edges (propagator assignments):
    2 → 3:  nt1 → dv2  [G_hat[0,3]]
    2 → 5:  nt1 → dv2  [G_hat[0,3]]
    3 → 0:  nt2 → dn2  [G_hat[1,1]]
    4 → 3:  nt1 → dv2  [G_hat[0,3]]
    4 → 5:  nt1 → dv2  [G_hat[0,3]]
    5 → 1:  nt2 → dn1  [G_hat[1,0]]

=====
Diagram 1L-956
=====

External legs:
    leaf 0 ← dn2
    leaf 1 ← dn1
Vertex assignments:
    v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

coeff =  $\frac{1}{2} nstar_1$ 

    v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

coeff =  $\frac{1}{2} \varphi_2''$ 

    v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

coeff =  $\frac{1}{2} nstar_1$ 

    v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

coeff =  $\frac{1}{2} \varphi_2''$ 

Edges (propagator assignments):
    2 → 3:  nt1 → dv2  [G_hat[0,3]]
    2 → 5:  nt1 → dv2  [G_hat[0,3]]
    3 → 0:  nt2 → dn2  [G_hat[1,1]]
    4 → 3:  nt1 → dv2  [G_hat[0,3]]
    4 → 5:  nt1 → dv2  [G_hat[0,3]]
    5 → 1:  nt2 → dn1  [G_hat[1,0]]

```

=====

Diagram 1L-957

=====

External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

Edges (propagator assignments):

2 → 3: nt1 → dv2 [G_hat[0,3]]

2 → 5: nt1 → dv2 [G_hat[0,3]]

3 → 0: nt2 → dn2 [G_hat[1,1]]

4 → 3: nt1 → dv2 [G_hat[0,3]]

4 → 5: nt1 → dv2 [G_hat[0,3]]

5 → 1: nt2 → dn1 [G_hat[1,0]]

=====

Diagram 1L-958

=====

External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

```

    v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

    coeff =  $\frac{1}{2} nstar_1$ 

    v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

    coeff =  $\frac{1}{2} \varphi_2''$ 

    Edges (propagator assignments):
      2 → 3:  nt1 → dv2  [G_hat[0,3]]
      2 → 5:  nt1 → dv2  [G_hat[0,3]]
      3 → 0:  nt2 → dn2  [G_hat[1,1]]
      4 → 3:  nt1 → dv2  [G_hat[0,3]]
      4 → 5:  nt1 → dv2  [G_hat[0,3]]
      5 → 1:  nt2 → dn1  [G_hat[1,0]]

=====
Diagram 1L-959
=====

    External legs:
      leaf 0 ← dn2
      leaf 1 ← dn1
    Vertex assignments:
      v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

      coeff =  $\frac{1}{2} nstar_1$ 

      v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

      coeff =  $\frac{1}{2} \varphi_2''$ 

      v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

      coeff =  $\frac{1}{2} nstar_1$ 

      v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

      coeff =  $\frac{1}{2} \varphi_2''$ 

    Edges (propagator assignments):
      2 → 3:  nt1 → dv2  [G_hat[0,3]]
      2 → 5:  nt1 → dv2  [G_hat[0,3]]
      3 → 0:  nt2 → dn2  [G_hat[1,1]]
      4 → 3:  nt1 → dv2  [G_hat[0,3]]
      4 → 5:  nt1 → dv2  [G_hat[0,3]]
      5 → 1:  nt2 → dn1  [G_hat[1,0]]

```

=====

Diagram 1L-960

=====

External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

Edges (propagator assignments):

2 → 3: nt1 → dv2 [G_hat[0,3]]

2 → 5: nt1 → dv2 [G_hat[0,3]]

3 → 0: nt2 → dn2 [G_hat[1,1]]

4 → 3: nt1 → dv2 [G_hat[0,3]]

4 → 5: nt1 → dv2 [G_hat[0,3]]

5 → 1: nt2 → dn1 [G_hat[1,0]]

=====

Diagram 1L-961

=====

External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

```

    v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

    coeff =  $\frac{1}{2} nstar_2$ 

    v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

    coeff =  $\frac{1}{2} \varphi_1''$ 

    Edges (propagator assignments):
      2 → 3:  nt1 → dv1  [G_hat[0,2]]
      2 → 5:  nt1 → dv1  [G_hat[0,2]]
      3 → 0:  nt1 → dn2  [G_hat[0,1]]
      4 → 3:  nt2 → dv1  [G_hat[1,2]]
      4 → 5:  nt2 → dv1  [G_hat[1,2]]
      5 → 1:  nt1 → dn1  [G_hat[0,0]]

=====
Diagram 1L-962
=====

    External legs:
      leaf 0 ← dn2
      leaf 1 ← dn1
    Vertex assignments:
      v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

      coeff =  $\frac{1}{2} nstar_1$ 

      v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

      coeff =  $\frac{1}{2} \varphi_1''$ 

      v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

      coeff =  $\frac{1}{2} nstar_2$ 

      v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

      coeff =  $\frac{1}{2} \varphi_1''$ 

    Edges (propagator assignments):
      2 → 3:  nt1 → dv1  [G_hat[0,2]]
      2 → 5:  nt1 → dv1  [G_hat[0,2]]
      3 → 0:  nt1 → dn2  [G_hat[0,1]]
      4 → 3:  nt2 → dv1  [G_hat[1,2]]
      4 → 5:  nt2 → dv1  [G_hat[1,2]]
      5 → 1:  nt1 → dn1  [G_hat[0,0]]

```

=====

Diagram 1L-963

=====

External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

Edges (propagator assignments):

2 → 3: nt1 → dv1 [G_hat[0,2]]

2 → 5: nt1 → dv1 [G_hat[0,2]]

3 → 0: nt1 → dn2 [G_hat[0,1]]

4 → 3: nt2 → dv1 [G_hat[1,2]]

4 → 5: nt2 → dv1 [G_hat[1,2]]

5 → 1: nt1 → dn1 [G_hat[0,0]]

=====

Diagram 1L-964

=====

External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

```

    v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

    coeff =  $\frac{1}{2} nstar_2$ 

    v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

    coeff =  $\frac{1}{2} \varphi_1''$ 

    Edges (propagator assignments):
      2 → 3:  nt1 → dv1  [G_hat[0,2]]
      2 → 5:  nt1 → dv1  [G_hat[0,2]]
      3 → 0:  nt1 → dn2  [G_hat[0,1]]
      4 → 3:  nt2 → dv1  [G_hat[1,2]]
      4 → 5:  nt2 → dv1  [G_hat[1,2]]
      5 → 1:  nt1 → dn1  [G_hat[0,0]]

=====
Diagram 1L-965
=====

    External legs:
      leaf 0 ← dn2
      leaf 1 ← dn1
    Vertex assignments:
      v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

      coeff =  $\frac{1}{2} nstar_1$ 

      v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

      coeff =  $\frac{1}{2} \varphi_1''$ 

      v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

      coeff =  $\frac{1}{2} nstar_2$ 

      v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

      coeff =  $\frac{1}{2} \varphi_1''$ 

    Edges (propagator assignments):
      2 → 3:  nt1 → dv1  [G_hat[0,2]]
      2 → 5:  nt1 → dv1  [G_hat[0,2]]
      3 → 0:  nt1 → dn2  [G_hat[0,1]]
      4 → 3:  nt2 → dv1  [G_hat[1,2]]
      4 → 5:  nt2 → dv1  [G_hat[1,2]]
      5 → 1:  nt1 → dn1  [G_hat[0,0]]

```


=====

Diagram 1L-966

=====

External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

Edges (propagator assignments):

2 → 3: nt1 → dv1 [G_hat[0,2]]

2 → 5: nt1 → dv1 [G_hat[0,2]]

3 → 0: nt1 → dn2 [G_hat[0,1]]

4 → 3: nt2 → dv1 [G_hat[1,2]]

4 → 5: nt2 → dv1 [G_hat[1,2]]

5 → 1: nt1 → dn1 [G_hat[0,0]]

=====

Diagram 1L-967

=====

External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

```

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

coeff =  $\frac{1}{2} nstar_2$ 

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

coeff =  $\frac{1}{2} \varphi_1''$ 

Edges (propagator assignments):
  2 → 3:  nt1 → dv1  [G_hat[0,2]]
  2 → 5:  nt1 → dv1  [G_hat[0,2]]
  3 → 0:  nt1 → dn2  [G_hat[0,1]]
  4 → 3:  nt2 → dv1  [G_hat[1,2]]
  4 → 5:  nt2 → dv1  [G_hat[1,2]]
  5 → 1:  nt1 → dn1  [G_hat[0,0]]

=====
Diagram 1L-968
=====

External legs:
  leaf 0 ← dn2
  leaf 1 ← dn1
Vertex assignments:
  v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

coeff =  $\frac{1}{2} nstar_1$ 

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

coeff =  $\frac{1}{2} \varphi_1''$ 

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

coeff =  $\frac{1}{2} nstar_2$ 

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

coeff =  $\frac{1}{2} \varphi_1''$ 

Edges (propagator assignments):
  2 → 3:  nt1 → dv1  [G_hat[0,2]]
  2 → 5:  nt1 → dv1  [G_hat[0,2]]
  3 → 0:  nt1 → dn2  [G_hat[0,1]]
  4 → 3:  nt2 → dv1  [G_hat[1,2]]
  4 → 5:  nt2 → dv1  [G_hat[1,2]]
  5 → 1:  nt1 → dn1  [G_hat[0,0]]

```

=====

Diagram 1L-969

=====

External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

Edges (propagator assignments):

2 → 3: nt1 → dv1 [G_hat[0,2]]

2 → 5: nt1 → dv1 [G_hat[0,2]]

3 → 0: nt1 → dn2 [G_hat[0,1]]

4 → 3: nt2 → dv1 [G_hat[1,2]]

4 → 5: nt2 → dv1 [G_hat[1,2]]

5 → 1: nt1 → dn1 [G_hat[0,0]]

=====

Diagram 1L-970

=====

External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

```

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

coeff =  $\frac{1}{2} nstar_2$ 

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

coeff =  $\frac{1}{2} \varphi_1''$ 

Edges (propagator assignments):
  2 → 3:  nt1 → dv1  [G_hat[0,2]]
  2 → 5:  nt1 → dv1  [G_hat[0,2]]
  3 → 0:  nt1 → dn2  [G_hat[0,1]]
  4 → 3:  nt2 → dv1  [G_hat[1,2]]
  4 → 5:  nt2 → dv1  [G_hat[1,2]]
  5 → 1:  nt1 → dn1  [G_hat[0,0]]

=====
Diagram 1L-971
=====

External legs:
  leaf 0 ← dn2
  leaf 1 ← dn1
Vertex assignments:
  v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

coeff =  $\frac{1}{2} nstar_1$ 

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

coeff =  $\frac{1}{2} \varphi_1''$ 

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

coeff =  $\frac{1}{2} nstar_2$ 

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

coeff =  $\frac{1}{2} \varphi_1''$ 

Edges (propagator assignments):
  2 → 3:  nt1 → dv1  [G_hat[0,2]]
  2 → 5:  nt1 → dv1  [G_hat[0,2]]
  3 → 0:  nt1 → dn2  [G_hat[0,1]]
  4 → 3:  nt2 → dv1  [G_hat[1,2]]
  4 → 5:  nt2 → dv1  [G_hat[1,2]]
  5 → 1:  nt1 → dn1  [G_hat[0,0]]

```

=====

Diagram 1L-972

=====

External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

Edges (propagator assignments):

2 → 3: nt1 → dv1 [G_hat[0,2]]

2 → 5: nt1 → dv1 [G_hat[0,2]]

3 → 0: nt1 → dn2 [G_hat[0,1]]

4 → 3: nt2 → dv1 [G_hat[1,2]]

4 → 5: nt2 → dv1 [G_hat[1,2]]

5 → 1: nt1 → dn1 [G_hat[0,0]]

=====

Diagram 1L-973

=====

External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

```

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

coeff =  $\frac{1}{2} nstar_2$ 

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

coeff =  $\frac{1}{2} \varphi_1''$ 

Edges (propagator assignments):
  2 → 3:  nt1 → dv1  [G_hat[0,2]]
  2 → 5:  nt1 → dv1  [G_hat[0,2]]
  3 → 0:  nt1 → dn2  [G_hat[0,1]]
  4 → 3:  nt2 → dv1  [G_hat[1,2]]
  4 → 5:  nt2 → dv1  [G_hat[1,2]]
  5 → 1:  nt1 → dn1  [G_hat[0,0]]

=====
Diagram 1L-974
=====

External legs:
  leaf 0 ← dn2
  leaf 1 ← dn1
Vertex assignments:
  v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

coeff =  $\frac{1}{2} nstar_1$ 

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

coeff =  $\frac{1}{2} \varphi_1''$ 

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

coeff =  $\frac{1}{2} nstar_2$ 

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

coeff =  $\frac{1}{2} \varphi_1''$ 

Edges (propagator assignments):
  2 → 3:  nt1 → dv1  [G_hat[0,2]]
  2 → 5:  nt1 → dv1  [G_hat[0,2]]
  3 → 0:  nt1 → dn2  [G_hat[0,1]]
  4 → 3:  nt2 → dv1  [G_hat[1,2]]
  4 → 5:  nt2 → dv1  [G_hat[1,2]]
  5 → 1:  nt1 → dn1  [G_hat[0,0]]

```

=====

Diagram 1L-975

=====

External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

Edges (propagator assignments):

2 → 3: nt1 → dv1 [G_hat[0,2]]

2 → 5: nt1 → dv1 [G_hat[0,2]]

3 → 0: nt1 → dn2 [G_hat[0,1]]

4 → 3: nt2 → dv1 [G_hat[1,2]]

4 → 5: nt2 → dv1 [G_hat[1,2]]

5 → 1: nt1 → dn1 [G_hat[0,0]]

=====

Diagram 1L-976

=====

External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

```

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

coeff =  $\frac{1}{2} nstar_2$ 

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

coeff =  $\frac{1}{2} \varphi_1''$ 

Edges (propagator assignments):
  2 → 3:  nt1 → dv1  [G_hat[0,2]]
  2 → 5:  nt1 → dv1  [G_hat[0,2]]
  3 → 0:  nt1 → dn2  [G_hat[0,1]]
  4 → 3:  nt2 → dv1  [G_hat[1,2]]
  4 → 5:  nt2 → dv1  [G_hat[1,2]]
  5 → 1:  nt1 → dn1  [G_hat[0,0]]

=====
Diagram 1L-977
=====

External legs:
  leaf 0 ← dn2
  leaf 1 ← dn1
Vertex assignments:
  v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

coeff =  $\frac{1}{2} nstar_1$ 

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

coeff =  $\frac{1}{2} \varphi_1''$ 

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

coeff =  $\frac{1}{2} nstar_2$ 

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

coeff =  $\frac{1}{2} \varphi_2''$ 

Edges (propagator assignments):
  2 → 3:  nt1 → dv1  [G_hat[0,2]]
  2 → 5:  nt1 → dv2  [G_hat[0,3]]
  3 → 0:  nt1 → dn2  [G_hat[0,1]]
  4 → 3:  nt2 → dv1  [G_hat[1,2]]
  4 → 5:  nt2 → dv2  [G_hat[1,3]]
  5 → 1:  nt2 → dn1  [G_hat[1,0]]

```


=====

Diagram 1L-978

=====

External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

Edges (propagator assignments):

2 → 3: nt1 → dv1 [G_hat[0,2]]

2 → 5: nt1 → dv2 [G_hat[0,3]]

3 → 0: nt1 → dn2 [G_hat[0,1]]

4 → 3: nt2 → dv1 [G_hat[1,2]]

4 → 5: nt2 → dv2 [G_hat[1,3]]

5 → 1: nt2 → dn1 [G_hat[1,0]]

=====

Diagram 1L-979

=====

External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

```

    v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

    coeff =  $\frac{1}{2} nstar_2$ 

    v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

    coeff =  $\frac{1}{2} \varphi_2''$ 

    Edges (propagator assignments):
      2 → 3:  nt1 → dv1  [G_hat[0,2]]
      2 → 5:  nt1 → dv2  [G_hat[0,3]]
      3 → 0:  nt1 → dn2  [G_hat[0,1]]
      4 → 3:  nt2 → dv1  [G_hat[1,2]]
      4 → 5:  nt2 → dv2  [G_hat[1,3]]
      5 → 1:  nt2 → dn1  [G_hat[1,0]]

=====
Diagram 1L-980
=====

    External legs:
      leaf 0 ← dn2
      leaf 1 ← dn1
    Vertex assignments:
      v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

      coeff =  $\frac{1}{2} nstar_1$ 

      v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

      coeff =  $\frac{1}{2} \varphi_1''$ 

      v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

      coeff =  $\frac{1}{2} nstar_2$ 

      v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

      coeff =  $\frac{1}{2} \varphi_2''$ 

    Edges (propagator assignments):
      2 → 3:  nt1 → dv1  [G_hat[0,2]]
      2 → 5:  nt1 → dv2  [G_hat[0,3]]
      3 → 0:  nt1 → dn2  [G_hat[0,1]]
      4 → 3:  nt2 → dv1  [G_hat[1,2]]
      4 → 5:  nt2 → dv2  [G_hat[1,3]]
      5 → 1:  nt2 → dn1  [G_hat[1,0]]

```

=====

Diagram 1L-981

=====

External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

Edges (propagator assignments):

2 → 3: nt1 → dv1 [G_hat[0,2]]

2 → 5: nt1 → dv2 [G_hat[0,3]]

3 → 0: nt1 → dn2 [G_hat[0,1]]

4 → 3: nt2 → dv1 [G_hat[1,2]]

4 → 5: nt2 → dv2 [G_hat[1,3]]

5 → 1: nt2 → dn1 [G_hat[1,0]]

=====

Diagram 1L-982

=====

External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

```

    v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

    coeff =  $\frac{1}{2} nstar_2$ 

    v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

    coeff =  $\frac{1}{2} \varphi_2''$ 

    Edges (propagator assignments):
      2 → 3:  nt1 → dv1  [G_hat[0,2]]
      2 → 5:  nt1 → dv2  [G_hat[0,3]]
      3 → 0:  nt1 → dn2  [G_hat[0,1]]
      4 → 3:  nt2 → dv1  [G_hat[1,2]]
      4 → 5:  nt2 → dv2  [G_hat[1,3]]
      5 → 1:  nt2 → dn1  [G_hat[1,0]]

=====
Diagram 1L-983
=====

    External legs:
      leaf 0 ← dn2
      leaf 1 ← dn1
    Vertex assignments:
      v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

      coeff =  $\frac{1}{2} nstar_1$ 

      v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

      coeff =  $\frac{1}{2} \varphi_1''$ 

      v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

      coeff =  $\frac{1}{2} nstar_2$ 

      v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

      coeff =  $\frac{1}{2} \varphi_2''$ 

    Edges (propagator assignments):
      2 → 3:  nt1 → dv1  [G_hat[0,2]]
      2 → 5:  nt1 → dv2  [G_hat[0,3]]
      3 → 0:  nt1 → dn2  [G_hat[0,1]]
      4 → 3:  nt2 → dv1  [G_hat[1,2]]
      4 → 5:  nt2 → dv2  [G_hat[1,3]]
      5 → 1:  nt2 → dn1  [G_hat[1,0]]

```

=====

Diagram 1L-984

=====

External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

Edges (propagator assignments):

2 → 3: nt1 → dv1 [G_hat[0,2]]

2 → 5: nt1 → dv2 [G_hat[0,3]]

3 → 0: nt1 → dn2 [G_hat[0,1]]

4 → 3: nt2 → dv1 [G_hat[1,2]]

4 → 5: nt2 → dv2 [G_hat[1,3]]

5 → 1: nt2 → dn1 [G_hat[1,0]]

=====

Diagram 1L-985

=====

External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

```

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

coeff =  $\frac{1}{2} nstar_2$ 

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

coeff =  $\frac{1}{2} \varphi_2''$ 

Edges (propagator assignments):
  2 → 3:  nt1 → dv1  [G_hat[0,2]]
  2 → 5:  nt1 → dv2  [G_hat[0,3]]
  3 → 0:  nt1 → dn2  [G_hat[0,1]]
  4 → 3:  nt2 → dv1  [G_hat[1,2]]
  4 → 5:  nt2 → dv2  [G_hat[1,3]]
  5 → 1:  nt2 → dn1  [G_hat[1,0]]

=====
Diagram 1L-986
=====

External legs:
  leaf 0 ← dn2
  leaf 1 ← dn1
Vertex assignments:
  v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

coeff =  $\frac{1}{2} nstar_1$ 

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

coeff =  $\frac{1}{2} \varphi_1''$ 

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

coeff =  $\frac{1}{2} nstar_2$ 

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

coeff =  $\frac{1}{2} \varphi_2''$ 

Edges (propagator assignments):
  2 → 3:  nt1 → dv1  [G_hat[0,2]]
  2 → 5:  nt1 → dv2  [G_hat[0,3]]
  3 → 0:  nt1 → dn2  [G_hat[0,1]]
  4 → 3:  nt2 → dv1  [G_hat[1,2]]
  4 → 5:  nt2 → dv2  [G_hat[1,3]]
  5 → 1:  nt2 → dn1  [G_hat[1,0]]

```

=====

Diagram 1L-987

=====

External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

Edges (propagator assignments):

2 → 3: nt1 → dv1 [G_hat[0,2]]

2 → 5: nt1 → dv2 [G_hat[0,3]]

3 → 0: nt1 → dn2 [G_hat[0,1]]

4 → 3: nt2 → dv1 [G_hat[1,2]]

4 → 5: nt2 → dv2 [G_hat[1,3]]

5 → 1: nt2 → dn1 [G_hat[1,0]]

=====

Diagram 1L-988

=====

External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

```

    v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

    coeff =  $\frac{1}{2} nstar_2$ 

    v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

    coeff =  $\frac{1}{2} \varphi_2''$ 

    Edges (propagator assignments):
      2 → 3:  nt1 → dv1  [G_hat[0,2]]
      2 → 5:  nt1 → dv2  [G_hat[0,3]]
      3 → 0:  nt1 → dn2  [G_hat[0,1]]
      4 → 3:  nt2 → dv1  [G_hat[1,2]]
      4 → 5:  nt2 → dv2  [G_hat[1,3]]
      5 → 1:  nt2 → dn1  [G_hat[1,0]]

=====
Diagram 1L-989
=====

    External legs:
      leaf 0 ← dn2
      leaf 1 ← dn1
    Vertex assignments:
      v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

      coeff =  $\frac{1}{2} nstar_1$ 

      v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

      coeff =  $\frac{1}{2} \varphi_1''$ 

      v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

      coeff =  $\frac{1}{2} nstar_2$ 

      v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

      coeff =  $\frac{1}{2} \varphi_2''$ 

    Edges (propagator assignments):
      2 → 3:  nt1 → dv1  [G_hat[0,2]]
      2 → 5:  nt1 → dv2  [G_hat[0,3]]
      3 → 0:  nt1 → dn2  [G_hat[0,1]]
      4 → 3:  nt2 → dv1  [G_hat[1,2]]
      4 → 5:  nt2 → dv2  [G_hat[1,3]]
      5 → 1:  nt2 → dn1  [G_hat[1,0]]

```


=====

Diagram 1L-990

=====

External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

Edges (propagator assignments):

2 → 3: nt1 → dv1 [G_hat[0,2]]

2 → 5: nt1 → dv2 [G_hat[0,3]]

3 → 0: nt1 → dn2 [G_hat[0,1]]

4 → 3: nt2 → dv1 [G_hat[1,2]]

4 → 5: nt2 → dv2 [G_hat[1,3]]

5 → 1: nt2 → dn1 [G_hat[1,0]]

=====

Diagram 1L-991

=====

External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

```

    v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

    coeff =  $\frac{1}{2} nstar_2$ 

    v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

    coeff =  $\frac{1}{2} \varphi_2''$ 

    Edges (propagator assignments):
      2 → 3:  nt1 → dv1  [G_hat[0,2]]
      2 → 5:  nt1 → dv2  [G_hat[0,3]]
      3 → 0:  nt1 → dn2  [G_hat[0,1]]
      4 → 3:  nt2 → dv1  [G_hat[1,2]]
      4 → 5:  nt2 → dv2  [G_hat[1,3]]
      5 → 1:  nt2 → dn1  [G_hat[1,0]]

=====
Diagram 1L-992
=====

    External legs:
      leaf 0 ← dn2
      leaf 1 ← dn1
    Vertex assignments:
      v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

      coeff =  $\frac{1}{2} nstar_1$ 

      v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

      coeff =  $\frac{1}{2} \varphi_1''$ 

      v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

      coeff =  $\frac{1}{2} nstar_2$ 

      v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

      coeff =  $\frac{1}{2} \varphi_2''$ 

    Edges (propagator assignments):
      2 → 3:  nt1 → dv1  [G_hat[0,2]]
      2 → 5:  nt1 → dv2  [G_hat[0,3]]
      3 → 0:  nt1 → dn2  [G_hat[0,1]]
      4 → 3:  nt2 → dv1  [G_hat[1,2]]
      4 → 5:  nt2 → dv2  [G_hat[1,3]]
      5 → 1:  nt2 → dn1  [G_hat[1,0]]

```

=====

Diagram 1L-993

=====

External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

Edges (propagator assignments):

2 → 3: nt1 → dv2 [G_hat[0,3]]

2 → 5: nt1 → dv1 [G_hat[0,2]]

3 → 0: nt2 → dn2 [G_hat[1,1]]

4 → 3: nt2 → dv2 [G_hat[1,3]]

4 → 5: nt2 → dv1 [G_hat[1,2]]

5 → 1: nt1 → dn1 [G_hat[0,0]]

=====

Diagram 1L-994

=====

External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

```

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

coeff =  $\frac{1}{2} nstar_2$ 

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

coeff =  $\frac{1}{2} \varphi_1''$ 

Edges (propagator assignments):
  2 → 3:  nt1 → dv2  [G_hat[0,3]]
  2 → 5:  nt1 → dv1  [G_hat[0,2]]
  3 → 0:  nt2 → dn2  [G_hat[1,1]]
  4 → 3:  nt2 → dv2  [G_hat[1,3]]
  4 → 5:  nt2 → dv1  [G_hat[1,2]]
  5 → 1:  nt1 → dn1  [G_hat[0,0]]

=====
Diagram 1L-995
=====

External legs:
  leaf 0 ← dn2
  leaf 1 ← dn1
Vertex assignments:
  v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

coeff =  $\frac{1}{2} nstar_1$ 

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

coeff =  $\frac{1}{2} \varphi_2''$ 

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

coeff =  $\frac{1}{2} nstar_2$ 

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

coeff =  $\frac{1}{2} \varphi_1''$ 

Edges (propagator assignments):
  2 → 3:  nt1 → dv2  [G_hat[0,3]]
  2 → 5:  nt1 → dv1  [G_hat[0,2]]
  3 → 0:  nt2 → dn2  [G_hat[1,1]]
  4 → 3:  nt2 → dv2  [G_hat[1,3]]
  4 → 5:  nt2 → dv1  [G_hat[1,2]]
  5 → 1:  nt1 → dn1  [G_hat[0,0]]

```

=====

Diagram 1L-996

=====

External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

Edges (propagator assignments):

2 → 3: nt1 → dv2 [G_hat[0,3]]

2 → 5: nt1 → dv1 [G_hat[0,2]]

3 → 0: nt2 → dn2 [G_hat[1,1]]

4 → 3: nt2 → dv2 [G_hat[1,3]]

4 → 5: nt2 → dv1 [G_hat[1,2]]

5 → 1: nt1 → dn1 [G_hat[0,0]]

=====

Diagram 1L-997

=====

External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

```

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

coeff =  $\frac{1}{2} nstar_2$ 

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

coeff =  $\frac{1}{2} \varphi_1''$ 

Edges (propagator assignments):
  2 → 3:  nt1 → dv2  [G_hat[0,3]]
  2 → 5:  nt1 → dv1  [G_hat[0,2]]
  3 → 0:  nt2 → dn2  [G_hat[1,1]]
  4 → 3:  nt2 → dv2  [G_hat[1,3]]
  4 → 5:  nt2 → dv1  [G_hat[1,2]]
  5 → 1:  nt1 → dn1  [G_hat[0,0]]

=====
Diagram 1L-998
=====

External legs:
  leaf 0 ← dn2
  leaf 1 ← dn1
Vertex assignments:
  v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

coeff =  $\frac{1}{2} nstar_1$ 

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

coeff =  $\frac{1}{2} \varphi_2''$ 

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

coeff =  $\frac{1}{2} nstar_2$ 

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

coeff =  $\frac{1}{2} \varphi_1''$ 

Edges (propagator assignments):
  2 → 3:  nt1 → dv2  [G_hat[0,3]]
  2 → 5:  nt1 → dv1  [G_hat[0,2]]
  3 → 0:  nt2 → dn2  [G_hat[1,1]]
  4 → 3:  nt2 → dv2  [G_hat[1,3]]
  4 → 5:  nt2 → dv1  [G_hat[1,2]]
  5 → 1:  nt1 → dn1  [G_hat[0,0]]

```

=====

Diagram 1L-999

=====

External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

Edges (propagator assignments):

2 → 3: nt1 → dv2 [G_hat[0,3]]

2 → 5: nt1 → dv1 [G_hat[0,2]]

3 → 0: nt2 → dn2 [G_hat[1,1]]

4 → 3: nt2 → dv2 [G_hat[1,3]]

4 → 5: nt2 → dv1 [G_hat[1,2]]

5 → 1: nt1 → dn1 [G_hat[0,0]]

=====

Diagram 1L-1000

=====

External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

```

    v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

    coeff =  $\frac{1}{2} nstar_2$ 

    v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

    coeff =  $\frac{1}{2} \varphi_1''$ 

    Edges (propagator assignments):
      2 → 3:  nt1 → dv2  [G_hat[0,3]]
      2 → 5:  nt1 → dv1  [G_hat[0,2]]
      3 → 0:  nt2 → dn2  [G_hat[1,1]]
      4 → 3:  nt2 → dv2  [G_hat[1,3]]
      4 → 5:  nt2 → dv1  [G_hat[1,2]]
      5 → 1:  nt1 → dn1  [G_hat[0,0]]

=====
Diagram 1L-1001
=====

    External legs:
      leaf 0 ← dn2
      leaf 1 ← dn1

    Vertex assignments:
      v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

      coeff =  $\frac{1}{2} nstar_1$ 

      v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

      coeff =  $\frac{1}{2} \varphi_2''$ 

      v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

      coeff =  $\frac{1}{2} nstar_2$ 

      v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

      coeff =  $\frac{1}{2} \varphi_1''$ 

    Edges (propagator assignments):
      2 → 3:  nt1 → dv2  [G_hat[0,3]]
      2 → 5:  nt1 → dv1  [G_hat[0,2]]
      3 → 0:  nt2 → dn2  [G_hat[1,1]]
      4 → 3:  nt2 → dv2  [G_hat[1,3]]
      4 → 5:  nt2 → dv1  [G_hat[1,2]]
      5 → 1:  nt1 → dn1  [G_hat[0,0]]

```



```

=====
Diagram 1L-1002
=====

External legs:
  leaf 0 ← dn2
  leaf 1 ← dn1
Vertex assignments:
  v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

  coeff =  $\frac{1}{2} nstar_1$ 

  v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

  coeff =  $\frac{1}{2} \varphi_2''$ 

  v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

  coeff =  $\frac{1}{2} nstar_2$ 

  v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

  coeff =  $\frac{1}{2} \varphi_1''$ 

Edges (propagator assignments):
  2 → 3:  nt1 → dv2  [G_hat[0,3]]
  2 → 5:  nt1 → dv1  [G_hat[0,2]]
  3 → 0:  nt2 → dn2  [G_hat[1,1]]
  4 → 3:  nt2 → dv2  [G_hat[1,3]]
  4 → 5:  nt2 → dv1  [G_hat[1,2]]
  5 → 1:  nt1 → dn1  [G_hat[0,0]]

```

```

=====
Diagram 1L-1003
=====

External legs:
  leaf 0 ← dn2
  leaf 1 ← dn1
Vertex assignments:
  v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

  coeff =  $\frac{1}{2} nstar_1$ 

  v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

  coeff =  $\frac{1}{2} \varphi_2''$ 

```

```

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

coeff =  $\frac{1}{2} nstar_2$ 

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

coeff =  $\frac{1}{2} \varphi_1''$ 

Edges (propagator assignments):
  2 → 3:  nt1 → dv2  [G_hat[0,3]]
  2 → 5:  nt1 → dv1  [G_hat[0,2]]
  3 → 0:  nt2 → dn2  [G_hat[1,1]]
  4 → 3:  nt2 → dv2  [G_hat[1,3]]
  4 → 5:  nt2 → dv1  [G_hat[1,2]]
  5 → 1:  nt1 → dn1  [G_hat[0,0]]

=====
Diagram 1L-1004
=====

External legs:
  leaf 0 ← dn2
  leaf 1 ← dn1
Vertex assignments:
  v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

coeff =  $\frac{1}{2} nstar_1$ 

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

coeff =  $\frac{1}{2} \varphi_2''$ 

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

coeff =  $\frac{1}{2} nstar_2$ 

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

coeff =  $\frac{1}{2} \varphi_1''$ 

Edges (propagator assignments):
  2 → 3:  nt1 → dv2  [G_hat[0,3]]
  2 → 5:  nt1 → dv1  [G_hat[0,2]]
  3 → 0:  nt2 → dn2  [G_hat[1,1]]
  4 → 3:  nt2 → dv2  [G_hat[1,3]]
  4 → 5:  nt2 → dv1  [G_hat[1,2]]
  5 → 1:  nt1 → dn1  [G_hat[0,0]]

```

```

=====
Diagram 1L-1005
=====

External legs:
  leaf 0 ← dn2
  leaf 1 ← dn1
Vertex assignments:
  v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

  coeff =  $\frac{1}{2} nstar_1$ 

  v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

  coeff =  $\frac{1}{2} \varphi_2''$ 

  v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

  coeff =  $\frac{1}{2} nstar_2$ 

  v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

  coeff =  $\frac{1}{2} \varphi_1''$ 

Edges (propagator assignments):
  2 → 3:  nt1 → dv2  [G_hat[0,3]]
  2 → 5:  nt1 → dv1  [G_hat[0,2]]
  3 → 0:  nt2 → dn2  [G_hat[1,1]]
  4 → 3:  nt2 → dv2  [G_hat[1,3]]
  4 → 5:  nt2 → dv1  [G_hat[1,2]]
  5 → 1:  nt1 → dn1  [G_hat[0,0]]

```

```

=====
Diagram 1L-1006
=====

External legs:
  leaf 0 ← dn2
  leaf 1 ← dn1
Vertex assignments:
  v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

  coeff =  $\frac{1}{2} nstar_1$ 

  v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

  coeff =  $\frac{1}{2} \varphi_2''$ 

```

```

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

coeff =  $\frac{1}{2} nstar_2$ 

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

coeff =  $\frac{1}{2} \varphi_1''$ 

Edges (propagator assignments):
  2 → 3:  nt1 → dv2  [G_hat[0,3]]
  2 → 5:  nt1 → dv1  [G_hat[0,2]]
  3 → 0:  nt2 → dn2  [G_hat[1,1]]
  4 → 3:  nt2 → dv2  [G_hat[1,3]]
  4 → 5:  nt2 → dv1  [G_hat[1,2]]
  5 → 1:  nt1 → dn1  [G_hat[0,0]]

=====
Diagram 1L-1007
=====

External legs:
  leaf 0 ← dn2
  leaf 1 ← dn1
Vertex assignments:
  v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

coeff =  $\frac{1}{2} nstar_1$ 

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

coeff =  $\frac{1}{2} \varphi_2''$ 

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

coeff =  $\frac{1}{2} nstar_2$ 

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

coeff =  $\frac{1}{2} \varphi_1''$ 

Edges (propagator assignments):
  2 → 3:  nt1 → dv2  [G_hat[0,3]]
  2 → 5:  nt1 → dv1  [G_hat[0,2]]
  3 → 0:  nt2 → dn2  [G_hat[1,1]]
  4 → 3:  nt2 → dv2  [G_hat[1,3]]
  4 → 5:  nt2 → dv1  [G_hat[1,2]]
  5 → 1:  nt1 → dn1  [G_hat[0,0]]

```

=====

Diagram 1L-1008

=====

External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

Edges (propagator assignments):

2 → 3: nt1 → dv2 [G_hat[0,3]]

2 → 5: nt1 → dv1 [G_hat[0,2]]

3 → 0: nt2 → dn2 [G_hat[1,1]]

4 → 3: nt2 → dv2 [G_hat[1,3]]

4 → 5: nt2 → dv1 [G_hat[1,2]]

5 → 1: nt1 → dn1 [G_hat[0,0]]

=====

Diagram 1L-1009

=====

External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

```

    v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

    coeff =  $\frac{1}{2} nstar_2$ 

    v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

    coeff =  $\frac{1}{2} \varphi_2''$ 

    Edges (propagator assignments):
      2 → 3:  nt1 → dv2  [G_hat[0,3]]
      2 → 5:  nt1 → dv2  [G_hat[0,3]]
      3 → 0:  nt2 → dn2  [G_hat[1,1]]
      4 → 3:  nt2 → dv2  [G_hat[1,3]]
      4 → 5:  nt2 → dv2  [G_hat[1,3]]
      5 → 1:  nt2 → dn1  [G_hat[1,0]]

=====
Diagram 1L-1010
=====

    External legs:
      leaf 0 ← dn2
      leaf 1 ← dn1
    Vertex assignments:
      v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

      coeff =  $\frac{1}{2} nstar_1$ 

      v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

      coeff =  $\frac{1}{2} \varphi_2''$ 

      v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

      coeff =  $\frac{1}{2} nstar_2$ 

      v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

      coeff =  $\frac{1}{2} \varphi_2''$ 

    Edges (propagator assignments):
      2 → 3:  nt1 → dv2  [G_hat[0,3]]
      2 → 5:  nt1 → dv2  [G_hat[0,3]]
      3 → 0:  nt2 → dn2  [G_hat[1,1]]
      4 → 3:  nt2 → dv2  [G_hat[1,3]]
      4 → 5:  nt2 → dv2  [G_hat[1,3]]
      5 → 1:  nt2 → dn1  [G_hat[1,0]]

```

```

=====
Diagram 1L-1011
=====

External legs:
  leaf 0 ← dn2
  leaf 1 ← dn1
Vertex assignments:
  v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

  coeff =  $\frac{1}{2} nstar_1$ 

  v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

  coeff =  $\frac{1}{2} \varphi_2''$ 

  v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

  coeff =  $\frac{1}{2} nstar_2$ 

  v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

  coeff =  $\frac{1}{2} \varphi_2''$ 

Edges (propagator assignments):
  2 → 3:  nt1 → dv2  [G_hat[0,3]]
  2 → 5:  nt1 → dv2  [G_hat[0,3]]
  3 → 0:  nt2 → dn2  [G_hat[1,1]]
  4 → 3:  nt2 → dv2  [G_hat[1,3]]
  4 → 5:  nt2 → dv2  [G_hat[1,3]]
  5 → 1:  nt2 → dn1  [G_hat[1,0]]

```

```

=====
Diagram 1L-1012
=====

External legs:
  leaf 0 ← dn2
  leaf 1 ← dn1
Vertex assignments:
  v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

  coeff =  $\frac{1}{2} nstar_1$ 

  v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

  coeff =  $\frac{1}{2} \varphi_2''$ 

```

```

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

coeff =  $\frac{1}{2} nstar_2$ 

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

coeff =  $\frac{1}{2} \varphi_2''$ 

Edges (propagator assignments):
  2 → 3:  nt1 → dv2  [G_hat[0,3]]
  2 → 5:  nt1 → dv2  [G_hat[0,3]]
  3 → 0:  nt2 → dn2  [G_hat[1,1]]
  4 → 3:  nt2 → dv2  [G_hat[1,3]]
  4 → 5:  nt2 → dv2  [G_hat[1,3]]
  5 → 1:  nt2 → dn1  [G_hat[1,0]]

=====
Diagram 1L-1013
=====

External legs:
  leaf 0 ← dn2
  leaf 1 ← dn1
Vertex assignments:
  v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

coeff =  $\frac{1}{2} nstar_1$ 

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

coeff =  $\frac{1}{2} \varphi_2''$ 

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

coeff =  $\frac{1}{2} nstar_2$ 

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

coeff =  $\frac{1}{2} \varphi_2''$ 

Edges (propagator assignments):
  2 → 3:  nt1 → dv2  [G_hat[0,3]]
  2 → 5:  nt1 → dv2  [G_hat[0,3]]
  3 → 0:  nt2 → dn2  [G_hat[1,1]]
  4 → 3:  nt2 → dv2  [G_hat[1,3]]
  4 → 5:  nt2 → dv2  [G_hat[1,3]]
  5 → 1:  nt2 → dn1  [G_hat[1,0]]

```


=====

Diagram 1L-1014

=====

External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

Edges (propagator assignments):

2 → 3: nt1 → dv2 [G_hat[0,3]]

2 → 5: nt1 → dv2 [G_hat[0,3]]

3 → 0: nt2 → dn2 [G_hat[1,1]]

4 → 3: nt2 → dv2 [G_hat[1,3]]

4 → 5: nt2 → dv2 [G_hat[1,3]]

5 → 1: nt2 → dn1 [G_hat[1,0]]

=====

Diagram 1L-1015

=====

External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

```

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

coeff =  $\frac{1}{2} nstar_2$ 

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

coeff =  $\frac{1}{2} \varphi_2''$ 

Edges (propagator assignments):
  2 → 3:  nt1 → dv2  [G_hat[0,3]]
  2 → 5:  nt1 → dv2  [G_hat[0,3]]
  3 → 0:  nt2 → dn2  [G_hat[1,1]]
  4 → 3:  nt2 → dv2  [G_hat[1,3]]
  4 → 5:  nt2 → dv2  [G_hat[1,3]]
  5 → 1:  nt2 → dn1  [G_hat[1,0]]

=====
Diagram 1L-1016
=====

External legs:
  leaf 0 ← dn2
  leaf 1 ← dn1
Vertex assignments:
  v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

coeff =  $\frac{1}{2} nstar_1$ 

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

coeff =  $\frac{1}{2} \varphi_2''$ 

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

coeff =  $\frac{1}{2} nstar_2$ 

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

coeff =  $\frac{1}{2} \varphi_2''$ 

Edges (propagator assignments):
  2 → 3:  nt1 → dv2  [G_hat[0,3]]
  2 → 5:  nt1 → dv2  [G_hat[0,3]]
  3 → 0:  nt2 → dn2  [G_hat[1,1]]
  4 → 3:  nt2 → dv2  [G_hat[1,3]]
  4 → 5:  nt2 → dv2  [G_hat[1,3]]
  5 → 1:  nt2 → dn1  [G_hat[1,0]]

```

```

=====
Diagram 1L-1017
=====

External legs:
  leaf 0 ← dn2
  leaf 1 ← dn1
Vertex assignments:
  v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

  coeff =  $\frac{1}{2} nstar_1$ 

  v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

  coeff =  $\frac{1}{2} \varphi_2''$ 

  v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

  coeff =  $\frac{1}{2} nstar_2$ 

  v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

  coeff =  $\frac{1}{2} \varphi_2''$ 

Edges (propagator assignments):
  2 → 3:  nt1 → dv2  [G_hat[0,3]]
  2 → 5:  nt1 → dv2  [G_hat[0,3]]
  3 → 0:  nt2 → dn2  [G_hat[1,1]]
  4 → 3:  nt2 → dv2  [G_hat[1,3]]
  4 → 5:  nt2 → dv2  [G_hat[1,3]]
  5 → 1:  nt2 → dn1  [G_hat[1,0]]

```

```

=====
Diagram 1L-1018
=====

External legs:
  leaf 0 ← dn2
  leaf 1 ← dn1
Vertex assignments:
  v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

  coeff =  $\frac{1}{2} nstar_1$ 

  v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

  coeff =  $\frac{1}{2} \varphi_2''$ 

```

```

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

coeff =  $\frac{1}{2} nstar_2$ 

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

coeff =  $\frac{1}{2} \varphi_2''$ 

Edges (propagator assignments):
  2 → 3:  nt1 → dv2  [G_hat[0,3]]
  2 → 5:  nt1 → dv2  [G_hat[0,3]]
  3 → 0:  nt2 → dn2  [G_hat[1,1]]
  4 → 3:  nt2 → dv2  [G_hat[1,3]]
  4 → 5:  nt2 → dv2  [G_hat[1,3]]
  5 → 1:  nt2 → dn1  [G_hat[1,0]]

=====
Diagram 1L-1019
=====

External legs:
  leaf 0 ← dn2
  leaf 1 ← dn1
Vertex assignments:
  v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

coeff =  $\frac{1}{2} nstar_1$ 

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

coeff =  $\frac{1}{2} \varphi_2''$ 

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

coeff =  $\frac{1}{2} nstar_2$ 

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

coeff =  $\frac{1}{2} \varphi_2''$ 

Edges (propagator assignments):
  2 → 3:  nt1 → dv2  [G_hat[0,3]]
  2 → 5:  nt1 → dv2  [G_hat[0,3]]
  3 → 0:  nt2 → dn2  [G_hat[1,1]]
  4 → 3:  nt2 → dv2  [G_hat[1,3]]
  4 → 5:  nt2 → dv2  [G_hat[1,3]]
  5 → 1:  nt2 → dn1  [G_hat[1,0]]

```

```

=====
Diagram 1L-1020
=====
External legs:
  leaf 0 ← dn2
  leaf 1 ← dn1
Vertex assignments:
  v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]
  coeff =  $\frac{1}{2} nstar_1$ 
  v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]
  coeff =  $\frac{1}{2} \varphi_2''$ 
  v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]
  coeff =  $\frac{1}{2} nstar_2$ 
  v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]
  coeff =  $\frac{1}{2} \varphi_2''$ 
Edges (propagator assignments):
  2 → 3:  nt1 → dv2  [G_hat[0,3]]
  2 → 5:  nt1 → dv2  [G_hat[0,3]]
  3 → 0:  nt2 → dn2  [G_hat[1,1]]
  4 → 3:  nt2 → dv2  [G_hat[1,3]]
  4 → 5:  nt2 → dv2  [G_hat[1,3]]
  5 → 1:  nt2 → dn1  [G_hat[1,0]]

```

```

=====
Diagram 1L-1021
=====
External legs:
  leaf 0 ← dn2
  leaf 1 ← dn1
Vertex assignments:
  v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]
  coeff =  $\frac{1}{2} nstar_1$ 
  v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]
  coeff =  $\frac{1}{2} \varphi_2''$ 

```

```

    v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

    coeff =  $\frac{1}{2} nstar_2$ 

    v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

    coeff =  $\frac{1}{2} \varphi_2''$ 

    Edges (propagator assignments):
      2 → 3:  nt1 → dv2  [G_hat[0,3]]
      2 → 5:  nt1 → dv2  [G_hat[0,3]]
      3 → 0:  nt2 → dn2  [G_hat[1,1]]
      4 → 3:  nt2 → dv2  [G_hat[1,3]]
      4 → 5:  nt2 → dv2  [G_hat[1,3]]
      5 → 1:  nt2 → dn1  [G_hat[1,0]]

=====
Diagram 1L-1022
=====

    External legs:
      leaf 0 ← dn2
      leaf 1 ← dn1
    Vertex assignments:
      v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

      coeff =  $\frac{1}{2} nstar_1$ 

      v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

      coeff =  $\frac{1}{2} \varphi_2''$ 

      v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

      coeff =  $\frac{1}{2} nstar_2$ 

      v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

      coeff =  $\frac{1}{2} \varphi_2''$ 

    Edges (propagator assignments):
      2 → 3:  nt1 → dv2  [G_hat[0,3]]
      2 → 5:  nt1 → dv2  [G_hat[0,3]]
      3 → 0:  nt2 → dn2  [G_hat[1,1]]
      4 → 3:  nt2 → dv2  [G_hat[1,3]]
      4 → 5:  nt2 → dv2  [G_hat[1,3]]
      5 → 1:  nt2 → dn1  [G_hat[1,0]]

```

=====

Diagram 1L-1023

=====

External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

Edges (propagator assignments):

2 → 3: nt1 → dv2 [G_hat[0,3]]

2 → 5: nt1 → dv2 [G_hat[0,3]]

3 → 0: nt2 → dn2 [G_hat[1,1]]

4 → 3: nt2 → dv2 [G_hat[1,3]]

4 → 5: nt2 → dv2 [G_hat[1,3]]

5 → 1: nt2 → dn1 [G_hat[1,0]]

=====

Diagram 1L-1024

=====

External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

```

    v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

    coeff =  $\frac{1}{2} nstar_2$ 

    v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

    coeff =  $\frac{1}{2} \varphi_2''$ 

    Edges (propagator assignments):
      2 → 3:  nt1 → dv2  [G_hat[0,3]]
      2 → 5:  nt1 → dv2  [G_hat[0,3]]
      3 → 0:  nt2 → dn2  [G_hat[1,1]]
      4 → 3:  nt2 → dv2  [G_hat[1,3]]
      4 → 5:  nt2 → dv2  [G_hat[1,3]]
      5 → 1:  nt2 → dn1  [G_hat[1,0]]

=====
Diagram 1L-1025
=====

    External legs:
      leaf 0 ← dn2
      leaf 1 ← dn1
    Vertex assignments:
      v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

      coeff =  $\frac{1}{2} nstar_2$ 

      v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

      coeff =  $\frac{1}{2} \varphi_1''$ 

      v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

      coeff =  $\frac{1}{2} nstar_1$ 

      v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

      coeff =  $\frac{1}{2} \varphi_1''$ 

    Edges (propagator assignments):
      2 → 3:  nt2 → dv1  [G_hat[1,2]]
      2 → 5:  nt2 → dv1  [G_hat[1,2]]
      3 → 0:  nt1 → dn2  [G_hat[0,1]]
      4 → 3:  nt1 → dv1  [G_hat[0,2]]
      4 → 5:  nt1 → dv1  [G_hat[0,2]]
      5 → 1:  nt1 → dn1  [G_hat[0,0]]

```


=====

Diagram 1L-1026

=====

External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

Edges (propagator assignments):

2 → 3: nt2 → dv1 [G_hat[1,2]]

2 → 5: nt2 → dv1 [G_hat[1,2]]

3 → 0: nt1 → dn2 [G_hat[0,1]]

4 → 3: nt1 → dv1 [G_hat[0,2]]

4 → 5: nt1 → dv1 [G_hat[0,2]]

5 → 1: nt1 → dn1 [G_hat[0,0]]

=====

Diagram 1L-1027

=====

External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

```

    v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

    coeff =  $\frac{1}{2} nstar_1$ 

    v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

    coeff =  $\frac{1}{2} \varphi_1''$ 

    Edges (propagator assignments):
      2 → 3:  nt2 → dv1  [G_hat[1,2]]
      2 → 5:  nt2 → dv1  [G_hat[1,2]]
      3 → 0:  nt1 → dn2  [G_hat[0,1]]
      4 → 3:  nt1 → dv1  [G_hat[0,2]]
      4 → 5:  nt1 → dv1  [G_hat[0,2]]
      5 → 1:  nt1 → dn1  [G_hat[0,0]]

=====
Diagram 1L-1028
=====

    External legs:
      leaf 0 ← dn2
      leaf 1 ← dn1
    Vertex assignments:
      v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

      coeff =  $\frac{1}{2} nstar_2$ 

      v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

      coeff =  $\frac{1}{2} \varphi_1''$ 

      v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

      coeff =  $\frac{1}{2} nstar_1$ 

      v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

      coeff =  $\frac{1}{2} \varphi_1''$ 

    Edges (propagator assignments):
      2 → 3:  nt2 → dv1  [G_hat[1,2]]
      2 → 5:  nt2 → dv1  [G_hat[1,2]]
      3 → 0:  nt1 → dn2  [G_hat[0,1]]
      4 → 3:  nt1 → dv1  [G_hat[0,2]]
      4 → 5:  nt1 → dv1  [G_hat[0,2]]
      5 → 1:  nt1 → dn1  [G_hat[0,0]]

```

=====

Diagram 1L-1029

=====

External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

Edges (propagator assignments):

2 → 3: nt2 → dv1 [G_hat[1,2]]

2 → 5: nt2 → dv1 [G_hat[1,2]]

3 → 0: nt1 → dn2 [G_hat[0,1]]

4 → 3: nt1 → dv1 [G_hat[0,2]]

4 → 5: nt1 → dv1 [G_hat[0,2]]

5 → 1: nt1 → dn1 [G_hat[0,0]]

=====

Diagram 1L-1030

=====

External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

```

    v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

    coeff =  $\frac{1}{2} nstar_1$ 

    v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

    coeff =  $\frac{1}{2} \varphi_1''$ 

    Edges (propagator assignments):
      2 → 3:  nt2 → dv1  [G_hat[1,2]]
      2 → 5:  nt2 → dv1  [G_hat[1,2]]
      3 → 0:  nt1 → dn2  [G_hat[0,1]]
      4 → 3:  nt1 → dv1  [G_hat[0,2]]
      4 → 5:  nt1 → dv1  [G_hat[0,2]]
      5 → 1:  nt1 → dn1  [G_hat[0,0]]

=====
Diagram 1L-1031
=====

    External legs:
      leaf 0 ← dn2
      leaf 1 ← dn1

    Vertex assignments:
      v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

      coeff =  $\frac{1}{2} nstar_2$ 

      v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

      coeff =  $\frac{1}{2} \varphi_1''$ 

      v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

      coeff =  $\frac{1}{2} nstar_1$ 

      v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

      coeff =  $\frac{1}{2} \varphi_1''$ 

    Edges (propagator assignments):
      2 → 3:  nt2 → dv1  [G_hat[1,2]]
      2 → 5:  nt2 → dv1  [G_hat[1,2]]
      3 → 0:  nt1 → dn2  [G_hat[0,1]]
      4 → 3:  nt1 → dv1  [G_hat[0,2]]
      4 → 5:  nt1 → dv1  [G_hat[0,2]]
      5 → 1:  nt1 → dn1  [G_hat[0,0]]

```

=====

Diagram 1L-1032

=====

External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

Edges (propagator assignments):

2 → 3: nt2 → dv1 [G_hat[1,2]]

2 → 5: nt2 → dv1 [G_hat[1,2]]

3 → 0: nt1 → dn2 [G_hat[0,1]]

4 → 3: nt1 → dv1 [G_hat[0,2]]

4 → 5: nt1 → dv1 [G_hat[0,2]]

5 → 1: nt1 → dn1 [G_hat[0,0]]

=====

Diagram 1L-1033

=====

External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

```

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

coeff =  $\frac{1}{2} nstar_1$ 

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

coeff =  $\frac{1}{2} \varphi_1''$ 

Edges (propagator assignments):
  2 → 3:  nt2 → dv1  [G_hat[1,2]]
  2 → 5:  nt2 → dv1  [G_hat[1,2]]
  3 → 0:  nt1 → dn2  [G_hat[0,1]]
  4 → 3:  nt1 → dv1  [G_hat[0,2]]
  4 → 5:  nt1 → dv1  [G_hat[0,2]]
  5 → 1:  nt1 → dn1  [G_hat[0,0]]

=====
Diagram 1L-1034
=====

External legs:
  leaf 0 ← dn2
  leaf 1 ← dn1
Vertex assignments:
  v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

coeff =  $\frac{1}{2} nstar_2$ 

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

coeff =  $\frac{1}{2} \varphi_1''$ 

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

coeff =  $\frac{1}{2} nstar_1$ 

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

coeff =  $\frac{1}{2} \varphi_1''$ 

Edges (propagator assignments):
  2 → 3:  nt2 → dv1  [G_hat[1,2]]
  2 → 5:  nt2 → dv1  [G_hat[1,2]]
  3 → 0:  nt1 → dn2  [G_hat[0,1]]
  4 → 3:  nt1 → dv1  [G_hat[0,2]]
  4 → 5:  nt1 → dv1  [G_hat[0,2]]
  5 → 1:  nt1 → dn1  [G_hat[0,0]]

```

```

=====
Diagram 1L-1035
=====
External legs:
  leaf 0 ← dn2
  leaf 1 ← dn1
Vertex assignments:
  v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

  coeff =  $\frac{1}{2} nstar_2$ 

  v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

  coeff =  $\frac{1}{2} \varphi_1''$ 

  v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

  coeff =  $\frac{1}{2} nstar_1$ 

  v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

  coeff =  $\frac{1}{2} \varphi_1''$ 

Edges (propagator assignments):
  2 → 3:  nt2 → dv1  [G_hat[1,2]]
  2 → 5:  nt2 → dv1  [G_hat[1,2]]
  3 → 0:  nt1 → dn2  [G_hat[0,1]]
  4 → 3:  nt1 → dv1  [G_hat[0,2]]
  4 → 5:  nt1 → dv1  [G_hat[0,2]]
  5 → 1:  nt1 → dn1  [G_hat[0,0]]

```

```

=====
Diagram 1L-1036
=====
External legs:
  leaf 0 ← dn2
  leaf 1 ← dn1
Vertex assignments:
  v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

  coeff =  $\frac{1}{2} nstar_2$ 

  v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

  coeff =  $\frac{1}{2} \varphi_1''$ 

```

```

    v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

    coeff =  $\frac{1}{2} nstar_1$ 

    v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

    coeff =  $\frac{1}{2} \varphi_1''$ 

    Edges (propagator assignments):
      2 → 3:  nt2 → dv1  [G_hat[1,2]]
      2 → 5:  nt2 → dv1  [G_hat[1,2]]
      3 → 0:  nt1 → dn2  [G_hat[0,1]]
      4 → 3:  nt1 → dv1  [G_hat[0,2]]
      4 → 5:  nt1 → dv1  [G_hat[0,2]]
      5 → 1:  nt1 → dn1  [G_hat[0,0]]

=====
Diagram 1L-1037
=====

    External legs:
      leaf 0 ← dn2
      leaf 1 ← dn1

    Vertex assignments:
      v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

      coeff =  $\frac{1}{2} nstar_2$ 

      v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

      coeff =  $\frac{1}{2} \varphi_1''$ 

      v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

      coeff =  $\frac{1}{2} nstar_1$ 

      v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

      coeff =  $\frac{1}{2} \varphi_1''$ 

    Edges (propagator assignments):
      2 → 3:  nt2 → dv1  [G_hat[1,2]]
      2 → 5:  nt2 → dv1  [G_hat[1,2]]
      3 → 0:  nt1 → dn2  [G_hat[0,1]]
      4 → 3:  nt1 → dv1  [G_hat[0,2]]
      4 → 5:  nt1 → dv1  [G_hat[0,2]]
      5 → 1:  nt1 → dn1  [G_hat[0,0]]

```


=====

Diagram 1L-1038

=====

External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

Edges (propagator assignments):

2 → 3: nt2 → dv1 [G_hat[1,2]]

2 → 5: nt2 → dv1 [G_hat[1,2]]

3 → 0: nt1 → dn2 [G_hat[0,1]]

4 → 3: nt1 → dv1 [G_hat[0,2]]

4 → 5: nt1 → dv1 [G_hat[0,2]]

5 → 1: nt1 → dn1 [G_hat[0,0]]

=====

Diagram 1L-1039

=====

External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

```

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

coeff =  $\frac{1}{2} nstar_1$ 

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

coeff =  $\frac{1}{2} \varphi_1''$ 

Edges (propagator assignments):
  2 → 3:  nt2 → dv1  [G_hat[1,2]]
  2 → 5:  nt2 → dv1  [G_hat[1,2]]
  3 → 0:  nt1 → dn2  [G_hat[0,1]]
  4 → 3:  nt1 → dv1  [G_hat[0,2]]
  4 → 5:  nt1 → dv1  [G_hat[0,2]]
  5 → 1:  nt1 → dn1  [G_hat[0,0]]

=====
Diagram 1L-1040
=====

External legs:
  leaf 0 ← dn2
  leaf 1 ← dn1
Vertex assignments:
  v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

coeff =  $\frac{1}{2} nstar_2$ 

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

coeff =  $\frac{1}{2} \varphi_1''$ 

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

coeff =  $\frac{1}{2} nstar_1$ 

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

coeff =  $\frac{1}{2} \varphi_1''$ 

Edges (propagator assignments):
  2 → 3:  nt2 → dv1  [G_hat[1,2]]
  2 → 5:  nt2 → dv1  [G_hat[1,2]]
  3 → 0:  nt1 → dn2  [G_hat[0,1]]
  4 → 3:  nt1 → dv1  [G_hat[0,2]]
  4 → 5:  nt1 → dv1  [G_hat[0,2]]
  5 → 1:  nt1 → dn1  [G_hat[0,0]]

```

=====

Diagram 1L-1041

=====

External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

Edges (propagator assignments):

2 → 3: nt2 → dv1 [G_hat[1,2]]

2 → 5: nt2 → dv2 [G_hat[1,3]]

3 → 0: nt1 → dn2 [G_hat[0,1]]

4 → 3: nt1 → dv1 [G_hat[0,2]]

4 → 5: nt1 → dv2 [G_hat[0,3]]

5 → 1: nt2 → dn1 [G_hat[1,0]]

=====

Diagram 1L-1042

=====

External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

```

    v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

    coeff =  $\frac{1}{2} nstar_1$ 

    v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

    coeff =  $\frac{1}{2} \varphi_2''$ 

    Edges (propagator assignments):
      2 → 3:  nt2 → dv1  [G_hat[1,2]]
      2 → 5:  nt2 → dv2  [G_hat[1,3]]
      3 → 0:  nt1 → dn2  [G_hat[0,1]]
      4 → 3:  nt1 → dv1  [G_hat[0,2]]
      4 → 5:  nt1 → dv2  [G_hat[0,3]]
      5 → 1:  nt2 → dn1  [G_hat[1,0]]

=====
Diagram 1L-1043
=====

    External legs:
      leaf 0 ← dn2
      leaf 1 ← dn1
    Vertex assignments:
      v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

      coeff =  $\frac{1}{2} nstar_2$ 

      v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

      coeff =  $\frac{1}{2} \varphi_1''$ 

      v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

      coeff =  $\frac{1}{2} nstar_1$ 

      v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

      coeff =  $\frac{1}{2} \varphi_2''$ 

    Edges (propagator assignments):
      2 → 3:  nt2 → dv1  [G_hat[1,2]]
      2 → 5:  nt2 → dv2  [G_hat[1,3]]
      3 → 0:  nt1 → dn2  [G_hat[0,1]]
      4 → 3:  nt1 → dv1  [G_hat[0,2]]
      4 → 5:  nt1 → dv2  [G_hat[0,3]]
      5 → 1:  nt2 → dn1  [G_hat[1,0]]

```

```

=====
Diagram 1L-1044
=====
External legs:
  leaf 0 ← dn2
  leaf 1 ← dn1
Vertex assignments:
  v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

  coeff =  $\frac{1}{2} nstar_2$ 

  v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]
1)]

  coeff =  $\frac{1}{2} \varphi_1''$ 

  v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

  coeff =  $\frac{1}{2} nstar_1$ 

  v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]
2)]

  coeff =  $\frac{1}{2} \varphi_2''$ 

Edges (propagator assignments):
  2 → 3:  nt2 → dv1  [G_hat[1,2]]
  2 → 5:  nt2 → dv2  [G_hat[1,3]]
  3 → 0:  nt1 → dn2  [G_hat[0,1]]
  4 → 3:  nt1 → dv1  [G_hat[0,2]]
  4 → 5:  nt1 → dv2  [G_hat[0,3]]
  5 → 1:  nt2 → dn1  [G_hat[1,0]]

```

```

=====
Diagram 1L-1045
=====
External legs:
  leaf 0 ← dn2
  leaf 1 ← dn1
Vertex assignments:
  v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

  coeff =  $\frac{1}{2} nstar_2$ 

  v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]
1)]

  coeff =  $\frac{1}{2} \varphi_1''$ 

```

```

    v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

    coeff =  $\frac{1}{2} nstar_1$ 

    v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

    coeff =  $\frac{1}{2} \varphi_2''$ 

    Edges (propagator assignments):
      2 → 3:  nt2 → dv1  [G_hat[1,2]]
      2 → 5:  nt2 → dv2  [G_hat[1,3]]
      3 → 0:  nt1 → dn2  [G_hat[0,1]]
      4 → 3:  nt1 → dv1  [G_hat[0,2]]
      4 → 5:  nt1 → dv2  [G_hat[0,3]]
      5 → 1:  nt2 → dn1  [G_hat[1,0]]

=====
Diagram 1L-1046
=====

    External legs:
      leaf 0 ← dn2
      leaf 1 ← dn1

    Vertex assignments:
      v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

      coeff =  $\frac{1}{2} nstar_2$ 

      v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

      coeff =  $\frac{1}{2} \varphi_1''$ 

      v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

      coeff =  $\frac{1}{2} nstar_1$ 

      v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

      coeff =  $\frac{1}{2} \varphi_2''$ 

    Edges (propagator assignments):
      2 → 3:  nt2 → dv1  [G_hat[1,2]]
      2 → 5:  nt2 → dv2  [G_hat[1,3]]
      3 → 0:  nt1 → dn2  [G_hat[0,1]]
      4 → 3:  nt1 → dv1  [G_hat[0,2]]
      4 → 5:  nt1 → dv2  [G_hat[0,3]]
      5 → 1:  nt2 → dn1  [G_hat[1,0]]

```

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Diagram 1L-1047

=====

External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

Edges (propagator assignments):

2 → 3: nt2 → dv1 [G_hat[1,2]]

2 → 5: nt2 → dv2 [G_hat[1,3]]

3 → 0: nt1 → dn2 [G_hat[0,1]]

4 → 3: nt1 → dv1 [G_hat[0,2]]

4 → 5: nt1 → dv2 [G_hat[0,3]]

5 → 1: nt2 → dn1 [G_hat[1,0]]

=====

Diagram 1L-1048

=====

External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

```

    v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

    coeff =  $\frac{1}{2} nstar_1$ 

    v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

    coeff =  $\frac{1}{2} \varphi_2''$ 

    Edges (propagator assignments):
      2 → 3:  nt2 → dv1  [G_hat[1,2]]
      2 → 5:  nt2 → dv2  [G_hat[1,3]]
      3 → 0:  nt1 → dn2  [G_hat[0,1]]
      4 → 3:  nt1 → dv1  [G_hat[0,2]]
      4 → 5:  nt1 → dv2  [G_hat[0,3]]
      5 → 1:  nt2 → dn1  [G_hat[1,0]]

=====
Diagram 1L-1049
=====

    External legs:
      leaf 0 ← dn2
      leaf 1 ← dn1

    Vertex assignments:
      v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

      coeff =  $\frac{1}{2} nstar_2$ 

      v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

      coeff =  $\frac{1}{2} \varphi_1''$ 

      v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

      coeff =  $\frac{1}{2} nstar_1$ 

      v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

      coeff =  $\frac{1}{2} \varphi_2''$ 

    Edges (propagator assignments):
      2 → 3:  nt2 → dv1  [G_hat[1,2]]
      2 → 5:  nt2 → dv2  [G_hat[1,3]]
      3 → 0:  nt1 → dn2  [G_hat[0,1]]
      4 → 3:  nt1 → dv1  [G_hat[0,2]]
      4 → 5:  nt1 → dv2  [G_hat[0,3]]
      5 → 1:  nt2 → dn1  [G_hat[1,0]]

```


=====

Diagram 1L-1050

=====

External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

Edges (propagator assignments):

2 → 3: nt2 → dv1 [G_hat[1,2]]

2 → 5: nt2 → dv2 [G_hat[1,3]]

3 → 0: nt1 → dn2 [G_hat[0,1]]

4 → 3: nt1 → dv1 [G_hat[0,2]]

4 → 5: nt1 → dv2 [G_hat[0,3]]

5 → 1: nt2 → dn1 [G_hat[1,0]]

=====

Diagram 1L-1051

=====

External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

```

    v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

    coeff =  $\frac{1}{2} nstar_1$ 

    v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

    coeff =  $\frac{1}{2} \varphi_2''$ 

    Edges (propagator assignments):
      2 → 3:  nt2 → dv1  [G_hat[1,2]]
      2 → 5:  nt2 → dv2  [G_hat[1,3]]
      3 → 0:  nt1 → dn2  [G_hat[0,1]]
      4 → 3:  nt1 → dv1  [G_hat[0,2]]
      4 → 5:  nt1 → dv2  [G_hat[0,3]]
      5 → 1:  nt2 → dn1  [G_hat[1,0]]

=====
Diagram 1L-1052
=====

    External legs:
      leaf 0 ← dn2
      leaf 1 ← dn1
    Vertex assignments:
      v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

      coeff =  $\frac{1}{2} nstar_2$ 

      v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

      coeff =  $\frac{1}{2} \varphi_1''$ 

      v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

      coeff =  $\frac{1}{2} nstar_1$ 

      v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

      coeff =  $\frac{1}{2} \varphi_2''$ 

    Edges (propagator assignments):
      2 → 3:  nt2 → dv1  [G_hat[1,2]]
      2 → 5:  nt2 → dv2  [G_hat[1,3]]
      3 → 0:  nt1 → dn2  [G_hat[0,1]]
      4 → 3:  nt1 → dv1  [G_hat[0,2]]
      4 → 5:  nt1 → dv2  [G_hat[0,3]]
      5 → 1:  nt2 → dn1  [G_hat[1,0]]

```

=====

Diagram 1L-1053

=====

External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

Edges (propagator assignments):

2 → 3: nt2 → dv1 [G_hat[1,2]]

2 → 5: nt2 → dv2 [G_hat[1,3]]

3 → 0: nt1 → dn2 [G_hat[0,1]]

4 → 3: nt1 → dv1 [G_hat[0,2]]

4 → 5: nt1 → dv2 [G_hat[0,3]]

5 → 1: nt2 → dn1 [G_hat[1,0]]

=====

Diagram 1L-1054

=====

External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

```

    v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

    coeff =  $\frac{1}{2} nstar_1$ 

    v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

    coeff =  $\frac{1}{2} \varphi_2''$ 

    Edges (propagator assignments):
      2 → 3:  nt2 → dv1  [G_hat[1,2]]
      2 → 5:  nt2 → dv2  [G_hat[1,3]]
      3 → 0:  nt1 → dn2  [G_hat[0,1]]
      4 → 3:  nt1 → dv1  [G_hat[0,2]]
      4 → 5:  nt1 → dv2  [G_hat[0,3]]
      5 → 1:  nt2 → dn1  [G_hat[1,0]]

=====
Diagram 1L-1055
=====

    External legs:
      leaf 0 ← dn2
      leaf 1 ← dn1
    Vertex assignments:
      v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

      coeff =  $\frac{1}{2} nstar_2$ 

      v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

      coeff =  $\frac{1}{2} \varphi_1''$ 

      v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

      coeff =  $\frac{1}{2} nstar_1$ 

      v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

      coeff =  $\frac{1}{2} \varphi_2''$ 

    Edges (propagator assignments):
      2 → 3:  nt2 → dv1  [G_hat[1,2]]
      2 → 5:  nt2 → dv2  [G_hat[1,3]]
      3 → 0:  nt1 → dn2  [G_hat[0,1]]
      4 → 3:  nt1 → dv1  [G_hat[0,2]]
      4 → 5:  nt1 → dv2  [G_hat[0,3]]
      5 → 1:  nt2 → dn1  [G_hat[1,0]]

```

=====

Diagram 1L-1056

=====

External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

Edges (propagator assignments):

2 → 3: nt2 → dv1 [G_hat[1,2]]

2 → 5: nt2 → dv2 [G_hat[1,3]]

3 → 0: nt1 → dn2 [G_hat[0,1]]

4 → 3: nt1 → dv1 [G_hat[0,2]]

4 → 5: nt1 → dv2 [G_hat[0,3]]

5 → 1: nt2 → dn1 [G_hat[1,0]]

=====

Diagram 1L-1057

=====

External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

```

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

coeff =  $\frac{1}{2} nstar_1$ 

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

coeff =  $\frac{1}{2} \varphi_1''$ 

Edges (propagator assignments):
  2 → 3:  nt2 → dv2  [G_hat[1,3]]
  2 → 5:  nt2 → dv1  [G_hat[1,2]]
  3 → 0:  nt2 → dn2  [G_hat[1,1]]
  4 → 3:  nt1 → dv2  [G_hat[0,3]]
  4 → 5:  nt1 → dv1  [G_hat[0,2]]
  5 → 1:  nt1 → dn1  [G_hat[0,0]]

=====
Diagram 1L-1058
=====

External legs:
  leaf 0 ← dn2
  leaf 1 ← dn1
Vertex assignments:
  v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

coeff =  $\frac{1}{2} nstar_2$ 

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

coeff =  $\frac{1}{2} \varphi_2''$ 

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

coeff =  $\frac{1}{2} nstar_1$ 

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

coeff =  $\frac{1}{2} \varphi_1''$ 

Edges (propagator assignments):
  2 → 3:  nt2 → dv2  [G_hat[1,3]]
  2 → 5:  nt2 → dv1  [G_hat[1,2]]
  3 → 0:  nt2 → dn2  [G_hat[1,1]]
  4 → 3:  nt1 → dv2  [G_hat[0,3]]
  4 → 5:  nt1 → dv1  [G_hat[0,2]]
  5 → 1:  nt1 → dn1  [G_hat[0,0]]

```

=====

Diagram 1L-1059

=====

External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

Edges (propagator assignments):

2 → 3: nt2 → dv2 [G_hat[1,3]]

2 → 5: nt2 → dv1 [G_hat[1,2]]

3 → 0: nt2 → dn2 [G_hat[1,1]]

4 → 3: nt1 → dv2 [G_hat[0,3]]

4 → 5: nt1 → dv1 [G_hat[0,2]]

5 → 1: nt1 → dn1 [G_hat[0,0]]

=====

Diagram 1L-1060

=====

External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

```

    v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

    coeff =  $\frac{1}{2} nstar_1$ 

    v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

    coeff =  $\frac{1}{2} \varphi_1''$ 

    Edges (propagator assignments):
      2 → 3:  nt2 → dv2  [G_hat[1,3]]
      2 → 5:  nt2 → dv1  [G_hat[1,2]]
      3 → 0:  nt2 → dn2  [G_hat[1,1]]
      4 → 3:  nt1 → dv2  [G_hat[0,3]]
      4 → 5:  nt1 → dv1  [G_hat[0,2]]
      5 → 1:  nt1 → dn1  [G_hat[0,0]]

=====
Diagram 1L-1061
=====

    External legs:
      leaf 0 ← dn2
      leaf 1 ← dn1
    Vertex assignments:
      v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

      coeff =  $\frac{1}{2} nstar_2$ 

      v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

      coeff =  $\frac{1}{2} \varphi_2''$ 

      v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

      coeff =  $\frac{1}{2} nstar_1$ 

      v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

      coeff =  $\frac{1}{2} \varphi_1''$ 

    Edges (propagator assignments):
      2 → 3:  nt2 → dv2  [G_hat[1,3]]
      2 → 5:  nt2 → dv1  [G_hat[1,2]]
      3 → 0:  nt2 → dn2  [G_hat[1,1]]
      4 → 3:  nt1 → dv2  [G_hat[0,3]]
      4 → 5:  nt1 → dv1  [G_hat[0,2]]
      5 → 1:  nt1 → dn1  [G_hat[0,0]]

```



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=====
Diagram 1L-1062
=====

External legs:
  leaf 0 ← dn2
  leaf 1 ← dn1
Vertex assignments:
  v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

  coeff =  $\frac{1}{2} nstar_2$ 

  v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

  coeff =  $\frac{1}{2} \varphi_2''$ 

  v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

  coeff =  $\frac{1}{2} nstar_1$ 

  v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

  coeff =  $\frac{1}{2} \varphi_1''$ 

Edges (propagator assignments):
  2 → 3:  nt2 → dv2  [G_hat[1,3]]
  2 → 5:  nt2 → dv1  [G_hat[1,2]]
  3 → 0:  nt2 → dn2  [G_hat[1,1]]
  4 → 3:  nt1 → dv2  [G_hat[0,3]]
  4 → 5:  nt1 → dv1  [G_hat[0,2]]
  5 → 1:  nt1 → dn1  [G_hat[0,0]]

```

```

=====
Diagram 1L-1063
=====

External legs:
  leaf 0 ← dn2
  leaf 1 ← dn1
Vertex assignments:
  v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

  coeff =  $\frac{1}{2} nstar_2$ 

  v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

  coeff =  $\frac{1}{2} \varphi_2''$ 

```

```

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

coeff =  $\frac{1}{2} nstar_1$ 

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

coeff =  $\frac{1}{2} \varphi_1''$ 

Edges (propagator assignments):
  2 → 3:  nt2 → dv2  [G_hat[1,3]]
  2 → 5:  nt2 → dv1  [G_hat[1,2]]
  3 → 0:  nt2 → dn2  [G_hat[1,1]]
  4 → 3:  nt1 → dv2  [G_hat[0,3]]
  4 → 5:  nt1 → dv1  [G_hat[0,2]]
  5 → 1:  nt1 → dn1  [G_hat[0,0]]

=====
Diagram 1L-1064
=====

External legs:
  leaf 0 ← dn2
  leaf 1 ← dn1
Vertex assignments:
  v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

coeff =  $\frac{1}{2} nstar_2$ 

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

coeff =  $\frac{1}{2} \varphi_2''$ 

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

coeff =  $\frac{1}{2} nstar_1$ 

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

coeff =  $\frac{1}{2} \varphi_1''$ 

Edges (propagator assignments):
  2 → 3:  nt2 → dv2  [G_hat[1,3]]
  2 → 5:  nt2 → dv1  [G_hat[1,2]]
  3 → 0:  nt2 → dn2  [G_hat[1,1]]
  4 → 3:  nt1 → dv2  [G_hat[0,3]]
  4 → 5:  nt1 → dv1  [G_hat[0,2]]
  5 → 1:  nt1 → dn1  [G_hat[0,0]]

```

=====

Diagram 1L-1065

=====

External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

Edges (propagator assignments):

2 → 3: nt2 → dv2 [G_hat[1,3]]

2 → 5: nt2 → dv1 [G_hat[1,2]]

3 → 0: nt2 → dn2 [G_hat[1,1]]

4 → 3: nt1 → dv2 [G_hat[0,3]]

4 → 5: nt1 → dv1 [G_hat[0,2]]

5 → 1: nt1 → dn1 [G_hat[0,0]]

=====

Diagram 1L-1066

=====

External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

```

    v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

    coeff =  $\frac{1}{2} nstar_1$ 

    v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

    coeff =  $\frac{1}{2} \varphi_1''$ 

    Edges (propagator assignments):
      2 → 3:  nt2 → dv2  [G_hat[1,3]]
      2 → 5:  nt2 → dv1  [G_hat[1,2]]
      3 → 0:  nt2 → dn2  [G_hat[1,1]]
      4 → 3:  nt1 → dv2  [G_hat[0,3]]
      4 → 5:  nt1 → dv1  [G_hat[0,2]]
      5 → 1:  nt1 → dn1  [G_hat[0,0]]

=====
Diagram 1L-1067
=====

    External legs:
      leaf 0 ← dn2
      leaf 1 ← dn1
    Vertex assignments:
      v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

      coeff =  $\frac{1}{2} nstar_2$ 

      v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

      coeff =  $\frac{1}{2} \varphi_2''$ 

      v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

      coeff =  $\frac{1}{2} nstar_1$ 

      v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

      coeff =  $\frac{1}{2} \varphi_1''$ 

    Edges (propagator assignments):
      2 → 3:  nt2 → dv2  [G_hat[1,3]]
      2 → 5:  nt2 → dv1  [G_hat[1,2]]
      3 → 0:  nt2 → dn2  [G_hat[1,1]]
      4 → 3:  nt1 → dv2  [G_hat[0,3]]
      4 → 5:  nt1 → dv1  [G_hat[0,2]]
      5 → 1:  nt1 → dn1  [G_hat[0,0]]

```

=====

Diagram 1L-1068

=====

External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

Edges (propagator assignments):

2 → 3: nt2 → dv2 [G_hat[1,3]]

2 → 5: nt2 → dv1 [G_hat[1,2]]

3 → 0: nt2 → dn2 [G_hat[1,1]]

4 → 3: nt1 → dv2 [G_hat[0,3]]

4 → 5: nt1 → dv1 [G_hat[0,2]]

5 → 1: nt1 → dn1 [G_hat[0,0]]

=====

Diagram 1L-1069

=====

External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

```

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

coeff =  $\frac{1}{2} nstar_1$ 

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

coeff =  $\frac{1}{2} \varphi_1''$ 

Edges (propagator assignments):
  2 → 3:  nt2 → dv2  [G_hat[1,3]]
  2 → 5:  nt2 → dv1  [G_hat[1,2]]
  3 → 0:  nt2 → dn2  [G_hat[1,1]]
  4 → 3:  nt1 → dv2  [G_hat[0,3]]
  4 → 5:  nt1 → dv1  [G_hat[0,2]]
  5 → 1:  nt1 → dn1  [G_hat[0,0]]

=====
Diagram 1L-1070
=====

External legs:
  leaf 0 ← dn2
  leaf 1 ← dn1
Vertex assignments:
  v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

coeff =  $\frac{1}{2} nstar_2$ 

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

coeff =  $\frac{1}{2} \varphi_2''$ 

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

coeff =  $\frac{1}{2} nstar_1$ 

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

coeff =  $\frac{1}{2} \varphi_1''$ 

Edges (propagator assignments):
  2 → 3:  nt2 → dv2  [G_hat[1,3]]
  2 → 5:  nt2 → dv1  [G_hat[1,2]]
  3 → 0:  nt2 → dn2  [G_hat[1,1]]
  4 → 3:  nt1 → dv2  [G_hat[0,3]]
  4 → 5:  nt1 → dv1  [G_hat[0,2]]
  5 → 1:  nt1 → dn1  [G_hat[0,0]]

```

=====

Diagram 1L-1071

=====

External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

Edges (propagator assignments):

2 → 3: nt2 → dv2 [G_hat[1,3]]

2 → 5: nt2 → dv1 [G_hat[1,2]]

3 → 0: nt2 → dn2 [G_hat[1,1]]

4 → 3: nt1 → dv2 [G_hat[0,3]]

4 → 5: nt1 → dv1 [G_hat[0,2]]

5 → 1: nt1 → dn1 [G_hat[0,0]]

=====

Diagram 1L-1072

=====

External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

```

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

coeff =  $\frac{1}{2} nstar_1$ 

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

coeff =  $\frac{1}{2} \varphi_1''$ 

Edges (propagator assignments):
  2 → 3:  nt2 → dv2  [G_hat[1,3]]
  2 → 5:  nt2 → dv1  [G_hat[1,2]]
  3 → 0:  nt2 → dn2  [G_hat[1,1]]
  4 → 3:  nt1 → dv2  [G_hat[0,3]]
  4 → 5:  nt1 → dv1  [G_hat[0,2]]
  5 → 1:  nt1 → dn1  [G_hat[0,0]]

=====
Diagram 1L-1073
=====

External legs:
  leaf 0 ← dn2
  leaf 1 ← dn1
Vertex assignments:
  v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

coeff =  $\frac{1}{2} nstar_2$ 

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

coeff =  $\frac{1}{2} \varphi_2''$ 

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

coeff =  $\frac{1}{2} nstar_1$ 

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

coeff =  $\frac{1}{2} \varphi_2''$ 

Edges (propagator assignments):
  2 → 3:  nt2 → dv2  [G_hat[1,3]]
  2 → 5:  nt2 → dv2  [G_hat[1,3]]
  3 → 0:  nt2 → dn2  [G_hat[1,1]]
  4 → 3:  nt1 → dv2  [G_hat[0,3]]
  4 → 5:  nt1 → dv2  [G_hat[0,3]]
  5 → 1:  nt2 → dn1  [G_hat[1,0]]

```


=====

Diagram 1L-1074

=====

External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

Edges (propagator assignments):

2 → 3: nt2 → dv2 [G_hat[1,3]]

2 → 5: nt2 → dv2 [G_hat[1,3]]

3 → 0: nt2 → dn2 [G_hat[1,1]]

4 → 3: nt1 → dv2 [G_hat[0,3]]

4 → 5: nt1 → dv2 [G_hat[0,3]]

5 → 1: nt2 → dn1 [G_hat[1,0]]

=====

Diagram 1L-1075

=====

External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

```

    v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

    coeff =  $\frac{1}{2} nstar_1$ 

    v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

    coeff =  $\frac{1}{2} \varphi_2''$ 

    Edges (propagator assignments):
      2 → 3:  nt2 → dv2  [G_hat[1,3]]
      2 → 5:  nt2 → dv2  [G_hat[1,3]]
      3 → 0:  nt2 → dn2  [G_hat[1,1]]
      4 → 3:  nt1 → dv2  [G_hat[0,3]]
      4 → 5:  nt1 → dv2  [G_hat[0,3]]
      5 → 1:  nt2 → dn1  [G_hat[1,0]]

=====
Diagram 1L-1076
=====

    External legs:
      leaf 0 ← dn2
      leaf 1 ← dn1
    Vertex assignments:
      v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

      coeff =  $\frac{1}{2} nstar_2$ 

      v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

      coeff =  $\frac{1}{2} \varphi_2''$ 

      v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

      coeff =  $\frac{1}{2} nstar_1$ 

      v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

      coeff =  $\frac{1}{2} \varphi_2''$ 

    Edges (propagator assignments):
      2 → 3:  nt2 → dv2  [G_hat[1,3]]
      2 → 5:  nt2 → dv2  [G_hat[1,3]]
      3 → 0:  nt2 → dn2  [G_hat[1,1]]
      4 → 3:  nt1 → dv2  [G_hat[0,3]]
      4 → 5:  nt1 → dv2  [G_hat[0,3]]
      5 → 1:  nt2 → dn1  [G_hat[1,0]]

```

=====

Diagram 1L-1077

=====

External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

Edges (propagator assignments):

2 → 3: nt2 → dv2 [G_hat[1,3]]

2 → 5: nt2 → dv2 [G_hat[1,3]]

3 → 0: nt2 → dn2 [G_hat[1,1]]

4 → 3: nt1 → dv2 [G_hat[0,3]]

4 → 5: nt1 → dv2 [G_hat[0,3]]

5 → 1: nt2 → dn1 [G_hat[1,0]]

=====

Diagram 1L-1078

=====

External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

```

    v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

    coeff =  $\frac{1}{2} nstar_1$ 

    v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

    coeff =  $\frac{1}{2} \varphi_2''$ 

    Edges (propagator assignments):
      2 → 3:  nt2 → dv2  [G_hat[1,3]]
      2 → 5:  nt2 → dv2  [G_hat[1,3]]
      3 → 0:  nt2 → dn2  [G_hat[1,1]]
      4 → 3:  nt1 → dv2  [G_hat[0,3]]
      4 → 5:  nt1 → dv2  [G_hat[0,3]]
      5 → 1:  nt2 → dn1  [G_hat[1,0]]

=====
Diagram 1L-1079
=====

    External legs:
      leaf 0 ← dn2
      leaf 1 ← dn1
    Vertex assignments:
      v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

      coeff =  $\frac{1}{2} nstar_2$ 

      v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

      coeff =  $\frac{1}{2} \varphi_2''$ 

      v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

      coeff =  $\frac{1}{2} nstar_1$ 

      v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

      coeff =  $\frac{1}{2} \varphi_2''$ 

    Edges (propagator assignments):
      2 → 3:  nt2 → dv2  [G_hat[1,3]]
      2 → 5:  nt2 → dv2  [G_hat[1,3]]
      3 → 0:  nt2 → dn2  [G_hat[1,1]]
      4 → 3:  nt1 → dv2  [G_hat[0,3]]
      4 → 5:  nt1 → dv2  [G_hat[0,3]]
      5 → 1:  nt2 → dn1  [G_hat[1,0]]

```

```

=====
Diagram 1L-1080
=====

External legs:
  leaf 0 ← dn2
  leaf 1 ← dn1
Vertex assignments:
  v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

  coeff =  $\frac{1}{2} nstar_2$ 

  v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

  coeff =  $\frac{1}{2} \varphi_2''$ 

  v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

  coeff =  $\frac{1}{2} nstar_1$ 

  v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

  coeff =  $\frac{1}{2} \varphi_2''$ 

Edges (propagator assignments):
  2 → 3:  nt2 → dv2  [G_hat[1,3]]
  2 → 5:  nt2 → dv2  [G_hat[1,3]]
  3 → 0:  nt2 → dn2  [G_hat[1,1]]
  4 → 3:  nt1 → dv2  [G_hat[0,3]]
  4 → 5:  nt1 → dv2  [G_hat[0,3]]
  5 → 1:  nt2 → dn1  [G_hat[1,0]]

```

```

=====
Diagram 1L-1081
=====

External legs:
  leaf 0 ← dn2
  leaf 1 ← dn1
Vertex assignments:
  v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

  coeff =  $\frac{1}{2} nstar_2$ 

  v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

  coeff =  $\frac{1}{2} \varphi_2''$ 

```

```

    v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

    coeff =  $\frac{1}{2} nstar_1$ 

    v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

    coeff =  $\frac{1}{2} \varphi_2''$ 

    Edges (propagator assignments):
      2 → 3:  nt2 → dv2  [G_hat[1,3]]
      2 → 5:  nt2 → dv2  [G_hat[1,3]]
      3 → 0:  nt2 → dn2  [G_hat[1,1]]
      4 → 3:  nt1 → dv2  [G_hat[0,3]]
      4 → 5:  nt1 → dv2  [G_hat[0,3]]
      5 → 1:  nt2 → dn1  [G_hat[1,0]]

=====
Diagram 1L-1082
=====

    External legs:
      leaf 0 ← dn2
      leaf 1 ← dn1
    Vertex assignments:
      v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

      coeff =  $\frac{1}{2} nstar_2$ 

      v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

      coeff =  $\frac{1}{2} \varphi_2''$ 

      v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

      coeff =  $\frac{1}{2} nstar_1$ 

      v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

      coeff =  $\frac{1}{2} \varphi_2''$ 

    Edges (propagator assignments):
      2 → 3:  nt2 → dv2  [G_hat[1,3]]
      2 → 5:  nt2 → dv2  [G_hat[1,3]]
      3 → 0:  nt2 → dn2  [G_hat[1,1]]
      4 → 3:  nt1 → dv2  [G_hat[0,3]]
      4 → 5:  nt1 → dv2  [G_hat[0,3]]
      5 → 1:  nt2 → dn1  [G_hat[1,0]]

```

=====

Diagram 1L-1083

=====

External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

Edges (propagator assignments):

2 → 3: nt2 → dv2 [G_hat[1,3]]

2 → 5: nt2 → dv2 [G_hat[1,3]]

3 → 0: nt2 → dn2 [G_hat[1,1]]

4 → 3: nt1 → dv2 [G_hat[0,3]]

4 → 5: nt1 → dv2 [G_hat[0,3]]

5 → 1: nt2 → dn1 [G_hat[1,0]]

=====

Diagram 1L-1084

=====

External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

```

    v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

    coeff =  $\frac{1}{2} nstar_1$ 

    v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

    coeff =  $\frac{1}{2} \varphi_2''$ 

    Edges (propagator assignments):
      2 → 3:  nt2 → dv2  [G_hat[1,3]]
      2 → 5:  nt2 → dv2  [G_hat[1,3]]
      3 → 0:  nt2 → dn2  [G_hat[1,1]]
      4 → 3:  nt1 → dv2  [G_hat[0,3]]
      4 → 5:  nt1 → dv2  [G_hat[0,3]]
      5 → 1:  nt2 → dn1  [G_hat[1,0]]

=====
Diagram 1L-1085
=====

    External legs:
      leaf 0 ← dn2
      leaf 1 ← dn1
    Vertex assignments:
      v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

      coeff =  $\frac{1}{2} nstar_2$ 

      v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

      coeff =  $\frac{1}{2} \varphi_2''$ 

      v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

      coeff =  $\frac{1}{2} nstar_1$ 

      v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

      coeff =  $\frac{1}{2} \varphi_2''$ 

    Edges (propagator assignments):
      2 → 3:  nt2 → dv2  [G_hat[1,3]]
      2 → 5:  nt2 → dv2  [G_hat[1,3]]
      3 → 0:  nt2 → dn2  [G_hat[1,1]]
      4 → 3:  nt1 → dv2  [G_hat[0,3]]
      4 → 5:  nt1 → dv2  [G_hat[0,3]]
      5 → 1:  nt2 → dn1  [G_hat[1,0]]

```


=====

Diagram 1L-1086

=====

External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

$$\text{coeff} = \frac{1}{2} nstar_1$$

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

Edges (propagator assignments):

2 → 3: nt2 → dv2 [G_hat[1,3]]

2 → 5: nt2 → dv2 [G_hat[1,3]]

3 → 0: nt2 → dn2 [G_hat[1,1]]

4 → 3: nt1 → dv2 [G_hat[0,3]]

4 → 5: nt1 → dv2 [G_hat[0,3]]

5 → 1: nt2 → dn1 [G_hat[1,0]]

=====

Diagram 1L-1087

=====

External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

```

    v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

    coeff =  $\frac{1}{2} nstar_1$ 

    v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

    coeff =  $\frac{1}{2} \varphi_2''$ 

    Edges (propagator assignments):
      2 → 3:  nt2 → dv2  [G_hat[1,3]]
      2 → 5:  nt2 → dv2  [G_hat[1,3]]
      3 → 0:  nt2 → dn2  [G_hat[1,1]]
      4 → 3:  nt1 → dv2  [G_hat[0,3]]
      4 → 5:  nt1 → dv2  [G_hat[0,3]]
      5 → 1:  nt2 → dn1  [G_hat[1,0]]

=====
Diagram 1L-1088
=====

    External legs:
      leaf 0 ← dn2
      leaf 1 ← dn1
    Vertex assignments:
      v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

      coeff =  $\frac{1}{2} nstar_2$ 

      v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

      coeff =  $\frac{1}{2} \varphi_2''$ 

      v4 (SourceType): bigrade=(2, 0), resp=[('nt', 1), ('nt', 1)]

      coeff =  $\frac{1}{2} nstar_1$ 

      v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

      coeff =  $\frac{1}{2} \varphi_2''$ 

    Edges (propagator assignments):
      2 → 3:  nt2 → dv2  [G_hat[1,3]]
      2 → 5:  nt2 → dv2  [G_hat[1,3]]
      3 → 0:  nt2 → dn2  [G_hat[1,1]]
      4 → 3:  nt1 → dv2  [G_hat[0,3]]
      4 → 5:  nt1 → dv2  [G_hat[0,3]]
      5 → 1:  nt2 → dn1  [G_hat[1,0]]

```

=====

Diagram 1L-1089

=====

External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

Edges (propagator assignments):

2 → 3: nt2 → dv1 [G_hat[1,2]]

2 → 5: nt2 → dv1 [G_hat[1,2]]

3 → 0: nt1 → dn2 [G_hat[0,1]]

4 → 3: nt2 → dv1 [G_hat[1,2]]

4 → 5: nt2 → dv1 [G_hat[1,2]]

5 → 1: nt1 → dn1 [G_hat[0,0]]

=====

Diagram 1L-1090

=====

External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

```

    v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

    coeff =  $\frac{1}{2} nstar_2$ 

    v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

    coeff =  $\frac{1}{2} \varphi_1''$ 

    Edges (propagator assignments):
      2 → 3:  nt2 → dv1  [G_hat[1,2]]
      2 → 5:  nt2 → dv1  [G_hat[1,2]]
      3 → 0:  nt1 → dn2  [G_hat[0,1]]
      4 → 3:  nt2 → dv1  [G_hat[1,2]]
      4 → 5:  nt2 → dv1  [G_hat[1,2]]
      5 → 1:  nt1 → dn1  [G_hat[0,0]]

=====
Diagram 1L-1091
=====

    External legs:
      leaf 0 ← dn2
      leaf 1 ← dn1
    Vertex assignments:
      v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

      coeff =  $\frac{1}{2} nstar_2$ 

      v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

      coeff =  $\frac{1}{2} \varphi_1''$ 

      v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

      coeff =  $\frac{1}{2} nstar_2$ 

      v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

      coeff =  $\frac{1}{2} \varphi_1''$ 

    Edges (propagator assignments):
      2 → 3:  nt2 → dv1  [G_hat[1,2]]
      2 → 5:  nt2 → dv1  [G_hat[1,2]]
      3 → 0:  nt1 → dn2  [G_hat[0,1]]
      4 → 3:  nt2 → dv1  [G_hat[1,2]]
      4 → 5:  nt2 → dv1  [G_hat[1,2]]
      5 → 1:  nt1 → dn1  [G_hat[0,0]]

```

=====

Diagram 1L-1092

=====

External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

Edges (propagator assignments):

2 → 3: nt2 → dv1 [G_hat[1,2]]

2 → 5: nt2 → dv1 [G_hat[1,2]]

3 → 0: nt1 → dn2 [G_hat[0,1]]

4 → 3: nt2 → dv1 [G_hat[1,2]]

4 → 5: nt2 → dv1 [G_hat[1,2]]

5 → 1: nt1 → dn1 [G_hat[0,0]]

=====

Diagram 1L-1093

=====

External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

```

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

coeff =  $\frac{1}{2} nstar_2$ 

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

coeff =  $\frac{1}{2} \varphi_1''$ 

Edges (propagator assignments):
  2 → 3:  nt2 → dv1  [G_hat[1,2]]
  2 → 5:  nt2 → dv1  [G_hat[1,2]]
  3 → 0:  nt1 → dn2  [G_hat[0,1]]
  4 → 3:  nt2 → dv1  [G_hat[1,2]]
  4 → 5:  nt2 → dv1  [G_hat[1,2]]
  5 → 1:  nt1 → dn1  [G_hat[0,0]]

=====
Diagram 1L-1094
=====

External legs:
  leaf 0 ← dn2
  leaf 1 ← dn1
Vertex assignments:
  v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

coeff =  $\frac{1}{2} nstar_2$ 

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

coeff =  $\frac{1}{2} \varphi_1''$ 

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

coeff =  $\frac{1}{2} nstar_2$ 

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

coeff =  $\frac{1}{2} \varphi_1''$ 

Edges (propagator assignments):
  2 → 3:  nt2 → dv1  [G_hat[1,2]]
  2 → 5:  nt2 → dv1  [G_hat[1,2]]
  3 → 0:  nt1 → dn2  [G_hat[0,1]]
  4 → 3:  nt2 → dv1  [G_hat[1,2]]
  4 → 5:  nt2 → dv1  [G_hat[1,2]]
  5 → 1:  nt1 → dn1  [G_hat[0,0]]

```

=====

Diagram 1L-1095

=====

External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

Edges (propagator assignments):

2 → 3: nt2 → dv1 [G_hat[1,2]]

2 → 5: nt2 → dv1 [G_hat[1,2]]

3 → 0: nt1 → dn2 [G_hat[0,1]]

4 → 3: nt2 → dv1 [G_hat[1,2]]

4 → 5: nt2 → dv1 [G_hat[1,2]]

5 → 1: nt1 → dn1 [G_hat[0,0]]

=====

Diagram 1L-1096

=====

External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

```

    v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

    coeff =  $\frac{1}{2} nstar_2$ 

    v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

    coeff =  $\frac{1}{2} \varphi_1''$ 

    Edges (propagator assignments):
      2 → 3:  nt2 → dv1  [G_hat[1,2]]
      2 → 5:  nt2 → dv1  [G_hat[1,2]]
      3 → 0:  nt1 → dn2  [G_hat[0,1]]
      4 → 3:  nt2 → dv1  [G_hat[1,2]]
      4 → 5:  nt2 → dv1  [G_hat[1,2]]
      5 → 1:  nt1 → dn1  [G_hat[0,0]]

=====
Diagram 1L-1097
=====

    External legs:
      leaf 0 ← dn2
      leaf 1 ← dn1
    Vertex assignments:
      v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

      coeff =  $\frac{1}{2} nstar_2$ 

      v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

      coeff =  $\frac{1}{2} \varphi_1''$ 

      v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

      coeff =  $\frac{1}{2} nstar_2$ 

      v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

      coeff =  $\frac{1}{2} \varphi_1''$ 

    Edges (propagator assignments):
      2 → 3:  nt2 → dv1  [G_hat[1,2]]
      2 → 5:  nt2 → dv1  [G_hat[1,2]]
      3 → 0:  nt1 → dn2  [G_hat[0,1]]
      4 → 3:  nt2 → dv1  [G_hat[1,2]]
      4 → 5:  nt2 → dv1  [G_hat[1,2]]
      5 → 1:  nt1 → dn1  [G_hat[0,0]]

```


=====

Diagram 1L-1098

=====

External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

Edges (propagator assignments):

2 → 3: nt2 → dv1 [G_hat[1,2]]

2 → 5: nt2 → dv1 [G_hat[1,2]]

3 → 0: nt1 → dn2 [G_hat[0,1]]

4 → 3: nt2 → dv1 [G_hat[1,2]]

4 → 5: nt2 → dv1 [G_hat[1,2]]

5 → 1: nt1 → dn1 [G_hat[0,0]]

=====

Diagram 1L-1099

=====

External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

```

    v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

    coeff =  $\frac{1}{2} nstar_2$ 

    v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

    coeff =  $\frac{1}{2} \varphi_1''$ 

    Edges (propagator assignments):
      2 → 3:  nt2 → dv1  [G_hat[1,2]]
      2 → 5:  nt2 → dv1  [G_hat[1,2]]
      3 → 0:  nt1 → dn2  [G_hat[0,1]]
      4 → 3:  nt2 → dv1  [G_hat[1,2]]
      4 → 5:  nt2 → dv1  [G_hat[1,2]]
      5 → 1:  nt1 → dn1  [G_hat[0,0]]

=====
Diagram 1L-1100
=====

    External legs:
      leaf 0 ← dn2
      leaf 1 ← dn1
    Vertex assignments:
      v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

      coeff =  $\frac{1}{2} nstar_2$ 

      v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

      coeff =  $\frac{1}{2} \varphi_1''$ 

      v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

      coeff =  $\frac{1}{2} nstar_2$ 

      v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

      coeff =  $\frac{1}{2} \varphi_1''$ 

    Edges (propagator assignments):
      2 → 3:  nt2 → dv1  [G_hat[1,2]]
      2 → 5:  nt2 → dv1  [G_hat[1,2]]
      3 → 0:  nt1 → dn2  [G_hat[0,1]]
      4 → 3:  nt2 → dv1  [G_hat[1,2]]
      4 → 5:  nt2 → dv1  [G_hat[1,2]]
      5 → 1:  nt1 → dn1  [G_hat[0,0]]

```

=====

Diagram 1L-1101

=====

External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

Edges (propagator assignments):

2 → 3: nt2 → dv1 [G_hat[1,2]]

2 → 5: nt2 → dv1 [G_hat[1,2]]

3 → 0: nt1 → dn2 [G_hat[0,1]]

4 → 3: nt2 → dv1 [G_hat[1,2]]

4 → 5: nt2 → dv1 [G_hat[1,2]]

5 → 1: nt1 → dn1 [G_hat[0,0]]

=====

Diagram 1L-1102

=====

External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

```

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

coeff =  $\frac{1}{2} nstar_2$ 

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

coeff =  $\frac{1}{2} \varphi_1''$ 

Edges (propagator assignments):
  2 → 3:  nt2 → dv1  [G_hat[1,2]]
  2 → 5:  nt2 → dv1  [G_hat[1,2]]
  3 → 0:  nt1 → dn2  [G_hat[0,1]]
  4 → 3:  nt2 → dv1  [G_hat[1,2]]
  4 → 5:  nt2 → dv1  [G_hat[1,2]]
  5 → 1:  nt1 → dn1  [G_hat[0,0]]

=====
Diagram 1L-1103
=====

External legs:
  leaf 0 ← dn2
  leaf 1 ← dn1
Vertex assignments:
  v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

coeff =  $\frac{1}{2} nstar_2$ 

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

coeff =  $\frac{1}{2} \varphi_1''$ 

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

coeff =  $\frac{1}{2} nstar_2$ 

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

coeff =  $\frac{1}{2} \varphi_1''$ 

Edges (propagator assignments):
  2 → 3:  nt2 → dv1  [G_hat[1,2]]
  2 → 5:  nt2 → dv1  [G_hat[1,2]]
  3 → 0:  nt1 → dn2  [G_hat[0,1]]
  4 → 3:  nt2 → dv1  [G_hat[1,2]]
  4 → 5:  nt2 → dv1  [G_hat[1,2]]
  5 → 1:  nt1 → dn1  [G_hat[0,0]]

```

=====

Diagram 1L-1104

=====

External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

Edges (propagator assignments):

2 → 3: nt2 → dv1 [G_hat[1,2]]

2 → 5: nt2 → dv1 [G_hat[1,2]]

3 → 0: nt1 → dn2 [G_hat[0,1]]

4 → 3: nt2 → dv1 [G_hat[1,2]]

4 → 5: nt2 → dv1 [G_hat[1,2]]

5 → 1: nt1 → dn1 [G_hat[0,0]]

=====

Diagram 1L-1105

=====

External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

```

    v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

    coeff =  $\frac{1}{2} nstar_2$ 

    v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

    coeff =  $\frac{1}{2} \varphi_2''$ 

    Edges (propagator assignments):
      2 → 3:  nt2 → dv1  [G_hat[1,2]]
      2 → 5:  nt2 → dv2  [G_hat[1,3]]
      3 → 0:  nt1 → dn2  [G_hat[0,1]]
      4 → 3:  nt2 → dv1  [G_hat[1,2]]
      4 → 5:  nt2 → dv2  [G_hat[1,3]]
      5 → 1:  nt2 → dn1  [G_hat[1,0]]

=====
Diagram 1L-1106
=====

    External legs:
      leaf 0 ← dn2
      leaf 1 ← dn1
    Vertex assignments:
      v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

      coeff =  $\frac{1}{2} nstar_2$ 

      v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

      coeff =  $\frac{1}{2} \varphi_1''$ 

      v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

      coeff =  $\frac{1}{2} nstar_2$ 

      v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

      coeff =  $\frac{1}{2} \varphi_2''$ 

    Edges (propagator assignments):
      2 → 3:  nt2 → dv1  [G_hat[1,2]]
      2 → 5:  nt2 → dv2  [G_hat[1,3]]
      3 → 0:  nt1 → dn2  [G_hat[0,1]]
      4 → 3:  nt2 → dv1  [G_hat[1,2]]
      4 → 5:  nt2 → dv2  [G_hat[1,3]]
      5 → 1:  nt2 → dn1  [G_hat[1,0]]

```

=====

Diagram 1L-1107

=====

External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

Edges (propagator assignments):

2 → 3: nt2 → dv1 [G_hat[1,2]]

2 → 5: nt2 → dv2 [G_hat[1,3]]

3 → 0: nt1 → dn2 [G_hat[0,1]]

4 → 3: nt2 → dv1 [G_hat[1,2]]

4 → 5: nt2 → dv2 [G_hat[1,3]]

5 → 1: nt2 → dn1 [G_hat[1,0]]

=====

Diagram 1L-1108

=====

External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

```

    v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

    coeff =  $\frac{1}{2} nstar_2$ 

    v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

    coeff =  $\frac{1}{2} \varphi_2''$ 

    Edges (propagator assignments):
      2 → 3:  nt2 → dv1  [G_hat[1,2]]
      2 → 5:  nt2 → dv2  [G_hat[1,3]]
      3 → 0:  nt1 → dn2  [G_hat[0,1]]
      4 → 3:  nt2 → dv1  [G_hat[1,2]]
      4 → 5:  nt2 → dv2  [G_hat[1,3]]
      5 → 1:  nt2 → dn1  [G_hat[1,0]]

=====
Diagram 1L-1109
=====

    External legs:
      leaf 0 ← dn2
      leaf 1 ← dn1
    Vertex assignments:
      v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

      coeff =  $\frac{1}{2} nstar_2$ 

      v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

      coeff =  $\frac{1}{2} \varphi_1''$ 

      v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

      coeff =  $\frac{1}{2} nstar_2$ 

      v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

      coeff =  $\frac{1}{2} \varphi_2''$ 

    Edges (propagator assignments):
      2 → 3:  nt2 → dv1  [G_hat[1,2]]
      2 → 5:  nt2 → dv2  [G_hat[1,3]]
      3 → 0:  nt1 → dn2  [G_hat[0,1]]
      4 → 3:  nt2 → dv1  [G_hat[1,2]]
      4 → 5:  nt2 → dv2  [G_hat[1,3]]
      5 → 1:  nt2 → dn1  [G_hat[1,0]]

```



```

=====
Diagram 1L-1110
=====
External legs:
  leaf 0 ← dn2
  leaf 1 ← dn1
Vertex assignments:
  v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

  coeff =  $\frac{1}{2} nstar_2$ 

  v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

  coeff =  $\frac{1}{2} \varphi_1''$ 

  v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

  coeff =  $\frac{1}{2} nstar_2$ 

  v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

  coeff =  $\frac{1}{2} \varphi_2''$ 

Edges (propagator assignments):
  2 → 3:  nt2 → dv1  [G_hat[1,2]]
  2 → 5:  nt2 → dv2  [G_hat[1,3]]
  3 → 0:  nt1 → dn2  [G_hat[0,1]]
  4 → 3:  nt2 → dv1  [G_hat[1,2]]
  4 → 5:  nt2 → dv2  [G_hat[1,3]]
  5 → 1:  nt2 → dn1  [G_hat[1,0]]

```

```

=====
Diagram 1L-1111
=====
External legs:
  leaf 0 ← dn2
  leaf 1 ← dn1
Vertex assignments:
  v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

  coeff =  $\frac{1}{2} nstar_2$ 

  v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

  coeff =  $\frac{1}{2} \varphi_1''$ 

```

```

    v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

    coeff =  $\frac{1}{2} nstar_2$ 

    v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

    coeff =  $\frac{1}{2} \varphi_2''$ 

    Edges (propagator assignments):
      2 → 3:  nt2 → dv1  [G_hat[1,2]]
      2 → 5:  nt2 → dv2  [G_hat[1,3]]
      3 → 0:  nt1 → dn2  [G_hat[0,1]]
      4 → 3:  nt2 → dv1  [G_hat[1,2]]
      4 → 5:  nt2 → dv2  [G_hat[1,3]]
      5 → 1:  nt2 → dn1  [G_hat[1,0]]

=====
Diagram 1L-1112
=====

    External legs:
      leaf 0 ← dn2
      leaf 1 ← dn1
    Vertex assignments:
      v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

      coeff =  $\frac{1}{2} nstar_2$ 

      v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

      coeff =  $\frac{1}{2} \varphi_1''$ 

      v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

      coeff =  $\frac{1}{2} nstar_2$ 

      v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

      coeff =  $\frac{1}{2} \varphi_2''$ 

    Edges (propagator assignments):
      2 → 3:  nt2 → dv1  [G_hat[1,2]]
      2 → 5:  nt2 → dv2  [G_hat[1,3]]
      3 → 0:  nt1 → dn2  [G_hat[0,1]]
      4 → 3:  nt2 → dv1  [G_hat[1,2]]
      4 → 5:  nt2 → dv2  [G_hat[1,3]]
      5 → 1:  nt2 → dn1  [G_hat[1,0]]

```

```

=====
Diagram 1L-1113
=====
External legs:
  leaf 0 ← dn2
  leaf 1 ← dn1
Vertex assignments:
  v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

  coeff =  $\frac{1}{2} nstar_2$ 

  v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

  coeff =  $\frac{1}{2} \varphi_1''$ 

  v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

  coeff =  $\frac{1}{2} nstar_2$ 

  v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

  coeff =  $\frac{1}{2} \varphi_2''$ 

Edges (propagator assignments):
  2 → 3:  nt2 → dv1  [G_hat[1,2]]
  2 → 5:  nt2 → dv2  [G_hat[1,3]]
  3 → 0:  nt1 → dn2  [G_hat[0,1]]
  4 → 3:  nt2 → dv1  [G_hat[1,2]]
  4 → 5:  nt2 → dv2  [G_hat[1,3]]
  5 → 1:  nt2 → dn1  [G_hat[1,0]]

```

```

=====
Diagram 1L-1114
=====
External legs:
  leaf 0 ← dn2
  leaf 1 ← dn1
Vertex assignments:
  v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

  coeff =  $\frac{1}{2} nstar_2$ 

  v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

  coeff =  $\frac{1}{2} \varphi_1''$ 

```

```

    v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

    coeff =  $\frac{1}{2} nstar_2$ 

    v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

    coeff =  $\frac{1}{2} \varphi_2''$ 

    Edges (propagator assignments):
      2 → 3:  nt2 → dv1  [G_hat[1,2]]
      2 → 5:  nt2 → dv2  [G_hat[1,3]]
      3 → 0:  nt1 → dn2  [G_hat[0,1]]
      4 → 3:  nt2 → dv1  [G_hat[1,2]]
      4 → 5:  nt2 → dv2  [G_hat[1,3]]
      5 → 1:  nt2 → dn1  [G_hat[1,0]]

=====
Diagram 1L-1115
=====

    External legs:
      leaf 0 ← dn2
      leaf 1 ← dn1
    Vertex assignments:
      v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

      coeff =  $\frac{1}{2} nstar_2$ 

      v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

      coeff =  $\frac{1}{2} \varphi_1''$ 

      v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

      coeff =  $\frac{1}{2} nstar_2$ 

      v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

      coeff =  $\frac{1}{2} \varphi_2''$ 

    Edges (propagator assignments):
      2 → 3:  nt2 → dv1  [G_hat[1,2]]
      2 → 5:  nt2 → dv2  [G_hat[1,3]]
      3 → 0:  nt1 → dn2  [G_hat[0,1]]
      4 → 3:  nt2 → dv1  [G_hat[1,2]]
      4 → 5:  nt2 → dv2  [G_hat[1,3]]
      5 → 1:  nt2 → dn1  [G_hat[1,0]]

```

```

=====
Diagram 1L-1116
=====
External legs:
  leaf 0 ← dn2
  leaf 1 ← dn1
Vertex assignments:
  v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]
  coeff =  $\frac{1}{2} nstar_2$ 
  v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]
  coeff =  $\frac{1}{2} \varphi_1''$ 
  v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]
  coeff =  $\frac{1}{2} nstar_2$ 
  v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]
  coeff =  $\frac{1}{2} \varphi_2''$ 
Edges (propagator assignments):
  2 → 3:  nt2 → dv1  [G_hat[1,2]]
  2 → 5:  nt2 → dv2  [G_hat[1,3]]
  3 → 0:  nt1 → dn2  [G_hat[0,1]]
  4 → 3:  nt2 → dv1  [G_hat[1,2]]
  4 → 5:  nt2 → dv2  [G_hat[1,3]]
  5 → 1:  nt2 → dn1  [G_hat[1,0]]

```

```

=====
Diagram 1L-1117
=====
External legs:
  leaf 0 ← dn2
  leaf 1 ← dn1
Vertex assignments:
  v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]
  coeff =  $\frac{1}{2} nstar_2$ 
  v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]
  coeff =  $\frac{1}{2} \varphi_1''$ 

```

```

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

coeff =  $\frac{1}{2} nstar_2$ 

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

coeff =  $\frac{1}{2} \varphi_2''$ 

Edges (propagator assignments):
  2 → 3:  nt2 → dv1  [G_hat[1,2]]
  2 → 5:  nt2 → dv2  [G_hat[1,3]]
  3 → 0:  nt1 → dn2  [G_hat[0,1]]
  4 → 3:  nt2 → dv1  [G_hat[1,2]]
  4 → 5:  nt2 → dv2  [G_hat[1,3]]
  5 → 1:  nt2 → dn1  [G_hat[1,0]]

=====
Diagram 1L-1118
=====

External legs:
  leaf 0 ← dn2
  leaf 1 ← dn1
Vertex assignments:
  v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

coeff =  $\frac{1}{2} nstar_2$ 

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

coeff =  $\frac{1}{2} \varphi_1''$ 

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

coeff =  $\frac{1}{2} nstar_2$ 

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

coeff =  $\frac{1}{2} \varphi_2''$ 

Edges (propagator assignments):
  2 → 3:  nt2 → dv1  [G_hat[1,2]]
  2 → 5:  nt2 → dv2  [G_hat[1,3]]
  3 → 0:  nt1 → dn2  [G_hat[0,1]]
  4 → 3:  nt2 → dv1  [G_hat[1,2]]
  4 → 5:  nt2 → dv2  [G_hat[1,3]]
  5 → 1:  nt2 → dn1  [G_hat[1,0]]

```

=====

Diagram 1L-1119

=====

External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

Edges (propagator assignments):

2 → 3: nt2 → dv1 [G_hat[1,2]]

2 → 5: nt2 → dv2 [G_hat[1,3]]

3 → 0: nt1 → dn2 [G_hat[0,1]]

4 → 3: nt2 → dv1 [G_hat[1,2]]

4 → 5: nt2 → dv2 [G_hat[1,3]]

5 → 1: nt2 → dn1 [G_hat[1,0]]

=====

Diagram 1L-1120

=====

External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

```

    v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

    coeff =  $\frac{1}{2} nstar_2$ 

    v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

    coeff =  $\frac{1}{2} \varphi_2''$ 

    Edges (propagator assignments):
      2 → 3:  nt2 → dv1  [G_hat[1,2]]
      2 → 5:  nt2 → dv2  [G_hat[1,3]]
      3 → 0:  nt1 → dn2  [G_hat[0,1]]
      4 → 3:  nt2 → dv1  [G_hat[1,2]]
      4 → 5:  nt2 → dv2  [G_hat[1,3]]
      5 → 1:  nt2 → dn1  [G_hat[1,0]]

=====
Diagram 1L-1121
=====

    External legs:
      leaf 0 ← dn2
      leaf 1 ← dn1
    Vertex assignments:
      v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

      coeff =  $\frac{1}{2} nstar_2$ 

      v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

      coeff =  $\frac{1}{2} \varphi_2''$ 

      v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

      coeff =  $\frac{1}{2} nstar_2$ 

      v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

      coeff =  $\frac{1}{2} \varphi_1''$ 

    Edges (propagator assignments):
      2 → 3:  nt2 → dv2  [G_hat[1,3]]
      2 → 5:  nt2 → dv1  [G_hat[1,2]]
      3 → 0:  nt2 → dn2  [G_hat[1,1]]
      4 → 3:  nt2 → dv2  [G_hat[1,3]]
      4 → 5:  nt2 → dv1  [G_hat[1,2]]
      5 → 1:  nt1 → dn1  [G_hat[0,0]]

```



```

=====
Diagram 1L-1122
=====
External legs:
  leaf 0 ← dn2
  leaf 1 ← dn1
Vertex assignments:
  v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]
  coeff =  $\frac{1}{2} nstar_2$ 
  v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]
  coeff =  $\frac{1}{2} \varphi_2''$ 
  v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]
  coeff =  $\frac{1}{2} nstar_2$ 
  v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]
  coeff =  $\frac{1}{2} \varphi_1''$ 
Edges (propagator assignments):
  2 → 3:  nt2 → dv2  [G_hat[1,3]]
  2 → 5:  nt2 → dv1  [G_hat[1,2]]
  3 → 0:  nt2 → dn2  [G_hat[1,1]]
  4 → 3:  nt2 → dv2  [G_hat[1,3]]
  4 → 5:  nt2 → dv1  [G_hat[1,2]]
  5 → 1:  nt1 → dn1  [G_hat[0,0]]

```

```

=====
Diagram 1L-1123
=====
External legs:
  leaf 0 ← dn2
  leaf 1 ← dn1
Vertex assignments:
  v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]
  coeff =  $\frac{1}{2} nstar_2$ 
  v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]
  coeff =  $\frac{1}{2} \varphi_2''$ 

```

```

    v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

    coeff =  $\frac{1}{2} nstar_2$ 

    v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

    coeff =  $\frac{1}{2} \varphi_1''$ 

    Edges (propagator assignments):
      2 → 3:  nt2 → dv2  [G_hat[1,3]]
      2 → 5:  nt2 → dv1  [G_hat[1,2]]
      3 → 0:  nt2 → dn2  [G_hat[1,1]]
      4 → 3:  nt2 → dv2  [G_hat[1,3]]
      4 → 5:  nt2 → dv1  [G_hat[1,2]]
      5 → 1:  nt1 → dn1  [G_hat[0,0]]

=====
Diagram 1L-1124
=====

    External legs:
      leaf 0 ← dn2
      leaf 1 ← dn1
    Vertex assignments:
      v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

      coeff =  $\frac{1}{2} nstar_2$ 

      v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

      coeff =  $\frac{1}{2} \varphi_2''$ 

      v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

      coeff =  $\frac{1}{2} nstar_2$ 

      v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

      coeff =  $\frac{1}{2} \varphi_1''$ 

    Edges (propagator assignments):
      2 → 3:  nt2 → dv2  [G_hat[1,3]]
      2 → 5:  nt2 → dv1  [G_hat[1,2]]
      3 → 0:  nt2 → dn2  [G_hat[1,1]]
      4 → 3:  nt2 → dv2  [G_hat[1,3]]
      4 → 5:  nt2 → dv1  [G_hat[1,2]]
      5 → 1:  nt1 → dn1  [G_hat[0,0]]

```

```

=====
Diagram 1L-1125
=====
External legs:
  leaf 0 ← dn2
  leaf 1 ← dn1
Vertex assignments:
  v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

  coeff =  $\frac{1}{2} nstar_2$ 

  v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

  coeff =  $\frac{1}{2} \varphi_2''$ 

  v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

  coeff =  $\frac{1}{2} nstar_2$ 

  v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

  coeff =  $\frac{1}{2} \varphi_1''$ 

Edges (propagator assignments):
  2 → 3:  nt2 → dv2  [G_hat[1,3]]
  2 → 5:  nt2 → dv1  [G_hat[1,2]]
  3 → 0:  nt2 → dn2  [G_hat[1,1]]
  4 → 3:  nt2 → dv2  [G_hat[1,3]]
  4 → 5:  nt2 → dv1  [G_hat[1,2]]
  5 → 1:  nt1 → dn1  [G_hat[0,0]]

```

```

=====
Diagram 1L-1126
=====
External legs:
  leaf 0 ← dn2
  leaf 1 ← dn1
Vertex assignments:
  v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

  coeff =  $\frac{1}{2} nstar_2$ 

  v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

  coeff =  $\frac{1}{2} \varphi_2''$ 

```

```

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

coeff =  $\frac{1}{2} nstar_2$ 

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

coeff =  $\frac{1}{2} \varphi_1''$ 

Edges (propagator assignments):
  2 → 3:  nt2 → dv2  [G_hat[1,3]]
  2 → 5:  nt2 → dv1  [G_hat[1,2]]
  3 → 0:  nt2 → dn2  [G_hat[1,1]]
  4 → 3:  nt2 → dv2  [G_hat[1,3]]
  4 → 5:  nt2 → dv1  [G_hat[1,2]]
  5 → 1:  nt1 → dn1  [G_hat[0,0]]

=====
Diagram 1L-1127
=====

External legs:
  leaf 0 ← dn2
  leaf 1 ← dn1
Vertex assignments:
  v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

coeff =  $\frac{1}{2} nstar_2$ 

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

coeff =  $\frac{1}{2} \varphi_2''$ 

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

coeff =  $\frac{1}{2} nstar_2$ 

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

coeff =  $\frac{1}{2} \varphi_1''$ 

Edges (propagator assignments):
  2 → 3:  nt2 → dv2  [G_hat[1,3]]
  2 → 5:  nt2 → dv1  [G_hat[1,2]]
  3 → 0:  nt2 → dn2  [G_hat[1,1]]
  4 → 3:  nt2 → dv2  [G_hat[1,3]]
  4 → 5:  nt2 → dv1  [G_hat[1,2]]
  5 → 1:  nt1 → dn1  [G_hat[0,0]]

```

=====

Diagram 1L-1128

=====

External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)]

$$\text{coeff} = \frac{1}{2} \varphi_1''$$

Edges (propagator assignments):

2 → 3: nt2 → dv2 [G_hat[1,3]]

2 → 5: nt2 → dv1 [G_hat[1,2]]

3 → 0: nt2 → dn2 [G_hat[1,1]]

4 → 3: nt2 → dv2 [G_hat[1,3]]

4 → 5: nt2 → dv1 [G_hat[1,2]]

5 → 1: nt1 → dn1 [G_hat[0,0]]

=====

Diagram 1L-1129

=====

External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

```

    v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

    coeff =  $\frac{1}{2} nstar_2$ 

    v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

    coeff =  $\frac{1}{2} \varphi_1''$ 

    Edges (propagator assignments):
      2 → 3:  nt2 → dv2  [G_hat[1,3]]
      2 → 5:  nt2 → dv1  [G_hat[1,2]]
      3 → 0:  nt2 → dn2  [G_hat[1,1]]
      4 → 3:  nt2 → dv2  [G_hat[1,3]]
      4 → 5:  nt2 → dv1  [G_hat[1,2]]
      5 → 1:  nt1 → dn1  [G_hat[0,0]]

=====
Diagram 1L-1130
=====

    External legs:
      leaf 0 ← dn2
      leaf 1 ← dn1
    Vertex assignments:
      v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

      coeff =  $\frac{1}{2} nstar_2$ 

      v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

      coeff =  $\frac{1}{2} \varphi_2''$ 

      v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

      coeff =  $\frac{1}{2} nstar_2$ 

      v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

      coeff =  $\frac{1}{2} \varphi_1''$ 

    Edges (propagator assignments):
      2 → 3:  nt2 → dv2  [G_hat[1,3]]
      2 → 5:  nt2 → dv1  [G_hat[1,2]]
      3 → 0:  nt2 → dn2  [G_hat[1,1]]
      4 → 3:  nt2 → dv2  [G_hat[1,3]]
      4 → 5:  nt2 → dv1  [G_hat[1,2]]
      5 → 1:  nt1 → dn1  [G_hat[0,0]]

```

```

=====
Diagram 1L-1131
=====
External legs:
  leaf 0 ← dn2
  leaf 1 ← dn1
Vertex assignments:
  v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

  coeff =  $\frac{1}{2} nstar_2$ 

  v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

  coeff =  $\frac{1}{2} \varphi_2''$ 

  v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

  coeff =  $\frac{1}{2} nstar_2$ 

  v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

  coeff =  $\frac{1}{2} \varphi_1''$ 

Edges (propagator assignments):
  2 → 3:  nt2 → dv2  [G_hat[1,3]]
  2 → 5:  nt2 → dv1  [G_hat[1,2]]
  3 → 0:  nt2 → dn2  [G_hat[1,1]]
  4 → 3:  nt2 → dv2  [G_hat[1,3]]
  4 → 5:  nt2 → dv1  [G_hat[1,2]]
  5 → 1:  nt1 → dn1  [G_hat[0,0]]

```

```

=====
Diagram 1L-1132
=====
External legs:
  leaf 0 ← dn2
  leaf 1 ← dn1
Vertex assignments:
  v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

  coeff =  $\frac{1}{2} nstar_2$ 

  v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

  coeff =  $\frac{1}{2} \varphi_2''$ 

```

```

    v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

    coeff =  $\frac{1}{2} nstar_2$ 

    v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

    coeff =  $\frac{1}{2} \varphi_1''$ 

    Edges (propagator assignments):
      2 → 3:  nt2 → dv2  [G_hat[1,3]]
      2 → 5:  nt2 → dv1  [G_hat[1,2]]
      3 → 0:  nt2 → dn2  [G_hat[1,1]]
      4 → 3:  nt2 → dv2  [G_hat[1,3]]
      4 → 5:  nt2 → dv1  [G_hat[1,2]]
      5 → 1:  nt1 → dn1  [G_hat[0,0]]

=====
Diagram 1L-1133
=====

    External legs:
      leaf 0 ← dn2
      leaf 1 ← dn1
    Vertex assignments:
      v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

      coeff =  $\frac{1}{2} nstar_2$ 

      v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

      coeff =  $\frac{1}{2} \varphi_2''$ 

      v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

      coeff =  $\frac{1}{2} nstar_2$ 

      v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

      coeff =  $\frac{1}{2} \varphi_1''$ 

    Edges (propagator assignments):
      2 → 3:  nt2 → dv2  [G_hat[1,3]]
      2 → 5:  nt2 → dv1  [G_hat[1,2]]
      3 → 0:  nt2 → dn2  [G_hat[1,1]]
      4 → 3:  nt2 → dv2  [G_hat[1,3]]
      4 → 5:  nt2 → dv1  [G_hat[1,2]]
      5 → 1:  nt1 → dn1  [G_hat[0,0]]

```



```

=====
Diagram 1L-1134
=====
External legs:
  leaf 0 ← dn2
  leaf 1 ← dn1
Vertex assignments:
  v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

  coeff =  $\frac{1}{2} nstar_2$ 

  v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

  coeff =  $\frac{1}{2} \varphi_2''$ 

  v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

  coeff =  $\frac{1}{2} nstar_2$ 

  v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

  coeff =  $\frac{1}{2} \varphi_1''$ 

Edges (propagator assignments):
  2 → 3:  nt2 → dv2  [G_hat[1,3]]
  2 → 5:  nt2 → dv1  [G_hat[1,2]]
  3 → 0:  nt2 → dn2  [G_hat[1,1]]
  4 → 3:  nt2 → dv2  [G_hat[1,3]]
  4 → 5:  nt2 → dv1  [G_hat[1,2]]
  5 → 1:  nt1 → dn1  [G_hat[0,0]]

```

```

=====
Diagram 1L-1135
=====
External legs:
  leaf 0 ← dn2
  leaf 1 ← dn1
Vertex assignments:
  v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

  coeff =  $\frac{1}{2} nstar_2$ 

  v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

  coeff =  $\frac{1}{2} \varphi_2''$ 

```

```

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

coeff =  $\frac{1}{2} nstar_2$ 

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

coeff =  $\frac{1}{2} \varphi_1''$ 

Edges (propagator assignments):
  2 → 3:  nt2 → dv2  [G_hat[1,3]]
  2 → 5:  nt2 → dv1  [G_hat[1,2]]
  3 → 0:  nt2 → dn2  [G_hat[1,1]]
  4 → 3:  nt2 → dv2  [G_hat[1,3]]
  4 → 5:  nt2 → dv1  [G_hat[1,2]]
  5 → 1:  nt1 → dn1  [G_hat[0,0]]

=====
Diagram 1L-1136
=====

External legs:
  leaf 0 ← dn2
  leaf 1 ← dn1
Vertex assignments:
  v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

coeff =  $\frac{1}{2} nstar_2$ 

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

coeff =  $\frac{1}{2} \varphi_2''$ 

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

coeff =  $\frac{1}{2} nstar_2$ 

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 1)], phys=[('dv', 1), ('dv',
1)]

coeff =  $\frac{1}{2} \varphi_1''$ 

Edges (propagator assignments):
  2 → 3:  nt2 → dv2  [G_hat[1,3]]
  2 → 5:  nt2 → dv1  [G_hat[1,2]]
  3 → 0:  nt2 → dn2  [G_hat[1,1]]
  4 → 3:  nt2 → dv2  [G_hat[1,3]]
  4 → 5:  nt2 → dv1  [G_hat[1,2]]
  5 → 1:  nt1 → dn1  [G_hat[0,0]]

```

=====

Diagram 1L-1137

=====

External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

Edges (propagator assignments):

2 → 3: nt2 → dv2 [G_hat[1,3]]

2 → 5: nt2 → dv2 [G_hat[1,3]]

3 → 0: nt2 → dn2 [G_hat[1,1]]

4 → 3: nt2 → dv2 [G_hat[1,3]]

4 → 5: nt2 → dv2 [G_hat[1,3]]

5 → 1: nt2 → dn1 [G_hat[1,0]]

=====

Diagram 1L-1138

=====

External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

```

    v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

    coeff =  $\frac{1}{2} nstar_2$ 

    v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

    coeff =  $\frac{1}{2} \varphi_2''$ 

    Edges (propagator assignments):
      2 → 3:  nt2 → dv2  [G_hat[1,3]]
      2 → 5:  nt2 → dv2  [G_hat[1,3]]
      3 → 0:  nt2 → dn2  [G_hat[1,1]]
      4 → 3:  nt2 → dv2  [G_hat[1,3]]
      4 → 5:  nt2 → dv2  [G_hat[1,3]]
      5 → 1:  nt2 → dn1  [G_hat[1,0]]

=====
Diagram 1L-1139
=====

    External legs:
      leaf 0 ← dn2
      leaf 1 ← dn1
    Vertex assignments:
      v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

      coeff =  $\frac{1}{2} nstar_2$ 

      v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

      coeff =  $\frac{1}{2} \varphi_2''$ 

      v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

      coeff =  $\frac{1}{2} nstar_2$ 

      v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

      coeff =  $\frac{1}{2} \varphi_2''$ 

      Edges (propagator assignments):
        2 → 3:  nt2 → dv2  [G_hat[1,3]]
        2 → 5:  nt2 → dv2  [G_hat[1,3]]
        3 → 0:  nt2 → dn2  [G_hat[1,1]]
        4 → 3:  nt2 → dv2  [G_hat[1,3]]
        4 → 5:  nt2 → dv2  [G_hat[1,3]]
        5 → 1:  nt2 → dn1  [G_hat[1,0]]

```

=====

Diagram 1L-1140

=====

External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

Edges (propagator assignments):

2 → 3: nt2 → dv2 [G_hat[1,3]]

2 → 5: nt2 → dv2 [G_hat[1,3]]

3 → 0: nt2 → dn2 [G_hat[1,1]]

4 → 3: nt2 → dv2 [G_hat[1,3]]

4 → 5: nt2 → dv2 [G_hat[1,3]]

5 → 1: nt2 → dn1 [G_hat[1,0]]

=====

Diagram 1L-1141

=====

External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

```

    v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

    coeff =  $\frac{1}{2} nstar_2$ 

    v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

    coeff =  $\frac{1}{2} \varphi_2''$ 

    Edges (propagator assignments):
      2 → 3:  nt2 → dv2  [G_hat[1,3]]
      2 → 5:  nt2 → dv2  [G_hat[1,3]]
      3 → 0:  nt2 → dn2  [G_hat[1,1]]
      4 → 3:  nt2 → dv2  [G_hat[1,3]]
      4 → 5:  nt2 → dv2  [G_hat[1,3]]
      5 → 1:  nt2 → dn1  [G_hat[1,0]]

=====
Diagram 1L-1142
=====

    External legs:
      leaf 0 ← dn2
      leaf 1 ← dn1
    Vertex assignments:
      v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

      coeff =  $\frac{1}{2} nstar_2$ 

      v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

      coeff =  $\frac{1}{2} \varphi_2''$ 

      v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

      coeff =  $\frac{1}{2} nstar_2$ 

      v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

      coeff =  $\frac{1}{2} \varphi_2''$ 

    Edges (propagator assignments):
      2 → 3:  nt2 → dv2  [G_hat[1,3]]
      2 → 5:  nt2 → dv2  [G_hat[1,3]]
      3 → 0:  nt2 → dn2  [G_hat[1,1]]
      4 → 3:  nt2 → dv2  [G_hat[1,3]]
      4 → 5:  nt2 → dv2  [G_hat[1,3]]
      5 → 1:  nt2 → dn1  [G_hat[1,0]]

```

=====

Diagram 1L-1143

=====

External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

Edges (propagator assignments):

2 → 3: nt2 → dv2 [G_hat[1,3]]

2 → 5: nt2 → dv2 [G_hat[1,3]]

3 → 0: nt2 → dn2 [G_hat[1,1]]

4 → 3: nt2 → dv2 [G_hat[1,3]]

4 → 5: nt2 → dv2 [G_hat[1,3]]

5 → 1: nt2 → dn1 [G_hat[1,0]]

=====

Diagram 1L-1144

=====

External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

```

    v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

    coeff =  $\frac{1}{2} nstar_2$ 

    v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

    coeff =  $\frac{1}{2} \varphi_2''$ 

    Edges (propagator assignments):
      2 → 3:  nt2 → dv2  [G_hat[1,3]]
      2 → 5:  nt2 → dv2  [G_hat[1,3]]
      3 → 0:  nt2 → dn2  [G_hat[1,1]]
      4 → 3:  nt2 → dv2  [G_hat[1,3]]
      4 → 5:  nt2 → dv2  [G_hat[1,3]]
      5 → 1:  nt2 → dn1  [G_hat[1,0]]

=====
Diagram 1L-1145
=====

    External legs:
      leaf 0 ← dn2
      leaf 1 ← dn1
    Vertex assignments:
      v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

      coeff =  $\frac{1}{2} nstar_2$ 

      v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

      coeff =  $\frac{1}{2} \varphi_2''$ 

      v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

      coeff =  $\frac{1}{2} nstar_2$ 

      v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

      coeff =  $\frac{1}{2} \varphi_2''$ 

    Edges (propagator assignments):
      2 → 3:  nt2 → dv2  [G_hat[1,3]]
      2 → 5:  nt2 → dv2  [G_hat[1,3]]
      3 → 0:  nt2 → dn2  [G_hat[1,1]]
      4 → 3:  nt2 → dv2  [G_hat[1,3]]
      4 → 5:  nt2 → dv2  [G_hat[1,3]]
      5 → 1:  nt2 → dn1  [G_hat[1,0]]

```


=====

Diagram 1L-1146

=====

External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

Edges (propagator assignments):

2 → 3: nt2 → dv2 [G_hat[1,3]]

2 → 5: nt2 → dv2 [G_hat[1,3]]

3 → 0: nt2 → dn2 [G_hat[1,1]]

4 → 3: nt2 → dv2 [G_hat[1,3]]

4 → 5: nt2 → dv2 [G_hat[1,3]]

5 → 1: nt2 → dn1 [G_hat[1,0]]

=====

Diagram 1L-1147

=====

External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

```

    v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

    coeff =  $\frac{1}{2} nstar_2$ 

    v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

    coeff =  $\frac{1}{2} \varphi_2''$ 

    Edges (propagator assignments):
      2 → 3:  nt2 → dv2  [G_hat[1,3]]
      2 → 5:  nt2 → dv2  [G_hat[1,3]]
      3 → 0:  nt2 → dn2  [G_hat[1,1]]
      4 → 3:  nt2 → dv2  [G_hat[1,3]]
      4 → 5:  nt2 → dv2  [G_hat[1,3]]
      5 → 1:  nt2 → dn1  [G_hat[1,0]]

=====
Diagram 1L-1148
=====

    External legs:
      leaf 0 ← dn2
      leaf 1 ← dn1
    Vertex assignments:
      v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

      coeff =  $\frac{1}{2} nstar_2$ 

      v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

      coeff =  $\frac{1}{2} \varphi_2''$ 

      v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

      coeff =  $\frac{1}{2} nstar_2$ 

      v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

      coeff =  $\frac{1}{2} \varphi_2''$ 

    Edges (propagator assignments):
      2 → 3:  nt2 → dv2  [G_hat[1,3]]
      2 → 5:  nt2 → dv2  [G_hat[1,3]]
      3 → 0:  nt2 → dn2  [G_hat[1,1]]
      4 → 3:  nt2 → dv2  [G_hat[1,3]]
      4 → 5:  nt2 → dv2  [G_hat[1,3]]
      5 → 1:  nt2 → dn1  [G_hat[1,0]]

```

=====

Diagram 1L-1149

=====

External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

Edges (propagator assignments):

2 → 3: nt2 → dv2 [G_hat[1,3]]

2 → 5: nt2 → dv2 [G_hat[1,3]]

3 → 0: nt2 → dn2 [G_hat[1,1]]

4 → 3: nt2 → dv2 [G_hat[1,3]]

4 → 5: nt2 → dv2 [G_hat[1,3]]

5 → 1: nt2 → dn1 [G_hat[1,0]]

=====

Diagram 1L-1150

=====

External legs:

leaf 0 ← dn2

leaf 1 ← dn1

Vertex assignments:

v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

$$\text{coeff} = \frac{1}{2} nstar_2$$

v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)]

$$\text{coeff} = \frac{1}{2} \varphi_2''$$

```

    v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

    coeff =  $\frac{1}{2} nstar_2$ 

    v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

    coeff =  $\frac{1}{2} \varphi_2''$ 

    Edges (propagator assignments):
      2 → 3:  nt2 → dv2  [G_hat[1,3]]
      2 → 5:  nt2 → dv2  [G_hat[1,3]]
      3 → 0:  nt2 → dn2  [G_hat[1,1]]
      4 → 3:  nt2 → dv2  [G_hat[1,3]]
      4 → 5:  nt2 → dv2  [G_hat[1,3]]
      5 → 1:  nt2 → dn1  [G_hat[1,0]]

=====
Diagram 1L-1151
=====

    External legs:
      leaf 0 ← dn2
      leaf 1 ← dn1
    Vertex assignments:
      v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

      coeff =  $\frac{1}{2} nstar_2$ 

      v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

      coeff =  $\frac{1}{2} \varphi_2''$ 

      v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]

      coeff =  $\frac{1}{2} nstar_2$ 

      v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]

      coeff =  $\frac{1}{2} \varphi_2''$ 

    Edges (propagator assignments):
      2 → 3:  nt2 → dv2  [G_hat[1,3]]
      2 → 5:  nt2 → dv2  [G_hat[1,3]]
      3 → 0:  nt2 → dn2  [G_hat[1,1]]
      4 → 3:  nt2 → dv2  [G_hat[1,3]]
      4 → 5:  nt2 → dv2  [G_hat[1,3]]
      5 → 1:  nt2 → dn1  [G_hat[1,0]]

```

```

=====
Diagram 1L-1152
=====
External legs:
  leaf 0 ← dn2
  leaf 1 ← dn1
Vertex assignments:
  v2 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]
coeff =  $\frac{1}{2} nstar_2$ 

  v3 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]
coeff =  $\frac{1}{2} \varphi_2''$ 

  v4 (SourceType): bigrade=(2, 0), resp=[('nt', 2), ('nt', 2)]
coeff =  $\frac{1}{2} nstar_2$ 

  v5 (VertexType): bigrade=(1, 2), resp=[('nt', 2)], phys=[('dv', 2), ('dv',
2)]
coeff =  $\frac{1}{2} \varphi_2''$ 

Edges (propagator assignments):
  2 → 3:  nt2 → dv2  [G_hat[1,3]]
  2 → 5:  nt2 → dv2  [G_hat[1,3]]
  3 → 0:  nt2 → dn2  [G_hat[1,1]]
  4 → 3:  nt2 → dv2  [G_hat[1,3]]
  4 → 5:  nt2 → dv2  [G_hat[1,3]]
  5 → 1:  nt2 → dn1  [G_hat[1,0]]

```

1.7 6. Diagram Isomorphism Check

Identify which typed diagrams are truly distinct. Two typed diagrams are **isomorphic** if they have the same: - External leg assignments (which field at each leaf) - Vertex type at each internal vertex (same coefficient, same legs) - Propagator assignment on every edge (same $\hat{G}_{ij}$ indices)

We build a canonical signature for each diagram and group duplicates into equivalence classes.

```

[13]: from collections import defaultdict

def diagram_signature(td):
    """
    Build a hashable canonical signature for a typed diagram.

```

Two diagrams with the same signature are identical up to the internal ordering of leg permutations - they represent the same Feynman diagram.

"""

External legs: sorted (leaf, field) pairs

ext = tuple(sorted(td.external_legs.items()))

Vertex assignments: sorted (vertex, type_key) pairs

verts = []

for v, vtype in sorted(td.vertex_assignments.items()):

 tname = type(vtype).__name__

 resp = tuple(vtype.response_legs)

 phys = tuple(vtype.physical_legs) if hasattr(vtype, 'physical_legs') else

↪ else ()

 verts.append((v, tname, str(vtype.coefficient), vtype.bigrade, resp,

↪ phys))

verts = tuple(verts)

Edge propagator assignments: sorted (edge, prop_indices) pairs

edges = tuple(sorted(
 ((u, v), td.propagator_indices[(u, v)])
 for (u, v) in td.edge_types

))

return (ext, verts, edges)

def find_equivalence_classes(typed_diagrams, label_prefix=''):

"""Group typed diagrams by their canonical signature."""

 sig_to_indices = defaultdict(list)

 signatures = []

 for i, td in enumerate(typed_diagrams):

 sig = diagram_signature(td)

 signatures.append(sig)

 sig_to_indices[sig].append(i)

 classes = list(sig_to_indices.values())

 print(f'{len(typed_diagrams)} typed diagrams → {len(classes)} unique ↪
↪ diagrams')

 print()

 for ci, members in enumerate(classes, 1):

 member_labels = [f'{label_prefix}{m+1}' for m in members]

 multiplicity = len(members)

 print(f' Class {ci} (multiplicity {multiplicity}): diagrams ↪
↪ {member_labels}')

```

    # Show the representative diagram
    td = typed_diagrams[members[0]]
    print(f'    External legs: ', end='')
    print(', '.join(f'leaf {lf}←{f[0]}{f[1]}' for lf, f in sorted(td.
↪external_legs.items()))
    print(f'    Edges: ', end='')
    print(', '.join(
        f'{u}→{v} [G_hat[{td.propagator_indices[(u,v)][0]}, {td.
↪propagator_indices[(u,v)][1]}]]'
        for (u, v) in sorted(td.edge_types)
    ))
    if td.vertex_assignments:
        for v, vtype in sorted(td.vertex_assignments.items()):
            tname = type(vtype).__name__
            print(f'    v{v} ({tname}): resp={vtype.response_legs}', end='')
            if hasattr(vtype, 'physical_legs'):
                print(f', phys={vtype.physical_legs}', end='')
            print(f', coeff={vtype.coefficient}')

    return classes

print('='*60)
print('TREE-LEVEL ISOMORPHISM ANALYSIS')
print('='*60)
tree_classes = find_equivalence_classes(typed_tree, label_prefix='Tree-')

print()
print('='*60)
print('1-LOOP ISOMORPHISM ANALYSIS')
print('='*60)
loop_classes = find_equivalence_classes(typed_1loop, label_prefix='1L-')

```

```

=====
TREE-LEVEL ISOMORPHISM ANALYSIS
=====
8 typed diagrams → 4 unique diagrams

```

```

Class 1 (multiplicity 2): diagrams ['Tree-1', 'Tree-2']
    External legs: leaf 0→dn1, leaf 1←dn2
    Edges: 2→0 [G_hat[0,0]], 2→1 [G_hat[0,1]]
    v2 (SourceType): resp=[('nt', 1), ('nt', 1)], coeff=1/2*nstar1
Class 2 (multiplicity 2): diagrams ['Tree-3', 'Tree-4']
    External legs: leaf 0→dn1, leaf 1←dn2
    Edges: 2→0 [G_hat[1,0]], 2→1 [G_hat[1,1]]
    v2 (SourceType): resp=[('nt', 2), ('nt', 2)], coeff=1/2*nstar2
Class 3 (multiplicity 2): diagrams ['Tree-5', 'Tree-6']

```

```

    External legs: leaf 0↔dn2, leaf 1↔dn1
    Edges: 2→0 [G_hat[0,1]], 2→1 [G_hat[0,0]]
    v2 (SourceType): resp=[('nt', 1), ('nt', 1)], coeff=1/2*nstar1
Class 4 (multiplicity 2): diagrams ['Tree-7', 'Tree-8']
    External legs: leaf 0↔dn2, leaf 1↔dn1
    Edges: 2→0 [G_hat[1,1]], 2→1 [G_hat[1,0]]
    v2 (SourceType): resp=[('nt', 2), ('nt', 2)], coeff=1/2*nstar2

=====
1-LOOP ISOMORPHISM ANALYSIS
=====

1152 typed diagrams → 80 unique diagrams

    Class 1 (multiplicity 8): diagrams ['1L-1', '1L-2', '1L-3', '1L-4', '1L-5',
    '1L-6', '1L-7', '1L-8']
        External legs: leaf 0↔dn1, leaf 1↔dn2
        Edges: 2→3 [G_hat[0,2]], 2→4 [G_hat[0,2]], 3→0 [G_hat[0,0]], 3→4
[G_hat[0,2]], 4→1 [G_hat[0,1]]
        v2 (SourceType): resp=[('nt', 1), ('nt', 1)], coeff=1/2*nstar1
        v3 (VertexType): resp=[('nt', 1), ('nt', 1)], phys=[('dv', 1)],
coeff=1/2*phi1_1
        v4 (VertexType): resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)],
coeff=1/2*phi2_1
    Class 2 (multiplicity 8): diagrams ['1L-9', '1L-10', '1L-11', '1L-12',
    '1L-13', '1L-14', '1L-15', '1L-16']
        External legs: leaf 0↔dn1, leaf 1↔dn2
        Edges: 2→3 [G_hat[0,2]], 2→4 [G_hat[0,3]], 3→0 [G_hat[0,0]], 3→4
[G_hat[0,3]], 4→1 [G_hat[1,1]]
        v2 (SourceType): resp=[('nt', 1), ('nt', 1)], coeff=1/2*nstar1
        v3 (VertexType): resp=[('nt', 1), ('nt', 1)], phys=[('dv', 1)],
coeff=1/2*phi1_1
        v4 (VertexType): resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)],
coeff=1/2*phi2_2
    Class 3 (multiplicity 8): diagrams ['1L-17', '1L-18', '1L-19', '1L-20',
    '1L-21', '1L-22', '1L-23', '1L-24']
        External legs: leaf 0↔dn1, leaf 1↔dn2
        Edges: 2→3 [G_hat[0,3]], 2→4 [G_hat[0,2]], 3→0 [G_hat[1,0]], 3→4
[G_hat[1,2]], 4→1 [G_hat[0,1]]
        v2 (SourceType): resp=[('nt', 1), ('nt', 1)], coeff=1/2*nstar1
        v3 (VertexType): resp=[('nt', 2), ('nt', 2)], phys=[('dv', 2)],
coeff=1/2*phi1_2
        v4 (VertexType): resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)],
coeff=1/2*phi2_1
    Class 4 (multiplicity 8): diagrams ['1L-25', '1L-26', '1L-27', '1L-28',
    '1L-29', '1L-30', '1L-31', '1L-32']
        External legs: leaf 0↔dn1, leaf 1↔dn2
        Edges: 2→3 [G_hat[0,3]], 2→4 [G_hat[0,3]], 3→0 [G_hat[1,0]], 3→4
[G_hat[1,3]], 4→1 [G_hat[1,1]]

```



```

    v2 (SourceType): resp=[('nt', 1), ('nt', 1)], coeff=1/2*nstar1
    v3 (VertexType): resp=[('nt', 2), ('nt', 2)], phys=[('dv', 2)],
coeff=1/2*phi1_2
    v4 (VertexType): resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)],
coeff=1/2*phi2_2
    Class 5 (multiplicity 8): diagrams ['1L-33', '1L-34', '1L-35', '1L-36',
'1L-37', '1L-38', '1L-39', '1L-40']
        External legs: leaf 0+dn1, leaf 1+dn2
        Edges: 2→3 [G_hat[1,2]], 2→4 [G_hat[1,2]], 3→0 [G_hat[0,0]], 3→4
[G_hat[0,2]], 4→1 [G_hat[0,1]]
        v2 (SourceType): resp=[('nt', 2), ('nt', 2)], coeff=1/2*nstar2
        v3 (VertexType): resp=[('nt', 1), ('nt', 1)], phys=[('dv', 1)],
coeff=1/2*phi1_1
        v4 (VertexType): resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)],
coeff=1/2*phi2_1
        Class 6 (multiplicity 8): diagrams ['1L-41', '1L-42', '1L-43', '1L-44',
'1L-45', '1L-46', '1L-47', '1L-48']
            External legs: leaf 0+dn1, leaf 1+dn2
            Edges: 2→3 [G_hat[1,2]], 2→4 [G_hat[1,3]], 3→0 [G_hat[0,0]], 3→4
[G_hat[0,3]], 4→1 [G_hat[1,1]]
            v2 (SourceType): resp=[('nt', 2), ('nt', 2)], coeff=1/2*nstar2
            v3 (VertexType): resp=[('nt', 1), ('nt', 1)], phys=[('dv', 1)],
coeff=1/2*phi1_1
            v4 (VertexType): resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)],
coeff=1/2*phi2_2
            Class 7 (multiplicity 8): diagrams ['1L-49', '1L-50', '1L-51', '1L-52',
'1L-53', '1L-54', '1L-55', '1L-56']
                External legs: leaf 0+dn1, leaf 1+dn2
                Edges: 2→3 [G_hat[1,3]], 2→4 [G_hat[1,2]], 3→0 [G_hat[1,0]], 3→4
[G_hat[1,2]], 4→1 [G_hat[0,1]]
                v2 (SourceType): resp=[('nt', 2), ('nt', 2)], coeff=1/2*nstar2
                v3 (VertexType): resp=[('nt', 2), ('nt', 2)], phys=[('dv', 2)],
coeff=1/2*phi1_2
                v4 (VertexType): resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)],
coeff=1/2*phi2_1
                Class 8 (multiplicity 8): diagrams ['1L-57', '1L-58', '1L-59', '1L-60',
'1L-61', '1L-62', '1L-63', '1L-64']
                    External legs: leaf 0+dn1, leaf 1+dn2
                    Edges: 2→3 [G_hat[1,3]], 2→4 [G_hat[1,3]], 3→0 [G_hat[1,0]], 3→4
[G_hat[1,3]], 4→1 [G_hat[1,1]]
                    v2 (SourceType): resp=[('nt', 2), ('nt', 2)], coeff=1/2*nstar2
                    v3 (VertexType): resp=[('nt', 2), ('nt', 2)], phys=[('dv', 2)],
coeff=1/2*phi1_2
                    v4 (VertexType): resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)],
coeff=1/2*phi2_2
                    Class 9 (multiplicity 8): diagrams ['1L-65', '1L-66', '1L-67', '1L-68',
'1L-69', '1L-70', '1L-71', '1L-72']
                        External legs: leaf 0+dn2, leaf 1+dn1

```

Edges: 2→3 [G_hat[0,2]], 2→4 [G_hat[0,2]], 3→0 [G_hat[0,1]], 3→4 [G_hat[0,2]], 4→1 [G_hat[0,0]]
v2 (SourceType): resp=[('nt', 1), ('nt', 1)], coeff=1/2*nstar1
v3 (VertexType): resp=[('nt', 1), ('nt', 1)], phys=[('dv', 1)],
coeff=1/2*phi1_1
v4 (VertexType): resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)],
coeff=1/2*phi2_1
Class 10 (multiplicity 8): diagrams ['1L-73', '1L-74', '1L-75', '1L-76', '1L-77', '1L-78', '1L-79', '1L-80']
External legs: leaf 0→dn2, leaf 1→dn1
Edges: 2→3 [G_hat[0,2]], 2→4 [G_hat[0,3]], 3→0 [G_hat[0,1]], 3→4 [G_hat[0,3]], 4→1 [G_hat[1,0]]
v2 (SourceType): resp=[('nt', 1), ('nt', 1)], coeff=1/2*nstar1
v3 (VertexType): resp=[('nt', 1), ('nt', 1)], phys=[('dv', 1)],
coeff=1/2*phi1_1
v4 (VertexType): resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)],
coeff=1/2*phi2_2
Class 11 (multiplicity 8): diagrams ['1L-81', '1L-82', '1L-83', '1L-84', '1L-85', '1L-86', '1L-87', '1L-88']
External legs: leaf 0→dn2, leaf 1→dn1
Edges: 2→3 [G_hat[0,3]], 2→4 [G_hat[0,2]], 3→0 [G_hat[1,1]], 3→4 [G_hat[1,2]], 4→1 [G_hat[0,0]]
v2 (SourceType): resp=[('nt', 1), ('nt', 1)], coeff=1/2*nstar1
v3 (VertexType): resp=[('nt', 2), ('nt', 2)], phys=[('dv', 2)],
coeff=1/2*phi1_2
v4 (VertexType): resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)],
coeff=1/2*phi2_1
Class 12 (multiplicity 8): diagrams ['1L-89', '1L-90', '1L-91', '1L-92', '1L-93', '1L-94', '1L-95', '1L-96']
External legs: leaf 0→dn2, leaf 1→dn1
Edges: 2→3 [G_hat[0,3]], 2→4 [G_hat[0,3]], 3→0 [G_hat[1,1]], 3→4 [G_hat[1,3]], 4→1 [G_hat[1,0]]
v2 (SourceType): resp=[('nt', 1), ('nt', 1)], coeff=1/2*nstar1
v3 (VertexType): resp=[('nt', 2), ('nt', 2)], phys=[('dv', 2)],
coeff=1/2*phi1_2
v4 (VertexType): resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)],
coeff=1/2*phi2_2
Class 13 (multiplicity 8): diagrams ['1L-97', '1L-98', '1L-99', '1L-100', '1L-101', '1L-102', '1L-103', '1L-104']
External legs: leaf 0→dn2, leaf 1→dn1
Edges: 2→3 [G_hat[1,2]], 2→4 [G_hat[1,2]], 3→0 [G_hat[0,1]], 3→4 [G_hat[0,2]], 4→1 [G_hat[0,0]]
v2 (SourceType): resp=[('nt', 2), ('nt', 2)], coeff=1/2*nstar2
v3 (VertexType): resp=[('nt', 1), ('nt', 1)], phys=[('dv', 1)],
coeff=1/2*phi1_1
v4 (VertexType): resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)],
coeff=1/2*phi2_1
Class 14 (multiplicity 8): diagrams ['1L-105', '1L-106', '1L-107', '1L-108',

```

'1L-109', '1L-110', '1L-111', '1L-112']
    External legs: leaf 0+dn2, leaf 1+dn1
    Edges: 2→3 [G_hat[1,2]], 2→4 [G_hat[1,3]], 3→0 [G_hat[0,1]], 3→4
[G_hat[0,3]], 4→1 [G_hat[1,0]]
    v2 (SourceType): resp=[('nt', 2), ('nt', 2)], coeff=1/2*nstar2
    v3 (VertexType): resp=[('nt', 1), ('nt', 1)], phys=[('dv', 1)],
coeff=1/2*phi1_1
    v4 (VertexType): resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)],
coeff=1/2*phi2_2
    Class 15 (multiplicity 8): diagrams ['1L-113', '1L-114', '1L-115', '1L-116',
'1L-117', '1L-118', '1L-119', '1L-120']
    External legs: leaf 0+dn2, leaf 1+dn1
    Edges: 2→3 [G_hat[1,3]], 2→4 [G_hat[1,2]], 3→0 [G_hat[1,1]], 3→4
[G_hat[1,2]], 4→1 [G_hat[0,0]]
    v2 (SourceType): resp=[('nt', 2), ('nt', 2)], coeff=1/2*nstar2
    v3 (VertexType): resp=[('nt', 2), ('nt', 2)], phys=[('dv', 2)],
coeff=1/2*phi1_2
    v4 (VertexType): resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)],
coeff=1/2*phi2_1
    Class 16 (multiplicity 8): diagrams ['1L-121', '1L-122', '1L-123', '1L-124',
'1L-125', '1L-126', '1L-127', '1L-128']
    External legs: leaf 0+dn2, leaf 1+dn1
    Edges: 2→3 [G_hat[1,3]], 2→4 [G_hat[1,3]], 3→0 [G_hat[1,1]], 3→4
[G_hat[1,3]], 4→1 [G_hat[1,0]]
    v2 (SourceType): resp=[('nt', 2), ('nt', 2)], coeff=1/2*nstar2
    v3 (VertexType): resp=[('nt', 2), ('nt', 2)], phys=[('dv', 2)],
coeff=1/2*phi1_2
    v4 (VertexType): resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)],
coeff=1/2*phi2_2
    Class 17 (multiplicity 16): diagrams ['1L-129', '1L-130', '1L-131', '1L-132',
'1L-133', '1L-134', '1L-135', '1L-136', '1L-137', '1L-138', '1L-139', '1L-140',
'1L-141', '1L-142', '1L-143', '1L-144']
    External legs: leaf 0+dn1, leaf 1+dn2
    Edges: 2→4 [G_hat[0,2]], 3→2 [G_hat[0,2]], 3→4 [G_hat[0,2]], 4→0
[G_hat[0,0]], 5→1 [G_hat[0,1]], 5→2 [G_hat[0,2]]
    v2 (VertexType): resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)],
coeff=1/2*phi2_1
    v3 (SourceType): resp=[('nt', 1), ('nt', 1)], coeff=1/2*nstar1
    v4 (VertexType): resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)],
coeff=1/2*phi2_1
    v5 (SourceType): resp=[('nt', 1), ('nt', 1)], coeff=1/2*nstar1
    Class 18 (multiplicity 16): diagrams ['1L-145', '1L-146', '1L-147', '1L-148',
'1L-149', '1L-150', '1L-151', '1L-152', '1L-153', '1L-154', '1L-155', '1L-156',
'1L-157', '1L-158', '1L-159', '1L-160']
    External legs: leaf 0+dn1, leaf 1+dn2
    Edges: 2→4 [G_hat[0,3]], 3→2 [G_hat[0,2]], 3→4 [G_hat[0,3]], 4→0
[G_hat[1,0]], 5→1 [G_hat[0,1]], 5→2 [G_hat[0,2]]
    v2 (VertexType): resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)],

```

```

coeff=1/2*phi2_1
  v3 (SourceType): resp=[('nt', 1), ('nt', 1)], coeff=1/2*nstar1
  v4 (VertexType): resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)],
coeff=1/2*phi2_2
  v5 (SourceType): resp=[('nt', 1), ('nt', 1)], coeff=1/2*nstar1
  Class 19 (multiplicity 16): diagrams ['1L-161', '1L-162', '1L-163', '1L-164',
'1L-165', '1L-166', '1L-167', '1L-168', '1L-169', '1L-170', '1L-171', '1L-172',
'1L-173', '1L-174', '1L-175', '1L-176']
  External legs: leaf 0+dn1, leaf 1+dn2
  Edges: 2+4 [G_hat[1,2]], 3+2 [G_hat[0,3]], 3+4 [G_hat[0,2]], 4+0
[G_hat[0,0]], 5+1 [G_hat[0,1]], 5+2 [G_hat[0,3]]
  v2 (VertexType): resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)],
coeff=1/2*phi2_2
  v3 (SourceType): resp=[('nt', 1), ('nt', 1)], coeff=1/2*nstar1
  v4 (VertexType): resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)],
coeff=1/2*phi2_1
  v5 (SourceType): resp=[('nt', 1), ('nt', 1)], coeff=1/2*nstar1
  Class 20 (multiplicity 16): diagrams ['1L-177', '1L-178', '1L-179', '1L-180',
'1L-181', '1L-182', '1L-183', '1L-184', '1L-185', '1L-186', '1L-187', '1L-188',
'1L-189', '1L-190', '1L-191', '1L-192']
  External legs: leaf 0+dn1, leaf 1+dn2
  Edges: 2+4 [G_hat[1,3]], 3+2 [G_hat[0,3]], 3+4 [G_hat[0,3]], 4+0
[G_hat[1,0]], 5+1 [G_hat[0,1]], 5+2 [G_hat[0,3]]
  v2 (VertexType): resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)],
coeff=1/2*phi2_2
  v3 (SourceType): resp=[('nt', 1), ('nt', 1)], coeff=1/2*nstar1
  v4 (VertexType): resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)],
coeff=1/2*phi2_2
  v5 (SourceType): resp=[('nt', 1), ('nt', 1)], coeff=1/2*nstar1
  Class 21 (multiplicity 16): diagrams ['1L-193', '1L-194', '1L-195', '1L-196',
'1L-197', '1L-198', '1L-199', '1L-200', '1L-201', '1L-202', '1L-203', '1L-204',
'1L-205', '1L-206', '1L-207', '1L-208']
  External legs: leaf 0+dn1, leaf 1+dn2
  Edges: 2+4 [G_hat[0,2]], 3+2 [G_hat[0,2]], 3+4 [G_hat[0,2]], 4+0
[G_hat[0,0]], 5+1 [G_hat[1,1]], 5+2 [G_hat[1,2]]
  v2 (VertexType): resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)],
coeff=1/2*phi2_1
  v3 (SourceType): resp=[('nt', 1), ('nt', 1)], coeff=1/2*nstar1
  v4 (VertexType): resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)],
coeff=1/2*phi2_1
  v5 (SourceType): resp=[('nt', 2), ('nt', 2)], coeff=1/2*nstar2
  Class 22 (multiplicity 16): diagrams ['1L-209', '1L-210', '1L-211', '1L-212',
'1L-213', '1L-214', '1L-215', '1L-216', '1L-217', '1L-218', '1L-219', '1L-220',
'1L-221', '1L-222', '1L-223', '1L-224']
  External legs: leaf 0+dn1, leaf 1+dn2
  Edges: 2+4 [G_hat[0,3]], 3+2 [G_hat[0,2]], 3+4 [G_hat[0,3]], 4+0
[G_hat[1,0]], 5+1 [G_hat[1,1]], 5+2 [G_hat[1,2]]
  v2 (VertexType): resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)],

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coeff=1/2*phi2_1
    v3 (SourceType): resp=[('nt', 1), ('nt', 1)], coeff=1/2*nstar1
    v4 (VertexType): resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)],
coeff=1/2*phi2_2
    v5 (SourceType): resp=[('nt', 2), ('nt', 2)], coeff=1/2*nstar2
    Class 23 (multiplicity 16): diagrams ['1L-225', '1L-226', '1L-227', '1L-228',
'1L-229', '1L-230', '1L-231', '1L-232', '1L-233', '1L-234', '1L-235', '1L-236',
'1L-237', '1L-238', '1L-239', '1L-240']
    External legs: leaf 0+dn1, leaf 1+dn2
    Edges: 2+4 [G_hat[1,2]], 3+2 [G_hat[0,3]], 3+4 [G_hat[0,2]], 4+0
[G_hat[0,0]], 5+1 [G_hat[1,1]], 5+2 [G_hat[1,3]]
    v2 (VertexType): resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)],
coeff=1/2*phi2_2
    v3 (SourceType): resp=[('nt', 1), ('nt', 1)], coeff=1/2*nstar1
    v4 (VertexType): resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)],
coeff=1/2*phi2_1
    v5 (SourceType): resp=[('nt', 2), ('nt', 2)], coeff=1/2*nstar2
    Class 24 (multiplicity 16): diagrams ['1L-241', '1L-242', '1L-243', '1L-244',
'1L-245', '1L-246', '1L-247', '1L-248', '1L-249', '1L-250', '1L-251', '1L-252',
'1L-253', '1L-254', '1L-255', '1L-256']
    External legs: leaf 0+dn1, leaf 1+dn2
    Edges: 2+4 [G_hat[1,3]], 3+2 [G_hat[0,3]], 3+4 [G_hat[0,3]], 4+0
[G_hat[1,0]], 5+1 [G_hat[1,1]], 5+2 [G_hat[1,3]]
    v2 (VertexType): resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)],
coeff=1/2*phi2_2
    v3 (SourceType): resp=[('nt', 1), ('nt', 1)], coeff=1/2*nstar1
    v4 (VertexType): resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)],
coeff=1/2*phi2_2
    v5 (SourceType): resp=[('nt', 2), ('nt', 2)], coeff=1/2*nstar2
    Class 25 (multiplicity 16): diagrams ['1L-257', '1L-258', '1L-259', '1L-260',
'1L-261', '1L-262', '1L-263', '1L-264', '1L-265', '1L-266', '1L-267', '1L-268',
'1L-269', '1L-270', '1L-271', '1L-272']
    External legs: leaf 0+dn1, leaf 1+dn2
    Edges: 2+4 [G_hat[0,2]], 3+2 [G_hat[1,2]], 3+4 [G_hat[1,2]], 4+0
[G_hat[0,0]], 5+1 [G_hat[0,1]], 5+2 [G_hat[0,2]]
    v2 (VertexType): resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)],
coeff=1/2*phi2_1
    v3 (SourceType): resp=[('nt', 2), ('nt', 2)], coeff=1/2*nstar2
    v4 (VertexType): resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)],
coeff=1/2*phi2_1
    v5 (SourceType): resp=[('nt', 1), ('nt', 1)], coeff=1/2*nstar1
    Class 26 (multiplicity 16): diagrams ['1L-273', '1L-274', '1L-275', '1L-276',
'1L-277', '1L-278', '1L-279', '1L-280', '1L-281', '1L-282', '1L-283', '1L-284',
'1L-285', '1L-286', '1L-287', '1L-288']
    External legs: leaf 0+dn1, leaf 1+dn2
    Edges: 2+4 [G_hat[0,3]], 3+2 [G_hat[1,2]], 3+4 [G_hat[1,3]], 4+0
[G_hat[1,0]], 5+1 [G_hat[0,1]], 5+2 [G_hat[0,2]]
    v2 (VertexType): resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)],

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coeff=1/2*phi2_1
    v3 (SourceType): resp=[('nt', 2), ('nt', 2)], coeff=1/2*nstar2
    v4 (VertexType): resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)],
coeff=1/2*phi2_2
    v5 (SourceType): resp=[('nt', 1), ('nt', 1)], coeff=1/2*nstar1
    Class 27 (multiplicity 16): diagrams ['1L-289', '1L-290', '1L-291', '1L-292',
'1L-293', '1L-294', '1L-295', '1L-296', '1L-297', '1L-298', '1L-299', '1L-300',
'1L-301', '1L-302', '1L-303', '1L-304']
    External legs: leaf 0+dn1, leaf 1+dn2
    Edges: 2+4 [G_hat[1,2]], 3+2 [G_hat[1,3]], 3+4 [G_hat[1,2]], 4+0
[G_hat[0,0]], 5+1 [G_hat[0,1]], 5+2 [G_hat[0,3]]
    v2 (VertexType): resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)],
coeff=1/2*phi2_2
    v3 (SourceType): resp=[('nt', 2), ('nt', 2)], coeff=1/2*nstar2
    v4 (VertexType): resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)],
coeff=1/2*phi2_1
    v5 (SourceType): resp=[('nt', 1), ('nt', 1)], coeff=1/2*nstar1
    Class 28 (multiplicity 16): diagrams ['1L-305', '1L-306', '1L-307', '1L-308',
'1L-309', '1L-310', '1L-311', '1L-312', '1L-313', '1L-314', '1L-315', '1L-316',
'1L-317', '1L-318', '1L-319', '1L-320']
    External legs: leaf 0+dn1, leaf 1+dn2
    Edges: 2+4 [G_hat[1,3]], 3+2 [G_hat[1,3]], 3+4 [G_hat[1,3]], 4+0
[G_hat[1,0]], 5+1 [G_hat[0,1]], 5+2 [G_hat[0,3]]
    v2 (VertexType): resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)],
coeff=1/2*phi2_2
    v3 (SourceType): resp=[('nt', 2), ('nt', 2)], coeff=1/2*nstar2
    v4 (VertexType): resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)],
coeff=1/2*phi2_2
    v5 (SourceType): resp=[('nt', 1), ('nt', 1)], coeff=1/2*nstar1
    Class 29 (multiplicity 16): diagrams ['1L-321', '1L-322', '1L-323', '1L-324',
'1L-325', '1L-326', '1L-327', '1L-328', '1L-329', '1L-330', '1L-331', '1L-332',
'1L-333', '1L-334', '1L-335', '1L-336']
    External legs: leaf 0+dn1, leaf 1+dn2
    Edges: 2+4 [G_hat[0,2]], 3+2 [G_hat[1,2]], 3+4 [G_hat[1,2]], 4+0
[G_hat[0,0]], 5+1 [G_hat[1,1]], 5+2 [G_hat[1,2]]
    v2 (VertexType): resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)],
coeff=1/2*phi2_1
    v3 (SourceType): resp=[('nt', 2), ('nt', 2)], coeff=1/2*nstar2
    v4 (VertexType): resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)],
coeff=1/2*phi2_1
    v5 (SourceType): resp=[('nt', 2), ('nt', 2)], coeff=1/2*nstar2
    Class 30 (multiplicity 16): diagrams ['1L-337', '1L-338', '1L-339', '1L-340',
'1L-341', '1L-342', '1L-343', '1L-344', '1L-345', '1L-346', '1L-347', '1L-348',
'1L-349', '1L-350', '1L-351', '1L-352']
    External legs: leaf 0+dn1, leaf 1+dn2
    Edges: 2+4 [G_hat[0,3]], 3+2 [G_hat[1,2]], 3+4 [G_hat[1,3]], 4+0
[G_hat[1,0]], 5+1 [G_hat[1,1]], 5+2 [G_hat[1,2]]
    v2 (VertexType): resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)],

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coeff=1/2*phi2_1
    v3 (SourceType): resp=[('nt', 2), ('nt', 2)], coeff=1/2*nstar2
    v4 (VertexType): resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)],
coeff=1/2*phi2_2
    v5 (SourceType): resp=[('nt', 2), ('nt', 2)], coeff=1/2*nstar2
    Class 31 (multiplicity 16): diagrams ['1L-353', '1L-354', '1L-355', '1L-356',
'1L-357', '1L-358', '1L-359', '1L-360', '1L-361', '1L-362', '1L-363', '1L-364',
'1L-365', '1L-366', '1L-367', '1L-368']
    External legs: leaf 0+dn1, leaf 1+dn2
    Edges: 2+4 [G_hat[1,2]], 3+2 [G_hat[1,3]], 3+4 [G_hat[1,2]], 4+0
[G_hat[0,0]], 5+1 [G_hat[1,1]], 5+2 [G_hat[1,3]]
    v2 (VertexType): resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)],
coeff=1/2*phi2_2
    v3 (SourceType): resp=[('nt', 2), ('nt', 2)], coeff=1/2*nstar2
    v4 (VertexType): resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)],
coeff=1/2*phi2_1
    v5 (SourceType): resp=[('nt', 2), ('nt', 2)], coeff=1/2*nstar2
    Class 32 (multiplicity 16): diagrams ['1L-369', '1L-370', '1L-371', '1L-372',
'1L-373', '1L-374', '1L-375', '1L-376', '1L-377', '1L-378', '1L-379', '1L-380',
'1L-381', '1L-382', '1L-383', '1L-384']
    External legs: leaf 0+dn1, leaf 1+dn2
    Edges: 2+4 [G_hat[1,3]], 3+2 [G_hat[1,3]], 3+4 [G_hat[1,3]], 4+0
[G_hat[1,0]], 5+1 [G_hat[1,1]], 5+2 [G_hat[1,3]]
    v2 (VertexType): resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)],
coeff=1/2*phi2_2
    v3 (SourceType): resp=[('nt', 2), ('nt', 2)], coeff=1/2*nstar2
    v4 (VertexType): resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)],
coeff=1/2*phi2_2
    v5 (SourceType): resp=[('nt', 2), ('nt', 2)], coeff=1/2*nstar2
    Class 33 (multiplicity 16): diagrams ['1L-385', '1L-386', '1L-387', '1L-388',
'1L-389', '1L-390', '1L-391', '1L-392', '1L-393', '1L-394', '1L-395', '1L-396',
'1L-397', '1L-398', '1L-399', '1L-400']
    External legs: leaf 0+dn2, leaf 1+dn1
    Edges: 2+4 [G_hat[0,2]], 3+2 [G_hat[0,2]], 3+4 [G_hat[0,2]], 4+0
[G_hat[0,1]], 5+1 [G_hat[0,0]], 5+2 [G_hat[0,2]]
    v2 (VertexType): resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)],
coeff=1/2*phi2_1
    v3 (SourceType): resp=[('nt', 1), ('nt', 1)], coeff=1/2*nstar1
    v4 (VertexType): resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)],
coeff=1/2*phi2_1
    v5 (SourceType): resp=[('nt', 1), ('nt', 1)], coeff=1/2*nstar1
    Class 34 (multiplicity 16): diagrams ['1L-401', '1L-402', '1L-403', '1L-404',
'1L-405', '1L-406', '1L-407', '1L-408', '1L-409', '1L-410', '1L-411', '1L-412',
'1L-413', '1L-414', '1L-415', '1L-416']
    External legs: leaf 0+dn2, leaf 1+dn1
    Edges: 2+4 [G_hat[0,3]], 3+2 [G_hat[0,2]], 3+4 [G_hat[0,3]], 4+0
[G_hat[1,1]], 5+1 [G_hat[0,0]], 5+2 [G_hat[0,2]]
    v2 (VertexType): resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)],

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coeff=1/2*phi2_1
  v3 (SourceType): resp=[('nt', 1), ('nt', 1)], coeff=1/2*nstar1
  v4 (VertexType): resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)],
coeff=1/2*phi2_2
  v5 (SourceType): resp=[('nt', 1), ('nt', 1)], coeff=1/2*nstar1
  Class 35 (multiplicity 16): diagrams ['1L-417', '1L-418', '1L-419', '1L-420',
'1L-421', '1L-422', '1L-423', '1L-424', '1L-425', '1L-426', '1L-427', '1L-428',
'1L-429', '1L-430', '1L-431', '1L-432']
  External legs: leaf 0+dn2, leaf 1+dn1
  Edges: 2+4 [G_hat[1,2]], 3+2 [G_hat[0,3]], 3+4 [G_hat[0,2]], 4+0
[G_hat[0,1]], 5+1 [G_hat[0,0]], 5+2 [G_hat[0,3]]
  v2 (VertexType): resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)],
coeff=1/2*phi2_2
  v3 (SourceType): resp=[('nt', 1), ('nt', 1)], coeff=1/2*nstar1
  v4 (VertexType): resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)],
coeff=1/2*phi2_1
  v5 (SourceType): resp=[('nt', 1), ('nt', 1)], coeff=1/2*nstar1
  Class 36 (multiplicity 16): diagrams ['1L-433', '1L-434', '1L-435', '1L-436',
'1L-437', '1L-438', '1L-439', '1L-440', '1L-441', '1L-442', '1L-443', '1L-444',
'1L-445', '1L-446', '1L-447', '1L-448']
  External legs: leaf 0+dn2, leaf 1+dn1
  Edges: 2+4 [G_hat[1,3]], 3+2 [G_hat[0,3]], 3+4 [G_hat[0,3]], 4+0
[G_hat[1,1]], 5+1 [G_hat[0,0]], 5+2 [G_hat[0,3]]
  v2 (VertexType): resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)],
coeff=1/2*phi2_2
  v3 (SourceType): resp=[('nt', 1), ('nt', 1)], coeff=1/2*nstar1
  v4 (VertexType): resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)],
coeff=1/2*phi2_2
  v5 (SourceType): resp=[('nt', 1), ('nt', 1)], coeff=1/2*nstar1
  Class 37 (multiplicity 16): diagrams ['1L-449', '1L-450', '1L-451', '1L-452',
'1L-453', '1L-454', '1L-455', '1L-456', '1L-457', '1L-458', '1L-459', '1L-460',
'1L-461', '1L-462', '1L-463', '1L-464']
  External legs: leaf 0+dn2, leaf 1+dn1
  Edges: 2+4 [G_hat[0,2]], 3+2 [G_hat[0,2]], 3+4 [G_hat[0,2]], 4+0
[G_hat[0,1]], 5+1 [G_hat[1,0]], 5+2 [G_hat[1,2]]
  v2 (VertexType): resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)],
coeff=1/2*phi2_1
  v3 (SourceType): resp=[('nt', 1), ('nt', 1)], coeff=1/2*nstar1
  v4 (VertexType): resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)],
coeff=1/2*phi2_1
  v5 (SourceType): resp=[('nt', 2), ('nt', 2)], coeff=1/2*nstar2
  Class 38 (multiplicity 16): diagrams ['1L-465', '1L-466', '1L-467', '1L-468',
'1L-469', '1L-470', '1L-471', '1L-472', '1L-473', '1L-474', '1L-475', '1L-476',
'1L-477', '1L-478', '1L-479', '1L-480']
  External legs: leaf 0+dn2, leaf 1+dn1
  Edges: 2+4 [G_hat[0,3]], 3+2 [G_hat[0,2]], 3+4 [G_hat[0,3]], 4+0
[G_hat[1,1]], 5+1 [G_hat[1,0]], 5+2 [G_hat[1,2]]
  v2 (VertexType): resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)],

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coeff=1/2*phi2_1
    v3 (SourceType): resp=[('nt', 1), ('nt', 1)], coeff=1/2*nstar1
    v4 (VertexType): resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)],
coeff=1/2*phi2_2
    v5 (SourceType): resp=[('nt', 2), ('nt', 2)], coeff=1/2*nstar2
    Class 39 (multiplicity 16): diagrams ['1L-481', '1L-482', '1L-483', '1L-484',
'1L-485', '1L-486', '1L-487', '1L-488', '1L-489', '1L-490', '1L-491', '1L-492',
'1L-493', '1L-494', '1L-495', '1L-496']
    External legs: leaf 0+dn2, leaf 1+dn1
    Edges: 2+4 [G_hat[1,2]], 3+2 [G_hat[0,3]], 3+4 [G_hat[0,2]], 4+0
[G_hat[0,1]], 5+1 [G_hat[1,0]], 5+2 [G_hat[1,3]]
    v2 (VertexType): resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)],
coeff=1/2*phi2_2
    v3 (SourceType): resp=[('nt', 1), ('nt', 1)], coeff=1/2*nstar1
    v4 (VertexType): resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)],
coeff=1/2*phi2_1
    v5 (SourceType): resp=[('nt', 2), ('nt', 2)], coeff=1/2*nstar2
    Class 40 (multiplicity 16): diagrams ['1L-497', '1L-498', '1L-499', '1L-500',
'1L-501', '1L-502', '1L-503', '1L-504', '1L-505', '1L-506', '1L-507', '1L-508',
'1L-509', '1L-510', '1L-511', '1L-512']
    External legs: leaf 0+dn2, leaf 1+dn1
    Edges: 2+4 [G_hat[1,3]], 3+2 [G_hat[0,3]], 3+4 [G_hat[0,3]], 4+0
[G_hat[1,1]], 5+1 [G_hat[1,0]], 5+2 [G_hat[1,3]]
    v2 (VertexType): resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)],
coeff=1/2*phi2_2
    v3 (SourceType): resp=[('nt', 1), ('nt', 1)], coeff=1/2*nstar1
    v4 (VertexType): resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)],
coeff=1/2*phi2_2
    v5 (SourceType): resp=[('nt', 2), ('nt', 2)], coeff=1/2*nstar2
    Class 41 (multiplicity 16): diagrams ['1L-513', '1L-514', '1L-515', '1L-516',
'1L-517', '1L-518', '1L-519', '1L-520', '1L-521', '1L-522', '1L-523', '1L-524',
'1L-525', '1L-526', '1L-527', '1L-528']
    External legs: leaf 0+dn2, leaf 1+dn1
    Edges: 2+4 [G_hat[0,2]], 3+2 [G_hat[1,2]], 3+4 [G_hat[1,2]], 4+0
[G_hat[0,1]], 5+1 [G_hat[0,0]], 5+2 [G_hat[0,2]]
    v2 (VertexType): resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)],
coeff=1/2*phi2_1
    v3 (SourceType): resp=[('nt', 2), ('nt', 2)], coeff=1/2*nstar2
    v4 (VertexType): resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)],
coeff=1/2*phi2_1
    v5 (SourceType): resp=[('nt', 1), ('nt', 1)], coeff=1/2*nstar1
    Class 42 (multiplicity 16): diagrams ['1L-529', '1L-530', '1L-531', '1L-532',
'1L-533', '1L-534', '1L-535', '1L-536', '1L-537', '1L-538', '1L-539', '1L-540',
'1L-541', '1L-542', '1L-543', '1L-544']
    External legs: leaf 0+dn2, leaf 1+dn1
    Edges: 2+4 [G_hat[0,3]], 3+2 [G_hat[1,2]], 3+4 [G_hat[1,3]], 4+0
[G_hat[1,1]], 5+1 [G_hat[0,0]], 5+2 [G_hat[0,2]]
    v2 (VertexType): resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)],

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coeff=1/2*phi2_1
    v3 (SourceType): resp=[('nt', 2), ('nt', 2)], coeff=1/2*nstar2
    v4 (VertexType): resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)],
coeff=1/2*phi2_2
    v5 (SourceType): resp=[('nt', 1), ('nt', 1)], coeff=1/2*nstar1
    Class 43 (multiplicity 16): diagrams ['1L-545', '1L-546', '1L-547', '1L-548',
'1L-549', '1L-550', '1L-551', '1L-552', '1L-553', '1L-554', '1L-555', '1L-556',
'1L-557', '1L-558', '1L-559', '1L-560']
    External legs: leaf 0+dn2, leaf 1+dn1
    Edges: 2+4 [G_hat[1,2]], 3+2 [G_hat[1,3]], 3+4 [G_hat[1,2]], 4+0
[G_hat[0,1]], 5+1 [G_hat[0,0]], 5+2 [G_hat[0,3]]
    v2 (VertexType): resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)],
coeff=1/2*phi2_2
    v3 (SourceType): resp=[('nt', 2), ('nt', 2)], coeff=1/2*nstar2
    v4 (VertexType): resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)],
coeff=1/2*phi2_1
    v5 (SourceType): resp=[('nt', 1), ('nt', 1)], coeff=1/2*nstar1
    Class 44 (multiplicity 16): diagrams ['1L-561', '1L-562', '1L-563', '1L-564',
'1L-565', '1L-566', '1L-567', '1L-568', '1L-569', '1L-570', '1L-571', '1L-572',
'1L-573', '1L-574', '1L-575', '1L-576']
    External legs: leaf 0+dn2, leaf 1+dn1
    Edges: 2+4 [G_hat[1,3]], 3+2 [G_hat[1,3]], 3+4 [G_hat[1,3]], 4+0
[G_hat[1,1]], 5+1 [G_hat[0,0]], 5+2 [G_hat[0,3]]
    v2 (VertexType): resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)],
coeff=1/2*phi2_2
    v3 (SourceType): resp=[('nt', 2), ('nt', 2)], coeff=1/2*nstar2
    v4 (VertexType): resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)],
coeff=1/2*phi2_2
    v5 (SourceType): resp=[('nt', 1), ('nt', 1)], coeff=1/2*nstar1
    Class 45 (multiplicity 16): diagrams ['1L-577', '1L-578', '1L-579', '1L-580',
'1L-581', '1L-582', '1L-583', '1L-584', '1L-585', '1L-586', '1L-587', '1L-588',
'1L-589', '1L-590', '1L-591', '1L-592']
    External legs: leaf 0+dn2, leaf 1+dn1
    Edges: 2+4 [G_hat[0,2]], 3+2 [G_hat[1,2]], 3+4 [G_hat[1,2]], 4+0
[G_hat[0,1]], 5+1 [G_hat[1,0]], 5+2 [G_hat[1,2]]
    v2 (VertexType): resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)],
coeff=1/2*phi2_1
    v3 (SourceType): resp=[('nt', 2), ('nt', 2)], coeff=1/2*nstar2
    v4 (VertexType): resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)],
coeff=1/2*phi2_1
    v5 (SourceType): resp=[('nt', 2), ('nt', 2)], coeff=1/2*nstar2
    Class 46 (multiplicity 16): diagrams ['1L-593', '1L-594', '1L-595', '1L-596',
'1L-597', '1L-598', '1L-599', '1L-600', '1L-601', '1L-602', '1L-603', '1L-604',
'1L-605', '1L-606', '1L-607', '1L-608']
    External legs: leaf 0+dn2, leaf 1+dn1
    Edges: 2+4 [G_hat[0,3]], 3+2 [G_hat[1,2]], 3+4 [G_hat[1,3]], 4+0
[G_hat[1,1]], 5+1 [G_hat[1,0]], 5+2 [G_hat[1,2]]
    v2 (VertexType): resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)],

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coeff=1/2*phi2_1
    v3 (SourceType): resp=[('nt', 2), ('nt', 2)], coeff=1/2*nstar2
    v4 (VertexType): resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)],
coeff=1/2*phi2_2
    v5 (SourceType): resp=[('nt', 2), ('nt', 2)], coeff=1/2*nstar2
    Class 47 (multiplicity 16): diagrams ['1L-609', '1L-610', '1L-611', '1L-612',
'1L-613', '1L-614', '1L-615', '1L-616', '1L-617', '1L-618', '1L-619', '1L-620',
'1L-621', '1L-622', '1L-623', '1L-624']
    External legs: leaf 0+dn2, leaf 1+dn1
    Edges: 2+4 [G_hat[1,2]], 3+2 [G_hat[1,3]], 3+4 [G_hat[1,2]], 4+0
[G_hat[0,1]], 5+1 [G_hat[1,0]], 5+2 [G_hat[1,3]]
    v2 (VertexType): resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)],
coeff=1/2*phi2_2
    v3 (SourceType): resp=[('nt', 2), ('nt', 2)], coeff=1/2*nstar2
    v4 (VertexType): resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)],
coeff=1/2*phi2_1
    v5 (SourceType): resp=[('nt', 2), ('nt', 2)], coeff=1/2*nstar2
    Class 48 (multiplicity 16): diagrams ['1L-625', '1L-626', '1L-627', '1L-628',
'1L-629', '1L-630', '1L-631', '1L-632', '1L-633', '1L-634', '1L-635', '1L-636',
'1L-637', '1L-638', '1L-639', '1L-640']
    External legs: leaf 0+dn2, leaf 1+dn1
    Edges: 2+4 [G_hat[1,3]], 3+2 [G_hat[1,3]], 3+4 [G_hat[1,3]], 4+0
[G_hat[1,1]], 5+1 [G_hat[1,0]], 5+2 [G_hat[1,3]]
    v2 (VertexType): resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)],
coeff=1/2*phi2_2
    v3 (SourceType): resp=[('nt', 2), ('nt', 2)], coeff=1/2*nstar2
    v4 (VertexType): resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)],
coeff=1/2*phi2_2
    v5 (SourceType): resp=[('nt', 2), ('nt', 2)], coeff=1/2*nstar2
    Class 49 (multiplicity 16): diagrams ['1L-641', '1L-642', '1L-643', '1L-644',
'1L-645', '1L-646', '1L-647', '1L-648', '1L-649', '1L-650', '1L-651', '1L-652',
'1L-653', '1L-654', '1L-655', '1L-656']
    External legs: leaf 0+dn1, leaf 1+dn2
    Edges: 2+3 [G_hat[0,2]], 2+5 [G_hat[0,2]], 3+0 [G_hat[0,0]], 4+3
[G_hat[0,2]], 4+5 [G_hat[0,2]], 5+1 [G_hat[0,1]]
    v2 (SourceType): resp=[('nt', 1), ('nt', 1)], coeff=1/2*nstar1
    v3 (VertexType): resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)],
coeff=1/2*phi2_1
    v4 (SourceType): resp=[('nt', 1), ('nt', 1)], coeff=1/2*nstar1
    v5 (VertexType): resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)],
coeff=1/2*phi2_1
    Class 50 (multiplicity 16): diagrams ['1L-657', '1L-658', '1L-659', '1L-660',
'1L-661', '1L-662', '1L-663', '1L-664', '1L-665', '1L-666', '1L-667', '1L-668',
'1L-669', '1L-670', '1L-671', '1L-672']
    External legs: leaf 0+dn1, leaf 1+dn2
    Edges: 2+3 [G_hat[0,2]], 2+5 [G_hat[0,3]], 3+0 [G_hat[0,0]], 4+3
[G_hat[0,2]], 4+5 [G_hat[0,3]], 5+1 [G_hat[1,1]]
    v2 (SourceType): resp=[('nt', 1), ('nt', 1)], coeff=1/2*nstar1

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    v3 (VertexType): resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)],
coeff=1/2*phi2_1
    v4 (SourceType): resp=[('nt', 1), ('nt', 1)], coeff=1/2*nstar1
    v5 (VertexType): resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)],
coeff=1/2*phi2_2
    Class 51 (multiplicity 16): diagrams ['1L-673', '1L-674', '1L-675', '1L-676',
'1L-677', '1L-678', '1L-679', '1L-680', '1L-681', '1L-682', '1L-683', '1L-684',
'1L-685', '1L-686', '1L-687', '1L-688']
        External legs: leaf 0+dn1, leaf 1+dn2
        Edges: 2+3 [G_hat[0,3]], 2+5 [G_hat[0,2]], 3+0 [G_hat[1,0]], 4+3
[G_hat[0,3]], 4+5 [G_hat[0,2]], 5+1 [G_hat[0,1]]
        v2 (SourceType): resp=[('nt', 1), ('nt', 1)], coeff=1/2*nstar1
        v3 (VertexType): resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)],
coeff=1/2*phi2_2
        v4 (SourceType): resp=[('nt', 1), ('nt', 1)], coeff=1/2*nstar1
        v5 (VertexType): resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)],
coeff=1/2*phi2_1
    Class 52 (multiplicity 16): diagrams ['1L-689', '1L-690', '1L-691', '1L-692',
'1L-693', '1L-694', '1L-695', '1L-696', '1L-697', '1L-698', '1L-699', '1L-700',
'1L-701', '1L-702', '1L-703', '1L-704']
        External legs: leaf 0+dn1, leaf 1+dn2
        Edges: 2+3 [G_hat[0,3]], 2+5 [G_hat[0,3]], 3+0 [G_hat[1,0]], 4+3
[G_hat[0,3]], 4+5 [G_hat[0,3]], 5+1 [G_hat[1,1]]
        v2 (SourceType): resp=[('nt', 1), ('nt', 1)], coeff=1/2*nstar1
        v3 (VertexType): resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)],
coeff=1/2*phi2_2
        v4 (SourceType): resp=[('nt', 1), ('nt', 1)], coeff=1/2*nstar1
        v5 (VertexType): resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)],
coeff=1/2*phi2_2
    Class 53 (multiplicity 16): diagrams ['1L-705', '1L-706', '1L-707', '1L-708',
'1L-709', '1L-710', '1L-711', '1L-712', '1L-713', '1L-714', '1L-715', '1L-716',
'1L-717', '1L-718', '1L-719', '1L-720']
        External legs: leaf 0+dn1, leaf 1+dn2
        Edges: 2+3 [G_hat[0,2]], 2+5 [G_hat[0,2]], 3+0 [G_hat[0,0]], 4+3
[G_hat[1,2]], 4+5 [G_hat[1,2]], 5+1 [G_hat[0,1]]
        v2 (SourceType): resp=[('nt', 1), ('nt', 1)], coeff=1/2*nstar1
        v3 (VertexType): resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)],
coeff=1/2*phi2_1
        v4 (SourceType): resp=[('nt', 2), ('nt', 2)], coeff=1/2*nstar2
        v5 (VertexType): resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)],
coeff=1/2*phi2_1
    Class 54 (multiplicity 16): diagrams ['1L-721', '1L-722', '1L-723', '1L-724',
'1L-725', '1L-726', '1L-727', '1L-728', '1L-729', '1L-730', '1L-731', '1L-732',
'1L-733', '1L-734', '1L-735', '1L-736']
        External legs: leaf 0+dn1, leaf 1+dn2
        Edges: 2+3 [G_hat[0,2]], 2+5 [G_hat[0,3]], 3+0 [G_hat[0,0]], 4+3
[G_hat[1,2]], 4+5 [G_hat[1,3]], 5+1 [G_hat[1,1]]
        v2 (SourceType): resp=[('nt', 1), ('nt', 1)], coeff=1/2*nstar1

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    v3 (VertexType): resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)],
coeff=1/2*phi2_1
    v4 (SourceType): resp=[('nt', 2), ('nt', 2)], coeff=1/2*nstar2
    v5 (VertexType): resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)],
coeff=1/2*phi2_2
    Class 55 (multiplicity 16): diagrams ['1L-737', '1L-738', '1L-739', '1L-740',
'1L-741', '1L-742', '1L-743', '1L-744', '1L-745', '1L-746', '1L-747', '1L-748',
'1L-749', '1L-750', '1L-751', '1L-752']
        External legs: leaf 0+dn1, leaf 1+dn2
        Edges: 2+3 [G_hat[0,3]], 2+5 [G_hat[0,2]], 3+0 [G_hat[1,0]], 4+3
[G_hat[1,3]], 4+5 [G_hat[1,2]], 5+1 [G_hat[0,1]]
        v2 (SourceType): resp=[('nt', 1), ('nt', 1)], coeff=1/2*nstar1
        v3 (VertexType): resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)],
coeff=1/2*phi2_2
        v4 (SourceType): resp=[('nt', 2), ('nt', 2)], coeff=1/2*nstar2
        v5 (VertexType): resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)],
coeff=1/2*phi2_1
    Class 56 (multiplicity 16): diagrams ['1L-753', '1L-754', '1L-755', '1L-756',
'1L-757', '1L-758', '1L-759', '1L-760', '1L-761', '1L-762', '1L-763', '1L-764',
'1L-765', '1L-766', '1L-767', '1L-768']
        External legs: leaf 0+dn1, leaf 1+dn2
        Edges: 2+3 [G_hat[0,3]], 2+5 [G_hat[0,3]], 3+0 [G_hat[1,0]], 4+3
[G_hat[1,3]], 4+5 [G_hat[1,3]], 5+1 [G_hat[1,1]]
        v2 (SourceType): resp=[('nt', 1), ('nt', 1)], coeff=1/2*nstar1
        v3 (VertexType): resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)],
coeff=1/2*phi2_2
        v4 (SourceType): resp=[('nt', 2), ('nt', 2)], coeff=1/2*nstar2
        v5 (VertexType): resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)],
coeff=1/2*phi2_2
    Class 57 (multiplicity 16): diagrams ['1L-769', '1L-770', '1L-771', '1L-772',
'1L-773', '1L-774', '1L-775', '1L-776', '1L-777', '1L-778', '1L-779', '1L-780',
'1L-781', '1L-782', '1L-783', '1L-784']
        External legs: leaf 0+dn1, leaf 1+dn2
        Edges: 2+3 [G_hat[1,2]], 2+5 [G_hat[1,2]], 3+0 [G_hat[0,0]], 4+3
[G_hat[0,2]], 4+5 [G_hat[0,2]], 5+1 [G_hat[0,1]]
        v2 (SourceType): resp=[('nt', 2), ('nt', 2)], coeff=1/2*nstar2
        v3 (VertexType): resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)],
coeff=1/2*phi2_1
        v4 (SourceType): resp=[('nt', 1), ('nt', 1)], coeff=1/2*nstar1
        v5 (VertexType): resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)],
coeff=1/2*phi2_1
    Class 58 (multiplicity 16): diagrams ['1L-785', '1L-786', '1L-787', '1L-788',
'1L-789', '1L-790', '1L-791', '1L-792', '1L-793', '1L-794', '1L-795', '1L-796',
'1L-797', '1L-798', '1L-799', '1L-800']
        External legs: leaf 0+dn1, leaf 1+dn2
        Edges: 2+3 [G_hat[1,2]], 2+5 [G_hat[1,3]], 3+0 [G_hat[0,0]], 4+3
[G_hat[0,2]], 4+5 [G_hat[0,3]], 5+1 [G_hat[1,1]]
        v2 (SourceType): resp=[('nt', 2), ('nt', 2)], coeff=1/2*nstar2

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    v3 (VertexType): resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)],
coeff=1/2*phi2_1
    v4 (SourceType): resp=[('nt', 1), ('nt', 1)], coeff=1/2*nstar1
    v5 (VertexType): resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)],
coeff=1/2*phi2_2
    Class 59 (multiplicity 16): diagrams ['1L-801', '1L-802', '1L-803', '1L-804',
'1L-805', '1L-806', '1L-807', '1L-808', '1L-809', '1L-810', '1L-811', '1L-812',
'1L-813', '1L-814', '1L-815', '1L-816']
        External legs: leaf 0+dn1, leaf 1+dn2
        Edges: 2+3 [G_hat[1,3]], 2+5 [G_hat[1,2]], 3+0 [G_hat[1,0]], 4+3
[G_hat[0,3]], 4+5 [G_hat[0,2]], 5+1 [G_hat[0,1]]
        v2 (SourceType): resp=[('nt', 2), ('nt', 2)], coeff=1/2*nstar2
        v3 (VertexType): resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)],
coeff=1/2*phi2_2
        v4 (SourceType): resp=[('nt', 1), ('nt', 1)], coeff=1/2*nstar1
        v5 (VertexType): resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)],
coeff=1/2*phi2_1
    Class 60 (multiplicity 16): diagrams ['1L-817', '1L-818', '1L-819', '1L-820',
'1L-821', '1L-822', '1L-823', '1L-824', '1L-825', '1L-826', '1L-827', '1L-828',
'1L-829', '1L-830', '1L-831', '1L-832']
        External legs: leaf 0+dn1, leaf 1+dn2
        Edges: 2+3 [G_hat[1,3]], 2+5 [G_hat[1,3]], 3+0 [G_hat[1,0]], 4+3
[G_hat[0,3]], 4+5 [G_hat[0,3]], 5+1 [G_hat[1,1]]
        v2 (SourceType): resp=[('nt', 2), ('nt', 2)], coeff=1/2*nstar2
        v3 (VertexType): resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)],
coeff=1/2*phi2_2
        v4 (SourceType): resp=[('nt', 1), ('nt', 1)], coeff=1/2*nstar1
        v5 (VertexType): resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)],
coeff=1/2*phi2_2
    Class 61 (multiplicity 16): diagrams ['1L-833', '1L-834', '1L-835', '1L-836',
'1L-837', '1L-838', '1L-839', '1L-840', '1L-841', '1L-842', '1L-843', '1L-844',
'1L-845', '1L-846', '1L-847', '1L-848']
        External legs: leaf 0+dn1, leaf 1+dn2
        Edges: 2+3 [G_hat[1,2]], 2+5 [G_hat[1,2]], 3+0 [G_hat[0,0]], 4+3
[G_hat[1,2]], 4+5 [G_hat[1,2]], 5+1 [G_hat[0,1]]
        v2 (SourceType): resp=[('nt', 2), ('nt', 2)], coeff=1/2*nstar2
        v3 (VertexType): resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)],
coeff=1/2*phi2_1
        v4 (SourceType): resp=[('nt', 2), ('nt', 2)], coeff=1/2*nstar2
        v5 (VertexType): resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)],
coeff=1/2*phi2_1
    Class 62 (multiplicity 16): diagrams ['1L-849', '1L-850', '1L-851', '1L-852',
'1L-853', '1L-854', '1L-855', '1L-856', '1L-857', '1L-858', '1L-859', '1L-860',
'1L-861', '1L-862', '1L-863', '1L-864']
        External legs: leaf 0+dn1, leaf 1+dn2
        Edges: 2+3 [G_hat[1,2]], 2+5 [G_hat[1,3]], 3+0 [G_hat[0,0]], 4+3
[G_hat[1,2]], 4+5 [G_hat[1,3]], 5+1 [G_hat[1,1]]
        v2 (SourceType): resp=[('nt', 2), ('nt', 2)], coeff=1/2*nstar2

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    v3 (VertexType): resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)],
coeff=1/2*phi2_1
    v4 (SourceType): resp=[('nt', 2), ('nt', 2)], coeff=1/2*nstar2
    v5 (VertexType): resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)],
coeff=1/2*phi2_2
    Class 63 (multiplicity 16): diagrams ['1L-865', '1L-866', '1L-867', '1L-868',
'1L-869', '1L-870', '1L-871', '1L-872', '1L-873', '1L-874', '1L-875', '1L-876',
'1L-877', '1L-878', '1L-879', '1L-880']
    External legs: leaf 0+dn1, leaf 1+dn2
    Edges: 2+3 [G_hat[1,3]], 2+5 [G_hat[1,2]], 3+0 [G_hat[1,0]], 4+3
[G_hat[1,3]], 4+5 [G_hat[1,2]], 5+1 [G_hat[0,1]]
    v2 (SourceType): resp=[('nt', 2), ('nt', 2)], coeff=1/2*nstar2
    v3 (VertexType): resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)],
coeff=1/2*phi2_2
    v4 (SourceType): resp=[('nt', 2), ('nt', 2)], coeff=1/2*nstar2
    v5 (VertexType): resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)],
coeff=1/2*phi2_1
    Class 64 (multiplicity 16): diagrams ['1L-881', '1L-882', '1L-883', '1L-884',
'1L-885', '1L-886', '1L-887', '1L-888', '1L-889', '1L-890', '1L-891', '1L-892',
'1L-893', '1L-894', '1L-895', '1L-896']
    External legs: leaf 0+dn1, leaf 1+dn2
    Edges: 2+3 [G_hat[1,3]], 2+5 [G_hat[1,3]], 3+0 [G_hat[1,0]], 4+3
[G_hat[1,3]], 4+5 [G_hat[1,3]], 5+1 [G_hat[1,1]]
    v2 (SourceType): resp=[('nt', 2), ('nt', 2)], coeff=1/2*nstar2
    v3 (VertexType): resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)],
coeff=1/2*phi2_2
    v4 (SourceType): resp=[('nt', 2), ('nt', 2)], coeff=1/2*nstar2
    v5 (VertexType): resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)],
coeff=1/2*phi2_2
    Class 65 (multiplicity 16): diagrams ['1L-897', '1L-898', '1L-899', '1L-900',
'1L-901', '1L-902', '1L-903', '1L-904', '1L-905', '1L-906', '1L-907', '1L-908',
'1L-909', '1L-910', '1L-911', '1L-912']
    External legs: leaf 0+dn2, leaf 1+dn1
    Edges: 2+3 [G_hat[0,2]], 2+5 [G_hat[0,2]], 3+0 [G_hat[0,1]], 4+3
[G_hat[0,2]], 4+5 [G_hat[0,2]], 5+1 [G_hat[0,0]]
    v2 (SourceType): resp=[('nt', 1), ('nt', 1)], coeff=1/2*nstar1
    v3 (VertexType): resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)],
coeff=1/2*phi2_1
    v4 (SourceType): resp=[('nt', 1), ('nt', 1)], coeff=1/2*nstar1
    v5 (VertexType): resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)],
coeff=1/2*phi2_1
    Class 66 (multiplicity 16): diagrams ['1L-913', '1L-914', '1L-915', '1L-916',
'1L-917', '1L-918', '1L-919', '1L-920', '1L-921', '1L-922', '1L-923', '1L-924',
'1L-925', '1L-926', '1L-927', '1L-928']
    External legs: leaf 0+dn2, leaf 1+dn1
    Edges: 2+3 [G_hat[0,2]], 2+5 [G_hat[0,3]], 3+0 [G_hat[0,1]], 4+3
[G_hat[0,2]], 4+5 [G_hat[0,3]], 5+1 [G_hat[1,0]]
    v2 (SourceType): resp=[('nt', 1), ('nt', 1)], coeff=1/2*nstar1

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    v3 (VertexType): resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)],
coeff=1/2*phi2_1
    v4 (SourceType): resp=[('nt', 1), ('nt', 1)], coeff=1/2*nstar1
    v5 (VertexType): resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)],
coeff=1/2*phi2_2
    Class 67 (multiplicity 16): diagrams ['1L-929', '1L-930', '1L-931', '1L-932',
'1L-933', '1L-934', '1L-935', '1L-936', '1L-937', '1L-938', '1L-939', '1L-940',
'1L-941', '1L-942', '1L-943', '1L-944']
    External legs: leaf 0+dn2, leaf 1+dn1
    Edges: 2+3 [G_hat[0,3]], 2+5 [G_hat[0,2]], 3+0 [G_hat[1,1]], 4+3
[G_hat[0,3]], 4+5 [G_hat[0,2]], 5+1 [G_hat[0,0]]
    v2 (SourceType): resp=[('nt', 1), ('nt', 1)], coeff=1/2*nstar1
    v3 (VertexType): resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)],
coeff=1/2*phi2_2
    v4 (SourceType): resp=[('nt', 1), ('nt', 1)], coeff=1/2*nstar1
    v5 (VertexType): resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)],
coeff=1/2*phi2_1
    Class 68 (multiplicity 16): diagrams ['1L-945', '1L-946', '1L-947', '1L-948',
'1L-949', '1L-950', '1L-951', '1L-952', '1L-953', '1L-954', '1L-955', '1L-956',
'1L-957', '1L-958', '1L-959', '1L-960']
    External legs: leaf 0+dn2, leaf 1+dn1
    Edges: 2+3 [G_hat[0,3]], 2+5 [G_hat[0,3]], 3+0 [G_hat[1,1]], 4+3
[G_hat[0,3]], 4+5 [G_hat[0,3]], 5+1 [G_hat[1,0]]
    v2 (SourceType): resp=[('nt', 1), ('nt', 1)], coeff=1/2*nstar1
    v3 (VertexType): resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)],
coeff=1/2*phi2_2
    v4 (SourceType): resp=[('nt', 1), ('nt', 1)], coeff=1/2*nstar1
    v5 (VertexType): resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)],
coeff=1/2*phi2_2
    Class 69 (multiplicity 16): diagrams ['1L-961', '1L-962', '1L-963', '1L-964',
'1L-965', '1L-966', '1L-967', '1L-968', '1L-969', '1L-970', '1L-971', '1L-972',
'1L-973', '1L-974', '1L-975', '1L-976']
    External legs: leaf 0+dn2, leaf 1+dn1
    Edges: 2+3 [G_hat[0,2]], 2+5 [G_hat[0,2]], 3+0 [G_hat[0,1]], 4+3
[G_hat[1,2]], 4+5 [G_hat[1,2]], 5+1 [G_hat[0,0]]
    v2 (SourceType): resp=[('nt', 1), ('nt', 1)], coeff=1/2*nstar1
    v3 (VertexType): resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)],
coeff=1/2*phi2_1
    v4 (SourceType): resp=[('nt', 2), ('nt', 2)], coeff=1/2*nstar2
    v5 (VertexType): resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)],
coeff=1/2*phi2_1
    Class 70 (multiplicity 16): diagrams ['1L-977', '1L-978', '1L-979', '1L-980',
'1L-981', '1L-982', '1L-983', '1L-984', '1L-985', '1L-986', '1L-987', '1L-988',
'1L-989', '1L-990', '1L-991', '1L-992']
    External legs: leaf 0+dn2, leaf 1+dn1
    Edges: 2+3 [G_hat[0,2]], 2+5 [G_hat[0,3]], 3+0 [G_hat[0,1]], 4+3
[G_hat[1,2]], 4+5 [G_hat[1,3]], 5+1 [G_hat[1,0]]
    v2 (SourceType): resp=[('nt', 1), ('nt', 1)], coeff=1/2*nstar1

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    v3 (VertexType): resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)],
coeff=1/2*phi2_1
    v4 (SourceType): resp=[('nt', 2), ('nt', 2)], coeff=1/2*nstar2
    v5 (VertexType): resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)],
coeff=1/2*phi2_2
    Class 71 (multiplicity 16): diagrams ['1L-993', '1L-994', '1L-995', '1L-996',
'1L-997', '1L-998', '1L-999', '1L-1000', '1L-1001', '1L-1002', '1L-1003',
'1L-1004', '1L-1005', '1L-1006', '1L-1007', '1L-1008']
        External legs: leaf 0+dn2, leaf 1+dn1
        Edges: 2+3 [G_hat[0,3]], 2+5 [G_hat[0,2]], 3+0 [G_hat[1,1]], 4+3
[G_hat[1,3]], 4+5 [G_hat[1,2]], 5+1 [G_hat[0,0]]
        v2 (SourceType): resp=[('nt', 1), ('nt', 1)], coeff=1/2*nstar1
        v3 (VertexType): resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)],
coeff=1/2*phi2_2
        v4 (SourceType): resp=[('nt', 2), ('nt', 2)], coeff=1/2*nstar2
        v5 (VertexType): resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)],
coeff=1/2*phi2_1
    Class 72 (multiplicity 16): diagrams ['1L-1009', '1L-1010', '1L-1011',
'1L-1012', '1L-1013', '1L-1014', '1L-1015', '1L-1016', '1L-1017', '1L-1018',
'1L-1019', '1L-1020', '1L-1021', '1L-1022', '1L-1023', '1L-1024']
        External legs: leaf 0+dn2, leaf 1+dn1
        Edges: 2+3 [G_hat[0,3]], 2+5 [G_hat[0,3]], 3+0 [G_hat[1,1]], 4+3
[G_hat[1,3]], 4+5 [G_hat[1,3]], 5+1 [G_hat[1,0]]
        v2 (SourceType): resp=[('nt', 1), ('nt', 1)], coeff=1/2*nstar1
        v3 (VertexType): resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)],
coeff=1/2*phi2_2
        v4 (SourceType): resp=[('nt', 2), ('nt', 2)], coeff=1/2*nstar2
        v5 (VertexType): resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)],
coeff=1/2*phi2_2
    Class 73 (multiplicity 16): diagrams ['1L-1025', '1L-1026', '1L-1027',
'1L-1028', '1L-1029', '1L-1030', '1L-1031', '1L-1032', '1L-1033', '1L-1034',
'1L-1035', '1L-1036', '1L-1037', '1L-1038', '1L-1039', '1L-1040']
        External legs: leaf 0+dn2, leaf 1+dn1
        Edges: 2+3 [G_hat[1,2]], 2+5 [G_hat[1,2]], 3+0 [G_hat[0,1]], 4+3
[G_hat[0,2]], 4+5 [G_hat[0,2]], 5+1 [G_hat[0,0]]
        v2 (SourceType): resp=[('nt', 2), ('nt', 2)], coeff=1/2*nstar2
        v3 (VertexType): resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)],
coeff=1/2*phi2_1
        v4 (SourceType): resp=[('nt', 1), ('nt', 1)], coeff=1/2*nstar1
        v5 (VertexType): resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)],
coeff=1/2*phi2_1
    Class 74 (multiplicity 16): diagrams ['1L-1041', '1L-1042', '1L-1043',
'1L-1044', '1L-1045', '1L-1046', '1L-1047', '1L-1048', '1L-1049', '1L-1050',
'1L-1051', '1L-1052', '1L-1053', '1L-1054', '1L-1055', '1L-1056']
        External legs: leaf 0+dn2, leaf 1+dn1
        Edges: 2+3 [G_hat[1,2]], 2+5 [G_hat[1,3]], 3+0 [G_hat[0,1]], 4+3
[G_hat[0,2]], 4+5 [G_hat[0,3]], 5+1 [G_hat[1,0]]
        v2 (SourceType): resp=[('nt', 2), ('nt', 2)], coeff=1/2*nstar2

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    v3 (VertexType): resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)],
coeff=1/2*phi2_1
    v4 (SourceType): resp=[('nt', 1), ('nt', 1)], coeff=1/2*nstar1
    v5 (VertexType): resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)],
coeff=1/2*phi2_2
    Class 75 (multiplicity 16): diagrams ['1L-1057', '1L-1058', '1L-1059',
'1L-1060', '1L-1061', '1L-1062', '1L-1063', '1L-1064', '1L-1065', '1L-1066',
'1L-1067', '1L-1068', '1L-1069', '1L-1070', '1L-1071', '1L-1072']
        External legs: leaf 0+dn2, leaf 1+dn1
        Edges: 2+3 [G_hat[1,3]], 2+5 [G_hat[1,2]], 3+0 [G_hat[1,1]], 4+3
[G_hat[0,3]], 4+5 [G_hat[0,2]], 5+1 [G_hat[0,0]]
        v2 (SourceType): resp=[('nt', 2), ('nt', 2)], coeff=1/2*nstar2
        v3 (VertexType): resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)],
coeff=1/2*phi2_2
        v4 (SourceType): resp=[('nt', 1), ('nt', 1)], coeff=1/2*nstar1
        v5 (VertexType): resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)],
coeff=1/2*phi2_1
    Class 76 (multiplicity 16): diagrams ['1L-1073', '1L-1074', '1L-1075',
'1L-1076', '1L-1077', '1L-1078', '1L-1079', '1L-1080', '1L-1081', '1L-1082',
'1L-1083', '1L-1084', '1L-1085', '1L-1086', '1L-1087', '1L-1088']
        External legs: leaf 0+dn2, leaf 1+dn1
        Edges: 2+3 [G_hat[1,3]], 2+5 [G_hat[1,3]], 3+0 [G_hat[1,1]], 4+3
[G_hat[0,3]], 4+5 [G_hat[0,3]], 5+1 [G_hat[1,0]]
        v2 (SourceType): resp=[('nt', 2), ('nt', 2)], coeff=1/2*nstar2
        v3 (VertexType): resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)],
coeff=1/2*phi2_2
        v4 (SourceType): resp=[('nt', 1), ('nt', 1)], coeff=1/2*nstar1
        v5 (VertexType): resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)],
coeff=1/2*phi2_2
    Class 77 (multiplicity 16): diagrams ['1L-1089', '1L-1090', '1L-1091',
'1L-1092', '1L-1093', '1L-1094', '1L-1095', '1L-1096', '1L-1097', '1L-1098',
'1L-1099', '1L-1100', '1L-1101', '1L-1102', '1L-1103', '1L-1104']
        External legs: leaf 0+dn2, leaf 1+dn1
        Edges: 2+3 [G_hat[1,2]], 2+5 [G_hat[1,2]], 3+0 [G_hat[0,1]], 4+3
[G_hat[1,2]], 4+5 [G_hat[1,2]], 5+1 [G_hat[0,0]]
        v2 (SourceType): resp=[('nt', 2), ('nt', 2)], coeff=1/2*nstar2
        v3 (VertexType): resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)],
coeff=1/2*phi2_1
        v4 (SourceType): resp=[('nt', 2), ('nt', 2)], coeff=1/2*nstar2
        v5 (VertexType): resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)],
coeff=1/2*phi2_1
    Class 78 (multiplicity 16): diagrams ['1L-1105', '1L-1106', '1L-1107',
'1L-1108', '1L-1109', '1L-1110', '1L-1111', '1L-1112', '1L-1113', '1L-1114',
'1L-1115', '1L-1116', '1L-1117', '1L-1118', '1L-1119', '1L-1120']
        External legs: leaf 0+dn2, leaf 1+dn1
        Edges: 2+3 [G_hat[1,2]], 2+5 [G_hat[1,3]], 3+0 [G_hat[0,1]], 4+3
[G_hat[1,2]], 4+5 [G_hat[1,3]], 5+1 [G_hat[1,0]]
        v2 (SourceType): resp=[('nt', 2), ('nt', 2)], coeff=1/2*nstar2

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    v3 (VertexType): resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)],
coeff=1/2*phi2_1
    v4 (SourceType): resp=[('nt', 2), ('nt', 2)], coeff=1/2*nstar2
    v5 (VertexType): resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)],
coeff=1/2*phi2_2
    Class 79 (multiplicity 16): diagrams ['1L-1121', '1L-1122', '1L-1123',
'1L-1124', '1L-1125', '1L-1126', '1L-1127', '1L-1128', '1L-1129', '1L-1130',
'1L-1131', '1L-1132', '1L-1133', '1L-1134', '1L-1135', '1L-1136']
        External legs: leaf 0+dn2, leaf 1+dn1
        Edges: 2+3 [G_hat[1,3]], 2+5 [G_hat[1,2]], 3+0 [G_hat[1,1]], 4+3
[G_hat[1,3]], 4+5 [G_hat[1,2]], 5+1 [G_hat[0,0]]
        v2 (SourceType): resp=[('nt', 2), ('nt', 2)], coeff=1/2*nstar2
        v3 (VertexType): resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)],
coeff=1/2*phi2_2
        v4 (SourceType): resp=[('nt', 2), ('nt', 2)], coeff=1/2*nstar2
        v5 (VertexType): resp=[('nt', 1)], phys=[('dv', 1), ('dv', 1)],
coeff=1/2*phi2_1
    Class 80 (multiplicity 16): diagrams ['1L-1137', '1L-1138', '1L-1139',
'1L-1140', '1L-1141', '1L-1142', '1L-1143', '1L-1144', '1L-1145', '1L-1146',
'1L-1147', '1L-1148', '1L-1149', '1L-1150', '1L-1151', '1L-1152']
        External legs: leaf 0+dn2, leaf 1+dn1
        Edges: 2+3 [G_hat[1,3]], 2+5 [G_hat[1,3]], 3+0 [G_hat[1,1]], 4+3
[G_hat[1,3]], 4+5 [G_hat[1,3]], 5+1 [G_hat[1,0]]
        v2 (SourceType): resp=[('nt', 2), ('nt', 2)], coeff=1/2*nstar2
        v3 (VertexType): resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)],
coeff=1/2*phi2_2
        v4 (SourceType): resp=[('nt', 2), ('nt', 2)], coeff=1/2*nstar2
        v5 (VertexType): resp=[('nt', 2)], phys=[('dv', 2), ('dv', 2)],
coeff=1/2*phi2_2

```